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Vitor Balestro
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Convexity from the Geometric Point of View: Exercises and Solutions

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Ralph Teixeira

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Vitor Balestro
Institute of Mathematics and Statistics
Fluminense Federal University
Niterói, Rio de Janeiro, Brazil

Horst Martini
Faculty of Mathematics
Chemnitz University of Technology
09107 Chemnitz, Germany

Ralph Teixeira
Institute of Mathematics and Statistics
Fluminense Federal University
Niterói, Rio de Janeiro, Brazil

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This book is dedicated to our wives Mariana, Angelika, and Isabel.

Preface

The present book collects the *solutions to all 347 exercises* contained in our recently published monograph [29] entitled *Convexity from the Geometric Point of View*. One should mention that first we did not plan to add exercises to [29] (see also below, at the end of the preface). But later we were asked by the publisher whether we could add exercises, and we agreed. So we wrote a large variety of exercises, and we were putting them and their solutions into the internet, as electronic addendum to [29]. Again later we observed that the length of this collecting text has book size, and so we decided to publish it in the same book form as the original monograph [29], and this idea was immediately also supported by Birkhäuser. It turned out that an additional reason came into the game, namely, the observation that there are analogous examples of "book pairs", even in the field of *convexity* studied by us in [29]. Namely, we mean here the convexity book [166] of Leonard and Lewis and its *solutions manual* [167], the latter presenting all solutions to the odd-numbered problems posed in [166]. Thus, [166] and [167] is indeed such a book pair. And there are more examples. Similar in nature, the famous geometry monograph [46] of Berger (containing also sections dedicated to convexity) is completed by the book [47] which contains the solutions of those exercises from [46] which are marked there with a star.

Besides discussing the solutions themselves and this book here, we will (in great parts of this preface) also mention some facts about our book [29], which contains (as already said) *all* the 347 exercises whose solutions we present here. Depending on the contents of [29], our book here consists of 11 chapters which contain all the analogously numbered exercises from [29]. Since Chapter 12 and the appendix of [29] have no exercises, analogues of these two parts of [29] do not occur here. Each of the 11 chapters below has also a final part for *further reading*, where we give background literature to selected notions and tools playing a role in the exercises and/or solutions of that chapter.

The idea of writing the book *Convexity from the Geometric Point of View* [29] (to which we refer here from now on also as the "main book") emerged in 2019, when we observed that someone interested in entering the mathematical field of *convexity* (also called *convex geometry*) would have to dig through a large and scattered amount of literature in order to acquire in an appropriate time a serious, broad, and general background knowledge. To the extent of our knowledge at that time, the main references from the field were for *beginners* either too dense or too deep, or they specifically aimed at some *subfield* of the already extensively grown theory; for these two aspects we mention, as examples, the two leading monographs [228] of Schneider and [117] of Gruber, as well as [24] of Artstein-Avidan, Giannopoulos, and V. Milman (on *asymptotic geometric analysis*), and [188] of Martini, Montejano, and Oliveros (only on *bodies of constant width*). Of course, there are many further important monographs in the field that one could mention here; we give only examples. Despite the fact that all of them are essential books, we still missed a text specifically written for newcomers, with a relatively broad selection of topics treated in great detail. The nice book [142] by Hug and Weil (which we became aware of when we were already writing our book [29]) is an important step to fill this gap. However, we assessed, at that time, that our approach showed also differences in style, still

justifying the publication of [29]. And the final contents of our book [29] (like also their structure) were not decided until we were fairly advanced in writing it.

Now we come to the present book here. As already said, its *Chapters 1 to 11* correspond in the same sequence with Chapters 1 to 11 of the main book [29], and Chapter 12 and the Appendix from there do not occur in the present book. In each chapter here, the *exercises* from [29] are repeated in the same sequence, and the *solution* to each exercise follows immediately after it. Every chapter starts with a *colored plate*, taken analogously from the main book (to which the references in the plates also refer). Before we start with the exercises, on the first page of each chapter a collection of keywords (corresponding with the contents of all its exercises) is given, which can be conveniently used for further reference in related encyclopedic works or in the internet. And after the exercises and solutions of each chapter, selected hints and references for *further reading* are given.

We took the decision to start with a chapter on *convex analysis* (see Chapter 1), which might be somehow strange for a geometry book. The original reason for this decision was that we wanted to have the corresponding "machinery" *before* introducing typical convexity notions like, e.g., *support function*. When we observed that we can succeed with covering everything that we need on *convex functions* in a relatively small chapter at the beginning, the above-mentioned decision was definitively taken. The exercises for this first chapter contain simple verifications of *properties of convex functions, operations of functions* (such as, e.g., the Asplund product), and related *inequalities*. Here are examples of more detailed notions occurring in (the exercises of) Chapter 1: *globally Lipschitz functions, lower semi-continuity, epigraphs, discrete Jensen inequality, sublinear functions, subdifferentials, indicator and conjugate functions, support functions, polar functions, Minkowski sums*, and the *Prékopa-Leindler inequality*.

In Chapter 2, we gather the "basic" theory on *convex sets*, comprising the elementary definitions and the "toolkit" for their investigation and combinations. These are separation notions, combinatorial results, support and gauge functions, the Hausdorff distance, and others. In Chapters 1 and 2 (and also in later chapters) a lot of results were somehow subsequently entered, namely, when we realized that they are needed in other parts of the whole text. Chapter 2 is the largest chapter of the main book and, correspondingly, it has the largest collection of exercises. In these exercises we discuss *elementary properties of convex sets* (and, in particular, of convex cones), results concerning the *Hausdorff metric* and exercises involving *symmetrization processes*. Notions and topics described in this chapter in a more detailed manner are: *convex hull* and *relative interior* etc., *extreme points, Helly-type theorems, valuations, Hausdorff distance, (constant) width, separation, polar sets,* and *Steiner* and *Minkowski symmetrization*.

At some moment we realized that the topic of *convex polytopes* should have a chapter for itself (see Chapter 3). In the main text of [29] and the corresponding exercises, we introduce and investigate the *face lattice of convex polytopes*, and the reader is also invited to study the *Ehrhart polynomial*. Examples of further notions, considered in the exercises of this chapter, are: *Minkowski addition of polytopes, diameters and projections of polytopes, lattice polytopes, Steiner symmetrals of polytopes*, and geometric properties as well as applications of *simplices*.

The question of understanding the equivalence between different ways of defining the *surface area of a convex body* gave the original inspiration to write Chapter 4. There are three more or less canonical ways in which the (total) surface area of the boundary of a convex body is defined: using *approximations by polytopes*, considering *Hausdorff measures*, or via *Minkowski contents*. As a matter of fact, we could not find an easy proof of this equivalence (i.e., a proof that does not require to dig a little too deep into geometric measure theory). In this chapter, it is proved that the first two ways are equivalent. The fact that they are equivalent to the third one is proved later, in Chapter 6. Moreover, in Chapter 4 we discuss ways to endow the boundary of a convex body with a *Borel measure* (this is a necessary step to define later the surface area measure of a convex body and, in more generality, the *mixed surface area measures*). In the exercises of this chapter, also these subjects and some notions related to *probability spaces* are considered, such as *concentration functions* and the *Chebyshev and*

Markov inequalities. Further on, interesting phenomena regarding *volume* and *surface area, centroids of convex bodies, centrally symmetric convex bodies, circum-* and *inradii, polar bodies*, and *Brunn's concavity principle* are discussed.

All this context led us also to write an *Appendix on measure theory* for the main book (the last part of [29]). Despite knowing that there are various quite good references on this topic, we planned to introduce the ideas and notation in precisely the same way we wanted to use them in [29]. Also, *Hausdorff measures* are not usually treated in an introductory course on measure and integration theory (at least not in Brazil, where two of the three authors come from), a further reason why we wanted to have an Appendix on that subject. Also we do not think that measure and integration theory are as ubiquitous as, e.g., linear algebra (again meant from the Brazilian perspective). However, since the Appendix of [29] contains no exercises, it is only shortly mentioned here.

At large, we wrote Chapter 5 aiming to introduce *inequalities between geometric quantities* and giving geometrically flavored proofs for them. We knew that some of the presented inequalities could be obtained as easy consequences of more sophisticated results (e.g., the *isoperimetric inequality* is a direct consequence of the *1st Minkowski inequality for mixed volumes*). Nevertheless, we believe that the reader might benefit from this more geometric approach (instead of going, in an opposed manner, straightforwardly to more sophisticated tools, loosing then the intuitive, geometric view). The exercises for this chapter also include *variations and extensions of the presented geometric inequalities*. Moreover, we introduce the *ℓ-norm*, the *Steiner point*, and the *Santaló point* of a convex body. Some *functional inequalities* are also dealt with, and further notions studied or used explicitly are *mean width, convolution bodies, Anderson's theorem, the isodiametric inequality*, and *extremal cross-polytopes contained in convex bodies*.

In the main book, the "core" of the *Brunn-Minkowski inequality* is developed in Chapters 6, 7, and 8. In the first two of them, we present *mixed volumes* and *mixed surface area measures*, respectively. In these chapters, we give special attention to provide a detailed and formal construction of the concepts. Since the proof of the *Alexandrov-Fenchel inequality* is quite difficult, and because this inequality has a lot of important consequences, we decided to dedicate an entire chapter to it (Chapter 8). Some partial results concerning the *characterization of the equality case in the Alexandrov-Fenchel inequality* are also discussed (the complete characterization is a famous open problem). We mention some further notions and tools from the exercises of these three chapters. The exercises of Chapter 6 contain *properties of mixed volumes* and also some interesting related *inequalities*. Further keywords playing a concrete role there are: *quermassintegrals* and *intrinsic volumes, mean width* and *constant width, Borel measures* and applications, *Mahler's conjecture, Urysohn's inequality*, and the *Wills functional*. In the exercises of Chapter 7, we introduce notions and tools such as *mixed surface area measures, mixed bodies*, and the *Blaschke addition* of convex bodies. Additional keywords presented there are *Gauss map, Minkowski symmetrization, projection body*, the *Kneser-Süss inequality* involving Blaschke sums, and *affine surface area*. And the exercises of Chapter 8 (clearly based on the Alexandrov-Fenchel inequality) treat also the *Brunn-Minkowski inequalities for quermassintegrals*, extensions of the *isoperimetric inequality*, and generalizations of the *Minkowski inequalities*.

Chapters 9 and 10 discuss *affine aspects of convex geometry*. In Chapter 9, we mainly study *positions of convex bodies* which have geometric meanings (such as the *John position* and *positions minimizing quermassintegrals*). The results proved in Section 9.3 of the main book are consequences of the more general results of Section 9.4, but the more geometrically flavored approach of Section 9.3 justifies this certain redundancy. The exercises part of Chapter 9 contains some consequences of the *geometric Brascamp-Lieb inequality*, and also a short introduction to the *Orlicz projection body* is given. Further notions basically treated there are: *Minkowski asymmetry, positions of John, of minimal mean width, and of minimal surface area, isotropic position*, and the *reverse isoperimetric inequality*.

In the exercises of Chapter 10, various properties of *zonotopes, zonoids, projection bodies*, and *centroid bodies* as *affine constructions* are discussed. We present also some *inequalities* involving these notions and mention some further notions studied there in the exercises: *mixed projection bodies*

and *mixed projection maps, Hausdorff metric, ellipsoids*, certain *positions* and *valuations, Blaschke bodies*, and *Brunn-Minkowski type inequalities for projection bodies*.

Chapter 11 is not that "structured" as the other chapters are, meaning that its sections are more independent of each other. Therefore we first describe the main contents of the sections in the main book, to give then basic keywords for the subjects treated in their exercises! Section 11.1 presents the concept of *dual mixed volumes* which, in some sense, is indeed a dual notion to mixed volumes. The L_p *mixed volumes* discussed in Section 11.2 are a further extension of mixed volumes. Both notions were introduced by Lutwak, and they inspired a lot of later and recent research. The aim of Section 11.3 is to give a detailed proof of *Hadwiger's theorem characterizing valuations*, presenting also certain properties of valuations. This section can be seen as an introduction to the vast and deep theory of *valuations on convex bodies*. The *Bourgain-Milman inequality* is stated and proved in Section 11.5, also as being introductory to the neighboring field named *asymptotic geometric analysis*. This choice reflects the fact that we were also influenced by the outstanding monographs [24] and [25] of Artstein-Avidan, Giannopoulos, and V. Milman. The results of Section 11.4 were originally introduced as auxiliary results in the proof of the Bourgain-Milman inequality, but the topic justifies a section for itself. Each of these sections has overlaps with the exercises part of Chapter 11. Important keywords regarding these exercises are: the L_p *Petty projection inequality* that we highlight here, the *Minkowski valuation property* (and, e.g., mixed surface area measures having it), the *difference body map*, L_p *moment bodies, rotation means, polytope dissections*, and the essential notions of *star bodies* and *radial sum* needed for dualizing concepts, to get for instance *dual mixed volumes* and the *dual Brunn-Minkowski inequality*. Finally, we highlight the *Orlicz projection body* and the *Wills functional*.

The last chapter of the main book (Chapter 12) is dedicated to *steps of the historical development of convexity* as a field of knowledge, and hence there are no exercises. Also various *relationships to and overlaps with other areas of mathematics* (such as discrete geometry, algebraic geometry, differential geometry, stochastic geometry, convex analysis, and other ones) are discussed there, but again without exercises. And since it is just auxiliary material for our presentations, the *Appendix* of the main book (on *measure and integration theory*) has no exercises, too.

Once more it is worth mentioning that our original plan for contents of the monograph [29] did not contain exercises. Adding them was suggested by the first editor of this book, Loretta Bartolini (to whom we address our thanks). Later the process was handled by Elizabeth Loew, who is also editing this present book (and to whom we also say thanks). In addition, we give our due acknowledgment to João Rodriguez Marcondes, who was our student at UFF. At the time we were writing [29], he was somehow the "first reader" of the material, providing important insights both on the theory and on the exercises. And finally we thank once more André Martini for producing the nice plates that one can find at the beginning of each chapter and before the references.

Niterói, Brazil — Vitor Balestro
Chemnitz, Germany — Horst Martini
Niterói, Brazil — Ralph Teixeira

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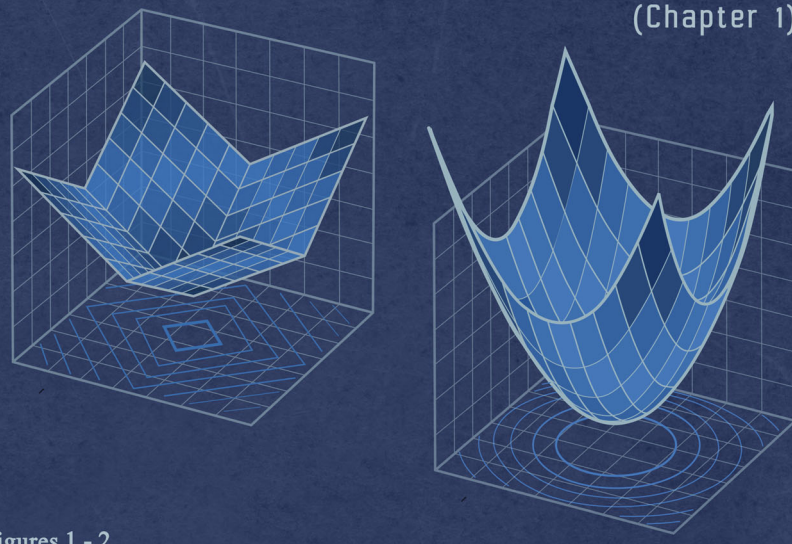
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PLATE 1

Convex functions

Two types of convex functions

(Chapter 1)



Figures 1 - 2

Multifocal ellipses as level curves of convex functions

(Fermat-Torricelli problem, see 2.9.1.6)

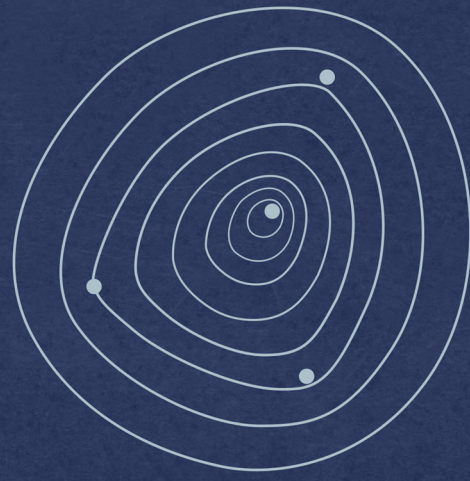


Figure 3

Convex functions ► (see Chapter 1 and 12.4.6)

Chapter 1
Convex Functions



"The solution of every problem is another problem."

— Johann Wolfgang von Goethe

In the exercises and solutions corresponding with Chapter 1 (here and in [29]), the following notions, tools, and topics are investigated or used: *continuous and noncontinuous functions, upper and lower semi-continuous functions and hulls, level sets and sublevel sets, numerous special types of convex functions, epigraph, closure and interior, probability measure and space, discrete Jensen's inequality, effective domains, subdifferential, cyclically monotonic sets (Rockafellar's theorem), infimal convolution of functions, Minkowski sum and difference, proper functions, Asplund product, log-concave functions, Prékopa-Leindler inequality, Brunn-Minkowski inequality, support and polar functions, integrable functions, global minimum, difference body and central symmetral*, and the *Rogers-Shephard inequality*.

Plate 1:

Our Plate 1 here shows two known types of graphs of *convex functions*, namely, a piecewise linear and a strictly convex one (Figures 1 and 2). Figure 3 presents *multifocal* ellipses as level curves of such functions studied for a well-known location problem. The references given in Plate 1 (and this holds analogously for the other plates in this book here) refer to corresponding parts of our main book [29], namely, Chapter 1 and subsection 12.4.6.

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Exercises and Solutions

Let $D \subseteq \mathbb{R}^n$ be a *convex set*, i.e., a point set such that any line segment between two points from it is completely contained it. A function $f\colon D \to (-\infty, +\infty]$ is said to be *convex* if

$$f((1-\lambda)x + \lambda y) \leq (1-\lambda)f(x) + \lambda f(y)$$

for any $x, y \in D$ and $\lambda \in [0, 1]$.

Exercise 1.1. *Let $D \subseteq \mathbb{R}^n$ be a non-empty convex set, and assume that $f\colon D \to (-\infty, +\infty]$ is a convex function. Prove that the extension $\bar{f}\colon \mathbb{R}^n \to (-\infty, +\infty]$ of f given by*

$$\bar{f}(x) = \begin{cases} f(x) & \text{if } x \in D \\ +\infty & \text{if } x \in \mathbb{R}^n \setminus D \end{cases}$$

is convex.

Solution 1.1. We have to prove that

$$\bar{f}((1-\lambda)x + \lambda y) \leq (1-\lambda)\bar{f}(x) + \lambda \bar{f}(y)$$

for any $\lambda \in (0, 1)$ and $x, y \in \mathbb{R}^n$. Since $f = \bar{f}$ in D and f is convex in D, we already have that this holds for $x, y \in D$. If this is not the case (i.e., if one of the points is not in D), then the right-hand side of the above is $+\infty$, and hence the inequality holds trivially.

□

Exercise 1.2. *Give an example of a convex function $f\colon \mathbb{R} \to (-\infty, +\infty]$ that is not continuous when restricted to its effective domain.*

Remark 1.1. Remember that a convex function is always continuous in the *interior* of its effective domain.

Solution 1.2. Let $f\colon \mathbb{R} \to (-\infty, +\infty]$ be given as

$$f(x) = \begin{cases} 0 & \text{if } x < 0 \\ 1 & \text{if } x = 0 \\ +\infty & \text{if } x > 0 \end{cases}.$$

Then f is convex, and it is not continuous at $x = 0$.

□

Exercise 1.3. *Provide an example of a continuous convex function that is not globally Lipschitz.*

A function is said to be *globally Lipschitz* if it is Lipschitz in its entire domain.

Solution 1.3. Let $f\colon \mathbb{R} \to \mathbb{R}$ be simply given as $f(x) = x^2$. To verify that f is convex, we notice that for any $x, z \in \mathbb{R}$ and every $\lambda \in (0, 1)$ we have

$$\begin{aligned} f((1-\lambda)x + \lambda z) - (1-\lambda)f(x) - \lambda f(z) = \\ = (1-\lambda)^2 x^2 + 2xz\lambda(1-\lambda) + \lambda^2 z^2 - (1-\lambda)x^2 - \lambda z^2 = \\ = (1-\lambda)x((1-\lambda)x + z\lambda - x) + \lambda z((1-\lambda)x + \lambda z - z) = \\ = (1-\lambda)x\lambda(z-x) + \lambda z(1-\lambda)(x-z) = \\ = (1-\lambda)\lambda(z-x)(x-z) = -(1-\lambda)\lambda(z-x)^2 \leq 0. \end{aligned}$$

The fact that f is not Lipschitz in $\mathbb{R}$ comes immediately from the equality

$$|f(x) - f(z)| = |x^2 - z^2| = |x + z||x - z|.$$

In fact, for any $c \in \mathbb{R}$, choosing $x, z \in \mathbb{R}$ such that $|x + z| > c$, we get $|f(x) - f(z)| > c|x - z|$.

□

For what follows, we need a new definition. A function $f\colon \mathbb{R}^n \to [-\infty, +\infty)$ is said to be *upper semi-continuous* if the function $-f$ is lower semi-continuous. See the main book for the definition of *lower semi-continuity*.

Exercise 1.4. *Let $f\colon \mathbb{R}^n \to [-\infty, +\infty)$. Prove that the following statements are equivalent:*

(i) *f is upper semi-continuous.*

(ii) *For each $t \in \mathbb{R}$, the* upper level set

$$\{f \geq t\} := \{x \in \mathbb{R}^n : f(x) \geq t\}$$

is closed.

(iii) *For every $x \in \mathbb{R}^n$ and any $\varepsilon > 0$, there exists a neighborhood V of x such that $f(y) \leq f(x) + \varepsilon$ for any $y \in V$.*

(iv) *For each $x \in \mathbb{R}^n$ and every sequence $(x_k)_{k\in\mathbb{N}}$ converging to x, it holds that*

$$\limsup_{k\to\infty} f(x_k) \leq f(x).$$

Solution 1.4. We use Proposition 1.1.3 from the main book for the function $-f$. First, we have that $-f$ is lower semi-continuous if and only if $\{-f \leq t\}$ is closed for any $t \in \mathbb{R}$. Since we always have the equality $\{-f \leq t\} = \{f \geq -t\}$, we get that **(i)** and **(ii)** are equivalent.

Moving forward, the inequality $-f(y) \geq -f(x) - \varepsilon$ is equivalent to $f(y) \leq f(x) + \varepsilon$. By item **(c)** of Proposition 1.1.3 from the main book, we get that **(i)** and **(iii)** are equivalent.

Finally, the equivalence between **(i)** and **(iv)** follows from the fact that $\liminf_{k\to\infty} -f(x_k) \geq -f(x)$ if and only if $\limsup_{k\to\infty} f(x_k) \leq f(x)$. The proof is complete.

□

Exercise 1.5. *Let $f\colon \mathbb{R}^n \to (-\infty, +\infty]$ be a convex function. Prove that if there exists a number $c \in \mathbb{R}$ such that $f(x) \leq c$ for all $x \in \mathbb{R}^n$, then f is constant.*

Solution 1.5. We proceed by contradiction. Assume that $f(x) \leq c$ for every $x \in \mathbb{R}^n$ and that there exist points $x_0, y_0 \in \mathbb{R}^n$ such that $f(x_0) \neq f(y_0)$. Without loss of generality, we assume also that $f(y_0) > f(x_0)$. For an arbitrarily fixed number $t > 0$, let $z_t = y_0 + t(y_0 - x_0)$. A simple calculation shows that

$$y_0 = (1 - \lambda)x_0 + \lambda z_t$$

for $\lambda = \frac{1}{1+t}$. Thus, the convexity of f yields

$$f(y_0) = f((1 - \lambda)x_0 + \lambda z_t) \leq (1 - \lambda)f(x_0) + \lambda f(z_t) = \frac{t}{1 + t} f(x_0) + \frac{1}{1 + t} f(z_t).$$

We rewrite this inequality as

$$f(z_t) \geq (1+t)f(y_0) - tf(x_0) = t(f(y_0) - f(x_0)) + f(y_0).$$

Therefore, if we choose $t \in \mathbb{R}$ such that

$$t > \frac{c - f(y_0)}{f(y_0) - f(x_0)},$$

we obtain that $f(z_t) > c$, and this contradiction concludes the proof.

□

The *epigraph* of a function $f\colon \mathbb{R}^n \to [-\infty, +\infty]$ is the set $\operatorname{epi}(f) := \{(x,t) \in \mathbb{R}^{n+1} : t \geq f(x)\}$.

Exercise 1.6. *Let $f\colon \mathbb{R}^n \to [-\infty, +\infty]$. Prove that*

$$f(x) = \inf\left\{t \in \mathbb{R} : (x,t) \in \operatorname{epi}(f)\right\}$$

for each $x \in \mathbb{R}^n$, where we use that $\inf \emptyset = +\infty$ and $\inf(\mathbb{R}) = -\infty$. As a consequence, show that if $f, g\colon \mathbb{R}^n \to [-\infty, +\infty]$ are such that $\operatorname{epi}(f) = \operatorname{epi}(g)$, then $f = g$.

Remark 1.2. The original definition of epigraph was given for functions onto $(-\infty, +\infty]$, but this definition clearly extends without any modification to the case of functions onto $[-\infty, +\infty]$.

Solution 1.6. If $f(x) = +\infty$, then $(x,t) \notin \operatorname{epi}(f)$ for every $t \in \mathbb{R}$. Hence,

$$\inf\left\{t \in \mathbb{R} : (x,t) \in \operatorname{epi}(f)\right\} = \inf \emptyset = +\infty = f(x).$$

On the other hand, if $f(x) = -\infty$, then $(x,t) \in \operatorname{epi}(f)$ for every $t \in \mathbb{R}$. Thus,

$$\inf\left\{t \in \mathbb{R} : (x,t) \in \operatorname{epi}(f)\right\} = \inf \mathbb{R} = -\infty = f(x).$$

Finally, if $f(x) \in \mathbb{R}$, then $(x,t) \in \operatorname{epi}(f)$ whenever $t \geq f(x)$, and $(x,t) \notin \operatorname{epi}(f)$ if $t < f(x)$. Therefore,

$$\inf\left\{t \in \mathbb{R} : (x,t) \in \operatorname{epi}(f)\right\} = f(x).$$

Now, if f and g have the same epigraph, then using the equality that we just have proved, it follows that

$$f(x) = \inf\left\{t \in \mathbb{R} : (x,t) \in \operatorname{epi}(f)\right\} = \inf\left\{t \in \mathbb{R} : (x,t) \in \operatorname{epi}(g)\right\} = g(x)$$

for each $x \in \mathbb{R}^n$. Thus, we are done.

□

In the following, we use also the topological notion of *closure*. The closure of a set $A \subseteq \mathbb{R}^n$ in the usual topology is denoted by $\operatorname{cl}(A)$ (recall that a point $x \in \mathbb{R}^n$ is in the closure of A if and only if it is the limit of a sequence of points of A).

Exercise 1.7. *Let $f\colon \mathbb{R}^n \to (-\infty, +\infty]$ be a function. Prove that there exists a unique function $\operatorname{cl} f\colon \mathbb{R}^n \to [-\infty, +\infty]$ whose epigraph is $\operatorname{cl}(\operatorname{epi}(f))$. Verify that $\operatorname{cl} f$ is lower semi-continuous.*

Remark 1.3. The function $\mathrm{cl}f$ is called the *lower semi-continuous hull* of f.

Solution 1.7. Define $\mathrm{cl}f\colon \mathbb{R}^n \to [-\infty, +\infty]$ by

$$\mathrm{cl}f(x) := \inf\big\{t \in \mathbb{R} : (x,t) \in \mathrm{cl}(\mathrm{epi}(f))\big\}.$$

All we have to prove is that $\mathrm{epi}(\mathrm{cl}f) = \mathrm{cl}(\mathrm{epi}(f))$. For that sake, first let $(x, t_0) \in \mathrm{epi}(\mathrm{cl}f)$. Then,

$$t_0 \geq \mathrm{cl}f(x) = \inf\big\{t \in \mathbb{R} : (x,t) \in \mathrm{cl}(\mathrm{epi}(f))\big\},$$

from where we get that there exists $-\infty < t_1 \leq t_0$ such that $(x, t_1) \in \mathrm{cl}(\mathrm{epi}(f))$. We let $(x_j, s_j)_{j\in\mathbb{N}}$ be a sequence of points of $\mathrm{epi}(f)$ such that $(x_j, s_j) \to (x, t_1)$ as $j \to \infty$. Since $t_0 - t_1 \geq 0$, we have that

$$s_j + t_0 - t_1 \geq s_j \geq f(x_j)$$

for each $j \in \mathbb{N}$. Therefore, the sequence $(x_j, s_j + t_0 - t_1)_{j\in\mathbb{N}}$ is a sequence of points of $\mathrm{epi}(f)$. Since it is clear that it converges to (x, t_0), we get that $(x, t_0) \in \mathrm{cl}(\mathrm{epi}(f))$, and this gives the inclusion

$$\mathrm{epi}(\mathrm{cl}f) \subseteq \mathrm{cl}(\mathrm{epi}(f)).$$

To prove the reverse inclusion, let $(x, t_0) \in \mathrm{cl}(\mathrm{epi}(f))$. Thus,

$$t_0 \geq \inf\big\{t \in \mathbb{R} : (x,t) \in \mathrm{cl}(\mathrm{epi}(f))\big\} = \mathrm{cl}f(x),$$

and this means that $(x, t_0) \in \mathrm{epi}(\mathrm{cl}f)$. It follows that $\mathrm{cl}(\mathrm{epi}(f)) \subseteq \mathrm{epi}(\mathrm{cl}f)$.

Hence we have proved that $\mathrm{cl}f$ as defined above is a function whose epigraph is $\mathrm{cl}(\mathrm{epi}(f))$. The uniqueness of a function having this set as epigraph comes from the fact that two functions with the same epigraph are equal (this is proved in Exercise 1.6). Finally, we have that $\mathrm{cl}f$ is lower semi-continuous because its epigraph is a closed subset of $\mathbb{R}^{n+1}$.

□

Exercise 1.8. *Let $f\colon \mathbb{R}^n \to (-\infty, +\infty]$ be a convex function. Show that* $\mathrm{cl}(\mathrm{epi}(f))$ *is a convex subset of $\mathbb{R}^{n+1}$. In particular, conclude that if* $\mathrm{cl}f > -\infty$ *in $\mathbb{R}^n$, then* $\mathrm{cl}f$ *is convex.*

Remark 1.4. The condition $\mathrm{cl}f > -\infty$ appears here because according to our definition, only functions $f\colon \mathbb{R}^n \to (-\infty, +\infty]$ may be convex. This also happens in some of the next exercises.

Solution 1.8. We prove that the closure of a convex set is always a convex set. For that sake, let $K \subseteq \mathbb{R}^{n+1}$ be a convex set. Let $x, y \in \mathrm{cl}(K)$ and $\lambda \in (0, 1)$ be arbitrarily given. Choose sequences $(x_j)_{j\in\mathbb{N}}$ and $(y_j)_{j\in\mathbb{N}}$ in K such that $x_j \to x$ and $y_j \to y$ as $j \to \infty$. For each $j \in \mathbb{N}$, the convexity of K guarantees that $(1-\lambda)x_j + \lambda y_j \in K$. Since it is clear that

$$(1-\lambda)x_j + \lambda y_j \to (1-\lambda)x + \lambda y$$

as $j \to \infty$, we get that $(1-\lambda)x + \lambda y \in \mathrm{cl}(K)$. Thus, $\mathrm{cl}(K)$ is convex. Having this in mind, recall that if f is convex, then $\mathrm{epi}(f)$ is convex, and hence, $\mathrm{cl}(\mathrm{epi}(f))$ is convex.

As it was proved in Exercise 1.7, the epigraph of $\mathrm{cl}f$ is $\mathrm{cl}(\mathrm{epi}(f))$. Hence, if f is convex, then $\mathrm{epi}(\mathrm{cl}f)$ is convex. If, in addition, we have that $\mathrm{cl}f > -\infty$, then $\mathrm{cl}f$ is a convex function.

□

Now, we will use the symbols ∂A and $\text{int}(A)$ for (topological) *boundary* and *interior* of a set $A \subseteq \mathbb{R}^n$ respectively (see Chapter 1 of the main book).

Exercise 1.9. *Let $f\colon \mathbb{R}^n \to \mathbb{R}$ be a convex function. Prove that*

$$\partial(\text{epi}(f)) = \big\{(x, f(x)) : x \in \mathbb{R}^n\big\} \text{ and}$$
$$\text{int}(\text{epi}(f)) = \big\{(x,t) : x \in \mathbb{R}^n \text{ and } t > f(x)\big\}.$$

Remark 1.5. We do not need convexity here: these equalities hold for any continuous function $f\colon \mathbb{R}^n \to \mathbb{R}$. Recall that if $f\colon \mathbb{R}^n \to \mathbb{R}$ is convex, then f is continuous in $\mathbb{R}^n$ because the interior of the effective domain of f is $\mathbb{R}^n$.

Solution 1.9. First of all, since f is convex and $\text{dom}(f) = \mathbb{R}^n$, we have that f is continuous. Thus, in particular, f is lower semi-continuous, and it follows that $\text{epi}(f)$ is closed. Therefore,

$$\text{epi}(f) = \text{int}(\text{epi}(f)) \cup \partial(\text{epi}(f)).$$

The inclusion

$$\big\{(x, f(x)) : x \in \mathbb{R}^n\big\} \subseteq \partial(\text{epi}(f))$$

is clear, because for each $x \in \mathbb{R}^n$, any neighborhood of the point $(x, f(x))$ contains points having the form (x,t) for $t < f(x)$. Hence, any neighborhood of $(x, f(x))$ has points that are not in $\text{epi}(f)$. On the other hand, any such neighborhood contains points of $\text{epi}(f)$, since in particular, we have $(x, f(x)) \in \text{epi}(f)$.

Now, let $(x,t) \in \partial(\text{epi}(f))$ be such that $t > f(x)$. Let $c > 0$ be small enough such that $t - c > f(x)$. By continuity, there exists a number $\varepsilon > 0$ such that $f(y) < t - c$ for each $y \in B(x, 2\varepsilon)$. Hence,

$$s_0 := \max\{f(y) : y \in \text{cl}(B(x,\varepsilon))\} \leq t - c < t.$$

Consider the set

$$R = B(x,\varepsilon) \times \{s > s_0\} \subseteq \mathbb{R}^{n+1}.$$

Since R is clearly open and $(x,t) \in R$, there exists a number $\delta > 0$ such that $B((x,t),\delta) \subseteq R$. Let $(y,s) \in B((x,t),\delta)$. Then, $(y,s) \in R$, and this leads to $y \in B(x,\varepsilon)$ and $s > s_0$. Consequently,

$$f(y) \leq s_0 < s,$$

and hence, $(y,s) \in \text{epi}(f)$. It follows that $B((x,t),\delta) \subseteq \text{epi}(f)$, and this shows that $(x,t) \in \text{int}(\text{epi}(f))$. Thus,

$$\big\{(x,t) : x \in \mathbb{R}^n \text{ and } t > f(x)\big\} \subseteq \text{int}(\text{epi}(f)).$$

Since all points of $\text{epi}(f)$ have the form $(x, f(x))$ or (x,t) for $t > f(x)$, and because the sets $\text{int}(\text{epi}(f))$ and $\partial(\text{epi}(f))$ are disjoint, we have the desired equalities.

□

Remark 1.6. Notice that this proof only uses the continuity of f; the convexity was not needed.

Exercise 1.10. *Let $D \subseteq \mathbb{R}^n$ be a convex set, and let $f\colon D \to \mathbb{R}$ be a continuous function. Assume that for each $x \in D$, there exists an affine function $g\colon \mathbb{R}^n \to \mathbb{R}$ and an open set $U \subseteq \mathbb{R}^n$ containing x such that $f(x) = g(x)$ and $f(z) \geq g(z)$ for any $z \in U \cap D$. Prove that f is convex.*

Solution 1.10. To prove that $f\colon D \to \mathbb{R}$ is convex, it is obviously enough to see that for each $x, y \in D$, the function $h\colon (0,1) \to \mathbb{R}$ given by

$$h(\lambda) = f((1-\lambda)x + \lambda y)$$

is convex. Let $x_0 = (1-\lambda_0)x + \lambda_0 y$ be arbitrarily given for some $0 < \lambda_0 < 1$. By the hypothesis, there exist an affine function $g\colon \mathbb{R}^n \to \mathbb{R}$ and an open neighborhood U of x_0 such that $g(x_0) = f(x_0)$ and $g(x) \leq f(x)$ for every $x \in U \cap D$. Of course, we may assume that U is an open ball centered at x (because U contains such an open ball), and hence, we write $U = B(x_0, \rho)$, with $\rho > 0$. Also, we write

$$g(\cdot) = \langle v, \cdot \rangle - \alpha$$

for some vector $v \in \mathbb{R}^n$ and some number $\alpha \in \mathbb{R}$. Now, let $0 < \lambda_1 < \lambda_0 < \lambda_2 < 1$ be such that

$$(1-\lambda)x + \lambda y \in B(x_0, \rho) \cap D$$

for each $\lambda_1 < \lambda < \lambda_2$. Notice that the existence of such numbers λ_1 and λ_2 is guaranteed because x_0 is clearly a point of the interior of a segment of D and of a segment of $B(x_0, \rho)$. Now, let $g_0\colon \mathbb{R} \to \mathbb{R}$ be the affine function given as

$$g_0(t) := g((1-t)x + ty) = \langle v, (1-t)x + ty \rangle - \alpha = t\langle v, y - x \rangle + \langle v, x \rangle - \alpha$$

for each $t \in \mathbb{R}$. Of course, we have that $g((1-\lambda)x + \lambda y) = g_0(\lambda)$ for every $\lambda \in (0,1)$. In particular,

$$g_0(\lambda_0) = g(x_0) = f(x_0) = h(\lambda_0) \text{ and}$$
$$g_0(\lambda) = g((1-\lambda)x + \lambda y) \leq f((1-\lambda)x + \lambda y) = h(\lambda) \text{ for any } \lambda \in (\lambda_1, \lambda_2).$$

It follows that the problem reduces to the one-dimensional case: all we have to prove is that if $h\colon (0,1) \to \mathbb{R}$ is a continuous function with the property that for every $\lambda_0 \in (0,1)$ there exist numbers $0 < \lambda_1 < \lambda_0 < \lambda_2 < 1$ and an affine function $g_0\colon \mathbb{R} \to \mathbb{R}$ such that $h(\lambda_0) = g_0(\lambda_0)$ and $g_0(\lambda) \leq h(\lambda)$ for every $\lambda_1 < \lambda < \lambda_2$, then h is convex.

Suppose that $h\colon (0,1) \to \mathbb{R}$ is a function that satisfies this property, but is not convex. Then, there exist numbers $t_1, t_2, \mu_0 \in (0,1)$ such that $t_1 < t_2$ and

$$h((1-\mu_0)t_1 + \mu_0 t_2) > (1-\mu_0)h(t_1) + \mu_0 h(t_2).$$

Let $g\colon \mathbb{R} \to \mathbb{R}$ be the affine function given as

$$g(t) = \frac{h(t_2) - h(t_1)}{t_2 - t_1} t + \frac{h(t_1)t_2 - h(t_2)t_1}{t_2 - t_1}$$

That is, g is the affine function passing through the points $(t_1, h(t_1))$ and $(t_2, h(t_2))$, in such a way that we have $g(t_1) = h(t_1)$ and $g(t_2) = h(t_2)$. From the inequality above writing $(1-\mu_0)t_1 + \mu_0 t_2$, we obviously have $t_1 < z < t_2$ and

$$h(z) > (1-\mu_0)h(t_1) + \mu_0 h(t_2) = g((1-\mu_0)t_1 + \mu_0(t_2)) = g(z).$$

Notice that this inequality has the obvious geometric interpretation: the number $h(z)$ is above the line segment connecting $(t_1, h(t_1))$ and $(t_2, h(t_2))$. This is a consequence of the assumed non-convexity of h. Of course, this is a consequence of the assumed non-convexity of h.

Since h is continuous, we may let $c := \max\{h(t) - g(t) : t \in [t_1, t_2]\}$, and from the fact that $h(z) - g(z) > 0$, we get that $c > 0$. Let $t_0 \in (t_1, t_2)$ be a point where this maximum is attained (observe that it cannot be attained for t_1 or t_2 because $h - g$ vanishes in these points). By our assumption on h, there exists an affine function $g_0 \colon \mathbb{R} \to \mathbb{R}$ and numbers $0 < \lambda_1 < t_0 < \lambda_2 < 1$ such that $g_0(t_0) = h(t_0)$ and $g(t) \leq h(t)$ for each $\lambda_1 < t < \lambda_2$. Now, observe that $g_1(\cdot) := g(\cdot) + c$ is an affine function for which it holds that

$$g_1(t_0) = g(t_0) + c = h(t_0) = g_0(t_0) \text{ and}$$
$$g_1(t) = g(t) + c \geq h(t) \geq g_0(t) \text{ for any } t \in (\lambda_1, \lambda_2).$$

Consequently, g_1 and g_0 are affine functions that coincide in t_0 and such that $g_1 \geq g_0$ is an open neighborhood around t_0. It follows that $g_1 = g_0$, and hence, we also have that $g_1(t) = h(t) = g_0(t)$ for every $t \in (\lambda_1, \lambda_2)$. Thus,

$$h(t) - g(t) = c$$

for every $t \in (\lambda_1, \lambda_2)$. Since t_0 is any point in $[t_1, t_2]$ such that $h(t_0) - g(t_0) = c$, we get that the set

$$C := \{t \in [t_1, t_2] : h(t) - g(t) = c\}$$

is open (and non-empty) in $[t_1, t_2]$. On the other hand, using continuity again, we get that C is also closed in $[t_1, t_2]$. Hence, $C = [t_1, t_2]$, and this leads to $h(t_1) - g(t_1) = c > 0$. This is a contradiction to the fact that $h(t_1) = g(t_1)$, and hence, h is indeed convex.

□

We remind the reader that a *probability space* is a measure space (X, Σ, μ) such that $\mu(X) = 1$, with μ as the corresponding *probability measure*.

Exercise 1.11. *Exhibit a probability space (X, Σ, μ), an interval D, a function $f \colon X \to D$ that is not constant μ-almost everywhere, and a non-affine convex function $\varphi \colon D \to (-\infty, +\infty]$ such that both sides of (1.2.8) are $+\infty$.*

Solution 1.11. Let $\varphi \colon [0, 2] \to (-\infty, +\infty]$ be given as

$$\varphi(x) := \begin{cases} 0, & \text{if } 0 \leq x \leq 1 \\ +\infty, & \text{if } 1 < x \leq 2 \end{cases},$$

and $f \colon [0, 1] \to [0, 2]$ given as $f(x) = 2\sqrt{x}$. We let the probability measure μ be simply the Lebesgue measure of $\mathbb{R}$ restricted to $[0, 1]$. It is easy to see that both sides of (1.2.8) in the main book are $+\infty$ for these functions and measure.

□

Jensen's inequality, whose discrete version we consider below, relates the value of a convex function f of an integral to the integral of the convex function, thus generalizing the fact that the secant of f lies above the graph of f (see Theorem 1.2.2 in [29]).

Exercise 1.12. *Use the discrete Jensen inequality to prove the* inequality of arithmetic and geometric means*, which states that for any $c_1, \ldots, c_m$ such that $c_1 + \ldots + c_m = 1$, and for every $t_1, \ldots, t_m > 0$, it holds that*

$$c_1 t_1 + \ldots + c_m t_m \geq t_1^{c_1} \ldots t_m^{c_m}. \tag{1.1}$$

Hint: Consider the logarithm function.

Solution 1.12. Since the logarithm is a concave function, and because $c_1 + \ldots + c_m = 1$, we may write

$$\log(c_1 t_1 + \ldots + c_m t_m) \geq c_1 \log(t_1) + \ldots + c_m \log(t_m) = \log\left(t_1^{c_1} \ldots t_m^{c_m}\right).$$

Since the logarithm is an increasing function, we get that

$$c_1 t_1 + \ldots c_m t_m \geq t_1^{c_1} \ldots t_m^{c_m}.$$

□

In the following, we use the symbol dom for the *effective domain* of a function (i.e., the set of points where it assumes a real value), and max, min, sup, and inf for maximum, minimum, supremum, and infimum, respectively.

Exercise 1.13. *Let $f\colon \mathbb{R}^n \to (-\infty, +\infty]$ be a proper convex function, and let $x \in \operatorname{int}(\operatorname{dom}(f))$ be such that $c = f(x)$ is not a global minimum of f. Show that for any $u \in \partial f(x)$,*

(a) *$f'_+(x, u) > 0$ and*

(b) *$\sup\left\{f'_-(x, u+v) : v \in u^\perp\right\} \leq |u|^2 \leq \inf\left\{f'_+(x, u+v) : v \in u^\perp\right\}$, where $u^\perp := \mathbf{h}(u, 0)$.*

Solution 1.13. (a) From Theorem 1.3.1 in the main book, we have

$$f'_+(x, u) = \max\{\langle v, u\rangle : v \in \partial f(x)\} \geq \langle u, u\rangle = |u|^2.$$

Thus, we have to prove that u is non-zero. This comes from the fact that c is not a global minimum of f. Indeed, if $0_{\mathbb{R}^n} \in \partial f(x)$, then

$$f(x + v) \geq f(x) + \langle 0_{\mathbb{R}^n}, v\rangle = f(x)$$

for every $v \in \mathbb{R}^n$. This gives that $c = f(x) \leq f(z)$ for every $z \in \mathbb{R}^n$, and this is a contradiction to the fact that c is not a global minimum.

(b) For any $v \in u^\perp$, we use Theorem 1.3.1 from the main book again to get

$$f'_+(x, u + v) = \max\{\langle z, u + v\rangle : z \in \partial f(x)\} \geq \langle u, u + v\rangle = |u|^2.$$

Thus, $|u|^2 \leq f'_+(x, u + v)$ for every $v \in u^\perp$. Passing to the infimum on the right-hand side, we get

$$|u|^2 \leq \inf\left\{f'_+(x, u + v) : v \in u^\perp\right\}.$$

To prove the other inequality, we use the fact that $f'_-(x, w) = -f'_+(x, -w)$ for any $w \in \mathbb{R}^n$. Then, if $v \in u^\perp$, we have

$$f'_-(x, u + v) = -f'_+(x, -u - v) = -\max\{\langle z, -u - v\rangle : z \in \partial f(x)\} \leq -\langle u, -u - v\rangle = |u|^2.$$

Therefore, passing to the supremum of $f'_+(x, u+v)$ for $v \in u^\perp$, we get what we wanted to get.

□

A function $g\colon \mathbb{R}^n \to \mathbb{R}$ is said to be *sublinear* if $g(x+y) \le g(x) + g(y)$ and $g(\lambda x) = \lambda g(x)$ for any $x, y \in \mathbb{R}^n$ and $\lambda \ge 0$.

Exercise 1.14. *Let $g\colon \mathbb{R}^n \to \mathbb{R}$ be a sublinear function, and assume that there exists $x \in \mathbb{R}^n$ such that $g(x) = c \neq 0$.*

(a) *Prove that $\{g > c\} \neq \emptyset$ and $\{g < c\} \neq \emptyset$.*

(b) *Show that $\{g \le c\}$ is a convex set.*

(c) *Prove that $x \in \partial\{g \le c\}$.*

Remark 1.7. In Chapter 2 of the main book, it is proved that every proper convex set of $C \subseteq \mathbb{R}^n$ has non-empty boundary ∂C and that C is supported by a hyperplane at each point of ∂C. As a consequence of the previous exercise, we have that $\{g \le c\}$ is supported by some hyperplane at x whenever g is sublinear and $g(x) = c \neq 0$. The reader should keep this in mind for the next exercise.

Solution 1.14. (a) If $c < 0$, since $g(0_{\mathbb{R}^n}) = 0$, we have that $0_{\mathbb{R}^n} \in \{g > c\}$. Now, letting $t > 1$ we have $g(tx) = tg(x) < g(x) = c$, from where we get that $tx \in \{g < c\}$. If $c > 0$, then $g(0_{\mathbb{R}^n}) = 0 < c$. Also, choosing any $t > 1$, we have $g(tx) = tg(x) = tc > c$.

(b) This follows from the fact that every sublinear function is convex. If $x, y \in \{g \le c\}$ and $\lambda \in (0,1)$, then

$$g((1-\lambda)x + \lambda y) \le (1-\lambda)g(x) + \lambda g(y) \le (1-\lambda)c + \lambda c = c,$$

and hence, $(1-\lambda)x + \lambda y \in \{g \le c\}$.

(c) To prove that $x \in \partial\{g \le c\}$, we have to show that every neighborhood of x has points of $\{g \le c\}$ and of $\mathbb{R}^n \setminus \{g \le c\} = \{g > c\}$. For that sake, let $\varepsilon > 0$ be arbitrarily given. Then,

$$\{g \le c\} \cap B(x, \varepsilon) \neq \emptyset,$$

because $x \in \{g \le c\} \cap B(x, \varepsilon)$. Hence, we have to show that $B(x, \varepsilon)$ contains a point of $\{g > c\}$. If $c > 0$, let

$$z = \left(1 + \frac{\varepsilon}{2|x|}\right) x,$$

where we notice carefully that $x \neq 0_{\mathbb{R}^n}$, since otherwise we would have $g(x) = 0$. Then $z \in B(x, \varepsilon)$, because

$$|z - x| = \frac{\varepsilon}{2},$$

and

$$g(z) = \left(1 + \frac{\varepsilon}{2|x|}\right) g(x) > c.$$

In the case where $c < 0$, choose $0 < \delta < \varepsilon$ such that we also have $\delta < 2|x|$, and let

$$z = \left(1 - \frac{\delta}{2|x|}\right) x.$$

It is clear that $z \in B(x, \varepsilon)$ and

$$g(z) = \left(1 - \frac{\delta}{2|x|}\right) g(x) > c.$$

This concludes the proof.

□

A *sublevel set* of a function f is a set having the form $\{x \in \mathbb{R}^n : f(x) \leq c\}$ for some $c \in \mathbb{R}^n$. The subdifferential of a function at a given domain point is a set of vectors that generalize the notion of *gradient vector* (see Chapter 1 in the main book for more details).

In what follows, recall that the *unit sphere* of $\mathbb{R}^n$ is the set $\mathbb{S}^{n-1} := \{x \in \mathbb{R}^n : |x| = 1\}$. The *unit ball* of $\mathbb{R}^n$ is the set $B^n := \{x \in \mathbb{R}^n : |x| \leq 1\}$.

Exercise 1.15. *Let $g : \mathbb{R}^n \to \mathbb{R}$ be a sublinear function, and let $x \in \mathbb{R}^n$ be a point such that $g(x) = c \neq 0$. Prove that g is differentiable at x if and only if the sublevel set $\{g \leq c\}$ is supported by a unique hyperplane at x.*

Solution 1.15. First, notice that $x \neq 0_{\mathbb{R}^n}$ since $g(x) = c \neq 0$. Assume that g is differentiable at x, and let $\mathbf{h} := \mathbf{h}(u, \alpha)$ be a supporting hyperplane of $\{g \leq c\}$ at x. Assume also that $\{g \leq c\} \subseteq \mathbf{h}^-$ and that $u \in \mathbb{S}^{n-1}$. We claim that

$$\mathbf{h} \cap \{g < c\} = \emptyset.$$

In fact, if $z \in \{g < c\}$, then there exists an open ball $B(z, \rho)$ such that $g(w) < c$ for every $w \in B(z, \rho)$. If we also have that $z \in \mathbf{h}$, then this ball has points of $\operatorname{int}(\mathbf{h}^+)$, and this contradicts the inclusion $\{g \leq c\} \subseteq \mathbf{h}^-$.

Let $v \in \mathbb{R}^n$ be any vector such that $\langle u, v \rangle = 0$. Then, for any $\lambda \in \mathbb{R}$, we have that $x + \lambda v \in \mathbf{h}$, because $\langle u, x + \lambda v \rangle = \langle u, x \rangle = \alpha$. Thus, we have the inequality

$$g(x + \lambda v) \geq c$$

for any $\lambda \in \mathbb{R}$. Since g is differentiable at x, we have that the derivative of g in the direction of v is given as $\langle \nabla g(x), v \rangle$, where $\nabla g(x)$ is the gradient vector of g at x. Therefore, using the fact that the derivative of a differentiable function at a given direction equals the one-sided derivatives at this same direction, we get

$$\langle \nabla g(x), v \rangle = \lim_{\lambda \to 0^+} \frac{g(x + \lambda v) - g(x)}{\lambda} = \lim_{\lambda \to 0^+} \frac{g(x + \lambda v) - c}{\lambda} \geq 0,$$

and

$$\langle \nabla g(x), v \rangle = \lim_{\lambda \to 0^-} \frac{g(x + \lambda v) - g(x)}{\lambda} = \lim_{\lambda \to 0^-} \frac{g(x + \lambda v) - c}{\lambda} \leq 0.$$

Then we get $\langle \nabla g(x), v \rangle = 0$ for every $v \in u^\perp$. Consequently, we have that $\nabla g(x) = \alpha u$ for some $\alpha \in \mathbb{R}$. On the other hand,

$$\langle \nabla g(x), x \rangle = \lim_{\lambda \to 0} \frac{g(x + \lambda x) - g(x)}{\lambda} = \lim_{\lambda \to 0} \frac{(1 + \lambda) g(x) - g(x)}{\lambda} = g(x) = c \neq 0.$$

Thus, $\nabla g(x)$ and u are non-zero vectors having the same direction. Hence, $\nabla g(x)$ determines the (unique) direction of an outer normal of $\{g \leq c\}$ at x.

Now, suppose that $\{g \leq c\}$ is supported by a unique hyperplane at x, and assume that $u \in \mathbb{S}^{n-1}$ is the unique outer normal unit vector of $\{g \leq c\}$ at x. To prove that g is differentiable at x, it suffices to prove that the subdifferential of g at x is a singleton (see Theorem 1.3.2 of the main book). For that sake, let $v \in \partial g(x)$. Since g is convex and c is not a global minimum of g (see item **(a)** of Exercise 1.14), we get from Proposition 1.3.3 in the main book that v is an outer normal of $\{g \leq c\}$ at x, and hence, $v = \alpha u$ for some $\alpha > 0$. Since g is sublinear, we have from Proposition 1.2.4 of the main book that

$$g'_+(x, x) = -g'_+(x, -x) = g(x) = c.$$

Recalling again that g is convex, by using Theorem 1.3.1 from the main book together with the equalities above and with the fact that $v \in \partial g(x)$, we get

$$\begin{aligned} c = g(x) = g'_+(x, x) \geq \langle v, x\rangle = \alpha\langle u, x\rangle \quad \text{and} \\ -c = -g(x) = g'_+(x, -x) \geq \langle v, -x\rangle = -\alpha\langle u, x\rangle, \end{aligned}$$

and these inequalities lead to $c = \alpha\langle u, x\rangle$. Since $c \neq 0$, we have that $\alpha \neq 0$ and $\langle u, x\rangle \neq 0$. This yields

$$\alpha = \frac{c}{\langle u, x\rangle}.$$

Hence, if we start with any vector $v \in \partial g(x)$, we arrive at the equality

$$v = \frac{c}{\langle u, x\rangle} u,$$

from where we get that $\partial g(x)$ is a singleton. This concludes the proof.

□

Exercise 1.16. *Give an example of a convex function f that admits a non-minimizer point $x \in \operatorname{int}(\operatorname{dom}(f))$ and a vector $u \in \partial f(x)$ such that the equality $f'_-(x, u) = 0$ holds.*

Solution 1.16. Let $f\colon \mathbb{R}^2 \to \mathbb{R}$ be given as $f(x, y) = |x| + |y|$. Then, $\operatorname{dom}(f) = \mathbb{R}^2$ and f is clearly convex. Let $x = (0, 1)$ and $u = (1, 1)$. For any $(\alpha, \beta) \in \mathbb{R}^2$, we have

$$f((0, 1) + (\alpha, \beta)) - f(0, 1) = |\alpha| + |\beta + 1| - 1 \geq |\alpha| + |\beta| \geq \alpha + \beta = \langle(\alpha, \beta), (1, 1)\rangle,$$

and this shows that $(1, 1) \in \partial f(x)$. Now, notice that for any $-1 < \lambda < 0$, we have

$$f(x + \lambda u) = f((0, 1) + (\lambda, \lambda)) = |\lambda| + |\lambda + 1| = -\lambda + \lambda + 1 = 1.$$

Thus,

$$f'_-(x, u) = \lim_{\lambda \to 0^-} \frac{f(x + \lambda u) - f(x)}{\lambda} = \lim_{\lambda \to 0^-} \frac{1 - 1}{\lambda} = 0,$$

and hence, we have the desired result.

□

Exercise 1.17 (Rockafellar's theorem)**.** *A set $C \subseteq \mathbb{R}^n \times \mathbb{R}^n$ is said to be* cyclically monotonic *if for any $m \in \mathbb{N}$ and every pair $(x_1, v_1), \ldots, (x_m, v_m) \in C$, we have*

$$\langle x_2 - x_1, v_1 \rangle + \langle x_3 - x_2, v_2 \rangle + \ldots + \langle x_m - x_{m-1}, v_{m-1} \rangle + \langle x_1 - x_m, v_m \rangle \leq 0.$$

Prove that a non-empty set $C \subseteq \mathbb{R}^n \times \mathbb{R}^n$ is cyclically monotonic if and only if there exists a proper convex function $f \colon \mathbb{R}^n \to (-\infty, +\infty]$ such that $C \subseteq \partial f$.

Solution 1.17. First, let $f \colon \mathbb{R}^n \to (-\infty, +\infty]$ be a convex function, and assume that $C \subseteq \partial f$. Let $(x_1, v_1), \ldots, (x_m, v_m) \in C$. Then

$$\begin{aligned} &\langle v_1, x_2 - x_1 \rangle + \langle v_2, x_3 - x_2 \rangle + \ldots + \langle v_{m-1}, x_m - x_{m-1} \rangle + \langle v_m, x_1 - x_m \rangle \leq \\ &\leq f(x_2) - f(x_1) + f(x_3) - f(x_2) + \ldots + f(x_m) - f(x_{m-1}) + f(x_1) - f(x_m) = 0, \end{aligned}$$

and hence, C is cyclically monotonic.

Conversely, let $C \subseteq \mathbb{R}^n \times \mathbb{R}^n$ be a cyclically monotonic non-empty set. We fix a point $(x_0, v_0) \in C$ and let $f \colon \mathbb{R}^n \to (-\infty, +\infty]$ be defined as

$$f(x) = \sup \big\{ \langle v_0, x_1 - x_0 \rangle + \langle v_1, x_2 - x_1 \rangle + \langle v_2, x_3 - x_2 \rangle + \ldots + \langle v_m, x - x_m \rangle \big\},$$

where the supremum is taken over all possible choices of $m \in \mathbb{N}$ and $(x_1, v_1), \ldots, (x_m, v_m) \in C$. Since f is the supremum of a collection of affine functions, we get that f is convex (see Proposition 1.1.1 in the main book). Also, because C is cyclically monotonic, we have that

$$\langle v_0, x_1 - x_0 \rangle + \langle v_1, x_2 - x_1 \rangle + \ldots + \langle v_m, x_0 - x_m \rangle \leq 0$$

for any pairs $(x_1, v_1), \ldots, (x_m, v_m) \in C$. This gives that $f(x_0) \leq 0$, and hence, f is proper. Notice that we actually have $f(x_0) = 0$, because $f(x_0) \geq \langle v_0, x_0 - x_0 \rangle = 0$ (indeed, we are calculating the expression for the single point $(x_0, v_0) \in C$).

We claim that $C \subseteq \partial f$. To prove that, we let $(x, v) \in C$, and all we have to show is that v is a subgradient of f at x. For this purpose, choose a number $\alpha < f(x)$. There exist pairs $(x_1, v_1), \ldots, (x_m, v_m) \in C$ such that

$$\langle v_0, x_1 - x_0 \rangle + \langle v_1, x_2 - x_1 \rangle + \ldots + \langle v_m, x - x_m \rangle > \alpha.$$

On the other hand, since $(x, v) \in C$, writing $(x, v) = (x_{m+1}, v_{m+1})$, we have from the definition of f that

$$\begin{aligned} f(y) &\geq \langle v_0, x_1 - x_0 \rangle + \langle v_1, x_2 - x_1 \rangle + \ldots + \langle v_m, x_{m+1} - x_m \rangle + \langle v_{m+1}, y - x_{m+1} \rangle = \\ &= \langle v_0, x_1 - x_0 \rangle + \langle v_1, x_2 - x_1 \rangle + \ldots + \langle v_m, x - x_m \rangle + \langle v, y - x \rangle \end{aligned}$$

for every $y \in \mathbb{R}^n$. These inequalities lead immediately to

$$f(y) > \alpha + \langle v, y - x \rangle.$$

Since this holds for any $\alpha < f(x)$ and every $y \in \mathbb{R}^n$, we have

$$f(y) \geq f(x) + \langle v, y - x \rangle$$

for every $y \in \mathbb{R}^n$. Hence, $v \in \partial f(x)$, and this concludes the proof.

□

We remind the reader that the *conjugate function* of a proper function $f\colon (-\infty,+\infty] \to \mathbb{R}$ is the function $f^*(x) := \sup\{\langle x, y\rangle - f(y)\colon y \in \mathbb{R}^n\}$.

Exercise 1.18. *Give an example of a real-valued, convex, and continuous function $f\colon \mathbb{R}^n \to \mathbb{R}$ such that f^* is not real-valued.*

Solution 1.18. Let $f\colon \mathbb{R} \to \mathbb{R}$ be given as $f(x) = |x|$. It is clear that f is real-valued, convex, and continuous. We claim that $f^*(x) = +\infty$ whenever $x < -1$ or $x > 1$. Indeed, if this is the case, then

$$f^*(x) = \sup\big\{xz - |z| : z \in \mathbb{R}\big\}.$$

We observe that for $z \geq 0$, we have

$$xz - |z| = xz - z = z(x-1),$$

and the latter goes to $+\infty$ as $z \to +\infty$ if $x > 1$. On the other hand, for $z < 0$, it holds that

$$xz - |z| = xz + z = z(x+1),$$

and the latter goes to $+\infty$ as $z \to -\infty$ if $x < -1$.

□

Exercise 1.19. *Let $f\colon \mathbb{R}^n \to (-\infty,+\infty]$ be any given function. Fix $x_0 \in \mathbb{R}^n$ and let $\bar{f}(x) := f(x - x_0)$ for each $x \in \mathbb{R}^n$. Show that*

$$\bar{f}^*(x) = \langle x, x_0\rangle + f^*(x)$$

for every $x \in \mathbb{R}^n$. Also, if $a \in \mathbb{R}$ and $\tilde{f}(x) := a + f(x)$, verify that

$$\tilde{f}^*(x) = -a + f^*(x)$$

for any $x \in \mathbb{R}^n$.

Solution 1.19. For any $x \in \mathbb{R}^n$, we just compute

$$\begin{aligned}\bar{f}^*(x) = \sup\big\{\langle x, y\rangle - \bar{f}(y) : y \in \mathbb{R}^n\big\} = \sup\big\{\langle x, y\rangle - f(y - x_0) : y \in \mathbb{R}^n\big\} = \\ = \sup\big\{\langle x, y - x_0 + x_0\rangle - f(y - x_0) : y \in \mathbb{R}^n\big\} = \sup\big\{\langle x, x_0\rangle + \langle x, z\rangle - f(z) : z \in \mathbb{R}^n\big\} = \\ = \langle x, x_0\rangle + f^*(x).\end{aligned}$$

For the remaining equality, we calculate

$$\tilde{f}^*(x) = \sup\big\{\langle x, y\rangle - a - f(y) : y \in \mathbb{R}^n\big\} = -a + f^*(x).$$

□

Exercise 1.20. *Let $A \subseteq \mathbb{R}^n$ be non-empty. The* indicator function *of A is the function $I_A^\infty\colon \mathbb{R}^n \to [0,+\infty]$ given as*

$$I_A^\infty(x) = \begin{cases} 0 & \text{if } x \in A \\ +\infty & \text{if } x \in \mathbb{R}^n \setminus A \end{cases}.$$

(a) *Prove that I_A^∞ is convex if and only if A is convex.*

(b) *Show that I_A^∞ is lower semi-continuous if and only if A is closed.*

(c) *Prove that*

$$(I_A^\infty)^*(v) = \sup\{\langle x, v\rangle : x \in A\}$$

for each $v \in \mathbb{R}^n$.

Remark 1.8. In other words, the conjugate function of the indicator function is the *support function*, which is intensively studied in Chapter 2 of the main book.

Solution 1.20. (a) Let $x, y \in \mathbb{R}^n$ and $\lambda \in (0,1)$. If $x \notin A$ or $y \notin A$, then $(1-\lambda)I_A^\infty(x) + \lambda I_A^\infty(y) = +\infty$. Hence, the inequality

$$I_A^\infty((1-\lambda)x + \lambda y) \leq (1-\lambda)I_A^\infty(x) + \lambda I_A^\infty(y)$$

holds trivially. If A is convex and $x, y \in A$, then $(1-\lambda)x + \lambda y \in A$, from where we get

$$I_A^\infty((1-\lambda)x + \lambda y) = 0 = (1-\lambda)I_A^\infty(x) + \lambda I_A^\infty(y),$$

and this shows that I_A^∞ is convex. On the other hand, if A is not convex, then we can choose two points $x, y \in A$ and a number $\lambda \in (0,1)$ such that $(1-\lambda)x + \lambda y \notin A$. Then,

$$I_A^\infty((1-\lambda)x + \lambda y) = +\infty > 0 = (1-\lambda)I_A^\infty(x) + \lambda I_A^\infty(y),$$

from where we conclude that I_A^∞ is not convex.

(b) First, observe that if $(z, s) \in \operatorname{epi}(I_A^\infty)$, then $I_A^\infty(z) \leq s < +\infty$, yielding that $z \in I_A^\infty$. Assume that A is closed, and let $(x_j, t_j)_{j\in\mathbb{N}}$ be a sequence of points in $\operatorname{epi}(I_A^\infty)$ such that $(x_j, t_j) \to (x, t) \in \mathbb{R}^n \times \mathbb{R}$ as $j \to \infty$. Then x_j is a sequence of points of A such that $x_j \to x$. Since A is closed, we get that $x \in A$. Because $(x_j, t_j) \in \operatorname{epi}(I_A^\infty)$, we have that $t_j \geq I_A^\infty(x_j) = 0$ for each $j \in \mathbb{N}$. Since $t_j \to t$, it follows that $t \geq 0 = I_A^\infty(x)$. This shows that $(x, t) \in \operatorname{epi}(I_A^\infty)$, and this proves that $\operatorname{epi}(I_A^\infty)$ is a closed subset of $\mathbb{R}^{n+1}$. Hence, I_A^∞ is lower semi-continuous.

Now, suppose that $A \subseteq \mathbb{R}^n$ is not closed. Then we may choose a sequence $(x_j)_{j\in\mathbb{N}}$ in A that converges to a point $x \in \mathbb{R}^n \setminus A$. In particular, by the first observation that we made, it follows that $(x, 0) \notin \operatorname{epi}(I_A^\infty)$. Since $(x_j, 0) \in \operatorname{epi}(I_A^\infty)$ for each $j \in \mathbb{N}$, we have that $(x_j, 0)_{j\in\mathbb{N}}$ is a sequence of points in $\operatorname{epi}(I_A^\infty)$ converging to the point $(x, 0)$ that does not belong to $\operatorname{epi}(I_A^\infty)$. Thus, I_A^∞ is not lower semi-continuous.

(c) For each $v \in \mathbb{R}^n$, we have

$$(I_A^\infty)^*(v) = \sup\left\{\langle v, x\rangle - I_A^\infty(x) : x \in \mathbb{R}^n\right\}.$$

Notice that $\langle v, x\rangle - I_A^\infty(x) = -\infty$ whenever $x \notin A$. Hence, the supremum can be taken over the points of A (we are assuming that A is non-empty; observe that if A is empty, then I_A^∞ equals $+\infty$ everywhere). Thus,

$$(I_A^\infty)^*(v) = \sup\left\{\langle v, x\rangle - I_A^\infty(x) : x \in A\right\} = \sup\left\{\langle v, x\rangle : x \in A\right\},$$

and this concludes the proof.

□

Exercise 1.21. *Let $f\colon \mathbb{R}^n \to (-\infty,+\infty]$ be a proper convex function. Show that*

$$(\lambda f)^*(\cdot) = \lambda f^*\left(\frac{\cdot}{\lambda}\right)$$

for every $\lambda > 0$.

Solution 1.21. Let $\lambda > 0$. For each $v \in \mathbb{R}^n$, we have

$$\begin{aligned}(\lambda f)^*(v) = \sup\{\langle v,x\rangle - \lambda f(x) : x\in\mathbb{R}^n\} = \sup\left\{\lambda\left\langle\frac{v}{\lambda},x\right\rangle - \lambda f(x) : x\in\mathbb{R}^n\right\} = \\ = \lambda\sup\left\{\left\langle\frac{v}{\lambda},x\right\rangle - f(x) : x\in\mathbb{R}^n\right\} = \lambda f^*\left(\frac{v}{\lambda}\right).\end{aligned}$$

□

Most of the exercises below were taken from (or inspired by) some recent papers dealing with functional versions of geometric inequalities, such as [55], [80] and [81]. Some of them were taken from [228]. More references in this direction are given in Remark 1.19 below.

Exercise 1.22. *Let $f,g\colon \mathbb{R}^n \to (-\infty,+\infty]$ be functions. The* infimal convolution *of f and g is the function $f\square g\colon \mathbb{R}^n \to [-\infty,+\infty]$ defined as*

$$f\square g(x) = \inf\left\{f(z) + g(x-z) : z\in\mathbb{R}^n\right\}$$

for each $x \in \mathbb{R}^n$. We also define the Minkowski sum *(or simply* sum*) of two sets $A, B \subseteq \mathbb{R}^n$ as*

$$A + B := \{a + b : a \in A \text{ and } b \in B\}.$$

(a) *Assume that $f\square g > -\infty$ in $\mathbb{R}^n$. Prove that if f and g are convex, then $f\square g$ is convex.*

(b) *For any subsets $A, B \subseteq \mathbb{R}^n$, prove that $I_A^\infty \square I_B^\infty = I_{A+B}^\infty$.*

(c) *Give an example of functions $f,g\colon \mathbb{R}^n \to (-\infty,+\infty]$ that are proper, convex, and closed, but are such that $f\square g$ is proper and convex, but not closed.*

Remark 1.9. The Minkowski sum of sets is a very important concept in convex geometry. It will be better investigated in the next chapter.

Solution 1.22. (a) Let $x, y \in \mathbb{R}^n$ and $\lambda \in (0,1)$. Then

$$\begin{aligned}(1-\lambda)f\square g(x) + \lambda f\square g(y) = \\ = (1-\lambda)\inf\{f(z)+g(x-z) : z\in\mathbb{R}^n\} + \lambda\inf\{f(w)+g(y-w) : w\in\mathbb{R}^n\} = \\ = \inf\{(1-\lambda)f(z)+(1-\lambda)g(x-z) : z\in\mathbb{R}^n\} + \inf\{\lambda f(w)+\lambda g(y-w) : w\in\mathbb{R}^n\} \geq \\ \geq \inf\{(1-\lambda)f(z)+(1-\lambda)g(x-z)+\lambda f(w)+\lambda g(y-w) : z,w\in\mathbb{R}^n\} = \\ = \inf\{(1-\lambda)f(z)+\lambda f(w)+(1-\lambda)g(x-z)+\lambda g(y-w) : z,w\in\mathbb{R}^n\} \geq \\ \geq \inf\{f((1-\lambda)z+\lambda w)+g((1-\lambda)(x-z)+\lambda(y-w)) : z\in\mathbb{R}^n\} = \\ = \inf\{f((1-\lambda)z+\lambda w))+g((1-\lambda)x+\lambda y-((1-\lambda)z+\lambda w)) : z,w\in\mathbb{R}^n\} = \\ = \inf\{f(a)+g((1-\lambda)x+\lambda y-a) : a\in\mathbb{R}^n\} = \\ = f\square g((1-\lambda)x+\lambda y),\end{aligned}$$

and this shows that $f\square g$ is convex. Observe that we used the fact that for any sets $A, B \subseteq \mathbb{R}$, we have

$$\inf\{a+b : a \in A \text{ and } b \in B\} \leq \inf A + \inf B.$$

Moreover, we replaced $(1-\lambda)z + \lambda w$ by a, and $(1-\lambda)z + \lambda w$ ranges through the entire $\mathbb{R}^n$ when z, w ranges through $\mathbb{R}^n$.

(b) We have

$$I_A^\infty \square I_B^\infty(x) = \inf\left\{I_A^\infty(z) + I_B^\infty(x-z) : z \in \mathbb{R}^n\right\} \geq 0.$$

We divide our argumentation into two cases:

(i) Assume that $x \in A + B$, and write $x = z_0 + y_0$ with $z_0 \in A$ and $y_0 \in B$. Then

$$I_A^\infty \square I_B^\infty(x) \leq I_A^\infty(z_0) + I_B^\infty(x - z_0) = 0 + I_B^\infty(y_0) = 0.$$

Consequently,

$$I_A^\infty \square I_B^\infty(x) = 0 = I_{A+B}^\infty(x).$$

(ii) Let $x \notin A+B$. In this case, for any $z \in \mathbb{R}^n$, we cannot have simultaneously that $z \in A$ and $x - z \in B$. Thus, we have $I_A^\infty(z) = +\infty$ or $I_B^\infty(x-z) = +\infty$, and this gives

$$I_A^\infty \square I_B^\infty(x) = +\infty = I_{A+B}^\infty(x),$$

and hence, we have $I_A^\infty \square I_B^\infty = I_{A+B}^\infty$.

(c) Let

$$A = \{(x,y) \in \mathbb{R}^2 : y \geq 1/x \text{ and } x > 0\} \text{ and}$$
$$B = \{(x,y) \in \mathbb{R}^2 : y \geq -1/x \text{ and } x < 0\}.$$

We claim that A and B are closed and convex. We prove it for A. If $(x_j, y_j)_{j\in\mathbb{N}}$ is a sequence in A such that $(x_j, y_j) \to (x, y)$, then $x \geq 0$ and $x_j y_j \geq 1$ for each $j \in \mathbb{N}$. Hence, $xy \geq 1$. It follows that $x \neq 0$, from where we get that $x > 0$. This shows that $(x, y) \in A$.

For the convexity, let (x_1, y_1) and (x_2, y_2) be points of A, and let $\lambda \in (0, 1)$. Then, x_1, x_2, y_1, y_2 are strictly positive, and $x_1 y_1 \geq 1$ and $x_2 y_2 \geq 1$. From the arithmetic-geometric inequality, we have

$$x_1 y_2 + x_2 y_1 \geq 2\sqrt{x_1 y_2 x_2 y_1} \geq 2.$$

Therefore, writing

$$(1-\lambda)(x_1, y_1) + \lambda(x_2, y_2) = ((1-\lambda)x_1 + \lambda x_2, (1-\lambda)y_1 + \lambda y_2),$$

we calculate

$$[(1-\lambda)x_1 + \lambda x_2][(1-\lambda)y_1 + \lambda y_2] = (1-\lambda)^2 x_1 y_1 + (1-\lambda)\lambda(x_1 y_2 + x_2 y_1) + \lambda^2 x_2 y_2 \geq$$
$$\geq (1-\lambda)^2 + 2(1-\lambda)\lambda + \lambda^2 = 1.$$

This, together with the obvious fact that $(1-\lambda)x_1 + \lambda x_2 > 0$, gives that $(1-\lambda)(x_1, y_1) + \lambda(x_2, y_2) \in A$.

To prove that B is convex and closed, simply notice that B is the image of A under the linear transformation $T\colon \mathbb{R}^2 \to \mathbb{R}^2$ defined as

$$T(x, y) = (-x, y).$$

Since T is clearly an isomorphism, we have that B is closed. Also, it is readily seen that any linear transformation preserves convexity, and hence, B is convex.

It follows that the functions $f := I_A^\infty$ and $g := I_B^\infty$ are proper, closed, and convex, and we already know that $f\square g = I_A^\infty \square I_B^\infty = I_{A+B}^\infty$. Therefore, $f\square g > -\infty$ in $\mathbb{R}^n$, and hence, from item **(a)**, we get that $f\square g$ is convex. We also have that $f\square g$ is proper, because $A + B \neq \emptyset$, and for any $x \in A + B$, we have $f\square g(x) = I_{A+B}^\infty(x) = 0$.

Now, we claim that

$$A + B = \{(x, y) \in \mathbb{R}^2 : y > 0\}.$$

To prove that, first let $(x_1, y_1) \in A$ and $(x_2, y_2) \in B$ be arbitrarily given. Then, $x_1 > 0$, $x_2 < 0$, $y_1 \geq 1/x_1$ and $y_2 \geq -1/x_2$. Therefore,

$$y_1 + y_2 \geq \frac{1}{x_1} - \frac{1}{x_2} > 0,$$

and this gives the inclusion $A + B \subseteq \{(x, y) \in \mathbb{R}^2 : y > 0\}$. To prove the reverse inclusion, let $(x, y) \in \mathbb{R}^n$ be such that $y > 0$, and choose any $z > 0$ large enough such that

$$\begin{gathered} \frac{y}{2} \geq 1/z, \\ x - z < 0, \quad \text{and} \\ \frac{y}{2} \geq -\frac{1}{x - z}. \end{gathered}$$

Then we have that $(z, y/2) \in A$ and $(x - z, y/2) \in B$. Also,

$$(x, y) = \left(z, \frac{y}{2}\right) + \left(x - z, \frac{y}{2}\right) \in A + B.$$

This gives $\{(x, y) \in \mathbb{R}^2 : y > 0\} \subseteq A + B$.

It follows that $A + B$ is not a closed set, and hence, the function $f\square g = I_{A+B}^\infty$ is not closed. Summarizing, f and g are proper, closed, and convex, and $f\square g$ is proper and convex, but not closed. Hence, we have the desired example.

□

Exercise 1.23. *Let $f, g\colon \mathbb{R}^n \to (-\infty, +\infty]$ be functions such that $f\square g(x) > -\infty$ for each $x \in \mathbb{R}^n$. Prove that*

$$\operatorname{dom}(f\square g) = \operatorname{dom}(f) + \operatorname{dom}(g).$$

Solution 1.23. If one of the functions is not proper, then $\operatorname{dom}(f) + \operatorname{dom}(g) = \emptyset$ (indeed, the Minkowski sum of the empty set with any set is the empty set). Also, in this case, we clearly have that $f\square g = +\infty$ everywhere, and hence, $\operatorname{dom}(f\square g) = \emptyset$.

Now, assume that both functions are proper. Let $x \in \mathrm{dom}(f\square g)$. Then, since

$$f\square g(x) = \inf\{f(z) + g(x-z) : z \in \mathbb{R}^n\} < +\infty,$$

we get that there exists a point $z_0 \in \mathbb{R}^n$ such that

$$f(z_0) + g(x - z_0) < +\infty.$$

In particular, it follows that $f(z_0) < +\infty$ and $g(x-z_0) < +\infty$. Hence, $z_0 \in \mathrm{dom}(f)$ and $x-z_0 \in \mathrm{dom}(g)$. Thus,

$$x = z_0 + x - z_0 \in \mathrm{dom}(f) + \mathrm{dom}(g).$$

Now, let $x \in \mathrm{dom}(f)$ and $y \in \mathrm{dom}(g)$ be arbitrarily given. We have

$$f\square g(x+y) = \inf\{f(z) + g(x+y-z) : z \in \mathbb{R}^n\} \leq f(x) + g(x+y-x) = f(x) + g(y) < \infty,$$

and hence, we get that $x + y \in \mathrm{dom}(f\square g)$. The proof is complete.

□

Exercise 1.24. *Let $f, g\colon \mathbb{R}^n \to (-\infty, +\infty]$ be proper functions. Prove that*

$$\mathrm{epi}(f\square g) = \mathrm{epi}(f) + \mathrm{epi}(g).$$

Verify that if one of the functions is not proper, then both sides of the above are empty.

Remark 1.10. This result has a clear geometric interpretation: if $f, g\colon \mathbb{R}^n \to (-\infty, +\infty]$ are proper, then $f\square g\colon \mathbb{R}^n \to [-\infty, \infty]$ is the (unique) function whose epigraph is $\mathrm{epi}(f) + \mathrm{epi}(g)$. Because of this, the infimal convolution is sometimes called *epi-sum*.

Solution 1.24. Since f and g are proper, we have that their respective epigraphs are non-empty. Let $(x,t) \in \mathrm{epi}(f)$ and $(y,s) \in \mathrm{epi}(g)$ be arbitrarily given. Then, $f(x) \leq t$ and $f(y) \leq s$. Thus,

$$f\square g(x+y) = \inf\{f(z) + g(x+y-z) : z \in \mathbb{R}^n\} \leq f(x) + g(y) \leq t + s,$$

and hence, $(x+y, t+s) \in \mathrm{epi}(f\square g)$. This gives the inclusion

$$\mathrm{epi}(f) + \mathrm{epi}(g) \subseteq \mathrm{epi}(f\square g).$$

Now, let $(x,t) \in \mathrm{epi}(f\square g)$. Then $f\square g(x) \leq t$, that is, $\inf\{f(z) + g(x-z) : z \in \mathbb{R}^n\} \leq t$. Then there exists a point $z_0 \in \mathbb{R}^n$ such that

$$f(z_0) + g(x - z_0) \leq t.$$

Therefore, we have that $(x - z_0, t - f(z_0)) \in \mathrm{epi}(g)$. Since $(z_0, f(z_0)) \in \mathrm{epi}(f)$, and writing

$$(x,t) = (z_0, f(z_0)) + (x - z_0, t - f(z_0)),$$

we get that $(x,t) \in \mathrm{epi}(f) + \mathrm{epi}(g)$. Hence, $\mathrm{epi}(f\square g) \subseteq \mathrm{epi}(f) + \mathrm{epi}(g)$, and we have the desired equality.

Last, assume that one of the functions (f, say) is not proper. Then, $f(x) = +\infty$ for all $x \in \mathbb{R}^n$. In this case, we have that $\operatorname{epi}(f) = \emptyset$, and this gives

$$\operatorname{epi}(f) + \operatorname{epi}(g) = \{p + q : p \in \operatorname{epi}(f) \text{ and } q \in \operatorname{epi}(g)\} = \emptyset.$$

On the other hand, since $g(y) > -\infty$ for every $y \in \mathbb{R}^n$, we have

$$f\square g(x) = \inf\{f(z) + g(x - z) : z \in \mathbb{R}^n\} = \inf\{+\infty + g(x - z) : z \in \mathbb{R}^n\} = +\infty$$

for any $x \in \mathbb{R}^n$. Thus, $\operatorname{epi}(f\square g) = \emptyset$.

□

Exercise 1.25. *Let $f, g\colon \mathbb{R}^n \to (-\infty, +\infty]$ be proper functions. Prove that if $f\square g(x) > -\infty$ for every $x \in \mathbb{R}^n$, then $(f\square g)^* = f^* + g^*$.*

Solution 1.25. Here it is useful to observe that

$$f\square g(x) = \inf\{f(x_1) + g(x_2) : x_1 + x_2 = x\},$$

for each $x \in \mathbb{R}^n$. Since $f\square g > -\infty$, we have that the conjugate $(f\square g)^*$ is well-defined. Hence, for any $v \in \mathbb{R}^n$, we have

$$\begin{aligned}
(f\square g)^*(v) &= \sup\left\{\langle v, x\rangle - f\square g(x) : x \in \mathbb{R}^n\right\} = \\
&= \sup\left\{\langle v, x\rangle - \inf\{f(x_1) + g(x_2) : x_1 + x_2 = x\} : x \in \mathbb{R}^n\right\} = \\
&= \sup\left\{\langle v, x\rangle + \sup\{-f(x_1) - g(x_2) : x_1 + x_2 = x\} : x \in \mathbb{R}^n\right\} = \\
&= \sup\left\{\sup\{\langle v, x_1 + x_2\rangle - f(x_1) - g(x_2) : x_1 + x_2 = x\} : x \in \mathbb{R}^n\right\} = \\
&= \sup\left\{\langle v, x_1 + x_2\rangle - f(x_1) - g(x_2) : x_1, x_2 \in \mathbb{R}^n\right\} = \\
&= \sup\{\langle v, x_1\rangle - f(x_1) : x_1 \in \mathbb{R}^n\} + \sup\{\langle v, x_2\rangle - g(x_2) : x_2 \in \mathbb{R}^n\} = \\
&= f^*(v) + g^*(v),
\end{aligned}$$

and this is the desired equality.

□

Exercise 1.26. *The* Asplund product *of two functions $f, g\colon \mathbb{R}^n \to (-\infty, +\infty]$ is the function $f \star g\colon \mathbb{R}^n \to (-\infty, +\infty]$ defined as*

$$f \star g(x) := \sup\{f(z)g(x - z) : z \in \mathbb{R}^n\}$$

for each $x \in \mathbb{R}^n$.

(a) *The* characteristic function *of a subset $A \subseteq \mathbb{R}^n$ is the function $1_A\colon \mathbb{R}^n \to \{0, 1\}$ given as*

$$1_A(x) = \begin{cases} 1 & \text{if } x \in A \\ 0 & \text{if } x \in \mathbb{R}^n \setminus A \end{cases}.$$

Show that $1_A \star 1_B = 1_{A+B}$ for any sets $A, B \subseteq \mathbb{R}^n$.

(b) *Assume that $f, g\colon \mathbb{R}^n \to (-\infty, +\infty]$ are non-negative functions. Prove that if f is convex or g is convex, then $f \star g$ is convex.*

(c) *Let $\varphi, \psi : \mathbb{R}^n \to (-\infty, +\infty]$ be functions, and consider the functions $f = e^{-\varphi}$ and $g = e^{-\psi}$. Prove that*

$$f \star g = e^{-\varphi \square \psi}.$$

Conclude that if f and g are log-concave and $\varphi \square \psi > -\infty$ in $\mathbb{R}^n$, then $f \star g$ is also log-concave.

Remark 1.11. We also have convexity for $f \star g$ if one of the functions is convex, the other is non-negative, and $f \star g > -\infty$.

Solution 1.26. (a) Let $x \in A + B$. Then, we may write $x = a + b$ with $a \in A$ and $b \in B$. Consequently,

$$1_A \star 1_B(x) = \sup\{1_A(z)1_B(x - z) : z \in \mathbb{R}\} \geq 1_A(a)1_B(x - a) = 1_B(b) = 1.$$

On the other hand, since $1_A 1_B \leq 1$ in $\mathbb{R}^n$, we have that $1_A \star 1_B(x) \leq 1$. Thus,

$$1_A \star 1_B(x) = 1 = 1_{A+B}(x).$$

Now, assume that $x \notin A + B$. In this case, if $z \in A$, then $x - z \notin B$. It follows that $1_A(z)1_B(x - z) = 0$ for every $z \in A$. Since this equality clearly holds if $z \notin A$ (because in this case, $1_A(z) = 0$), we get that

$$1_A \star 1_B(x) = \sup\{1_A(z)1_B(x - z) : z \in \mathbb{R}^n\} = 0 = 1_{A+B}(x),$$

and hence we have the desired equality.

(b) It is immediate that $f \star g = g \star f$, and hence, we may assume that g is convex. For any $x, y, z \in \mathbb{R}^n$ and every $\lambda \in (0, 1)$, we have

$$g((1 - \lambda)x + \lambda y - z) = g((1 - \lambda)(x - z) + \lambda(y - z)) \leq (1 - \lambda)g(x - z) + \lambda g(y - z).$$

Therefore, since f is non-negative, we have

$$f(z)g((1 - \lambda)x + \lambda y - z) \leq (1 - \lambda)f(z)g(x - z) + \lambda f(z)g(y - z).$$

Thus,

$$\begin{aligned} f \star g((1 - \lambda)x + \lambda y) &= \sup\{f(z)g((1 - \lambda)x + \lambda y - z) : z \in \mathbb{R}^n\} \leq \\ &\leq \sup\{(1 - \lambda)f(z)g(x - z) + \lambda f(z)g(y - z) : z \in \mathbb{R}^n\} \leq \\ &\leq (1 - \lambda)\sup\{f(z)g(x - z) : z \in \mathbb{R}^n\} + \lambda \sup\{f(z)g(y - z) : z \in \mathbb{R}^n\} = \\ &= (1 - \lambda)f \star g(x) + \lambda f \star g(y). \end{aligned}$$

Since f and g are non-negative, we also have that $f \star g(x) > -\infty$ for every $x \in \mathbb{R}^n$. Hence, $f \star g$ is convex. Notice that the proof of the claim in Remark 1.11 above also follows immediately.

(c) The result comes directly from the following calculation:

$$\begin{aligned} f \star g(x) &= \sup\left\{f(z)g(x - z) : z \in \mathbb{R}^n\right\} = \sup\left\{e^{-\varphi(z)}e^{-\psi(x-z)} : z \in \mathbb{R}^n\right\} = \\ &= \sup\left\{e^{-\varphi(z)-\psi(x-z)} : z \in \mathbb{R}^n\right\} = \exp\left(\sup\{-\varphi(z) - \psi(x - z) : z \in \mathbb{R}^n\}\right) = \\ &= \exp\left(-\inf\{\varphi(z) + \psi(x - z) : z \in \mathbb{R}^n\}\right) = e^{-\varphi \square \psi(x)}. \end{aligned}$$

We warn the reader that we used the notation $e^x = \exp(x)$ to avoid superscripts that are too extensive. Finally, we notice that if f and g are log-concave, then φ and ψ are convex. This, together with the fact that $\varphi \square \psi > -\infty$, gives that $\varphi \square \psi$ is convex (see item **(a)** of Exercise 1.22). Thus, in this case, we have that $f \star g$ is log-concave.

□

Exercise 1.27. *Remembering the convention that $e^{-\infty} = 0$, verify that $1_A = e^{-I_A^\infty}$. Prove that 1_A is a log-concave function if and only if A is a convex set. Also, use this equality to give a solution to item* **(a)** *of Exercise 1.26.*

Solution 1.27. We have that $e^{-I_A^\infty(x)} = 0$ if and only if $I_A^\infty(x) = +\infty$, and this happens if and only if $x \notin A$. On the other hand, $e^{-I_A^\infty(x)} = 1$ if and only if $I_A^\infty(x) = 0$, which occurs if and only if $x \in A$. Hence, we have the equality $1_A = e^{-I_A^\infty}$.

From the definition, we have that 1_A is log-concave if and only if I_A^∞ is convex, and from Exercise 1.20, it follows that I_A^∞ is convex if and only if A is convex.

Finally, to give a proof of item **(a)** of Exercise 1.26, we use item **(c)** of the same exercise together with item **(b)** of Exercise 1.22 and write

$$1_A \star 1_B = e^{-I_A^\infty \square I_B^\infty} = e^{-I_{A+B}^\infty} = 1_{A+B},$$

which concludes the solution of the exercise.

□

Exercise 1.28. *For a non-negative function $f\colon \mathbb{R}^n \to [0, +\infty]$ and a number $\lambda > 0$, define the function $\lambda \cdot f\colon \mathbb{R}^n \to [0, +\infty]$ given as*

$$\lambda \cdot f(x) = f\left(\frac{x}{\lambda}\right)^\lambda$$

for each $x \in \mathbb{R}^n$.

(a) *For any set $A \subseteq \mathbb{R}^n$, prove that $\lambda \cdot 1_A = 1_{\lambda A}$.*

(b) *For any non-negative functions $f, g\colon \mathbb{R}^n \to [0, +\infty]$ and any number $\lambda \in (0, 1)$, show that*

$$[\lambda \cdot f] \star [(1-\lambda) \cdot g](\lambda x + (1-\lambda)y) \geq f(x)^\lambda g(y)^{1-\lambda}$$

holds for any $x, y \in \mathbb{R}^n$.

(c) *Show that if f and g are integrable and $[\lambda \cdot f] \star [(1-\lambda) \cdot g]$ is measurable, then*

$$\int_{\mathbb{R}^n} [\lambda \cdot f] \star [(1-\lambda) \cdot g]\, dx \geq \left(\int_{\mathbb{R}^n} f\, dx\right)^\lambda \left(\int_{\mathbb{R}^n} g\, dx\right)^{1-\lambda}$$

for every $\lambda \in (0, 1)$.

Remark 1.12. Notice that item **(c)** leads to an interpretation of the Prékopa-Leindler inequality (cf. Chapter 1 of the main book) as an inequality for a type of convolution.

Solution 1.28. **(a)** We have that

$$\lambda \cdot 1_A(x) = 1_A\left(\frac{x}{\lambda}\right)^\lambda.$$

Since $x/\lambda \in A$ if and only if $x \in \lambda A$, we have that $\lambda \cdot 1_A(x) = 1$ if $x \in \lambda A$ and $\lambda \cdot 1_A(x) = 0$ if $x \notin A$. Hence, we have the desired equality.

(b) For any $x, y \in \mathbb{R}^n$ and $\lambda \in (0,1)$, we have

$$[\lambda \cdot f](\lambda x) = f(x)^\lambda \text{ and}$$
$$[(1-\lambda)\cdot g](\lambda x + (1-\lambda)y - \lambda x) = g(y)^{1-\lambda}$$

Now,

$$\begin{aligned}[\lambda \cdot f] \star [(1-\lambda)\cdot g](\lambda x + (1-\lambda)y) = \\ = \sup\left\{\big([\lambda \cdot f](z)\big)\big([(1-\lambda)\cdot g](\lambda x + (1-\lambda)y - z)\big) : z \in \mathbb{R}^n\right\} \geq \\ \geq \big([\lambda \cdot f](\lambda x)\big)\big([(1-\lambda)\cdot g](\lambda x + (1-\lambda)y - \lambda x)\big) = f(x)^\lambda g(y)^{1-\lambda},\end{aligned}$$

where the inequality justifies because the supremum for $z \in \mathbb{R}^n$ is obviously greater than or equal to the particular case where $z = \lambda x$.

(c) From the inequality proved in item **(b)**, the result follows immediately from the Prékopa-Leindler inequality (see again Chapter 1 of the main book).

□

Exercise 1.29. *Let $f \in \mathrm{LC}(\mathbb{R}^n)$. Prove that for any $\alpha, \beta > 0$, it holds that*

$$(\alpha \cdot f) \star (\beta \cdot f) = (\alpha + \beta) \cdot f$$

Solution 1.29. Let $x \in \mathbb{R}^n$. We write

$$[(\alpha \cdot f) \star (\beta \cdot f)](x) = \sup\left\{ f\left(\frac{z}{\alpha}\right)^\alpha f\left(\frac{x-z}{\beta}\right)^\beta : z \in \mathbb{R}^n \right\}.$$

Since f is log-concave, from Lemma 1.5.3 applied for $\lambda := \alpha/(\alpha+\beta)$, we get the bound

$$f\left(\frac{z}{\alpha}\right)^{\frac{\alpha}{\alpha+\beta}} f\left(\frac{x-z}{\beta}\right)^{\frac{\beta}{\alpha+\beta}} \leq f\left(\frac{\alpha}{\alpha+\beta}\frac{z}{\alpha} + \frac{\beta}{\alpha+\beta}\frac{x-z}{\beta}\right) = f\left(\frac{x}{\alpha+\beta}\right).$$

Elevating both sides to $\alpha+\beta$ and passing to the supremum as $z \in \mathbb{R}^n$, we get that

$$(\alpha \cdot f) \star (\beta \cdot f) \leq f\left(\frac{x}{\alpha+\beta}\right)^{\alpha+\beta} = [(\alpha+\beta)\cdot f](x).$$

On the other hand, letting $z_0 = \alpha x/(\alpha+\beta)$, we observe that

$$\begin{aligned}\sup\left\{ f\left(\frac{z}{\alpha}\right)^\alpha f\left(\frac{x-z}{\beta}\right)^\beta : z \in \mathbb{R}^n \right\} \geq \\ \geq f\left(\frac{z_0}{\alpha}\right)^\alpha f\left(\frac{x-z_0}{\beta}\right)^\beta = f\left(\frac{x}{\alpha+\beta}\right)^\alpha f\left(\frac{x}{\alpha+\beta}\right)^\beta = \\ = f\left(\frac{x}{\alpha+\beta}\right)^{\alpha+\beta} = [(\alpha+\beta)\cdot f](x).\end{aligned}$$

That is,

$$[(\alpha \cdot f) \star (\beta \cdot f)](x) \geq [(\alpha + \beta) \cdot f](x).$$

Therefore, we have the desired equality.

□

Exercise 1.30. *Let $\varphi\colon \mathbb{R}^n \to (-\infty, +\infty]$ be a function. Prove that φ is lower semi-continuous if and only if $e^{-\varphi}$ is upper semi-continuous. Conclude that the functions of* $\mathrm{LC}(\mathbb{R}^n)$ *are upper semi-continuous.*

Solution 1.30. Of course, for any $t > 0$, we have

$$\{e^{-\varphi} \geq t\} = \{-\varphi \geq \log(t)\} = \{\varphi \leq -\log(t)\}.$$

If $e^{-\varphi}$ is upper semi-continuous, then the left-hand side of the above is always closed, and hence, all sublevel sets of φ are closed (recall that $-\log$ is a bijection from $[0, +\infty)$ onto $\mathbb{R}$).

On the other hand, if φ is lower semi-continuous, then the equalities above already guarantee that the upper level sets $\{e^{-\varphi} \geq t\}$ are closed for $t > 0$. In the case where $t \leq 0$, it is clear that

$$\{e^{-\varphi} \geq t\} = \mathbb{R}^n.$$

It follows that all upper level sets of $e^{-\varphi}$ are closed, and hence, $e^{-\varphi}$ is upper semi-continuous.

Any function of $\mathrm{LC}(\mathbb{R}^n)$ is of the form $e^{-\varphi}$, where φ is lower semi-continuous (among other properties). Hence, every function of $\mathrm{LC}(\mathbb{R}^n)$ is upper semi-continuous.

□

In what follows, we recall that, from the definition of the integral of simple functions, we have

$$\mathrm{vol}_n(A) = \int_{\mathbb{R}^n} 1_A \, dx,$$

for any Borel (or even Lebesgue) measurable set $A \subseteq \mathbb{R}^n$, where $\mathrm{vol}_n(\cdot)$ denotes the usual Lebesgue measure and dx stands for the integration with respect to that measure (these notations will **always** be adopted in the main book and here). The reader who is not familiar with basic measure and integration theory may find an introduction to this subject in Appendix A of the main book.

Exercise 1.31. *Using Exercise 1.28, show that*

$$\mathrm{vol}_n(\lambda A + (1-\lambda)B) \geq \mathrm{vol}_n(A)^{\lambda} \mathrm{vol}_n(B)^{1-\lambda} \tag{1.2}$$

for any compact sets $A, B \subseteq \mathbb{R}^n$ and every number $\lambda \in (0, 1)$.

Remark 1.13. Inequality (1.2) is the classical *Brunn-Minkowski inequality*. This is the most important inequality in convex geometry, and it is intensively studied in Section 5.1 of the main book.

Solution 1.31. From item **(a)** of Exercise 1.26 and item **(a)** of Exercise 1.28, we have

$$[\lambda \cdot 1_A] \star [(1-\lambda) \cdot 1_B] = 1_{\lambda A} \star 1_{(1-\lambda)B} = 1_{\lambda A + (1-\lambda)B}$$

for any $\lambda \in (0,1)$ and any subsets $A, B \subseteq \mathbb{R}^n$. Since we are assuming that A and B are compact, we have that 1_A and 1_B are integrable. Moreover, it is not difficult to see that $\lambda A + (1-\lambda)B$ is also compact, from where we get that $1_{\lambda A+(1-\lambda)B}$ is measurable (integrable, in fact). Thus, using item **(c)** of Exercise 1.28, we get

$$\begin{aligned}\mathrm{vol}_n(\lambda A + (1-\lambda)B) = \int_{\mathbb{R}^n} 1_{\lambda A+(1-\lambda)B}\, dx = \int_{\mathbb{R}^n} [\lambda \cdot 1_A] \star [(1-\lambda)\cdot 1_B]\, dx \geq \\ \left(\int_{\mathbb{R}^n} 1_A\, dx\right)^{\lambda} \left(\int_{\mathbb{R}^n} 1_B\, dx\right)^{1-\lambda} = \mathrm{vol}_n(A)^{\lambda}\mathrm{vol}_n(B)^{1-\lambda},\end{aligned}$$

and this concludes the proof.

□

Remark 1.14. Notice that we do not need A and B to be compact, but only measurable with finite Lebesgue measure.

Exercise 1.32. *Prove that the function*

$$\varphi(x) = \frac{|x|^2}{2}$$

is the unique proper function from $\mathbb{R}^n$ *to* $(-\infty, +\infty]$ *such that* $\varphi^* = \varphi$.

Solution 1.32. We first prove that $\varphi(x) = |x|^2/2$ is such that $\varphi^* = \varphi$, and after that φ is the unique proper function with this property. For each $y \in \mathbb{R}^n$, we write

$$\varphi^*(y) = \sup\left\{\langle x, y\rangle - \frac{|x|^2}{2} : x \in \mathbb{R}^n\right\} = \sup\left\{\left\langle x, y - \frac{x}{2}\right\rangle : x \in \mathbb{R}^n\right\}.$$

Hence, to calculate $\varphi^*(y)$, we have to maximize the function $F\colon \mathbb{R}^n \to \mathbb{R}$ given as

$$F(x) = \left\langle x, y - \frac{x}{2}\right\rangle.$$

This function is clearly differentiable, and writing (in coordinates) $x = (x_1, \ldots, x_n)$ and $y = (y_1, \ldots, y_n)$, we get immediately that

$$\begin{aligned}\nabla F(x) = (y_1 - x_1, \ldots, y_n - x_n) \text{ and} \\ \mathrm{hess} F(x) = -\mathrm{Id}_{\mathbb{R}^n}.\end{aligned}$$

Therefore, the *Hessian matrix* $\mathrm{hess}F$ of F is negative definite at each point of $\mathbb{R}^n$, from where we get that F is concave. Consequently, any local maximum of F is a global maximum (see Remark 1.1.2 of the main book). Now, we observe that $\nabla F(y) = 0$, and $\mathrm{hess}F(y)$ is negative definite. Thus, y is a local maximum of F, and hence a global maximum. It follows that

$$\varphi^*(y) = F(y) = \left\langle y, y - \frac{y}{2}\right\rangle = \frac{|y|^2}{2} = \varphi(y)$$

for any $y \in \mathbb{R}^n$, as we wanted to prove.

Now, we prove that $\varphi(x) = |x|^2/2$ is the unique proper function such that $\varphi^* = \varphi$. For that sake, assume that ψ is a proper function such that $\psi^* = \psi$. Hence, for any $y \in \mathbb{R}^n$,

$$\psi(y) = \psi^*(y) = \sup\left\{\langle x, y\rangle - \psi(x) : x \in \mathbb{R}^n\right\} \geq \langle y, y\rangle - \psi(y),$$

from where we get the inequality

$$\psi(y) \geq \frac{|y|^2}{2}$$

for each $y \in \mathbb{R}^n$. Having this inequality in mind, we observe that

$$\langle x, y\rangle - \psi(x) \leq \langle x, y\rangle - \frac{|x|^2}{2} = \left\langle x, y - \frac{x}{2}\right\rangle$$

for any $x, y \in \mathbb{R}^n$. Thus,

$$\psi^*(y) = \sup\left\{\langle x, y\rangle - \psi(x) : x \in \mathbb{R}^n\right\} \leq \sup\left\{\left\langle x, y - \frac{x}{2}\right\rangle : x \in \mathbb{R}^n\right\},$$

and we already know that the right-hand side supremum is attained for $x = y$. Then,

$$\psi^*(y) \leq \left\langle y, y - \frac{y}{2}\right\rangle = \frac{|y|^2}{2},$$

and the proof is complete.

□

Exercise 1.33. *Let $\varphi\colon \mathbb{R}^n \to (-\infty, +\infty]$ be a function, and assume that $f = e^{-\varphi}$ does not vanish identically. We define the* support function *of f to be the function $h_f\colon \mathbb{R}^n \to (-\infty, +\infty]$ given as*

$$h_f := (-\log f)^*.$$

(a) *Show that we indeed have $h_f > -\infty$ everywhere.*

(b) *Prove that*

$$h_{\lambda\cdot f} = \lambda h_f$$

for any number $\lambda > 0$, where $\lambda \cdot f$ stands for the scalar multiplication defined in Exercise 1.28.

(c) *Let $A \subseteq \mathbb{R}^n$. Show that*

$$h_{1_A}(u) = \sup\{\langle u, x\rangle : x \in A\}$$

for any $u \in \mathbb{R}^n$.

(d) *Let $\varphi, \psi\colon \mathbb{R}^n \to (-\infty, +\infty]$ be proper functions such that $\varphi\square\psi > -\infty$ in $\mathbb{R}^n$. Letting $f = e^{-\varphi}$ and $g = e^{-\psi}$, prove that*

$$h_{f\star g} = h_f + h_g.$$

Solution 1.33. **(a)** Since f does not vanish everywhere, there exists $x_0 \in \mathbb{R}^n$ such that $f(x_0) \neq 0$. Hence, $\varphi(x_0) < \infty$. Therefore, for every $v \in \mathbb{R}^n$, we have

$$h_f(v) = (-\log f)^*(v) = \varphi^*(v) = \sup\{\langle v, x\rangle - \varphi(x) : x \in \mathbb{R}^n\} \geq \langle v, x_0\rangle - \varphi(x_0) > -\infty.$$

(b) We have

$$\lambda \cdot f(x) = f\left(\frac{x}{\lambda}\right)^{\lambda} = e^{-\lambda\varphi(x/\lambda)},$$

for each $x \in \mathbb{R}^n$. Then,

$$h_{\lambda\cdot f}(v) = (-\log(\lambda \cdot f))^*(v) = (\lambda\varphi(\cdot/\lambda))^*(v) = \sup\{\langle v, x\rangle - \lambda\varphi(x/\lambda) : x \in \mathbb{R}^n\} =$$
$$= \lambda \sup\{\langle v, x/\lambda\rangle - \varphi(x/\lambda) : x \in \mathbb{R}^n\} = \lambda \sup\{\langle v, z\rangle - \varphi(z) : z \in \mathbb{R}^n\} = \lambda\varphi^*(v) = \lambda h_f(v)$$

for each $v \in \mathbb{R}^n$. Notice that since $\lambda > 0$, we can replace x/λ by z to take the supremum.

(c) We have that $1_A = e^{-I_A^\infty}$. Thus, for every $u \in \mathbb{R}^n$, we have

$$h_{1_A}(u) = (-\log 1_A)^*(u) = (I_A^\infty)^*(u) = \sup\{\langle u, x\rangle : x \in A\},$$

where the last equality comes from item **(c)** of Exercise 1.20.

(d) From item **(d)** of Exercise 1.26, we have $f \star g = e^{-\varphi\square\psi}$. Using Exercise 1.25, since $\varphi\square\psi > -\infty$, we get that $(\varphi\square\psi)^* = \varphi^* + \psi^*$. Thus,

$$h_{f\star g} = \left(-\log e^{-\varphi\square\psi}\right)^* = (\varphi\square\psi)^* = \varphi^* + \psi^* = h_f + h_g,$$

and this concludes the proof.

□

Exercise 1.34. *Let $\varphi\colon \mathbb{R}^n \to (-\infty, +\infty]$ be a lower semi-continuous convex function such that*

$$\lim_{|x|\to+\infty} \varphi(x) = +\infty,$$

meaning that for every number $c \in \mathbb{R}$, there exists a number $\rho > 0$ such that $\varphi(x) > c$ whenever $|x| > \rho$.

(a) *Prove that there exist constants $a > 0$ and $b \in \mathbb{R}$ such that*

$$\varphi(x) \geq a|x| + b$$

for every $x \in \mathbb{R}^n$.

(b) *Let $f = e^{-\varphi}$. Show that*

$$\int_{\mathbb{R}^n} f\,dx < +\infty.$$

Remark 1.15. A log-concave function is always measurable because for any $a \in \mathbb{R}$, the set $\{f \geq a\}$ is convex (and hence Lebesgue measurable; see Corollary A.4.3 of the main book). It is relevant to know when the integral of a log-concave function f is finite (i.e., f is *integrable*; see Appendix A in the main book), since when one studies log-concave functions from the geometric perspective, the total integral "plays the role" of volume. For more in this direction, we refer the reader to Colesanti [79].

Solution 1.34. (a) First, we will prove that φ is bounded from below. Let $r > 0$ be such that $\varphi(x) > 1$ whenever $|x| > r$. The closed ball rB^n is compact, and since φ is lower semi-continuous, it follows from item **(iii)** of Proposition 1.1.4 of the main book that there exists a point $z_0 \in rB^n$ such that

$$\varphi(z_0) = \inf\{\varphi(x) : x \in rB^n\}.$$

Therefore, $\varphi(x) \geq \min\{\varphi(z_0), 1\}$ for every $x \in \mathbb{R}^n$. Thus, we have that φ is bounded from below, and we denote

$$m := \inf\{\varphi(x) : x \in \mathbb{R}^n\}.$$

Moving forward, we consider the particular case where $0_{\mathbb{R}^n} \in \operatorname{dom}(f)$. In this case, we let $\rho > 0$ be such that $\varphi(x) > \varphi(0_{\mathbb{R}^n}) + 1$ if $|x| \geq \rho$. For any $x \in \mathbb{R}^n$ such that $|x| > \rho$, we have

$$\begin{aligned}\varphi\left(\frac{\rho}{|x|}x\right) = \varphi\left(\frac{\rho}{|x|}x + \left(1 - \frac{\rho}{|x|}\right)0_{\mathbb{R}^n}\right) &\leq \frac{\rho}{|x|}\varphi(x) + \left(1 - \frac{\rho}{|x|}\right)\varphi(0_{\mathbb{R}^n}) = \\ &= \varphi(0_{\mathbb{R}^n}) + \frac{\rho}{|x|}\big(\varphi(x) - \varphi(0_{\mathbb{R}^n})\big).\end{aligned}$$

Rearranging the terms and using the fact that $\varphi(\rho x/|x|) > \varphi(0_{\mathbb{R}^n}) + 1$ (because $|\rho x/|x|| = \rho$), we get

$$\begin{aligned}\varphi(x) \geq \frac{|x|}{\rho}\left(\varphi\left(\frac{\rho}{|x|}x\right) - \varphi(0_{\mathbb{R}^n})\right) + \varphi(0_{\mathbb{R}^n}) &\geq \frac{|x|}{\rho}\big((\varphi(0_{\mathbb{R}^n}) + 1 - \varphi(0_{\mathbb{R}^n})\big) + \varphi(0_{\mathbb{R}^n}) = \\ &= \frac{|x|}{\rho} + \varphi(0_{\mathbb{R}^n}) \geq \frac{|x|}{\rho} + m > \frac{|x|}{\rho} + m - 1.\end{aligned}$$

On the other hand, if $|x| \leq \rho$, then

$$\frac{|x|}{\rho} + m - 1 \leq m \leq \varphi(x).$$

Therefore, letting $a = 1/\rho$ and $b = m - 1$, we have

$$\varphi(x) \geq a|x| + b$$

for any $x \in \mathbb{R}^n$.

(b) Let $a > 0$ and $b \in \mathbb{R}$ be as before. Then, since $\varphi(x) \geq a|x| + b$, we have

$$f(x) \leq e^{-a|x|-b}.$$

Hence, using integration by polar coordinates, we get

$$\begin{aligned}\int_{\mathbb{R}^n} f(x)\,dx \leq e^{-b}\int_{\mathbb{R}^n} e^{-a|x|}\,dx = e^{-b}\int_0^{+\infty}\int_{\mathbb{S}^{n-1}} e^{-as}s^{n-1}\,d\mathcal{H}^{n-1}ds = \\ = \frac{e^{-b}\mathcal{H}^{n-1}(\mathbb{S}^{n-1})}{a^n}\int_0^{+\infty} e^{-t}t^{n-1}\,dt = \frac{e^{-b}\mathcal{H}^{n-1}(\mathbb{S}^{n-1})}{a^n}(n-1)! < +\infty.\end{aligned}$$

We use the fact that

$$\int_0^{+\infty} e^{-t} t^{n-1}\, dt = (n-1)!$$

is an easy calculus exercise. Indeed, this integral is an expression of the usual Gamma function, which is used in Section 4.3 of the main book. For the convenience of the reader, we do the calculations. It suffices to prove that

$$\int_0^{+\infty} e^{-t} t\, dt = 1 \quad \text{and}$$

$$\int_0^{+\infty} e^{-t} t^n\, dt = n \int_0^{\infty} e^{-t} t^{n-1}\, dt, \quad \text{for each integer } n \geq 2.$$

For the first equality, we simply notice that $-e^{-t}t - e^{-t}$ is a primitive of $e^{-t}t$. Hence,

$$\int_0^{+\infty} e^{-t} t\, dt = \left(e^{-t}t - e^{-t}\right)\Big|_0^{+\infty} = 1.$$

For the second equality, we use integration by parts to obtain

$$\int_0^{+\infty} e^{-t} t^n\, dt = -e^{-t}t^n\Big|_0^{+\infty} + \int_0^{+\infty} e^{-t} n t^{n-1}\, dt = n \int_0^{+\infty} e^{-t} t^{n-1}\, dt.$$

These equalities together with an induction argument lead immediately to

$$\int_0^{+\infty} e^{-t} t^{n-1}\, dt = (n-1)!$$

for every natural number $n \geq 1$, as we have claimed.

□

Now that we gave a sufficient condition under which a log-concave function has "finite volume", we investigate when this "volume" is positive. For that sake, we define the *support* of a function $f\colon \mathbb{R}^n \to (-\infty, +\infty]$ to be the set

$$\operatorname{spt}(f) := \left\{x \in \mathbb{R}^n : f(x) \neq 0\right\}.$$

Notice that for any functions $\varphi\colon \mathbb{R}^n \to (-\infty, +\infty]$ and $f = e^{-\varphi}$, we have that $\operatorname{spt}(f) = \operatorname{dom}(\varphi)$, because $f(x) \neq 0$ if and only if $\varphi(x) \neq +\infty$.

Exercise 1.35. *Let $\varphi\colon \mathbb{R}^n \to (-\infty, +\infty]$ be a convex function, and let $f = e^{-\varphi}$. Using the fact that any convex set with empty interior has zero Lebesgue measure, prove that*

$$\int_{\mathbb{R}^n} f\, dx = 0$$

if and only if $\operatorname{int}(\operatorname{dom}(\varphi)) = \emptyset$ *(or, equivalently, if and only if* $\operatorname{int}(\operatorname{spt}(f)) = \emptyset$*).*

Remark 1.16. Since $\operatorname{dom}(\varphi)$ is convex, it holds that $\operatorname{int}(\operatorname{dom}(\varphi)) = \emptyset$ if and only if $\operatorname{dom}(\varphi)$ is contained in some hyperplane of $\mathbb{R}^n$. This is clarified in Chapter 2 of the main book. It turns out that any subset of a hyperplane of $\mathbb{R}^n$ has zero Lebesgue measure; see, again in the main book, Corollary A.3.2 and Remark A.3.1.

Solution 1.35. First, observe that

$$\int_{\mathbb{R}^n} f\,dx = \int_{\mathrm{spt}(f)} f\,dx = \int_{\mathrm{dom}(\varphi)} f\,dx.$$

If $\mathrm{int}(\mathrm{dom}(\varphi)) = \emptyset$, then the Lebesgue measure of $\mathrm{dom}(\varphi)$ is zero. In this case, the right-hand side of the equality above vanishes, and hence, the integral of f over $\mathbb{R}^n$ is 0.

Conversely, if the interior of $\mathrm{dom}(\varphi)$ is non-empty, then $\mathrm{int}(\mathrm{dom}(\varphi))$ is an open and non-empty set. Thus, it contains some open ball, and then it also contains a (nondegenerate) closed ball B. Since any closed ball is a compact set and because φ is continuous in $\mathrm{int}(\mathrm{dom}(\varphi))$ (and hence in B), we get that there exists $m \in \mathbb{R}$ such that

$$\varphi(x) \leq m$$

for each $x \in B$. This leads to the inequality

$$f(x) = e^{-\varphi(x)} \geq e^{-m}$$

for any $x \in B$. Finally, since $f \geq 0$ in $\mathbb{R}^n$, we obtain

$$\int_{\mathbb{R}^n} f\,dx \geq \int_B f\,dx \geq \int_B e^{-m}\,dx = e^{-m}\mathrm{vol}_n(B) > 0,$$

and the proof is complete.

□

Now, we define the following families of functions:

$$\begin{aligned}
\mathrm{Cvx}(\mathbb{R}^n) &:= \Big\{\varphi\colon \mathbb{R}^n \to (-\infty,+\infty] : \varphi \text{ is closed, convex and proper}\Big\},\\
\mathrm{Cvx}_\infty(\mathbb{R}^n) &:= \Big\{\varphi\colon \mathbb{R}^n \to (-\infty,+\infty] : \varphi \in \mathrm{Cvx}(\mathbb{R}^n) \text{ and } \lim_{|x|\to+\infty} \varphi(x) = +\infty\Big\},\\
\mathrm{LC}(\mathbb{R}^n) &:= \Big\{f\colon \mathbb{R}^n \to [0,+\infty) : f = e^{-\varphi} \text{ for some } \varphi \in \mathrm{Cvx}(\mathbb{R}^n)\Big\} \text{ and}\\
\mathrm{LC}_\infty(\mathbb{R}^n) &:= \Big\{f\colon \mathbb{R}^n \to [0,+\infty) : f = e^{-\varphi} \text{ for some } \varphi \in \mathrm{Cvx}_\infty(\mathbb{R}^n)\Big\}.
\end{aligned}$$

Notice that, in particular, Exercise 1.34 gives that any function of $\mathrm{LC}_\infty(\mathbb{R}^n)$ is integrable (measurability comes from Remark 1.15).

Exercise 1.36. *Let $\varphi\colon \mathbb{R}^n \to (-\infty,+\infty]$ be proper and lower semi-continuous, and assume that*

$$\lim_{|x|\to+\infty} \varphi(x) = +\infty,$$

in the same sense as in Exercise 1.34. Prove that φ attains a global minimum. Show that every $\varphi \in \mathrm{Cvx}_\infty(\mathbb{R}^n)$ attains a global minimum and that every function $f \in \mathrm{LC}_\infty(\mathbb{R}^n)$ attains a global maximum.

Solution 1.36. We first show that φ is bounded from below. We follow the same argument as in item **(a)** of Exercise 1.34. Let $r > 0$ be such that $\varphi(x) > 1$ whenever $|x| > r$. Also, since φ is lower semi-continuous, we have that φ attains a minimum (m, say) over the compact set rB^n (see item **(iii)** of Proposition 1.1.4 in the main book). Then, the number $\min\{1, m\}$ is a global lower bound for φ.

Let $m_0 := \inf_{\mathbb{R}^n} \varphi$, and let $\rho > 0$ be such that $\varphi(x) > m_0 + 1$ whenever $|x| > \rho$. Using again item **(iii)** of Proposition 1.1.4 from the main book, we see that φ attains a minimum over the compact set ρB^n. That is,

$$\varphi(x_0) = \inf\left\{\varphi(x) : x \in \rho B^n\right\}$$

for some $x_0 \in \rho B^n$. Because $\varphi(x) > m_0 + 1$ if $x \notin \rho B^n$, we get that

$$\varphi(x_0) = \inf\left\{\varphi(x) : x \in \rho B^n\right\} = \inf\left\{\varphi(x) : x \in \mathbb{R}^n\right\} = m_0,$$

and this gives the desired result.

The claims for functions of $\mathrm{Cvx}_\infty(\mathbb{R}^n)$ and of $\mathrm{LC}_\infty(\mathbb{R}^n)$ follow straightforwardly.

□

Exercise 1.37. **(a)** *Verify that* $\mathrm{Cvx}(\mathbb{R}^n)$ *is closed under conjugation; that is, if* $\varphi \in \mathrm{Cvx}(\mathbb{R}^n)$*, then* $\varphi^* \in \mathrm{Cvx}(\mathbb{R}^n)$.

(b) *Show that* $\mathrm{Cvx}_\infty(\mathbb{R}^n)$ *is not closed under conjugation; that is, give an example of a function* $\varphi \in \mathrm{Cvx}_\infty(\mathbb{R}^n)$ *such that* $\varphi^* \notin \mathrm{Cvx}_\infty(\mathbb{R}^n)$.

(c) *Prove that* $\mathrm{Cvx}_\infty(\mathbb{R}^n)$ *is closed for infimal convolution; that is, show that if* $\varphi, \psi \in \mathrm{Cvx}_\infty(\mathbb{R}^n)$*, then* $\varphi\square\psi \in \mathrm{Cvx}_\infty(\mathbb{R}^n)$. *Check that this is not true for* $\mathrm{Cvx}(\mathbb{R}^n)$.

(d) *Verify that* $\mathrm{LC}_\infty(\mathbb{R}^n)$ *is closed for Asplund products.*

Remark 1.17. If we denote by $F : \mathrm{Cvx}(\mathbb{R}^n) \to \mathrm{Cvx}(\mathbb{R}^n)$ the map that associates each $\varphi \in \mathrm{Cvx}(\mathbb{R}^n)$ to its conjugate φ^*, then from Theorem 1.4.1 in the main book, we have that F is an *involution*, meaning that F is a bijection such that $F \circ F = \mathrm{Id}_{\mathrm{Cvx}(\mathbb{R}^n)}$.

Solution 1.37. **(a)** This result is proved in Theorem 1.4.1 of the main book.

(b) Consider $\varphi = I^\infty_{\{0_{\mathbb{R}^n}\}}$. From item **(c)** of Exercise 1.20, we have

$$\varphi^* = \left(I^\infty_{\{0_{\mathbb{R}^n}\}}\right)^* = \sup\left\{\langle \cdot, x\rangle : x \in \{0_{\mathbb{R}^n}\}\right\} = 0,$$

and hence, we do not have that $\lim_{|x|\to\infty} \varphi^*(x) = +\infty$, and then $\varphi^* \notin \mathrm{Cvx}_\infty(\mathbb{R}^n)$.

(c) From Exercise 1.34, we know that φ, ψ are bounded from below. Thus,

$$\varphi\square\psi(x) = \inf\left\{\varphi(z) + \psi(x - z) : z \in \mathbb{R}^n\right\} \geq \inf_{\mathbb{R}^n} \varphi + \inf_{\mathbb{R}^n} \psi > -\infty.$$

Thus, the co-domain of $\varphi\square\psi$ is $(-\infty, +\infty]$, and from item **(a)** of Exercise 1.22, we get that $\varphi\square\psi$ is convex.

Now, since φ and ψ are proper, we may choose $x_0, y_0 \in \mathbb{R}^n$ such that $\varphi(x_0) < +\infty$ and $\psi(y_0) < +\infty$. Then,

$$\varphi\square\psi(x_0 + y_0) \leq \varphi(x_0) + \psi(y_0) < +\infty,$$

and hence, $\varphi\square\psi$ is proper.

Now, we prove that $\varphi\square\psi$ is closed. For that sake, we show that $\mathrm{epi}(\varphi\square\psi)$ is a closed subset of $\mathbb{R}^{n+1}$. Let (x_m, t_m) be a sequence in $\mathrm{epi}(\varphi\square\psi)$ such that $(x_m, t_m) \to (x, t)$ as $m \to \infty$, and up to discarding

finitely many terms, assume that $t_m < t + 1$ for every $m \in \mathbb{N}$. We know that, for every $m \in \mathbb{N}$, it holds that

$$\varphi \square \psi(x_m) = \inf \left\{ \varphi(z) + \psi(x_m - z) : z \in \mathbb{R}^n \right\} \leq t_m.$$

Then, for each $m \in \mathbb{N}$, there exists a point $z_m \in \mathbb{R}^n$ such that

$$\varphi(z_m) + \psi(x_m - z_m) \leq t_m + \frac{1}{m}. \tag{1.3}$$

We claim that there exists $\rho > 0$ such that $z_m \in \rho B^n$ for every $m \in \mathbb{N}$. Indeed, since $\varphi \in \mathrm{Cvx}_\infty(\mathbb{R}^n)$, there exists $\rho > 0$ such that $|z| > \rho$ implies

$$\varphi(z) > -\inf_{\mathbb{R}^n} \psi + t + 2.$$

Then, if $|z| > \rho$, we have

$$\varphi(z) + \psi(x_m - z) > -\inf_{\mathbb{R}^n} \psi + t + 2 + \inf_{\mathbb{R}^n} \psi = t + 2 > t_m + \frac{1}{m},$$

from where we cannot have $|z_m| > \rho$. Therefore, since ρB^n is compact, passing to a subsequence if necessary, we may assume that $z_m \to z_0 \in \mathbb{R}^n$ as $m \to \infty$. Because φ and ψ are lower semi-continuous, we get from item **(d)** of Proposition 1.1.3 in the main book that

$$\varphi(z_0) + \psi(x - z_0) \leq \liminf_{m \to \infty} \left[\varphi(z_m) + \psi(x_m - z_m) \right] \leq \liminf_{m \to \infty} \left(t_m + \frac{1}{m} \right) = t.$$

Finally, this leads to

$$\varphi \square \psi(x) = \inf \left\{ \varphi(z) + \psi(x - z) : z \in \mathbb{R}^n \right\} \leq \varphi(z_0) + \psi(x - z_0) \leq t,$$

and hence, $(x, t) \in \mathrm{epi}(\varphi \square \psi)$. This shows that $\varphi \square \psi$ is lower semi-continuous.

It only remains to prove that

$$\lim_{|y| \to \infty} \varphi \square \psi(y) = +\infty.$$

For that sake, let

$$m = \min \left\{ \inf_{\mathbb{R}^n} \varphi, \inf_{\mathbb{R}^n} \psi \right\},$$

and for any number $c \in \mathbb{R}$, let $\rho > 0$ be such that $|x| > \rho$ implies $\varphi(x) > c - m$ and $\psi(x) > c - m$. Let $y \in \mathbb{R}^n$ be any vector such that $|y| > 2\rho$. If $z \in \mathbb{R}^n$ is such that $|z| > \rho$, then

$$\varphi(z) + \psi(y - z) > c - m + m = c.$$

On the other hand, if $|z| \leq \rho$, then we have

$$|y - z| \geq |y| - |z| > 2\rho - \rho = \rho,$$

from where we get

$$\varphi(z) + \psi(y - z) > c - m + m = c.$$

Thus, for any $|y| > 2\rho$, we have

$$\varphi\square\psi(y) = \inf\left\{\varphi(z) + \psi(y - z) : z \in \mathbb{R}^n\right\} \geq c,$$

and this concludes the proof. To show that $\mathrm{Cvx}(\mathbb{R}^n)$ is not closed for infimal convolution, simply use the same example as constructed in item **(c)** of Exercise 1.22.

(d) Since $f, g \in \mathrm{LC}_\infty(\mathbb{R}^n)$, we get that $\varphi \in \mathrm{Cvx}_\infty(\mathbb{R}^n)$ and $\psi \in \mathrm{Cvx}_\infty(\mathbb{R}^n)$. From the previous item, it follows that $\varphi\square\psi \in \mathrm{Cvx}_\infty(\mathbb{R}^n)$. From item **(c)** of Exercise 1.26, we also have

$$f \star g = e^{-\varphi\square\psi},$$

and hence, $f \star g \in \mathrm{LC}_\infty(\mathbb{R}^n)$.

□

Exercise 1.38. *Verify that $h_f \in \mathrm{Cvx}(\mathbb{R}^n)$ for every $f \in \mathrm{LC}(\mathbb{R}^n)$. Let $H\colon \mathrm{LC}(\mathbb{R}^n) \to \mathrm{Cvx}(\mathbb{R}^n)$ be the mapping that associates each $f \in \mathrm{LC}(\mathbb{R}^n)$ to its support function $H(f) := h_f \in \mathrm{Cvx}(\mathbb{R}^n)$. Prove that H is a bijection.*

Solution 1.38. If $f \in \mathrm{LC}(\mathbb{R}^n)$, then $f = e^{-\varphi}$ for some $\varphi \in \mathrm{Cvx}(\mathbb{R}^n)$. Then, $h_f = \varphi^*$, and from item **(a)** of Exercise 1.37, we have that $\varphi^* \in \mathrm{Cvx}(\mathbb{R}^n)$.

Now, let $f, g \in \mathrm{LC}(\mathbb{R}^n)$ with $f = e^{-\varphi}$ and $g = e^{-\psi}$. If $h_f = h_g$, then $\varphi^* = \psi^*$. Therefore, $\varphi^{**} = \psi^{**}$. Since φ and ψ are convex, proper and closed, Theorem 1.4.1 in the main book gives that $\varphi^{**} = \varphi$ and $\psi^{**} = \psi$, from where we get that $\varphi = \psi$. Thus, $f = g$, and this shows that H is injective.

It remains to prove that H is surjective. Let $\varphi \in \mathrm{Cvx}(\mathbb{R}^n)$. Then, $\varphi^* \in \mathrm{Cvx}(\mathbb{R}^n)$, and hence, $f = e^{-\varphi^*} \in \mathrm{LC}(\mathbb{R}^n)$. Using Theorem 1.4.1 again, we get that $\varphi^{**} = \varphi$. Thus,

$$H(f) = h_f = (\varphi^*)^* = \varphi,$$

and this concludes the proof.

□

Exercise 1.39. *Let $f\colon \mathbb{R}^n \to [0, +\infty)$, and write $f = e^{-\varphi}$. We define the* polar function *of f to be the function*

$$f^\circ := e^{-\varphi^*} = e^{-h_f}.$$

(a) *Prove that the map that associates each function $f \in \mathrm{LC}(\mathbb{R}^n)$ to its polar function is an involution of $\mathrm{LC}(\mathbb{R}^n)$.*

(b) *Show that for any $f \in \mathrm{LC}(\mathbb{R}^n)$, we have the equality*

$$f^\circ(y) = \inf\left\{\frac{e^{-\langle x,y\rangle}}{f(x)} : x \in \mathbb{R}^n\right\}$$

for every $y \in \mathbb{R}^n$, where the quotient must be interpreted as $+\infty$ whenever $f(x) = 0$.

(c) *Prove that for* $f = e^{-|x|^2/2}$*, we have* $f^\circ = f$*. Even more, show that this function is the unique function of* $\mathrm{LC}(\mathbb{R}^n)$ *having this property.*

Solution 1.39. (a) First, notice that since $\varphi^* \in \mathrm{Cvx}(\mathbb{R}^n)$ whenever $\varphi \in \mathrm{Cvx}(\mathbb{R}^n)$, we have that $f^\circ \in \mathrm{LC}(\mathbb{R}^n)$ for every $f \in \mathrm{LC}(\mathbb{R}^n)$. If $f, g \in \mathrm{LC}(\mathbb{R}^n)$ are such that $f^\circ = g^\circ$, then writing $f = e^{-\varphi}$ and $g = e^{-\psi}$, we get $\varphi^* = \psi^*$, and hence, $\varphi = \psi$. This leads to $f = g$, and hence, the map that associates each function of $\mathrm{LC}(\mathbb{R}^n)$ to its polar function is injective. Also, if $f = e^{-\varphi}$, then $f^\circ = e^{-\varphi^*}$ and

$$(f^\circ)^\circ = e^{-\varphi^{**}} = e^{-\varphi} = f.$$

Since this implies, in particular, that the map $f \mapsto f^\circ$ is surjective, we have the desired equality.

(b) Write $f = e^{-\varphi}$. Then,

$$f^\circ(y) = \exp\big(-\sup\{\langle y, x\rangle - \varphi(x) : x \in \mathbb{R}^n\}\big) = \exp\big(\inf\{-\langle x, y\rangle + \varphi(x) : x \in \mathbb{R}^n\}\big) =$$
$$= \inf\left\{e^{-\langle x,y\rangle+\varphi(x)} : x \in \mathbb{R}^n\right\} = \inf\left\{\frac{e^{-\langle x,y\rangle}}{e^{-\varphi(x)}} : x \in \mathbb{R}^n\right\} = \inf\left\{\frac{e^{-\langle x,y\rangle}}{f(x)} : x \in \mathbb{R}^n\right\}.$$

(c) If $\varphi \in \mathrm{Cvx}(\mathbb{R}^n)$ and $f = e^{-\varphi}$, then we have that $f^\circ = f$ if and only if $\varphi^* = \varphi$. Hence, the result follows from Exercise 1.32.

□

Exercise 1.40. *Let* $f\colon \mathbb{R}^n \to [0, +\infty]$ *be a non-negative function, and put* $\bar{f}(x) = f(-x)$ *for each* $x \in \mathbb{R}^n$*. Define the function* Δf *by*

$$\Delta f := \left(\frac{1}{2} \cdot f\right) \star \left(\frac{1}{2} \cdot \bar{f}\right).$$

(a) *Prove that*

$$\Delta f(x) = \sup\left\{\sqrt{f(z)f(-y)} : z, y \in \mathbb{R}^n,\ x = \frac{z+y}{2}\right\}$$

for each $x \in \mathbb{R}^n$*. Conclude that* Δf *is even.*

(b) *Verify that if* $f \in \mathrm{LC}_\infty(\mathbb{R}^n)$*, then* $\bar{f} \in \mathrm{LC}_\infty(\mathbb{R}^n)$*.*

(c) *Show that if* $f \in \mathrm{LC}_\infty(\mathbb{R}^n)$*, then* $\Delta f \in \mathrm{LC}_\infty(\mathbb{R}^n)$*.*

(d) *For any sets* $A, B \subseteq \mathbb{R}^n$*, denote by* $A - B$ *the* Minkowski difference *(or simply* difference*) of* A *and* B*, which is the Minkowski sum of* A *and* $-B$*. Show that*

$$\Delta 1_A = 1_{\frac{A-A}{2}}$$

for any set $A \subseteq \mathbb{R}^n$*.*

Remark 1.18. The subcase $A - A$ is called the *difference body*, and $\frac{1}{2}(A - A)$ the *central symmetral*, of A. These concepts are investigated in Chapters 2 and 10 of the main book.

Solution 1.40. **(a)** We have

$$\Delta f(x) = \left(\frac{1}{2}\cdot f\right) \star \left(\frac{1}{2}\cdot \bar{f}\right)(x) = \sup\left\{f(2w)^{\frac{1}{2}}\bar{f}(2(x-w))^{\frac{1}{2}} : w \in \mathbb{R}^n\right\} =$$
$$= \sup\left\{\sqrt{f(z)\bar{f}(y)} : z, y \in \mathbb{R}^n,\ x = \frac{y+z}{2}\right\} = \sup\left\{\sqrt{f(z)f(-y)} : z, y \in \mathbb{R}^n,\ x = \frac{y+z}{2}\right\}.$$

Notice that for the second equality, we simply wrote $z = 2w$, which gives a one-to-one correspondence of $\mathbb{R}^n$ (hence, taking the supremum as $w \in \mathbb{R}^n$ is the same as taking the supremum as $z \in \mathbb{R}^n$). In addition, we wrote $2x - 2w = y$, from where we obtain $x = (y+z)/2$.

Now, we argue how this expression gives that Δf is an even function. Indeed,

$$\Delta f(-x) = \sup\left\{\sqrt{f(z)f(-y)} : -x = \frac{y+z}{2}\right\} = \sup\left\{\sqrt{f(-(-z))f(-y)} : x = \frac{-y-z}{2}\right\} =$$
$$= \Delta f(x),$$

where we get the last equality simply by replacing z and y in the expression for $\Delta f(x)$ proved previously by $-y$ and $-z$, respectively.

(b) Since $f \in \mathrm{LC}_\infty(\mathbb{R}^n)$ and $f = e^{-\varphi}$, then $\varphi \in \mathrm{Cvx}_\infty(\mathbb{R}^n)$. Letting $\bar{\varphi}(x) = \varphi(-x)$, it is immediate that $\bar{\varphi}$ is closed and proper. To verify that $\bar{\varphi}$ is convex, notice that for any $x, y \in \mathbb{R}^n$ and $\lambda \in (0,1)$, we have

$$\bar{\varphi}((1-\lambda)x + \lambda y) = \varphi((1-\lambda)(-x) + \lambda(-y)) \le (1-\lambda)\varphi(-x) + \lambda\bar{\varphi}(-y) = (1-\lambda)\bar{\varphi}(x) + \lambda\bar{\varphi}(y).$$

Moreover, since $|x| = |-x|$ for every $x \in \mathbb{R}^n$, we get

$$\lim_{|x|\to\infty} \bar{\varphi}(x) = +\infty,$$

and hence, $\bar{\varphi} \in \mathrm{Cvx}_\infty(\mathbb{R}^n)$. Thus, $\bar{f} \in \mathrm{LC}_\infty(\mathbb{R}^n)$.

(c) Notice first that

$$\frac{1}{2}\cdot f(x) = \exp\left(-\frac{1}{2}\varphi(2x)\right),$$

and since it is clear that

$$x \mapsto \frac{1}{2}\varphi(2x)$$

is a function of $\mathrm{Cvx}_\infty(\mathbb{R}^n)$, we get that $\frac{1}{2}\cdot f \in \mathrm{LC}_\infty(\mathbb{R}^n)$. By the same argument, we have also that $\frac{1}{2}\cdot \bar{f} \in \mathrm{LC}_\infty(\mathbb{R}^n)$, and item **(d)** of Exercise 1.37 gives that $\mathrm{LC}_\infty(\mathbb{R}^n)$ is closed for Asplund product. Thus, $\Delta f \in \mathrm{LC}_\infty(\mathbb{R}^n)$.

(d) From item **(a)**, we have

$$\Delta 1_A(x) = \sup\left\{\sqrt{1_A(z)1_A(-y)} : x = \frac{z+y}{2}\right\},$$

and the value of $1_A(z)1_A(-y)$ is always 0 or 1. Consequently, we either have that $\Delta 1_A(x) = 0$ or $\Delta 1_A(x) = 1$. Also, observe that

$$1_A(z)1_A(-y) = \begin{cases} 1, & \text{if } z \in A \text{ and } y \in -A \\ 0, & \text{otherwise} \end{cases}.$$

Therefore, $\Delta 1_A(x) = 1$ if and only if there exist points $z \in A$ and $y \in -A$ such that $x = (z + y)/2$. This happens if and only if $x \in (A - A)/2$. This gives the desired equality.

□

For the next exercise, observe that if $f \in \mathrm{LC}_\infty(\mathbb{R}^n)$, then Δf is integrable because it is also a function of $\mathrm{LC}_\infty(\mathbb{R}^n)$. See the comment right above Exercise 1.37. See also the comment above Exercise 1.35 for the definition of the *support* of a function (denoted by spt).

Exercise 1.41. **(a)** *If f is a non-negative function, show that*

$$\mathrm{spt}(\Delta f) = \frac{\mathrm{spt}(f) - \mathrm{spt}(f)}{2}.$$

(b) *Letting $f \in \mathrm{LC}_\infty(\mathbb{R}^n)$, prove that*

$$\int_{\mathbb{R}^n} f\,dx = 0$$

if and only if

$$\int_{\mathbb{R}^n} \Delta f\,dx = 0.$$

Hint: Using item **(a)** and Remark 1.16, prove that the interior of $\mathrm{spt}(\Delta f)$ is empty if and only if the interior of $\mathrm{spt}(f)$ is empty.

(c) *Assuming again that $f \in \mathrm{LC}_\infty(\mathbb{R}^n)$, prove the inequalities*

$$\int_{\mathbb{R}^n} f\,dx \le \int_{\mathbb{R}^n} \Delta f\,dx \le 2^n \int_{\mathbb{R}^n} f\,dx.$$

Remark 1.19. The right-hand inequality above was originally proved by Colesanti in [78], and it can be seen as a functional version of the geometric *Rogers-Shephard inequality*, which is studied in Section 5.3 of the main book. A weaker variant of this geometric inequality is given in the next exercise. Recently, a lot of nice functional inequalities related to classical convex geometry inequalities have been studied. In this direction, we refer the reader to [9], [15], [12], [16], [14], [126], [127], and [128], among others.

Solution 1.41. **(a)** We have that $x \in \mathrm{spt}(\Delta f)$ if and only if there exist $z, y \in \mathbb{R}^n$ such that $x = (z + y)/2$, $f(z) \neq 0$ and $f(-y) \neq 0$. Now, observe that $f(z) \neq 0$ if and only if $z \in \mathrm{spt}(f)$ and $f(-y) \neq 0$ is equivalent to $y \in -\mathrm{spt}(f)$.

Therefore, $x \in \mathrm{spt}(\Delta f)$ if and only if we may write $x = (z + y)/2$ for some $z \in \mathrm{spt}(f)$ and some $y \in -\mathrm{spt}(f)$, and this is equivalent to

$$x \in \frac{\mathrm{spt}(f) - \mathrm{spt}(f)}{2}.$$

(b) From Exercise 1.35, it is enough to prove that the interior of $\mathrm{spt}(\Delta f)$ is empty if and only if the interior of $\mathrm{spt}(f)$ is empty.

If $\mathrm{int}(\mathrm{spt}(f)) \neq \emptyset$, then there exist a point $x \in \mathrm{spt}(f)$ and a number $r > 0$ such that $B(x, 2r) \subseteq \mathrm{spt}(f)$. Then, we clearly have that

$$B\Big(\frac{x}{2}, r\Big) \subseteq \frac{1}{2}\mathrm{spt}(f).$$

Letting z be any point in $-\mathrm{spt}(f)/2$, we get that

$$B\Big(\frac{x}{2} + z, r\Big) = z + B\Big(\frac{x}{2}, r\Big) \subseteq \frac{\mathrm{spt}(f)}{2} + z \subseteq \frac{\mathrm{spt}(f) - \mathrm{spt}(f)}{2} = \mathrm{spt}(\Delta f),$$

from where we get that the interior of $\mathrm{spt}(\Delta f)$ is non-empty.

Conversely, assume that $\mathrm{int}(\mathrm{spt}(f)) = \emptyset$. From Remark 1.16, we have that $\mathrm{spt}(f)$ is contained in some hyperplane $\mathbf{h}(u, \beta)$. Hence, we have immediately that $-\mathrm{spt}(f) \subseteq \mathbf{h}(u, -\beta)$. Letting $x \in \mathrm{spt}(\Delta f)$ be arbitrarily given, we may write $x = (z + y)/2$ for some $z \in \mathrm{spt}(f)$ and some $y \in -\mathrm{spt}(f)$. Thus,

$$\langle x, u\rangle = \Big\langle \frac{z}{2}, u\Big\rangle + \Big\langle \frac{y}{2}, u\Big\rangle = \frac{\beta}{2} - \frac{\beta}{2} = 0,$$

and this shows that $\mathrm{spt}(\Delta f) \subseteq \mathbf{h}(u, 0)$. Using Remark 1.16 again, we obtain that the interior of $\mathrm{spt}(\Delta f)$ is empty, and hence, we have the desired result.

(c) The case where the integral of f is 0 is an immediate consequence of item **(b)**. Thus, we may assume that

$$I(f) := \int_{\mathbb{R}^n} f\,dx = 0.$$

The left-hand side inequality is a consequence of item **(c)** of Exercise 1.28:

$$\int_{\mathbb{R}^n} \Delta f\,dx = \int_{\mathbb{R}^n} \Big(\frac{1}{2}\cdot f\Big) \star \Big(\frac{1}{2}\cdot \bar{f}\Big)\,dx \geq \Big(\int_{\mathbb{R}^n} f\,dx\Big)^{\frac{1}{2}} \Big(\int_{\mathbb{R}^n} \bar{f}\,dx\Big)^{\frac{1}{2}} = \int_{\mathbb{R}^n} f\,dx.$$

Now, we prove the other inequality. Observe that despite being defined as a supremum, Δf is finite (in the sense that $\Delta f(x) < +\infty$ for every $x \in \mathbb{R}^n$) because it is log-concave. In view of the expression for $\Delta f(x)$ obtained in item **(c)**, let $(z_j)_{j\in\mathbb{N}}$ and $(y_j)_{j\in\mathbb{N}}$ be sequences of points of $\mathbb{R}^n$ such that

$$x = \frac{z_j + y_j}{2} \quad \text{for each } j \in \mathbb{N} \text{ and}$$
$$\Delta f(x) = \lim_{j\to\infty} \sqrt{f(z_j) f(-y_j)}.$$

Since f is log-concave, for each $z \in \mathbb{N}$, we have

$$f(z - x) = f\left(\frac{1}{2}(2z - z_j) + \frac{1}{2}(-y_j)\right) \geq \sqrt{f(2z - z_j) f(-y_j)}, \tag{1.4}$$

for each $j \in \mathbb{N}$. Similarly,

$$f(z) = f\left(\frac{1}{2}(2z - z_j) + \frac{1}{2} z_j\right) \geq \sqrt{f(2z - z_j) f(z_j)}, \tag{1.5}$$

for every $j \in \mathbb{N}$. Now, we define the function $F\colon \mathbb{R}^n \to [0, +\infty)$ as

$$F(x) := \bar{f} * f(x) = \int_{\mathbb{R}^n} f(z-x) f(z)\, dz$$

for each $x \in \mathbb{R}^n$, where we remind the reader that $\bar{f} * f$ denotes the usual convolution of $\bar{f}$ and f. We justify why $F(x) < +\infty$ for any $x \in \mathbb{R}^n$. Indeed, we have $f = e^{-\varphi}$ and $\varphi \in \mathrm{Cvx}_\infty(\mathbb{R}^n)$, and we know that φ is bounded from below (by $m \in \mathbb{R}$, say). Then, $f(z-x) \le e^{-m}$ for every $z \in \mathbb{R}^n$, and hence,

$$F(x) = \bar{f} * f(x) = \int_{\mathbb{R}^n} f(z-x) f(z)\, dz \le e^{-m} \int_{\mathbb{R}^n} f(z)\, dz < +\infty,$$

because f is integrable (in fact, $f \in \mathrm{LC}_\infty(\mathbb{R}^n)$). Since f and $\bar{f}$ are integrable log-concave functions, we get from Proposition 1.5.1 in the main book that F is log-concave, and hence, F is measurable. Then we may integrate F and use Fubini's theorem to obtain

$$\int_{\mathbb{R}^n} F(x)\, dx = \int_{\mathbb{R}^n} \int_{\mathbb{R}^n} f(z-x) f(z)\, dz dx = \int_{\mathbb{R}^n} \left(\int_{\mathbb{R}^n} f(z-x)\, dx \right) dz =$$
$$= \int_{\mathbb{R}^n} \left(\int_{\mathbb{R}^n} f(y)\, dy \right) f(z)\, dz = \int_{\mathbb{R}^n} f(y)\, dy \int_{\mathbb{R}^n} f(z)\, dz = \left(\int_{\mathbb{R}^n} f(z)\, dz \right)^2 = I(f)^2.$$

On the other hand, by (1.4) and (1.5), we have

$$F(x) = \int_{\mathbb{R}^n} f(z-x) f(z)\, dz \ge \int_{\mathbb{R}^n} \sqrt{f(2z-z_j) f(-y_j)} \sqrt{f(2z-z_j) f(z_j)}\, dz =$$
$$= \sqrt{f(z_j) f(-y_j)} \int_{\mathbb{R}^n} f(2z-z_j)\, dz = \frac{\sqrt{f(z_j) f(-y_j)}}{2^n} \int_{\mathbb{R}^n} f(y)\, dy$$

for each $j \in \mathbb{N}$, where in the last equality, we did the change of variables $y = 2z - z_j$. Letting $j \to \infty$, we get

$$F(x) \ge \frac{\Delta f(x)}{2^n} \int_{\mathbb{R}^n} f(y)\, dy = \frac{\Delta f(x)}{2^n} I(f).$$

Thus, integrating both sides and using the equality that we derived above for the integral of F, we get

$$I(f)^2 = \int_{\mathbb{R}^n} F(x)\, dx \ge \int_{\mathbb{R}^n} \frac{\Delta f(x)}{2^n} I(f)\, dx = \frac{1}{2^n} I(f) \int_{\mathbb{R}^n} \Delta f(x)\, dx,$$

and finally, we divide both sides by $I(f)$ to get the desired equality (recall that we are assuming that $I(f) \neq 0$).

□

Exercise 1.42. *Let $K \subseteq \mathbb{R}^n$ be a non-empty compact convex set. Prove that*

$$\mathrm{vol}_n(K - K) \le 4^n \mathrm{vol}_n(K).$$

Solution 1.42. Since K is non-empty, compact, and convex, we have that $1_K \in \mathrm{LC}_\infty(\mathbb{R}^n)$. Thus,

$$\int_{\mathbb{R}^n} \Delta 1_K \, dx \leq 2^n \int_{\mathbb{R}^n} 1_K \, dx = 2^n \mathrm{vol}_n(K).$$

On the other hand, from item **(d)** of Exercise 1.40, we know that $\Delta 1_K = 1_{(K-K)/2}$. Thus,

$$\int_{\mathbb{R}^n} \Delta 1_K \, dx = \int_{\mathbb{R}^n} 1_{\frac{K-K}{2}} \, dx = \mathrm{vol}_n\left(\frac{K-K}{2}\right) = \frac{1}{2^n}\mathrm{vol}_n(K-K),$$

where in the last equality, we used the fact that the n-dimensional Lebesgue volume is homogeneous of degree n. Replacing this expression for the integral of $\Delta 1_K$ in the inequality above yields the desired result.

For Further Reading

Fitting to Chapter 1 of the main book [29] and the exercises above, we subdivide the text for further reading, belonging to the first chapter, into the two parts "convex functions" and "the Prékopa-Leindler inequality."

Convex Functions

In this first part of Chapter 1, the reader finds exercises and solutions treating *properties of convex functions and their domains and epigraphs*. These properties (e.g., lower semi-continuity or log-concavity) are essential for, and in most cases directly needed in, our main book. In a natural way, this goes along with the presentation of *special types of convex functions* (for instance, sublinear, log-concave, or support functions), convex functions that are *associated* with given functions (via notions like infimal convolutions, Asplund products, and being conjugate), and fundamental related *inequalities* (as those of Jensen, Rogers-Shephard, and Prékopa-Leindler). Further needed notions and tools discussed here are Minkowski sum and difference, or central symmetral. Nice examples of convex functions are shown in Plate 1 of the main book, which is presented here once more (but the references there refer to the main book).

The study of convex functions has its own *history*. Perhaps one can say that Jensen [145] was the first who emphasized this notion as an important one by itself. The developments of later decades were comprehensively described by Beckenbach [42] and Green [110]. Further on, the systematic study of convex functions and, later, of their role in *convex analysis* is presented in steps by Fenchel [93], Moreau [199], clearly Rockafellar (see the famous book [214] and also [215]), Ioffe and Tikhomirov [143], Laurent [159], and Ekeland and Teman [87]. This growth created or influenced many further areas, like *convex analysis, optimization, and inequalities*; see, e.g., Stoer and Witzgall [242], Hardy et al. [136], Bellman [43], and again [214].

The *theory of convex functions* started to develop strongly (also as a useful tool), and monographs paying more and more attention also to teaching aspects came to be. Examples are Roberts and Varberg [212], suitable for undergraduates, graduates, and also mathematicians being newcomers in the field; Barbu and Precupanu [35] with special emphasis on Banach spaces, deeply studying the needed

properties of convex functions; the booklet [248] of van Tiel containing carefully prepared material on convex functions, similar concepts, and related Banach space theory; and Liu [169], painstakingly using Rockafellar's monograph [214] as scientific model and containing a large portion of classical material on convex functions, such as *conjugate functions, sub-gradients, and differentiability*. The latter is also an important topic of Borwein and Vanderwerff in their book [64] on convex functions, also partially based on the monographs [214] and [212]. Besides differentiability, e.g., the following topics are presented: polyhedral convex functions, extensions to Banach spaces, duality relations and semi-definite optimization, various convexity notions and corresponding smoothness concepts, deeper properties of the subdifferential operator, convex functions on lattices, and characterizations of finite-dimensional Banach spaces with the help of function properties. More than 600 exercises on different difficulty levels are given. A further book, strongly related to differentiability of convex functions, is due to Giles [105], and results on the differentiability and extendibility of convex functions on Banach spaces are also surveyed by Borwein, Montesinos, and Vanderwerff in [65]. On the other hand, results on *non-differentiability points* of convex functions in n-dimensional space are collected by Ambrosio in [19].

Another deep and essential monograph on convex functions is the book [205] of Phelps, collecting differentiability properties of such functions and Lipschitzian operators on infinite- dimensional spaces. It is strongly related to fields like *nonlinear functional analysis* and topics like monotone operators, differentiability of vector measures, variational principles, and *Asplund spaces*. The author describes the duality between such spaces and the spaces with the Radon-Nikodym property. The monograph [205] is on the level of researchers and graduates, and it is closely related to the abovementioned book of Giles [105], where Asplund spaces are also treated.

Various further viewpoints and topics around convex functions are discussed in our main book [29], namely, in Chapter 1 there and particularly in the notes 1.7.1 to this chapter. These refer to *convex functions on manifolds* and are combined with (generalized) *convexity notions, functional analysis, approximation theory*, and the broad field of *inequalities*. Furthermore, the reader might also find parts of subsection 12.4.6 of [29] (on convex analysis) inspiring in this direction.

The Prékopa-Leindler Inequality

Now, we shortly present some facts about the *Prékopa-Leindler inequality*, which is comprehensively discussed in Section 1.5 and subsection 1.7.2 of the main book [29]. This integral inequality (Theorem 1.5.1 in the main book) goes back to the papers [208] and [165] of Prékopa and Leindler, respectively. On the one hand, the *Brunn-Minkowski inequality* has the Prékopa-Leindler inequality as a functional counterpart; on the other hand, Brascamb and Lieb [69] extended the Prékopa-Leindler inequality also to strengthenings of itself and of the Brunn-Minkowski inequality. This led finally to the famous *Brascamb-Lieb inequality*, also discussed in various monographs treating convexity; see, for example, Chapter 1 of Brazitikos et al. [70] and Chapter 14 of Simon [236].

Ball, in Lecture 5 of [33], shows nicely how the Brunn-Minkowski inequality, taken as integral inequality, turns out to be a special case of the Prékopa-Leindler inequality. See also the comprehensive and important survey [96] of Gardner, where the relations between the Brunn-Minkowski theorem and the inequalities of Prékopa-Leindler, Brascamp-Lieb, and the reverse of the latter are nicely presented. This *reverse Brascamp-Lieb inequality* is due to Barthe [36]. In this paper, he also reproved the multidimensional Brascamp-Lieb inequality and gave complete solutions regarding the equality cases. Bobkov and Ledoux [54] deal with probability measures, looking also for an analogue of the Brascamp-Lieb inequality involving entropy instead of variance. They come to a *logarithmic Sobolev inequality* for uniformly convex bodies, encompassing a concentration inequality for unit spheres due to Gromov

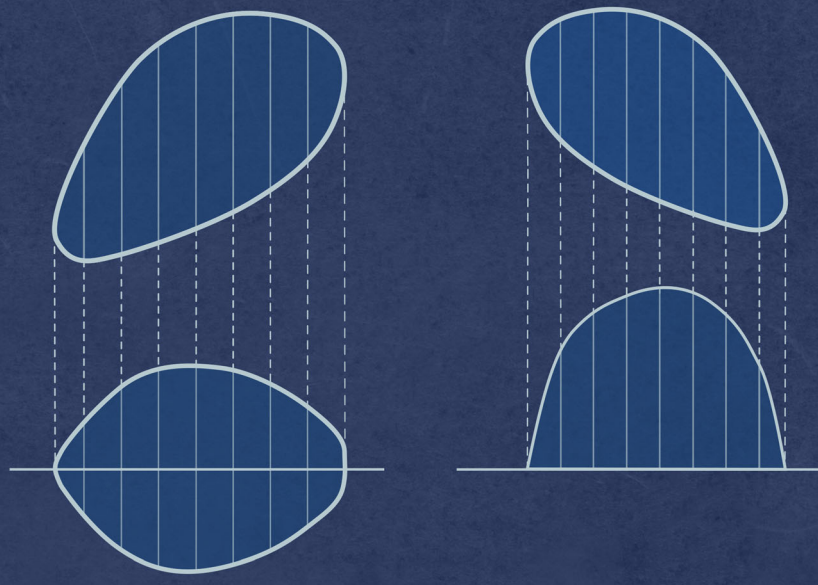
and Milman. Since [54] is in parts also a survey, an interesting variety of related topics and applications of the Brunn-Minkowski inequality in the Prékopa-Leindler form are derived. Further aspects around the Prékopa-Leindler inequality that are discussed in subsection 1.7.2 of the main book are relations to other inequalities from convexity, the inequality on manifolds, related applications, stability versions, and functional versions.

PLATE 2

Symmetrizations and similar procedures

Steiner symmetrization

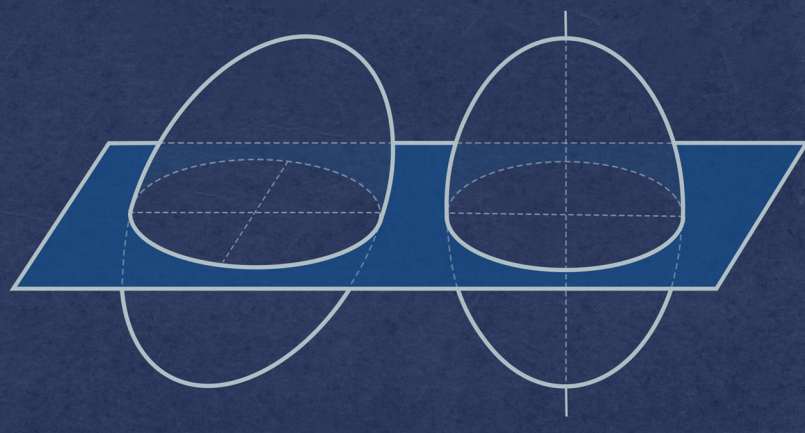
Shakedown (or X-ray)



Figures 4 - 5

Figure 6

Schwarz symmetrization



Symmetrizations and similar procedures ► (see, e.g., 2.7 and 2.9.3)

Chapter 2
Convex Sets



"One of the endlessly alluring aspects of mathematics is that its thorniest paradoxes have a way of blooming into beautiful theories."

— Philip J. Davis

The collection of exercises from Chapter 2 of the main book presents a colored bunch of flowers in the form of *fundamental, mainly geometric notions, topics, and methods*, which are typical and basic for the field of convexity. For example, these are *elementary and topological notions* (such as *convex or affine hull, positive hull, relative interior, extreme and exposed points or faces*), and also *elegant, applicable definitions* (like those of *supporting hyperplane, support function*, or the *polarity concept*) perfectly fitting with the aims of this part of mathematics. Further on, there one can find *Helly-type theorems*, e.g., Carathéodory's (colorful) theorem, properties of *centrally symmetric and centered convex bodies, separation and supporting properties, Minkowski addition* (combined with set unions and set intersections), *symmetrizations* (of Steiner and Minkowski), and *metrical properties and notions referring to convex bodies* (such as *in- and circumball, in- and circumradius, the width function, sets of constant width, and metric projections*). Also types of related useful functions (*support functions, sublinear functions, globally Lipschitz functions* etc.), *subdifferentials, convex cones*, results related to the *Hausdorff metric*, and *valuations* are discussed.

Plate 2:

Plate 2 shows a *Steiner symmetral* (Figure 4) and the strongly related (shakedown or) *X-ray* (Figure 5) of a convex set. And the so-called *Schwarz symmetrization* shown in Figure 6 generates a body of revolution. The Steiner symmetrization is explicitly discussed in this chapter here. Also in Plate 2, the literature hints refer to the main book. The shakedown procedure is discussed at the end of subsection 4.7.3 in the main book, and for the Schwarz symmetrization, we refer to subsection 5.7.2 there.

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V. Balestro et al., *Convexity from the Geometric Point of View: Exercises and Solutions*, Cornerstones, https://doi.org/10.1007/978-3-031-83937-5_2

Exercises and Solutions

For the first exercise below, recall that the *convex hull* of a set A is the set of all convex combinations of points of A or, equivalently, the smallest (regarding inclusion) convex set containing A. In what follows, we denote the convex hull of a set A by $\operatorname{conv}(A)$ (or $\operatorname{conv}A$).

Exercise 2.1. *Let $A \subseteq \mathbb{R}^n$ be a non-empty set. Prove that* $\operatorname{conv}A$ *is the union of all simplices whose vertices are points of A.*

Solution 2.1. Let S be any simplex whose vertices are points of A. Then, any point of S is a convex combination of points of A, from where we get that $S \subseteq A$. Thus,

$$\bigcup \big\{\text{simplices whose vertices are points of } A\big\} \subseteq \operatorname{conv}(A).$$

To prove the reverse inclusion, we use Carathéodory's theorem (Theorem 2.1.2 in the main book): any given point $x \in \operatorname{conv}(A)$ can be written as a convex combination of affinely independent points of A ($x_1, \ldots, x_m$, say). Then, $S := \operatorname{conv}\{x_1, \ldots, x_m\}$ is a simplex whose vertices are points of A and $x \in S$. This concludes the proof.

□

Exercise 2.2. *Let $x_1, \ldots, x_m \in \mathbb{R}^n$, and let $\mathbf{h}$ be a supporting hyperplane of* $\operatorname{conv}\{x_1, \ldots, x_m\}$. *Prove that*

$$\big(\operatorname{conv}\{x_1, \ldots, x_m\}\big) \cap \mathbf{h} = \operatorname{conv}\big(\{x_1, \ldots, x_m\} \cap \mathbf{h}\big).$$

Solution 2.2. This is proved in Lemma 3.1.1 of the main book.

□

Exercise 2.3 (Carathéodory's colorful theorem)**.** *Let $A_1, \ldots, A_{n+1}$ be non-empty finite subsets of $\mathbb{R}^n$ such that $0_{\mathbb{R}^n}$ lies in the convex hull of each one of them. Prove that there exist points $x_1 \in A_1$, $x_2 \in A_2, \ldots, x_{n+1} \in A_{n+1}$ such that $0_{\mathbb{R}^n} \in \operatorname{conv}\{x_1, \ldots, x_{n+1}\}$.*

Remark 2.1. We interpret each set A_j as a "color." Hence, the origin can be written as a combination having all "colors."

Solution 2.3. For simplicity, we call any set of the form $\operatorname{conv}\{z_1, \ldots, z_{n+1}\}$ where $z_j \in A_j$ for each $1 \leq j \leq n+1$ a *rainbow polytope*. Since each A_j is a finite set, there are finitely many rainbow polytopes. Hence, the number

$$d = \min\big\{\operatorname{dist}(0_{\mathbb{R}^n}, P) : P \text{ is a rainbow polytope}\big\}$$

is attained for some rainbow polytope P_0, which we write as $P_0 := \operatorname{conv}\{x_1, \ldots, x_{n+1}\}$ where $x_j \in A_j$ for each $1 \leq j \leq x_{n+1}$. Hence, it is clearly enough to prove that $0_{\mathbb{R}^n} \in P_0$. Since P_0 is compact, we have that $0_{\mathbb{R}^n} \in P_0$ if and only if $d = 0$. Thus, assume by contradiction that $d > 0$, and let $z \in P_0$ be the metric projection of $0_{\mathbb{R}^n}$ onto z. Then,

$$d = \operatorname{dist}(0_{\mathbb{R}^n}, P_0) = |z - 0_{\mathbb{R}^n}| = |z|.$$

By Lemma 2.2.4 from the main book, the direction $0_{\mathbb{R}^n} - z = -z$ is an outer normal of P_0 at z. This means that the hyperplane $\mathbf{h} := \mathbf{h}(-z, -|z|^2)$ supports P_0 at z with $P_0 \subseteq \mathbf{h}^-$. Observe also that since $\langle 0_{\mathbb{R}^n}, -z\rangle = 0 > -|z|^2$, we have $0_{\mathbb{R}^n} \notin \mathbf{h}^-$.

From Exercise 2.2, we have

$$P_0 \cap \mathbf{h} = \text{conv}(\{x_1, \ldots, x_{n+1}\} \cap \mathbf{h}).$$

By Carathéodory's theorem (Theorem 2.1.2 in the main book), each point of $\text{conv}(\{x_1, \ldots, x_{n+1}\} \cap \mathbf{h})$ can be written as a convex combination of affinely independent points of $\{x_1, \ldots, x_{n+1}\} \cap \mathbf{h}$. Since the affine dimension of $\mathbf{h}$ is $n-1$, the set $\{x_1, \ldots, x_{n+1}\} \cap \mathbf{h}$ has at most n affinely independent points. Therefore, since $z \in P_0 \cap \mathbf{h} = \text{conv}(\{x_1, \ldots, x_{n+1}\} \cap \mathbf{h})$, we can write z as a convex combination of n points chosen among $x_1, \ldots, x_{n+1}$. Renumbering the points if necessary, we may assume that

$$z = \lambda_1 x_1 + \ldots + \lambda_n x_n, \tag{2.1}$$

where $\lambda_j \geq 0$ for every $1 \leq j \leq n$ and $\sum_{j=1}^{n} \lambda_j = 1$.

Now, we claim that the set A_{n+1} has a point of $\text{int}(\mathbf{h}^+)$. In fact, if $A_{n+1} \subseteq \mathbf{h}^-$, then $\text{conv}A_{n+1} \subseteq \mathbf{h}^-$, and this contradicts the fact that $0_{\mathbb{R}^n} \in \text{conv}A_{n+1}$ (recall that $0_{\mathbb{R}^n} \notin \mathbf{h}^-$). Thus, let $y \in A_{n+1} \cap \text{int}(\mathbf{h}^+)$, and let $P_1 := \text{conv}\{x_1, \ldots, x_n, y\}$. Then, P_1 is a rainbow polytope, obtained simply by replacing $x_{n+1} \in A_{n+1}$ by $y \in A_{n+1}$ in P_0. From (2.1), we get that $z \in P_1$ (indeed, P_1 obviously contains all convex combinations of the points $x_1, \ldots, x_n$). Consequently, from the convexity of P_1, we get that the segment $\text{seg}[z, y] \subseteq P_1$. We claim that this segment (and hence P_1) has a point closer to $0_{\mathbb{R}^n}$ than z, thus contradicting the hypothesis that P_0 is the rainbow polytope which is closest to $0_{\mathbb{R}^n}$. To prove this claim, notice that we may assume that $|y| \geq |z|$, because otherwise y is itself the desired point of $\text{seg}[z, y]$, which is closer to $0_{\mathbb{R}^n}$ than z. Since $y \in \text{int}(\mathbf{h}^+)$, we have

$$\langle z, y \rangle < |z|^2.$$

Let

$$\lambda := \frac{|z|^2 - \langle z, y \rangle}{|z - y|^2}.$$

From the previous inequality, we have that $\lambda > 0$. Also, from $|y| \geq |z|$, we get

$$|z - y|^2 = \langle z - y, z - y \rangle = |z|^2 - 2\langle z, y \rangle + |y|^2 \geq 2(|z|^2 - \langle z, y \rangle),$$

and hence, $\lambda \leq 1/2 < 1$. Let $w = (1 - \lambda)z + \lambda y$. Then, $w \in \text{seg}(z, y)$, and a direct calculation gives that

$$\langle w, z - w \rangle = 0.$$

Thus,

$$|z|^2 = |w|^2 + |z - w|^2 > |w|^2,$$

from where we get that $|w| < |z|$. This concludes the proof.

□

Exercise 2.4. *Let $K \subseteq \mathbb{R}^n$ be a non-empty convex set, and let $\mathcal{A} = \{A_1, \ldots, A_m\}$ be any finite family of non-empty convex subsets of $\mathbb{R}^n$. Prove that if any fixed $n+1$ elements of $\mathcal{A}$ are simultaneously intersected by some translate of K, then there exists a translate of K that intersects all sets of $\mathcal{A}$.*

Solution 2.4. Observe that for any $x \in \mathbb{R}^n$ and any set $A \subseteq \mathbb{R}^n$, we have that $(x + K) \cap A \neq \emptyset$ if and only if $x \in A - K$. Therefore, any given $n + 1$ elements of the family $\{A_1 - K, \ldots, A_m - K\}$ have non-empty intersection. It follows from Helly's theorem (Theorem 2.1.5 in the main book) that there exists a point

$$x \in (A_1 - K) \cap \ldots (A_m - K),$$

and hence, $(x + K) \cap A_j \neq \emptyset$ for each $1 \leq j \leq m$.

□

Exercise 2.5. **(a)** *Let $C, D \subseteq \mathbb{R}^n$ be closed, convex, and non-empty, and assume that $C \cup D$ is convex. Prove that for each $c \in C$ and $d \in D$, there exists a number $\lambda \in [0, 1]$ such that*

$$(1 - \lambda)c + \lambda d \in C \cap D.$$

Conclude, in particular, that $C \cap D \neq \emptyset$.

(b) *Give an example of a pair of non-empty convex sets $C, D \subseteq \mathbb{R}^n$ such that C is closed, D is not closed, $C \cup D$ is convex, and $C \cap D = \emptyset$.*

Solution 2.5. **(a)** Let $c \in C$ and $d \in D$ be arbitrarily given. If $d \in C$ or $c \in D$, then there is nothing to prove. Hence, we may assume that $d \notin C$ and $c \notin D$. Let

$$\lambda_0 = \sup\{\lambda \in [0, 1] : (1 - \lambda)c + \lambda d \in C\},$$

and let $x = (1 - \lambda_0)c + \lambda_0 d$. There is a sequence of numbers $\lambda_m \in [0, 1]$ such that $(1 - \lambda_m)c + \lambda_m d \in C$ for each $m \in \mathbb{N}$, and $\lambda_m \to \lambda_0$. Hence, x is the limit of a sequence of points of C, and then since C is closed, we have that $x \in C$. Moreover, by convexity and from the definition of λ_0, we clearly have that $(1 - \lambda)c + \lambda d \in C$ if and only if $0 \leq \lambda \leq \lambda_0$. Now, let

$$\lambda_1 = \inf\{\lambda \in [0, 1] : (1 - \lambda)c + \lambda d \in D\},$$

and let $y = (1 - \lambda_1)c + \lambda_1 d$. By the same argument that we have used to show that $x \in C$, we prove that $y \in D$, and we also have obviously that $(1 - \lambda)c + \lambda d \in D$ if and only if $\lambda_1 \leq \lambda \leq 1$. Assume by contradiction that $x \notin D$. Then, $\lambda_0 < \lambda_1$, and consequently for any $\lambda_0 < \lambda < \lambda_1$, it holds that

$$(1 - \lambda)c + \lambda d \notin C \cup D.$$

And since $c, d \in C \cup D$, this contradicts the fact that $C \cup D$ is convex. Therefore, $x \in D$, and because we already knew that $x \in C$, it follows that $x \in C \cap D$.

(b) Let $D = B(0, 1)$ and $C = \{x_0\}$ where x_0 is any point such that $|x_0| = 1$. Then, C is closed, and it is also immediate that $C \cap D = \emptyset$, because all points of D have norm strictly smaller than 1. To check that $C \cup D$ is convex, we only need to show that the segment connecting x_0 to any point of D is contained in $C \cup D$ (indeed, since D is convex, we already have that any segment connecting two points of D is in D and hence in $C \cup D$). For any $y \in D$ and $\lambda \in (0, 1)$, we have

$$|(1 - \lambda)x_0 + \lambda y| \leq (1 - \lambda)|x_0| + \lambda|y| < 1 - \lambda + \lambda = 1,$$

from where we get that $(1 - \lambda)x_0 + \lambda y \in D$.

□

From now on, we denote by CC^n the collection of all *non-empty, compact, and convex subsets* of $\mathbb{R}^n$. The collection of *convex bodies* of $\mathbb{R}^n$ is denoted by $\mathcal{K}^n$. By $\mathcal{C}^n$, we denote the collection of all *non-empty compact subsets* of $\mathbb{R}^n$.

Exercise 2.6. *Let $K, L \subseteq \mathbb{R}^n$ be closed, convex, and non-empty. Prove that if $K \cup L$ is convex, then*

$$d(K \cup L, x) = \min\{d(K, x), d(L, x)\} \text{ and}$$
$$d(K \cap L, x) = \max\{d(K, x), d(L, x)\}$$

for any fixed $x \in \mathbb{R}^n$.

Remark 2.2. In particular, if $K \cup L$ is convex, we have that

$$d(K \cup L, x) + d(K \cap L, x) = d(K, x) + d(L, x).$$

We say that the function $CC^n \ni K \mapsto d(K, x)$ has the *valuation property*. A function $\varphi\colon CC^n \to \mathbb{R}$ has the *valuation property* if $\varphi(K \cup L) + \varphi(K \cap L) = \varphi(K) + \varphi(L)$ whenever $K, L \in CC^n$ are such that $K \cup L \in CC^n$.

Solution 2.6. The first equality is easy: since $K, L \subseteq K \cup L$, we have that $d(K, x) \geq d(K \cup L, x)$ and $d(L, x) \geq d(K \cup L, x)$. If $d(K \cup L, x)$ is attained by a point $z \in K$, then $d(K \cup L, x) = |x - z| \geq d(K, x)$, from where we get that $d(K \cup L, x) = d(K, x)$. Also, it is clear that $d(L, x) \geq d(K, x)$, since otherwise $d(K \cup L, x)$ would be attained by a point of L. Hence, $d(K \cup L, x) = \min\{d(K, x), d(L, x)\}$. The case where $d(K \cup L, x)$ is attained by a point of L is completely analogous.

Now, we prove the other equality. Since $K \cap L \subseteq K$ and $K \cap L \subseteq L$, we have that $d(K \cap L, x) \geq d(K, x)$ and $d(K \cap L, x) \geq d(L, x)$. Thus,

$$d(K \cap L, x) \geq \max\{d(K, x), d(L, x)\}.$$

Assume, by contradiction, that the inequality is strict. Then, we have that $|x - z| > \max\{d(K, x), d(L, x)\}$. Moreover, in this case, $d(K, x)$ must be attained for a point $x_0 \in K \setminus L$, and $d(L, x)$ must be attained for a point $y_0 \in L \setminus K$. By Exercise 2.5, there is a $\lambda \in [0, 1]$ such that $z_0 = (1 - \lambda)x_0 + \lambda y_0 \in K \cap L$. For this point, we have

$$\begin{aligned} |x - z_0| &= |x - (1 - \lambda)x_0 - \lambda y_0| = |(1 - \lambda)(x - x_0) + \lambda(x - y_0)| \leq \\ &\leq (1 - \lambda)|x - x_0| + \lambda|y - y_0| = \\ &= (1 - \lambda)d(K, x) + \lambda d(L, x) \leq \max\{d(K, x), d(L, x)\}. \end{aligned}$$

This contradicts the fact that $|x - z| > \max\{d(K, x), d(L, x)\}$ for every $z \in K \cap L$. This concludes the proof.

□

Exercise 2.7. **(a)** *Verify that the Minkowski sum of bounded sets is bounded.*

(b) *Prove that the Minkowski sum of a closed set with a compact set is closed.*

(c) *Let K be a convex body and L be a non-empty compact and convex set. Show that $K + L$ is a convex body.*

(d) *Give an example of two closed sets whose sum is not closed.*

Solution 2.7. **(a)** Assume that $A, B \subseteq \mathbb{R}^n$ are bounded sets. Then, there exist $m_1, m_2 > 0$ such that $|x| < m_1$ for every $x \in A$ and $|y| < m_2$ for any $y \in B$. Thus, by the triangle inequality, we have $|z| < m_1 + m_2$ for each $z \in A + B$.

(b) Assume that $A \subseteq \mathbb{R}^n$ is closed and $B \subseteq \mathbb{R}^n$ is compact. Of course, we may assume that both are non-empty, since otherwise their sum would be the empty set. Let $(z_m)_{m\in\mathbb{N}}$ be a sequence in $A + B$ that converges to a point $z \in \mathbb{R}^n$. For each $m \in \mathbb{N}$, write $z_m = x_m + y_m$ with $x_m \in A$ and $y_m \in B$. Because B is compact, we may assume that there is a subsequence $(y_{m_k})_{k\in\mathbb{N}}$ of (y_m) converging to a point $y \in B$. Since the sequence (z_{m_k}) converges to z, we get that $x_{m_k} = z_{m_k} - y_{m_k}$ converges to $z - y$. Since A is closed and (x_{m_k}) is a sequence of points of A, we get that $z - y \in A$. Hence, $z = (z - y) + y \in A + B$, and this proves that $A + B$ is closed.

(c) Let K and L be convex bodies. Then, from the previous items, we have that $K + L$ is closed and bounded (hence, compact). The convexity of $K + L$ comes from Remark 2.1.6 in the main book. It only remains to verify that $K + L$ contains a ball. For that sake, choose any ball $x + \rho B^n \subseteq K$, any point $z \in L$, and notice that $x + z + \rho B^n \subseteq K + L$.

(d) Consider the sets A and B in item **(c)** of Exercise 1.22.

□

Exercise 2.8. *Let $A, B, C \subseteq \mathbb{R}^n$ be arbitrary non-empty sets. Prove that*

$$(A \cup B) + C = (A + C) \cup (B + C) \text{ and}$$
$$(A \cap B) + C \subseteq (A + C) \cap (B + C).$$

Show that in the second line, equality holds if we assume that A, B are closed convex sets such that $A \cup B$ is convex and that C is convex.

Solution 2.8. Any point $z \in (A \cup B) + C$ can be written as $z = x + c$ for some $x \in A \cup B$ and $c \in C$. If $x \in A$, then $z \in A + C$, and if $x \in B$, we have that $z \in B + C$. This proves that

$$(A \cup B) + C \subseteq (A + C) \cup (B + C).$$

The reverse inclusion comes easily since it is straightforward to see that $A + C \subseteq (A \cup B) + C$, and the same holds for $B + C$.

Now, let $z \in (A \cap B) + C$, and write $z = x + c$ for $x \in A \cap B$ and $c \in C$. Since $x \in A$, we have that $z \in A + C$, and because $x \in B$, we also have that $z \in B + C$. Thus, $z \in (A + C) \cap (B + C)$. This shows that $(A \cap B) + C \subseteq (A + C) \cap (B + C)$.

Finally, assume that A, B and $A \cup B$ are closed and convex sets such that $A \cup B$ is convex and that C is convex. Let $x \in (A + C) \cap (B + C)$. Hence, we may write $x = a + c_1 = b + c_2$ for some $a \in A$, $b \in B$ and $c_1, c_2 \in C$. Since $A \cup B$ is convex, from Exercise 2.5 we obtain that there exists a number $\lambda \in [0, 1]$ such that

$$(1 - \lambda)a + \lambda b \in A \cap B.$$

Therefore, using this fact and also the convexity of C, we get

$$x = (1 - \lambda)x + \lambda x = (1 - \lambda)(a + c_1) + \lambda(b + c_2) = (1 - \lambda)a + \lambda b + (1 - \lambda)c_1 + \lambda c_2 \in (A \cap B) + C,$$

and this shows that $(A + C) \cap (B + C) \subseteq (A \cap B) + C$. The proof is complete.

□

Exercise 2.9. *Let $A, B \subseteq \mathbb{R}^n$ be non-empty sets. Verify that*

$$(A \cup B) + (A \cap B) \subseteq A + B.$$

Prove that equality holds in the case where A and B are closed and convex and $A \cup B$ is convex.

Solution 2.9. Let $x + y \in (A \cup B) + (A \cap B)$, with $x \in A \cup B$ and $y \in A \cap B$. If $x \in A$, then observing that $y \in B$, we have that $x + y \in A + B$. If $x \notin A$, then we must have $x \in B$, and since $y \in A$, we get that $x + y \in A + B$. This shows that $(A \cup B) + (A \cap B) \subseteq A + B$.

Now, we assume that A and B are closed and convex and that they are such that $A \cup B$ is convex. Let $a \in A$ and $b \in B$ be arbitrarily given. From Exercise 2.5, we have that there exists a number $\lambda \in [0, 1]$ such that

$$z := (1 - \lambda)a + \lambda b \in A \cap B.$$

We write $a + b = z + (a + b - z)$, and

$$a + b - z = a + b - (1 - \lambda)a - \lambda b = \lambda a + (1 - \lambda)b,$$

and since $A \cup B$ is convex, we obtain that $a + b - z \in A \cup B$. Hence,

$$a + b = z + (a + b - z) \in (A \cap B) + (A \cup B),$$

and this gives the reverse inclusion $A + B \subseteq (A \cup B) + (A \cap B)$.

□

Exercise 2.10. *Let K be a convex set. Prove that a point $x \in K$ is extreme if and only if $K \setminus \{x\}$ is convex.*

Solution 2.10. If x is an extreme point, then x is not in the interior of any segment of K. Hence, if $y, z \in K \setminus \{x\}$, it follows that $\text{seg}[y, z] \subseteq K \setminus \{x\}$, since $x \notin \text{seg}(y, z)$. In the other direction, assume that $K \setminus \{x\}$ is convex. Then, a given segment of K cannot contain x as an interior point, since, otherwise, this segment would not be contained in $K \setminus \{x\}$. It follows that x is an extreme point of K.

□

Exercise 2.11. *Let $A, B \subseteq \mathbb{R}^n$ be non-empty, compact, and convex. Prove that any extreme point of $A + B$ is a sum of an extreme point of A and an extreme point of B.*

Solution 2.11. Let $x \in A + B$ be an extreme point, and write $x = a + b$ for some $a \in A$ and $b \in B$. Suppose that $a \notin \text{ext}(A)$. Then, we may write $a = \lambda z_1 + (1 - \lambda)z_2$ for some $\lambda \in (0, 1)$ and some $z_1, z_2 \in A$. Thus, we have

$$x = \lambda z_1 + (1 - \lambda)z_2 + b = \lambda(z_1 + b) + (1 - \lambda)(z_2 + b),$$

and since $z_1 + b$ and $z_2 + b$ are points of $A + B$, we get that x lies in the interior of a segment of $A + B$, contradicting the fact that x is an extreme point. This contradiction gives that $a \in \text{ext}(A)$. Analogously, we prove that $b \in \text{ext}(B)$.

□

Let $C \subseteq \mathbb{R}^n$ be a non-empty closed and convex set, and let $x \in C$. A *cap of C around x* is a set having the form $A \cap \mathbf{h}^+$, where $\mathbf{h}^+$ is a closed half-space containing x in its interior.

Exercise 2.12. *If $C \subseteq \mathbb{R}^n$ is non-empty, closed, and convex, prove that $x \in C$ is an extreme point of C if and only if every open ball of $\mathbb{R}^n$ centered at x contains a cap of C around x.*

Solution 2.12. Assume first that $x \in C$ is not an extreme point. Then, there exist $y, z \in C$ such that $x \in \text{seg}(y, z)$. It is immediate to notice that any cap of C around x must contain y or z. Hence, any open ball B centered at x whose radius is small enough such that $y, z \notin B$ cannot contain a cap of C around x.

Now, assume that $x \in \text{ext}C$, and let B be any open ball centered at x. Let $D := \text{conv}(C \setminus B)$. Since x is an extreme point of C, we have that x is not a convex combination of points of $C \setminus B$. Consequently, $x \notin D$. We claim that, even better, we have that $x \notin \text{cl}(D)$. In fact, if x is the limit of a sequence (z_j) of points of D, then using Carathéodory's theorem, we may write each z_j as a convex combination

$$z_j = \lambda_1^j x_1^j + \ldots + \lambda_{n+1}^j x_{n+1}^j$$

of points $x_1^j, \ldots, x_{n+1}^j$ of $C \setminus B$. Because $z_j \to x$, we have that the sequences $(\lambda_k^j)_j$ and $(x_k^j)_j$ converge as $j \to \infty$. It is immediate to see that the limit (x_k, say) of each sequence $(x_k^j)_j$ must be in $C \setminus B$ (this set is clearly closed), and hence, we may write

$$x = \lambda_1 x_1 + \ldots + \lambda_{n+1} x_{n+1},$$

which is a convex combination of points of $C \setminus B$. This contradicts the fact that x is an extreme point of C.

Because $x \notin \text{cl}(D)$, it follows that x and $\text{cl}(D)$ can be strongly separated by a hyperplane $\mathbf{h}$. Assuming that $\mathbf{h}^+$ is the closed half-space that contains x in its interior, it follows straightforwardly that $C \cap \mathbf{h}^+ \subseteq B$ (because $C \setminus B$ is contained in D, and D lies within the interior of the other half-space determined by $\mathbf{h}$, of course). The proof is complete.

□

Exercise 2.13. *Let $A \subseteq \mathbb{R}^n$ be bounded. We define the* circumradius *of A to be the number*

$$r_{\text{out}}(A) := \inf \left\{ r > 0 : \exists\, x \in \mathbb{R}^n \text{ such that } A \subseteq x + rB^n \right\},$$

that is, the circumradius of A is the infimum of the radius of the closed balls containing A. If $\text{int}(A) \neq \emptyset$, then we define the inradius *of A as*

$$r_{\text{in}}(A) := \sup \left\{ r > 0 : \exists\, x \in \mathbb{R}^n \text{ such that } x + rB^n \subseteq A \right\}.$$

In other words, the inradius of A is the supremum of the radius of the closed balls contained in A. If $\text{int}(A) = \emptyset$, then we simply define $r_{\text{in}}(A) = 0$.

(a) *Assume that $A \subseteq \mathbb{R}^n$ is bounded and contains at least two points. Prove that the circumradius of A is positive and that it is attained, that is, show that there exists a point $x \in \mathbb{R}^n$ such that $A \subseteq x + r_{\text{out}}(A)B^n$.*

(b) *Give an example of a bounded set $A \subseteq \mathbb{R}^n$ with non-empty interior whose inradius is not attained, that is, such that there is no point $x \in \mathbb{R}^n$ for which $x + r_{\text{in}}(A)B^n \subseteq A$.*

(c) *On the other hand, prove that the inradius is attained for any compact set having non-empty interior.*

Remark 2.3. If $A \subseteq \mathbb{R}^n$ is a bounded set containing at least two points, then any ball containing A and whose radius is the circumradius of A is called a *circumball* of A.

Solution 2.13. (a) The fact that the circumradius of A is finite comes immediately from the boundedness of A. To prove that this is positive, let $x, y \in A$ be distinct. Then, A is not contained in any ball whose radius is less than $r_0 = |x - y|/2$, since otherwise any two points of A would be at distance less than $|x - y|$. Therefore,

$$r_{\text{out}}(A) \geq \frac{|x - y|}{2} > 0.$$

Let $(r_m)_{m\in\mathbb{N}}$ be a sequence of positive numbers such that for each $m \in \mathbb{N}$, there exists a point $x_m \in \mathbb{R}^n$ with $A \subseteq x_m + r_m B^n$, and $r_m \to r_{\text{out}}(A)$. For the sake of simplicity, we write $r_{\text{out}}(A) = r$, and notice that $r < r_m$ for each $m \in \mathbb{N}$. We may assume without loss of generality that $r_m < r + 1$ for all $m \in \mathbb{N}$. Fixing any $a \in A$, we have for each $m \in \mathbb{N}$ that $a = x_m + r_m v$ for some $v \in B^n$. Thus, writing $x_m = a + r_m(-v)$ and observing that $-v \in B^n$, we get

$$x_m \in A + r_m B^n \subseteq (r + 1)B^n,$$

and this shows that $(x_m)_{m\in\mathbb{N}}$ is a bounded sequence. Therefore, passing to a subsequence if necessary, we may assume that x_m converges to a point $x \in \mathbb{R}^n$. We claim that $A \subseteq x + rB^n$. Indeed, suppose that there is a point $a \in A$ such that $a \notin x + rB^n$. Then,

$$d := \operatorname{dist}(a, x + rB^n) = \inf\{|a - z| : z \in x + rB^n\} > 0.$$

Choose $m_0 \in \mathbb{N}$ such that we have

$$|x - x_{m_0}| < \frac{d}{2} \text{ and}$$
$$r_{m_0} - r < \frac{d}{2}.$$

Since $a \in x_{m_0} + r_{m_0} B^n$, we get

$$|a - x| \leq |a - x_{m_0}| + |x_{m_0} - x| < r_{m_0} + \frac{d}{2} < r + d.$$

On the other hand, since it is clear that

$$x + r\frac{a - x}{|a - x|} \in x + rB^n,$$

the distance of this point to a must be greater than or equal to d. Moreover, we have that $|a - x| > r$, since otherwise we would have $a \in x + rB^n$. Thus,

$$d \leq \left|a - x - r\frac{a - x}{|a - x|}\right| = |a - x|\left|1 - \frac{r}{|a - x|}\right| = |a - x|\left(1 - \frac{r}{|a - x|}\right) <$$
$$< (r + d)\left(1 - \frac{r}{r + d}\right) = d,$$

and this contradiction gives that $A \subseteq x + rB^n$. This concludes the proof.

(b) Consider $B := B(0_{\mathbb{R}^n}, 1)$, the open unit ball with center at $0_{\mathbb{R}^n}$. Of course, we have that $\rho B^n \subseteq B$ for every $\rho < 1$, from where we get that $r_{\text{in}}(B) \geq 1$. It is also easy to see that no closed ball with radius greater than or equal to 1 is contained in B, since otherwise we would have points of B at distance equal to 2. Thus, $r_{\text{in}}(B) = 1$, but no closed ball of radius 1 is contained in B.

(c) Assume that $A \subseteq \mathbb{R}^n$ is compact and has non-empty interior. In particular, A contains a closed ball, and hence, $r_{\text{in}}(A) > 0$. Let $(r_m)_{m\in\mathbb{N}}$ be a sequence of positive numbers such that for each $m \in \mathbb{N}$, there exists a point $x_m \in \mathbb{R}^n$ for which $x_m + r_m B^n \subseteq A$, and with $r_m \to r_{\text{in}}(A)$. Denote $r := r_{\text{in}}(A)$, and observe that we have $r_m < r$ for each $m \in \mathbb{N}$. Since A is bounded and $x_m \in A$ for every $m \in \mathbb{N}$, passing to a subsequence if necessary, we may assume that (x_m) converges to some point $x \in \mathbb{R}^n$. We will prove that $x + rB^n \subseteq A$. Assume that this is not true, and let $z \in (x + rB^n) \setminus A$. Since A is compact, we get that

$$d := \operatorname{dist}(z, A) > 0,$$

and from the definition of the distance of a point to a set, we also have that $|z - a| \geq d$ for every $a \in A$. Choose $m_0 \in \mathbb{N}$ such that

$$|x_{m_0} - x| < \frac{d}{2} \text{ and}$$
$$r - r_m < \frac{d}{2}.$$

Because $x_{m_0} + r_{m_0} B^n \subseteq A$, we have that $z \notin x_{m_0} + r_{m_0} B^n$. Hence, the segment connecting x_m and z intersects the boundary of $x_{m_0} + r_{m_0} B^n$ at some point (y, say). In particular, since $y \in x_{m_0} + r_{m_0} B^n$, we obtain that $y \in A$. The points x_{m_0}, y, and z are aligned, with y between x_{m_0} and z. Thus,

$$|x_{m_0} - z| = |x_{m_0} - y| + |y - z| = r_{m_0} + |y - z|.$$

On the other hand,

$$|x_{m_0} - z| \leq |x_{m_0} - x| + |x - z| < \frac{d}{2} + r,$$

where the inequality $|x - z| < r$ justifies because $z \in x + rB^n$. From this inequality together with the previous equality, we get

$$|y - z| = |x_{m_0} - x| - r_{m_0} < r - r_{m_0} + \frac{d}{2} < d,$$

and since $y \in A$, this leads to a contradiction.

□

Exercise 2.14. **(a)** *Let $A \subseteq \mathbb{R}^n$ be a bounded set containing at least two points. Prove that there is a unique circumball of A.*

(b) *Give an example of a compact set $A \subseteq \mathbb{R}^n$ having non-empty interior that contains more than one closed ball whose radius is $r_{\text{in}}(A)$.*

Solution 2.14. **(a)** Denote $r := r_{\text{out}}(A)$ and assume that A has two distinct circumballs $x_1 + rB^n$ and $x_2 + rB^n$. Then, we have that

$$A \subseteq (x_1 + rB^n) \cap (x_2 + rB^n),$$

and hence, $d := |x_1 - x_2| < 2r$. Indeed, if $d > 2r$, then the intersection above is empty, and if $d = 2r$, then this intersection has a unique point (thus, contradicting the fact that A has at least two points). Let

$$r_0 = \sqrt{r^2 - \left(\frac{d}{2}\right)^2} \text{ and}$$
$$x = \frac{x_1 + x_2}{2}.$$

Then, since $r > d/2$, we have that $r_0 > 0$, and it is clear that $r_0 < r$. The closed ball $x + r_0 B^n$ contains the intersection $(x_1 + rB^n) \cap (x_2 + rB^n)$, and hence, it contains A. This contradicts the fact that r is the circumradius of A and the proof is complete.

(b) Consider any rectangle in $\mathbb{R}^2$, for example.

□

Exercise 2.15. *Let $K \subseteq \mathbb{R}^n$ be a non-empty compact convex set. Prove that the center of the circumball of K lies in K.*

Solution 2.15. Let $r = r_{\text{out}}(K)$, and assume that $x_0 + rB^n$ is the circumball of K. If $x_0 \notin K$, then $\delta := \text{dist}(x_0, K) > 0$, and this distance is attained in some point $z_0 \in \partial K$ (the metric projection of x_0 onto K). Hence, the vector

$$u := \frac{x_0 - z_0}{|x_0 - z_0|}$$

is an outer normal vector of K at z_0, from where we get that $K \subseteq \mathbf{h}^-(u, \langle u, z_0\rangle) =: \mathbf{h}$. Since we also have $K \subseteq x_0 + rB^n$, we get that K is contained in the "cap"

$$C := (x_0 + rB^n) \cap \mathbf{h}^-(u, \langle u, z_0\rangle);$$

see Figure 2.1.

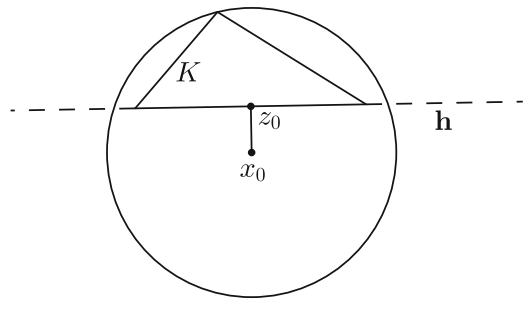


Fig. 2.1 K is contained in the "cap" $(x_0 + rB^n) \cap \mathbf{h}$

It is easy to see that the greater distance (d, say) from a point of the cap C to z_0 is attained for the points of the intersection of $\mathbf{h}$ and the boundary of $x_0 + rB^n$ (this intersection is a "copy" of $\mathbb{S}^{n-2}$, of course). By the classical Pythagorean theorem, this greater distance d is given by

$$d^2 = r^2 - \delta^2.$$

Thus, for any $x \in C$, we have

$$|x - z_0| \leq d = \sqrt{r^2 - \delta^2}.$$

Because $K \subseteq C$, this inequality holds for every point of K. Consequently, K is contained in the ball $z_0 + \sqrt{r^2 - \delta^2}B^n$, and the radius of this ball is strictly smaller than r. This contradicts the fact that $r = r_{\text{out}}(K)$. Therefore, we have that $x_0 \in K$.

□

For the next exercise, recall that the *width* of a set $K \in CC^n$ in a given direction $u \in \mathbb{S}^{n-1}$ is the number $w_K(u) := h_K(u) + h_K(-u)$. See Lemma 2.3.3 in the main book for the geometric meaning of this definition.

Exercise 2.16. **(a)** *Let $K \subseteq \mathbb{R}^n$ be compact and convex. Prove that*

$$r_{\text{out}}(K) = \inf\left\{r > 0 : \exists\, x \in \mathbb{R}^n \text{ such that } h_{K-x}(u) \leq r \text{ for all } u \in \mathbb{S}^{n-1}\right\}.$$

(b) *Assume that $K \subseteq \mathbb{R}^n$ is a convex body. Show that*

$$r_{\text{in}}(K) = \sup\left\{r > 0 : \exists\, x \in \mathbb{R}^n \text{ such that } h_{K-x}(u) \geq r \text{ for all } u \in \mathbb{S}^{n-1}\right\}.$$

(c) *Let $K \subseteq \mathbb{R}^n$ be a convex body, and recall that $w_K(u)$ stands for the width of K in the direction of $u \in \mathbb{S}^{n-1}$. Prove that*

$$2r_{\text{in}}(K) \leq w_K(u) \leq 2r_{\text{out}}(K)$$

for every $u \in \mathbb{S}^{n-1}$.

Solution 2.16. **(a)** It is easy to see that for any $r > 0$, we have that $K \subseteq x + rB^n$ if and only if $K - x \subseteq rB^n$. From Proposition 2.3.2 in the main book, we have that the latter occurs if and only if $h_{K-x}(u) \leq h_{rB^n}(u)$ for every $u \in \mathbb{S}^{n-1}$. Since $h_{rB^n} = r$ in $\mathbb{S}^{n-1}$, we get what we wanted to get.

(b) Similar to the argument in item **(a)**, we have that $x + rB^n \subseteq K$ if and only if $rB^n \subseteq K - x$, and this is equivalent to $h_{rB^n} \leq h_{K-x}$. The result follows.

(c) Let $x_0 + rB^n$ be the circumball of K. Then, $r = r_{\text{out}}(K)$, and we have from item **(a)** that $h_{K-x_0}(u) \leq r$ for every $u \in \mathbb{S}^{n-1}$. Thus,

$$\begin{aligned} w_K(u) &= h_K(u) + h_K(-u) = k_K(u) - \langle u, x_0\rangle + h_K(-u) - \langle -u, x_0\rangle \\ &= h_{K-x_0}(u) + h_{K-x_0}(-u) \leq \\ &\leq 2r = 2r_{\text{out}}(K) \end{aligned}$$

for each $u \in \mathbb{S}^{n-1}$. The proof of the other inequality is analogous.

□

Exercise 2.17. *Let $K \in CC^n$ be arbitrarily given.*

(a) *Prove that*

$$2r_{\text{out}}(K) \geq \max\left\{w_K(u) : u \in \mathbb{S}^{n-1}\right\}$$

and that equality holds if K is centrally symmetric. Give an example of a convex body for which equality does not hold.

(b) *Show that*

$$r_{\text{out}}(K - K) \leq 2r_{\text{out}}(K)$$

and that equality holds if K is symmetric.

Solution 2.17. (a) Let $r := r_{\text{out}}(K)$, and assume that $K \subseteq x + rB^n$. Then, for every $u \in \mathbb{S}^{n-1}$, we must have

$$w_K(u) = h_K(u) + h_K(-u) \leq h_{x+rB^n}(u) + h_{x+rB^n}(-u) = 2r,$$

from where we get that

$$\max\left\{w_K(u) : u \in \mathbb{S}^{n-1}\right\} \leq 2r.$$

If K is symmetric, then it is clearly contained in the ball whose center is the symmetry center of K and whose radius is half the maximum width of K. This leads to

$$2r \leq \max\left\{w_K(u) : u \in \mathbb{S}^{n-1}\right\}.$$

A counterexample is the "kite" in $\mathbb{R}^2$ with vertices $(2, 0)$, $(-2, 0)$, $(0, 1)$, and $(0, -3)$.

(b) For each $u \in \mathbb{S}^{n-1}$, we have

$$\begin{aligned} w_{K-K}(u) = h_{K-K}(u) + h_{K-K}(-u) = h_K(u) + h_{-K}(u) + h_K(-u) + h_{-K}(-u) = \\ = 2\big[h_K(u) + h_K(-u)\big] = 2w_K(u). \end{aligned}$$

Therefore, since $K - K$ is centrally symmetric,

$$2r_{\text{out}}(K - K) = \max\left\{w_{K-K}(u) : u \in \mathbb{S}^{n-1}\right\} = \max\left\{2w_K(u) : u \in \mathbb{S}^{n-1}\right\} \leq 4r_{\text{out}}(K).$$

Notice that if K is centrally symmetric, then the last inequality holds with equality. The proof is complete.

□

A convex body $K \subseteq \mathbb{R}^n$ is said to be of *constant width* if its width function $w_K(u)$ is constant for $u \in \mathbb{S}^{n-1}$. In other words, we say that K is a body of constant width when it has the same width in every direction. As an alternative terminology, in this case, we also say that K **has** constant width.

Exercise 2.18. (a) *Prove that a convex body $K \subseteq \mathbb{R}^n$ has constant width if and only if $K - K$ is a ball (centered at the origin, of course). In this case, verify also that the radius of $K - K$ equals the width of K.*

(b) *Conclude that any centrally symmetric convex body of constant width is a ball.*

Solution 2.18. (a) Assume that the constant width of K is c. Then,

$$c = h_K(u) + h_K(-u) = h_K(u) + h_{-K}(u) = h_{K-K}(u)$$

for every $u \in \mathbb{S}^{n-1}$. On the other hand, we also have that

$$c = h_{cB^n}(u)$$

for any $u \in \mathbb{S}^{n-1}$. It follows that $K - K = cB^n$. The proof of the converse is similar. Just notice that if $K - K = cB^n$, then

$$c = h_{K-K}(u) = h_K(u) + h_{-K}(u) = h_K(u) + h_K(-u)$$

for all $u \in \mathbb{S}^{n-1}$. This means that K has constant width c.

(b) Since K has constant width, we already know that $K - K$ is a ball. On the other hand, because K is centrally symmetric, we get that $K - x_0 = -K + x_0$ for some $x_0 \in \mathbb{R}^n$, and hence, $K - K = 2K - 2x_0$. Thus,

$$K = \frac{1}{2}(K - K) + x_0,$$

from where it follows that K is a ball.

□

In what follows, recall that the *polar set* of a subset $A \subseteq \mathbb{R}^n$ is the set

$$A^\circ := \big\{z \in \mathbb{R}^n : \langle z, x\rangle \leq 1 \ \forall\ x \in A\big\}.$$

Exercise 2.19. *Let $K \subseteq \mathbb{R}^n$ be a convex body symmetric with respect to the origin.*

(a) *Show that the circumball of K is centered at the origin and that the inradius of K is attained for some ball centered at the origin.*

(b) *Prove that*

$$r_{\text{in}}(K^\circ) = \frac{1}{r_{\text{out}}(K)},$$

and derive an expression for $r_{\text{out}}(K^\circ)$ in terms of $r_{\text{in}}(K)$.

Solution 2.19. (a) Let $z + rB^n$ be the circumball of K. Since K is symmetric with respect to the origin, we have that $K = -K$, and hence, the inclusion $K \subseteq z + rB^n$ gives $K = -K \subseteq -(z + rB^n) = -z + rB^n$. Therefore, if $z \neq 0_{\mathbb{R}^n}$, then $z + rB^n$ and $-z + rB^n$ are distinct circumballs of K. Since the circumball is unique, we get that $z = 0_{\mathbb{R}^n}$.

Now, assume that the inradius (r, say) of K is attained for a ball $y + rB^n \subseteq K$. Then, $-y + rB^n = -(y + rB^n) \subseteq -K = K$. It follows that

$$\text{conv}\big((y + rB^n) \cup (-y + rB^n)\big) \subseteq K,$$

and we claim that

$$rB^n \subseteq \operatorname{conv}\big((y + rB^n) \cup (-y + rB^n)\big).$$

In fact, for any $u \in B^n$, we write

$$ru = \frac{1}{2}(y + ru) + \frac{1}{2}(-y + ru),$$

and the right-hand side is a convex combination of points of $(y + rB^n) \cup (-y + rB^n)$. It follows that $rB^n \subseteq K$, and this shows that the inradius of K is attained for some ball centered at the origin.

(b) Let $r = r_{\text{in}}(K^\circ)$. Then, we know from item **(a)** that $rB^n \subseteq K^\circ$. Since $(rB^n)^\circ = (1/r)B^n$, and because polarity reverses inclusion, we get

$$K = K^{\circ\circ} \subseteq \frac{1}{r}B^n.$$

Consequently, we have the inequality

$$r_{\text{out}}(K) \leq \frac{1}{r}.$$

If this inequality is strict, then the circumball of K has radius $\rho < 1/r$, and hence, $K \subseteq \rho B^n$ (from item **(a)**, the circumball of K is centered at the origin). This leads to

$$\frac{1}{\rho}B^n \subseteq K^\circ,$$

and since $1/\rho > r$, this contradicts the fact that r is the inradius of K°. Hence, the desired equality follows.

For the circumradius of K°, we have the expression

$$r_{\text{out}}(K^\circ) = \frac{1}{r_{\text{in}}(K)}.$$

Indeed, this follows from applying the equality that we have proved for K°:

$$r_{\text{in}}(K) = r_{\text{in}}(K^{\circ\circ}) = \frac{1}{r_{\text{out}}(K^\circ)},$$

and this concludes the solution of the exercise.

□

For what follows, remember that the *Hausdorff distance* (or *metric*) in the space $\mathcal{C}^n$ of compact and non-empty subsets of $\mathbb{R}^n$ is the distance defined by

$$d_{\mathcal{H}}(L, C) = \inf\{\lambda \geq 0 \colon L \subseteq C + \lambda B^n \text{ and } C \subseteq L + \lambda B^n\},$$

for $C, L \in \mathcal{C}^n$, where B^n denotes the Euclidean unit ball, as usual.

Exercise 2.20. *Regard the inradius and the circumradius as functionals defined over the family $\mathcal{CC}^n$ of non-empty compact convex subsets of $\mathbb{R}^n$. Prove that these functionals are continuous in the Hausdorff metric.*

Solution 2.20. We first prove the case of the circumradius. Let $(K_m)_{m\in\mathbb{N}}$ be a sequence in CC^n converging in the Hausdorff metric to $K \in CC^n$. Denote $r_m := r_{\text{out}}(K_m)$ for each $m \in \mathbb{N}$ and $r := r_{\text{out}}(K)$, and let $x + rB^n$ be the circumball of K, and $x_m + r_m B^n$ be the circumball of K_m for each $m \in \mathbb{N}$. For any given $\varepsilon > 0$, let $m_0 \in \mathbb{N}$ be such that $d_{\mathcal{H}}(K_m, K) < \varepsilon$ whenever $m > m_0$. Then, in this case, we have

$$K_m \subseteq K + \varepsilon B^n \subseteq x + (r + \varepsilon)B^n,$$

from where we get that $r_m \leq r + \varepsilon$ for any $m > m_0$. On the other hand, it also holds that

$$K \subseteq K_m + \varepsilon B^n \subseteq x_m + (r_m + \varepsilon)B^n,$$

which leads to $r \leq r_m + \varepsilon$ whenever $m > m_0$. Hence, we have that

$$r - \varepsilon \leq r_m \leq r + \varepsilon$$

for each $m > m_0$, from where it follows that $r_m \to r$. This proves the continuity of the circumradius.

Now, we consider the case of the inradius. We first let $K \in CC^n$ be a convex body, hence having positive inradius. If $(K_m)_{m\in\mathbb{N}}$ is a sequence in CC^n converging to K, then it is clear that to sufficiently large m, we have that K_m is a convex body (a way to verify this is to remember that the Lebesgue volume is continuous in the Hausdorff metric). Hence, we may assume without loss of generality that K_m is a convex body for any $m \in \mathbb{N}$. As before, denote the inradius of K and K_m by r and r_m, respectively, and fix balls $x + rB^n \subseteq K$ and $x_m + r_m B^n \subseteq K_m$ (for each $m \in \mathbb{N}$). For a given $\varepsilon > 0$, let $m_0 \in \mathbb{N}$ be such that $d_{\mathcal{H}}(K, K_m) < 2\varepsilon$ whenever $m > m_0$. We claim that

$$x + (r - \varepsilon)B^n \subseteq K_m$$

for every $m > m_0$. In fact, fix any $m > m_0$ and suppose that there exists a point $z \in x + (r - \varepsilon)B^n$ such that $z \notin K_m$. Since $K_m \subseteq K + 2\varepsilon B^n$, we get that $z \notin K + 2\varepsilon B^n$, which means that

$$|z - y| > 2\varepsilon$$

for every $y \in K$. Write $z = x + (r - \varepsilon)u$ for $u \in B^n$. Then, $y := x + ru \in x + rB^n \subseteq K$, and

$$|z - y| = \varepsilon|u| \leq \varepsilon,$$

which is a contradiction. Consequently, we have that $r_m \geq r - \varepsilon$ for each $m \in \mathbb{N}$. Interchanging the roles of K_m and K, we have that

$$x_m + (r_m - \varepsilon)B^n \subseteq K$$

for each $m > m_0$. Thus, $r \geq r_m - \varepsilon$ for every $m > m_0$, and this shows that $r_m \to r$.

It remains to consider the case where K is not a convex body or, equivalently (since $K \in CC^n$), $r_{\text{in}}(K) = 0$. There are two ways of tackling this case. The first one is to recall that the n-dimensional volume is continuous in the Hausdorff metric. Then, if $(K_m)_{m\in\mathbb{N}}$ is a sequence in CC^n, which converges to K, we have $\text{vol}_n(K_m) \to 0$. On the other hand, whenever K_m is a convex body, it contains a closed ball whose radius is $r_m := r_{\text{in}}(K_m)$. Thus, in this case it holds that

$$0 < \text{vol}_n(r_m B^n) \leq \text{vol}_n(K_m),$$

from where we get

$$0 < (r_m)^n \leq \frac{\mathrm{vol}_n(K_m)}{\mathrm{vol}_n(B^n)},$$

and the right-hand side goes to 0 as $m \to \infty$. Since we have additionally that $r_m = 0$ whenever K_m is not a convex body, we get that $r_m \to 0$, and the proof is done.

An alternative way is to notice that, if K is not a convex body, then K is contained in a hyperplane $\mathbf{h} := \mathbf{h}(u, \alpha)$. For any given $\varepsilon > 0$, if $d_{\mathcal{H}}(K_m, K) < \varepsilon$, then $K_m \subseteq K + \varepsilon B^n \subseteq \mathbf{h} + \varepsilon B^n$. The latter is the "slab" between the hyperplanes $\mathbf{h}(u, \alpha - \varepsilon)$ and $\mathbf{h}(u, \alpha + \varepsilon)$. It follows that, for sufficiently large m, the set K_m cannot contain any ball whose radius is greater than ε, which gives $r_m \leq \varepsilon$. This proves that $r_m \to 0$.

□

Exercise 2.21. **(a)** *Let $K \subseteq \mathbb{R}^n$ be a non-empty compact and convex set that is symmetric with respect to a hyperplane $\mathbf{h}(u, \beta)$. Prove that for any $u \in \mathbb{S}^{n-1}$, the Steiner symmetral $S_u(K)$ of K with respect to u is a translate of K.*

(b) *Conclude that any Steiner symmetral of any ball is a ball having the same radius.*

(c) *Prove that the inradius and the circumradius are respectively non-decreasing and non-increasing with respect to Steiner symmetrization.*

Remark 2.4. There is an illustration of Steiner symmetrization in Plate 2 (Figure 4).

Solution 2.21. **(a)** Let $x \in \mathrm{proj}_{u^\perp}(K)$, and assume that

$$\lambda_1 := \min\{t \in \mathbb{R} : x + tu \in K\} \text{ and}$$
$$\lambda_2 := \max\{t \in \mathbb{R} : x + tu \in K\}.$$

Since K is symmetric with respect to $\mathbf{h} = \mathbf{h}(u, \beta)$, we have that the midpoint of the segment $\mathrm{seg}[x + \lambda_1 u, x + \lambda_2 u]$ lies in $\mathbf{h} = \mathbf{h}(u, \beta)$. Hence,

$$\beta = \left\langle u, x + \frac{\lambda_1 + \lambda_2}{2} u \right\rangle = \frac{\lambda_1 + \lambda_2}{2}.$$

It is clear that the length of $\mathrm{seg}[x + \lambda_1 u, x + \lambda_2 u]$ is $\lambda_2 - \lambda_1$, and then if we write

$$S_u(K) \cap \{tu : t \in \mathbb{R}\} = \mathrm{seg}[x - \alpha u, x + \alpha u],$$

we have

$$\alpha = \frac{\lambda_2 - \lambda_1}{2}.$$

Thus, $x + \lambda_1 u = x - \alpha u + \beta u$ and $x + \lambda_2 u = x + \alpha u + \beta u$. In other words,

$$K \cap \{x + tu : t \in \mathbb{R}\} = \mathrm{seg}[x + \lambda_1 u, x + \lambda_2 u] = \beta u + \mathrm{seg}[x - \alpha u, x + \alpha u] =$$
$$= \beta u + S_u(K) \cap \{x + tu : t \in \mathbb{R}\}.$$

Since this holds for any $x \in \mathrm{proj}_{u^\perp}(K) =: K_u$, we get

$$K = \bigcup_{x \in K_u} K \cap (x + \mathbb{R}u) = \bigcup_{x \in K_u} \big(\beta u + S_u(K) \cap (x + \mathbb{R}u)\big) =$$
$$= \beta u + \bigcup_{x \in K_u} S_u(K) \cap (x + \mathbb{R}u) = \beta u + S_u(K),$$

and hence, $S_u(K) = -\beta u + K$.

(b) It is clear that any ball is symmetric with respect to any hyperplane passing through its center. Thus, any Steiner symmetral of any given ball B is a translate of B, and hence a ball having the same radius of B.

(c) If $K \subseteq x_0 + rB^n =: B$, then $S_u(B)$ is a ball of radius r containing K (recall that inclusion is preserved under Steiner symmetrizations). Hence, the circumradius is non-increasing under Steiner symmetrizations. The argument for the inradius is analogous.

□

Exercise 2.22. **(a)** *Verify that the Minkowski sum of two closed balls B_1 and B_2 is a closed ball whose radius equals the sum of the radii of B_1 and B_2.*

(b) *Let $A, B \subseteq \mathbb{R}^n$ be bounded sets. Prove that*

$$r_{\mathrm{out}}(A + B) \le r_{\mathrm{out}}(A) + r_{\mathrm{out}}(B) \quad \text{and}$$
$$r_{\mathrm{in}}(A + B) \ge r_{\mathrm{in}}(A) + r_{\mathrm{in}}(B).$$

Provide examples where the inequalities are strict.

Solution 2.22. **(a)** Let $B_1 := x_1 + r_1 B^n$ and $B_2 := x_2 + r_2 B^n$. We may assume that $r_1, r_2 > 0$, since otherwise the Minkowski sum gives just a translation. We will prove that

$$B_1 + B_2 = x_1 + x_2 + (r_1 + r_2)B^n.$$

First of all, if $z \in x_1 + x_2 + (r_1 + r_2)B^n$, then there exists $v \in \mathbb{R}^n$ such that $z = x_1 + x_2 + (r_1 + r_2)v$. Then,

$$z = x_1 + x_2 + (r_1 + r_2)B^n = (x_1 + r_1 B^n) + (x_2 + r_2 B^n)v \in B_1 + B_2,$$

and this gives the "$\supseteq$" inclusion. Now, let $y \in B_1 + B_2$. Then, there exist $u, v \in B^n$ such that $y = x_1 + r_1 u + x_2 + r_2 v$. Thus,

$$|y - (x_1 + x_2)| = |r_1 u + r_2 v| \le r_1|u| + r_2|v| \le r_1 + r_2,$$

from where we get that $y \in x_1 + x_2 + (r_1 + r_2)B^n$. This gives the "$\subseteq$" inclusion.

(b) It is straightforward that if $A \subseteq C$ and $B \subseteq D$, then $A + B \subseteq C + D$. Let $x_0 + r_{\mathrm{out}}(A)B^n$ be the circumball of A and $x_1 + r_{\mathrm{out}}(B)B^n$ be the circumball of B. Then,

$$A + B \subseteq x_0 + r_{\mathrm{out}}(A)B^n + x_1 + r_{\mathrm{out}}(B)B^n = x_0 + x_1 + (r_{\mathrm{out}}(A) + r_{\mathrm{out}}(B))B^n,$$

and hence, $A + B$ is contained in a closed ball whose radius is $r_{\mathrm{out}}(A) + r_{\mathrm{out}}(B)$. This gives the first inequality. The proof of the second inequality is completely analogous.

To give examples where the inequalities are strict, we consider sets in $\mathbb{R}^2$. Let $A = \text{seg}[0, e_1]$ and $B = \text{seg}[0, e_2]$. Then, we clearly have that $r_{\text{out}}(A) = r_{\text{out}}(B) = 1/2$. On the other hand, $A + B$ is a square whose sides have length 1. Therefore, $r_{\text{out}}(A + B) = \sqrt{2}/2 > 1/2$. This same example holds also for the inradius: we have that $r_{\text{in}}(A) = r_{\text{in}}(B) = 0$, and $r_{\text{in}}(A + B) = 1$.

□

In what follows, recall that the *relative interior* (*boundary*) of a set $A \subseteq \mathbb{R}^n$ is its interior (boundary) in the topology of its affine hull (denoted as $\text{aff}(A)$ and defined before the next exercise). We denote the relative interior (boundary) of A as $\text{relint}(A)$ ($\text{relbd}(A)$).

Exercise 2.23. *Let $K \subseteq \mathbb{R}^n$ be a non-empty convex set. Prove that*

(a) $\text{relint}(K) = \text{relint}(\text{cl}(K))$ *and*

(b) $\text{cl}(K) = \text{cl}(\text{relint}(K))$.

Solution 2.23. (a) Because $K \subseteq \text{cl}(K)$, we have the inclusion

$$\text{relint}K \subseteq \text{relint}(\text{cl}(K)).$$

Now, let $x \in \text{relint}(\text{cl}(K))$, and suppose that $\varepsilon > 0$ is such that $B(x, 2\varepsilon) \subseteq \text{cl}(K)$. Since the relative interior of any non-empty convex set is non-empty, we may choose a point $y \in \text{relint}K$, and we may assume that $y \neq x$ (otherwise, we already have that $x \in \text{relint}K$). Then,

$$z := x + \varepsilon \frac{x - y}{|x - y|} \in B(x, 2\varepsilon) \subseteq \text{cl}(K),$$

and we clearly have that $x \in \text{seg}[y, z)$. From Lemma 2.1.4 in the main book, it follows that $x \in \text{relint}K$. This shows that $\text{relint}(\text{cl}(K)) \subseteq \text{relint}K$.

(b) Since $\text{relint}K \subseteq K$, we have trivially that

$$\text{cl}(\text{relint}K) \subseteq \text{cl}(K).$$

Now, let $x \in \text{cl}(K)$, and choose $y \in \text{relint}(K)$. Again, we may assume that $y \neq x$ because otherwise we would already have $x \in \text{cl}(\text{relint}K)$. Using Lemma 2.1.4 one more time, we get that

$$\text{seg}[y, x) \subseteq \text{relint}K,$$

and it is clear that x is the limit of a sequence of points of $\text{seg}[y, x)$. This yields that $x \in \text{cl}(\text{relint}K)$, and the proof is complete.

□

In what follows, we use the notation $\text{aff}(A)$ to refer to the *affine hull* of a set $A \subseteq \mathbb{R}^n$. This is the smallest affine subspace containing A.

Exercise 2.24. (a) *Prove that for any affine subspace $S \subseteq \mathbb{R}^n$, $x \in S$ and $\varepsilon > 0$, we have*

$$\text{aff}(B(x, \varepsilon) \cap S) = S.$$

(b) *For any $A \subseteq \mathbb{R}^n$, show that the equality* $\text{relint}(\text{relint}A) = \text{relint}A$ *holds (in other words,* $\text{relint}A$ *is relatively open).*

(c) *Let $K \subseteq \mathbb{R}^n$ be convex. Prove that*

$$\text{relbd}(\text{relint}(K)) = \text{relbd}(K) = \text{relbd}(\text{cl}(K)).$$

Solution 2.24. (a) Given an affine subspace $S \subseteq \mathbb{R}^n$, a point $x \in S$, and a number $\varepsilon > 0$, we have

$$S = V + x$$

for some vector subspace $V \subseteq \mathbb{R}^n$ of dimension k, say. Hence, we may fix a basis $\{v_1, \ldots, v_k\}$ of V such that $|v_j| < \varepsilon$ for each $1 \leq j \leq k$. Thus, $x + v_j \in B(x, \varepsilon) \cap S$ for each $1 \leq j \leq k$, and we claim that this leads to

$$V + x \subseteq \text{aff}(B(x, \varepsilon) \cap S).$$

Indeed, any element $x + \alpha_1 v_1 + \ldots + \alpha_k v_k$ of $V + x$ can be written as

$$x + \alpha_1 v_1 + \ldots + \alpha_k v_k = \alpha_1(x + v_1) + \ldots + \alpha_k(x + v_k) + [1 - (\alpha_1 + \ldots + \alpha_k)]x,$$

and the latter is an affine combination of points of $B(x, \varepsilon) \cap S$. Since $S = V + x$, it follows that $S \subseteq \text{aff}(B(x, \varepsilon) \cap S)$. The reverse inclusion follows trivially from the facts that $B(x, \varepsilon) \cap S \subseteq S$ and $S = \text{aff}(S)$.

(b) If $\text{relint}A = \emptyset$, then there is nothing to prove. Therefore, we assume that $\text{relint}A \neq \emptyset$. Let $x \in \text{relint}A$. Then, there exists a number $\varepsilon > 0$ such that

$$B(x, 2\varepsilon) \cap \text{aff}(A) \subseteq A.$$

If $y \in B(x, \varepsilon) \cap \text{aff}(A)$, then we clearly have

$$B(y, \varepsilon) \cap \text{aff}(A) \subseteq B(x, 2\varepsilon) \cap \text{aff}(A) \subseteq A,$$

from where $y \in \text{relint}(A)$. Then, $B(x, \varepsilon) \cap \text{aff}(A) \subseteq \text{relint}A$. From item **(a)**, we have

$$\text{aff}(A) = \text{aff}(B(x, \varepsilon) \cap \text{aff}(A)) \subseteq \text{aff}(\text{relint}A) \subseteq \text{aff}(A),$$

where the last inclusion follows from the fact that $\text{relint}A \subseteq A$. Thus, $\text{aff}(A) = \text{aff}(\text{relint}A)$. Then, we may write the inclusion $B(x, \varepsilon) \cap \text{aff}(A) \subseteq \text{relint}A$, which we obtained before as

$$B(x, \varepsilon) \cap \text{aff}(\text{relint}A) \subseteq \text{relint}A,$$

from where we get that $x \in \text{relint}(\text{relint}A)$. This proves the inclusion

$$\text{relint}A \subseteq \text{relint}(\text{relint}A).$$

The reverse inclusion follows trivially from the fact that $\text{relint}A \subseteq A$.

(c) Because $\text{cl}(K)$ is a closed set, we have that

$$\text{cl}(K) = \text{relint}(\text{cl}(K)) \cup \text{relbd}(\text{cl}(K)),$$

and this union is clearly disjoint. Hence, we may write

$$\text{relbd}(\text{cl}(K)) = \text{cl}(K) \setminus \text{relint}(\text{cl}(K)).$$

From item **(a)** of Exercise 2.23, we may replace relint(cl(K)) by relint$K$ to get

$$\text{relbd}(\text{cl}(K)) = \text{cl}(K) \setminus \text{relint}K.$$

Since it is easy to see that cl(K) \ relint$K$ = relbd$K$, we get the equality relbd(cl(K)) = relbd$K$.

To prove the remaining equality, we observe that cl(relint$K$) is a closed set, and hence,

$$\text{relbd}(\text{relint}(K)) = \text{cl}(\text{relint}K) \setminus \text{relint}(\text{relint}K).$$

Now, we use the fact that relint(relint$K$) = relint$K$ and item **(b)** of Exercise 2.23 to write

$$\text{relbd}(\text{relint}K) = \text{cl}(K) \setminus \text{relint}K = \text{relbd}K,$$

and the proof is complete.

□

Exercise 2.25. *Let $K \subseteq \mathbb{R}^n$ be a compact convex set. Show that $K = \text{conv}(\text{relbd}K)$.*

Solution 2.25. Since K is closed, we have that $K = \text{int}(K) \cup \text{relbd}K$, and in particular, we have relbd$K \subseteq K$. Then,

$$\text{conv}(\text{relbd}K) \subseteq \text{conv}(K) = K.$$

Now, let $x \in K$. If $x \in \text{relbd}K$, then it is clear that we have $x \in \text{conv}(\text{relbd}K)$. In the case where $x \in \text{int}(K)$, we fix any vector $u \in \mathbb{S}^{n-1}$, which lies in the vector subspace associated (by means of a translation) with aff(K). We claim that the ray $x + \mathbb{R}_+ u$ intersects relbd$K$ at a point y. Indeed, let

$$t_0 := \sup\left\{t > 0 : x + tu \in K\right\}.$$

Then, we have that $t_0 > 0$ because x is an interior point of K, and $t_0 < +\infty$ since K is compact. Now, we verify that the point $y := x + t_0 u$ is a point of relbd$K$. In fact, it is clear that

$$y = \lim_{t \to t_0^-} (x + tu),$$

and hence, y is the limit of a sequence of points in K. Since K is closed, we have $y \in K$. As a consequence, any neighborhood of y contains a point of K (y itself). To verify that any neighborhood of y has also a point of aff(K) \ K, simply notice that for every $\varepsilon > 0$, we have that

$$x + \left(t_0 + \frac{\varepsilon}{2}\right) u \in \text{aff}(K) \setminus K,$$

since otherwise t_0 would not be the supremum of $t > 0$ such that $x + tu \in K$. Thus, we indeed have that $y \in \text{relbd}K$.

Analogously, the ray $x + \mathbb{R}_+(-u)$ intersects ∂K at a point z. It follows that $x \in \text{seg}[y, z]$, and since $y, z \in \text{relbd}K$, we get that $x \in \text{conv}(\text{relbd}K)$. This shows the inclusion $K \subseteq \text{conv}(\text{relbd}K)$.

□

Exercise 2.26. **(a)** *Let $A, B \subseteq \mathbb{R}^n$ be non-empty. Prove that*

$$\text{aff}(A + B) = \text{aff}(A) + \text{aff}(B).$$

(b) *Verify that for any $x \in \mathbb{R}^n$, we have* $\text{aff}(x+A) = x+\text{aff}(A)$. *Conclude that* $\text{relint}(x+A) = x+\text{relint}(A)$.

Solution 2.26. **(a)** A point $x \in \text{aff}(A + B)$ has the form

$$x = \lambda_1(a_1 + b_1) + \ldots + \lambda_m(a_m + b_m),$$

where $\lambda_1 + \ldots + \lambda_m = 1$, and $a_j \in A$ and $b_j \in B$ for every $1 \leq j \leq m$. Hence, we may rewrite

$$x = (\lambda_1 a_1 + \ldots + \lambda_m a_m) + (\lambda_1 b_1 + \ldots + \lambda_m b_m) \in \text{aff}(A) + \text{aff}(B),$$

and this gives the inclusion $\text{aff}(A + B) \subseteq \text{aff}(A) + \text{aff}(B)$.

To prove the reverse inclusion, assume that $\dim A = k$ and $\dim B = m$. Then, we may choose $k + 1$ affinely independent points $a_0, \ldots, a_k \in A$ and $m + 1$ affinely independent points $b_0, \ldots, b_m \in B$, such that the vector subspaces associated with A and B are spanned by $\{a_1 - a_0, \ldots, a_k - a_0\}$ and $\{b_1 - b_0, \ldots, b_m - b_0\}$, respectively. Thus, any point $x \in \text{aff} A$ can be written as

$$x = a_0 + \alpha_1(a_1 - a_0) + \ldots + \alpha_k(a_k - a_0),$$

for $\alpha_1, \ldots, \alpha_k \in \mathbb{R}$, and any point $y \in \text{aff}(B)$ can be written as

$$y = b_0 + \beta_1(b_1 - b_0) + \ldots + \beta_m(b_m - b_0),$$

for $\beta_1, \ldots, \beta_m \in \mathbb{R}$. Hence, any point $x + y \in \text{aff}(A) + \text{aff}(B)$ can be written as

$$\begin{aligned} x + y = a_0 + \alpha_1(a_1 - a_0) + \ldots + \alpha_k(a_k - a_0) + b_0 + \beta_1(b_1 - b_0) + \ldots + \beta_m(b_m - b_0) = \\ = a_0 + b_0 + \alpha_1(a_1 + b_0 - a_0 - b_0) + \ldots + \alpha_k(a_k + b_0 - a_0 - b_0) + \\ + \beta_1(b_1 + a_0 - a_0 - b_0) + \ldots + \beta_m(b_m + a_0 - a_0 - b_0) = \\ = \alpha_1(a_1 + b_0) + \ldots + \alpha_k(a_k + b_0) + \beta_1(b_1 + a_0) + \ldots + \beta_m(b_m + a_0) + \\ + \left[1 - \sum_{j=1}^{k} \alpha_j - \sum_{j=1}^{m} \beta_j\right](a_0 + b_0), \end{aligned}$$

and the latter is clearly an affine combination of points of $A + B$. This proves that $\text{aff}(A) + \text{aff}(B) \subseteq \text{aff}(A + B)$.

(b) The first claim is straightforward from item **(a)**. Indeed, since $\text{aff}(x) = \{x\}$, we have

$$\text{aff}(x + A) = \text{aff}(x) + \text{aff}(A) = x + \text{aff}(A).$$

Now, let $y \in \text{relint}(x + A)$. Then, there exists a number $\varepsilon > 0$ such that

$$B(y, \varepsilon) \cap \text{aff}(x + A) \subseteq x + A.$$

Thus,

$$-x + B(y, \varepsilon) \cap \text{aff}(x + A) \subseteq A,$$

since it is obvious that inclusion is preserved under translation, and also it is clear that $x + A - x = A$. Finally,

$$A \supseteq -x + B(y, \varepsilon) \cap \operatorname{aff}(x + A) = \big(B(y, \varepsilon) - x\big) \cap \big(x + \operatorname{aff}(A) - x\big) = B(y - x, \varepsilon) \cap \operatorname{aff}(A),$$

and hence, we have that $y - x \in \operatorname{relint} A$. Now, we simply write $y = x + (y - x) \in x + \operatorname{relint}(A)$, and the inclusion $\operatorname{relint}(x + A) \subseteq x + \operatorname{relint}(A)$ is proved.

For the reverse inclusion, assume that $z \in x + \operatorname{relint}(A)$, and let $a \in \operatorname{relint}(A)$ be such that $z = x + a$. Let $\varepsilon > 0$ be such that $B(a, \varepsilon) \cap \operatorname{aff}(A) \subseteq A$. Then,

$$A + x \supseteq x + B(a, \varepsilon) \cap \operatorname{aff}(A) = B(x + a, \varepsilon) \cap (x + \operatorname{aff}(A)) = B(z, \varepsilon) \cap \operatorname{aff}(A + x),$$

and this shows that $z \in \operatorname{relint}(A + x)$. The proof is complete.

□

Exercise 2.27. **(a)** *Let $A, B \subseteq \mathbb{R}^n$ be non-empty sets. Prove that A and B can be properly separated if and only if $A - B$ and $\{0_{\mathbb{R}^n}\}$ can be properly separated.*

(b) *Assume that $x \in \operatorname{relint}(A)$. Show that x and A cannot be properly separated.*

(c) *If A and B are convex, show that*

$$\operatorname{relint}(A + B) \subseteq \operatorname{relint}(A) + \operatorname{relint}(B).$$

Solution 2.27. **(a)** Assume that A and B are properly separated by the hyperplane $\mathbf{h} := \mathbf{h}(u, \alpha)$. We may assume without loss of generality that $A \subseteq \mathbf{h}^-$, $B \subseteq \mathbf{h}^+$ and that there exists a point $a_0 \in A \cap \operatorname{int}(\mathbf{h}^-)$. Hence, $\langle u, a_0\rangle < \alpha$. We claim that $A - B$ and $\{0_{\mathbb{R}^n}\}$ are properly separated by the hyperplane $\mathbf{h}(u, 0)$. Indeed, we obviously have that $\{0_{\mathbb{R}^n}\} \in \mathbf{h}^+(u, 0)$. On the other hand, for any $a - b \in A - B$, since $\langle u, a\rangle \leq \alpha$ and $\langle u, b\rangle \geq \alpha$, we get

$$\langle u, a - b\rangle \leq \alpha - \alpha = 0,$$

from where we get that $A - B \subseteq \mathbf{h}^-(u, 0)$. Hence, we already have that $A - B$ and $\{0_{\mathbb{R}^n}\}$ are separated by $\mathbf{h}(u, 0)$. To verify that the separation is proper, observe that for any $b \in B$, we have

$$\langle u, a_0 - b\rangle = \langle u, a_0\rangle - \langle u, b\rangle < \alpha - \alpha = 0,$$

from where we get that $a_0 - b \in \operatorname{int}(\mathbf{h}^-(u, 0))$.

Now, assume that $A - B$ and $\{0_{\mathbb{R}^n}\}$ are properly separated by a hyperplane $\mathbf{h}(u, \beta)$, and assume without loss of generality that $A - B \subseteq \mathbf{h}^-(u, \beta)$ and $\{0_{\mathbb{R}^n}\} \in \mathbf{h}^+(u, \beta)$. We claim that $A - B$ and $\{0_{\mathbb{R}^n}\}$ are properly separated by $\mathbf{h}(u, 0)$. Indeed, we have that $\{0_{\mathbb{R}^n}\} \subseteq \mathbf{h}^+(u, 0)$, and $0 = \langle u, 0_{\mathbb{R}^n}\rangle \geq \beta$, from where it follows that $\langle u, a - b\rangle \leq \beta \leq 0$ for any $a - b \in A - B$. Thus, $A - B \subseteq \mathbf{h}^-(u, \beta)$. This shows that $\{0_{\mathbb{R}^n}\}$ and $A - B$ are separated by $\mathbf{h}(u, 0)$, but we still have to show that the separation is proper. If $\beta = 0$, then there is nothing to prove, because in this case, $\mathbf{h}(u, \beta) = \mathbf{h}(u, 0)$ and the hypothesis states that $\mathbf{h}(u, \beta)$ properly separates $\{0_{\mathbb{R}^n}\}$ and $A - B$. If $\beta < 0$, then $\mathbf{h}^-(u, \beta) \subseteq \operatorname{int}(\mathbf{h}(u, 0))$, and then $A - B \subseteq \operatorname{int}(\mathbf{h}^-(u, 0))$. Observe carefully that in both cases, we may assume that $(A - B) \cap \operatorname{int}(\mathbf{h}^-(u, 0)) \neq \emptyset$. Therefore, we get that $\langle u, a - b\rangle \leq 0$ for every $a - b \in A - B$, and also there exists a point $a_0 - b_0 \in A - B$ such that $\langle u, a_0 - b_0\rangle < 0$. Let

$$\alpha := \sup\{\langle u, a\rangle : a \in A\}.$$

This is finite, because for any $b \in B$, the number $\langle u, b\rangle$ is an upper bound of the set $\{\langle u, a\rangle : a \in A\}$. By the definition of α, we have that $A \subseteq \mathbf{h}^-(u, \alpha)$. Also, since

$$\langle u, a\rangle \leq \langle u, b\rangle$$

for any $a \in A$ and $b \in B$, we obtain that

$$\langle u, b\rangle \geq \sup\{\langle u, a\rangle : a \in A\} = \alpha,$$

and hence, $B \subseteq \mathbf{h}^+(u, \alpha)$. Thus, the hyperplane $\mathbf{h}(u, \alpha)$ separates A and B. To verify that the separation is proper, observe that $\langle u, a_0\rangle < \langle u, b_0\rangle$, and this yields that $a_0 \in \operatorname{int}(\mathbf{h}^-(u, \alpha))$ or $b_0 \in \operatorname{int}(\mathbf{h}^+(u, \alpha))$.

(b) If $x \in \operatorname{relint}(A)$, then there exists a number $\varepsilon > 0$ such that $B(x, 2\varepsilon) \cap \operatorname{aff}(A) \subseteq A$. Assume that $\mathbf{h} := \mathrm{h}(u, \alpha)$ is a hyperplane that properly separates x and A. Without loss of generality, suppose that $A \subseteq \mathbf{h}^-$. By doing so, we have that $x \in \mathbf{h}^+$, and since $x \in A \subseteq \mathbf{h}^-$, we get that $x \in \mathbf{h}^+ \cap \mathbf{h}^- = \mathbf{h}$. Since the separation is proper, we cannot have $A \subseteq \mathbf{h}$, and hence, there is a point $z \in A \cap \operatorname{int}(\mathbf{h}^-)$. It follows that $\langle u, z\rangle < \alpha$. Letting

$$v = \frac{z - x}{|z - x|},$$

we have that $x - \varepsilon v \in B(x, 2\varepsilon) \cap \operatorname{aff}(A)$, because

$$x - \varepsilon v = \left(1 - \frac{\varepsilon}{|z - x|}\right) x + \frac{\varepsilon}{|z - x|} z,$$

and the right-hand side is an affine combination of $x, z \in A$. Also,

$$\langle u, x - \varepsilon v\rangle = \alpha - \varepsilon \left\langle u, \frac{z - x}{|z - x|}\right\rangle = \alpha + \frac{\varepsilon\alpha}{|z - x|} - \frac{\varepsilon}{|z - x|}\langle u, z\rangle > \alpha,$$

and hence, $x - \varepsilon v \in \operatorname{int}(\mathbf{h}^+)$. This is a contradiction because

$$x - \varepsilon v \in B(x, 2\varepsilon) \cap \operatorname{aff}(A) \subseteq A \subseteq \mathbf{h}^-,$$

and the proof is complete.

(c) Let $x \in \operatorname{relint}(A + B)$. From item **(b)** of Exercise 2.26, we may assume that $x = 0_{\mathbb{R}^n}$. Since $0_{\mathbb{R}^n} \in \operatorname{relint}(A + B)$, we have from item **(b)** that $\{0_{\mathbb{R}^n}\}$ and $A + B$ cannot be properly separated. Thus, from item **(a)**, A and $-B$ cannot be separated, and hence, Theorem 2.2.3 gives that there is a point

$$z \in \operatorname{relint}(A) \cap \operatorname{relint}(-B).$$

Therefore, $-z \in \operatorname{relint}(B)$, and $0_{\mathbb{R}^n} = z + (-z) \in \operatorname{relint}(A) + \operatorname{relint}(B)$. This concludes the proof.

□

The *metric projection* of a point $x \in \mathbb{R}^n$ onto a set A is the point of A of minimal distance to x (this always exists and is unique if A is non-empty, closed, and convex; see the main book [29]).

Exercise 2.28. *Let $A \subseteq \mathbb{R}^n$ be a closed non-empty set. Show that for each $x \notin A$, one has*

$$\operatorname{dist}(x, A) = \sup\left\{r > 0 : (x + rB^n) \cap A = \emptyset\right\}.$$

Verify also that the set of metric projections of x onto A is the intersection of A with the ball $x + \operatorname{dist}(x, A)B^n$.

Solution 2.28. Let $z \in A$ be a point such that $|x - z| = \operatorname{dist}(x, A)$, and write $d = \operatorname{dist}(x, A)$. Then, $z \in (x + dB^n) \cap A$, and hence,

$$\operatorname{dist}(x, A) \leq \sup\left\{r > 0 : (x + rB^n) \cap A = \emptyset\right\}.$$

Now, let $r > 0$ be such that $(x+rB^n)\cap A = \emptyset$, and still assume that $z \in A$ is such that $\operatorname{dist}(x, A) = |x-z|$. We have that $z \notin x + rB^n$ (otherwise, the intersection would not be empty), and this gives that $\operatorname{dist}(x, A) = |x - z| > r$. Since this holds for every $r > 0$ such that $(x + rB^n) \cap A = \emptyset$, we get that

$$\operatorname{dist}(x, A) \geq \sup\left\{r > 0 : (x + rB^n) \cap A = \emptyset\right\},$$

and the equality is proved. The other claim is trivial.

□

Exercise 2.29. *Let $A \subseteq \mathbb{R}^n$ be a closed set having non-empty interior. Assume that for each boundary point x of A, there exists a hyperplane supporting A at x. Prove that A is convex. Also, show that the hypothesis that A has non-empty interior cannot be dropped.*

Solution 2.29. The case where $n = 1$ is trivial, and hence, we assume that $n \geq 2$. If A is not convex, then there are (distinct) points $x, y \in A$ such that the segment $\operatorname{seg}[x, y]$ has a point z that is not in A. Since A has non-empty interior, we get that A contains an open ball, and hence, we may choose a point $a \in \operatorname{int}(A)$ such that x, y and a are affinely independent. Because $a \in A$ and $z \notin A$, there is a point $w \in \partial A \cap \operatorname{seg}[a, z)$, and by the hypothesis, we have that A is supported by a hyperplane $\mathbf{h}$ at w. Of course, we have that $a \notin \mathbf{h}$ (indeed, a is an interior point of A), and hence, the points x, y and a are not simultaneously in $\mathbf{h}$. It follows that the intersection of $\mathbf{h}$ and $\operatorname{aff}\{x, y, a\}$ is a line ℓ, and of course, ℓ contains w. The points a and z are in opposite sides of ℓ (with respect to $\operatorname{aff}\{x, y, a\}$), and hence, either x or y is not in the same side as a. This contradicts the fact that $\mathbf{h}$ supports A.

□

Exercise 2.30. *The objective of this exercise is to prove that any closed set of $\mathbb{R}^n$ for which metric projections are unique is convex. We divide it into steps. In what follows, $A \subseteq \mathbb{R}^n$ is a closed non-empty set.*

(a) *Show that if A is not convex, then there exist points $x, y \in A$ such that $A \cap \operatorname{seg}[x, y] = \{x, y\}$.*

(b) *Verify that if x and y are given as in item* **(a)**, *then there exists a number $\rho > 0$ such that*

$$\left(\frac{x + y}{2} + \rho B^n\right) \cap A = \emptyset.$$

(c) *Letting x, y and ρ be given as in items* **(a)** *and* **(b)**, *write*

$$B_0 := \frac{x + y}{2} + \rho B^n,$$

and let

$$\mathcal{B} := \left\{B : B \text{ is a closed ball such that } B \supseteq B_0 \text{ and } (\operatorname{int} B) \cap A = \emptyset\right\}.$$

Prove that $\mathcal{B}$ contains a ball B_1 with maximal radius and that this radius is greater than ρ.

(d) *Verify that the intersection $\partial B_0 \cap \partial B_1$ is either empty or a singleton.*

Now, assume that A has the property that for each point $x \in \mathbb{R}^n$, there exists a unique point $y \in A$ such that $|x - y| = \operatorname{dist}(x, A)$.

(e) *If B_1 is the closed ball as defined in item* **(c)***, show that $B_1 \cap A$ is non-empty and consists of a unique point w, say.*

(f) *Let w be the point given in item* **(e)***. If $\partial B_0 \cap \partial B_1 \neq \emptyset$, having item* **(d)** *in mind, we let z be the unique point of this intersection. Otherwise, we let $z = (x + y)/2$. Prove that in both cases, there exists a number $\varepsilon > 0$ such that $B_1 + \varepsilon(z - w) \subseteq \mathcal{B}$.*

(g) *Show that $\mathcal{B}$ contains a ball whose radius is greater than the radius of B_1. Conclude that this contradiction yields the convexity of A.*

Solution 2.30. (a) If A is not convex, then we may let $a, b \in A$ be so that there exists $\lambda_0 \in (0, 1)$ such that $(1 - \lambda_0)a + \lambda_0 b \notin A$. Thus, we let

$$\lambda_1 = \inf\{\lambda > 0 : (1 - \lambda)a + \lambda b \notin A\},$$

and set $x = (1 - \lambda_1)a + \lambda_1 b$. Since A is closed, we clearly have that $x \in A$. Now, let

$$\lambda_2 = \sup\{\lambda > \lambda_1 : (1 - \delta)a + \delta b \notin A \text{ for all } \sigma \in (\lambda_1, \lambda)\},$$

and put $y = (1 - \lambda_2)a + \lambda_2 b$. Since A is closed, we have that $\mathbb{R}^n$ is open, and hence, $y \in A$ (otherwise, we would be able to slightly enlarge the supremum in the definition of λ_2). From the definitions of x and y, we have that $A \cap \operatorname{seg}[x, y] = \{x, y\}$.

(b) It is clear that $(x + y)/2 \notin A$, and $\mathbb{R}^n \setminus A$ is open. The result follows.

(c) Let

$$r_0 = \sup\big\{r : r \text{ is the radius of a ball in } \mathcal{B}\big\},$$

and let D_m be a sequence in $\mathcal{B}$ whose respective radii r_m are such that $r_m \to r_0$. Using Blaschke's selection theorem, we may assume that $D_m \to B_1$ for some ball B_1 of radius r_0. Since B_0 is compact, A is closed, and B_0 and A are disjoint, we get that $\mathcal{B}$ contains a ball whose radius is greater than ρ. It follows that $r_0 > \rho$.

(d) This follows from the facts that $B_0 \subseteq B_1$ and their radii are different.

(e) If $B_1 \cap A = \emptyset$, then, since B_1 is compact and A is closed, we may enlarge the radius of B_1 while still having $\operatorname{int}(B_1) \cap A = \emptyset$. This contradicts the definition of r_0. Hence, $B_1 \cap A \neq \emptyset$. If this intersection has more than one point, then there are two points in A realizing the distance for the center of B_1. This contradicts the assumption on the uniqueness of such a point.

(f) It is clear that, in both cases, if ε is small enough, then $B_1 + \varepsilon(z - w)$ contains B_0. To verify that ε can be taken small enough such that $\operatorname{int}(B_1 + \varepsilon(z - w)) \cap A = \emptyset$, consider a small (open) cap around w. The complement of this cap has positive distance to A, and hence, we do not need to worry about it. For the small cap, observe that we are moving "away" from w, and hence, for small ε, the translated

cap will not have an interior point lying in A.

(g) The ball $B_1 + \varepsilon(z - w)$ does not intersect A, and hence, we may enlarge its radius (which is r_0, of course) without intersecting A. It follows that $\mathcal{B}$ contains a ball whose radius is greater than r_0, a contradiction.

□

Exercise 2.31. *Let $h\colon \mathbb{R}^n \to \mathbb{R}$ be a sublinear function.*

(a) *Prove that h is the support function of a convex body that contains the origin as an interior point if and only if $h(u) > 0$ for every $u \in \mathbb{S}^{n-1}$.*

(b) *Show that h is the support function of a convex body if and only if $h(u)+h(-u) > 0$ for each $u \in \mathbb{S}^{n-1}$.*

(c) *Let K be the non-empty convex compact set whose support function is h. Verify that K is centered at the origin if and only if h is even.*

Solution 2.31. (a) We already know that $h = h_K$ for some $K \in CC^n$. If K is a convex body having the origin as an interior point, then there exists a number $\rho > 0$ such that $\rho B^n \subseteq K$. Hence,

$$h(u) = h_K(u) \geq h_{\rho B^n}(u) = \rho > 0$$

for any $u \in \mathbb{S}^{n-1}$. Conversely, if $h_K(u) = h(u) > 0$ for any $u \in \mathbb{S}^{n-1}$, then by the continuity of h and the compactness of $\mathbb{S}^{n-1}$, we get that there exists a number $\beta > 0$ with the property that $h(u) \geq \beta$ (namely, let β be simply the minimum of h over $\mathbb{S}^{n-1}$). Thus, $h_K(u) \geq \beta = h_{\beta B^n}(u)$ for each $u \in \mathbb{S}^{n-1}$, and it follows that $\beta B^n \subseteq K$. This shows that K is a convex body having the origin in its interior.

(b) Again, we assume that $h = h_K$ for some $K \in CC^n$. If K is a convex body, then we may let $B := x_0 + \rho B^n \subseteq K$. For any $u \in \mathbb{S}^{n-1}$, we have

$$h(u) + h(-u) = h_K(u) + h_K(-u) \geq h_B(u) + h_B(-u) = \langle x_0, u\rangle + \rho + \langle x_0, -u\rangle + \rho = 2\rho > 0.$$

On the other hand, if K is not a convex body, then $K \subseteq \mathbf{h}(u, \beta)$ for some $u \in \mathbb{S}^{n-1}$ and $\beta \in \mathbb{R}$. Then, $\langle z, u\rangle = \beta$ for any $z \in K$, from where we get that $h_K(u) = \beta$ and $h_K(-u) = -\beta$. It follows that $h(u) + h(-u) = \beta - \beta = 0$, and the proof is complete.

(c) This result is a consequence of the fact that $h_A(-x) = h_{-A}(x)$ for any non-empty set $A \subseteq \mathbb{R}^n$ and every point $x \in \mathbb{R}^n$. This equality can be immediately derived from the definition of the support function or, alternatively, as a consequence of (2.6.5) (in the main book) for $T = -\mathrm{Id}_{\mathbb{R}^n}$.

If K is centered at the origin, then $K = -K$. It follows that $h_K(x) = h_{-K}(x) = h_K(-x)$ for any $x \in \mathbb{R}^n$. Thus, h_K is even. For the converse, we simply notice that if h_K is even, then $h_K(x) = h_K(-x) = h_{-K}(x)$ for every $x \in \mathbb{R}^n$. This gives that $K = -K$, and hence, K is centered at the origin.

□

Exercise 2.32. *Prove that if $K \in CC^n$ is centered at a point $x \in K$, then K is not centered with respect to any other point. Verify that if K is centered at the origin, then a translate of K by a non-zero vector is not centered at the origin.*

Solution 2.32. If K is centered at x, then we may write $K = x + K_0$, where $K_0 := K - x$ is centered at the origin. By item **(c)** of Exercise 2.31, it follows that the support function of K_0 is even. Assume

that K is symmetric with respect to a point $z \neq x$. Then, $K - z$ is centered at the origin, and hence, its support function is also even. Writing $K - z = K_0 + (x - z)$, we have that

$$h_{K-z}(y) = h_{K_0}(y) + \langle x - z, y \rangle.$$

Because h_{K_0} is even and the function $y \mapsto \langle x - z, y \rangle$ is not even (because $x - z \neq 0_{\mathbb{R}^n}$), we get that the right-hand side of the equality above is not an even function, and this contradiction gives that K is not symmetric with respect to another point than x.

For the other claim, just notice that if both K and $K + x$ are centered at the origin (for some $x \neq 0_{\mathbb{R}^n}$), then $K + x$ is symmetric with respect to $0_{\mathbb{R}^n}$ and to x, which contradicts what we just have proved.

□

Exercise 2.33. *Let $K \subseteq \mathbb{R}^n$ be a convex body having the origin as an interior point, and let γ_K be the gauge function of K (see Section 2.3 of the main book). Prove that*

$$\partial \gamma_K(0_{\mathbb{R}^n}) = K^\circ.$$

Conclude that γ_K is not differentiable at the origin.

Solution 2.33. By definition, we have that $v \in \partial \gamma_K(0_{\mathbb{R}^n})$ if and only if

$$\gamma_K(y) \geq \langle v, y \rangle$$

for any $y \in \mathbb{R}^n$. Since both sides are positively homogeneous and every $y \in \mathbb{R}^n$ can be written as $y = tz$ for some $z \in K$ and some $t \geq 0$, this condition is equivalent to

$$\langle v, z \rangle \leq \gamma_K(z) \leq 1$$

for any $z \in K$. Since this characterizes the polar body K°, it follows that $\partial \gamma_K(0_{\mathbb{R}^n}) = K^\circ$. Because K° is not a singleton, we get that γ_K is not differentiable at $0_{\mathbb{R}^n}$.

□

Exercise 2.34. *Let $C \subseteq \mathbb{R}^n$ be a non-empty compact convex set.*

(a) *Verify that there exists $u_0 \in \mathbb{S}^{n-1}$ such that*

$$h_C(u_0) = \sup \left\{ h_C(u) : u \in \mathbb{S}^{n-1} \right\}.$$

Check also that this supremum is positive if $C \neq \{0_{\mathbb{R}^n}\}$.

(b) *Let u_0 be as in item* **(a)**, *and assume that u_0 is an outer normal of C at $z_0 \in \partial C$. Prove that $z_0 = h_C(u_0)u_0$.*

(c) *Let z_0 be as in item* **(b)**. *Show that z_0 is an exposed point of C.*

Solution 2.34. **(a)** The existence of such a vector $u_0 \in \mathbb{S}^{n-1}$ follows immediately from the continuity of h_C and the compactness of $\mathbb{S}^{n-1}$. If $C \neq \{0_{\mathbb{R}^n}\}$, then there exists $u \in \mathbb{S}^{n-1}$ such that $h_C(u) \neq 0$. If $h_C(u) > 0$, then we are done. If $h_C(u) < 0$, then the sublinearity of h_C gives

$$h_C(u) + h_C(-u) \geq h_C(u - u) = h_C(0_{\mathbb{R}^n}) = 0,$$

and hence, $h_C(-u) \geq -h_C(u) > 0$.

(b) Let

$$\beta_0 := \sup\left\{h_C(u) : u \in \mathbb{S}^{n-1}\right\} = h_C(u_0).$$

Then, $h_C(u) \leq \beta_0 = h_{\beta_0 B^n}(u)$ for any $u \in \mathbb{S}^{n-1}$. It follows that $C \subseteq \beta_0 B^n$, and in particular, we have that $z_0 \in \beta_0 B^n$. This gives the inequality $|z_0| \leq \beta_0$. On the other hand, since u_0 supports C at z_0, we have that $\beta_0 = h_C(u_0) = \langle u_0, z_0\rangle$. Therefore, the Cauchy-Schwarz inequality gives

$$\beta_0 = \langle u_0, z_0\rangle \leq |z_0||u_0| = |z_0| \leq \beta_0,$$

and hence, we have that

$$|z_0||u_0| = \langle u_0, z_0\rangle = \beta_0.$$

The equality case of the Cauchy-Schwarz inequality gives that $z_0 = \alpha u_0$ for some $\alpha \in \mathbb{R}$. By the second equality above, we get that $\alpha = h_C(u_0)$.

(c) We claim that

$$\mathbf{h}(u_0, \langle u_0, z_0\rangle) \cap C = \{z_0\}.$$

Indeed, let $z \in \mathbf{h}(u_0, \langle u_0, z_0\rangle) \cap \partial C$. Then, since $\mathbf{h}(u_0, \langle u_0, z_0\rangle)$ is a supporting hyperplane of C, we get that z is a boundary point of C where u_0 is an outer normal. By item **(b)**, we get that $z = h_C(u_0)u_0 = z_0$, and hence, $\mathbf{h}(u_0, \langle u_0, z_0\rangle) \cap C$ only contains the point z_0.

□

For the next exercise, recall that f^* denotes the *conjugate function* of f (see Section 1.4 of the main book for definitions and properties).

Exercise 2.35. *Let $K \subseteq \mathbb{R}^n$ be a non-empty closed and convex set.*

(a) *Verify that $K = \mathrm{dom}(h_K^*)$.*

(b) *Prove that $\partial h_K(0_{\mathbb{R}^n}) = K$, where we remind the reader that $\partial h_K(0_{\mathbb{R}^n})$ denotes the subdifferential of h_K at $0_{\mathbb{R}^n}$.*

Solution 2.35. (a) From item **(c)** of Exercise 1.20, we have that $(I_K^\infty)^* = h_K$. Because we are assuming that K is closed and convex, we get that the indicator function I_K^∞ is closed and convex. Therefore, from Theorem 1.4.1 in the main book, we get that

$$I_K^\infty = (I_K^\infty)^{**} = h_K^*.$$

Thus,

$$\mathrm{dom}(h_K^*) = \mathrm{dom}(I_K^\infty) = K,$$

where the last equality follows directly from the definition of the indicator function.

(b) First, observe that if $x \in K$, then

$$h_K(y) = \sup\left\{\langle y, z\rangle : z \in K\right\} \geq \langle x, y\rangle$$

for any $y \in \mathbb{R}^n$. Thus, since $h_K(0_{\mathbb{R}^n}) = 0$, we get that

$$h_K(y) - h_K(0_{\mathbb{R}^n}) \geq \langle x, y \rangle$$

for every $y \in \mathbb{R}^n$. This means that $x \in \partial h_K(0_{\mathbb{R}^n})$. Hence, $K \subseteq \partial h_K(0_{\mathbb{R}^n})$.

On the other hand, assume that there exists a point $x_0 \in \partial h_K(0_{\mathbb{R}^n})$ such that $x_0 \notin K$. Because K is closed, we have that $\delta := \operatorname{dist}(x_0, K) > 0$, and since K is convex, we may assume that this distance is attained for a point $z_0 \in \partial K$. Letting

$$u = \frac{x_0 - z_0}{|x_0 - z_0|},$$

we have that u is an outer normal of K at z_0. Thus, $h_K(u) = \langle u, z_0 \rangle$, and since $x_0 = z_0 + |x_0 - z_0|u$, we get that

$$\langle u, x_0 \rangle = \langle u, z_0 \rangle + |x_0 - z_0| > \langle u, z_0 \rangle = h_K(u).$$

But, since $x_0 \in \partial h_K(0_{\mathbb{R}^n})$, we have the inequality

$$h_K(u) \geq \langle u, x_0 \rangle,$$

and hence, we have a contradiction. It follows that $x_0 \in K$, and this shows the remaining inclusion $\partial h_K(0_{\mathbb{R}^n}) \subseteq K$.

□

Exercise 2.36. *Let $K \subseteq \mathbb{R}^n$ be a non-empty, compact, and convex set. For each non-zero $u \in \mathbb{R}^n$, let*

$$F(K, u) := \{x \in K : \langle x, u \rangle = h_K(u)\}.$$

(a) *Assume that* $\mathbf{h}$ *is the supporting hyperplane of K whose outer normal is u. Prove that* $\mathbf{h} \cap K = F(K, u)$.

(b) *Show that $F(K, u)$ is a face of K for any non-zero vector $u \in \mathbb{R}^n$.*

(c) *Prove that $\partial h_K(u) = F(K, u)$ for each non-zero vector $u \in \mathbb{R}^n$.*

Solution 2.36. (a) Let $z_0 \in K$ be a point where u is an outer normal of K. Then, K is supported by the hyperplane $\mathbf{h} := \mathbf{h}(u, \langle u, z_0 \rangle)$, and $K \subseteq \mathbf{h}^-$. It follows that $h_K(u) = \langle u, z_0 \rangle$. Thus,

$$\mathbf{h} = \{x \in \mathbb{R}^n : \langle u, x \rangle = h_K(u)\},$$

and this leads immediately to the equality $K \cap \mathbf{h} = F(K, u)$.

(b) From item **(a)** and Proposition 2.4.7 in the main book, we have that $F(K, u)$ is an exposed face, hence a face.

(c) First, let $z \in F(K, u)$. Then, $\langle u, z \rangle = h_K(u)$. Thus, $z \in K$, and this leads to $h_K(y) \geq \langle y, z \rangle$ for every $y \in \mathbb{R}^n$. It follows that

$$h_K(y) - h_K(u) = h_K(y) - \langle u, z \rangle \geq \langle y, z \rangle - \langle u, z \rangle = \langle z, y - u \rangle,$$

and this inequality gives that $z \in \partial h_K(u)$. This proves the inclusion $K \subseteq \partial h_K(u)$.

Now, prove the reverse inclusion. For this sake, let $w \in \partial h_K(u)$, and observe that we have

$$h_K(y) - h_K(u) \geq \langle w, y - u \rangle$$

for every $y \in \mathbb{R}^n$. On the other hand, since h_K is a sublinear function, we have

$$h_K(y) = h_K(u + y - u) \leq h_K(u) + h_K(y - u), \tag{2.2}$$

from where we get (using also the previous inequality)

$$h_K(y - u) \geq h_K(y) - h_K(u) \geq \langle w, y - u \rangle$$

for every $y \in \mathbb{R}^n$. Writing $y = v + u$, we get

$$h_K(v) \geq \langle w, v \rangle$$

for any $v \in \mathbb{R}^n$. This gives that $w \in \partial h_K(0_{\mathbb{R}^n})$, and from item **(b)** of Exercise 2.35, we obtain that $w \in K$. In particular, it holds that

$$\langle w, u \rangle \leq h_K(u).$$

On the other hand, letting $y = 0_{\mathbb{R}^n}$ in (2.2), we get

$$-h_K(u) \geq \langle w, -u \rangle,$$

which leads to $h_K(u) \leq \langle w, u \rangle$. Summarizing, we have that $w \in K$ and $\langle w, u \rangle = h_K(u)$. This means that $w \in F(K, u)$, and then the inclusion $\partial h_K(u) \subseteq F(K, u)$ is proved.

□

Exercise 2.37. *Let $C \in CC^n$. Prove that*

$$(h_C)'_+(u, v) = h_{F(C,u)}(v)$$

for any $u, v \in \mathbb{S}^{n-1}$. In other words, the right-sided derivative of h_C at u in the direction of v equals the support function of the exposed face $F(C, u)$ evaluated at v.

Solution 2.37. See the proof of Lemma 3.3.2 in the main book.

□

Exercise 2.38. *Let $f : \mathbb{R}^n \to \mathbb{R}$ be a convex function. Define*

$$K_f := \mathrm{cl}(\mathrm{dom}(f^*)),$$

and notice that K_f is non-empty, closed, and convex.

(a) *Prove that K_f is bounded if and only if f is globally Lipschitz.*

(b) *Show that*

$$f(y) - f(x) \leq h_{K_f}(y - x)$$

for any $x, y \in \mathbb{R}^n$.

Hint. For item **(a)**, use Exercise 2.23 to show that K_f is bounded if and only if relint(dom(f^*)) is bounded. Then, use Remark 1.4.2 in the main book and Exercise 1.9 to conclude that if f is Lipschitz, then relint(dom(f^*)) is bounded.

Solution 2.38. **(a)** It is enough to prove that if f is globally Lipschitz, then relint(dom(f^*)), because by Exercise 2.23,

$$K_f = \text{cl}(\text{dom}(f^*)) = \text{cl}(\text{relint}(\text{dom}(f^*))),$$

and a set is bounded if and only if its closure is bounded.

Assume that f is globally Lipschitz with

$$|f(x) - f(y)| \leq c|x - y|$$

for any $x, y \in \mathbb{R}^n$. Let $u \in \text{relint}(\text{dom}(f^*))$. By Remark 1.4.2 from the main book, the hyperplane $\mathbf{h} := \mathbf{h}((u, -1), f^*(u))$ supports epi(f) in some boundary point, with epi(f) $\subseteq \mathbf{h}^-$. From Exercise 1.9, all boundary points of epi(f) have the form $(x, f(x))$, and hence, we may assume that

$$(x, f(x)) \in \text{epi}(f) \cap \mathbf{h}.$$

Therefore,

$$\langle (x, f(x)), (u, -1) \rangle = f^*(u),$$

and this leads to

$$\langle x, u \rangle = f(x) + f^*(u).$$

On the other hand, $(x + u, f(x + u)) \in \text{epi}(f) \subseteq \mathbf{h}^-$. Then,

$$\langle (x + u, f(x + u)), (u, -1) \rangle \leq f^*(u),$$

and this yields

$$\langle x, u \rangle + \langle u, u \rangle \leq f(x + u) + f^*(u).$$

Replacing $\langle x, u \rangle$ by $f(x) + f^*(u)$, we obtain

$$|u|^2 \leq f(x + u) - f(x) \leq |f(x + u) - f(x)| \leq c|u|,$$

from where we get that $|u| \leq c$. Since this holds for any $u \in \text{relint}(\text{dom}(f^*))$, we get that this set is bounded.

Conversely, assume that K_f is bounded. Then, dom(f^*) is bounded, and we may assume that $|v| < c$ for every $v \in \text{dom}(f^*)$. Let $x, y \in \mathbb{R}^n$ be given arbitrarily, and assume without loss of generality that $f(y) \geq f(x)$. Recall that since f is convex, we have $\partial f(y) \neq \emptyset$. Choose $v \in \partial f(y)$, and remember that this means that for any $z \in \mathbb{R}^n$, it holds that

$$f(z) - f(y) \geq \langle v, z - y \rangle, \tag{2.3}$$

which can be rewritten as

$$\langle v, z \rangle - f(z) \leq \langle v, y \rangle - f(y).$$

Therefore,

$$f^*(v) = \sup\{\langle v, z\rangle - f(z) : z \in \mathbb{R}^n\} \leq \langle v, y\rangle - f(y) < +\infty,$$

and hence, $v \in \mathrm{dom}(f^*)$. Thus, $|v| < c$. Setting $z = x$ in (2.3), multiplying both sides by -1, and remembering that $f(y) \geq f(x)$, we get

$$|f(y) - f(x)| = f(y) - f(x) \leq \langle v, y - x\rangle \leq |\langle v, y - x\rangle| \leq c|y - x|,$$

where the last inequality is just the classical Cauchy-Schwarz inequality. This proves that f is Lipschitz.

(b) Let $x, y \in \mathbb{R}^n$. As before, assume that $f(y) \geq f(x)$, and let $v \in \partial f(y)$. In item **(a)**, we proved that $v \in \mathrm{dom}(f^*)$, and hence, $v \in K_f$. Then,

$$f(y) - f(x) \leq \langle v, y - x\rangle \leq \sup\{\langle u, y - x\rangle : u \in K_f\} = h_{K_f}(y - x),$$

where we warn the reader that the first inequality was also proved in item **(a)**. In fact, this follows immediately from the fact that $v \in \partial f(y)$.

□

Exercise 2.39. *Let $f\colon \mathbb{R}^n \to \mathbb{R}$ be a convex and globally Lipschitz function, and let K_f be as in Exercise 2.38 .*

(a) *Prove that*

$$h_{K_f}(u) = \lim_{t\to\infty} \frac{f(x + tu)}{t}$$

for any $x, u \in \mathbb{R}^n$.

(b) *Show that if f is differentiable, then*

$$\nabla f(\mathbb{R}^n) = \mathrm{int}(K_f),$$

where $\nabla f(\mathbb{R}^n)$ denotes the image of the gradient map $\nabla f\colon \mathbb{R}^n \to \mathbb{R}^n$.

Solution 2.39. (a) Since f is globally Lipschitz, we may let $c > 0$ be such that $|f(x) - f(y)| \leq c|x - y|$ for every $x, y \in \mathbb{R}^n$. Let $x, u \in \mathbb{R}^n$ be arbitrarily given, and assume that $u \neq 0_{\mathbb{R}^n}$ (the case where u is zero is trivial). Consider the function $t \mapsto f(x + tu)$. We claim that this function is convex. Indeed, for any $t_0, t_1 \in \mathbb{R}$ and every $\lambda \in (0, 1)$, we have

$$\begin{aligned} f(x + ((1 - \lambda)t_0 + \lambda t_1)u) &= f((1 - \lambda)(x + t_0u) + \lambda(x + t_1u)) \leq \\ &\leq (1 - \lambda)f(x + t_0u) + \lambda f(x + t_1u). \end{aligned}$$

Because of that, the function $g\colon \mathbb{R} \to \mathbb{R}$ given as

$$g_x(t) := \frac{f(x + tu) - f(x)}{t}$$

is non-decreasing for $t > 0$. Indeed, if $t_1 > t_0 > 0$, then we may write

$$x + t_0u = (1 - \lambda)x + \lambda(x + t_1u)$$

for $\lambda = t_0/t_1$. Then, from the convexity of $t \mapsto f(x+tu)$, we get

$$f(x+t_0u) \leq \left(1-\frac{t_0}{t_1}\right) f(x) + \frac{t_0}{t_1} f(x+t_1u),$$

and rearranging the terms, we get

$$\frac{f(x+t_0u)-f(x)}{t_0} \leq \frac{f(x+t_1u)-f(x)}{t_1}.$$

This is the same as $g_x(t_1) \geq g_x(t_0)$. On the other hand, from item **(b)** of Exercise 2.38, we have

$$g_x(t) = \frac{f(x+tu)-f(x)}{t} \leq \frac{h_{K_f}(tu)}{t} = h_{K_f}(u)$$

for each $t > 0$. Consequently, $\lim_{t\to\infty} g_x(t) =: c_0$ exists, and $c_0 \leq h_{K_f}(u)$. Since x was taken arbitrarily in $\mathbb{R}^n$, repeating the argument for any other point $z \in \mathbb{R}^n$ and defining

$$g_z(t) := \frac{f(z+tu)-f(z)}{t},$$

we also have that $\lim_{t\to\infty} g_z(t)$ exists. We claim that

$$\lim_{t\to\infty} g_z(t) = \lim_{t\to\infty} g_x(t) = c_0.$$

To verify this, observe that for any $t > 0$, we may use the fact that f is globally Lipschitz to derive the inequality

$$\begin{aligned}\left|\frac{f(x+tu)-f(x)}{t} - \frac{f(z+tu)-f(z)}{t}\right| &= \frac{1}{t}|f(x+tu)-f(z+tu)+f(z)-f(x)| \leq \\ &\leq \frac{1}{t}\Big(|f(x+tu)-f(z+tu)| + |f(z)-f(x)|\Big) \leq \frac{2c|z-x|}{t},\end{aligned}$$

and the latter goes to 0 as $t \to \infty$. Summarizing, we have that

$$\lim_{t\to\infty} g_z(t) = c_0$$

for any $z \in \mathbb{R}^n$. Now, choose any point $v \in \operatorname{relint}(\operatorname{dom}(f^*))$. Then, $v \in K_f$, and from the same argument used in item **(a)** of Exercise 2.38, we may let $z_0 \in \mathbb{R}^n$ be such that the hyperplane $\mathbf{h} := \mathbf{h}((v,-1), f^*(v))$ supports $\operatorname{epi}(f)$ at $(z_0, f(z_0))$, with $\operatorname{epi}(f) \subseteq \mathbf{h}^-$. In this case, in the solution of the mentioned exercise, we proved that we have $\langle z_0, v\rangle = f(z_0) + f^*(v)$, and since $(z_0+u, f(z_0+u)) \in \operatorname{epi}(f) \subseteq \mathbf{h}^-$, we get

$$\langle (z_0+u, f(z_0+u)), (v,-1)\rangle \leq f^*(v) = \langle z_0, v\rangle - f(z_0),$$

and hence,

$$\langle u, v\rangle \leq f(z_0+u) - f(z_0) = g_{z_0}(1) \leq c_0 \leq h_{K_f}(u).$$

Because this holds for any $v \in \operatorname{relint}(\operatorname{dom}(f^*))$, we get

$$\sup\{\langle u, v\rangle : v \in \operatorname{relint}(\operatorname{dom}(f^*))\} \leq c_0 \leq h_{K_f}(u)$$

Write $A := \operatorname{relint}(\operatorname{dom}(f^*))$. Then, the left-hand side of the above is $h_A(u)$. From Exercise 2.23, we have

$$K_f = \operatorname{cl}(\operatorname{dom}(f^*)) = \operatorname{cl}(\operatorname{relint}(\operatorname{dom}(f^*)) = \operatorname{cl}(A),$$

and remembering that $h_A = h_{\operatorname{cl}(A)}$ (see Proposition 2.3.3 from the main book), we get

$$h_{K_f}(u) = h_A(u) \leq c_0 \leq h_{K_f}(u).$$

Therefore, since $f(x)/t \to 0$ as $t \to \infty$, we get

$$h_{K_f}(u) = c_0 = \lim_{t\to\infty} g_x(t) = \lim_{t\to\infty} \frac{f(x+tu) - f(x)}{t} = \lim_{t\to\infty} \frac{f(x+tu)}{t},$$

and the proof is complete.

□

Exercise 2.40. *Prove that if $f, g\colon \mathbb{R}^n \to \mathbb{R}$ are convex and globally Lipschitz, then*

$$K_{\alpha f+\beta g} = \alpha K_f + \beta K_g$$

for any $\alpha, \beta \geq 0$.

Solution 2.40. It is clear that $\alpha f + \beta g$ is convex and globally Lipschitz. Then, $K_{\alpha f+\beta g}$ is a non-empty compact and convex set. Thus, to prove that $K_{\alpha f+\beta g} = \alpha K_f + \beta K_g$, it suffices to prove that

$$h_{K_{\alpha f+\beta g}} = h_{\alpha K_f+\beta K_g}.$$

From Exercise 2.39, we have

$$h_{K_{\alpha f+\beta g}}(u) = \lim_{t\to\infty} \frac{(\alpha f+\beta g)(x+tu)}{t} = \alpha \lim_{t\to\infty} \frac{f(x+tu)}{t} + \beta \lim_{t\to\infty} \frac{g(x+tu)}{t} =$$
$$= \alpha h_{K_f}(u) + \beta h_{K_g}(u) = h_{\alpha K_f+\beta K_g}(u)$$

for any $u \in \mathbb{R}^n$ (and $x \in \mathbb{R}^n$). This concludes the proof.

□

Exercise 2.41. **(a)** *Let $K, L \subseteq \mathbb{R}^n$ be convex bodies such that $L \subseteq K$ and assume that $x \in \partial K \cap \partial L$. Show that any supporting hyperplane of K at x is also a supporting hyperplane of L at x.*

(b) *Let $K \subseteq \mathbb{R}^n$ be a convex body and assume that $rB^n \subseteq K$. Prove that if $x \in r\mathbb{S}^{n-1} \cap \partial K$, then $x/|x|$ is the unique unit outer normal vector of K at x.*

(c) *Under the same hypotheses as in item* **(b)**, *prove that*

$$\frac{x}{r|x|} \in \frac{1}{r}\mathbb{S}^{n-1} \cap \partial K^\circ.$$

Solution 2.41. **(a)** Let $\mathbf{h} := \mathbf{h}(u, \alpha)$ be a supporting hyperplane of K at x with $K \subseteq \mathbf{h}^-$. Then, $x \in \mathbf{h}$ (and $x \in \partial L$) and $L \subseteq \mathbf{h}^-$. Hence, $\mathbf{h}$ supports L at x.

(b) By item **(a)**, any supporting hyperplane of K at x is also a supporting hyperplane of rB^n at x. Because K has a supporting hyperplane at x and the unique supporting hyperplane of rB^n at x is the hyperplane $\mathbf{h}(x/|x|, r)$, we get the desired result.

(c) It is immediate that

$$\frac{x}{r|x|} \in \frac{1}{r}\mathbb{S}^{n-1}.$$

On the other hand, we have from item **(b)** that $\mathbf{h} := \mathbf{h}(x/|x|, r)$ is a supporting hyperplane of K at x, and it clearly holds that $K \subseteq \mathbf{h}^-$. Thus, for every $z \in K$, we have

$$\left\langle \frac{x}{|x|}, z \right\rangle \leq r,$$

which can be written as

$$\left\langle \frac{x}{r|x|}, z \right\rangle \leq 1.$$

Then, $x/r|x| \in K^\circ$. Moreover, since the equality is attained for $z = x \in K$, we get that $x/r|x|$ is a boundary point of K°.

□

Exercise 2.42. *Assume that $K \subseteq \mathbb{R}^n$ is a convex body having the origin as an interior point, and let $x_0 \in \partial K$ and $y_0 \in \partial K^\circ$. Prove that y_0 is an outer normal of K at x_0 if and only if x_0 is an outer normal of K° at y_0.*

Solution 2.42. Assume that y_0 is an outer normal of K at x_0. Then, from Proposition 2.3.6 in the main book, we have that

$$\langle x_0, y_0 \rangle = ||x_0||_K ||y_0||_{K^\circ} = 1,$$

where the last equality follows from the facts that $x_0 \in \partial K$ and $y_0 \in \partial K^\circ$, of course. For any $z \in K^\circ$, we have

$$\langle z, x_0 \rangle \leq 1 = \langle x_0, y_0 \rangle,$$

and this shows that x_0 is an outer normal of K° at y_0.

Conversely, assume that x_0 is an outer normal of K° at y_0. Then, since $K = K^{\circ\circ}$, we have that $x_0 \in \partial K^{\circ\circ}$ and, by what we have proved, y_0 is an outer normal of $K^{\circ\circ}$ $(= K)$ at x_0. This concludes the proof.

□

Exercise 2.43. **(a)** *Let $L \in \mathcal{K}_0^n$ be such that $L = L^\circ$. Prove that $L = B^n$.*

(b) *Verify that if $K \in \mathcal{K}_0^n$ is such that $K^\circ = \alpha K$ for some $\alpha > 0$, then K is a ball centered at the origin.*

Solution 2.43. **(a)** Let $x \in L$. Then, $x \in L^\circ$, and it gives that $\langle x, x \rangle \leq 1$. As a consequence, we obtain that $|x| \leq 1$ for every $x \in L$, and it follows that $L \subseteq B^n$. Because polarity reverses inclusion, we get that $L = L^\circ \supseteq B^n$, and hence, $L = B^n$.

(b) Let $L = \sqrt{\alpha}K$. Then we have

$$L^\circ = \frac{1}{\sqrt{\alpha}}K^\circ = \sqrt{\alpha}K = L,$$

and hence, from item **(a)**, we have that $L = B^n$. Hence, K is a ball centered at the origin.

□

For what follows, remember that the *positive hull* of a set $A \subseteq \mathbb{R}^n$ is the set of all (finite) positive combinations of the points of A.

Exercise 2.44. **(a)** *Let $A, B \subseteq \mathbb{R}^n$ be non-empty sets. Verify that* $\text{pos}(A + B) \subseteq \text{pos}(A) + \text{pos}(B)$.

(b) *Show that* $\text{pos}(A) = \text{pos}(\text{conv}(A))$ *for any non-empty set $A \subseteq \mathbb{R}^n$.*

(c) *Check that the positive hull of any set is convex. Conclude that the positive hull of any set is a convex cone.*

Solution 2.44. **(a)** Any point of $\text{pos}(A + B)$ can be written as

$$\lambda_1(a_1 + b_1) + \ldots + \lambda_m(a_m + b_m)$$

for $a_1, \ldots, a_m \in A$, $b_1, \ldots, b_m \in B$, and $\lambda_1, \ldots, \lambda_m > 0$. Thus, we just have to rewrite

$$\lambda_1(a_1 + b_1) + \ldots + \lambda_m(a_m + b_m) = \lambda_1 a_1 + \ldots + \lambda_m a_m + \lambda_1 b_1 + \ldots + \lambda_m b_m,$$

and since $\lambda_1 a_1 + \ldots + \lambda_m a_m \in \text{pos}(A)$ and $\lambda_1 b_1 + \ldots + \lambda_m b_m \in \text{pos}(B)$, we have what we wanted to get.

(b) Because $A \subseteq \text{conv}(A)$, then the inclusion $\text{pos}(A) \subseteq \text{pos}(\text{conv}(A))$ follows trivially. The reverse inclusion is also obvious: every positive combination of points of $\text{conv}(A)$ is clearly a positive combination of points of A.

(c) Let $x := \alpha_1 a_1 + \ldots + \alpha_m a_m$ and $y := \lambda_1 z_1 + \ldots + \lambda_k z_k$ be arbitrary points of $\text{pos}(A)$, where all the coefficients are positive and $a_1, \ldots, a_m, z_1, \ldots, z_k \in A$. For any $\lambda \in (0, 1)$, we have

$$(1 - \lambda)x + \lambda y = (1 - \lambda)\alpha_1 a_1 + \ldots + (1 - \lambda)\alpha_m a_m + \lambda\lambda_1 z_1 + \ldots + \lambda\lambda_k z_k,$$

and the latter is obviously a positive combination of points of A. Thus, $(1 - \lambda)x + \lambda y \in \text{pos}(A)$, and this proves that $\text{pos}(A)$ is convex. Since it is immediate to verify that $\lambda x \in \text{pos}(A)$ for every $x \in \text{pos}(A)$ and any $\lambda \geq 0$, we conclude that $\text{pos}(A)$ is a convex cone.

□

Exercise 2.45. *Let $A \subseteq \mathbb{R}^n$ be a non-empty set. Prove that the positive hull of A is the intersection of all convex cones of $\mathbb{R}^n$, which contain A.*

Solution 2.45. Let C be a convex cone containing A. We claim that $\text{pos}(A) \subseteq C$. Let $x \in \text{pos}(A)$ be non-zero (observe that $0_{\mathbb{R}^n}$ is clearly a point of both sets), and write

$$x = \lambda_1 a_1 + \ldots + \lambda_m a_m$$

for $a_1, \ldots, a_m \in A$ and $\lambda_1, \ldots, \lambda_m \geq 0$. Since $x \neq 0_{\mathbb{R}^n}$, the sum $\lambda := \lambda_1 + \ldots + \lambda_m$ is strictly positive. Thus,

$$\frac{x}{\lambda} = \frac{\lambda_1}{\lambda} a_1 + \ldots + \frac{\lambda_m}{\lambda} a_m$$

is a convex combination of points of C. Since C is convex, we get that $x/\lambda \in C$. Finally, because C is a cone, we have that

$$x = \lambda \left(\frac{x}{\lambda} \right) \in C,$$

and this shows that $\operatorname{pos}(A)$ is contained in any convex cone containing A (and hence also in the intersection of these convex cones).

On the other hand, it is clear that $\operatorname{pos}(A)$ is a convex cone containing A. Therefore, the intersection of all these convex cones is contained in $\operatorname{pos}(A)$, and then we have the desired equality.

□

Exercise 2.46. **(a)** *Let $C \subseteq \mathbb{R}^n$ be a convex cone (not necessarily closed). Verify that its polar set C° is a closed convex cone.*

(b) *Let $C \subseteq \mathbb{R}^n$ be a non-empty, closed, and convex cone. Show that $C^{\circ\circ} = C$.*

(c) *Let A be a non-empty set. Prove the following expression for the polar set of* $\operatorname{pos}(A)$*:*

$$\operatorname{pos}(A)^\circ = \{u \in \mathbb{R}^n : \langle u, x \rangle \leq 0 \text{ for all } x \in A\}.$$

Solution 2.46. **(a)** Let $C \subseteq \mathbb{R}^n$ be a closed convex cone. Then, by Remark 2.4.2 in the main book, its polar C° is given as

$$C^\circ = \{u \in \mathbb{R}^n : \langle u, x \rangle \leq 0 \text{ for all } x \in C\}.$$

It is immediate to verify that C° is a convex cone. To verify that this is closed, let $(u_m)_{m \in \mathbb{N}}$ be a sequence on C° converging to $u \in \mathbb{R}^n$. By the continuity of the inner product, since $\langle u_m, x \rangle \leq 0$ for each $m \in \mathbb{R}^n$ and every $x \in C$, we have that $\langle u, x \rangle \leq 0$ for each $x \in C$. Notice that to prove that C° is closed and convex, we also can use Lemma 2.3.6 from the main book straightforwardly.

(b) First of all, consider the case where $C = \{0_{\mathbb{R}^n}\}$. In this case, it is immediate to check that $C^\circ = \mathbb{R}^n$, and $C^{\circ\circ} = \{0_{\mathbb{R}^n}\} = C$. Thus, from now on, we only consider the case where C has a non-zero vector.

Let $x \in C$. Then, $\langle u, x \rangle \leq 0$ for every $u \in C^\circ$, and this gives that $x \in C^{\circ\circ}$. Thus, we have the inclusion $C \subseteq C^{\circ\circ}$.

For the reverse inclusion, let $z \in C^{\circ\circ}$ and assume that $z \notin C$. Since C is closed and convex, we have that z and C can be strongly separated by a hyperplane $\mathbf{h} := \mathbf{h}(u_0, \beta)$ with $C \subseteq \mathbf{h}^-(u_0, \beta - \varepsilon)$ and $z \in \mathbf{h}^+(u_0, \beta + \varepsilon)$ for some $\varepsilon > 0$, say. Observe that, in particular, we also have the inclusion $C \subseteq \mathbf{h}^-(u_0, \beta)$. This inclusion has two consequences: first, since $0_{\mathbb{R}^n} \in C$, we have that $\beta \geq \langle 0_{\mathbb{R}^n}, u_0 \rangle = 0$. The second is that

$$\lambda \langle u_0, x \rangle = \langle u_0, \lambda x \rangle \leq \beta$$

for each $x \in C$ and $\lambda > 0$ (because $\lambda x \in C$). Therefore, $\langle u_0, x \rangle \leq \beta/\lambda$ for each $x \in C$ and $\lambda > 0$. Thus, $\langle u_0, x \rangle \leq 0$ for every $x \in C$. It follows that $u_0 \in C^\circ$. Since $z \in C^{\circ\circ}$, we have that $\langle z, u_0 \rangle \leq 0$. On the other hand, $\langle z, u_0 \rangle \geq \beta + \varepsilon \geq 0 + \varepsilon > 0$, a contradiction. Hence, $C^{\circ\circ} \subseteq C$, and the proof is complete.

(c) We already know that

$$\text{pos}(A)^\circ = \{u \in \mathbb{R}^n : \langle u, x\rangle \leq 0 \text{ for all } x \in \text{pos}(A)\}.$$

If $\langle u, x\rangle \leq 0$ for every $x \in \text{pos}(A)$, then, in particular, we have $\langle u, x\rangle \leq 0$ for every $x \in A$, because $A \subseteq \text{pos}(A)$. On the other hand, if $\langle u, x\rangle \leq 0$ for every $x \in A$, then we claim that $\langle u, z\rangle \leq 0$ for every $z \in \text{pos}(A)$. Indeed, writing $z = \lambda_1 x_1 + \ldots + \lambda_m x_m$ as a positive combination of points of A, we get

$$\langle u, z\rangle = \lambda_1 \langle u, x_1\rangle + \ldots + \lambda_m \langle u, x_m\rangle,$$

and the latter is a sum of nonpositive numbers since $\lambda_j \geq 0$ and $\langle u, x_j\rangle \leq 0$ for each $1 \leq j \leq m$. Summarizing, $\langle u, z\rangle \leq 0$ for every $z \in \text{pos}(A)$ if and only if $\langle u, x\rangle \leq 0$ for every $x \in A$. This gives that

$$\text{pos}(A)^\circ = \{u \in \mathbb{R}^n : \langle u, x\rangle \leq 0 \text{ for all } x \in \text{pos}(A)\} = \{u \in \mathbb{R}^n : \langle u, x\rangle \leq 0 \text{ for all } x \in A\}.$$

□

Exercise 2.47. *Let $A \subseteq \mathbb{R}^n$ be a compact non-empty set with the property that $0_{\mathbb{R}^n} \notin \text{conv}A$. Prove that* pos$(A)$ *is closed. Give an example of a closed set F such that* pos(F) *is not closed.*

Solution 2.47. Since pos(A) = pos(conv(A)), and since the convex hull of a compact set is compact, it is sufficient to prove the result for the case where we replace A by a non-empty compact convex set C that does not contain the origin.

Let $(z_m)_{m\in\mathbb{N}}$ be a sequence in pos(C) such that $z_m \to z$ for some $z \in \mathbb{R}^n$. Since C is convex, we have that pos$(C) = \{\lambda x : x \in C \text{ and } \lambda \geq 0\}$. Hence, we deduce that for each $m \in \mathbb{N}$, there exists a point $x_m \in C$ and a number $\lambda_m \geq 0$ such that

$$z_m = \lambda_m x_m.$$

Since C is compact, passing to a subsequence if necessary, we may assume that the sequence (x_m) converges to a point $x \in C$. Because C does not contain the origin, we get that $x \neq 0_{\mathbb{R}^n}$.

We claim that (λ_m) is a bounded sequence. Indeed, choose any $0 < \varepsilon < |x|$. For sufficiently large $m \in \mathbb{N}$, we have $|x_m| > |x| - \varepsilon := c > 0$. If (λ_m) is not bounded, then we may assume that there is a subsequence (λ_{m_k}) such that $\lambda_{m_k} \to \infty$ (recall that $\lambda_m \geq 0$ for every m). Thus, for k sufficiently large, we have $|x_{m_k}| > |x| - \varepsilon$, and this leads to

$$|z_{m_k}| = \lambda_{m_k} |x_{m_k}| > \lambda_{m_k} c,$$

and since $c > 0$, the latter goes to ∞ as $k \to \infty$. This is a contradiction to the fact that (z_m) converges.

Because (λ_m) is a bounded sequence, passing to a subsequence if necessary, we may assume that λ_m converges to a number $\lambda \in \mathbb{R}$. Therefore, $\lambda_m x_m$ converges to λx, and since $\lambda_m x_m = z_m \to z$, we get that $\lambda x = z$. Since $\lambda \geq 0$ (because it is the limit of a sequence of non-negative numbers) and $x \in C$, we get that $z \in \text{pos}(C)$. This shows that pos(C) is closed.

□

Exercise 2.48. *Let $K \subseteq \mathbb{R}^n$ be a non-empty compact convex set, and let $h_K : \mathbb{R}^n \to \mathbb{R}$ be its support function. Define the set*

$$C_K := \{\lambda(x, -1) \in \mathbb{R}^n \times \mathbb{R} : x \in K \text{ and } \lambda \geq 0\}.$$

Verify that C_K is a closed convex cone, and prove that epi$(h_K) = C_K^\circ$.

Solution 2.48. We first claim that the set $C = \{(x, -1) : x \in K\}$ is convex. Indeed, for $x, z \in K$ and $\lambda \in (0, 1)$, we have that

$$(1 - \lambda)(x, -1) + \lambda(z, -1) = ((1 - \lambda)x + \lambda z, -1),$$

and since K is convex, we have that $(1-\lambda)x+\lambda z \in K$. Therefore, $(1-\lambda)(x, -1)+\lambda(y, -1) \in C$. Observe also that C is clearly compact (because K is compact) and does not contain the origin of $\mathbb{R}^{n+1}$, because the last coordinate of each point of C is -1. Since C is convex, it follows from Proposition 2.4.1 in the main book that

$$C_K = \{\lambda(x, -1) : \lambda \geq 0 \text{ and } x \in K\} = \operatorname{pos}(C).$$

Thus, C_K is the positive hull of a non-empty compact set that does not contain the origin. Hence, C_K is a convex cone, because the positive hull of any set is a convex cone. The fact that C_K is closed comes straightforwardly from Exercise 2.47 (which states that the positive hull of a compact set that does not contain the origin is closed).

Now, we move forward to prove that $\operatorname{epi} h_K = C_K^\circ$. Let $(u, t) \in \operatorname{epi}(h_K)$. Then, $t \geq h_K(u)$. Also, by the definition of the support function, we see that for any $x \in K$, we have $\langle u, x\rangle \leq h_K(u)$. Observe that

$$\langle (u, t), (x, -1)\rangle = \langle u, x\rangle - t \leq h_K(u) - t \leq t - t = 0$$

for every $x \in K$. From item **(c)** of Exercise 2.46, we get that $(u, t) \in \operatorname{pos}(C)^\circ$, and since $C_K = \operatorname{pos}(C)$, it follows that $(u, t) \in C_K^\circ$. An alternative way of showing this (without using the mentioned exercise) is to notice that we may write

$$\langle (u, t), \lambda(x, -1)\rangle = \lambda\langle (u, t), (x, -1)\rangle \leq 0$$

for each $x \in K$ and any $\lambda \geq 0$. Since all points of C_K have the form $\lambda(x, -1)$ for $\lambda \geq 0$ and $x \in K$, it follows that $(u, t) \in C_K^\circ$. This gives the inclusion $\operatorname{epi}(h_K) \subseteq C_K^\circ$.

Now, let $(u, t) \in C_K^\circ$. For every $x \in K$, we have

$$0 \geq \langle (u, t), (x, -1)\rangle = \langle u, x\rangle - t,$$

and this leads to

$$h_K(u) = \sup\{\langle u, x\rangle : x \in K\} \leq t.$$

Then, $(u, t) \in \operatorname{epi}(h_K)$. This gives the reverse inclusion $C_K^\circ \subseteq \operatorname{epi}(h_K)$, and the proof is complete.

□

Exercise 2.49. *Let $h\colon \mathbb{R}^n \to \mathbb{R}$ be a sublinear function.*

(a) *Show that* $\operatorname{epi}(h)$ *is a convex cone.*

(b) *Letting $(x_1, \ldots, x_{n+1})$ be coordinates in $\mathbb{R}^{n+1}$, and $T\colon \mathbb{R}^{n+1} \to \mathbb{R}^n$ be the map defined as $T(x_1, \ldots, x_n, x_{n+1}) = (x_1, \ldots, x_n)$, define*

$$K := T\big(\operatorname{epi}(h)^\circ \cap \{x_{n+1} = -1\}\big).$$

Prove that K is a non-empty, compact, and convex set.

(c) *Show that h is the support function of K.*

Solution 2.49. (a) The epigraph of h is a convex set, because h is convex in particular. Let $(x,t) \in \operatorname{epi}(h)$. Then, $t \geq h(x)$. For any $\lambda > 0$, we have that $\lambda t \geq \lambda h(x) = h(\lambda x)$, and hence,

$$\lambda(x,t) = (\lambda x, \lambda t) \in \operatorname{epi}(h).$$

For $\lambda = 0$, we have that $\lambda(x,t) = (0,0) = 0_{\mathbb{R}^{n+1}}$, and this point is in $\operatorname{epi}(h)$ because $h(0_{\mathbb{R}^n}) = 0$. Then, $\operatorname{epi}(h)$ is a convex set with the property that $\lambda z \in \operatorname{epi}(h)$ whenever $\lambda \geq 0$ and $z \in \operatorname{epi}(h)$.

(b) It is enough to prove that $\operatorname{epi}(h)^\circ \cap \{x_{n+1} = -1\}$ is non-empty, compact, and convex, because T is linear and these properties are preserved under linear maps. The convexity is straightforward since the intersection of convex sets is always convex.

Now, we prove that $\operatorname{epi}(h)^\circ \cap \{x_{n+1} = -1\}$ is non-empty. Observe that $\operatorname{dom}(h) = \mathbb{R}^n$, and hence, $\operatorname{int}(\operatorname{dom}(f)) = \mathbb{R}^n$. Since h is (in particular) convex, we obtain that h is continuous. Therefore, $\operatorname{epi}(h)$ is closed. Also, observe that $0_{\mathbb{R}^{n+1}}$ is a boundary point of $\operatorname{epi}(h)$, because every neighborhood of $0_{\mathbb{R}^{n+1}}$ contains points having the form $(0_{\mathbb{R}^n}, \varepsilon)$ for $\varepsilon > 0$ and $\varepsilon < 0$. Since $h(0_{\mathbb{R}^n}) = 0$, if $\varepsilon > 0$, then $(0_{\mathbb{R}^n}, \varepsilon) \in \operatorname{epi}(h)$, and if $\varepsilon < 0$, then $(0_{\mathbb{R}^n}, \varepsilon) \notin \operatorname{epi}(h)$. Since $\operatorname{epi}(h)$ is closed and convex, it admits a supporting hyperplane at $0_{\mathbb{R}^{n+1}}$. Letting $\mathbf{h} := \mathbf{h}((v,\alpha),\beta)$, we have immediately that $\beta = 0$ because $0_{\mathbb{R}^{n+1}} \in \mathbf{h}$. We may assume that $\operatorname{epi}(h) \subseteq \mathbf{h}^-$, and this leads to

$$\langle (x,t), (v,\alpha) \rangle \leq 0$$

for every $(x,t) \in \operatorname{epi}(h)$. Thus, $\langle v, x \rangle \leq -\alpha t$ for each $x \in \mathbb{R}^n$ and $t \geq h(x)$. As a consequence, we get that $\alpha < 0$. Indeed, fixing any $x \in \mathbb{R}^n$, the inequality

$$\alpha \leq \frac{-\langle v, x \rangle}{t}$$

holds for any $t \geq h(x)$, and hence, it holds for arbitrarily large t. Thus, $\alpha \leq 0$. On the other hand, if $\alpha = 0$, then we have $\langle v, x \rangle \leq 0$ for every $x \in \mathbb{R}^n$, from which we deduce that $v = 0_{\mathbb{R}^n}$. But if this is true, then $\mathbf{h}$ is not a hyperplane. We claim that

$$w := \left(\frac{v}{-\alpha}, -1 \right) \in \operatorname{epi}(h)^\circ \cap \{x_{n+1} = -1\}.$$

Since the last coordinate of w is indeed -1, it suffices to prove that it lies in $\operatorname{epi}(h)^\circ$. For that sake, let $(x,t) \in \operatorname{epi}(h)$. Then, we already know that $\langle v, x \rangle \leq -\alpha t$, and this leads to

$$\left\langle \left(\frac{v}{-\alpha}, -1 \right), (x,t) \right\rangle = \frac{\langle v, x \rangle}{-\alpha} - t \leq \frac{-\alpha t}{-\alpha} - t = 0.$$

This proves that $w \in \operatorname{epi}(h)^\circ$.

It remains to prove that $\operatorname{epi}(h)^\circ \cap \{x_{n+1} = -1\}$ is compact. It is immediate that this set is closed, because it is the intersection of two closed sets (recall that the polar of any convex cone is closed), and hence, we only have to prove boundedness. Assume that there is a sequence $(z_m, -1) \in \operatorname{epi}(h)^\circ \cap \{x_{n+1} = -1\}$ such that

$$|(z_m, -1)| \to \infty$$

as $m \to \infty$. Hence, it is clear that $|z_m| \to \infty$ as $m \to \infty$ (as points of $\mathbb{R}^n$). Because $(z_m, h(z_m)) \in \operatorname{epi}(h)$ and $(z_m, -1) \in \operatorname{epi}(h)^\circ$ for each $m \in \mathbb{N}$, we have that

$$\langle (z_m, -1), (z_m, h(z_m)) \rangle \leq 0,$$

and this leads to $h(z_m) \geq |z_m|^2$ for every $m \in \mathbb{N}$. We clearly may assume that all z_m's are non-zero, and from sublinearity, we get that

$$h\left(\frac{z_m}{|z_m|}\right) \geq |z_m|$$

for each $m \in \mathbb{N}$. Consequently, h is not bounded in B^n. This is a contradiction to the fact that h is continuous in $\mathbb{R}^n$, and the proof is complete.

(c) From Theorem 2.3.1 in the main book, we have that h is the support function of a unique non-empty, compact, and convex set K_0, say. Let C_{K_0} be the closed convex cone given in Exercise 2.48. Since $h = h_{K_0}$, we obtain

$$\operatorname{epi}(h) = \operatorname{epi}(h_K) = C_{K_0}^{\circ},$$

from where we get that

$$\operatorname{epi}(h)^{\circ} = C_{K_0}^{\circ\circ} = C_{K_0}.$$

The last equality comes from the fact that C_{K_0} is a closed convex cone (see item **(b)** of Exercise 2.46). Therefore,

$$K = T\big(\operatorname{epi}(h)^{\circ} \cap \{x_{n+1} = -1\}\big) = T\big(C_{K_0} \cap \{x_{n+1} = -1\}\big) = K_0,$$

where the last equality comes straightforwardly from the definition of C_{K_0}. It follows that h is the support function of K.

□

Exercise 2.50. *Assume that $C_m \to C$ in CC^n with respect to the Hausdorff metric. Prove that for any linear map $T\colon \mathbb{R}^n \to \mathbb{R}^n$, we have that $TC_m \to TC$.*

Solution 2.50. This follows from the fact that

$$TB^n \subseteq ||T||_{\mathrm{L}(n)} B^n.$$

Thus, if $C \subseteq D + \varepsilon B^n$, then

$$TC \subseteq TD + T\varepsilon B^n \subseteq TD + \varepsilon ||T||_{\mathrm{L}(n)} B^n,$$

and this shows that if $d_{\mathcal{H}}(C, C_m) \to 0$, then $d_{\mathcal{H}}(TC, TC_m) \to 0$.

□

Exercise 2.51. *Prove that if $(C_m)_{m\in\mathbb{N}}$ is a sequence of centrally symmetric sets in CC^n converging in the Hausdorff distance to a set $C \in CC^n$, then C is also centrally symmetric. Moreover, show that if all C_m are symmetric with respect to a same point $z_0 \in \mathbb{R}^n$, then z_0 is also the center of C.*

Solution 2.51. Assume that x_m is the center of C_m (for each $m \in \mathbb{N}$). Because $C_m \to C$ in the Hausdorff metric, we have that (C_m) is bounded with respect to $d_{\mathcal{H}}$, and by Proposition 2.5.3 in the main book, this means that the family (C_m) is bounded in $\mathbb{R}^n$. As a consequence, we get that $(x_m)_{m\in\mathbb{N}}$ is a bounded sequence in $\mathbb{R}^n$. Passing to a subsequence if necessary, we may assume that $x_m \to x_0$ for

some $x_0 \in \mathbb{R}^n$. We prove that C is symmetric with respect to x_0. To do so, we compare the support functions of $C - x_0$ and $-(C - x_0)$. For any $u \in \mathbb{S}^{n-1}$, we have

$$\begin{aligned}
\left|h_{C-x_0}(u) - h_{-(C-x_0)}(u)\right| &\le \left|h_{C-x_0}(u) - h_{C_m-x_m}(u)\right| + \left|h_{C_m-x_m}(u) - h_{-(C-x_0)}(u)\right| = \\
&= \left|h_{C-x_0}(u) - h_{C_m-x_m}(u)\right| + \left|h_{-(C_m-x_m)}(u) - h_{-(C-x_0)}(u)\right| = \\
&= \left|h_C(u) - h_{C_m}(u) + \langle x_0 - x_m, u\rangle\right| + \left|h_{-C_m}(u) - h_{-C}(u) + \langle x_m - x_0, u\rangle\right| \le \\
&\le \left|h_C(u) - h_{C_m}(u)\right| + \left|h_{C_m}(-u) - h_C(-u)\right| + 2|\langle x_m - x_0, u\rangle| \le \\
&\le 2\sup\left\{\left|h_C(v) - h_{C_m}(v)\right| : v \in \mathbb{S}^{n-1}\right\} + 2|x_m - x_0| = 2d_{\mathcal{H}}(C, C_m) + 2|x_m - x_0|.
\end{aligned}$$

Because $C_m \to C$ in the Hausdorff metric and $x_m \to x_0$ in $\mathbb{R}^n$, it follows that the latter becomes arbitrarily small as $m \to \infty$. Hence, $h_{C-x_0}(u) = h_{-(C-x_0)}(u)$ for every $u \in \mathbb{S}^{n-1}$. This shows that $C - x_0 = -(C - x_0)$.

Observe that we proved that the center of C is the limit of a subsequence of the centers of the C_m's. Therefore, if all C_m's are centered at z_0, then C is also centered at z_0.

□

Let $u \in \mathbb{S}^{n-1}$ be arbitrarily given. The orthogonal complement $u^\perp$ is a hyperplane of $\mathbb{R}^n$, and it can be seen as normed vector space with the induced norm of $\mathbb{R}^n$. It is clear that the unit ball of this norm is $B^n \cap u^\perp =: B_{u^\perp}$. Hence, we may introduce a Hausdorff metric in $u^\perp$ by letting

$$d_{\mathcal{H},u^\perp}(C, L) := \inf\left\{\lambda > 0 : C \subseteq L + \lambda B_{u^\perp} \text{ and } L \subseteq C + \lambda B_{u^\perp}\right\}.$$

The objective of the next exercise is, roughly speaking, to check that convergence in the Hausdorff metric of $\mathbb{R}^n$ implies convergence of the projections on $u^\perp$ in the Hausdorff metric of $u^\perp$.

Exercise 2.52. *Let $(C_m)_{m\in\mathbb{N}}$ be a sequence in $\mathcal{C}^n$ converging in the Hausdorff metric of $\mathbb{R}^n$ to $C \in \mathcal{C}^n$, and let $u \in \mathbb{S}^{n-1}$. Prove that the sequence $(\mathrm{proj}_{u^\perp}(C_m))_{m\in\mathbb{N}}$ converges to $\mathrm{proj}_{u^\perp}(C)$ in the Hausdorff metric of $u^\perp$.*

Solution 2.52. This is an immediate consequence of the obvious equality $B^n \cap u^\perp = \mathrm{proj}_{u^\perp}(B^n)$. Indeed, by the linearity of the projection, if $C, D \in \mathcal{C}^n$ are such that $C \subseteq D + \varepsilon B^n$ for some $\varepsilon > 0$, then

$$\mathrm{proj}_{u^\perp}(C) \subseteq \mathrm{proj}_{u^\perp}(D + \varepsilon B^n) = \mathrm{proj}_{u^\perp}(D) + \varepsilon B_{u^\perp}.$$

The desired result follows from this fact.

□

Exercise 2.53. *Let $L, C \subseteq \mathbb{R}^n$ be non-empty compact sets. Prove that indeed we have*

$$0 \le d_{\mathcal{H}}(L, C) < \infty.$$

Also, writing $d = d_{\mathcal{H}}(L, C)$, show that

$$L \subseteq C + dB^n \text{ and } C \subseteq L + dB^n.$$

Remark 2.5. This justifies why the "infimum" can be replaced by the "minimum" in the definition of the Hausdorff metric given in (2.5.1), in Section 2.5 of the main book.

Solution 2.53. Write $d = d_{\mathcal{H}}(L, C)$. Then d is, by definition, the infimum of a set of non-negative numbers. Hence, $d \ge 0$. To prove that $d < +\infty$, it is sufficient to show that there exists a number $\lambda > 0$

such that $L \subseteq C + \lambda B^n$ and $C \subseteq L + \lambda B^n$. To do that, fix points $x_0 \in L$ and $y_0 \in C$. Since $C - x_0$ and $L - y_0$ are compact sets, we may choose a number $\lambda > 0$ such that $C - x_0 \in \lambda B^n$ and $L - y_0 \in \lambda B^n$. Then

$$C = x_0 + (C - x_0) \subseteq x_0 + \lambda B^n \subseteq L + \lambda B^n,$$

where the last inclusion comes from the fact that $x_0 \in L$. Analogously, we get that $L \subseteq C + \lambda B^n$. Therefore, we have $d_{\mathcal{H}}(L, C) \leq \lambda < +\infty$.

Now, assume that $C \nsubseteq L + dB^n$ or $L \nsubseteq C + dB^n$. Of course, we may assume without loss of generality that the first one holds. Then, we may choose a point $x \in C$ such that $x \notin L + dB^n$, and by the compactness of L, this means that $\delta := \operatorname{dist}(x, L) > d$. Choose a number $\varepsilon > 0$ such that $\delta > d - \varepsilon$. Thus, $\delta + \varepsilon > d$, and hence, $\delta + \varepsilon$ cannot be a lower bound of the set $\{\lambda \geq 0 : C \subseteq L + \lambda B^n \text{ and } C \subseteq L + \lambda B^n\}$. Thus, there exists a number $d \leq d_1 < \delta + \varepsilon$ such that $C \subseteq L + d_1 B^n$ and $L \subseteq C + d_1 B^n$. The first of these inclusions leads to $x \in L + d_1 B^n$, and since $d \leq d_1$, this gives that $x \in L + dB^n$, which is a contradiction.

□

Exercise 2.54. **(a)** *Let K and L be non-empty compact sets. Prove that*

$$d_{\mathcal{H}}(K, L) = \max\Big\{ \max\big\{\operatorname{dist}(x, L) : x \in K\big\},\ \max\{\operatorname{dist}(y, K) : y \in L\}\Big\}.$$

(b) *Now, assume that K and L are convex bodies. Show that*

$$d_{\mathcal{H}}(K, L) = \max\Big\{ \max\big\{\operatorname{dist}(x, \partial L) : x \in \partial K\big\},\ \max\{\operatorname{dist}(y, \partial K) : y \in \partial L\}\Big\}.$$

(c) *Let K be a compact convex set, and let B be its circumball. Prove that*

$$d_{\mathcal{H}}(K, B) \leq r_{\text{out}}(K).$$

Give an example where equality holds.

Solution 2.54. **(a)** Observe first that the maximum

$$\max\big\{\operatorname{dist}(x, L) : x \in K\big\}$$

exists because $\operatorname{dist}(\cdot, L)$ is a continuous function and K is compact. Of course, the same holds for $\max\big\{\operatorname{dist}(y, K) : y \in L\big\}$.

Denote $d = d_{\mathcal{H}}(K, L)$. Then, $K \subseteq L + dB^n$, and this gives that

$$\max\big\{\operatorname{dist}(x, L) : x \in K\big\} \leq d.$$

In fact, if there is a point $x \in K$ such that $\operatorname{dist}(x, L) > d$, then $x \notin L + dB^n$, because $L + dB^n$ is the set of all points of $\mathbb{R}^n$ whose distance to L is less than or equal to d. Similarly, we have that

$$\max\big\{\operatorname{dist}(y, K) : y \in L\big\} \leq d.$$

Therefore,

$$\delta := \max\Big\{ \max\big\{\operatorname{dist}(x, L) : x \in K\big\},\ \max\{\operatorname{dist}(y, K) : y \in L\}\Big\} \leq d.$$

For the reverse inequality, observe that $\operatorname{dist}(x, L) \leq \delta$ for any $x \in K$. Consequently, we have that $K \subseteq L + \delta B^n$. Similarly, we get that $L \subseteq K + \delta B^n$. Hence, $d = d_{\mathcal{H}}(K, L) \leq \delta$. Thus, $d = \delta$.

(b) Now, assume that K and L are convex bodies. Then, ∂K and ∂L are compact sets, and hence, from item **(a)**, we have

$$d_{\mathcal{H}}(\partial K, \partial L) = \max\left\{\max\left\{\operatorname{dist}(x, \partial L) : x \in \partial K\right\},\ \max\{\operatorname{dist}(y, \partial K) : y \in \partial L\}\right\}.$$

On the other hand, from item **(iii)** of Proposition 2.5.2 of the main book, we have that $d_{\mathcal{H}}(K, L) = d_{\mathcal{H}}(\partial K, \partial L)$. Hence, the desired equality follows.

(c) Since $K \subseteq B \subseteq B + r_{\text{out}}(K)B^n$, it suffices to prove that $B \subseteq K + r_{\text{out}}(K)B^n$. Due to Exercise 2.15, the center of B (x_0, say) is a point of K. Then,

$$B = x_0 + r_{\text{out}}(K)B^n \subseteq K + r_{\text{out}}(K)B^n,$$

and hence, we have that $d_{\mathcal{H}}(K, B) \leq r_{\text{out}}(K)$.

To give an example of a case where equality holds, consider the segment $K := \operatorname{seg}[-e_1, e_1]$ in $\mathbb{R}^2$. It is clear that the circumball of K is the unit ball B^2 of $\mathbb{R}^2$. Since $K \subseteq B^2$, we have that

$$\max\left\{\operatorname{dist}(x, B^2) : x \in K\right\} = 0,$$

and it is also immediate to observe that the number

$$\max\left\{\operatorname{dist}(y, K) : y \in B^2\right\}$$

is attained for the points $e_2 \in B$ and $0_{\mathbb{R}^2} \in K$. Hence, from item **(a)**, we have that $d_{\mathcal{H}}(K, B) = |e_2 - 0_{\mathbb{R}^2}| = 1 = r_{\text{out}}(K)$.

□

Remark 2.6. Item **(a)** is essentially the same as item **(i)** of Proposition 2.5.2 in the main book.

Exercise 2.55. *Let $\mathbb{B} \subseteq \mathbb{R}^n$ be a convex body containing the origin as an interior point, and define the* relative Hausdorff metric with respect to $\mathbb{B}$ *as*

$$d_{\mathcal{H},\mathbb{B}}(L, C) := \inf\{\lambda \geq 0 : C \subseteq L + \lambda\mathbb{B} \text{ and } L \subseteq C + \lambda\mathbb{B}\},$$

for $L, C \in \mathcal{C}^n$. Prove that this distance induces the same topology on $\mathcal{C}^n$ as the usual Hausdorff metric $d_{\mathcal{H}}$.

Solution 2.55. We indeed have that the metrics $d_{\mathcal{H}}$ and $d_{\mathcal{H},\mathbb{B}}$ are *equivalent* in the sense that there exist numbers $\alpha, \beta > 0$ such that

$$\alpha d_{\mathcal{H}}(L, C) \leq d_{\mathcal{H},\mathbb{B}}(L, C) \leq \beta d_{\mathcal{H}}(L, C)$$

for any $L, C \in \mathcal{C}^n$. To verify this, observe that since $\mathbb{B}$ is compact, there exists a number $c_1 > 0$ such that $\mathbb{B} \subseteq c_1 B^n$. On the other hand, since $\mathbb{B}$ contains the origin as an interior point, there exists a number $c_2 > 0$ such that $c_2 B^n \subseteq \mathbb{B}$. The latter can be rewritten as

$$B^n \subseteq \frac{1}{c_2}\mathbb{B}.$$

Writing $d = d_{\mathcal{H}}(L, C)$, we have that

$$C \subseteq L + dB^n \subseteq L + \frac{d}{c_2}\mathbb{B} \text{ and}$$
$$L \subseteq C + dB^n \subseteq C + \frac{d}{c_2}\mathbb{B},$$

from where we get the inequality

$$d_{\mathcal{H},\mathbb{B}}(L, C) \leq \frac{d}{c_2} = \frac{1}{c_2} d_{\mathcal{H}}(L, C),$$

and we only have to let $\beta = 1/c_2$. It is clear that β does not depend on L and C. Now, assume that $C \subseteq L + \delta\mathbb{B}$ and $L \subseteq C + \delta\mathbb{B}$. This leads to

$$C \subseteq L + \delta c_1 B^n \text{ and } C \subseteq L + \delta c_1 B^n,$$

and passing to the infimum of the numbers δ with the property that $C \subseteq L + \delta\mathbb{B}$ and $L \subseteq C + \delta\mathbb{B}$, we arrive at

$$c_1 d_{\mathcal{H}}(L, C) \leq d_{\mathcal{H},\mathbb{B}}(L, C).$$

Thus, setting $\alpha = c_1$, we conclude that $d_{\mathcal{H}}$ and $d_{\mathcal{H},\mathbb{B}}$ are equivalent metrics in C^n. It is well-known from the standard theory of metric spaces that equivalent metrics provide the same topology.

□

Exercise 2.56. *Let $K \subseteq \mathbb{R}^n$ be a convex body. Prove that the Steiner symmetral of K in the direction $u = e_n$ can be expressed as*

$$S_{e_n}(K) = \left\{\left(x, \frac{t-s}{2}\right) : (x, t),\ (x, s) \in K\right\},$$

Solution 2.56. Let $z \in \text{proj}_{e_n^\perp}(K)$ be arbitrarily given, and assume that the line $t \mapsto z + te_n$ touches the boundary of K at $z + \lambda_1 e_n$ and $z + \lambda_2 e_n$, with $\lambda_1 \leq \lambda_2$. For any points $z + te_n, z + se_n$ in K (with $t < s$, say), we have $\lambda_1 \leq t, s \leq \lambda_2$. Thus,

$$\lambda_1 - \lambda_2 \leq t - s \leq s - t \leq \lambda_2 - \lambda_1.$$

Dividing both sides by 2 and writing the points of $\mathbb{R}^n$ as (x, t) for $x \in \mathbb{R}^{n-1}$, we get what we wanted to have.

□

Exercise 2.57. *Notice that the definition of Steiner symmetrization can be extended verbatim for sets of CC^n. Fixing $u \in \mathbb{S}^{n-1}$, we may see the Steiner symmetrization in $u^\perp$ as a map $S_u : CC^n \to CC^n$. Show that this map does not need to be continuous in the Hausdorff metric.*

Hint: Consider the segments $\text{seg}[(0, 0), (1/m, 1)]$ in $\mathbb{R}^2$, and take symmetrizations in the x-axis (which is $e_2^\perp$, of course).

Solution 2.57. For each $m \in \mathbb{N}$, let $C_m = \text{seg}[(0, 0), (1/m, 1)]$ in $\mathbb{R}^2$. It is clear that

$$C_m \to \text{seg}[(0, 0), (0, 1)]$$

in the Hausdorff metric as $m \to +\infty$. On the other hand, for each $m \in \mathbb{N}$, we have

$$S_{e_2}\big(\mathrm{seg}[(0,0),(1/m,1)]\big) = \mathrm{seg}[(0,0),(1/m,0)],$$

and this clearly converges to the origin in the Hausdorff metric as $m \to +\infty$. Since,

$$S_{e_2}\big(\mathrm{seg}[(0,0),(0,1)]\big) = \mathrm{seg}[(0,-1/2),(0,1/2)],$$

we get the desired counterexample.

□

Exercise 2.58. *In this exercise, $K \subseteq \mathbb{R}^n$ is an arbitrary set of CC^n, and $u \in \mathbb{S}^{n-1}$ is any unit vector. The* Minkowski symmetral *of K in the direction of u is defined as the set*

$$M_u(K) := \frac{1}{2}\big(\mathrm{ref}_{u^\perp}(K) + K\big).$$

(a) *Verify that $M_u(K)$ is symmetric with respect to $u^\perp$.*

(b) *Show that the procedure of Minkowski symmetrization is* idempotent*, that is, we always have*

$$M_u(M_u(K)) = M_u(K).$$

(c) *Check that the Minkowski symmetrization is monotone for inclusion and homogeneous for scalar multiplication.*

(d) *Prove that the Minkowski symmetrization is translation invariant in the following sense: for each $x \in \mathbb{R}^n$, the symmetral $M_u(x+K)$ of $x+K$ is a translate of the symmetral $M_u(K)$ of K.*

(e) *Show that*

$$h(\mathrm{ref}_{u^\perp}(K), x) = h(K, \mathrm{ref}_{u^\perp}(x))$$

for every $x \in \mathbb{R}^n$. Conclude that $h_{M_u(K)}(v) = h_K(v)$ whenever $v \in u^\perp$.

(f) *Prove that we always have $S_u(K) \subseteq M_u(K)$ if K is a convex body and $u \in \mathbb{S}^{n-1}$.*

Solution 2.58. **(a)** This follows immediately from the fact that

$$\mathrm{ref}_{u^\perp} \circ \mathrm{ref}_{u^\perp} = \mathrm{Id}_{\mathbb{R}^n}.$$

(b) This also follows from the equality above. Indeed,

$$\begin{aligned} M_u(M_u(K)) &= \frac{1}{2}\big(\mathrm{ref}_{u^\perp}(M_u(K)) + M_u(K)\big) = \\ &= \frac{1}{2}\left(\mathrm{ref}_{u^\perp}\left(\frac{1}{2}\mathrm{ref}_{u^\perp}(K) + K\right) + \frac{1}{2}\big(\mathrm{ref}_{u^\perp}(K) + K\big)\right) = \\ &= \frac{1}{2}\left(\frac{1}{2}K + \frac{1}{2}\mathrm{ref}_{u^\perp}(K) + \frac{1}{2}\mathrm{ref}_{u^\perp}(K) + \frac{1}{2}K\right) = \frac{1}{2}\big(\mathrm{ref}_{u^\perp}(K) + K\big) = M_u(K). \end{aligned}$$

(c) This is a consequence of the trivial fact that $\mathrm{ref}_{u^\perp}(K)$ is monotone for inclusion and homogeneous for scalar multiplication.

(d) Simply notice that $\mathrm{ref}_{u^\perp}(x_0 + K) = \mathrm{ref}_{u^\perp}(x_0) + \mathrm{ref}_{u^\perp}(K)$.

(e) It is clear that $\mathrm{ref}_{u^\perp}$ is a linear isomorphism, and we also know from (2.6.5) in the main book that

$$h(\mathrm{ref}_{u^\perp}(K), x) = h(K, (\mathrm{ref}_{u^\perp})^* x).$$

Hence, it suffices to show that $\mathrm{ref}_{u^\perp}$ is self-adjoint. To do that, we just write

$$\begin{aligned}\langle x, \mathrm{ref}_{u^\perp}(y)\rangle = \langle x, y - 2\langle y, u\rangle u\rangle = \langle x, y\rangle - 2\langle y, u\rangle\langle x, u\rangle = \langle x, y\rangle - 2\langle y, \langle x, u\rangle u\rangle = \\ = \langle x - 2\langle x, u\rangle u, y\rangle = \langle \mathrm{ref}_{u^\perp}(x), y\rangle.\end{aligned}$$

This equality shows that $\mathrm{ref}_{u^\perp}$ is self-adjoint.

(f) Any point of $S_u(K)$ comes from translating a segment $z + \mathrm{seg}[\alpha u, \beta u]$, which is the intersection of K with a line in the direction u and passing through a point z of the orthogonal projection of K onto $u^\perp$. The corresponding segment in $S_u(K)$ is

$$z + \mathrm{seg}\left[\frac{\alpha - \beta}{2}u, \frac{\beta - \alpha}{2}u\right].$$

It is clear that we may write

$$\mathrm{seg}\left[\frac{\alpha - \beta}{2}u, \frac{\beta - \alpha}{2}u\right] = \frac{1}{2}\big(z + \mathrm{seg}[\alpha u, \beta u] + z + \mathrm{seg}[-\beta u, -\alpha u]\big).$$

Since $z + \mathrm{seg}[\alpha u, \beta u] \subseteq K$ and because it is clear that $z + \mathrm{seg}[-\beta u, -\alpha u] \subseteq \mathrm{ref}_{u^\perp}(K)$, we get that the latter is contained in $M_u(K)$. This gives the desired result.

□

Exercise 2.59. *Let $(K_j)_{j\in\mathbb{N}}$ be a sequence of convex bodies such that $K_j \to K$ for some $K \in CC^n$ in the Hausdorff metric. If $u \in \mathbb{S}^{n-1}$ is so that the Steiner symmetrals $S_u(K_j)$ converge to some $L \in CC^n$, prove that $L \subseteq S_u(K)$.*

Solution 2.59. Let $x \in L$. If $S_u(K_j) \to L$, then from item **(a)** of Theorem 2.6.1 from the main book, we get that there exists a sequence $(x_j)_{j\in\mathbb{N}}$ such that $x_j \in S_u(K_j)$ for each $j \in \mathbb{N}$ and $x_j \to x$. For each $j \in \mathbb{N}$, we denote the reflection of x_j across $u^\perp$ by $\mathrm{ref}_{u^\perp}(x_j)$, and we let y_j, z_j be the "first" and "last" intersections of the line $\{x_j + tu : t \in \mathbb{R}\}$ with K_j (i.e., they correspond to the minimum and maximum values of t such that $x_j + tu \in K_j$, respectively). Hence, $|y_j - z_j| \geq |x_j - \mathrm{ref}_{u^\perp}(x_j)|$. Passing to subsequences if necessary, we may assume that $y_j \to y$ and $z_j \to z$, and from item **(b)** of Theorem 2.6.1, we get that $y, z \in K$. Since for each $j \in \mathbb{N}$, we may write $y_j = x_j + t_j u$ and $z_j = x_j + s_j u$ for $t_j, s_j \in \mathbb{R}$, we get also that $y, z \in \{x + tu : t \in \mathbb{R}\}$. Even more, passing to the limit in the inequality previously derived, we obtain $|y - z| \geq |x - \mathrm{ref}_{u^\perp}(x)|$, and this clearly yields that $x \in S_u(K)$.

□

For Further Reading

Since the collection of keywords listed at the beginning of this chapter here shows in great parts the *geometric and elementary character* of that book part and its exercises, we restrict ourselves here to the

citation of books of related type, namely, being geometric in nature and suitable for a broad introduction into the field *mainly via undergraduate courses in convexity*; we mainly follow a chronological order.

Due to this philosophy, essential monographs also close to the research front (e.g., those of Thompson [243], Groemer [116], Schneider [228], Gardner [97], Gruber [117], Martini et al. [188], and Hug and Weil [142]) will be discussed in connection with other chapters.

A Basic Reference

Clearly, the monograph [58] of Bonnesen and Fenchel (i.e., the first comprehensive overview to the field) was the main reference in the first half of the twentieth century, showing the strong growth of the conceptual apparatus and of the literature within convexity already in those days. With this monograph, the knowledge about the field and its geometric ideas became widespread, also regarding applications in various other branches of mathematics. A (non-complete) list of scientists whose pioneering work helped to establish the field successfully (roughly until 1970) is the following: Minkowski, Helly, Radon, Blaschke, Bonnesen and Fenchel, A. D. Alexandrov, Coxeter, Busemann, Hadwiger, Klee, Grünbaum, Boltyanski, and L. Fejes Tóth. Of course, many further brilliant mathematicians enriched the field, and some of them gave basic contributions even coming from neighboring disciplines (e.g., from discrete geometry). And many famous researchers followed in the decades after 1970.

Related Booklets

In a motivating way, this time of growth was supported by a number of nice book(let)s, presenting basic (and mainly geometric) ideas in a short, refreshing, and didactically brilliant way. We give some examples here. First, there is Blaschke's famous booklet *Kreis und Kugel* [52] containing also *various new methods* for studying the isoperimetric problem in two and three dimensions and the Brunn-Minkowski theorem. All this is based on clearly defined notions and tools like his selection theorem or Steiner symmetrization. The book [261] of Yaglom and Boltyanski gives an excellent *introduction to planar convexity*, with many problems extendable also to higher dimensions. The second book part presents solutions, and to support the geometric understanding, 310 figures are given. Many general properties of convex figures are discussed. For instance, Helly-type theorems and problems that might be treated with continuity arguments are studied; Minkowski addition of convex figures, isoperimetric-like problems, sets of constant width, and rotors are further topics discussed in this classical book. Suitable as "completion" to [261] is the elementary monograph [182] of Lyusternik. It contains characterizations of constant width sets and ellipsoids, theorems from lattice geometry, and various results on convex polytopes. Eggleston published the little monograph [86] treating the *convexity property in general*, and based on this, he presents Helly-type theorem (including also Horn's theorem), support and distance functions, Blaschke's selection theorem, mixed volumes, the Brunn-Minkowski theorem, bodies of constant width, and various further topics of the field, helping to develop respective notions. Hadwiger [130] summarizes many parts of the field developed in the first half of the century, also by him. For example, he discusses the theory of volume, surface area and the mean curvature integral in 3-space, isoperimetry, Minkowski addition, inner and outer parallel bodies, Steiner symmetrization, integral geometry, and various further notions and tools. The *combinatorial geometry of convex bodies* is excellently treated in the booklets [134] and [56] of Hadwiger, Debrunner, Boltyanski, and Gohberg, respectively. For example, there one can find a *variety of interesting covering problems*.

Monographs

We continue with (larger) monographs that satisfy the criteria described above. Their union covers an impressively broad part of the whole field *convex geometry*.

Valentine [247] gives an introduction to the theory of convex sets, emphasizing characterizations of convexity, local convexity, duality, types of boundary points, Helly-type theorems (including that of Krasnosel'skii), and various further themes. In Benson's book [44] elementary fundaments from measure theory, equidecomposability, Helly-type theorems, sets of constant width, transformations of convex sets (also duality), Minkowski addition, Steiner symmetrization, extremum problems, and tools like Blaschke's selection theorem are discussed. In both books [247] and [44], also some *Minkowski geometry* (i.e., the geometry of finite-dimensional real normed spaces) is studied, in those days a rarity.

Guggenheimer [120] presents topics like Minkowski addition, Hausdorff metric, Blaschke's selection theorem, distance functions and polarity, and a section on integral geometry; the part on Helly-type theorems includes also Klee's theorem, and Steiner symmetrization is applied to the Brunn-Minkowski theorem. The books of Kelly and Weiss [150] and also of Lay [160] present brilliantly prepared undergraduate courses, with clearly presented proofs and inspiring examples and exercises. No special prior knowledge is necessary. The first book treats the basic structure and geometry of convex bodies and surfaces, types of hulls, linear combinations of sets, and properties of the Hausdorff metric. Many classical theorems (like Helly-type ones) are added. And Lay studies separation and support, Kirchberger-type theorems, sets of constant width, universal covers, three concepts of duality (also with relations to linear programming), isoperimetry, convex sets in finite matrix games, and also a little introduction to convex functions.

The monograph [256] of Webster is fundamental for *undergraduate or graduate courses on convexity and on polytopes* (with connections to linear programming). In the part on polytopes, he treats cyclic polytopes, the Dehn-Sommerville equations, and also Gale transforms, and in the general part the author discusses, for example, convex functions, support and gauge functions, mixed volumes, the Brunn-Minkowski theorem, isoperimetric and isodiametric inequalities, Steiner and Schwarz symmetrizations, Helly-type theorems, some axiomatic convexity, convexity-preserving mappings, and sets of constant width. Convex programming is presented as essential application field. Regarding *axiomatic convexity*, also the interesting booklet [238] of V. Soltan should be consulted.

The monograph [201] of Moszyńska can be basically used for seminars dedicated to convexity. Themes studied therein are fundamental properties of convex sets, polytopes, and star-shaped sets. They refer to the space of convex bodies and extensions of it, curvature and surface area notions, selectors for convex bodies, and (where it makes sense) fruitful extensions to star bodies, such as star duality, star metrics, and star body selectors. The Busemann-Petty problem is discussed, too. Here is possibly also the right place to mention the comprehensive survey [135] on star-shaped sets in the broadest sense.

The monograph [240] of V. Soltan comprehensively offers many useful *algebraic and topological properties of convex sets* in a very detailed manner, being gap-filling in a wide sense (e.g., in view of needs in convex analysis). The author gives also various proofs of "folklore statements" (whose confirmations are often not easy to find). These proofs are very elegant, but sometimes also unavoidably difficult.

In a sense being a model for our book here, [167] is a solutions manual for the monograph [166] of Leonard and Lewis on convex sets, containing many inspiring problems and exercises. It collects the solutions of all odd-numbered problems posed in [166], and additionally this is so nicely done that from there also a good approach to the even-numbered problems is possible. In the main book [166], the following topics are, among others, discussed in a detailed way and on the undergraduate level: basic properties of convex sets and convex hulls, Helly-type theorems, the art-gallery theorem, and sets of constant width. A large variety of carefully selected examples and counterexamples clearly shows the background of mathematical notions and concepts needed in convexity.

Despite our restricting remark given at the beginning of this part for further reading, we mention, regarding more modern (and mainly graduate) courses, also once more the two monographs [188] and [142], both written in the style of graduate textbooks.

For the sake of completeness, we finally mention here also Chapter 2 of the main book.

Research Lecture Notes

In the very recent lecture notes [26], edited by A. Colesanti and M. Ludwig, developments of modern convexity are collected and interestingly presented, among them valuations (on function spaces), geometric and analytic inequalities, duality concepts, symmetrizations, etc.; these courses, yielding the chapters of [26], are mainly addressed to PhD students and postdoctoral students and show the recent state regarding methods and results within the field of convexity.

PLATE 3

Schlegel diagrams - a look into 4-space

Schlegel diagram of the 4-cube

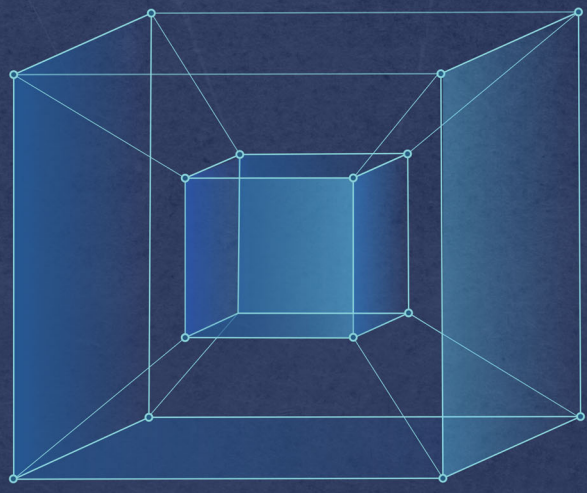
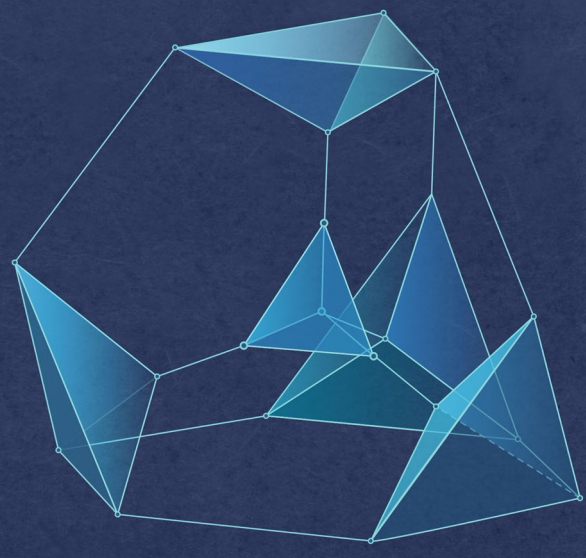


Figure 7

Figure 8

Schlegel diagram of a truncated 4-simplex



Schlegel diagrams - a look into 4-space ▸ (see, e.g., 2.9.1.4)

Chapter 3
A First Look into Polytopes



"When intuition and logic agree, you are always right."

— Blaise Pascal

In our third chapter, exercises treating various properties of n-dimensional *convex polytopes* (i.e., of n-dimensional convex hulls of finite point sets) are presented. Partial topics and basic notions referring to polytope theory and presented here are *Minkowski sums of polytopes, polyhedral sets* (polyhedra), *diameters* of polytopes, their *face lattice*, the role of *partially ordered sets*, (orthogonal) *projections, lattice polytopes, Steiner symmetrals* of polytopes, geometric properties of *simplices*, and *decompositions* of general polytopes into simplices.

Plate 3:

Plate 3 presents *Schlegel diagrams* of two four-dimensional convex polytopes, namely, of the 4-cube and of a truncated 4-simplex. Such diagrams are suitable central projections of a given 4-polytope into 3-space through a center outside of (but sufficiently close to) one of the facets of the projected 4-polytope. This yields a polytopal subdivision of that facet in 3-space, which, together with the original facet, is combinatorially equivalent to the original 4-polytope. Further *special types of polytopes* we see in Plate 10 (special types of *zonotopes*), Plate 11 (*regular-faced 3-polytopes*, whose definition is a weakening of that of Platonic solids), Plate 5 (as polyhedral *difference bodies* of convex 3-polytopes), and finally Plate 12 before the references (showing the *Platonic solids*).

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Exercises and Solutions

Exercise 3.1. *Let $T\colon \mathbb{R}^n \to \mathbb{R}^m$ be a linear transformation. Verify that for any polytope $P \subseteq \mathbb{R}^n$, we have that $T(P)$ is also a polytope.*

Solution 3.1. Let $P = \operatorname{conv}\{x_1, \ldots, x_k\}$. By linearity, it is immediate to see that

$$T(P) = \operatorname{conv}\{T(x_1), \ldots, T(x_k)\}.$$

Hence, $T(P)$ is a polytope.

□

Exercise 3.2. *Let $\pi\colon \mathbb{R}^{2n} \to \mathbb{R}^n$ be given as*

$$\pi(x, z) = x + z,$$

for any $x, z \in \mathbb{R}^n$. Verify that for any sets $A, B \subseteq \mathbb{R}^n$, it holds that

$$\pi(A \times B) = A + B.$$

Use this fact to give another proof of the fact that the Minkowski sum of polytopes is a polytope.

Solution 3.2. The fact that $\pi(A \times B) = A + B$ is immediate. If A and B are polytopes, then the characterization of polytopes as bounded intersections of half-spaces yields immediately that $A \times B$ is a polytope. The linearity of π and the characterization of polytopes as convex hulls of finitely many points give that the linear image of a polytope is a polytope. Thus, we get that $A + B = \pi(A \times B)$ is a polytope.

□

Exercise 3.3. *Let $P \subseteq \mathbb{R}^n$ be a proper polytope having vertices $\{x_1, \ldots, x_k\}$. Prove that there is a number $\rho > 0$ with the property that for any $j = 1, \ldots, k$, the ball $B(x_j, \rho)$ only touches the facets of P that contain x_j.*

Solution 3.3. Let $j \in \{1, \ldots, k\}$. Each one of the (finitely many) facets that does not contain x_j has a positive distance to x_j (two disjoint compact sets have positive distance). Let ρ_j be the half of the smallest of these distances. Then, $B(x_j, \rho_j)$ does not intersect any facet that does not contain x_j. Now, we let ρ be the minimum among $\rho_1, \ldots, \rho_k$, and we have the desired result.

□

An intersection of finitely many closed half-spaces in $\mathbb{R}^n$ is called a *polyhedron*. In view of Corollary 3.1.6 in the main book, the difference between polyhedra and polytopes is that the first ones are not necessarily bounded. The exercise below gives a useful characterization of polyhedra in terms of standard linear algebra.

Exercise 3.4. *Verify that a set $P \subseteq \mathbb{R}^n$ is a polyhedron if and only if there exists a number $m \in \mathbb{N}$, an $m \times n$ matrix M, and a vector $v \in \mathbb{R}^m$ such that*

$$P = \left\{x \in \mathbb{R}^n : Mx \leq v\right\},$$

where the inequality is taken componentwise.

Solution 3.4. A polyhedron in $\mathbb{R}^n$ is given as

$$\bigcap_{j=1}^{m} \mathbf{h}^-(u_j, \alpha_j)$$

for some $u_1, \ldots, u_m \in \mathbb{R}^n$ and $\alpha_1, \ldots, \alpha_m \in \mathbb{R}$. This corresponds directly to $Mx \leq v$ (componentwise) for M being the matrix whose lines are the vectors u_j (in coordinates, of course) and v is the vector (of $\mathbb{R}^m$) with entries $\alpha_1, \ldots, \alpha_m$.

□

Exercise 3.5. *Let $K, L \in CC^n$. Prove that for any $x \in \mathrm{relbd}(K)$ and every $y \in \mathrm{relbd}(L)$, the following equality holds:*

$$\mathrm{NC}_{K+L}(x + y) = \mathrm{NC}_K(x) \cap \mathrm{NC}_L(y).$$

Solution 3.5. Let $u \in \mathrm{NC}_K(x) \cap \mathrm{NC}_L(y)$. Then, $\langle u, z\rangle \leq \langle u, x\rangle$ for all $z \in K$ and $\langle u, w\rangle \leq \langle u, y\rangle$ for any $w \in L$. It follows that

$$\langle u, z + w\rangle \leq \langle u, x + y\rangle$$

for any $z \in K$ and every $w \in L$. This means that u is an outer normal of $K + L$ at $x + y$. Hence, we have the inclusion

$$\mathrm{NC}_K(x) \cap \mathrm{NC}_L(y) \subseteq \mathrm{NC}_{K+L}(x + y).$$

To prove the reverse inclusion, now assume that $u \in \mathrm{NC}_{K+L}(x + y)$. Then, $\langle u, z + w\rangle \leq \langle u, x + y\rangle$ for any $z \in K$ and every $w \in L$. Letting $w = y$, we get that $\langle u, z\rangle \leq \langle u, x\rangle$ for all $z \in K$. This means that $u \in \mathrm{NC}_K(x)$. Analogously, letting $z = x$ yields $u \in \mathrm{NC}_L(y)$.

□

The *diameter* of a non-empty set $A \subseteq \mathbb{R}^n$ is defined as $\mathrm{diam}(A) = \max\{|x - y| \colon x, y \in A\}$.

Exercise 3.6. *Let $K \subseteq \mathbb{R}^n$ be compact and convex. Prove that*

$$\mathrm{diam}(K) = \max\{|x - z| : x, z \in \mathrm{ext}K\}.$$

Conclude that the diameter of a polytope is attained by a pair of vertices.

Solution 3.6. Let $x_0, y_0 \in K$ and assume that one of them (x_0, say) is not an extremal point. Thus, there exist $x, z \in K$ such that $x_0 \in \mathrm{seg}(x, z)$. Hence, from basic planar geometry, we have that either $|y_0 - x|$ or $|y_0 - z|$ is greater than $|y_0 - x_0|$. This shows that

$$\mathrm{diam}(K) \geq \max\{|x - z| : x, z \in \mathrm{ext}K\} \geq \max\{|x - z| : x, z \in K\} = \mathrm{diam}(K)$$

and the proof is complete. The claim for polytopes comes immediately from the fact that the vertices are precisely the extremal points.

□

Exercise 3.7. *Let $P \subseteq \mathbb{R}^n$ be a polytope, and let F, G be faces (see p. 120 in the main book) of P. Prove that if $G \subseteq F$, then G is a face of F.*

Solution 3.7. If $G = \emptyset$ or $G = F$, there is nothing to prove. Thus, we assume that G is a proper subset of F. Let $\mathbf{h}$ and $\mathbf{g}$ be hyperplanes such that $F = \mathbf{h} \cap P$ and $G = \mathbf{g} \cap P$. Since $G \neq F$, we have that $\mathbf{h} \neq \mathbf{g}$, and hence, $\mathbf{h} \cap \mathbf{g}$ has co-dimension 1 in $\mathbf{h}$. Since $\mathbf{g}$ supports P, we have that F is contained in one of the closed half-spaces determined by $\mathbf{h} \cap \mathbf{g}$ in $\mathbf{h}$. Thus, $\mathbf{h} \cap \mathbf{g}$ supports F (in $\mathbf{h}$). Now, notice that $G = F \cap G \subseteq \mathbf{h} \cap \mathbf{g}$, and consequently

$$G = \mathbf{h} \cap \mathbf{g} \cap P = \mathbf{g} \cap (\mathbf{h} \cap P) = \mathbf{g} \cap F = (\mathbf{g} \cap \mathbf{h}) \cap F,$$

where in the last equality, we used the fact that $F = \mathbf{h} \cap F$. Since $\mathbf{g} \cap \mathbf{h}$ is a hyperplane of $\mathbf{h}$, the equality above gives that G is a face of F.

□

A *lattice* is an algebraic structure consisting of a set X and two binary commutative and associative operations $\vee$ (*join*) and $\wedge$ (*intersection*) such that

(i) $x \vee (x \wedge y) = x$ and

(ii) $x \wedge (x \vee y) = x$

for any $x, y \in X$. The objective of the next exercise is to verify that the faces of a polytope form a lattice with suitable operations.

Exercise 3.8. *Let $P \subseteq \mathbb{R}^n$ be a polytope, and denote by $\mathcal{F}(P)$ the family of the faces of P. For any $F, G \in \mathcal{F}(P)$, define the binary operations* intersection *and* join *respectively by*

$$F \wedge G := F \cap G \quad \text{and}$$
$$F \vee G := \bigcap \big\{H \in \mathcal{F}(P) : F, G \subseteq H\big\}.$$

(a) *Check that $\big(\mathcal{F}(P), \wedge, \vee\big)$ is a lattice. This is called the* face lattice of P.

(b) *Verify that $\emptyset$ and P satisfy $\emptyset \vee F = F$ and $P \wedge F = F$ for any $F \in \mathcal{F}(P)$. That is, the sets $\emptyset$ and P are* identity elements *of the join and of the intersection operations, respectively.*

Remark 3.1. A *bounded lattice* is a lattice that contains identity elements for the join and intersection operations. In view of **(b)**, the face lattice of a polytope is always a bounded lattice. The identity elements of the join and intersection operations are called the *bottom of the lattice* and the *top of the lattice*, respectively.

Solution 3.8. (a) The binary operations $\vee$ and $\wedge$ are obviously commutative and associative. Since it is clear that F is the intersection of the faces of P containing F (this is obvious if $F = P$ or $F = \emptyset$; see Corollary 3.2.3 from the main book for the case of the proper faces), and because $F \subseteq H$ implies $F \cap G \subseteq H$, we have

$$F \vee (F \wedge G) = F \vee (F \cap G) = \bigcap \big\{H \in \mathcal{F}(P) : F, F \cap G \subseteq H\big\} =$$
$$= \bigcap \big\{H \in \mathcal{F}(P) : F \subseteq H\big\} = F.$$

On the other hand, it is clear that $F \vee G$ contains F, because it is an intersection of sets containing F. Thus, $F \wedge (F \vee G)$ is the intersection of F with a set containing F. It follows that

$$F \wedge (F \vee G) = F.$$

(b) It is clear that $F \wedge P = F \cap P = F$ for any $F \in \mathcal{F}(P)$. On the other hand, we also have for every F that

$$F \vee \emptyset = \bigcap \{H \in \mathcal{F}(P) : F, \emptyset \subseteq H\} = \bigcap \{H \in \mathcal{F}(P) : F \subseteq H\} = F.$$

The last equality follows from the fact that $F \in \mathcal{F}(P)$ and $F \subseteq F$.

□

Exercise 3.9. *Let $P \subseteq \mathbb{R}^n$ be a polytope, and let $F, G \subseteq \mathbb{R}$ be such that $G \subseteq F$. Assume that* $\dim(G) = j$ *and* $\dim(F) = k$. *Prove that there are faces $F_{j+1}, \ldots, F_{k-1}$ of P such that* $\dim(F_i) = i$ *for each $i = j + 1, \ldots, k - 1$, and*

$$G \subseteq F_{j+1} \subseteq \ldots \subseteq F_{k-1} \subseteq F.$$

Solution 3.9. From Exercise 3.7, we know that G is a face of F. From Corollary 3.1.5 (and, of course, recalling that F itself is a polytope), it follows that G is contained in some facet of F. We let this facet be F_{k-1}. If $G \neq F_{k-1}$, we repeat the argument replacing F by F_{k-1} and obtain a facet F_{k-2} of F_{k-1} such that $G \subseteq F_{k-2}$. We repeat this process until $k - m = \dim(G)$. Then, G is a face of the polytope F_{k-m} whose dimension is $\dim(G)$. It follows that $G = F_{k-m}$.

□

Recall that a *partial order* in a set X is a binary relation (which, for simplicity, we will denote by $\leq$), which is reflexive, antisymmetric, and transitive, meaning respectively that:

(i) $x \leq x$ for all $x \in X$,

(ii) for any given $x, y \in X$, $x \leq y$ and $y \leq x$ imply $x = y$, and

(iii) given $x, y, z \in X$, if $x \leq y$ and $y \leq z$, then $x \leq z$.

A set X endowed with a partial order is called a *partially ordered set* (or a *poset*).

Exercise 3.10. *Let $(X, \wedge, \vee)$ be a lattice. Define a binary operation in X writing $x \leq y$ whenever $x = x \wedge y$. Prove that this binary operation is a partial order in X. Show also that for the face lattice of a polytope, this partial order is simply the inclusion.*

Solution 3.10. For any arbitrarily given $x, y \in X$, we have that

$$x = x \vee (x \wedge y) \text{ and}$$
$$x = x \wedge (x \vee y).$$

Letting $y = x$ in the first equality, we arrive at

$$x = x \vee (x \wedge x),$$

and setting $y = x \wedge x$ in the second equality, we get

$$x = x \wedge (x \vee (x \wedge x)) = x \wedge x.$$

Hence, $x = x \wedge x$, which means that $x \leq x$. That is, the partial order $\leq$ in X is reflexive.

Now, let $x, y \in X$ be such that $x \leq y$ and $y \leq x$. Thus, by the definition of $\leq$, we have that $x = y = x \wedge y$. It follows that

$$x = x \vee (x \wedge y) = x \vee y,$$

and

$$y = y \vee (y \wedge x) = y \vee x = x \vee y.$$

Therefore, it holds that $x = y$. This shows that $\leq$ is antisymmetric.

It only remains to show that $\leq$ is transitive. For that sake, let $x, y, z \in X$ be such that $x \leq y$ and $y \leq z$. This yields $x = x \wedge y$ and $y = y \wedge z$. Thus,

$$x \wedge z = (x \wedge y) \wedge z = x \wedge (y \wedge z) = x \wedge y = x,$$

and this is the same as saying that $x \leq z$. This proves that $\leq$ is a partial order in X. The remaining inclusion is trivial: let $P \subseteq \mathbb{R}^n$ be a polytope and let $F, G \in \mathcal{F}(P)$. Then, $G = F \wedge G$ means that $G = G \cap F$. This happens if and only if $G \subseteq F$.

□

Let $(X, \wedge, \vee)$ be a bounded lattice (see Remark 3.1). As it is usual in the literature, we will denote the identity element of the join by 0 and the identity element of the intersection by 1 (in the case of the face lattice of a polytope P, we have that $0 = \emptyset$ and $1 = P$, or course). An element $x \in X \setminus \{0\}$ is said to be an *atom* of X if there is no element $z \in X \setminus \{0, x\}$ such that $0 \leq z$ and $z \leq x$. Moreover, an element $a \in X$ is called a *co-atom* of X if there is no $b \in X \setminus \{a, 1\}$ such that $a \leq b$ and $b \leq 1$.

A bounded lattice is said to be *atomic* if each of its elements is the join of finitely many atoms, and *co-atomic* if each of its elements is the intersection of finitely many co-atoms.

Exercise 3.11. *Let $P \subseteq \mathbb{R}^n$ be any given proper polytope.*

(a) *Show that the atoms of $\mathcal{F}(P)$ are precisely the vertices of P and that the co-atoms of $\mathcal{F}(P)$ are exactly the facets of P.*

(b) *Prove that the face lattice of P is atomic and co-atomic.*

Solution 3.11. (a) Since the partial order in $\mathcal{F}(P)$ is the usual inclusion, the atoms are the faces that do not contain a proper non-empty face. It follows that the vertices of P are the atoms of $\mathcal{F}(P)$.

It is also immediate to notice that the co-atoms of $\mathcal{F}(P)$ are the faces that are not properly contained in a face different from P. Since any proper face of P is contained in some facet of P, and because a facet cannot be properly contained in any proper face of P (due to a dimensional argument), we get that the co-atoms of $\mathcal{F}(P)$ are exactly the facets of P.

(b) Let $F \in \mathcal{F}(P)$ be any non-empty face of P. Then, F is the join of all vertices of P, which lie in F. In fact, it is clear that F is the intersection of all faces containing all of the vertices of P, which lie in F. This shows that $\mathcal{F}(P)$ is atomic. We have that $\mathcal{F}(P)$ is co-atomic as a consequence of the fact that each face of P is the intersection of the facets at which it lies.

□

We say that a bounded lattice $(X, \wedge, \vee)$ is *complemented* if for each $x \in X$, there exists an element $y \in X$ such that $x \wedge y = 0$ and $x \vee y = 1$.

Exercise 3.12. *Let $P \subseteq \mathbb{R}^n$ be a proper polytope. Show that $\mathcal{F}(P)$ is complemented.*

Solution 3.12. Just consider $\emptyset$ and P. We have $F \wedge \emptyset = \emptyset$ and $F \vee P = P$ for any $F \in \mathcal{F}(P)$.

□

Recall that C^n stands for the family of non-empty compact subsets of $\mathbb{R}^n$, and let $\mathbb{Z}^n := \{(z_1, \ldots, z_n) : z_1, \ldots, z_n \in \mathbb{Z}\}$ be the *grid* of *integer points* of $\mathbb{R}^n$. For each $A \in C^n$, let

$$L(A) := \sum_{x \in A \cap \mathbb{Z}^n} 1.$$

That is, $L(A)$ is the number of integer points of $\mathbb{R}^n$ that lie in A.

Exercise 3.13. **(a)** *Let $L : C^n \to \mathbb{R}$ be defined as above. Check that L has the* valuation property*: for any $A, B \in C^n$, it holds that*

$$L(A \cup B) + L(A \cap B) = L(A) + L(B).$$

(b) *Prove that*

$$L(m[0,1]^n) = (m+1)^n$$

for any $m \in \mathbb{N}$.

Remark 3.2. A *lattice polytope* is a polytope whose vertices are integer points. For any lattice polytope P, the number of integer points of mP is polynomial on $m \in \mathbb{N}$. Such a polynomial is known as the *Ehrhart polynomial* of P. In item **(b)** above, we are computing precisely the Ehrhart polynomial of the unit cube of $\mathbb{R}^n$. For more in this direction, we refer the reader to the survey [62].

Solution 3.13. **(a)** Write $A \cup B$ as the disjoint union

$$A \cup B = (A \setminus B) \cup (A \cap B) \cup (B \setminus A).$$

Then, write $A = (A \setminus B) \cup (A \cap B)$ and $B = (B \setminus A) \cup (A \cap B)$, and notice that they are also disjoint unions. Now, just notice that L is additive for disjoint unions.

(b) The integer points of $m[0,1]^n$ are precisely all the points whose entries are in the set $\{0, 1, \ldots, m\}$. Hence, each entry can be filled in $m+1$ ways, and from (very) basic combinatorics, we get that $m[0,1]^n$ has $(m+1)^n$ integer points.

□

Let m, n be the given natural numbers, P be a proper polytope in $\mathbb{R}^n$, and Q be a proper polytope in $\mathbb{R}^m$. A map $\pi : P \to Q$ is said to be a *projection of polytopes* if

(i) π is the restriction to P of a map having the form $x \mapsto Ax + b$, where $A : \mathbb{R}^n \to \mathbb{R}^m$ is a linear transformation and $b \in \mathbb{R}^m$;

(ii) $\pi(P) = Q$.

Exercise 3.14. *Let $\pi : P \to Q$ be a projection of polytopes. Prove that if F is a face of Q, then $\pi^{-1}(F)$ is a face of P.*

Solution 3.14. We just use Lemma 2.4.5. Let $x, y \in P$ be such that $(x + y)/2 \in \pi^{-1}(F)$. Then, $\pi(x), \pi(y) \in Q$ and

$$\frac{\pi(x) + \pi(y)}{2} \in F.$$

Since F is a face of Q, we get that $\pi(x), \pi(y) \in F$. Hence, $x, y \in \pi^{-1}(F)$. This shows that $\pi^{-1}(F)$ is a face of P.

□

For the next exercise, we introduce a terminology that will be also used later in this book. A set having the form

$$H := \mathbf{h}^-(u, 0) \cap \mathbb{S}^{n-1}$$

for some $u \in \mathbb{S}^{n-1}$ is called a *closed hemisphere* of $\mathbb{S}^{n-1}$.

Exercise 3.15. *Let $E = \{u_1, \ldots, u_m\} \subseteq \mathbb{S}^{n-1}$ be a non-empty finite set that is not contained in a closed hemisphere of $\mathbb{S}^{n-1}$.*

(a) *Show that* $\operatorname{pos}(E) = \mathbb{R}^n$.

(b) *Prove that for any numbers $\alpha_1, \ldots, \alpha_m > 0$, it holds that the polyhedron*

$$P := \bigcap_{j=1}^{m} \mathbf{h}^-(u_j, \alpha_j)$$

is a polytope.

Hint: In item **(a)**, verify that if $\operatorname{pos}(E) \neq \mathbb{R}^n$, then $\operatorname{pos}(E)$ has a supporting hyperplane (see Theorem 2.2.1).

Remark 3.3. The construction in item **(b)** is a *Wulff shape*. This type of construction is studied in Section 8.1 of the main book.

Solution 3.15. (a) Since E is finite, it is immediate to check that $\operatorname{pos}(E)$ is closed, and we also know that $\operatorname{pos}(E)$ is convex. Hence, if $\operatorname{pos}(E)$ is not $\mathbb{R}^n$, then it has a supporting hyperplane $\mathbf{h}$ (with $\operatorname{pos}(E) \subseteq \mathbf{h}^-$, say). Thus,

$$E \subseteq \operatorname{pos}(E) \cap \mathbb{S}^{n-1} \subseteq \mathbf{h}^- \cap \mathbb{S}^{n-1}.$$

This contradicts the hypothesis that E is not contained in a closed hemisphere.

(b) Assume that P is not bounded. Then, there is a sequence (x_j) in P such that $|x_j| \to +\infty$, and of course, we may assume that all x_j's are non-zero. For each $j \in \mathbb{R}$, we let

$$z_j := \frac{x_j}{|x_j|}.$$

Notice that (z_j) is a sequence in $\mathbb{S}^{n-1}$, and hence, we may assume that it converges to some $z \in \mathbb{S}^{n-1}$ (passing to a subsequence, if necessary). Since $\langle x_j, u_k \rangle \leq \alpha_k$ for each $j \in \mathbb{N}$ and every $k \in \{1, \ldots, m\}$, it follows that

$$\langle z_j, u_k \rangle \leq \frac{\alpha_k}{|x_j|}.$$

Letting $j \to +\infty$, we obtain

$$\langle z, u_k \rangle \leq 0$$

for every $k \in \{1, \ldots, m\}$. This means that $u_k \in \mathbf{h}^-(z, 0)$ for all k, which leads to $E \subseteq \mathbf{h}^-(z, 0) \cap \mathbb{S}^{n-1}$. This contradicts the fact that E is not contained in a closed hemisphere.

□

We say that a proper polytope $P \subseteq \mathbb{R}^n$ is *circumscribed* about a convex body $K \subseteq \mathbb{R}^n$ if $K \subseteq P$ and K intersects each facet of P.

Exercise 3.16. *Let $u_1, \ldots, u_m \in \mathbb{S}^{n-1}$ be mutually distinct vectors such that $\operatorname{pos}\{u_1, \ldots, u_m\} = \mathbb{R}^n$. Prove that there exists a polytope $P \subseteq \mathbb{R}^n$ circumscribed about B^n whose facets have precisely $u_1, \ldots, u_m$ as unit outer normal vectors.*

Solution 3.16. Let

$$P = \bigcap_{j=1}^{m} \mathbf{h}^-(u_j, 1).$$

It is clear that P is a polytope (see Exercise 3.15), and it is also immediate that P contains the unit ball B^n (in particular, P is a convex body). Since the u_j's are mutually distinct, we have that $u_k \in \operatorname{int}(\mathbf{h}^-(u_j, 1))$ whenever $k \neq j$. Hence, each u_j is a boundary point of P belonging to a unique supporting hyperplane. It follows that $u_1, \ldots, u_m$ are unit outer normal vectors of the facets of P. Of course, by the definition of P, no additional vector can be the unit outer normal vector of a facet of P.

□

Exercise 3.17. *Prove that any Steiner symmetral of a polytope is a polytope.*

Remark 3.4. We thank João Rodriguez Marcondes for giving the main idea of the solution below.

Solution 3.17. Let P be the original polytope and $Q = S_u(P)$ be its Steiner symmetral. Our strategy is to show that Q has a finite number of extreme points; from Minkowski's theorem (Theorem 2.4.3 of the main book), we conclude that Q must be a polytope.

Now, one must be careful since the extreme points of Q do not necessarily correspond directly to vertices of P. As a quick example, consider in $\mathbb{R}^3$ the tetrahedron P with vertices $A(1, 0, -1)$, $B(-1, 0, -1)$, $C(0, -1, 1)$, and $D(0, 1, 1)$ and take $u = (0, 0, 1)$. When symmetrizing P, its four vertices are shifted toward the plane $z = 0$ (so $Q = S_u(P)$ will have vertices at $(\pm 1, 0, 0)$ and $(0, \pm 1, 0)$). However, *Q will be an octahedra, with two "extra" vertices at* $(0, 0, \pm 1)$. In a sense, these two new vertices are "generated" because the projections of the edges AB and CD onto the horizontal plane do have a transversal intersection at the origin.

In higher dimensions, things clearly get more complicated—every time the $u-$projection of two faces of P intersect at a single point, a pair of extreme points might be generated in the Steiner symmetral $Q = S_u(P)$. But here is the good news: we affirm that every extreme point of Q (or every symmetrical *pair* of extreme points) corresponds to a unique pair of faces of P in a way to be described below. So, if P has a faces (counting here faces of all possible dimensions, including 0), we can conclude that Q has at most $2a^2$ extreme points (usually a lot less), which is enough for our purposes.

To establish this correspondence, let us fix some notation: for every "vertical fiber" $[p_1, p_2]$ with endpoints p_1, p_2 on ∂P, we can associate the corresponding segment $[q_1, q_2]$ with endpoints q_1, q_2 on ∂Q (such that $\operatorname{proj}_{u^\perp}(p_1) = \operatorname{proj}_{u^\perp}(p_2) = \operatorname{proj}_{u^\perp}(q_1) = \operatorname{proj}_{u^\perp}(q_2)$). From now on, we may abuse notation and write $q_i = S_u(p_i)$ in a situation like this. Now, let P_1 and P_2 be the smallest faces of P that contain p_1 and p_2, respectively. In particular, Lemma 3.2.1 of the main book states that these

are the only faces of P that satisfy $p_1 \in \text{relint}(P_1)$ and $p_2 \in \text{relint}(P_2)$. Finally, consider the map that takes the segment $[q_1, q_2]$ to the pair of faces (P_1, P_2). We shall now prove that, when restricted to segments whose endpoints are extreme points of Q, this map must be injective.

Indeed, if $[q_1, q_2]$ and $[q_3, q_4]$ were to map to the same pair of faces (P_1, P_2), we would have corresponding points on ∂P, which would satisfy $p_1, p_3 \in \text{relint}(P_1)$ and $p_2, p_4 \in \text{relint}(P_2)$. If either P_1 or P_2 were a single point, that would mean $p_1 = p_3$ or $p_2 = p_4$, forcing the equality between the fibers $[p_1, p_2] = [p_3, p_4]$ and hence $[q_1, q_2] = [q_3, q_4]$. So, we are left with the case where both P_1 and P_2 are at least 1−dimensional. But in that case, the segments $[p_1, p_3] \subseteq \text{relint}(P_1)$ and $[p_2, p_4]$ could be extended to strictly larger segments $[z_1, z_3] \subseteq \text{relint}(P_1)$ and $[z_2, z_4] \subseteq \text{relint}(P_2)$ where $[z_1, z_2]$ and $[z_3, z_4]$ are also "vertical fibers" in P (note that this is possible since $[p_1, p_2]$ is parallel to $[p_3, p_4]$, forcing all these points to be in the same 2-dimensional plane, as well as z_1, z_2, z_3 and z_4). But since $p_1 \in \text{relint}([z_1, z_3])$, we would have $q_1 \in \text{relint}([S_u(z_1), S_u(z_3)])$, and therefore, q_1 would not be an extreme point of Q; the same argument can be used to show that q_2, q_3 and q_4 cannot be extreme points either.

So, every vertical segment $[q_1, q_2]$ of extreme points of Q maps uniquely to a pair of faces (P_1, P_2) of P, and we are done.

□

Exercise 3.18. *Let $P \subseteq \mathbb{R}^n$ be a polytope having at least two vertices and whose circumball is $x_0 + rB^n$. Show that $x_0 \in \text{conv}\big((\text{ext}P) \cap \partial(x_0 + rB^n)\big)$. In particular, conclude that $x_0 \in P$.*

Remark 3.5. In other words, the center of the circumball of a polytope is in the convex hull of the vertices of P that are in the boundary of the circumball.

Solution 3.18. Let $\text{ext}P = \{x_1, \ldots, x_k\}$. We first prove that $(\text{ext}P) \cap \partial(x_0 + rB^n)$ is non-empty. Indeed, if this is not true, then we would have $x_j \in \text{int}(x_0 + rB^n)$ for each $1 \leq j \leq k$. Since $\partial(x_0 + rB^n)$ is compact, we have that $\text{dist}(x_j, \partial(x_0 + rB^n)) > 0$ for each j. Let

$$d = \min\big\{\text{dist}(x_j, \partial(x_0 + rB^n)) : 1 \leq j \leq k\big\}.$$

This number is clearly positive, because it is the minimum of a finite set of positive numbers. It is also immediate that $d < r$, because P has at least two vertices, and hence, at least one of them is not x_0. We claim that

$$x_j \in x_0 + (r - d)B^n$$

for each $1 \leq j \leq k$. This is clear if $x_j = x_0$, and otherwise, we let

$$z = \frac{r}{|x_j - x_0|}(x_j - x_0).$$

Then $z \in \partial(x_0 + rB^n)$, which leads to

$$|x_j - z| \geq d.$$

Since z is a multiple of $x_j - x_0$, we have that the points x_0, x_j, and z are aligned. Also, since $x_j \in \text{int}(x_0 + rB^n)$, we have $|x_j - x_0| < r$, and it is clear that $|z - x_0| = r$. Hence x_j lies (strictly) between z and x_0, yielding

$$|x_j - x_0| = |z - x_0| - |z - x_j| \leq r - d,$$

from where we get that $x_j \in x_0 + (r-d)B^n$. Since this holds for each $1 \leq j \leq k$, and because $x_0 + rB^n$ is a convex set, we get

$$P = \operatorname{conv}(\operatorname{ext} P) = \operatorname{conv}\{x_1, \ldots, x_k\} \subseteq x_0 + (r-d)B^n,$$

and this contradicts the fact that r is the circumradius of P. This shows that $\operatorname{ext} P \cap \partial(x_0 + rB^n) \neq \emptyset$.

Assume that $\operatorname{ext} P \cap \partial(x_0 + rB^n) = \{x_1, \ldots, x_m\}$. Thus, $x_j \in \operatorname{int}(x_0 + rB^n)$ for each $m+1 \leq j \leq k$, and hence, we may let

$$\varepsilon = \min\left\{\operatorname{dist}(x_j, \partial(x_0 + rB^n)) : m+1 \leq j \leq k\right\},$$

and, as it was shown before, we have $0 < \varepsilon < r$. Notice that by the previous argument, we have

$$x_j \in x_0 + (r-\varepsilon)B^n \subseteq x_0 + (r-\delta)B^n \tag{3.1}$$

for each $m+1 \leq j \leq k$ and any $0 < \delta \leq \varepsilon$.

Suppose that $x_0 \notin \operatorname{conv}\{x_1, \ldots, x_m\}$. Let $\mathbf{h} := \mathbf{h}(u, \langle u, x_0\rangle)$ be any hyperplane through x_0 such that

$$\operatorname{conv}\{x_1, \ldots, x_m\} \subseteq \operatorname{int}(\mathbf{h}^+),$$

and assume that $u \in \mathbb{S}^{n-1}$. In case of any doubt regarding the existence of such a hyperplane, just let $z \in \operatorname{conv}\{x_1, \ldots, x_m\}$ be a point where $\operatorname{dist}(x_0, \operatorname{conv}\{x_1, \ldots, x_m\})$ is attained, and consider u to be the unit vector in the direction of $z - x_0$. We claim that

$$x_j \in x_0 + \sigma u + (r - \delta + \sigma)B^n \tag{3.2}$$

for each $m+1 \leq j \leq k$ and every $0 < \sigma < \delta \leq \varepsilon$. This comes immediately from (3.1) together with the triangle inequality:

$$|x_j - x_0 - \sigma u| \leq |x_j - x_0| + |\sigma u| \leq r - \delta + \sigma.$$

We move forward to prove that we can choose $0 < \sigma < \delta < \varepsilon$ small enough such that (3.2) holds also for $1 \leq j \leq m$. For each of these j's, we have $x_j \in \partial(x_0 + rB^n)$, from where $|x_j - x_0| = r$. Then,

$$|x_j - x_0 - \sigma u|^2 = r^2 - 2\langle x_j - x_0, \sigma u\rangle + \sigma^2$$

for every $\sigma > 0$. Let $d := \max\{\langle x_j - x_0, u\rangle : 1 \leq j \leq m\}$, and observe that $d > 0$ because $x_j \in \operatorname{int}(\mathbf{h}^+)$ for each $1 \leq j \leq m$. Thus,

$$|x_j - x_0 - \sigma u|^2 \leq r^2 - 2\sigma d + \sigma^2$$

for each $1 \leq j \leq m$. The function $\sigma \mapsto -2\sigma d + \sigma^2$ is negative for every $0 < \sigma < d$, and its limit is 0 when $\sigma \to 0$. Hence, we may choose a number $0 < \sigma_0 < \varepsilon/2$ such that $-2\sigma_0 d + \sigma_0^2 < 0$ and

$$r^2 - 2\sigma_0 d + \sigma_0^2 = (r - \alpha)^2$$

with $0 < \alpha < \varepsilon/2$. Letting $\delta_0 := \alpha + \sigma_0$, we have $\sigma_0 < \delta_0 < \varepsilon$ (hence, we guarantee that (3.2) holds for $m+1 \leq j \leq k$ when $\sigma = \sigma_0$ and $\delta = \delta_0$) and

$$|x_j - x_0 - \sigma_0 u| \leq \sqrt{r - 2\sigma_0 d + \sigma_0^2} = r - \alpha = r - \delta_0 + \sigma_0$$

for every $1 \leq j \leq m$. It follows that (3.2) also holds for $1 \leq j \leq m$ if we let $\sigma = \sigma_0$ and $\delta = \delta_0$. Since any ball is convex, we obtain

$$P = \operatorname{conv}(\operatorname{ext} P) = \operatorname{conv}\{x_1, \ldots, x_k\} \subseteq x_0 + \sigma_0 u + (r - \delta_0 + \sigma_0)B^n,$$

and since $-\delta_0 + \sigma_0 = -\alpha < 0$, we get that the right-hand side is a ball containing P whose radius is smaller than r. This contradicts the fact that r is the circumradius of P, and hence, we have that $x_0 \in \operatorname{conv}\big((\operatorname{ext} P) \cap \partial(x_0 + rB^n)\big)$. Last, it is clear that

$$\operatorname{conv}\big((\operatorname{ext} P) \cap \partial(x_0 + rB^n)\big) \subseteq \operatorname{conv}(\operatorname{ext} P) = P,$$

from which we get that $x_0 \in P$. This concludes the solution of the exercise.

□

Exercise 3.19. *Let $P \subseteq \mathbb{R}^n$ be a proper polytope whose facets are $F_1, \ldots, F_m$ with corresponding supporting hyperplanes $\mathbf{h}_1, \ldots, \mathbf{h}_m$. Let x be an interior point of P, and assume that*

$$\min\big\{\operatorname{dist}(x, \mathbf{h}_j) : 1 \leq j \leq m\big\} = \operatorname{dist}(x, \mathbf{h}_k)$$

for some $1 \leq k \leq m$. Show that the orthogonal projection of x onto $\mathbf{h}_k$ lies in the relative interior of F_k.

Solution 3.19. Let p be the projection of x onto $\mathbf{h}_k$. The segment $\operatorname{seg}[x, p]$ cuts the boundary of P in some point (z, say). The point z is in some facet of P (F_i, say). If p is not in F_k, then since z is obviously closer to x than p, we have that the closest facet to x is F_i, and not F_k. This contradiction gives that p is a point of F_k. If p is in the relative boundary of F_k, then p also lies in another facet of P (F_r, say). The orthogonal projection q of x onto $\mathbf{h}_r$ cannot be p (the direction of the vector $p - x$ is not orthogonal to $\mathbf{h}_r$, because it is orthogonal to $\mathbf{h}_k$, which is a concurrent hyperplane). It follows that $|q - x| < |p - x|$, and hence, $\mathbf{h}_r$ is closer to x than h_k.

□

Exercise 3.20. *Let $x_1, \ldots, x_k \in r\mathbb{S}^{n-1}$ for some $r > 0$, and let $P = \operatorname{conv}\{x_1, \ldots, x_k\}$. Prove that $r_{\mathrm{out}}(P) = r$ if and only if $0_{\mathbb{R}^n} \in P$.*

Solution 3.20. Assume first that $0_{\mathbb{R}^n} \notin P$. Then, we may let $\mathbf{h}$ be a hyperplane strongly separating them, with outer normal u, say. Let p be the projection of the origin onto $\mathbf{h}$. It is not difficult to see that p is closer to the vertices of P than $0_{\mathbb{R}^n}$, and hence, there is a ball centered in p containing P whose radius is smaller than r.

Now assume that $0_{\mathbb{R}^n} \in P$. If $r_{\mathrm{out}}(P) < r$, then we have a ball $z_0 + \rho B^n$ containing P with $\rho < r$. It is clear that z_0 is not $0_{\mathbb{R}^n}$, because the norms of the vertices of P are r (hence, greater than ρ). Let $\mathbf{h}$ be the hyperplane through the origin whose outer normal points in the direction of z_0. Since $0_{\mathbb{R}^n}$ is a point of P, there must be a vertex x_j of P that lies in the opposite (closed) half-space determined by $\mathbf{h}$ as z_0. It is easy to see that the distance from x_j to z_0, which is ρ, is greater than the distance from x_j to the origin, which is r. This is a contradiction, and the proof is complete.

□

Exercise 3.21. *Let $P \subseteq \mathbb{R}^n$ be a polytope having at most $n + 1$ vertices. Prove that*

$$r_{\mathrm{out}}(P) \leq \left(\frac{n}{2n+2}\right)^{1/2} \operatorname{diam}(P). \tag{3.3}$$

Remark 3.6. In particular, from Exercise 3.6, we have that (3.3) holds also if we replace P by any finite set containing at most $n+1$ points (observe that the circumradius is clearly monotone under inclusion).

Solution 3.21. Let the vertex set of P be $\mathrm{ext}P = \{x_1, \ldots, x_k\}$ (hence, we have $k \leq n+1$). Up to a translation, we may assume that the center of the circumball of P is the origin. Denoting the circumradius of P by r, we have then that rB^n is the circumball of P. By reordering the vertices if necessary, we may write

$$(\mathrm{ext}P) \cap rB^n = \{x_1, \ldots, x_m\}.$$

By Exercise 3.18, we have that $0_{\mathbb{R}^n} \in \mathrm{conv}\{x_1, \ldots, x_m\}$, and hence, we may write

$$0_{\mathbb{R}^n} = \lambda_1 x_1 + \ldots + \lambda_m x_m$$

where $\lambda_1, \ldots, \lambda_m \geq 0$ and $\lambda_1 + \ldots + \lambda_m = 1$. For each $1 \leq j \leq m$, we have

$$1 - \lambda_j = \sum_{i=1,\, i\neq j}^{m} \lambda_i \geq \sum_{i=1}^{m} \lambda_i \frac{|x_i - x_j|^2}{\mathrm{diam}(P)^2} = \sum_{i=1}^{m} \lambda_i \frac{|x_i|^2 + |x_j|^2 - 2\langle x_i, x_j\rangle}{\mathrm{diam}(P)^2} =$$

$$= \frac{1}{\mathrm{diam}(P)^2}\left(2r^2 - 2\sum_{i=1}^{m} \lambda_i \langle x_i, x_j\rangle\right) = \frac{1}{\mathrm{diam}(P)^2}\left(2r^2 - 2\left\langle \sum_{i=1}^{m} \lambda_i x_i, x_j\right\rangle\right) =$$

$$= \frac{1}{\mathrm{diam}(P)^2}\left(2r^2 - 2\langle 0_{\mathbb{R}^n}, x_j\rangle\right) = \frac{2r^2}{\mathrm{diam}(P)^2}.$$

Summing over $j = 1, \ldots, m$, we get

$$m - 1 \geq \frac{2mr^2}{\mathrm{diam}(P)^2},$$

and this leads to

$$r \leq \left(\frac{m-1}{2m}\right)^{1/2} \mathrm{diam}(P) \leq \left(\frac{n}{2n+2}\right)^{1/2} \mathrm{diam}(P).$$

The last inequality is justified because $m \leq k \leq n+1$ and the map $x \mapsto (x-1)/2x$ is non-decreasing in the natural numbers. Then, the proof is complete.

□

Exercise 3.22. *Show that (3.3) actually holds for any bounded set $A \subseteq \mathbb{R}^n$.*

Hint. Use Theorem 2.1.5 and Remark 2.1.7 from the main book.

Solution 3.22. The case where A is finite and contains at most $n+1$ points is discussed in Remark 3.6, so we may assume that A is either infinite or is finite and has more than $n+1$ points. Let

$$r := \left(\frac{n}{2n+2}\right)^{1/2} \mathrm{diam}(A)$$

and consider the family

$$\mathcal{A} := \{x + rB^n : x \in A\}$$

of closed balls. For any $x_1, \ldots, x_{n+1} \in A$, the circumradius of $\{x_1, \ldots, x_{n+1}\}$ is smaller than or equal to r: this follows immediately from Remark 3.6 and from the fact that the diameter of $\{x_1, \ldots, x_{n+1}\}$ is smaller than or equal to $\operatorname{diam}(A)$. In particular, there exists a point $z_0 \in \mathbb{R}^n$ such that $\{x_1, \ldots, x_{n+1}\} \subseteq z_0 + rB^n$. It follows that

$$\bigcap_{j=1}^{n+1} (x_j + rB^n) \neq \emptyset,$$

because for each $1 \leq j \leq n+1$, we have that $x_j \in z_0 + rB^n$, and hence, $|z_0 - x_j| < r$ for each $1 \leq j \leq n+1$. This means that $z_0 \in x_j + rB^n$.

From Theorem 2.1.5 in the main book (for the case where A is finite and contains more than $n+1$ points) and Remark 2.1.7 from the main book (for the case where A is infinite), we have that the intersection of all sets of $\mathcal{A}$ is non-empty. Fix any point y_0 in this intersection. Then, for every $x \in A$, the point y_0 is in the ball $x + rB^n$, and this gives that x is a point of the ball $y_0 + rB^n$ for any $x \in A$. In other words, $A \subseteq y_0 + rB^n$. Hence,

$$r_{\text{out}}(A) \leq r = \left(\frac{n}{2n+2}\right)^{1/2} \operatorname{diam}(A),$$

and the proof is complete.

□

Exercise 3.23. *Let X be an n-simplex in $\mathbb{R}^n$. Prove that there exists a unique ball $B \subseteq X$ touching every facet of X.*

Solution 3.23. Let $F_1, \ldots, F_{n+1}$ be the facets of the simplex X, and let $u^1, \ldots, u^{n+1}$ be their respective unit outer normal vectors. For each $j \in \{1, \ldots, n+1\}$, let $\alpha_j \in \mathbb{R}$ be the number such that $F_j \subseteq \mathbf{h}(u^j, \alpha_j)$. In particular, we may express X as

$$X = \bigcap_{j=1}^{n+1} \mathbf{h}^-(u^j, \alpha_j).$$

Our problem clearly resumes to prove that there exist a point $x \in \mathbb{R}^n$ and a number $r > 0$ such that

$$\langle x, u^j \rangle = \alpha_j - r \tag{3.4}$$

for each $j \in \{1, \ldots, n+1\}$. Indeed, if this holds, then x has the same distance r to each of the hyperplanes $\mathbf{h}(u^j, \alpha_j)$, and hence, the feet of the perpendiculars drawn from x to each of these hyperplanes lie in a same ball centered at x. Moreover, since $r > 0$, we get that $x \in \operatorname{int}(\mathbf{h}^-(u^j, \alpha_j))$, and this gives that $x \in \operatorname{int} X$.

Writing in coordinates $x = (x_1, \ldots, x_n)$ and $u^j = (u^j_1, \ldots, u^j_n)$, the equalities (3.4) can be described as the linear system (in the matricial form):

$$\begin{bmatrix} u^1_1 & u^1_2 & \ldots & u^1_n & 1 \\ u^2_1 & u^2_2 & \ldots & u^2_n & 1 \\ \vdots & \vdots & \ldots & \vdots & \vdots \\ u^n_1 & u^n_2 & \ldots & u^n_n & 1 \\ u^{n+1}_1 & u^{n+1}_2 & \ldots & u^{n+1}_n & 1 \end{bmatrix} \cdot \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \\ r \end{bmatrix} = \begin{bmatrix} \alpha_1 \\ \alpha_2 \\ \vdots \\ \alpha_n \\ \alpha_{n+1} \end{bmatrix}.$$

Since $\operatorname{pos}\{u_1, \ldots, u_{n+1}\} = \mathbb{R}^n$, we get that $u_1, \ldots, u_{n+1}$ are affinely independent. We claim that this gives that the rows of the matrix of the system above are linearly independent vectors in $\mathbb{R}^{n+1}$. To verify

that, denote these rows naturally by $(u^j, 1)$ and assume that there exist numbers $\beta_1, \ldots, \beta_{n+1} \in \mathbb{R}$, not all simultaneously zero, such that

$$\beta_1(u^1, 1) + \ldots + \beta_{n+1}(u^{n+1}, 1) = 0_{\mathbb{R}^{n+1}}.$$

Then, we have immediately that

$$\begin{aligned} \beta_1 u^1 + \ldots + \beta_{n+1} u^{n+1} &= 0_{\mathbb{R}^n} \quad \text{and} \\ \beta_1 + \ldots + \beta_{n+1} &= 0, \end{aligned}$$

and this contradicts the fact that the vectors $u_1, \ldots, u_{n+1}$ are affinely independent. Now that we know that the rows of the matrix of the system are linearly independent, we get that there a unique solution exists. Hence, it only remains to prove that $r > 0$. It is immediate to notice that $r \neq 0$, because otherwise we would have that

$$x \in \bigcap_{j=1}^{n+1} \mathbf{h}(u^j, \alpha_j),$$

but this intersection is empty since X is n-dimensional. Assume (by contradiction) that $r < 0$. Then, $\langle x, u^j \rangle > \alpha_j$ for each $j \in \{1, \ldots, n+1\}$ and, in particular $x \notin X$. Fix any point $y \in \text{int}(X)$. Then, $\langle y, u^j \rangle < \alpha_j$ for every j, and the line through x and y intersects the boundary of X at two points. One of them (z, say) is such that y lies between x and z. Hence, we may write

$$x = z + \lambda(y - z)$$

for some $\lambda > 0$. Because z is a boundary point of X, we have that z lies in some facet, and by renumbering the facets if necessary, we may let $z \in F_1$. Then, $\langle z, u^1 \rangle = \alpha_1$, and

$$\langle x, u^1 \rangle = \langle z + \lambda(y - z), u^1 \rangle - \alpha_1 + \lambda \langle y - z, u^1 \rangle = \alpha_1 + \lambda\big(\langle y, u^1 \rangle - \langle z, u^1 \rangle\big).$$

Since $\langle y, u^1 \rangle < \alpha_1 = \langle z, u^1 \rangle$ and $\lambda > 0$, we arrive at

$$\langle x, u^1 \rangle < \alpha_1,$$

and this contradicts the assumption that $\langle x, u^j, \alpha_j \rangle > \alpha_j$ for every j. The proof is complete.

□

Exercise 3.24. *Let $n \geq 1$, and let $P \subseteq \mathbb{R}^n$ be a proper polytope. Show that we may decompose P as the union of n-dimensional simplices $X_1, \ldots, X_m$ such that for any $i \neq j$, we have that the intersection $X_i \cap X_j$ has a dimension smaller than n.*

Solution 3.24. We proceed by induction on the dimension. If $n = 1$, there is nothing to prove, since every proper polytope in $\mathbb{R}^1$ is a 1-simplex. Hence, we let $n > 1$ and assume that the result holds true for dimension $n - 1$. Let $P \subseteq \mathbb{R}^n$ be a proper polytope, and let $F_1, \ldots, F_m$ be its facets. Then, each F_j is an $(n-1)$-dimensional polytope. Fix any point $x \in \text{int}(P)$, and consider the pyramids $P_j := \text{conv}(F_j \cup \{z\})$, $j = \{1, \ldots, m\}$. We first claim that

$$P = P_1 \cup \ldots \cup P_m.$$

Indeed, if $x \in P$, then the half-line starting at z and passing through x intersects the boundary of P at some point (y, say). Because the boundary of P is the union of its facets, we have that $y \in F_j$ for some $j \in \{1, \ldots, m\}$. It follows immediately that $x \in P_j$.

Even more, we claim that any non-empty intersection between distinct P_j's is not full-dimensional. In fact, any intersection $P_i \cap P_j$ for $i \neq j$ is contained in $\operatorname{conv}((F_i \cap F_j) \cup \{z\})$. Since $F_1 \cap F_j$ is a polytope whose dimension is at most $n-2$ (because the intersection of distinct facets is contained in the intersection of their distinct supporting hyperplanes), it follows that the dimension of $(F_1 \cap F_2) \cup \{z\}$ is at most $n-1$.

As a consequence, we get that it suffices to prove that each pyramid P_j can be decomposed as the union of finitely many proper simplices whose intersections are either empty or not full-dimensional. To do that, we use the induction hypothesis. Since F_j is an $(n-1)$-dimensional polytope, we may let $Z_1, \ldots, Z_k$ be a decomposition of F_j by $(n-1)$-dimensional simplices having the desired properties. Then, we let $X_i := \operatorname{conv}(Z_i \cup \{z\})$ for each $i = \{1, \ldots, k\}$. It is clear that each X_i is an n-dimensional simplex and that this gives the desired decomposition.

□

Exercise 3.25. **(a)** *Let $T\colon \mathbb{R}^n \to \mathbb{R}^n$ be a linear isomorphism, and let $u \in \mathbb{S}^{n-1}$, $\alpha \in \mathbb{R}$. Verify that*

$$T(\mathbf{h}(u,\alpha)) = \mathbf{h}\big((T^{-1})^* u, \alpha\big).$$

(b) *Show that if a set $E \subseteq \mathbb{R}^n$ is supported by the hyperplane $\mathbf{h}(u,\alpha)$ at a point x, then TE is supported by $\mathbf{h}\big((T^{-1})^* u, \alpha\big)$ at the point Tx.*

(c) *Let $P \subseteq \mathbb{R}^n$ be a proper polytope whose facets are $F_1, \ldots, F_m$, having corresponding unit outer normal vectors $u_1, \ldots, u_m$. Prove that for any linear isomorphism $T\colon \mathbb{R}^n \to \mathbb{R}^n$, the polytope $T(P)$ has facets $T(F_1), \ldots, T(F_m)$ with corresponding unit outer normal vectors $(T^*)^{-1}u_1/|(T^*)^{-1}u_1|, \ldots, (T^*)^{-1}u_m/|(T^*)^{-1}u_m|$.*

Solution 3.25. **(a)** This is straightforward from

$$\begin{aligned} T(\mathbf{h}(u,\alpha)) &= \{Tx : \langle x, u\rangle = \alpha\} = \{z : \langle T^{-1}z, u\rangle = \alpha\} = \\ &= \{z : \langle z, (T^{-1})^* u\rangle = \alpha\} = \mathbf{h}\big((T^{-1})^* u, \alpha\big). \end{aligned}$$

(b) We know that $\langle u, x\rangle = \alpha$, and we may assume that $\langle u, z\rangle \leq \alpha$ for any $z \in E$. Any point of TE has the form Tz for some $z \in E$. For such a point, we have

$$\langle (T^{-1})^* u, Tz\rangle = \langle u, T^{-1}Tz\rangle = \langle u, z\rangle \leq \alpha,$$

and hence, $TE \subseteq \mathbf{h}^-((T^{-1})^* u, \alpha)$. The same calculation yields that $\langle (T^{-1})^* u, Tx\rangle = \alpha$.

(c) By item **(b)**, we have that a linear isomorphism preserves support relations, and hence, the faces of $T(P)$ are precisely the images of the faces of P under T (recall that all faces of a polytope are exposed and hence characterized in terms of supporting hyperplanes). Since a linear isomorphism also clearly preserves affine dimension, we get that the facets of $T(P)$ are exactly $T(F_1), \ldots, T(F_m)$. The characterization of their unit outer normal vectors comes immediately from item **(a)**.

□

Exercise 3.26. *Let $K, L \in CC^n$. We say that L is* locally embeddable *in K if for every point $x \in \partial K$, there exists a point $y \in L$ and a neighborhood U of y such that*

$$(L \cap U) + x - y \subseteq K.$$

Let $P \subseteq \mathbb{R}^n$ be a polytope. Prove that any positive scalar multiple of P is locally embeddable in P.

Solution 3.26. Let $x \in \partial P$ and $\lambda > 0$. Up to a translation, we may assume that $x = 0_{\mathbb{R}^n}$. Let $F_1, \ldots, F_m$ be the facets of P, with corresponding unit outer normal vectors $u_1, \ldots, u_m$, and assume that $0_{\mathbb{R}^n}$ belongs precisely to $F_1, \ldots, F_k$. Notice that for $r > 0$ small enough, the ball rB^n does not intersect any of the other facets. That is, the ball rB^n lies in the interior of the half-space determined by F_i, which contains P, for each $i = k+1, \ldots, m$. We claim that

$$(\lambda P) \cap rB^n \subseteq P.$$

In fact, if $z \in (\lambda P) \cap rB^n$, then $z/\lambda \in P$, and this gives that

$$\langle z, u_j \rangle = \lambda \left\langle \frac{z}{\lambda}, u_j \right\rangle \leq 0$$

for $j = 1, \ldots, k$. If $z \notin P$, then $\lambda z \notin \lambda P$, and hence, $\lambda z \in \lambda(rB^n \setminus P)$. This is clearly the same as saying $z \in rB^n \setminus P$. Since rB^n lies in the interior of the half-space determined by F_i, which contains P whenever $k+1 \leq i \leq m$, we must have $\langle z, u_j \rangle > 0$ for some $j \in \{1, \ldots, k\}$. This contradiction gives the desired statement.

□

The next exercise elucidates an interesting geometric property relating the zeros of a complex polynomial with the zeros of its derivative. No prior knowledge in complex analysis is needed, just simple differentiation rules.

Exercise 3.27. *Let p be a polynomial in one complex variable. Prove that the zeros of the derivative of p are contained in the convex polygon determined by the zeros of p.*

Solution 3.27. We may write

$$p(z) = \alpha \prod_{j=1}^{m} (z - a_j),$$

where $\alpha \in \mathbb{C}$ and $a_1, \ldots, a_m$ are the roots of p (not necessarily distinct). All common zeros of p and p' are obviously in the convex hull of the zeros of p, and hence, we only need to worry about the zeros of p', which are not zeros of p. We write

$$\frac{p'(z)}{p(z)} = \sum_{j=1}^{m} \frac{1}{z - a_j},$$

and this is the same as

$$\frac{p'(z)}{p(z)} = \sum_{j=1}^{m} \frac{\bar{z} - \bar{a}_j}{|z - a_j|^2}.$$

Thus, if $p'(z) = 0$ and $p(z) \neq 0$, we may write

$$\left(\sum_{j=1}^{m} \frac{1}{|z - a_j|^2} \right) \bar{z} = \sum_{j=1}^{m} \frac{1}{|z - a_j|^2} \bar{a}_j.$$

Taking conjugates in both sides, and writing $1/|z - a_j|^2 = \beta_j$ (which is a positive real number) for each j, we get

$$(\beta_1 + \ldots + \beta_m) z = \beta_1 a_1 + \ldots + \beta_m a_m,$$

and dividing both sides by $\beta_1 + \ldots + \beta_m$, we obtain z as a convex combination of $a_1, \ldots, a_m$.

□

For the next exercise, we use the fact that the graph of vertices and edges of any polytope is connected (for a proof, see [264]).

Exercise 3.28. *Let $P \subseteq \mathbb{R}^n$ be a proper polytope.*

(a) *Prove that the conjugate face of a vertex of P° is a facet of P and that the conjugate face of an edge of P° is an $(n-2)$-dimensional face of P.*

(b) *Let F, G be facets of P. Show that there exists a sequence $F_1, \ldots, F_r$ of facets of P such that $F_1 = F$, $F_r = G$, and $F_i \cap F_{i+1}$ is $(n-2)$-dimensional for any $i = 1, \ldots, r-1$.*

Solution 3.28. (a) This follows from the first claim in the proof of Theorem 3.2.1 in the main book.

(b) Simply take any path of edges between the vertices of P°, which correspond respectively to F and G. The endpoints of each edge correspond to facets of P whose intersection is an $(n-2)$-dimensional face of P.

□

For Further Reading

There are several basic monographs and surveys where the notions and topics mentioned at the beginning of this chapter here are broadly discussed, generalized, and modified. In particular, we mention first Grünbaum's classical book [119] (the second, extended version) and Ziegler's lectures [264] on general polytopes. Further, monographs dedicated to the theory of general convex n-polytopes are due to McMullen and Shephard [195] as well as Brøndsted [71].

Of course, there are also books that, as a whole, are dedicated to *more special aspects* of polytope theory, e.g., to themes like *integer points in polytopes* (see, e.g., Barvinok [38]) or *realization spaces* (cf. Richter-Gebert [211]). Here, we also mention the subfield of (abstract or geometric) *regular polytopes*. Three monographs in this direction are due to Coxeter, McMullen, and Schulte (see [83], [194], and [192], respectively), and the survey [186] collects material on polytope classes obtained by successive weakenings of definitions of regular polytopes. Similarly, parts 2.9.1.1, 2.9.1.2, 2.9.1.3, 2.9.1.4, and 2.9.1.11, as well as subsection 10.6.1 of the main book treat also types of highly symmetric polytopes.

On the other hand, there are various *fundamental books on general convexity* containing essential chapters or sections where in particular polytopes are studied. This can, of course, be seen in the most fundamental monographs of the field, like those of Schneider [228] and Gruber [117], and various further examples (such as Ewald [89], Webster [256], and Barvinok [37]) show, going this way, cross-connections to other fields (like algebraic geometry or linear and convex programming). General polytopes are also discussed in the main book, namely, in Chapter 3 and subsection 12.4.2 there.

Choosing, e.g., the topic *facial structure* of convex polytopes and polyhedral sets, we mention here some essential and interesting *surveys*. In their handbook article [41], Bayer and Lee survey *combinatorial properties of convex polytopes*, for example, shellings, interesting relations to commutative algebra and algebraic geometry, Gale transforms, and further related topics. Kalai [149] presents also nice topics from the combinatorial geometry of polytopes, namely, on the facial structure of mainly simplicial polytopes, e.g., referring to their k-dimensional skeleta. Ziegler [265] summarizes results on f-vectors of 4-polytopes, and the handbook article [139] of Henk, Richter-Gebert, and Ziegler

concentrates on the face lattices of low-dimensional polytopes and polytopes with few vertices. Bayer [40] surveys problems connected with the reconstruction of face lattices from certain informations, e.g., from their skeleta. She exposes a large variety of results and problems in this direction. Particularly, simplicial polytopes, nearly simple polytopes, cubical polytopes, zonotopes, cross-polytopes, and Eulerian posets are investigated, and various interesting problems are posed.

More information on topics treated in our exercises here can be found also in subsection 12.4.2 of our monograph [29].

PLATE 4

Cavalieri's principle

Cylinders with equal volumes

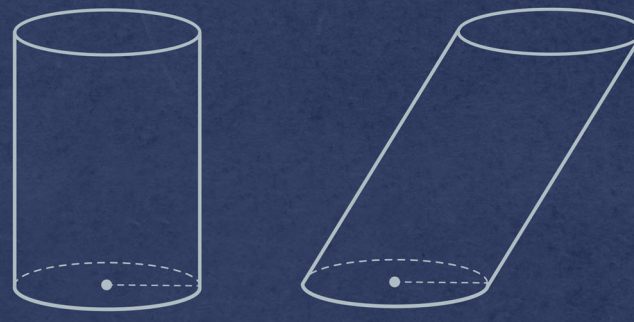
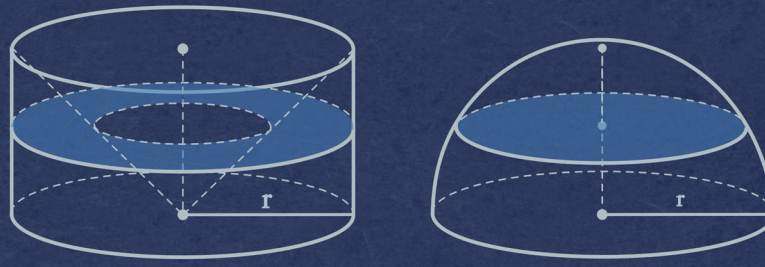


Figure 9

Figure 10

Archimedes



Cavalieri's principle ► (see, e.g., 4.2 and 4.7.1)

Chapter 4
Volume and Area



"Measure what is measurable, and make measurable what is not so."
— Galileo Galilei

The exercises in our Chapter 4 mainly treat the following notions and topics: *volume and surface area of special types of convex bodies* and their *invariance under orthogonal transformations, in- and circumradius* of (special types of) convex bodies, *general representations of volume and surface area*, the *gamma function*, properties of the *centroid* and of *isotropic convex bodies, volumes of sections, Markov's and Chebyshev's inequality*, and *concentration of measures*.

Plate 4:

Cavalieri's principle is well-known and says that if two geometric figures have the same height and the same two cross-sectional areas at each point of that height, then their volumes are equal. Plate 4 shows two examples; the second one is famous since one can find it in Archimedes' work *On the Sphere and Cylinder* (see [20]). In our main book, Cavalieri's principle is discussed in Section 4.2 and in subsection 4.7.1.

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V. Balestro et al., *Convexity from the Geometric Point of View: Exercises and Solutions*, Cornerstones, https://doi.org/10.1007/978-3-031-83937-5_4

Exercises and Solutions

Exercise 4.1. *Verify that the surface area of a convex body is invariant under orthogonal transformations and under translations. That is, show that if $S(\cdot)$ denotes the surface area, then $S(U(K) + x_0) = S(K)$ for any convex body $K \subseteq \mathbb{R}^n$, every $U \in \mathrm{O}(n)$ and all $x_0 \in \mathbb{R}^n$.*

Hint: Use the Minkowski surface area, which is defined in the main book under (4.1.1).

Remark 4.1. Geometrically, this confirms the intuitive idea that the surface area does not change under "rigid motions".

Remark 4.2. From now on, $S(K)$ always denotes the total surface area of a convex body K.

Solution 4.1. This follows immediately from the fact that $\mathrm{vol}_n(\cdot)$ is invariant under translations and $\mathrm{O}(n)$-invariant.

□

Exercise 4.2. *Recall that the* standard orthogonal n-simplex *of $\mathbb{R}^n$ is the set*

$$\Delta_n := \mathrm{conv}\{0_{\mathbb{R}^n}, e_1, \ldots, e_n\}.$$

Prove that the surface area of Δ_n is

$$S(\Delta_n) = \frac{n + \sqrt{n}}{(n-1)!}.$$

Solution 4.2. Of course, the facets of Δ_n are the sets $F_j := \mathrm{conv}\{0_{\mathbb{R}^n}, e_1, \ldots, e_{j-1}, e_{j+1}, \ldots, e_n\}$ for $1 \le j \le n$ and $F := \mathrm{conv}\{e_1, \ldots, e_n\}$. Each facet F_j is naturally identified with the standard orthogonal $(n-1)$-simplex, and hence,

$$\mathrm{vol}_{n-1}(F_j) = \mathrm{vol}_{n-1}(\Delta_{n-1}) = \frac{1}{(n-1)!}.$$

To calculate the area of the remaining facet F, we see Δ_n as a pyramid with apex at $0_{\mathbb{R}^n}$. Since $e_1, \ldots, e_n$ are affinely independent, their affine hull $\mathrm{aff}\{e_1, \ldots, e_n\}$ is a hyperplane, which we call $\mathbf{h}$. Let

$$u := \frac{e_1 + \ldots + e_n}{\sqrt{n}}.$$

Then, $u \in \mathbb{S}^{n-1}$ and for any affine combination $\lambda_1 e_1 + \ldots + \lambda_n e_n$, we have

$$\langle \lambda_1 e_1 + \ldots + \lambda_n e_n, u \rangle = \frac{1}{\sqrt{n}}(\lambda_1 + \ldots + \lambda_n) = \frac{1}{\sqrt{n}}.$$

Thus, $\mathbf{h} = \mathbf{h}(u, 1/\sqrt{n})$, and hence, the distance from $\mathbf{h}$ to $0_{\mathbb{R}^n}$ is $1/\sqrt{n}$. Therefore,

$$\frac{1}{n!} = \mathrm{vol}_n(\Delta_n) = \frac{\mathrm{dist}(0_{\mathbb{R}^n}, \mathbf{h})}{n}\mathrm{vol}_{n-1}(F) = \frac{1}{n\sqrt{n}}\mathrm{vol}_{n-1}(F).$$

It follows that

$$\mathrm{vol}_{n-1}(F) = \frac{\sqrt{n}}{(n-1)!},$$

and consequently,

$$S(\Delta_n) = \mathrm{vol}_{n-1}(F) + \sum_{j=1}^{n} \mathrm{vol}_{n-1}(F_j) = \frac{\sqrt{n}}{(n-1)!} + \frac{n}{(n-1)!} = \frac{n+\sqrt{n}}{(n-1)!},$$

as we wanted to prove.

□

Exercise 4.3. *The* regular n-dimensional cross-polytope *is the polytope of* $\mathbb{R}^n$ *given as*

$$P_\Delta^n := \mathrm{conv}\big\{ \pm e_1, \ldots, \pm e_n \big\}.$$

(a) *Show that* P_Δ^n *is the unit ball of the* ℓ_1-norm *of* $\mathbb{R}^n$*, which is defined as*

$$||(x_1, \ldots, x_n)||_1 := |x_1| + \ldots + |x_n|$$

for each $(x_1, \ldots, x_n) \in \mathbb{R}^n$.

(b) *Prove that the volume of* P_Δ^n *is given by*

$$\mathrm{vol}_n(P_\Delta^n) = \frac{2^n}{n!}.$$

(c) *Show that the surface area of* P_Δ^n *is*

$$S(P_\Delta^n) = \frac{2^n\sqrt{n}}{(n-1)!}.$$

Solution 4.3. (a) Denote the unit ball of the ℓ_1-norm of $\mathbb{R}^n$ as B_1^n. Since it is clear that $\pm e_1, \ldots, \pm e_n \in B_1^n$, from convexity, we get the inclusion

$$P_\Delta^n = \mathrm{conv}\{\pm e_1, \ldots, \pm e_n\} \subseteq B_1^n.$$

Now, let $(x_1, \ldots, x_n) = x_1 e_1 + \ldots + x_n e_n \in B_1^n$. Then, $|x_1| + \ldots + |x_n| \leq 1$. Denote the sign of x_j by $s(j)$ (hence, $s(j) = 1$ if x_j is positive, $s(j) = -1$ if x_j is negative and $s(j) = 0$ if x_j is 0). Then, writing

$$(x_1, \ldots, x_n) = |x_1|s(1)e_1 + \ldots + |x_n|s(n)e_n + \big(1 - |x_1| - \ldots - |x_n|\big)0_{\mathbb{R}^n},$$

we get that

$$(x_1, \ldots, x_n) \in \mathrm{conv}\{0_{\mathbb{R}^n}, s(1)e_1, \ldots, s(n)e_n\} \subseteq P_\Delta^n,$$

where the last inclusion holds because the P_Δ^n is convex and contains the points $0_{\mathbb{R}^n}, s(1)e_1, \ldots, s(n)e_n$. This shows that $B_1^n \subseteq P_\Delta^n$.

(b) It is clear that the intersection of P_Δ^n with the first orthant of $\mathbb{R}^n$ is the standard orthogonal n-simplex Δ_n. By symmetry, it follows that the intersection of P_Δ^n with each orthant of $\mathbb{R}^n$ is a "copy" of Δ_n. Hence, since $\mathbb{R}^n$ has 2^n orthants, we obtain

$$\mathrm{vol}_n(P_\Delta^n) = 2^n \mathrm{vol}_n(\Delta_n) = \frac{2^n}{n!}.$$

(c) It is readily seen that the facets of P_Δ^n are the sets of the form

$$\operatorname{conv}\{\square e_1, \ldots, \square e_n\},$$

where each box is filled with a sign + or −. Then P_Δ^n has 2^n facets, and it is also clear that each of these facets has the same area as $F := \operatorname{conv}\{e_1, \ldots, e_n\}$. In Exercise 4.2, we proved that the area of F is $\sqrt{n}/(n-1)!$. Hence,

$$S(P_\Delta^n) = 2^n \frac{\sqrt{n}}{(n-1)!} = \frac{2^n\sqrt{n}}{(n-1)!},$$

and the proof is complete

□

Exercise 4.4. *Prove that the circumradius and the inradius of the regular n-dimensional cross-polytope P_Δ^n are 1 and $1/\sqrt{n}$, respectively.*

Solution 4.4. We start with the circumradius. It is clear that $P_\Delta^n \subseteq B^n$, because P_Δ^n is a convex hull of points of B^n. Thus, $r_{\text{out}}(P_\Delta^n) \leq 1$. If the inequality is strict, then from item **(a)** of Exercise 2.19 (which states that the origin is the circumball of a body that is centered at the origin), we get that P_Δ^n is contained in some ball centered at the origin whose radius is strictly smaller than 1. This is a contradiction because e_1 is a point of P_Δ^n that is not contained in any of these balls. It follows that $r_{\text{out}}(P_\Delta^n) = 1$.

For the inradius, we use the equality $P_\Delta^n = \{(x_1, \ldots, x_n) : |x_1| + \ldots + |x_n| \leq 1\}$ obtained in item **(a)** of Exercise 4.3. We first prove that

$$\frac{1}{\sqrt{n}} B^n \subseteq P_\Delta^n.$$

Indeed, let $x = (x_1, \ldots, x_n) \in \frac{1}{\sqrt{n}} B^n$ and denote the sign of x_j by $s(j)$. It is immediate that $|(s(1), \ldots, s(n))| \leq \sqrt{n}$, and from the Cauchy-Schwarz inequality, we get

$$|x_1| + \ldots + |x_n| = \langle (x_1, \ldots, x_n), (s(1), \ldots, s(n)) \rangle \leq |(x_1, \ldots, x_n)||(s(1), \ldots, s(n)| \leq \frac{1}{\sqrt{n}}\sqrt{n} = 1,$$

from where we have $(x_1, \ldots, x_n) \in P_\Delta^n$. This shows the desired inclusion, and as a consequence, we have that

$$r_{\text{in}}(P_\Delta^n) \geq \frac{1}{\sqrt{n}}.$$

As another consequence of Exercise 2.19, since P_Δ^n is symmetric with respect to the origin, then we have that the inradius of P_Δ^n is attained for some ball that is centered at the origin. If the inequality above is strict, then this means that there is a number $r > 1/\sqrt{n}$ such that $rB^n \subseteq P_\Delta^n$. Consider the vector

$$v := \frac{r}{\sqrt{n}}(e_1 + \ldots + e_n) = \left(\frac{r}{\sqrt{n}}, \ldots, \frac{r}{\sqrt{n}}\right).$$

It is clear that $|v| = r$, from where $v \in rB^n$. Hence, we must have $v \in P_\Delta^n$, and this means that the sum of the absolute values of the coordinates of v must be smaller than or equal to 1. However, computing this sum and remembering that we are assuming that $r > 1/\sqrt{n}$, we get

$$\left|\frac{r}{\sqrt{n}}\right| + \ldots + \left|\frac{r}{\sqrt{n}}\right| = \frac{nr}{\sqrt{n}} > \frac{n}{\sqrt{n}}\frac{1}{\sqrt{n}} = 1,$$

which is a contradiction. Therefore, $r_{\text{in}}(P_\Delta^n) = 1/\sqrt{n}$ and the proof is complete.

□

Exercise 4.5. **(a)** *Show that the polar body of the regular cross-polytope P_Δ^n is the* cube

$$Q^n := [-1, 1]^n = \left\{(x_1, \ldots, x_n) : -1 \leq x_j \leq 1 \ \ \forall 1 \leq j \leq n\right\}.$$

(b) *Verify that Q^n is the unit ball of the ℓ_∞-*norm *of $\mathbb{R}^n$, which is defined as*

$$||(x_1, \ldots, x_n)||_\infty := \max\left\{|x_j| : 1 \leq j \leq n\right\}.$$

(c) *Prove that the surface area of Q^n is*

$$S(Q^n) = n2^n.$$

Solution 4.5. **(a)** Due to Exercise 4.3, a point $(v_1, \ldots, v_n)$ of P_Δ^n is characterized by $|v_1| + \ldots + |v_n| \leq 1$. Noticing that a point $(x_1, \ldots, x_n) \in Q^n$ is characterized by $|x_j| \leq 1$ for every $j = 1, \ldots, n$, we write

$$\begin{aligned}\langle (x_1, \ldots, x_n), (v_1, \ldots, v_n)\rangle = x_1v_1 + \ldots + x_nv_n \leq |x_1||v_1| + \ldots + |x_n||v_n| \leq |v_1| + \ldots \\ + |v_n| \leq 1.\end{aligned}$$

This gives the inclusion

$$Q^n \subseteq (P_\Delta^n)^\circ.$$

For the other inclusion, we proceed by contradiction. Suppose that $(x_1, \ldots, x_n) \in (P_\Delta^n)^\circ$ is such that $(x_1, \ldots, x_n) \notin Q^n$. Thus, there is $k \in \{1, \ldots, n\}$ such that $|x_k| > 1$. Assume first that $x_k > 1$. Then,

$$\langle (x_1, \ldots, x_n), e_k\rangle = x_k > 1,$$

and this contradicts the fact that $(x_1, \ldots, x_n) \in (P_\Delta^n)^\circ$. If $x_k < 1$, we simply repeat the argument replacing e_k by $-e_k$.

(b) Indeed, it is immediate from the definition of Q^n that $||(x_1, \ldots, x_n)||_\infty \leq 1$ if and only if $(x_1, \ldots, x_n) \in Q^n$.

(c) The facets of Q^n are of the form

$$\left\{(x_1, \ldots, x_{k-1}, \pm 1, x_k, \ldots, x_n) : |x_j| \leq 1 \ \ \forall\, j = 1, \ldots, k-1, k+1, \ldots, n\right\}.$$

Hence, each facet of Q^n is a copy of Q^{n-1}. Also, it is clear that there are $2n$ facets (since we only must have 1 or -1 in one of the n entries). Thus,

$$S(Q^n) = 2n\text{vol}_{n-1}(Q_{n-1}) = 2n\text{vol}_{n-1}\left([-1, 1]^{n-1}\right) = 2n(2^{n-1}) = n2^n.$$

□

For the next exercise, recall that $\Gamma\colon (0,+\infty)\to\mathbb{R}$ is the *Gamma function* defined in Proposition 4.3.1 in the main book. Remember also that κ_n denotes the volume of the unit ball of $\mathbb{R}^n$. From the mentioned proposition, we have an expression for κ_n in terms of the Gamma function.

Exercise 4.6. **(a)** *Verify that*

$$\Gamma(1/2)=\sqrt{\pi}.$$

(b) *Show that if n is even, then*

$$\kappa_n=\frac{\pi^{n/2}}{(n/2)!},$$

and that if n is odd, then writing $n=2k+1$, *we have*

$$\kappa_n=\frac{2k!(4\pi)^k}{(2k+1)!}.$$

Derive similar expressions for ω_n.

Solution 4.6. **(a)** We use the first identity in Propositon 4.3.1 of the main book. Notice that $\omega_1=\mathcal{H}^0(\mathbb{S}^0)$. Since $\mathcal{H}^0$ is the counting measure, and $\mathbb{S}^0$ is $\{-1,1\}\subseteq\mathbb{R}$, we get that

$$2=\frac{2\sqrt{\pi}}{\Gamma\left(\frac{1}{2}\right)},$$

and this gives the desired equality.

(b) If n is even, then writing $n=2k$, we have that

$$\Gamma\left(\frac{n}{2}+1\right)=\Gamma(k+1)=k!=\left(\frac{n}{2}\right)!,$$

and hence the first equality follows from Proposition 4.3.1 in the main book.

For the second one, it is easier if we proceed by induction on n (odd). The equality clearly holds for $n=1$, and hence, we assume that it also holds for an arbitrary odd $n>1$ and check what happens for $n+2$. Let $n=2k+1$. Thus, $2(k+1)+1=n+2$.

$$\kappa_{n+2}=\frac{2\pi^{\frac{n+2}{2}}}{\Gamma\left(\frac{n+2}{2}+1\right)}=\frac{2\pi^{\frac{n}{2}+1}}{\Gamma\left(\frac{n}{2}+1+1\right)}=\frac{\pi}{\frac{n}{2}+1}\frac{2\pi^{\frac{n}{2}}}{\Gamma\left(\frac{n}{2}+1\right)}=\frac{2\pi}{n+2}\kappa_n=$$

$$=\frac{2\pi}{2(k+1)+1}\frac{2k!(4\pi)^k}{(2k+1)!}=\frac{2\pi 2(k+1)2k!(4\pi)^k}{(2(k+1)+1)!}=\frac{2(k+1)!4\pi(4\pi)^k}{(2(k+1)+1)!}=\frac{2(k+1)!(4\pi)^{k+1}}{(2(k+1)+1)!},$$

and then we have the desired equality. The similar expressions for ω_n follow directly from the equality $\omega_n=n\kappa_n$.

□

Exercise 4.7. **(a)** *Show that*

$$\kappa_n=\frac{2\pi}{n}\kappa_{n-2}$$

for every $n\geq 2$.

(b) *Prove that*

$$\kappa_n \kappa_{n-1} = \frac{2^n \pi^{n-1}}{n!}$$

for each $n \geq 1$.

Solution 4.7. (a) Using the expression for κ_n given in Proposition 4.3.1 from the main book and recalling that $\Gamma(x+1) = x\Gamma(x)$ for any $x > 0$, we calculate

$$\kappa_n = \frac{\pi^{n/2}}{\Gamma\left(\frac{n}{2}+1\right)} = \frac{\pi^{n/2}}{\frac{n}{2}\Gamma\left(\frac{n}{2}\right)} = \frac{2\pi}{n}\frac{\pi^{(n-2)/2}}{\Gamma\left(\frac{n-2}{2}+1\right)} = \frac{2\pi}{n}\kappa_{n-2},$$

and this is what we wanted to show.

(b) First, observe that we may write

$$\kappa_n \kappa_{n-1} = \frac{\pi^{(2n-1)/2}}{\Gamma\left(\frac{n}{2}+1\right)\Gamma\left(\frac{n+1}{2}\right)}.$$

Let

$$\alpha(n) = \Gamma\left(\frac{n}{2}+1\right)\Gamma\left(\frac{n+1}{2}\right).$$

We claim that

$$\alpha(n) = \frac{n!\pi^{1/2}}{2^n}.$$

Assume that equality holds for some $n \in \mathbb{N}$. Then,

$$\begin{aligned}\alpha(n+2) = \Gamma\left(\frac{n+2}{2}+1\right)\Gamma\left(\frac{n+3}{2}\right) = \frac{n+2}{2}\Gamma\left(\frac{n}{2}+1\right)\frac{n+1}{2}\Gamma\left(\frac{n+1}{2}\right) = \\ = \frac{(n+2)(n+1)}{2^2}\alpha(n) = \frac{(n+2)!\pi^{1/2}}{2^{n+2}}.\end{aligned} \tag{4.1}$$

Now, notice that

$$\alpha(1) = \Gamma\left(\frac{1}{2}+1\right)\Gamma(1) = \frac{1}{2}\Gamma\left(\frac{1}{2}\right) = \frac{\sqrt{\pi}}{2},$$

and

$$\alpha(2) = \Gamma(2)\Gamma\left(1+\frac{1}{2}\right) = \frac{1}{2}\Gamma\left(\frac{1}{2}\right) = \frac{\sqrt{\pi}}{2} = \frac{2!\pi^{1/2}}{2^2}.$$

Therefore, by induction in the odd and even cases, we get the desired statement. Finally,

$$\kappa_n \kappa_{n-1} = \frac{\pi^{(2n-1)/2}}{\alpha(n)} = \frac{2^n \pi^{n-1}}{n!},$$

and the proof is complete.

□

Exercise 4.8. *Recalling that $\omega_n = n\kappa_n$, prove that $\kappa_n \to 0$ and $\omega_n \to 0$ as $n \to +\infty$.*

Solution 4.8. This follows immediately from the expressions derived in Exercise 4.6.

□

Exercise 4.9. *Let $u \in \mathbb{S}^{n-1}$ and let $E \subseteq u^\perp$ be a measurable set. Show that for any vector $v \notin u^\perp$, we have*

$$\mathrm{vol}_n\big(E + \mathrm{seg}[0_{\mathbb{R}^n}, v]\big) = |\langle u, v\rangle| \mathrm{vol}_{n-1}(E).$$

Remark 4.3. The objective of this exercise is to give a formal proof of the fact that the volume of the cylinder $E + \mathrm{seg}[0_{\mathbb{R}^n}, v]$ equals the "area of the basis multiplied by the height."

Solution 4.9. It is clear that the intersection of the cylinder $E + \mathrm{seg}[0_{\mathbb{R}^n}, v]$ with $u^\perp + tu$ is non-empty if and only if $0 \le t \le |\langle u, v\rangle|$. In this case, this intersection is just $tu + E$, that is, a translate of E. Hence, we may use Cavalieri's principle to calculate

$$\mathrm{vol}_n\big(E + \mathrm{seg}[0_{\mathbb{R}^n}, v]\big) = \int_0^{|\langle u,v\rangle|} \mathrm{vol}_{n-1}\big(E \cap \{u^\perp + tu\}\big)\, dt = \int_0^{|\langle u,v\rangle|} \mathrm{vol}_{n-1}(E)\, dt = \\ = |\langle u, v\rangle| \mathrm{vol}_{n-1}(E),$$

and this is what we wanted to prove.

□

The *centroid* of a convex body $K \subseteq \mathbb{R}^n$ is the point $c(K) \in \mathbb{R}^n$ defined as

$$c(K) := \frac{1}{\mathrm{vol}_n(K)} \int_K x\, dx,$$

where dx stands for integration with respect to the usual Lebesgue measure. The integration is taken in each coordinate, in the sense that

$$\int_K x\, dx = \left(\int_K x_1\, dx, \int_K x_2\, dx, \ldots, \int_K x_n\, dx \right).$$

Exercise 4.10. **(a)** *Let $K \subseteq \mathbb{R}^n$ be a convex body whose centroid is x_0. For any $z \in \mathbb{R}^n$, prove that the centroid of $z + K$ is $z + x_0$.*

(b) *Assume that $K \subseteq \mathbb{R}^n$ is a convex body whose centroid is $x_0 \in \mathbb{R}^n$. Show that $x_0 \in \mathrm{int}(K)$.*

(c) *Show that $c(TK) = T(c(K))$ for any $K \in \mathcal{K}^n$ and every $T \in \mathrm{GL}(n)$.*

Solution 4.10. **(a)** We just compute

$$c(z + K) = \frac{1}{\mathrm{vol}_n(z + K)} \int z + Ky\, dy = \frac{1}{\mathrm{vol}_n(K)} \int_K z + x\, dx = z + \frac{1}{\mathrm{vol}_n(K)} \int_K x\, dx = \\ = z + x_0.$$

(b) From item **(a)**, it clearly suffices to consider the case where $x_0 = 0_{\mathbb{R}^n}$. If $0_{\mathbb{R}^n}$ is not an interior point of K, then either $0_{\mathbb{R}^n} \in \partial K$ or $0_{\mathbb{R}^n} \notin K$. In both cases, one can find a vector $u \in \mathbb{S}^{n-1}$ such that $K \subseteq \mathbf{h}^+(u, 0)$. It follows that $\langle z, u\rangle \ge 0$ for any $z \in K$.

For simplicity, we write $\mathbf{h} = \mathbf{h}(u, 0)$. Of course, we may choose a ball $x_0 + rB^n \in \mathrm{int}(K) \cap \mathrm{int}(\mathbf{h}^+)$. From compactness and continuity, the function $v \mapsto \langle u, v\rangle$ attains a minimum on $x_0 + rB^n$, and since

this ball lies in the interior of $\mathbf{h}^+$, we get that this minimum has to be positive ($c > 0$, say). Thus, since the centroid of K is $0_{\mathbb{R}^n}$, we write

$$0 = \left\langle u, \int_K x\, dx \right\rangle = \int_K \langle u, x\rangle\, dx \geq \int_{x_0+rB^n} \langle u, x\rangle\, dx \geq c \int_{x_0+rB^n} dx = cr^n \mathrm{vol}_n(B^n) > 0,$$

a contradiction. It follows that the origin is an interior point of K.

(c) We just need to do a change of variables in the integration ($y = T^{-1}x$) and use the linearity of the integral:

$$c(TK) = \frac{1}{\mathrm{vol}_n(TK)} \int_{TK} x\, dx = \frac{1}{|\det(T)|\mathrm{vol}_n(K)} \int_K Ty|\det(T)|\, dy =$$
$$= \frac{1}{\mathrm{vol}_n(K)} \int_K Ty\, dy = T\left(\frac{1}{\mathrm{vol}_n(K)} \int_K y\, dy\right) = T(c(K)).$$

□

Exercise 4.11. *Let $K \subseteq \mathbb{R}^n$ be a centrally symmetric convex body with $K - x_0 = -(K - x_0)$. Prove that x_0 is the centroid of K.*

Solution 4.11. Since $K - x_0 = -K + x_0$, we get the equality

$$\frac{1}{\mathrm{vol}_n(K - x_0)} \int_{K-x_0} x\, dx = \frac{1}{\mathrm{vol}_n(-K + x_0)} \int_{-K+x_0} x\, dx,$$

and using standard change of variables, we arrive at

$$\frac{1}{\mathrm{vol}_n(K)} \int_K y - x_0\, dy = \frac{1}{\mathrm{vol}_n(K)} \int_K -y + x_0\, dy,$$

and this leads to

$$c(K) = \frac{1}{\mathrm{vol}_n(K)} \int_K y\, dy = \frac{1}{\mathrm{vol}_n(K)} \int_K x_0\, dy = x_0.$$

The proof is complete.

□

Exercise 4.12. *Let $c\colon \mathcal{K}^n \to \mathbb{R}^n$ be the map that associates each convex body $K \subseteq \mathbb{R}^n$ with its centroid $c(K)$. Prove that this map is continuous in the Hausdorff metric.*

Hint. Use the dominated convergence theorem (Theorem A.2.4 from the main book).

Solution 4.12. Let (K_m) be a sequence in $\mathcal{K}^n$ converging in the Hausdorff metric to a body $K \in \mathcal{K}^n$. Of course, we may write

$$c(K_m) = \frac{1}{\mathrm{vol}_n(K_m)} \int_{\mathbb{R}^n} 1_{K_m}(x) x\, dx.$$

We have that $\mathrm{vol}_n(K_m) \to \mathrm{vol}_n(K)$ as $m \to \infty$. Moreover, it is clear that 1_{K_m} converges pointwise to 1_K as $m \to +\infty$. In fact, any point of K belongs to K_m for m sufficiently large, and any point outside K is outside K_m for M large enough. Since this convergence is dominated (by $f = 1$, say), we only need to use Theorem A.2.4 in the main book.

□

Exercise 4.13. *Let $K \in \mathcal{K}^n$. Prove that the centroid $c(K)$ of K is the unique point $p \in \mathbb{R}^n$ with the property that*

$$f(p) = \frac{1}{\mathrm{vol}_n(K)} \int_K f(x)\, dx \tag{4.2}$$

for every affine function $f\colon \mathbb{R}^n \to \mathbb{R}$. In other words, the mean of any affine function over K equals the value of f at the centroid $c(K)$ of K, and $c(K)$ is the unique point of $\mathbb{R}^n$ having this property.

Remark 4.4. Actually, assuming that (4.2) holds for all **linear** functions $f\colon \mathbb{R}^n \to \mathbb{R}$ is sufficient to characterize p as the centroid of K.

Solution 4.13. Any affine function $f\colon \mathbb{R}^n \to \mathbb{R}$ can be written as

$$f(x) = \langle x, v_0 \rangle + b$$

for some fixed $v_0 \in \mathbb{R}^n$ and $b \in \mathbb{R}$. We have

$$f(c(K)) = \langle c(K), v_0 \rangle + b = \left\langle \frac{1}{\mathrm{vol}_n(K)} \int_K x\, dx, v_0 \right\rangle + b = \frac{1}{\mathrm{vol}_n(K)} \int_K \langle x, v_0 \rangle\, dx + b =$$
$$= \frac{1}{\mathrm{vol}_n(K)} \int_K \langle x, v_0 \rangle + b\, dx = \frac{1}{\mathrm{vol}_n(K)} \int_K f(x)\, dx.$$

Conversely, if $p \in \mathbb{R}^n$ is such that

$$f(p) = \frac{1}{\mathrm{vol}_n(K)} \int_K f(x)\, dx$$

for every affine function f, then

$$\langle p, v_0 \rangle + b = \frac{1}{\mathrm{vol}_n(K)} \int_K \langle x, v_0 \rangle + b\, dx$$

for any $v_0 \in \mathbb{R}^n$ and $b \in \mathbb{R}$. The equality above is obviously equivalent to

$$\langle p, v_0 \rangle = \left\langle \frac{1}{\mathrm{vol}_n(K)} \int_K x\, dx, v_0 \right\rangle.$$

It is clear that this holds for every $v_0 \in \mathbb{R}^n$ if and only if

$$p = \frac{1}{\mathrm{vol}_n(K)} \int_K x\, dx = c(K).$$

□

Exercise 4.14. *Parametrize $\mathbb{S}^1$ by $u(\theta) := (\cos\theta, \sin\theta)$, $\theta \in [0, 2\pi)$, and let $K \subseteq \mathbb{R}^2$ be a convex body having the origin as an interior point.*

(a) *Assume that K is strictly convex. Let $h_K(\theta) := h_K(u(\theta))$ be the restriction of h_K to $\mathbb{S}^1$. Prove that*

$$x(\theta) := h_K(\theta)(\cos\theta, \sin\theta) + h_K'(\theta)(-\sin\theta, \cos\theta), \quad \theta \in [0, 2\pi),$$

is a parametrization of ∂K. Check that for each $\theta \in [0, 2\pi)$, it holds that $(\cos\theta, \sin\theta)$ is the unit outer normal vector of K at the point $x(\theta)$.

(b) *Show that if h_K is C^2, then the area of K is given by*

$$\mathrm{vol}_2(K) := \frac{1}{2}\int_0^{2\pi} h_K(\theta)^2 - h_K'(\theta)^2\, d\theta,$$

where $d\theta$ stands for the usual Lebesgue measure in the interval $[0, 2\pi)$.

Hint: For item **(a)**, write the gradient ∇h_K in polar coordinates (ρ, θ) and use Proposition 2.3.9 of the main book. For item **(b)**, use the classical Green formula for the area of a parametrized curve.

Solution 4.14. (a) Write in polar coordinates $x_1 = \rho\cos\theta$ and $x_2 = \rho\sin\theta$, for $\rho \in [0, +\infty)$ and $\theta \in [0, 2\pi)$. Since $h_K(x_1, x_2) = h_K(\rho\cos\theta, \rho\sin\theta)$, we derive the gradient of h_K in polar coordinates by the following expressions given via the chain rule:

$$\frac{\partial h_K}{\partial \theta} = -\rho\sin\theta\frac{\partial h_K}{\partial x_1} + \rho\cos\theta\frac{\partial h_K}{\partial x_2} \quad \text{and}$$
$$\frac{\partial h_K}{\partial \rho} = \cos\theta\frac{\partial h_K}{\partial x_1} + \sin\theta\frac{\partial h_K}{\partial x_2}.$$

For points of $\mathbb{S}^1$, we have that $\rho = 1$. Thus, in this case, we may write

$$\frac{\partial h_K}{\partial x_1} = -\sin\theta\frac{\partial h_K}{\partial \theta} + \cos\theta\frac{\partial h_K}{\partial \rho} \quad \text{and}$$
$$\frac{\partial h_K}{\partial x_2} = \cos\theta\frac{\partial h_K}{\partial \theta} + \sin\theta\frac{\partial h_K}{\partial \rho}.$$

Now, we compute the derivative of h_K with respect to ρ (at a point of $\mathbb{S}^1$) using the positive homogeneity:

$$\frac{\partial h_K}{\partial \rho}(\cos\theta, \sin\theta) = \lim_{t\to 0}\frac{h_K((1+t)(\cos\theta, \sin\theta)) - h_K(\cos\theta, \sin\theta)}{t} =$$
$$= \lim_{t\to 0}\frac{(1+t)h_K(\cos\theta, \sin\theta) - h_K(\cos\theta, \sin\theta)}{t} = h_K(\cos\theta, \sin\theta).$$

For simplicity, we write $h_K'(\theta)$ to represent the usual derivative of the function $h_K(\theta) := h_K(u(\theta)) = h_K(\cos\theta, \sin\theta)$. From the equalities above, for any $u = (\cos\theta, \sin\theta) \in \mathbb{S}^1$, we have

$$x(\theta) := \nabla h_K(\cos\theta, \sin\theta) = h_K(\theta)(\cos\theta, \sin\theta) + h_K'(\theta)(-\sin\theta, \cos\theta)$$

for each $\theta \in [0, 2\pi)$. By Proposition 2.3.9 in the main book, the point $x(\theta)$ is the unique point of the convex curve ∂K where the unit outer normal vector is $(\cos\theta, \sin\theta)$. This concludes the proof.

(b) Since h_K is C^2, we may differentiate $x(\theta)$ to get

$$x'(\theta) = \big[h_K(\theta) + h_K''(\theta)\big](-\sin\theta, \cos\theta).$$

By the classical Green formula, we derive

$$\mathrm{vol}_2(K) = \frac{1}{2}\int_0^{2\pi}\det[x(\theta), x'(\theta)]\, d\theta = \frac{1}{2}\int_0^{2\pi} h_K(\theta)\big[h_K(\theta) + h_K''(\theta)\big]\, d\theta =$$
$$= \frac{1}{2}\int_0^{2\pi} h_K(\theta)^2 - h_K'(\theta)^2\, d\theta,$$

where the last equality follows from integration by parts (applied to rewrite the integral of the term $h_K h_K''$).

□

The objective of our next exercise is the so-called Brunn concavity principle, which was used to give the first proof of the Brunn-Minkowski inequality (see Section 5.1 and Exercise 5.17). As the main ingredient, the reader may use the fact that a modification in the proof of Theorem 2.7.1 of the main book (with the help of Exercise 2.59) leads to the following stronger result.

Theorem 4.1 (Steiner-Schwarz)**.** *Let $F \subseteq \mathbb{R}^n$ be a k-dimensional vector subspace, and let $K \subseteq \mathbb{R}^n$ be a convex body. There exists a sequence of successive Steiner symmetrizations of K in directions $u \in F$, which converge to a body $\tilde{K}$ with the property that for any $x \in F^\perp$, $\tilde{K} \cap (x + F)$ is a k-dimensional ball centered at x, and*

$$\mathrm{vol}_k(\tilde{K} \cap (x + F)) = \mathrm{vol}_k(K \cap (x + F)).$$

Exercise 4.15 (Brunn Concavity Principle)**.** *Let $F \subseteq \mathbb{R}^n$ be a k-dimensional vector subspace, and let $K \subseteq \mathbb{R}^n$ be a convex body. Prove that the function $f : F^\perp \to \mathbb{R}$ defined as*

$$f(x) = \mathrm{vol}_k(K \cap (x + F))^{1/k}$$

is concave on its support.

Solution 4.15. Let $\tilde{K}$ be as in Theorem 4.1. Since $\mathrm{vol}_k(K \cap (x + F)) = \mathrm{vol}_k(\tilde{K} \cap (x + F))$ for every $x \in F^\perp$, it suffices to show that the function

$$\tilde{f}(x) = \mathrm{vol}_k(\tilde{K} \cap (x + F))^{1/k}$$

defined for $x \in F^\perp$ is concave on its support.

For each x in the support of $\tilde{f}$ (denoted by $\mathrm{spt}(\tilde{f})$), the intersection $\tilde{K} \cap (x + F)$ is a nondegenerate k-dimensional ball, whose radius we denote by $r(x)$. In this case,

$$\tilde{f}(x) = \mathrm{vol}_k(\tilde{K} \cap (x + F))^{1/k} = (r(x)^k \kappa_k)^{1/k} = r(x)\kappa_k^{1/k}.$$

Hence, we have to prove that the function $r : \mathrm{spt}(\tilde{f}) \to (0, +\infty)$ is concave. Let $x, z \in \mathrm{spt}(\tilde{f})$, and let $\lambda \in (0, 1)$. Pick any unit vector $v \in F$ and, for the sake of simplicity, write $x_\lambda = (1 - \lambda)x + \lambda z$. We have that $x + r(x)v \in \tilde{K}$ and $z + r(z)v \in \tilde{K}$. Thus,

$$(1 - \lambda)(x + r(x)v) + \lambda(z + r(z)v) \in \tilde{K}.$$

Therefore, we get that

$$x_\lambda + \big[(1 - \lambda)r(x) + \lambda r(y)\big]v \in \tilde{K}.$$

Since this is clearly also a point of $x_\lambda + F$, we obtain that

$$r(x_\lambda) \geq (1 - \lambda)r(x) + \lambda r(y),$$

and hence, r is concave on the support of $\tilde{f}$. Notice that this argument also guarantees that $x_\lambda \in \mathrm{spt}(\tilde{f})$. The proof is complete.

□

Exercise 4.16. *Let $K \subseteq \mathbb{R}^n$ be a convex body symmetric with respect to the origin.*

(a) *Show that*

$$\max\left\{\mathrm{vol}_{n-1}(K \cap (u^\perp + tu)) : t \in \mathbb{R}\right\} = \mathrm{vol}_{n-1}(K \cap u^\perp)$$

for each $u \in \mathbb{S}^{n-1}$. In other words, the section of K parallel to $u^\perp$ having maximal area is the one that passes through the origin.

(b) *Verify that the inequality*

$$h_K(u) \geq \frac{\mathrm{vol}_n(K)}{2\mathrm{vol}_{n-1}(K \cap u^\perp)}$$

holds for each $u \in \mathbb{S}^{n-1}$.

(c) *Prove that*

$$h_K(u) \leq \frac{n\mathrm{vol}_n(K)}{2\mathrm{vol}_{n-1}(K \cap u^\perp)}$$

for all $u \in \mathbb{S}^{n-1}$.

(d) *Discuss the equality cases in the inequalities given in items* **(b)** *and* **(c)**.

Solution 4.16. (a) This is a consequence of the fact that $\mathrm{proj}_{u^\perp}(K \cap (u^\perp + tu)) \subseteq K \cap u^\perp$ for each $t \in \mathbb{R}$. In fact, let $x = z + tu \in K \cap (u^\perp + tu)$ be arbitrarily given, and notice that $z = \mathrm{proj}_{u^\perp}(x)$. Since K is centered at the origin, we get that $-x = -z - tu \in K \cap u^\perp - tu$. By convexity, the segment $\mathrm{seg}[-x, x]$ is contained in K, and since z is clearly a point of this segment, we obtain that $z \in K \cap u^\perp$.

The projection $\mathrm{proj}_{u^\perp}$ obviously acts on $K \cap (u^\perp + tu)$ as a translation, and hence, it does not modify the volume. It follows that

$$\mathrm{vol}_{n-1}(K \cap (u^\perp + tu)) \leq \mathrm{vol}_{n-1}(K \cap u^\perp),$$

as we wanted to prove.

(b) We compute the volume of K using Cavalieri's principle for the sections $K \cap (u^\perp + tu)$. This is clearly empty if $t < -h_K(-u)$ or $t > h_K(u)$. Moreover, since K is centered at the origin, we have that $h_K(-u) = h_K(u)$. This leads to

$$\mathrm{vol}_n(K) = \int_{-h_K(u)}^{h_K(u)} \mathrm{vol}_{n-1}(K \cap (u^\perp + tu))\, dt \leq$$
$$\leq \int_{-h_K(u)}^{h_K(u)} \mathrm{vol}_n(K \cap u^\perp)\, dt = 2h_K(u)\mathrm{vol}_{n-1}(K \cap u^\perp),$$

and rearranging the terms, we get the desired inequality.

(c) Let P be the pyramid whose base is $K \cap u^\perp$ and with apex at the point x of ∂K where u is the outer normal of K. Since this P is the union of segments connecting x to the points of $K \cap u^\perp$, we get that P is contained in the half of K determined by $u^\perp$. Thus,

$$\frac{1}{2}\mathrm{vol}_n(K) \geq \mathrm{vol}_n(P).$$

Since

$$\mathrm{vol}_n(P) = \frac{1}{n} h_K(u)\mathrm{vol}_{n-1}(K \cap u^\perp),$$

the desired inequality follows.

(d) If equality in **(b)** holds for a given $u \in \mathbb{S}^{n-1}$, then

$$\mathrm{vol}_{n-1}(K \cap (u^\perp + tu)) = \mathrm{vol}_{n-1}(K \cap u^\perp),$$

for each $-h_K(u) \leq t \leq h_K(u)$. This means that K is a cylinder whose axis is in the direction of u, and such that each orthogonal section is $K \cap u^\perp$.

On the other hand, if equality in **(c)** holds, then K is the union of two pyramids having opposite apices (with respect to the origin), and both having $K \cap u^\perp$ as basis.

□

The *Gaussian measure* of $\mathbb{R}^n$ is the Borel measure γ_n whose density is

$$G(x) = \frac{1}{(2\pi)^{n/2}} e^{-\frac{|x|^2}{2}}. \tag{4.3}$$

That is, for each Borel subset $A \subseteq \mathbb{R}^n$, the measure γ_n is defined by

$$\gamma_n(A) = \frac{1}{(2\pi)^{\frac{n}{2}}} \int_A e^{-\frac{|x|^2}{2}}\, dx.$$

It is clear from standard measure theory that the integral of a Borel measurable function $f\colon \mathbb{R}^n \to [-\infty, +\infty]$ with respect to γ_n is given by

$$\int_{\mathbb{R}^n} f(x)\, d\gamma_n(x) = \frac{1}{(2\pi)^{n/2}} \int_{\mathbb{R}^n} f(x) e^{-\frac{|x|^2}{2}}\, dx.$$

Exercise 4.17. **(a)** *Verify that the Gaussian measure is a probability measure.*

(b) *Verify that γ_n is invariant under orthogonal transformations.*

Solution 4.17. **(a)** This follows directly from a change of variables in (4.3.1) (main book): simply let $y = x/\sqrt{2}$, and write

$$\int_{\mathbb{R}^n} e^{-\frac{|x|^2}{2}}\, dx = \int_{\mathbb{R}^n} e^{-|y|^2} 2^{n/2}\, dy = (2\pi)^{n/2}.$$

(b) This is an immediate consequence of the $\mathrm{O}(n)$-invariance of the Lebesgue measure and of the fact that any orthogonal transformation is an isometry.

□

It is obvious that the Gaussian measure is not translation-invariant. However, we have a bound for the measure of translations of a body centered at the origin. This is in the exercise below.

Exercise 4.18. *Let $K \subseteq \mathbb{R}^n$ be a convex body symmetric with respect to the origin. Prove that for any $y \in \mathbb{R}^n$, the inequality*

$$\gamma_n(y + K) \geq e^{-|y|^2/2}\gamma_n(K)$$

holds.

Solution 4.18. This is proved in Lemma 11.4.6 of the main book.

□

Exercise 4.19 (Markov's Inequality)**.** *Let (X, μ) be a probability space, and let $f\colon X \to \mathbb{R}$ be a μ-measurable function. Prove that*

$$\mu\Big(\big\{x \in X : |f(x)| \geq \varepsilon\big\}\Big) \leq \frac{1}{\varepsilon}\int_X |f(x)|\, d\mu(x)$$

for every $\varepsilon > 0$.

Hint: Define $\phi\colon X \to \mathbb{R}$ as

$$\phi(x) = \begin{cases} \varepsilon, & \text{if } |f(x)| \geq \varepsilon \\ 0, & \text{if } |f(x)| < \varepsilon \end{cases}.$$

Notice that $\phi \leq |f|$ and integrate both with respect to μ.

Solution 4.19. We let

$$\phi(x) = \begin{cases} \varepsilon, & \text{if } |f(x)| \geq \varepsilon \\ 0, & \text{if } |f(x)| < \varepsilon \end{cases}.$$

Since $\phi \leq |f|$, we get

$$\int_X |f(x)|\, d\mu(x) \geq \int_X \phi(x)\, d\mu(x) = \int_{\{x \in X : |f(x)| \geq \varepsilon\}} \varepsilon\, d\mu(x) = \varepsilon\mu\Big(\big\{x \in X : |f(x)| \geq \varepsilon\big\}\Big),$$

and dividing both sides by ε yields the desired result.

□

Exercise 4.20. *Prove that the Gaussian measure γ_n is* log-concave *in the following sense: for any Borel subsets $A, B \subseteq \mathbb{R}^n$ and for every $\lambda \in [0, 1]$, we have that*

$$\gamma_n((1 - \lambda)A + \lambda B) \geq \gamma_n(A)^{1-\lambda}\gamma_n(B)^{\lambda}.$$

Solution 4.20. It is immediate to notice that

$$1_{(1-\lambda)A+\lambda B}((1 - \lambda)x + \lambda y) \geq 1_A(x)^{1-\lambda}1_B(y)^{\lambda}$$

for any $x, y \in \mathbb{R}^n$ and every $\lambda \in (0,1)$. Indeed, the right-hand side is 1 only if $x \in A$ and $y \in B$, and in this case, the left-hand side is also 1. Since the function $x \mapsto |x|^2/2$ is obviously convex, we also have

$$\frac{|(1-\lambda)x + \lambda y|^2}{2} \leq (1-\lambda)\frac{|x|^2}{2} + \lambda\frac{|y|^2}{2},$$

and this leads to

$$\exp\left(-\frac{|(1-\lambda)x+\lambda y|^2}{2}\right) \geq \exp\left(-(1-\lambda)\frac{|x|^2}{2}\right)\exp\left(-\lambda\frac{|y|^2}{2}\right) = \\ = \exp\left(-\frac{|x|^2}{2}\right)^{1-\lambda}\exp\left(-\frac{|y|^2}{2}\right)^{\lambda}.$$

That is, we verified once more that the function $x \mapsto e^{-|x|^2/2}$ is log-concave. As a consequence, we have that

$$1_{(1-\lambda)A+\lambda B}((1-\lambda)x+\lambda y)\exp\left(-\frac{|(1-\lambda)x+\lambda y|^2}{2}\right) \geq \\ \geq 1_A(x)^{1-\lambda}1_B(y)^{\lambda}\exp\left(-\frac{|x|^2}{2}\right)^{1-\lambda}\exp\left(-\frac{|y|^2}{2}\right)^{\lambda} = \left(1_A(x)e^{-\frac{|x|^2}{2}}\right)^{1-\lambda}\left(1_B(y)e^{-\frac{|y|^2}{2}}\right)^{\lambda}.$$

Therefore, we may apply the Prékopa-Leindler inequality (Theorem 1.5.1 in the main book) to get

$$\int_{\mathbb{R}^n} 1_{(1-\lambda)A+\lambda B}(x)e^{-\frac{|x|^2}{2}}\,dx \geq \left(\int_{\mathbb{R}^n} 1_A(x)e^{-\frac{|x|^2}{2}}\,dx\right)^{1-\lambda}\left(\int_{\mathbb{R}^n} 1_B(x)e^{-\frac{|x|^2}{2}}\,dx\right)^{\lambda},$$

and dividing both sides by $(2\pi)^{n/2}$, we arrive at

$$\frac{1}{(2\pi)^{n/2}}\int_{(1-\lambda)A+\lambda B} e^{-\frac{|x|^2}{2}}\,dx \geq \left(\frac{1}{(2\pi)^{n/2}}\int_A e^{-\frac{|x|^2}{2}}\,dx\right)^{1-\lambda}\left(\frac{1}{(2\pi)^{n/2}}\int_B e^{-\frac{|x|^2}{2}}\,dx\right)^{\lambda}.$$

This is the same as

$$\gamma_n((1-\lambda)A+\lambda B) \geq \gamma_n(A)^{1-\lambda}\gamma_n(B)^{\lambda}.$$

□

The density of the Gaussian measure is a log-concave function, and the reader may ask whether a log-concave density always yields a log-concave measure. This is true, in fact, and the reader is invited to prove that in the exercise below.

Exercise 4.21. *Let $\phi\colon \mathbb{R}^n \to [0,+\infty)$ be a log-concave function such that*

$$0 < I(\phi) := \int_{\mathbb{R}^n} \phi(x)\,dx < +\infty.$$

Let μ_ϕ be the Borel measure in $\mathbb{R}^n$ whose density with respect to the usual Lebesgue measure is $I(\phi)^{-1}\phi(\cdot)$. That is,

$$\mu_\phi(A) = \frac{1}{I(\phi)}\int_A \phi(x)\,dx$$

for each Borel set $A \subseteq \mathbb{R}^n$. Prove that μ_ϕ is a log-concave measure.

Solution 4.21. Let $A, B \subseteq \mathbb{R}^n$ be Borel sets, and let $\lambda \in (0,1)$. As it was proved in Exercise 4.20, we may write

$$1_{(1-\lambda)A+\lambda B}((1-\lambda)x+\lambda y) \geq 1_A(x)^{1-\lambda}1_B(y)^{\lambda}$$

for any $x, y \in \mathbb{R}^n$. Since ϕ is log-concave, this leads to

$$1_{(1-\lambda)A+\lambda B}((1-\lambda)x+\lambda y)\phi((1-\lambda)x+\lambda y) \geq \left[\phi(x)1_A(x)\right]^{1-\lambda}\left[\phi(y)1_B(y)\right]^{\lambda}$$

for every $x, y \in \mathbb{R}^n$. Therefore, the Prékopa-Leindler inequality yields

$$\int_{\mathbb{R}^n} 1_{(1-\lambda)A+\lambda B}(x)\phi(x)\,dx \geq \left(\int_{\mathbb{R}^n} \phi(x)1_A(x)\,dx\right)^{1-\lambda}\left(\int_{\mathbb{R}^n} \phi(x)1_B(x)\,dx\right)^{\lambda},$$

and this reads

$$\int_{(1-\lambda)A+\lambda B} \phi(x)\,dx \geq \left(\int_A \phi(x)\,dx\right)^{1-\lambda}\left(\int_B \phi(x)\,dx\right)^{\lambda}.$$

Dividing both sides by $I(\phi)$, we finally arrive at

$$\mu_\phi((1-\lambda)A+\lambda B) \geq \mu_\phi(A)^{1-\lambda}\mu_\phi(B)^{\lambda}.$$

□

Exercise 4.22. *Let $K \subseteq \mathbb{R}^n$ be a convex body symmetric with respect to the origin.*

(a) *Prove that for every $t > 1$, the inclusion*

$$\frac{2}{t+1}(\mathbb{R}^n \setminus tK) + \frac{t-1}{t+1}K \subseteq \mathbb{R}^n \setminus K$$

holds.

(b) *Write $\gamma_n(K) = c$. Use item* **(a)** *and the log-concavity of γ_n to show that*

$$\gamma_n(tK) \geq 1 - c\left(\frac{1-c}{c}\right)^{\frac{t+1}{2}}$$

for any $t > 1$.

Remark 4.5. The bound given in item **(b)** is only useful if $c > 1/2$, because if $c \leq 1/2$, then the trivial bound

$$\gamma_n(tK) \geq \gamma_n(K) = c$$

is sharper. In fact, it is easy to see that the inequality

$$c \geq 1 - c\left(\frac{1-c}{c}\right)^{\frac{t+1}{2}}$$

is equivalent to

$$(1-c)^{\frac{t-1}{2}} \geq c^{\frac{t-1}{2}},$$

which holds for $c \in (0,1)$ if and only if $c \leq 1/2$. Observe also that to have $\gamma_n(tK) \geq \gamma_n(K)$, we just need that $K \subseteq tK$. Hence, we do not need to have that K is origin-symmetric; it suffices to consider that K is any convex body containing the origin.

Solution 4.22. (a) Let $x \in \mathbb{R}^n \setminus (tK)$ and $y \in K$ be arbitrarily given. Assume by contradiction that

$$z := \frac{2}{t+1}x + \frac{t-1}{t+1}y \in K.$$

Since K is symmetric with respect to the origin, we have that $-y \in K$. Hence, letting $\alpha = (t+1)/2t$, we have that

$$\begin{aligned} K \ni \alpha z + (1-\alpha)(-y) = \frac{2\alpha}{t+1}x + \frac{\alpha(t-1)}{t+1}y - (1-\alpha)y = \\ = \frac{x}{t} + \frac{t-1}{2t}y - \frac{t-1}{2t}y = \frac{x}{t}. \end{aligned}$$

This gives that $x \in tK$, a contradiction.

(b) From the inclusion given in item **(a)** and since γ_n is log-concave, we may write

$$\gamma_n(\mathbb{R}^n \setminus K) \geq \gamma_n\left(\frac{2}{t+1}(\mathbb{R}^n \setminus tK) + \frac{t-1}{t+1}K\right) \geq \gamma_n(\mathbb{R}^n \setminus tK)^{\frac{2}{t+1}}\gamma_n(K)^{\frac{t-1}{t+1}}.$$

Thus,

$$1 - c \geq \gamma_n(\mathbb{R}^n \setminus tK)^{\frac{2}{t+1}} c^{\frac{t-1}{t+1}},$$

and hence,

$$(1-c)^{\frac{t+1}{2}} \geq (1-\gamma_n(tK))c^{\frac{t-1}{2}}.$$

Finally, this leads to

$$\gamma_n(tK) \geq 1 - \frac{(1-c)^{\frac{t+1}{2}}}{c^{\frac{t-1}{2}}} = 1 - c\left(\frac{1-c}{c}\right)^{\frac{t+1}{2}}.$$

□

Exercise 4.23. *Let $C \subseteq \mathbb{R}^n$ be a closed set such that $\gamma_n(C) > 0$. Prove that*

$$\int_{\mathbb{R}^n} e^{\frac{1}{4}d(x,C)^2}\, d\gamma_n(x) \leq \frac{1}{\gamma_n(C)},$$

where we remind the reader that $d(x,C) := \inf\{d(x,z) : z \in C\}$.

Hint: Use the Prékopa-Leindler inequality (Theorem 1.5.1 in the main book) for the functions $f(x) = e^{\frac{1}{4}d(x,C)^2}G(x)$, $g(x) = 1_C(x)G(x)$, $h(x) = G(x)$, and for $\lambda = 1/2$.

Solution 4.23. Let

$$\begin{aligned} f(x) &= e^{\frac{1}{4}d(x,C)^2}G(x), \\ g(x) &= 1_C(x)G(x) \text{ and} \\ h(x) &= G(x), \end{aligned}$$

where $G(x)$ is as defined in (4.3). We claim that

$$h\left(\frac{x+y}{2}\right)^2 \geq f(x)g(y)$$

for any $x, y \in \mathbb{R}^n$. To prove that, we may assume that $y \in C$, since otherwise $g(y) = 0$. In the case where $y \in C$, we have $d(x, C) \leq |x - y|$. Thus,

$$\begin{aligned} f(x)g(y) = e^{\frac{1}{4}d(x,C)^2}G(x)G(y) = \frac{1}{(2\pi)^n}e^{\frac{1}{4}d(x,C)^2-\frac{|x|^2}{2}-\frac{|y|^2}{2}} \leq \frac{1}{(2\pi)^n}e^{\frac{|x-y|^2}{4}-\frac{|x|^2}{2}-\frac{|y|^2}{2}} = \\ = \frac{1}{(2\pi)^n}e^{-|\frac{x+y}{2}|} = \left(\frac{1}{(2\pi)^{n/2}}e^{-\frac{1}{2}|\frac{x+y}{2}|}\right)^2 = G\left(\frac{x+y}{2}\right)^2 = h\left(\frac{x+y}{2}\right)^2. \end{aligned}$$

Therefore, we may use the Prékopa-Leindler inequality for $\lambda = 1/2$. This leads to

$$\begin{aligned} 1 = \left(\int_{\mathbb{R}^n} G(x)\,dx\right)^2 = \left(\int_{\mathbb{R}^n} h(x)\,dx\right)^2 \geq \int_{\mathbb{R}^n} e^{\frac{1}{4}d(x,C)^2}G(x)\,dx \int_{\mathbb{R}^n} 1_C(x)G(x)\,dx = \\ = \left(\int_{\mathbb{R}^n} e^{\frac{1}{4}d(x,C)^2}\,d\gamma_n(x)\right)\int_C d\gamma_n(x) = \gamma_n(C)\int_{\mathbb{R}^n} e^{\frac{1}{4}d(x,C)^2}\,d\gamma_n(x), \end{aligned}$$

and this gives the desired inequality.

□

Exercise 4.24. *Let $C \subseteq \mathbb{R}^n$ be a closed set such that $\gamma_n(C) \geq 1/2$, and set*

$$C_\varepsilon := \{x \in \mathbb{R}^n : d(x, C) \leq \varepsilon\}.$$

Prove that for any $\varepsilon \geq 0$, we have that

$$\gamma_n(C_\varepsilon) \geq 1 - 2e^{-\frac{\varepsilon^2}{4}}.$$

Remark 4.6. This can be seen as a *concentration inequality* since, in some sense, it states that the Gaussian measure is "concentrated around" any given closed set having measure at least $1/2$.

Solution 4.24. Since $\gamma_n(C) \geq 1/2$, from Exercise 4.23, we get

$$\int_{\mathbb{R}^n} e^{\frac{1}{4}d(x,C)^2}\,d\gamma_n(x) \leq 2.$$

Notice that $\mathbb{R}^n \setminus C_\varepsilon = \{x \in \mathbb{R}^n : d(x, C) > \varepsilon\}$. Now, we bound

$$\begin{aligned} \int_{\mathbb{R}^n} e^{\frac{1}{4}d(x,C)^2}\,d\gamma_n(x) \geq \int_{\mathbb{R}^n\setminus C_\varepsilon} e^{\frac{1}{4}d(x,C)^2}\,d\gamma_n(x) \geq \int_{\mathbb{R}^n\setminus C_\varepsilon} e^{\frac{\varepsilon^2}{4}}\,d\gamma_n(x) = e^{\frac{\varepsilon^2}{4}}\gamma_n(\mathbb{R}^n \setminus C_\varepsilon) = \\ = e^{\frac{\varepsilon^2}{4}}(1 - \gamma_n(C_\varepsilon)). \end{aligned}$$

Combining these inequalities, we get

$$2 \geq e^{\frac{\varepsilon^2}{4}}(1 - \gamma_n(C_\varepsilon)),$$

and rearranging the terms, we obtain the desired inequality.

□

We briefly investigate (abstract) concentration of measures below. For more in this direction, we refer the reader to [24] (which is the source of most exercises below). Let (X, d) be a metric space, and denote by $\mathcal{B}(X)$ its Borel σ-algebra. Assume that $\mu\colon \mathcal{B}(X) \to [0, 1]$ is a Borel probability measure on X. In this case, we say that (X, d, μ) is a *metric probability space*. The *concentration function* of (X, d, μ) is the function $\alpha_\mu\colon [0, +\infty) \to \mathbb{R}$ defined as

$$\alpha_\mu(t) := \sup\left\{1 - \mu(A_t) : A \in \mathcal{B}(X) \text{ and } \mu(A) \geq 1/2\right\},$$

where $A_t := \{x \in X : d(x, A) \leq t\}$.

Exercise 4.25. *Show that for any metric probability space (X, d, μ), the concentration function is non-increasing. Prove also that*

$$\lim_{t \to +\infty} \alpha_\mu(t) = 0.$$

Solution 4.25. If $0 \leq t_1 < t_2$, then for any $A \subseteq X$, we have $A_{t_1} \subseteq A_{t_2}$. Thus, for all $A \in \mathcal{B}(X)$, it holds that

$$1 - \mu(A_{t_1}) \geq 1 - \mu(A_{t_2}).$$

Passing to the supremum among the sets $A \in \mathcal{B}(X)$ such that $\mu(A) \geq 1/2$, we get

$$\alpha_\mu(t_1) \geq \alpha_\mu(t_2),$$

and this shows that α_μ is non-increasing.

To prove the other claim, let $x \in X$ be fixed, and let $0 < \varepsilon \leq 1/2$. Since $X = \bigcup_{m=1}^{\infty} B(x, m)$, we have from Lemma A.1.1 (main book) that

$$1 = \mu(X) = \lim_{m \to +\infty} \mu(B(x, m)).$$

It follows that there exists a number $m_0 \in \mathbb{N}$ for which

$$\mu(B(x, m_0)) > 1 - \varepsilon.$$

In particular, it holds that $\mu(B(x, m_0)) > 1/2$. As a consequence, for any $A \subseteq \mathcal{B}(X)$ with $\mu(A) \geq 1/2$, we must have $A \cap B(x, m_0) \neq \emptyset$. This means that there is a point $a \in A$ such that $d(x, a) \leq m_0$. From the triangle inequality, we get that $B(x, m_0) \subseteq A_{2m_0}$. Thus, for every $t \geq 2m_0$, we have

$$1 - \mu(A_t) \leq 1 - \mu(A_{2m_0}) \leq 1 - \mu(B(x, m_0)) < \varepsilon.$$

Since this holds for any $A \in \mathcal{B}(X)$ with $\mu(A) \geq 1/2$, passing to the supremum, we get that $\alpha_\mu(t) \leq \varepsilon$ for every $t \geq 2m_0$. This shows that $\alpha_\mu(t) \to 0$ as $t \to +\infty$.

□

Let (X, d, μ) be a metric probability space, and let $f\colon X \to \mathbb{R}$ be any measurable function. The *mean* (or *expected value*) of f is the number

$$\mathbb{E}(f) := \int_X f \, d\mu. \tag{4.4}$$

A *median* (or *Lévy mean*) of f is a number $\operatorname{med}(f)$ such that

$$\mu\big(\{f \geq \operatorname{med}(f)\}\big) \geq \frac{1}{2} \text{ and } \mu\big(\{f \leq \operatorname{med}(f)\}\big) \geq \frac{1}{2}.$$

It is not difficult to see that a median need not be unique, and its existence is also not completely trivial. In the next exercise, we provide some conditions that are sufficient to ensure the existence and uniqueness of the median.

Exercise 4.26. *Let (X, d, μ) be a metric probability space having the following properties:*

(i) *(X, d) is a connected metric space.*

(ii) *$\mu(A) > 0$ for every non-empty open set $A \subseteq X$.*

Prove that any continuous function $f\colon X \to \mathbb{R}$ has a unique median.

Solution 4.26. For the existence, we notice that the function $g\colon \mathbb{R} \to [0, 1]$ given by

$$g(t) = \mu\big(\{f > t\}\big)$$

is clearly continuous (this follows from Lemma A.1.1; we consider left and right continuity separately). Because

$$\emptyset = \bigcap_{m \in \mathbb{N}} \{f > m\},$$

we get (from Lemma A.1.1 again) that there exists $m_0 \in \mathbb{N}$ such that $g(m_0) < 1/2$. Similarly, since

$$X = \bigcup_{m \in \mathbb{N}} \{f > -m\},$$

we obtain that there exists $m_1 \in \mathbb{N}$ such that $g(-m_1) > 1/2$. Thus, from the intermediate value theorem, there exists $t_0 \in \mathbb{R}$ such that $g(t_0) = 1/2$. We claim that t_0 is a median of f. Indeed, simply notice that

$$\begin{aligned} \mu\big(\{f \leq t_0\}\big) = 1 - \mu\big(\{f > t_0\}\big) = 1 - g(t_0) = 1/2, \text{ and} \\ \mu\big(\{f \geq t_0\}\big) \geq \mu\big(\{f > t_0\}\big) = g(t_0) = 1/2. \end{aligned}$$

Now, we prove uniqueness. Assume that f has two medians m_1 and m_2, with $m_1 < m_2$, say. Since

$$\mu\big(\{f \geq m_2\}\big) \geq 1/2 \text{ and } \mu\big(\{f \leq m_1\}\big) \geq 1/2,$$

we have that the sets $\{f \geq m_2\}$ and $\{f \leq m_1\}$ are non-empty. That is, the image $\operatorname{im}(f)$ of f must contain a number greater than or equal to m_2 and a number smaller than or equal to m_1. Since X is connected, from standard analysis, we have that $\operatorname{im}(f)$ is an interval, and hence, we get that $(m_1, m_2) \subseteq \operatorname{im}(f)$. It follows that $f^{-1}((m_1, m_2))$ is a non-empty open subset of X, and from our hypothesis, we get that

$$\mu(f^{-1}((m_1, m_2))) > 0.$$

Writing X as the disjoint union of the sets $\{f \geq m_2\}$, $f^{-1}((m_1, m_2))$ and $\{f \leq m_1\}$, we get that

$$1 = \mu(X) = \mu\big(\{f \geq m_2\}\big) + \mu(f^{-1}((m_1, m_2))) + \mu\big(\{f \leq m_1\}\big) \geq 1 + \mu(f^{-1}((m_1, m_2)) > 1,$$

a contradiction. Hence, we get that f has a unique median.

□

For simplicity, from now on, we say that a function $f\colon (X, d) \to \mathbb{R}$ is 1-*Lipschitz* if $|f(x) - f(y)| \leq d(x, y)$ for any $x, y \in X$. In the exercise below, we relate these notions to the concentration function of (X, d, μ). Intuitively, we have that the value of $\alpha_\mu(t)$ "controls" the measure of the set of points of X for which f has distance greater than t to a median.

Exercise 4.27. *Let (X, d, μ) be a metric probability space, and assume that $f\colon X \to \mathbb{R}$ is a 1-Lipschitz function. Prove that*

$$\mu\Big(\big\{x \in X : |f(x) - \mathrm{med}(f)| > t\big\}\Big) \leq 2\alpha_\mu(t) \tag{4.5}$$

for any median $\mathrm{med}(f)$ *of f and all $t \geq 0$.*

Solution 4.27. Let

$$A = \big\{x \in X : f(x) \geq \mathrm{med}(f)\big\} \text{ and}$$
$$B = \big\{x \in X : f(x) \leq \mathrm{med}(f)\big\},$$

and notice that both sets are closed (since f is continuous, in particular) and non-empty (because they have positive measure). For any given $t \geq 0$, if $y \in A_t$, then there exists $x \in A$ such that $d(x, y) \leq t$. Since f is 1-Lipschitz, we get that

$$f(y) = f(y) - f(x) + f(x) \geq -d(x, y) + f(x) \geq -d(x, y) + \mathrm{med}(f) \geq -t + \mathrm{med}(f).$$

Similarly, if $y \in B_t$, then there exists $x \in B$ with $d(x, y) \leq t$, and hence,

$$f(y) = f(y) - f(x) + f(x) \leq d(x, y) + f(x) \leq d(x, y) + \mathrm{med}(f) \leq t + \mathrm{med}(f).$$

Summarizing, we have that

$$|f(y) - \mathrm{med}(f)| \leq t$$

for every $y \in A_t \cap B_t$. Thus,

$$\big\{x \in X : |f(x) - \mathrm{med}(f)| > t\big\} \subseteq X \setminus (A_t \cap B_t) = (X \setminus A_t) \cup (X \setminus B_t).$$

Notice also that $\mu(A_t) \geq \mu(A) \geq 1/2$ and $\mu(B_t) \geq \mu(B) \geq 1/2$, from where we get that $1 - \mu(A_t) \leq \alpha_\mu(t)$ and $1 - \mu(B_t) \leq \alpha_\mu(t)$. Putting these together, we arrive at

$$\mu\Big(\big\{x \in X : |f(x) - \mathrm{med}(f)| > t\big\}\Big) \leq \mu(X \setminus A_t) + \mu(X \setminus B_t) = 1 - \mu(A_t) + 1 - \mu(B_t) \leq$$
$$\leq 2\alpha_\mu(t),$$

as we wanted to prove.

□

Exercise 4.28. *Let (X, d, μ) be a metric probability space, and assume that there exist numbers $\alpha, t > 0$ such that*

$$\mu\Big(\big\{x \in X : |f(x) - \mathrm{med}(f)| > t\big\}\Big) \leq \alpha$$

for any 1-Lipschitz function $f : X \to \mathbb{R}$. Show that $\alpha_u(t) \leq \alpha$.

Hint: For a given Borel set $A \subseteq X$ such that $\mu(A) \geq 1/2$, consider the distance function $f(x) = d(x, A)$.

Solution 4.28. Let $A \subseteq X$ be a Borel subset such that $\mu(A) \geq 1/2$. Let $f(x) = d(x, A)$. From the triangle inequality, it is clear that f is 1-Lipschitz. Moreover, we claim that 0 is a median of f. Indeed, since $f(x) = 0$ for every $x \in A$, we have

$$\mu\big(\{f \leq 0\}\big) \geq \mu\big(\{f = 0\}\big) \geq \mu(A) \geq 1/2,$$

and since $f \geq 0$ in X, we get also that $\mu(\{f \geq 0\}) = 1$. It follows from the hypothesis that

$$\mu\big(\{x \in X : d(x, A) > t\}\big) \leq \alpha.$$

Since

$$\{x \in X : d(x, A) > t\} = X \setminus (A_t),$$

we get that

$$1 - \mu(A_t) \leq \alpha.$$

Since this holds for every Borel set $A \subseteq X$ such that $\mu(A) \geq 1/2$, passing to the supremum, we obtain that $\alpha_\mu(t) \leq \alpha$. The proof is complete.

□

Inequality (4.5) can be seen as a bound for the probability of *deviation from the median*. Our next objective is to derive an inequality for the *deviation from the mean* for 1-Lipschitz functions with respect to the Gaussian measure. This is made in Exercise 4.31. Before this, we bring Exercises 4.29 and 4.30 for the auxiliary results.

We recall that the *infimal convolution* of functions $f, g : \mathbb{R}^n \to (-\infty, +\infty]$ is the function $f \square g : \mathbb{R}^n \to [-\infty, +\infty]$ given as

$$f \square g(x) = \inf\big\{f(z) + g(x - z) : z \in \mathbb{R}^n\big\}.$$

We refer the reader to Exercises 1.22, 1.23, 1.24, and 1.25 for more on the infimal convolution.

Exercise 4.29. *Let $\varphi : \mathbb{R}^n \to \mathbb{R}$ be the function given as $\varphi(x) = |x|^2/4$, and let $f : \mathbb{R}^n \to \mathbb{R}$ be a measurable bounded function.*

(a) *Define the functions $\bar{f}, g, h : \mathbb{R}^n \to \mathbb{R}$ as*

$$\bar{f}(x) = f(x) + \frac{|x|^2}{2},\ g(x) = -f \square \varphi(x) + \frac{|y|^2}{2} \text{ and } h(x) = \frac{|x|^2}{2}.$$

Show that

$$h\left(\frac{x + y}{2}\right) \leq \frac{\bar{f}(x) + g(y)}{2}$$

for any $x, y \in \mathbb{R}^n$.

(b) *Prove that*

$$\left(\int_{\mathbb{R}^n} e^{f\square\varphi}\, d\gamma_n\right)\left(\int_{\mathbb{R}^n} e^{-f}\, d\gamma_n\right) \le 1.$$

Solution 4.29. (a) This follows from the equality

$$\frac{|x|^2}{2} + \frac{|y|^2}{2} - \frac{|x-y|^2}{4} = \frac{|x+y|^2}{4},$$

which clearly holds for any $x, y \in \mathbb{R}^n$. From the definition of the infimal convolution, we get

$$\frac{1}{2}(\bar{f}(x) + g(y)) = \frac{1}{2}\left(f(x) + \frac{|x|^2}{2} - \inf\left\{f(z) + \frac{|y-z|^2}{4} : z \in \mathbb{R}^n\right\} + \frac{|y|^2}{2}\right) \ge$$
$$\ge \frac{1}{2}\left(f(x) + \frac{|x|^2}{2} - f(x) - \frac{|y-x|^2}{4} + \frac{|y|^2}{2}\right) = \frac{1}{2}\frac{|x+y|^2}{4} = \frac{1}{2}\left|\frac{x+y}{2}\right|^2 = h\left(\frac{x+y}{2}\right).$$

(b) From item **(a)**, we have that

$$\exp\left(-h\left(\frac{x+y}{2}\right)\right) \ge \exp\left(-\bar{f}(x)\right)^{\frac{1}{2}} \exp\left(-g(y)\right)^{\frac{1}{2}}$$

for any $x, y \in \mathbb{R}^n$. Moreover, since f is bounded, it clearly follows that $f\square\varphi$ is also bounded, and hence, we have that $e^{-\bar{f}}$ and e^{-g} are Lebesgue integrable. Therefore, using the Prékopa-Leindler inequality, we obtain

$$\left(\int_{\mathbb{R}^n} e^{-h(x)}\, dx\right)^2 \ge \left(\int_{\mathbb{R}^n} e^{-\bar{f}(x)}\, dx\right)\left(\int_{\mathbb{R}^n} e^{-g(x)}\, dx\right).$$

Since $e^{-h(x)} = (2\pi)^{n/2} G(x)$ and $(2\pi)^{-n/2} e^{-|x|^2/2}$ is the density of γ_n, this reads

$$(2\pi)^n \ge \left((2\pi)^{n/2}\int_{\mathbb{R}^n} e^{f\square\varphi}\, d\gamma_n\right)\left((2\pi)^{n/2}\int_{\mathbb{R}^n} e^{-f}\, d\gamma_n\right).$$

Dividing both sides by $(2\pi)^n$, we get the desired inequality.

□

Exercise 4.30. *Let $f\colon \mathbb{R}^n \to \mathbb{R}$ be a bounded function, which is 1-Lipschitz with respect to the Euclidean distance of $\mathbb{R}^n$, and let $t > 0$ be arbitrarily given. Also, still assume that $\varphi\colon \mathbb{R}^n \to \mathbb{R}$ is the function given as $\varphi(x) = |x|^2/4$.*

(a) *Verify that the infimum in the definition of $(tf)\square\varphi(x)$ is attained. That is, check that for any $x \in \mathbb{R}^n$, there exists $y = y(x,t) \in \mathbb{R}^n$ such that $(tf)\square\varphi(x) = tf(y) + \varphi(x-y)$.*

(b) *Using the 1-Lipschitz hypothesis, show that*

$$(tf)\square\varphi(x) \ge tf(x) - t|x-y| + \frac{|x-y|^2}{4},$$

where y is as given in item **(a)**.

(c) *Studying the second-degree polynomial* $\alpha \mapsto \alpha^2/4 - t\alpha$*, conclude that*

$$(tf)\Box\varphi(x) \geq tf(x) - t^2$$

for any $x \in \mathbb{R}^n$.

(d) *Prove that*

$$\int_{\mathbb{R}^n} e^{tf}\, d\gamma_n \int_{\mathbb{R}^n} e^{-tf}\, d\gamma_n \leq e^{t^2}.$$

Solution 4.30. (a) Notice that $(tf)\Box\varphi$ is real-valued because f is bounded and $\varphi \geq 0$. For any fixed $x \in \mathbb{R}^n$, we clearly have that

$$\lim_{|z|\to+\infty} \big[tf(z) + \varphi(x-z)\big] = +\infty,$$

and hence, there exists a number $r > 0$ such that

$$\inf_{z\in\mathbb{R}^n} (tf(z) + \varphi(x-z)) = \inf_{z\in rB^n} (tf(z) + \varphi(x-z)).$$

From continuity, the latter is a minimum, and then it follows that for each $x \in \mathbb{R}^n$, there exists a $y = y(x,t) \in \mathbb{R}^n$ such that

$$(tf)\Box\varphi(x) = \inf\big\{tf(z) + \varphi(x-z) : z \in \mathbb{R}^n\big\} = tf(y) + \varphi(x-y).$$

(b) Because f is 1-Lipschitz, we get that $t(f(x) - f(y)) \leq t|x-y|$, and hence,

$$(tf)\Box\varphi(x) = tf(y) + \varphi(x-y) \geq tf(x) - t|x-y| + \frac{|x-y|^2}{4}.$$

(c) Letting $|x-y| = \alpha$, we have that the second-degree polynomial

$$\frac{\alpha^2}{4} - t\alpha$$

attains the minimum value $-t^2$ for $\alpha = 2t$. Thus,

$$(tf)\Box\varphi(x) \geq tf(x) - t|x-y| + \frac{|x-y|^2}{4} \geq tf(x) - t^2.$$

(d) We simply use the inequality proved in Exercise 4.29 together with the bound in item **(c)** to write

$$\int_{\mathbb{R}^n} e^{tf(x)-t^2}\, d\gamma_n(x) \int_{\mathbb{R}^n} e^{-tf(z)}\, d\gamma_n(z) \leq \int_{\mathbb{R}^n} e^{(tf)\Box\varphi(x)}\, d\gamma_n(x) \int_{\mathbb{R}^n} e^{-tf(z)}\, d\gamma_n(z) \leq 1.$$

Writing

$$\int_{\mathbb{R}^n} e^{tf(x)-t^2}\, d\gamma_n(x) = e^{-t^2} \int_{\mathbb{R}^n} e^{tf(x)}\, d\gamma_n(x),$$

we get the desired result.

□

Exercise 4.31. *Let $f\colon \mathbb{R}^n \to \mathbb{R}$ be a bounded 1-Lipschitz function.*

(a) *Use Exercise 4.30 and Jensen's inequality to prove that*

$$\mathbb{E}\big(\exp(t(f-\mathbb{E}(f))\big) \le e^{t^2}$$

for every $t > 0$, where we recall that $\mathbb{E}$ is the expected value *as defined in (4.4).*

(b) *Using item* **(a)***, show that for every $t, s > 0$, the inequality*

$$\gamma_n\big(\{x \in \mathbb{R}^n : f(x) - \mathbb{E}(f) \ge s\}\big) \le e^{t^2 - ts}$$

holds.

(c) *Prove that*

$$\gamma_n\big(\{x \in \mathbb{R}^n : |f(x) - \mathbb{E}(f)| \ge s\}\big) \le 2e^{-s^2/4}$$

for any $s > 0$.

Solution 4.31. (a) We first write

$$\mathbb{E}\big(\exp(t(f-\mathbb{E}(f))\big) = \int_{\mathbb{R}^n} e^{t(f-\mathbb{E}(f))}\, d\gamma_n = e^{-t\mathbb{E}(f)} \int_{\mathbb{R}^n} e^{tf}\, d\gamma_n.$$

Now, we apply Jensen's inequality (Theorem 1.2.2 in the main book) for the function $\alpha \mapsto e^{-t\alpha}$, which is clearly convex. This leads to

$$e^{-t\mathbb{E}(f)} \le \mathbb{E}(e^{-tf}) = \int_{\mathbb{R}^n} e^{-tf}\, d\gamma_n.$$

Thus, from Exercise 4.30, we get immediately the desired inequality.

(b) For every $s > 0$, we may write

$$\begin{aligned} e^{t^2} \ge \mathbb{E}\big(\exp(t(f-\mathbb{E}(f))\big) &= \int_{\mathbb{R}^n} e^{t(f-\mathbb{E}(f))}\, d\gamma_n \ge \int_{\{x\in\mathbb{R}^n : f(x)-\mathbb{E}(f)\ge s\}} e^{t(f-\mathbb{E}(f))}\, d\gamma_n \ge \\ &\ge \int_{\{x\in\mathbb{R}^n : f(x)-\mathbb{E}(f)\ge s\}} e^{ts}\, d\gamma_n = e^{ts}\gamma_n\big(\{x \in \mathbb{R}^n : f(x) - \mathbb{E}(f) \ge s\}\big), \end{aligned}$$

and dividing both sides by e^{ts}, we get what we wanted to show.

(c) Let $s > 0$ be fixed. The second-degree polynomial $t \mapsto t^2 - ts$ is clearly minimized for $t = s/2$, and the minimum attained value is $-s^2/4$. Therefore, letting $t = s/2$ in the inequality given in item **(b)**, we obtain

$$\gamma_n\big(\{x \in \mathbb{R}^n : f(x) - \mathbb{E}(f) \ge s\}\big) \le e^{-s^2/4}.$$

Repeating the same argument for $-f$, we obviously arrive at

$$\gamma_n\big(\{x \in \mathbb{R}^n : -f(x) - \mathbb{E}(-f) \ge s\}\big) \le e^{-s^2/4}.$$

Since $-\mathbb{E}(-f) = \mathbb{E}(f)$, and because

$$\{x \in \mathbb{R}^n : |f(x) - \mathbb{E}(f)| \geq s\} = \{x \in \mathbb{R}^n : f(x) - \mathbb{E}(f) \geq s\} \cup \{x \in \mathbb{R}^n : -f(x) + \mathbb{E}(f) \geq s\},$$

we get from the bounds above that

$$\gamma_n\big(\{x \in \mathbb{R}^n : |f(x) - \mathbb{E}(f)| \geq s\}\big) \leq 2e^{-s^2/4}.$$

The proof is complete.

□

For the sake of completeness, we introduce now some other concepts and some terminology from probability theory (as a very brief introduction). It is usual to use the symbol $\mathbb{P}$ to represent a probability measure, and hence, we denote a probability space by $(\Omega, \mathbb{P})$. In the language of probability theory, a measurable function $X\colon \Omega \to \mathbb{R}$ is also called a *random variable*. The *variance* of a random variable X is defined as

$$\mathrm{Var}[X] := \mathbb{E}\big((X - \mathbb{E}(X))^2\big).$$

The *standard deviation* of X, denoted by σ_X, is defined to be the square root of the variance. That is,

$$\sigma_X := \sqrt{\mathrm{Var}[X]}.$$

Exercise 4.32. *Let X be a random variable in a probability space $(\Omega, \mathbb{P})$.*

(a) *Show that*

$$\mathrm{Var}[X] = \mathbb{E}(X^2) - \mathbb{E}(X)^2.$$

(b) *Verify that for any constants $a, b \in \mathbb{R}$, we have*

$$\mathrm{Var}[aX + b] = a^2\mathrm{Var}[X].$$

Solution 4.32. **(a)** We simply calculate

$$\begin{aligned}\mathrm{Var}[X] &= \int_\Omega \left(X - \int_\Omega X\, d\mathbb{P}\right)^2 d\mathbb{P} = \\ &= \int_\Omega X^2\, d\mathbb{P} - 2\int_\Omega X\left(\int_\Omega X\right) d\mathbb{P} + \int_\Omega \left(\int_\Omega X\, d\mathbb{P}\right)^2 d\mathbb{P} = \\ &= \int_\Omega X^2\, d\mathbb{P} - \left(\int_\Omega X\, d\mathbb{P}\right)^2 = \mathbb{E}(X^2) - \mathbb{E}(X)^2.\end{aligned}$$

(b) It is immediate that $E(aX + b) = aE(X) + b$. Thus,

$$\begin{aligned}\mathrm{Var}[aX + b] &= \mathbb{E}\big((aX + b - a\mathbb{E}(X) - b)^2\big) = \mathbb{E}\big(a^2(X - \mathbb{E}(X)^2\big) = \\ &= a^2\mathbb{E}\big((X - \mathbb{E}(X))^2\big) = a^2\mathrm{Var}[X].\end{aligned}$$

□

Finally, let X be a random variable on a probability space $(\Omega, \mathbb{P})$, and assume that there is a measurable function $f_X : \mathbb{R} \to \mathbb{R}$ with the property that

$$\mathbb{P}\big(a \leq X \leq b\big) = \int_a^b f_X(t)\, dt$$

for any $a, b \in \mathbb{R}$ (where dt stands for integration with respect to the usual Lebesgue measure of $\mathbb{R}$). In this case, we say that f_X is the *probability density function* of X.

Exercise 4.33 (Chebyshev)**. (a)** *Let X be a random variable in a probability space $(\Omega, \mathbb{P})$. Prove that for every $k > 0$, the inequality*

$$\mathbb{P}\Big(\big\{x \in \Omega : |X(x) - \mathbb{E}(X)| \geq k\big\}\Big) \leq \frac{\mathrm{Var}[X]}{k^2}$$

holds. Notice that this can be seen as an inequality for the deviation from the mean.

(b) *There is a way of restating the inequality above in terms of standard deviations. Show that for every $c > 0$, we have*

$$\mathbb{P}\Big(\big\{x \in \Omega : |X(x) - \mathbb{E}(X)| \geq c\sigma_X\big\}\Big) \leq \frac{1}{c^2}.$$

Solution 4.33. (a) We just have to use Markov's inequality for the function $(X - \mathbb{E}(X))^2$:

$$\begin{aligned}\mathbb{P}\Big(\big\{x \in \Omega : |X(x) - \mathbb{E}(X)| \geq k\big\}\Big) &= \mathbb{P}\Big(\big\{x \in \Omega : (X(x) - \mathbb{E}(X))^2 \geq k^2\big\}\Big) \leq \\ &\leq \frac{1}{k^2} \int_\Omega (X(x) - \mathbb{E}(x))^2\, d\mathbb{P}(x) = \frac{\mathrm{Var}[X]}{k^2}.\end{aligned}$$

(b) Simply let $k = c\sigma_X$ in the inequality derived in item **(a)**. The desired result comes from the definition of σ_X.

□

A convex body $K \subseteq \mathbb{R}^n$ is said to be *isotropic* if it has its centroid at the origin and if there exists a constant $\rho \in \mathbb{R}$ such that

$$\int_K \langle x, u\rangle^2\, dx = \rho^2$$

for every $u \in \mathbb{S}^{n-1}$, where the integration is taken with respect to the usual Lebesgue measure of $\mathbb{R}^n$, of course. It is worth mentioning that most authors demand additionally that $\mathrm{vol}_n(K) = 1$, but we will not do it here. Hence, for us, the isotropy property is invariant under rescaling.

Exercise 4.34. *Let $K \subseteq \mathbb{R}^n$ be a convex body whose volume is* 1*. The restriction of the usual Lebesgue measure endows a probability measure on K, and hence, we regard (K, vol_n) as a probability space.*

(a) *For each $u \in \mathbb{S}^{n-1}$, let $X_u : K \to \mathbb{R}$ be the random variable defined as*

$$X_u(x) = \langle x, u\rangle$$

for each $x \in K$. Verify that X_u has probability density function $f_u : \mathbb{R} \to \mathbb{R}$ given as

$$f_u(t) = \mathrm{vol}_{n-1}\big(K \cap \{u^\perp + tu\}\big),$$

for $t \in \mathbb{R}$.

(b) *Show that if K is isotropic, then each random variable X_u as defined in item* **(a)** *has variance ρ^2.*

Solution 4.34. **(a)** We simply use Cavalieri's principle to write

$$\mathbb{P}(a \le X_u(x) \le b) = \int_{K \cap \{x : a \le \langle x,u\rangle \le b\}} d\mathrm{vol}_n = \int_a^b \int_{K \cap \{u + tu^\perp\}} d\mathcal{H}^{n-1}\, dt = \\ = \int_a^b \mathrm{vol}_{n-1}\big(K \cap \{u + tu^\perp\}\big)\, dt.$$

(b) Let $u \in \mathbb{S}^{n-1}$ be arbitrarily given. We compute:

$$\mathrm{Var}[X_u] = \int_K \left(X_u(x) - \int_K X_u(z)\, dz\right)^2 dx = \\ = \int_K \langle x,u\rangle^2 - 2\langle x,u\rangle \int_K \langle z,u\rangle\, dz + \left(\int_K \langle z,u\rangle\, dz\right)^2 dx = \\ = \int_K \langle x,u\rangle^2\, dx - 2\left(\int_K \langle x,u\rangle\, dx\right)\left(\int_K \langle z,u\rangle\, dz\right) + \int_K \left(\int_K \langle z,u\rangle\, dz\right)^2 dx = \\ = \rho^2 - \left(\int_K \langle x,u\rangle\, dx\right)^2.$$

Since K is isotropic, by definition, it follows that K has its centroid at the origin. Thus,

$$\int_K \langle x,u\rangle\, dx = 0.$$

Hence, we have the desired equality.

□

For Further Reading

For presenting at least some interesting related references, we choose the following themes from them treated in the exercises here: *volume and surface area of special types of convex bodies* and, further on, *interesting phenomena regarding volume and surface area*. With these restrictions, we are also close to the contents of Chapter 4 of the main book.

A natural topic strongly related to the latter keywords is given by the famous *isoperimetric problem* and also by the *isoperimetric inequality*, combining volume and surface area. Since this theme is handled in Chapter 5 of the main book, we do not discuss it here. This holds also for Cavalieri's principle (although it refers to Chapter 4 of [29]).

Volumes of Certain Types of Convex Bodies

We start with a few results around the *volume* of (special types of) convex bodies. It is well-known that for $n \to \infty$, the *volume of the unit ball* of Euclidean n-space goes down to zero and, simultaneously, concentrates on the thin shell near the bounding sphere. There are many different approaches to this fact and related phenomena. For example, Hayes [137] describes in one of the best mathematical writings of 2012 the behavior of an n-cube C of side length 2 and an inscribed n-ball B of radius 1 for increasing dimension, the *maximality of the volume of* B in dimension 5, and of its *surface area* in dimension 7. Since B remains inscribed to C, e.g., already in dimension 100, say, the part of C inside B is almost of measure zero, although B still touches each facet of C and no larger ball finds place inside B. See also Section 13.1 of the book [189] and Artin's booklet [21]. Rabiei [209] discusses related mass concentration phenomena in balls, and he gives also exact formulas for *ball caps* and upper bounds for related volume ratios; cf. also Chapter 13 in [189]. Two further related papers are [106], giving also a recursion formula for the ball volume, and [237], showing different derivations for the volume of the n-ball, for n even, and for n odd.

On Volumes of Convex Polytopes

We come now to some references that show the development of the geometrically natural approach to *polytope volumes via decompositions into simplices with disjoint interiors*. Hohenbalken [253] studies subdivisions of an arbitrary convex polytope P into a minimal number of simplices, having one vertex of P in common and yielding the volume of P and its centroid. Cohen and Hickey [77] use analogously, for n-polytopes P defined by linear systems of inequalities, simplices that are determined by a fixed vertex and n vertices from a suitable triangulation of the boundary of P. Based on *pyramids over faces* instead of simplices, Lasserre [158] gives a better algorithm for the volume of P, applicable even when P has redundant constraints, or empty interior. Similarly, Speevak [241] uses types of *generalized pyramids* to get the volume of P in a faster way. On the other hand, Schneider [230] uses partitions into special simplices called *orthoschemes* to get the volume of P as sum (depending on orientations and signs) of volumes of orthoschemes. For further such results, we refer to subsection 4.7.4 in the main book.

And we mention here Emert and Nelson [88]: they show that for a regular convex n-polytope, there is a nice relation between its *volume and surface area*, namely, that the surface area is a derivative of the volume with respect to the inner radius (i.e., the minimal distance from the center of it to the boundary). Conditions are given to get analogous results for arbitrary polytopes. In Chapters 4, 7, and 8 of the recent book [213] of Robins, Fourier transforms on polytopes and polytope volumes plus their discretizations via integer points are treated. Many nice examples, exercises, and solutions are presented.

Surface Areas of Polytopes

In a summarizing sense, we point out that there are two basic expository papers that show approaches to volumes of convex bodies and convex polytopes and various related problems in a comprehensive way. The first one is [112] by Gritzmann and Klee, giving various problems and results from computational convexity, particularly about the *computation of volumes and mixed volumes* of convex polytopes and more general convex bodies. For example, needed triangulations of polytopes, approximation methods, projections and sections, volumes of unions and intersections, discretizations via contained integer points, and various further aspects are broadly discussed. The second one is the handbook article [139] of Henk et al., in part 2 of which metric properties of polytopes (mainly around volume, surface area, mixed volumes, and intrinsic volumes) are presented.

We continue with a few results explicitly on *surface areas of convex polytopes* and start with Minkowski's existence theorem and Alexandrov's uniqueness theorem for polytopes. Up to translation, any convex n-polytope is uniquely determined by the outer normal vectors of all its m facets, in each case presenting the area of the corresponding facet by its length. On the other hand, every "vector star" (of cardinality m) not contained in a half-space and summing up to zero determines, via the described geometric correspondence, a unique translation class of convex n-polytopes with m facets. Thus, the *surface area of a convex polytope* is the *total length* of all these m vectors. These and further related questions (e.g., how can one construct a convex n-polytope from its "vector star") are nicely presented in the appendix (part A.4) of Gardner's book [97]; see also subsection 8.2.1 in Schneider's monograph [228]. In the paper [111], Gritzmann and Hufnagel treat the algorithmic complexity of this construction problem.

We give now a few references treating results on polytopes and their "vector stars." Let $q(1), \ldots, q(n)$ denote positive numbers satisfying $q(i_1) + \ldots + q(i_{n-1}) < q(i_n)$ for every permutation $i_1, i_2, \ldots, i_n$ of $1, 2, \ldots, n$. Larman [155] proves that then, for each dimension $n > 1$, the numbers $q(i)$ present the facet areas of some convex n-polytope. Continuing the work of A. D. Alexandrov and Zalgaller, Martinez-Maure [184] derives five conditions that are necessary for the existence of a convex 3-polytope whose unit outer normals and facet perimeters are prescribed. Their sufficiency is also shown, and to get these results, he uses a classical theorem of Steinitz, geometric duality, and Minkowski's related theorem (replacing perimeters by areas). Clearly, the next step would be the extension to higher dimensions.

L. Fejes Tóth [91] studies the following problem for an arbitrary convex n-polytope P with m facets: for every facet F of P, take the ratio of the $(n-1)$-volume of F to the maximal $(n-1)$-volume generated by a parallel hyperplane section of P. Writing $S_n(P)$ for the sum of these m ratios, he shows in [91] that $S_2(P)$ is bounded by $\sqrt{8}$ and 4; the lower bound is improved in [202] to 3. Vasin and Nikonorov [249] extend this problem suitably to n-dimensional convex bodies and give some estimates. They show that for any dimension, a convex body K exists whose analogously defined number $S_n(K)$ takes the largest (respectively, smallest) possible value. Abrosimov et al. [2] collect many historical facts and connections concerning problems of *maximizing and minimizing volumes of polytopes whose facet areas are prescribed*. Besides the many new results contained in [2], this paper can also be recommended as a comprehensive exposition of results from this subject.

PLATE 5

Difference bodies

The cuboctahedron as difference body of a tetrahedron

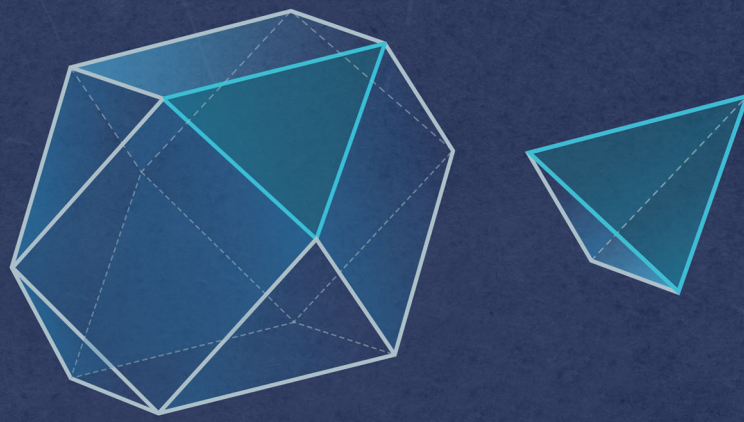
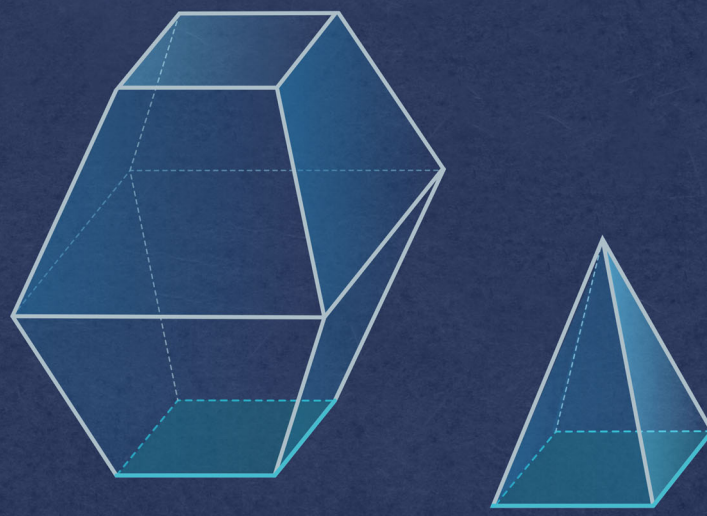


Figure 11

Figure 12

The difference body of a square pyramid



Difference bodies of convex polyhedra ► (see, e.g., 5.3 and 5.7.2)

Chapter 5
Classical Inequalities



"To create a healthy philosophy, you should renounce metaphysics but be a good mathematician."

— Bertrand Russell

The exercises from Chapter 5 of our main book refer to the following themes: *linear combinations of convex bodies, volume of sums and differences of convex bodies*, the *Rogers-Shephard inequality, volume of intersections with translates, Steiner points, Santaló points*, the *covariogram, volume and diameter* with the *isodiametric inequality, mean width*, contained *cross-polytopes of maximal volume, Mahler's and Urysohn's inequality* (also variants thereof), and *Asplund products*.

Plate 5:

Among various other topics, *difference bodies* are treated in Section 5.3 and subsection 5.7.3 of the main book (see also Plate 5, where two polyhedral examples are presented). These bodies associated with a given convex body have many applications, for instance, referring to generalizations of themselves (e.g., as *Orlicz difference bodies*), problems on *longest chords* of convex sets, *width functions* of convex bodies, in particular yielding *bodies of constant width* in normed spaces, and *covariograms*.

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V. Balestro et al., *Convexity from the Geometric Point of View: Exercises and Solutions*, Cornerstones, https://doi.org/10.1007/978-3-031-83937-5_5

Exercises and Solutions

We note that a *dilation* is a non-translational affine transform where the image of each line g is parallel to g. Geometric figures related by dilations are said to be *homothetic* (or homothets).

Exercise 5.1. *Let $A, B \subseteq \mathbb{R}^n$ be arbitrarily given non-empty compact sets. Prove that for every $\alpha, \beta > 0$, the inequality*

$$\mathrm{vol}_n(\alpha A + \beta B)^{1/n} \geq \alpha \mathrm{vol}_n(A)^{1/n} + \beta \mathrm{vol}_n(B)^{1/n}$$

holds. Show that if A and B are convex bodies, then equality holds for some $\alpha, \beta > 0$ if and only if A and B are homothetic, i.e., related by a dilation.

Solution 5.1. This follows directly from Theorem 5.1.2 in the main book, because the inequality in the exercise is clearly equivalent to the inequality

$$\mathrm{vol}_n\left(\frac{\alpha}{\alpha+\beta}A + \frac{\beta}{\alpha+\beta}B\right)^{1/n} \geq \frac{\alpha}{\alpha+\beta}\mathrm{vol}_n(A)^{1/n} + \frac{\beta}{\alpha+\beta}\mathrm{vol}_n(B)^{1/n}.$$

Thus, we just have to let $\lambda = \alpha/(\alpha+\beta)$, $K_0 = A$ and $K_1 = B$ in the aforementioned theorem.

□

Exercise 5.2. *Let K and L be convex bodies in $\mathbb{R}^n$. Prove that the equality*

$$\mathrm{vol}_n((1-\lambda)K + \lambda L) = \mathrm{vol}_n(K)^{1-\lambda}\mathrm{vol}_n(L)^{\lambda}$$

holds for some $\lambda \in (0,1)$ if and only if L is a translated copy of K.

Solution 5.2. The equality obviously holds if L is a translated copy of K, and hence, we just have to prove the converse. First, notice that Theorem 5.1.2 and Lemma 1.5.1 in the main book give that the inequality

$$\mathrm{vol}_n((1-\lambda)K + \lambda L)^{1/n} \geq (1-\lambda)\mathrm{vol}_n(K)^{1/n} + \lambda \mathrm{vol}_n(L)^{1/n} \geq \mathrm{vol}_n(K)^{\frac{1-\lambda}{n}}\mathrm{vol}_n(L)^{\frac{\lambda}{n}} \tag{5.1}$$

holds for any convex bodies $K, L \subseteq \mathbb{R}^n$ and every $\lambda \in (0,1)$. If $K, L \in \mathcal{K}^n$ and $\lambda \in (0,1)$ are such that

$$\mathrm{vol}_n((1-\lambda)K + \lambda L) = \mathrm{vol}_n(K)^{1-\lambda}\mathrm{vol}_n(L)^{\lambda},$$

then we have equality in both inequalities in (5.1). Thus, the equality case of Theorem 5.1.2 gives that K and L are homothetical. That is, we may write $L = x_0 + \beta K$ for some $x_0 \in \mathbb{R}^n$ and some $\beta > 0$. All we have to prove is that $\beta = 1$. Since the volume is translation-invariant, we may write

$$\mathrm{vol}_n((1-\lambda)K + \lambda\beta K) = \mathrm{vol}_n(K)^{1-\lambda}\mathrm{vol}_n(\beta K)^{\lambda},$$

and this leads immediately to the equality

$$1 - \lambda + \lambda\beta = \beta^{\lambda}.$$

We prove that this equality only holds for $\beta = 1$. In fact, consider the function

$$f(\beta) = \beta^{\lambda} - \lambda\beta + \lambda - 1,$$

defined for $\beta \geq 0$, and clearly continuous in $[0, +\infty)$ and differentiable in $(0, +\infty)$. It is clear that $f(0) = \lambda - 1 < 0$ (because $\lambda < 1$). The derivative of this function is given as

$$f'(\beta) = \lambda\beta^{\lambda-1} - \lambda = \lambda(\beta^{\lambda-1} - 1).$$

Since $0 < \lambda < 1$, we get that $f'(\beta) > 0$ for $\beta \in (0, 1)$ and $f'(\beta) < 0$ for $\beta \in (1, +\infty)$. Because $f(1) = 0$ we get that $f < 0$ for both $\beta \in [0, 1)$ and $\beta > 1$. That is, the unique root of f is $\beta = 1$. This gives the desired result.

□

Exercise 5.3. *Recall that $w(K)$ denotes the mean width of a set $K \in CC^n$. Verify that*

$$|w(K) - w(L)| \leq 2d_{\mathcal{H}}(K, L)$$

for any $K, L \in CC^n$. Conclude that the mean width is uniformly continuous in the Hausdorff metric.

Solution 5.3. This follows immediately from the fact that

$$d_{\mathcal{H}}(K, L) = ||h_K - h_L||_{C(\mathbb{S}^{n-1})}.$$

See item **(ii)** of Proposition 2.5.2 from the main book.

□

Exercise 5.4. *Prove that the Minkowski symmetrization introduced in Exercise 2.58 preserves the mean width. That is, show that for any $K \in CC^n$ and any $u \in \mathbb{S}^{n-1}$, we have that*

$$w(M_u(K)) = w(K).$$

Solution 5.4. We first notice that $\mathrm{ref}_{u^\perp}$ is volume-preserving. Indeed, let $\{u_1, \ldots, u_n\}$ be any orthonormal basis of $\mathbb{R}^n$ such that $u_n = u$. The matrix representation of $\mathrm{ref}_{u^\perp}$ in this basis is clearly the diagonal matrix $\mathrm{diag}(1, \ldots, 1, -1)$. Hence, $\det(\mathrm{ref}_{u^\perp}) = -1$.

Also, notice that the restriction of $\mathrm{ref}_{u^\perp}$ to $\mathbb{S}^{n-1}$ is clearly a bijection.

Among other characterizations, the standard probability measure σ on $\mathbb{S}^{n-1}$ is a scalar multiple of the conical measure $\nu_{\mathbb{S}^{n-1}}$ defined above Proposition 4.5.1 in the main book. It is immediate from the definition that $\nu_{\mathbb{S}^{n-1}}$ is invariant under volume-preserving linear isomorphisms that are bijections when restricted to $\mathbb{S}^{n-1}$. Therefore, we get

$$\sigma = (\mathrm{ref}_{u^\perp})_*\sigma.$$

In other words, σ is invariant under $\mathrm{ref}_{u^\perp}$.

Now, from the definition of the Minkowski symmetral $M_u(K)$, and as a consequence of item **(e)** of Exercise 2.58, we compute

$$w(M_u(K)) = 2\int_{\mathbb{S}^{n-1}} h(M_u(K), v)\, d\sigma(v) = 2\int_{\mathbb{S}^{n-1}} h\left(\frac{1}{2}\left(\mathrm{ref}_{u^\perp}(K) + K\right), v\right) d\sigma(v) =$$

$$= \int_{\mathbb{S}^{n-1}} h(K, v)\, d\sigma(v) + \int_{\mathbb{S}^{n-1}} h(\mathrm{ref}_{u^\perp}(K), v)\, d\sigma(v) = \frac{w(K)}{2} + \int_{\mathbb{S}^{n-1}} h(K, \mathrm{ref}_{u^\perp}(v))\, d\sigma(v).$$

To compute the latter integral, we just consider the change of variables $z = \mathrm{ref}_{u^\perp}(v)$, and because σ is invariant under $\mathrm{ref}_{u^\perp}$, we get

$$w(M_u(K)) = \frac{w(K)}{2} + \int_{\mathbb{S}^{n-1}} h(K, \mathrm{ref}_{u^\perp}(v))\, d\sigma(v) = \frac{w(K)}{2} + \int_{\mathbb{S}^{n-1}} h(K,z)\, d\sigma(z) = \\ = \frac{w(K)}{2} + \frac{w(K)}{2} = w(K),$$

as we wanted to prove.

□

Exercise 5.5. *The* ℓ-norm *of a convex body* $K \in \mathcal{K}_0^n$ *is defined as*

$$\ell(K) = \frac{1}{(2\pi)^{n/2}} \int_{\mathbb{R}^n} ||x||_K e^{-\frac{|x|^2}{2}}\, dx.$$

Prove that for any $K \in \mathcal{K}_0^n$, *we have the equality*

$$2\ell(K) = \ell(B^n) w(K^\circ).$$

Solution 5.5. Let us first express $\ell(B^n)$. Integrating by polar coordinates, we get

$$\ell(B^n) = \frac{1}{(2\pi)^{n/2}} \int_{\mathbb{R}^n} ||x||_{B^n} e^{-\frac{|x|^2}{2}}\, dx = \frac{1}{(2\pi)^{n/2}} \int_0^{+\infty} \int_{\mathbb{S}^{n-1}} t^n e^{-\frac{t^2}{2}}\, d\mathcal{H}^{n-1} dt = \\ = \frac{\omega_{n-1}}{(2\pi)^{n/2}} \int_0^{+\infty} t^n e^{-\frac{t^2}{2}}\, dt,$$

because $||\cdot||_{B^n} = |\cdot|$. Now, we integrate by polar coordinates again to write

$$2\ell(K) = \frac{2}{(2\pi)^{n/2}} \int_0^{+\infty} \int_{\mathbb{S}^{n-1}} t^n ||u||_K e^{-\frac{t^2}{2}}\, d\mathcal{H}^{n-1}(u) dt = \\ = \frac{2}{(2\pi)^{n/2}} \int_0^{+\infty} t^n e^{-\frac{t^2}{2}}\, dt \int_{\mathbb{S}^{n-1}} ||u||_K\, d\mathcal{H}^{n-1}(u) = \\ = \ell(B^n) \frac{2}{\omega_{n-1}} \int_{\mathbb{S}^{n-1}} h_{K^\circ}(u)\, d\mathcal{H}^{n-1}(u) = \ell(B^n) 2 \int_{\mathbb{S}^{n-1}} h_{K^\circ}(u)\, d\sigma(u) = \ell(B^n) w(K^\circ).$$

□

Exercise 5.6. *Let* $K \subseteq \mathbb{R}^n$ *be a convex body.*

(a) *Show that* $K - K$ *contains two copies of* $-K$ *(or* K*) that intersect only on their boundaries.*

(b) *Conclude from item* **(a)** *that*

$$\mathrm{vol}_n(K - K) \geq 2\mathrm{vol}_n(K),$$

which is a weaker bound than the one obtained by the Brunn-Minkowski inequality.

Solution 5.6. **(a)** Let $u \in \mathbb{S}^{n-1}$ be an unit outer normal vector of K at a point $y \in \partial K$, and assume that $-u$ is a unit outer normal of K at $z \in \partial K$. Translating K in order that the midpoint of $\mathrm{seg}[y, z]$ coincides with the origin, we may assume that u is a unit outer normal vector of K at x and $-u$ is a unit outer normal vector of K at $-x$. Hence,

$$\langle -x, u\rangle \leq \langle z, u\rangle \leq \langle x, u\rangle$$

for any $z \in K$. It follows that $\langle x - z, u\rangle \geq 0$ for every $z \in K$, and this gives that $x - K \in \mathbf{h}^+(u, 0)$. On the other hand, we also get from the above that $\langle -x - z, u\rangle \leq 0$ for all $z \in K$, and then it follows that $-x - K \subseteq \mathbf{h}^-(u, 0)$. Therefore, $x - K$ and $-x - K$ are translates of $-K$ contained in $K - K$ (because x and $-x$ are points of K), which only intersect at their boundaries.

(b) From the above, we have

$$\mathrm{vol}_n(K - K) \geq \mathrm{vol}_n((x - K) \cup (-x - K)) = 2\mathrm{vol}_n(K).$$

□

Exercise 5.7 (The Steiner point)**.** *We define the* Steiner point *of a given set $K \in CC^n$ as*

$$\mathrm{st}(K) := \frac{1}{\kappa_n} \int_{\mathbb{S}^{n-1}} h_K(u)u \, d\mathcal{H}^{n-1}(u).$$

Of course, this defines a map $\mathrm{st}\colon CC^n \to \mathbb{R}^n$*, which we call the* Steiner point map*.*

(a) *Show that the Steiner point map is* Minkowski linear*, meaning that* $\mathrm{st}(\lambda K + L) = \lambda \mathrm{st}(K) + \mathrm{st}(L)$ *for any $\lambda \in \mathbb{R}$ and every $K, L \in CC^n$.*

(b) *Prove that the Steiner point map is* translation-equivariant*, meaning that* $\mathrm{st}(x_0 + K) = x_0 + \mathrm{st}(K)$ *for any $x_0 \in \mathbb{R}^n$ and all $K \in CC^n$.*

(c) *Let $K \in CC^n$ be centrally symmetric with respect to $x_0 \in \mathbb{R}^n$. Prove that* $\mathrm{st}(K) = x_0$*.*

(d) *Verify that the Steiner point map is* $\mathrm{O}(n)$-equivariant*, meaning that* $\mathrm{st}(TK) = T(\mathrm{st}(K))$ *for any $T \in \mathrm{O}(n)$ and every $K \in CC^n$.*

(e) *Show that for any $K, L \in CC^n$, the estimate*

$$|\mathrm{st}(K) - \mathrm{st}(L)| \leq n d_{\mathcal{H}}(K, L)$$

holds. Conclude that the Steiner point map is uniformly continuous in the Hausdorff metric.

Remark 5.1. In [223], Schneider proved that (for dimension $n \geq 2$) the Steiner point map is the unique map $f\colon CC^n \to \mathbb{R}^n$, which is Minkowski linear, translation-equivariant, $\mathrm{SO}(n)$-equivariant, and continuous in the Hausdorff metric.

Solution 5.7. **(a)** For $\lambda \geq 0$, this follows directly from the fact that $h_{\lambda K+L} = \lambda h_K + h_L$. If $\lambda < 0$, we just need to use the fact that the spherical measure is invariant under $-\mathrm{Id}$ and notice that

$$\begin{aligned}\mathrm{st}(-K) = \frac{1}{\kappa_n} \int_{\mathbb{S}^{n-1}} h_{-K}(u)u \, d\mathcal{H}^{n-1}(u) = \frac{1}{\kappa_n} \int_{\mathbb{S}^{n-1}} h_K(-u)u \, d\mathcal{H}^{n-1}(u) = \\ = \frac{1}{\kappa_n} \int_{\mathbb{S}^{n-1}} h_K(v)(-v) \, d\mathcal{H}^{n-1}(v) = -\mathrm{st}(K).\end{aligned}$$

(b) We only need to show that $\mathrm{st}(x_0) = x_0$ for any $x_0 \in \mathbb{R}^n$, since then the desired follows immediately from item **(a)**. Observe that the map

$$x \mapsto \mathrm{st}(x) = \frac{1}{\kappa_n} \int_{\mathbb{S}^{n-1}} h_{\{x\}}(u) u \, d\mathcal{H}^{n-1}(u) = \frac{1}{\kappa_n} \int_{\mathbb{S}^{n-1}} \langle x, u \rangle u \, d\mathcal{H}^{n-1}(u)$$

is a linear transformation of $\mathbb{R}^n$. Notice that our objective is to prove that this linear transformation is the identity of $\mathbb{R}^n$. To do that, we just have to investigate st over the canonical basis $\{e_1, \ldots, e_n\}$ of $\mathbb{R}^n$. For simplicity, let $j = 1$ (the other cases are completely analogous). Writing in coordinates $u = (u_1, \ldots, u_n)$, we get

$$\mathrm{st}(e_1) = \frac{1}{\kappa_n} \int_{\mathbb{S}^{n-1}} \langle e_1, u \rangle u \, d\mathcal{H}^{n-1}(u) =$$

$$= \left(\frac{1}{\kappa_n} \int_{\mathbb{S}^{n-1}} u_1^2 \, d\mathcal{H}^{n-1}(u), \frac{1}{\kappa_n} \int_{\mathbb{S}^{n-1}} u_1 u_2 \, d\mathcal{H}^{n-1}(u), \ldots, \frac{1}{\kappa_n} \int_{\mathbb{S}^{n-1}} u_1 u_n \, d\mathcal{H}^{n-1}(u) \right).$$

Now, we compute the integrals. To the first one, we observe that

$$\int_{\mathbb{S}^{n-1}} u_j^2 \, d\mathcal{H}^{n-1}(u) = \int_{\mathbb{S}^{n-1}} u_k^2 \, d\mathcal{H}^{n-1}(u)$$

for any $j, k = 1, \ldots, n$. This follows from the $\mathrm{O}(n)$-invariance of $\mathcal{H}^{n-1}|_{\mathbb{S}^{n-1}}$ (as explained in Remark 4.5.2 of the main book). We simply let $T \in \mathrm{O}(n)$ be the orthogonal transformation that interchanges e_j and e_k and let all other vectors of the canonical basis unaltered. As a consequence, we get

$$n\kappa_n = \int_{\mathbb{S}^{n-1}} \langle u, u \rangle \, d\mathcal{H}^{n-1}(u) = \int_{\mathbb{S}^{n-1}} u_1^2 + \ldots + u_n^2 \, d\mathcal{H}^{n-1}(u) = n \int_{\mathbb{S}^{n-1}} u_1^2 \, d\mathcal{H}^{n-1}(u),$$

and then

$$\frac{1}{\kappa_n} \int_{\mathbb{S}^{n-1}} u_1^2 \, d\mathcal{H}^{n-1}(u) = 1.$$

In the case where $j \neq k$, let $T \in \mathrm{O}(n)$ be the orthogonal transformation such that $T(e_m) = e_m$ for $m \neq j$, and $T(e_j) = -e_j$. We have

$$\int_{\mathbb{S}^{n-1}} u_j u_k \, d\mathcal{H}^{n-1}(u) = \int_{\mathbb{S}^{n-1}} -u_j u_k \, d\mathcal{H}^{n-1}(u),$$

and this gives

$$\int_{\mathbb{S}^{n-1}} u_j u_k \, d\mathcal{H}^{n-1}(u) = 0$$

whenever $j \neq k$.

It follows from these calculations that $\mathrm{st}(e_1) = e_1$. As it was mentioned above, the argument is completely analogous for $j = 2, \ldots, n$, and this shows that st is the identity when restricted to singletons.

(c) We know that $K - x_0 = -(K - x_0) = -K + x_0$. Then,

$$\mathrm{st}(K) - x_0 = \mathrm{st}(K - x_0) = \mathrm{st}(-K + x_0) = \mathrm{st}(-K) + x_0 = -\mathrm{st}(K) + x_0,$$

and the desired statement follows.

(d) Since T is orthogonal, we have $T^* = T^{-1}$. We use Remark 4.5.2 from the main book to write

$$\begin{aligned}\mathrm{st}(TK) &= \frac{1}{\kappa_n}\int_{\mathbb{S}^{n-1}} h_{TK}(u)u\, d\mathcal{H}^{n-1}(u) = \frac{1}{\kappa_n}\int_{\mathbb{S}^{n-1}} h_K(T^*u)u\, d\mathcal{H}^{n-1}(u) = \\ &= \frac{1}{\kappa_n}\int_{\mathbb{S}^{n-1}} h_K(T^*Tu)Tu\, d\mathcal{H}^{n-1}(u) = \frac{1}{\kappa_n}\int_{\mathbb{S}^{n-1}} T(h_K(u)u)\, d\mathcal{H}^{n-1}(u) = \\ &= T\left(\frac{1}{\kappa_n}\int_{\mathbb{S}^{n-1}} h_K(u)u\, d\mathcal{H}^{n-1}(u)\right) = T(\mathrm{st}(K)).\end{aligned}$$

(e) We write

$$\begin{aligned}|\mathrm{st}(K) - \mathrm{st}(L)| &= \left|\frac{1}{\kappa_n}\int_{\mathbb{S}^{n-1}} h_K(u)u\, d\mathcal{H}^{n-1}(u) - \frac{1}{\kappa_n}\int_{\mathbb{S}^{n-1}} h_L(u)u\, d\mathcal{H}^{n-1}(u)\right| \leq \\ \leq \frac{1}{\kappa_n}\int_{\mathbb{S}^{n-1}} \left|(h_K(u) - h_L(u))u\right| d\mathcal{H}^{n-1}(u) &= \frac{1}{\kappa_n}\int_{\mathbb{S}^{n-1}} |h_K(u) - h_L(u)|\, d\mathcal{H}^{n-1}(u) \leq \\ \leq \frac{1}{\kappa_n}\int_{\mathbb{S}^{n-1}} ||h_K - h_L||_{C(\mathbb{S}^{n-1})}\, d\mathcal{H}^{n-1}(u) &= \frac{1}{\kappa_n} d_{\mathcal{H}}(K,L)\mathcal{H}^{n-1}(\mathbb{S}^{n-1}) = n d_{\mathcal{H}}(K,L).\end{aligned}$$

Notice that we used the equality $\mathcal{H}^{n-1}(\mathbb{S}^{n-1}) = n\kappa_n$ and also the fact that the Hausdorff distance between $K, L \in CC^n$ equals the distance between h_K and h_L in the maximum norm of $C(\mathbb{S}^{n-1})$.

□

Exercise 5.8. *Let $K, L \subseteq \mathbb{R}^n$ be convex bodies. Prove that the inequality*

$$\frac{\mathrm{vol}_n(K+L)}{\mathrm{vol}_n(K-L)} \leq \frac{1}{2^n}\binom{2n}{n}$$

holds.

Hint: Notice that $(K+L)-(K+L) = (K-L)-(K-L)$, and use the inequalities for the volume of difference bodies given in Section 5.3 of the main book.

Solution 5.8. We write

$$\begin{aligned}2^n \mathrm{vol}_n(K+L) &\leq \mathrm{vol}_n((K+L)-(K+L)) = \\ = \mathrm{vol}_n((K-L)-(K-L)) &\leq \binom{2n}{n}\mathrm{vol}_n(K-L).\end{aligned}$$

This leads directly to the desired inequality.

□

Let $K, L \subseteq \mathbb{R}^n$ be convex bodies. Define

$$M(K,L) := \sup_{x \in K+L} \mathrm{vol}_n(K \cap (x-L)).$$

Notice that the supremum is attained for some $x_0 \in \mathbb{R}^n$, since the function $x \mapsto \mathrm{vol}_n(K \cap (x-L))$ is clearly continuous and $K+L$ is compact. For $\theta \in [0,1]$, the *θ-convolution body* of K and L is the set

$$K +_\theta L := \left\{x \in K+L : \mathrm{vol}_n(K \cap (x-L)) \geq \theta M(K,L)\right\}.$$

The objective of the next exercises is to study early properties of the θ-convolution bodies. For more in this direction, we refer the reader to [18]. Moreover, some interesting generalizations can be found in [11]. We start by investigating some properties of $M(K, L)$.

Exercise 5.9. *Let $K, L \in \mathcal{K}^n$, $\lambda \geq 0$ and $z \in \mathbb{R}^n$. Show that*

(a) $M(K, L) = M(L, K)$,

(b) $M(z + K, L) = M(K, L)$,

(c) $M(\lambda K, \lambda L) = \lambda^n M(K, L)$.

Solution 5.9. **(a)** The n-dimensional volume is invariant under reflection through the origin and under translations. For any $x \in K + L$, we have

$$\begin{aligned}\mathrm{vol}_n(K \cap (x - L)) = \mathrm{vol}_n((L - x) \cap (-K)) = \mathrm{vol}_n(x + ((L - x) \cap (-K))) = \\ = \mathrm{vol}_n(L \cap (x - K)),\end{aligned}$$

and passing to the supremum on both sides, we get that $M(K, L) = M(L, K)$.

(b) Notice that $x \in z + K + L$ if and only if $y = x - z \in K + L$. Thus, using again the fact that the n-dimensional volume is translation-invariant, we arrive at

$$\begin{aligned}M(z + K, L) = \sup_{x \in z+K+L} \mathrm{vol}_n((z + K) \cap (x - L)) = \sup_{x-z \in K+L} \mathrm{vol}_n(K \cap ((x - z) - L)) = \\ = \sup_{y \in K+L} \mathrm{vol}_n(K \cap (y - L)) = M(K, L).\end{aligned}$$

(c) This is immediate due to the fact that $\lambda A \cap \lambda B = \lambda(A \cap B)$.

□

Exercise 5.10. *Let $K, L \subseteq \mathbb{R}^n$ be convex bodies. Prove that if $K = -L$ or if K and L are origin-symmetric, then*

$$M(K, L) = \mathrm{vol}_n(K \cap (-L)).$$

Solution 5.10. Assume first that $K = -L$. It is clear that

$$\mathrm{vol}_n(K \cap (x - L)) \leq \mathrm{vol}_n(K)$$

for any $x \in K + L$. This gives that $M(K, L) \leq \mathrm{vol}_n(K)$. If $K = -L$, then $0_{\mathbb{R}^n} \in K + L$, and $K \cap (-L) = K$. Thus,

$$M(K, L) \geq \mathrm{vol}_n(K \cap (0_{\mathbb{R}^n} - L)) = \mathrm{vol}_n(K).$$

This gives that $M(K, L) = \mathrm{vol}_n(K) = \mathrm{vol}_n(K \cap (-L))$.

Now, assume that K and L are origin-symmetric convex bodies. Let $x \in K + L$ be arbitrarily given. Since K and L are symmetric with respect to the origin, we have that

$$-[K \cap (x - L)] = (-K) \cap (-x + L) = K \cap (-x - L).$$

Also, it is easy to see that

$$K \cap (x - L) + K \cap (-x - L) \subseteq 2(K \cap (-L)).$$

Indeed, if $y \in K \cap (x - L)$ and $z \in K \cap (-x - L)$, we first notice that we have that $y + z \in 2K$ because $y, z \in K$. Now, we write $y = x - a$ and $z = -x - b$ for $a, b \in L$. Then, $y + z = -a - b \in -2L$. Using the Brunn-Minkowski inequality, we get

$$\begin{aligned} 2\mathrm{vol}_n(K \cap (-L))^{1/n} = \mathrm{vol}_n(2(K \cap (-L)))^{1/n} \geq \mathrm{vol}_n(K \cap (x - L) + K \cap (-x - L))^{1/n} \geq \\ \geq \mathrm{vol}_n(K \cap (x - L))^{1/n} + \mathrm{vol}_n(K \cap (-x - L))^{1/n} = 2\mathrm{vol}_n(K \cap (x - L))^{1/n}. \end{aligned}$$

Since this inequality holds for every $x \in K + L$, passing to the supremum, we get the desired result.

□

Exercise 5.11. *Let $K, L \in \mathcal{K}^n$, $\theta \in [0, 1]$, $x \in \mathbb{R}^n$ and $\lambda \geq 0$. Prove that*

(a) $K +_\theta L = L +_\theta K$,

(b) $(\lambda K) +_\theta (\lambda L) = \lambda(K +_\theta L)$,

(c) $(x + K) +_\theta L = x + (K +_\theta L)$.

Solution 5.11. **(a)** Since $M(K, L) = M(L, K)$ and

$$\begin{aligned} \mathrm{vol}_n(K \cap (x - L)) = \mathrm{vol}_n(-K \cap (L - x)) = \mathrm{vol}_n(x + (L - x) \cap (-K)) = \\ = \mathrm{vol}_n(L \cap (x - K)), \end{aligned}$$

we get that $x \in (\lambda K) + (\lambda L)$ if and only if $x \in L +_\theta K$.

(b) Let $x \in (\lambda K) + (\lambda L)$. We have that

$$\mathrm{vol}_n(\lambda K \cap (x - \lambda L)) \geq \theta M(\lambda K, \lambda L)$$

if and only if

$$\lambda^n \mathrm{vol}_n(K \cap (x/\lambda - L)) \geq \theta \lambda^n M(K, L),$$

and since $x/\lambda \in K + L$, this is clearly equivalent to $x/\lambda \in K +_\theta L$. Thus, $x \in (\lambda K) +_\theta (\lambda L)$ if and only if $x \in \lambda(K +_\theta L)$.

(c) This follows immediately from the translation-invariance of the volume. For every $z \in (x + K) + L$, it holds that

$$\mathrm{vol}_n((x + K) \cap (z - L)) = \mathrm{vol}_n(K \cap (z - x - L)),$$

and hence, $z \in (x + K) +_\theta L$ if and only if $z - x \in K +_\theta L$.

□

Exercise 5.12. **(a)** *Let $K, L \in \mathcal{K}^n$ and assume that*

$$\sup_{x \in K+L} \mathrm{vol}_n(K \cap (x - L)) = \mathrm{vol}_n(K \cap (x_0 - L))$$

for some $x_0 \in K + L$. Let $\tilde{L} := -x_0 + L$. Verify that

$$\sup_{x \in K+\tilde{L}} \mathrm{vol}_n(K \cap (x - \tilde{L})) = \mathrm{vol}_n(K \cap (-\tilde{L})).$$

(b) *Having item* **(a)** *in mind, prove that for any $\theta_1, \theta_2, \lambda_1, \lambda_2 \in [0, 1]$ such that $\lambda_1 + \lambda_2 \leq 1$, we have that*

$$\lambda_1(K +_{\theta_1} \tilde{L}) + \lambda_2(K +_{\theta_2} \tilde{L}) \subseteq K +_{\theta} \tilde{L},$$

where $\theta \in [0, 1]$ is such that $1 - \theta^{\frac{1}{n}} = \lambda_1(1 - \theta_1^{\frac{1}{n}}) + \lambda_2(1 - \theta_2^{\frac{1}{n}})$.

(c) *Show that for any $K, L \in \mathcal{K}^n$ and every $\theta \in [0, 1]$, the θ-convolution body $K +_\theta L$ is a convex set.*

Solution 5.12. (a) Assume that there is $z \in K + \tilde{L}$ such that

$$\mathrm{vol}_n(K \cap (z - \tilde{L})) > \mathrm{vol}_n(K \cap (-\tilde{L})).$$

Thus,

$$\mathrm{vol}_n(K \cap (z + x_0 - L)) > \mathrm{vol}_n(K \cap (x_0 - L)) = \sup_{x \in K+L} \mathrm{vol}_n(K \cap (x - L)).$$

Since $z \in K + \tilde{L}$, we have that $z \in K + L - x_0$, and hence, $z + x_0 \in K + L$. It follows that the inequality above is a contradiction. This gives what we wanted to have.

(b) From the convexity of $\tilde{L}$, we clearly have that

$$\lambda_1(-\tilde{L}) + \lambda_2(-\tilde{L}) + (1 - \lambda_1 - \lambda_2)(-\tilde{L}) = -\tilde{L},$$

and hence, for any $x_1 \in K +_{\theta_1} \tilde{L}$ and every $x_2 \in K +_{\theta_2} \tilde{L}$, it holds that

$$\begin{aligned} K \cap (\lambda_1 x_1 + \lambda_2 x_2 - \tilde{L}) = K \cap \big(\lambda_1(x_1 - \tilde{L}) + \lambda_2(x_2 - \tilde{L}) + (1 - \lambda_1 - \lambda_2)(-\tilde{L})\big) \supseteq \\ \supseteq \lambda_1(K \cap (x_1 - \tilde{L})) + \lambda_2(K \cap (x_2 - \tilde{L})) + (1 - \lambda_1 - \lambda_2)(K \cap (-\tilde{L})). \end{aligned}$$

Using the Brunn-Minkowski inequality, the inclusion above leads to

$$\begin{aligned} \mathrm{vol}_n(K \cap (\lambda_1 x_1 + \lambda_2 x_2 - \tilde{L}))^{\frac{1}{n}} \geq \\ \geq \lambda_1 \mathrm{vol}_n(K \cap (x_1 - \tilde{L}))^{\frac{1}{n}} + \lambda_2 \mathrm{vol}_n(K \cap (x_2 - \tilde{L}))^{\frac{1}{n}} + (1 - \lambda_1 - \lambda_2)\mathrm{vol}_n(K \cap (-\tilde{L}))^{\frac{1}{n}} \geq \\ \geq \lambda_1 \theta_1^{\frac{1}{n}} M(K, \tilde{L})^{\frac{1}{n}} + \lambda_2 \theta_2^{\frac{1}{n}} M(K, \tilde{L})^{\frac{1}{n}} + (1 - \lambda_1 - \lambda_2) M(K, \tilde{L})^{\frac{1}{n}}, \end{aligned} \tag{5.2}$$

where we are using the fact that $M(K, \tilde{L}) = \mathrm{vol}_n(K \cap (-\tilde{L}))$. It follows that

$$\begin{aligned} \mathrm{vol}_n(K \cap (\lambda_1 x_1 + \lambda_2 x_2 - \tilde{L})) \geq \big[\lambda_1 \theta_1^{1/n} + \lambda_2 \theta_2^{1/n} + 1 - \lambda_1 - \lambda_2\big]^n M(K, \tilde{L}) = \\ = \theta M(K, \tilde{L}). \end{aligned} \tag{5.3}$$

If $\theta > 0$, the left-hand side is positive, and hence, the intersection $K \cap (\lambda_1 x_1 + \lambda_2 x_2 - \tilde{L})$ is non-empty. This leads to $\lambda_1 x_1 + \lambda_2 x_2 \in K + \tilde{L}$, and hence, $\lambda_1 x_1 + \lambda_2 x_2 \in K +_\theta \tilde{L}$. This gives the inclusion.

If $\theta = 0$, then we must have $\theta_1 = \theta_2 = 0$ and $\lambda_1 + \lambda_2 = 1$. The sum $K +_0 \tilde{L}$ is clearly the same as the usual Minkowski sum $K + L$, and hence, the desired inclusion follows directly from convexity.

(c) First, consider the case where $M(K,\tilde{L}) = \mathrm{vol}_n(K \cap (-\tilde{L}))$. Let $\theta, \lambda \in [0,1]$. Letting $\theta_1 = \theta_2 = \theta$, $\lambda_1 = \lambda$ and $\lambda_1 + \lambda_2 = 1$ in item **(b)**, we arrive at

$$\lambda(K +_\theta \tilde{L}) + (1-\lambda)(K +_\theta \tilde{L}) \subseteq K +_\theta \tilde{L}.$$

This proves the convexity of $K +_\theta \tilde{L}$.

For the case where we do not necessarily have $M(K,L) = \mathrm{vol}_n(K \cap (-L))$, we simply let $x_0 \in K + L$ be such that $M(K,L) = \mathrm{vol}_n(K \cap (x_0 - L))$. Then, letting $\tilde{L} = -x_0 + L$, we have that $K +_\theta \tilde{L}$ is convex. Since

$$K +_\theta \tilde{L} = K +_\theta (-x_0 + L) = -x_0 + (K +_\theta L),$$

we obtain that $K +_\theta L$ is a translate of a convex set, and hence, it is convex itself.

□

Exercise 5.13. *Let $K, L \subseteq \mathbb{R}^n$ be convex bodies such that*

$$M(K,L) = \mathrm{vol}_n(K \cap (-L)).$$

Prove that for every $x \in K +_1 L$, the intersection $K \cap (x - L)$ is a translate of $K \cap (-L)$.

Solution 5.13. First, notice that if $x \in K +_1 L$, then $\mathrm{vol}_n(K \cap (x - L)) = M(K,L)$. In item **(b)** of Exercise 5.12, let $\theta_1 = \theta_2 = 1$. Thus, $\theta = 1$, and we have equality in (5.3). This leads to equality in (5.2), which gives that $K \cap (x_1 - L)$, $K \cap (x_2 - L)$ and $K \cap (-L)$ are homothetical. Since the volume of each of them equals $M(K,L)$, we have that they are translates. Since x_1, say, can be arbitrarily taken in $K +_1 L$, we get the wished result.

□

Exercise 5.14. *Verify that the quantity*

$$s(K) := \inf\{\mathrm{vol}_n(K)\mathrm{vol}_n(K - z)^\circ : z \in \mathrm{int}K\}$$

defined for any convex body $K \subseteq \mathbb{R}^n$ is indeed an affine invariant.

Solution 5.14. First, let $T \in \mathrm{GL}(n)$. For any $w \in \mathrm{int}(TK)$, we have that

$$\begin{aligned}\mathrm{vol}_n(TK)\mathrm{vol}_n(TK - w)^\circ &= \mathrm{vol}_n(TK)\mathrm{vol}_n(T(K - T^{-1}w))^\circ = \\ = \mathrm{vol}_n(TK)\mathrm{vol}_n\left((T^{-1})^*(K - T^{-1}w)^\circ\right) &= |\det(T)|\mathrm{vol}_n(K)|\det(T^{-1})^*|\mathrm{vol}_n(K - T^{-1}w)^\circ = \\ &= \mathrm{vol}_n(K)\mathrm{vol}_n(K - T^{-1}w)^\circ,\end{aligned}$$

where the second equality comes from Lemma 5.4.1 in the main book, and the last equality justifies since $\det(T^{-1})^* = \det(T^{-1}) = 1/\det(T)$. Since we have that $w \in \mathrm{int}(TK)$ if and only if $T^{-1}w \in \mathrm{int}K$ (indeed, T is, in particular, a homeomorphism), it follows that

$$\begin{aligned}s(TK) &= \inf\{\mathrm{vol}_n(TK)\mathrm{vol}_n(TK - w)^\circ : w \in \mathrm{int}(TK)\} = \\ &= \inf\{\mathrm{vol}_n(K)\mathrm{vol}_n(K - T^{-1}w)^\circ : w \in \mathrm{int}(TK)\} = \\ &= \inf\{\mathrm{vol}_n(K)\mathrm{vol}_n(K - z)^\circ : z \in \mathrm{int}(K)\} = s(K).\end{aligned}$$

It remains to prove that $s(\cdot)$ is invariant under translations. Let $x \in \mathbb{R}^n$ be arbitrarily given. Then, $w \in \operatorname{int}(K + x_0)$ if and only if $w - x_0 \in \operatorname{int}(K)$. For any such point, we have

$$\operatorname{vol}_n(K + x_0)\operatorname{vol}_n(K + x_0 - w)^\circ = \operatorname{vol}_n(K)\operatorname{vol}_n(K - (w - x_0))^\circ.$$

Thus,

$$\begin{aligned} s(K + x_0) &= \inf\{\operatorname{vol}_n(K + x_0)\operatorname{vol}_n(K + x_0 - w)^\circ : w \in \operatorname{int}(K + x_0)\} = \\ &= \inf\{\operatorname{vol}_n(K)\operatorname{vol}_n(K - z)^\circ : z \in \operatorname{int}(K)\} = s(K), \end{aligned}$$

and the proof is complete.

□

Exercise 5.15. *Let $a < 0 < b$, and consider the interval $K = [a, b] \subseteq \mathbb{R}$. Determine $z \in \mathbb{R}$ such that $(K - z)^\circ$ has the minimum length.*

Solution 5.15. We notice that $K - z$ is the interval $[a - z, b - z]$, and $x \in (K - z)^\circ$ if and only if $xy \leq 1$ for every $y \in [a - z, b - z]$. Hence, it is clear that if $z \notin (a, b)$, then $(K - z)^\circ$ is not bounded. Thus, to find the minimum length for $(K - z)^\circ$, we must consider $z \in (a, b)$. In this case, we see immediately that $xy \leq 1$ for every $y \in [a - z, b - z]$ if and only if

$$x \in \left[\frac{1}{(a - z)}, \frac{1}{(b - z)}\right].$$

That is,

$$(K - z)^\circ = \left[\frac{1}{(a - z)}, \frac{1}{(b - z)}\right].$$

Therefore, the length of $(K - z)^\circ$ is

$$\operatorname{length}(K - z)^\circ = \frac{1}{(b - z)} - \frac{1}{(a - z)} = \frac{(b - a)}{((z - a)(b - z))}.$$

The dependency on z occurs in the denominator, which is a second-degree polynomial. This polynomial is clearly maximized at $z = (a + b)/2$, and maximizing the denominator minimizes the fraction, of course.

□

Exercise 5.16 (The Santaló point)**.** *Let $K \subseteq \mathbb{R}^n$ be a convex body and define $f : \operatorname{int}(K) \to (0, +\infty)$ by*

$$f(z) = \operatorname{vol}_n((K - z)^\circ).$$

(a) *Prove that f is strictly convex.*

(b) *Show that f goes to infinity as z approaches the boundary of K in the following sense: for every $c > 0$, there exists $\varepsilon > 0$ such that if $z \in \operatorname{int}(K)$ and $\operatorname{dist}(z, \partial K) < \varepsilon$, then $f(z) > c$.*

(c) *Verify that there is a unique point $z_0 \in \operatorname{int}(K)$ such that*

$$\operatorname{vol}_n((K - z_0)^\circ) = \inf\left\{\operatorname{vol}_n((K - z)^\circ) : z \in \operatorname{int}(K)\right\}.$$

(d) *Prove also that f is differentiable and compute $f'(z, v)$ for $z \in \operatorname{int}(K)$ and $v \in \mathbb{R}^n$.*

Hint. Use equality (4.3.8) of the main book.

Solution 5.16. (a) By equality (4.3.8) in the main book, we have

$$f(z) = \kappa_n \int_{\mathbb{S}^{n-1}} h_{K-z}(u)^{-n}\, d\sigma(u).$$

The function $t \mapsto t^{-n}$ is strictly convex for $(0, +\infty)$, since its second derivative is strictly positive in this domain. Thus, for any $a, b > 0$ and every $\lambda \in (0, 1)$, we have

$$\big((1-\lambda)a + \lambda b\big)^{-n} \leq (1-\lambda)a^{-n} + \lambda b^{-n},$$

and the inequality is strict if $a \neq b$.

Therefore, for any distinct points $z, w \in \operatorname{int}(K)$, every $\lambda \in (0, 1)$, and each $u \in \mathbb{S}^{n-1}$, we have

$$\big((1-\lambda)h_{K-z}(u) + \lambda h_{K-w}(u)\big)^{-n} \leq (1-\lambda)h_{K-z}(u)^{-n} + \lambda h_{K-w}(u)^{-n},$$

and the inequality is strict for some $u \in \mathbb{S}^{n-1}$, because for $z \neq w$, we have that $h_{K-z}(u) \neq h_{K-w}(u)$ for some $u \in \mathbb{S}^{n-1}$. By continuity, this holds for some non-empty open set of $\mathbb{S}^{n-1}$, and hence,

$$\begin{aligned}
&\int_{\mathbb{S}^{n-1}} \big((1-\lambda)h_{K-z}(u) + \lambda h_{K-w}(u)\big)^{-n}\, d\sigma(u) < \\
&< \int_{\mathbb{S}^{n-1}} (1-\lambda)h_{K-z}(u)^{-n} + \lambda h_{K-w}(u)^{-n}\, d\sigma(u) = \\
&= (1-\lambda)\int_{\mathbb{S}^{n-1}} h_{K-z}(u)^{-n}\, d\sigma(u) + \lambda \int_{\mathbb{S}^{n-1}} h_{K-w}(u)^{-n}\, d\sigma(u).
\end{aligned}$$

Now, simply observe that $K = (1-\lambda)K + \lambda K$ and write $h_{K-(1-\lambda)z-\lambda w} = (1-\lambda)h_{K-z} + \lambda h_{K-w}$. Hence, from the inequality above, we arrive at

$$\begin{aligned}
f((1-\lambda)z + \lambda w) &= \kappa_n \int_{\mathbb{S}^{n-1}} h_{K-(1-\lambda)z-\lambda w}(u)^{-n}\, d\sigma(u) = \\
&= \kappa_n \int_{\mathbb{S}^{n-1}} \big((1-\lambda)h_{K-z}(u) + \lambda h_{K-w}(u)\big)^{-n}\, d\sigma(u) \leq \\
&\leq (1-\lambda)\kappa_n \int_{\mathbb{S}^{n-1}} h_{K-z}(u)^{-n}\, d\sigma(u) + \lambda\kappa_n \int_{\mathbb{S}^{n-1}} h_{K-w}(u)^{-n}\, d\sigma(u) = (1-\lambda)f(z) + \lambda f(w),
\end{aligned}$$

whenever $\lambda \in (0, 1)$ and $z, w \in \operatorname{int}(K)$ are distinct.

(b) Our strategy is to estimate the volume of the polar of a convex body $L \subseteq \mathbb{R}^n$ having the origin as an interior point in terms of the distance of $0_{\mathbb{R}^n}$ to the boundary of L. Let $d = \operatorname{dist}(0_{\mathbb{R}^n}, \partial L)$, and assume that this distance is attained for a point $x_0 \in \partial L$ (this points exists since ∂L is compact). Then, $dB^n \subseteq L$ (otherwise, we would have some point of ∂L closer to the origin than x_0), and $x_0 \in \partial L \cap d\mathbb{S}^{n-1}$. Hence, from item **(b)** of Exercise 2.41, we have that $x_0/|x_0|$ is a unit outer normal of both L and dB^n at x_0. Assume also that $\rho > 0$ is such that $L \subseteq \rho B^n$. Because polarity reverses inclusion, we obtain

$$\frac{1}{\rho}B^n \subseteq L^\circ \subseteq \frac{1}{d}B^n,$$

and from item **(c)** of Exercise 2.41, the point x_0 corresponds to a point

$$z_0 = \frac{x_0}{d|x_0|} \in \partial L^\circ \cap \frac{1}{d}\mathbb{S}^{n-1}.$$

In particular, $z_0 \in L^\circ$. Also, it is clear that

$$\frac{1}{\rho}B^n \cap z_0^\perp \subseteq L^\circ,$$

since the left-hand side is just a hyperplane section of a ball contained in L°. It follows from convexity that

$$M := \operatorname{conv}\left(\left(\frac{1}{\rho}B^n \cap z_0^\perp\right) \cup \{z_0\}\right) \subseteq L^\circ.$$

It turns out that the volume of M is easy to compute, because it is clearly a cone with apex at z_0. The distance from z_0 to $z_0^\perp$ is clearly $|z_0| = 1/d$, and using Cavalieri's principle (exactly in the same way as we did it in the proof of Theorem 4.4.2 in the main book), we get

$$\operatorname{vol}_n(L^\circ) \geq \operatorname{vol}_n(M) = \frac{1}{nd}\operatorname{vol}_n\left(\frac{1}{\rho}B^n \cap z_0^\perp\right) = \frac{\operatorname{vol}_{n-1}(B^{n-1})}{n\rho^{n-1}}\frac{1}{d},$$

where we also used the fact that $\frac{1}{\rho}B^n \cap z_0^\perp$ is a copy of the $(n-1)$-dimensional unit ball. Notice carefully that this estimates from below the volume of L° in terms of the inverse of the distance of the origin to the boundary of L up to a constant that only depends on a ball containing L (and on the dimension, of course).

Now, if $K \subseteq \mathbb{R}^n$ is any convex body, and $z \in \operatorname{int}(K)$ is such that $\operatorname{dist}(z, \partial K) < \varepsilon$, then $\operatorname{dist}(0_{\mathbb{R}^n}, \partial(K - z)) < \varepsilon$. Let $\rho > 0$ be such that $K - K \subseteq \rho B^n$. Hence, $K - z \subseteq \rho B^n$, and by the estimate that we have proved, it follows that

$$f(z) = \operatorname{vol}_n((K-z)^\circ) \geq \frac{\operatorname{vol}_{n-1}(B^{n-1})}{n\rho^{n-1}}\frac{1}{\varepsilon}.$$

Finally, for any $c > 0$, if

$$0 < \varepsilon < \frac{n\rho^{n-1}}{c\operatorname{vol}_{n-1}(B^{n-1})},$$

then $f(z) > c$ whenever $\operatorname{dist}(z, \partial K) < \varepsilon$.

(c) Let $z_1 \in \operatorname{int}(K)$ be any fixed point, and let $c := f(z_1)$. By item **(b)**, let $\varepsilon > 0$ be such that $f(z) > c$ whenever $\operatorname{dist}(z, \partial K) < \varepsilon$. Define

$$K_\varepsilon := \big\{z \in \operatorname{int}(K) : \operatorname{dist}(z, \partial K) \geq \varepsilon\}.$$

Then, K_ε is clearly compact, and $z_1 \in K_\varepsilon$ (since otherwise, we would have that $f(z_1) > c$). Since f is convex and finite in $\operatorname{int}(K)$, we have that f is continuous in $\operatorname{int}(K)$, and in particular, f is continuous over K_ε. Thus, f attains a minimum for some point $z_0 \in K_\varepsilon$. Because $z_1 \in K_\varepsilon$, we have $f(z_0) \leq f(z_1) = c$, and since $f(z) > c$ for every $z \in K \setminus K_\varepsilon$, we get that

$$f(z_0) = \min\{f(z) : z \in K\}.$$

This settles the existence of a minimizer for $f(z) = \mathrm{vol}_n((K-z)^\circ)$ as z ranges through $\mathrm{int}(K)$. The uniqueness comes immediately from the fact that f is strictly convex: if $y_0 \neq z_0$ is a point of $\mathrm{int}(K)$ where $f(y_0) = f(z_0)$, then for every $\lambda \in (0,1)$, we have

$$f(z_0) \leq f((1-\lambda)z_0 + \lambda y_0) \leq (1-\lambda)f(z_0) + \lambda f(y_0) = f(z_0),$$

where the first inequality comes from the fact that z_0 is a minimizer. This contradicts the fact that f is strictly convex and the proof is complete.

(d) Since f is convex, it suffices to prove that f has directional derivatives at each point of $\mathrm{int}(K)$. If $z \in \mathrm{int}(K)$ and $v \in \mathbb{S}^{n-1}$, then we have

$$f'(z,v) = \lim_{\varepsilon\to 0} \frac{1}{\varepsilon}\left(\kappa_n \int_{\mathbb{S}^{n-1}} h_{K-z-\varepsilon v}(u)^{-n}\, d\sigma(u) - \kappa_n \int_{\mathbb{S}^{n-1}} h_{K-z}(u)^{-n}\, d\sigma(u)\right) =$$
$$= \kappa_n \lim_{\varepsilon\to 0}\left(\int_{\mathbb{S}^{n-1}} \frac{h_{K-z-\varepsilon v}(u)^{-n} - h_{K-z}(u)^{-n}}{\varepsilon}\, d\sigma(u)\right),$$

and we must prove that this limit indeed exists. For simplicity, we consider first the case where $\varepsilon > 0$. For each fixed $u \in \mathbb{S}^{n-1}$, we define $f_u : \mathrm{int}(K) \to \mathbb{R}$ by

$$f_u(z) = h_{K-z}(u)^{-n} = \big(h_K(u) - \langle z, u\rangle\big)^{-n}.$$

It is clear that each function f_u is differentiable and that for every non-zero $v \in \mathbb{R}^n$, we have

$$f_u'(z,v) = n h_{K-z}(u)^{-n-1}\langle v, u\rangle.$$

For each $\varepsilon > 0$, we let $g_\varepsilon : \mathbb{S}^{n-1} \to \mathbb{R}$ be given as

$$g_\varepsilon(u) := \frac{h_{K-z-\varepsilon v}(u)^{-n} - h_{K-z}(u)^{-n}}{\varepsilon}.$$

Then, of course, for each fixed $u \in \mathbb{S}^{n-1}$, we have

$$\lim_{\varepsilon\to 0^+} g_\varepsilon(u) = \lim_{\varepsilon\to 0^+} \frac{h_{K-z-\varepsilon v}(u)^{-n} - h_{K-z}(u)^{-n}}{\varepsilon} = (f_u)'_+(z,v) = f_u'(z,v) =$$
$$= n h_{K-z}(u)^{-n-1}\langle v, u\rangle.$$

Hence, if $(\varepsilon_m)_{m\in\mathbb{N}}$ is a sequence of positive numbers converging to 0, we have that $(g_{\varepsilon_m})_{m\in\mathbb{N}}$ is a sequence of functions of $\mathbb{S}^{n-1}$ converging to the function $g : \mathbb{S}^{n-1} \to \mathbb{R}$ given as

$$g(u) := n h_{K-z}(u)^{-n-1}\langle v, u\rangle.$$

All of these functions are continuous, and hence, σ-integrable. Let $m_0 := \min\{h_{K-z}(u) : u \in \mathbb{S}^{n-1}\}$, and observe that since $K - z$ is a convex body having the origin as an interior point, we have $m_0 > 0$. Of course, since we want to compute a limit as $\varepsilon \to 0^+$, we may assume that

$$\varepsilon < \frac{m_0}{2} \ \ (< m_0),$$

and we write $a := h_{K-z}(u)$ and $b = \langle v, u\rangle$. Also, write $M_0 := \max\{h_{K-z}(u) : u \in \mathbb{S}^{n-1}\}$. Then, from the Cauchy-Schwarz inequality applied for $\langle v, u\rangle$, we have $b \leq |b| \leq 1$, and this leads to

$$a - \varepsilon b = h_{K-z}(u) - \varepsilon\langle v, u\rangle \geq h_{K-z}(u) - \varepsilon \geq m_0 - \frac{m_0}{2} = \frac{m_0}{2}.$$

We also have trivially from the definitions of m_0 and M_0 that

$$m_0 \le a \le M_0.$$

We estimate

$$\begin{aligned}
|g_\varepsilon(u)| &= \left|\frac{h_{K-z-\varepsilon v}(u)^{-n} - h_{K-z}(u)^{-n}}{\varepsilon}\right| = \frac{1}{\varepsilon}\left|(a-\varepsilon b)^{-n} - a^{-n}\right| = \\
&= \frac{1}{\varepsilon}\left|\frac{a^n - (a-\varepsilon b)^n}{a^n(a-\varepsilon b)^n}\right| \le \frac{1}{\varepsilon}\frac{2^n}{m_0^{2n}}\left|a^n - (a-\varepsilon b)^n\right| = \\
&= \frac{2^n}{m_0^{2n}}\frac{1}{\varepsilon}\left|a^n - a^n + \binom{n}{1}a^{n-1}\varepsilon b - \binom{n}{2}a^{n-2}(\varepsilon b)^2 + \ldots - (-\varepsilon b)^n\right| = \\
&= \frac{2^n}{m_0^{2n}}\left|\binom{n}{1}a^{n-1}b - \binom{n}{2}a^{n-2}b^2\varepsilon + \ldots - (-b)^n(\varepsilon)^{n-1}\right| \le \\
&\le \frac{2^n n!}{m_0^{2n}}\left(|a^{n-1}b| + \ldots |a^{n-2}b\varepsilon| + \ldots + |b^n\varepsilon^{n-1}|\right) \le \frac{2^n n!}{m_0^{2n}}\left(|a^{n-1}| + |a^{n-2}\varepsilon| + \ldots + |\varepsilon^{n-1}|\right) \le \\
&\le \frac{2^n n!}{m_0^{2n}}\left(M_0^{n-1} + M_0^{n-2}m_0 + \ldots + m_0^{n-1}\right) \le \frac{2^n n! n}{m_0^{2n}}M_0^{n-1},
\end{aligned}$$

and the latter does not depend on ε and u, but only on the dimension and on $K - z$. Consequently, the sequence (g_{ε_m}) is dominated by a constant (which can be seen as a σ-integrable function of $\mathbb{S}^{n-1}$, of course), and by the dominated convergence theorem, we have that

$$\int_{\mathbb{S}^{n-1}} n h_{K-z}(u)^{-n-1}\langle v, u\rangle \, d\sigma(u) = \int_{\mathbb{S}^{n-1}} \lim_{m\to+\infty} g_{\varepsilon_m}(u)\, d\sigma(u) = \lim_{m\to\infty}\int_{\mathbb{S}^{n-1}} g_{\varepsilon_m}(u)\, d\sigma(u).$$

Since this holds for every sequence $\varepsilon_m \to 0^+$, we get that

$$\begin{aligned}
\int_{\mathbb{S}^{n-1}} n h_{K-z}(u)^{-n-1}\langle v, u\rangle \, d\sigma(u) &= \lim_{\varepsilon\to 0^+}\int_{\mathbb{S}^{n-1}} g_\varepsilon(u)\, d\sigma(u) = \\
&= \lim_{\varepsilon\to 0^+}\int_{\mathbb{S}^{n-1}} \frac{h_{K-z-\varepsilon v}(u)^{-n} - h_{K-z}(u)^{-n}}{\varepsilon}\, d\sigma(u),
\end{aligned}$$

and hence,

$$f'_+(z, v) = \kappa_n \int_{\mathbb{S}^{n-1}} n h_{K-z}(u)^{-n-1}\langle v, u\rangle \, d\sigma(u).$$

Notice that we arrive at the same result if we repeat the argument for the case where $\varepsilon < 0$, because we always have that $(f_u)'_-(z, v) = (f_u)'_+(z, v) = f'_u(z, v)$. Hence, it follows that

$$f'(z, v) = \kappa_n \int_{\mathbb{S}^{n-1}} n h_{K-z}(u)^{-n-1}\langle v, u\rangle \, d\sigma(u),$$

for any $z \in \mathbb{S}^{n-1}$ and all $v \in \mathbb{S}^{n-1}$. This concludes the exercise.

□

Exercise 5.17. *Give a proof of Corollary 5.1.1 from the main book restricted to convex bodies based on the Brunn concavity principle, which was introduced in Exercise 4.15.*

Solution 5.17. Let $K, T \subseteq \mathbb{R}^n$ be convex bodies, and define $K_1 := K \times \{0\}$ and $T_1 := T \times \{1\}$, $K_1, T_1 \subseteq \mathbb{R}^{n+1}$. We warn the reader that we will write the points of $\mathbb{R}^{n+1}$ in the form (x,t), where $x \in \mathbb{R}^n$ and $t \in \mathbb{R}$. Let $L = \operatorname{conv}(K_1 \cup T_1)$, and observe that the points of L are of the form

$$\big(\lambda_1 x_1 + \ldots + \lambda_m x_m + \alpha_1 y_1 + \ldots + \alpha_k y_k, \alpha_1 + \ldots + \alpha_k\big),$$

where $x_1, \ldots, x_m \in K$, $y_1, \ldots, y_k \in T$, $\lambda_1, \ldots, \lambda_m, \alpha_1, \ldots, \alpha_k \geq 0$ and

$$\lambda_1 + \ldots + \lambda_m + \alpha_1 + \ldots + \alpha_k = 1.$$

For each $t \in [0,1]$, let

$$L(t) := \big\{x \in \mathbb{R}^n : (x,t) \in L\big\}.$$

By the form of the points of L described above, it is clear that $L(0) = K$, $L(1) = T$ and $L(1/2) = (K+T)/2$. Thus,

$$\begin{aligned} K \times \{0\} = L(0) \times \{0\} &= L \cap e_{n+1}^{\perp}, \\ \left(\frac{K+T}{2} \times \left\{\frac{1}{2}\right\}\right) = L\left(\frac{1}{2}\right) \times \left\{\frac{1}{2}\right\} &= L \cap \left(e_{n+1}^{\perp} + \frac{1}{2}e_{n+1}\right), \quad \text{and} \\ T \times \{1\} = L(1) \times \{1\} &= L \cap \left(e_{n+1}^{\perp} + e_{n+1}\right). \end{aligned}$$

Now, we just apply the Brunn concavity principle for $L \subseteq \mathbb{R}^{n+1}$, $F = e_{n+1}^{\perp}$ and the points $0_{\mathbb{R}^{n+1}}$, e_{n+1} and $(1/2)e_{n+1}$ of $F^{\perp}$. We get

$$\operatorname{vol}_n\left(L \cap \left(e_{n+1}^{\perp} + \frac{1}{2}e_{n+1}\right)\right)^{1/n} \geq \frac{1}{2}\operatorname{vol}_n\left(L \cap e_{n+1}^{\perp}\right)^{1/n} + \frac{1}{2}\operatorname{vol}_n\left(L \cap \left(e_{n+1}^{\perp} + e_{n+1}\right)\right)^{1/n}.$$

Since $\operatorname{vol}_n(K \times \{0\}) = \operatorname{vol}_n(K)$, and the same holds for the other Cartesian products above, we arrive at

$$\operatorname{vol}_n\left(\frac{K+T}{2}\right)^{1/n} \geq \frac{1}{2}\operatorname{vol}_n(K)^{1/n} + \frac{1}{2}\operatorname{vol}_n(T)^{1/n}.$$

The desired result follows.

□

The objective of the next exercise is to prove the *Grünbaum lemma*, which states, roughly speaking, that any hyperplane section through the centroid of a convex body divides it in two parts of "more or less" the same volume. As hints for this exercise, we suggest the reader to take a look at the proof of Theorem 5.1.2 from the main book and to have the Brunn concavity principle in mind.

Exercise 5.18 (Grünbaum)**.** *Let $K \subseteq \mathbb{R}^n$ be a convex body having volume 1 and whose centroid is the origin, and let $u \in \mathbb{S}^{n-1}$.*

(a) *Let $M > \max\{h_K(-u), h_K(u)\}$, and define $g : \mathbb{R} \to \mathbb{R}$ given as*

$$g(t) = \operatorname{vol}_n\big(K \cap \mathbf{h}^+(u,-t)\big).$$

Verify that $g(t) = 0$ for $t \leq -M$ and that $g(t) = 1$ for $t \geq M$.

(b) *Show that*

$$\int_{-M}^{M} tg'(t)\,dt = 0.$$

Using integration by parts, conclude that

$$\int_{-M}^{M} g(t)\,dt = M.$$

(c) *Prove that g is a log-concave non-decreasing function. As a consequence, derive the inequality*

$$g(t) \le g(0)e^{\alpha t},$$

where $\alpha = g'(0)/g(0)$.

(d) *Choosing M large enough such that $1/\alpha < M$, check that $g(0) \ge 1/e$. Conclude that*

$$\mathrm{vol}_n\big(K \cap \mathbf{h}^+(u,0)\big) \ge \frac{1}{e}.$$

Solution 5.18. (a) For any $x \in K$, we have

$$\langle x, -u\rangle \le h_K(-u) \le M,$$

and this leads to

$$\langle x, u\rangle \ge -M.$$

Hence,

$$K \subseteq \mathbf{h}^+(u,-M) \subseteq \mathbf{h}^+(u,-t)$$

whenever $-t \le -M$, that is, whenever $t \ge M$. Thus, for $t \ge M$, we get that

$$g(t) = \mathrm{vol}_n\big(K \cap \mathbf{h}^+(u,-t)\big) = \mathrm{vol}_n(K) = 1.$$

On the other hand, we also know that $\langle x, u\rangle \le h_K(u) < M$ for every $x \in K$. Thus, if $x \in \mathbb{R}^n$ is such that $\langle x, u\rangle \ge M$, then $x \notin K$. Let $-t \ge M$. If $\langle x, u\rangle \ge -t$, then $\langle x, u\rangle \ge M$, and hence, $x \notin K$. It follows that

$$K \cap \mathbf{h}^+(u,-t) = \emptyset$$

whenever $-t \ge M$.

(b) We use the fact that the centroid of K is the origin. This leads to

$$0 = \left\langle \int_K x\,dx, u \right\rangle = \int_K \langle x, u\rangle\,dx.$$

Now, we use Cavalieri's principle:

$$0 = \int_K \langle x, u\rangle\,dx = \int_{-M}^{M} \int_{K\cap\mathbf{h}(u,-t)} \langle z, u\rangle\,d\mathcal{H}^{n-1}(z)dt = \int_{-M}^{M} -t\,\mathrm{vol}_{n-1}\big(K \cap \mathbf{h}(u,-t)\big)\,dt.$$

Finally, from the proof of Theorem 5.1.2 (main book), we know that

$$g'(t) = -\mathrm{vol}_{n-1}\big(K \cap \mathbf{h}(u,-t)\big),$$

and replacing this in the equality above, we get that

$$\int_{-M}^{M} t g'(t)\,dt = 0.$$

For the remaining equality, we use integration by parts to get

$$0 = \int_{-M}^{M} t g'(t)\,dt = t g(t)\Big|_{-M}^{M} - \int_{-M}^{M} g(t)\,dt,$$

from where

$$\int_{-M}^{M} g(t)\,dt = M g(M) + M g(-M) = M,$$

since $g(M) = 1$ and $g(-M) = 0$.

(c) The fact that g is non-decreasing is obvious: we have $\mathbf{h}^+(u,-\alpha) \subseteq \mathbf{h}^+(u,-\beta)$ whenever $\alpha \le \beta$. To show that g is log-concave, we notice that

$$\big(-\log(g(t))\big)' = -\frac{g'(t)}{g(t)} = \frac{\mathrm{vol}_{n-1}\big(K \cap \mathbf{h}(u,-t)\big)}{\mathrm{vol}_n\big(K \cap \mathbf{h}^+(u,-t)\big)} \ge 0.$$

This means that $-\log(g(t))$ is convex, which is the same as saying that $g(t)$ is log-concave.

To prove the estimate for $g(t)$, we write $g(t) = e^{-\varphi(t)}$, for $\varphi(t) := -\log(g(t))$. The function φ is convex and differentiable, and hence, its epigraph is supported by the line passing through the point $(0,\varphi(0))$ and having $\varphi'(0)$ as angular coefficient. This leads to

$$\varphi(t) \ge \varphi(0) + \varphi'(0)t$$

for every $t \in \mathbb{R}$. Also, since $-\varphi(t) = \log(g(t))$, it is clear that $-\varphi'(0) = g'(0)/g(0) = \alpha$. Therefore, we arrive at

$$g(t) = e^{-\varphi(t)} \le e^{-\varphi(0)-\varphi'(0)t} = e^{-\varphi(0)} e^{\alpha t} = g(0) e^{\alpha t}.$$

(b) Since M is an arbitrary number greater than $\max\{h_K(-u), h_K(u)\}$, we may take M large enough such that $1/\alpha < M$. Thus,

$$1 = g(M) \le g(0) e^{\alpha M} \le g(0) e,$$

and hence,

$$\frac{1}{e} \le g(0) = \mathrm{vol}_n\big(K \cap \mathbf{h}^+(u,0)\big).$$

□

The objective of the next exercise is to provide a version of the Rogers-Shephard inequality for the case of differences of distinct bodies.

Exercise 5.19. *Let $K, L \subseteq \mathbb{R}^n$ be convex bodies, and recall that we write $K - L := K + (-L)$.*

(a) *Prove that*

$$K - L = \big\{x \in \mathbb{R}^n : K \cap (L + x) \neq \emptyset\big\}.$$

(b) *Check that for any $x, z \in K - L$ and every $\lambda \in [0, 1]$, the inclusion*

$$(1 - \lambda)\big(K \cap (L + x)\big) + \lambda\big(K \cap (L + z)\big) \subseteq K \cap \big(L + (1 - \lambda)x + \lambda z\big)$$

holds.

(c) *Verify that the support of the function*

$$g_{K,L}(x) := \mathrm{vol}_n(K \cap (L + x))^{1/n}$$

is contained in $K - L$. Show that $g_{K,L}$ is concave in $K - L$.

Remark 5.2. The function $g_{K,L} \colon K - L \to \mathbb{R}$ defined above is called the *cross covariogram* of K and L.

Solution 5.19. (a) Let $z \in K$ and $y \in L$. Then, it is clear that $z \in K \cap (L + z - y)$, from where we get that

$$K - L \subseteq \big\{x \in \mathbb{R}^n : K \cap (L + x) \neq \emptyset\big\}.$$

Now, let $x \in \mathbb{R}^n$ be such that $K \cap (L + x) \neq \emptyset$, and choose a point $z \in K \cap (L + x)$. Then, $z - x \in L$, and hence, writing $x = z - (z - x)$, we obtain that $x \in K - L$. This proves the reverse inclusion.

(b) Let $y \in K \cap (L + x)$ and $w \in K \cap (L + z)$. By the convexity of K, we have that $(1 - \lambda)y + \lambda w \in L$, and hence, all we have to prove is that this point also belongs to $L + (1 - \lambda)x + \lambda z$. Let $x_0, z_0 \in L$ be such that $y = x_0 + x$ and $w = z_0 + z$. Then,

$$(1 - \lambda)y + \lambda w = (1 - \lambda)(x_0 + x) + \lambda(z_0 + z) = (1 - \lambda)x_0 + \lambda z_0 + (1 - \lambda)x + \lambda z,$$

and since L is convex, we have that $(1 - \lambda)x_0 + \lambda z_0 \in L$. Therefore, it follows that $(1 - \lambda)y + \lambda w \in L + (1 - \lambda)x + \lambda z$, and this gives the desired inclusion.

(c) The support of $g_{K,L}$ is contained in $K - L$ because the intersection $K \cap (L + x)$ is empty if $x \notin K - L$. From the inclusion given in item **(b)** and using the Brunn-Minkowski inequality, we get

$$\begin{aligned} g_{K,L}((1 - \lambda)x + \lambda y) &= \mathrm{vol}_n\big(K \cap (L + (1 - \lambda)x + \lambda y)\big)^{1/n} \geq \\ &\geq \mathrm{vol}_n\big((1 - \lambda)(K \cap (L + x)) + \lambda(K \cap (L + y))\big)^{1/n} \geq \\ &\geq (1 - \lambda)\mathrm{vol}_n(K \cap (L + x))^{1/n} + \lambda\mathrm{vol}_n(K \cap (L + y))^{1/n} = (1 - \lambda)g_{K,L}(x) + \lambda g_{K,L}(y) \end{aligned}$$

for any $x, y \in K - L$ and every $\lambda \in [0, 1]$. Thus, $g_{K,L}$ is concave.

□

Exercise 5.20. *Let $K, L \subseteq \mathbb{R}^n$ be convex bodies, and let $\bar{g}_{K,L} : \mathbb{R}^n \to \mathbb{R}$ be defined as*

$$\bar{g}_{K,L}(x) := g_{K,L}(x)^n = \mathrm{vol}_n(K \cap (L + x)),$$

for each $x \in \mathbb{R}^n$. Show that $\bar{g}_{K,L}$ is log-concave.

Hint: Write $\bar{g}_{K,L}$ as a convolution of characteristic functions, and recall that characteristic functions are log-concave and that the convolution of log-concave functions is log-concave.

Solution 5.20. We write

$$\bar{g}_{K,L}(x) = \mathrm{vol}_n(K \cap (L + x)) = \int_{\mathbb{R}^n} 1_{K \cap (L+x)}(y)\, dy = \int_{\mathbb{R}^n} 1_K(y) 1_{L+x}(y)\, dy =$$
$$= \int_{\mathbb{R}^n} 1_K(y) 1_L(y - x)\, dy = \int_{\mathbb{R}^n} 1_K(y) 1_{-L}(x - y)\, dy = 1_K * 1_{-L}(x).$$

From Exercise 1.27, we have that the characteristic function of a convex set is log-concave. Hence, 1_K and 1_{-L} are log-concave. Finally, from Proposition 1.5.1 in the main book, we get that $1_K * 1_{-L}$ is log-concave.

□

A function $f : \mathbb{R}^n \to [0, +\infty)$ is said to be *unimodal* if for each $t \geq 0$, the level set $\{f \geq t\}$ is convex. The objective of the next exercise is to prove *Anderson's theorem*, which, roughly speaking, states that the integral of a non-negative, even, and unimodal function over a centrally symmetric convex body K does not decrease when K is translated toward the origin.

Exercise 5.21 (Anderson's Theorem)**. (a)** *Let $f : \mathbb{R}^n \to \mathbb{R}$ be a non-negative measurable function. Prove that*

$$f(x) = \int_0^{+\infty} 1_{\{f \geq t\}}(x)\, dt$$

for each $x \in \mathbb{R}^n$.

(b) *Let K and L be origin-symmetric convex bodies in $\mathbb{R}^n$. Show that*

$$\int_K 1_L(x + cy)\, dx \geq \int_K 1_L(x + y)\, dx$$

for any $y \in \mathbb{R}^n$ and every $0 \leq c \leq 1$.

(c) *Let $f : \mathbb{R}^n \to \mathbb{R}$ be a non-negative, even, unimodal, and integrable function, and let K be an origin-symmetric convex body in $\mathbb{R}^n$. Prove that*

$$\int_K f(x + cy)\, dx \geq \int_K f(x + y)\, dx$$

for each $y \in \mathbb{R}^n$ and all $0 \leq c \leq 1$.

Solution 5.21. (a) First, notice that for any $x \in \mathbb{R}^n$, we have

$$f(x) = \int_0^{f(x)} dt = \int_0^{+\infty} 1_{[0,f(x)]}(t)\, dt.$$

Now we observe that $1_{[0,f(x)]}(t) = 1_{\{f \geq t\}}(x)$ for any $t \geq 0$. Using this equality in the integral above, we get the desired property.

(b) For any $z \in \mathbb{R}^n$, we have

$$\int_K 1_L(x+z)\,dx = \int_K 1_{L-z}(x)\,dx = \mathrm{vol}_n(K \cap (L-z)) = \bar{g}_{K,L}(-z) = \bar{g}_{K,L}(z),$$

where the last equality follows immediately from the fact that L and K are origin-symmetric. We know from Exercise 5.20 that $\bar{g}_{K,L}$ is log-concave. Letting $\lambda = (1-c)/2$, we obtain

$$\begin{aligned}\bar{g}_{K,L}(cy) = \bar{g}_{K,L}((1-2\lambda)y) = \bar{g}_{K,L}((1-\lambda)y + \lambda(-y)) \geq \\ \geq \bar{g}_{K,L}(y)^{1-\lambda}\bar{g}_{K,L}(-y)^{\lambda} = \bar{g}_{K,L}(y)^{1-\lambda}\bar{g}_{K,L}(y)^{\lambda} = \bar{g}_{K,L}(y),\end{aligned}$$

and from the equality above applied for $z = y$ and for $z = cy$, we arrive at

$$\int_K 1_L(x+cy)\,dx = \bar{g}_{K,L}(cy) \geq \bar{g}_{K,L}(y) = \int_K 1_L(x+y)\,dx.$$

(c) From item **(a)** and Fubini's theorem, we may write

$$\int_K f(x+cy)\,dx = \int_K \int_0^{+\infty} 1_{\{f \geq t\}}(x+cy)\,dt dx = \int_0^{+\infty} \int_K 1_{\{f \geq t\}}(x+cy)\,dx dt.$$

Now, we have to investigate distinct cases for $\{f \geq t\}$, since we only know that these sets are convex (and origin-symmetric, because f is even). We have the following:

(i) If $\{f \geq t\}$ is a convex body, then from item **(b)**, we have

$$\int_K 1_{\{f \geq t\}}(x+cy)\,dx \geq \int_K 1_{\{f \geq t\}}(x)\,dx.$$

(ii) For $t = 0$, we have that $\{f \geq 0\} = \mathbb{R}^n$, and hence,

$$\int_K 1_{\{f \geq 0\}}(x+cy)\,dx = \int_K dx = \int_K 1_{\{f \geq 0\}}(x+y)\,dx.$$

(iii) If $t > 0$ and $\{f \geq t\}$ is non-empty and is not a convex body, then $\{f \geq t\}$ is not bounded. Since f is integrable, this yields that $\{f \geq t\}$ does not have maximal affine dimension, since otherwise it would have infinite volume. Thus, the volume of the intersection of K with any translation of $\{f \geq t\}$ is zero. This leads to

$$\begin{aligned}\int_K 1_{\{f \geq t\}}(x+cy)\,dx = \int_K 1_{\{f \geq t\}-cy}(x)\,dx = \mathrm{vol}_n(K \cap (\{f \geq t\} - cy)) = 0 = \\ = \mathrm{vol}_n(K \cap (\{f \geq t\} - y)) = \int_K 1_{\{f \geq t\}}(x+y)\,dx.\end{aligned}$$

Thus, we may write

$$\int_K 1_{\{f \geq t\}}(x+cy)\,dx \geq \int_K 1_{\{f \geq t\}}(x+y)\,dx$$

for every $t \geq 0$. This (and Fubini again) leads to

$$\int_K f(x+cy)\,dx = \int_0^{+\infty} \int_K 1_{\{f \geq t\}}(x+cy)\,dxdt \geq \int_0^{+\infty} \int_K 1_{\{f \geq t\}}(x+y)\,dxdt = \\ = \int_K \int_0^{+\infty} 1_{\{f \geq t\}}(x+y)\,dtdx = \int_K f(x+y)\,dx,$$

and the proof is complete.

□

Exercise 5.22. *Prove the following extension of the Rogers-Shephard inequality:*

$$\mathrm{vol}_n(K-L) \leq \binom{2n}{n} \frac{\mathrm{vol}_n(L)\mathrm{vol}_n(K)}{\max_{x \in K-L} \mathrm{vol}_n(K \cap (L+x))}$$

for any convex bodies $K, L \subseteq \mathbb{R}^n$.

Solution 5.22. Up to translating L (which does not modify volumes), we may assume that

$$\mathrm{vol}_n(K \cap L) = \max_{x \in K-L} \mathrm{vol}_n(K \cap (L+x)).$$

In this case, it is obvious that K and L have a common interior point, and hence, we get that $0_{\mathbb{R}^n} \in \mathrm{int}(K-L)$. We compute

$$\int_{K-L} \mathrm{vol}_n\big(K \cap (L+x)\big)\,dx = \int_{\mathbb{R}^n} \mathrm{vol}_n\big(K \cap (L+x)\big)\,dx = \int_{\mathbb{R}^n} \int_{\mathbb{R}^n} 1_{K \cap (L+x)}(z)\,dzdx = \\ = \int_{\mathbb{R}^n} \int_{\mathbb{R}^n} 1_K(z) 1_{L+x}(z)\,dzdx = \int_{\mathbb{R}^n} \int_{\mathbb{R}^n} 1_K(z) 1_{-L+z}(x)\,dzdx = \\ = \int_{\mathbb{R}^n} \left(\int_{\mathbb{R}^n} 1_{-L+z}(x)\,dx \right) 1_K(z)\,dz = \int_{\mathbb{R}^n} \mathrm{vol}_n(-L+z) 1_K(z)\,dz = \mathrm{vol}_n(L) \int_{\mathbb{R}^n} 1_K(z)\,dz = \\ = \mathrm{vol}_n(L)\mathrm{vol}_n(K).$$

Next, we use Lemma 5.3.1 of the main book for the functions $f(x) = g_{K,L}(x)$ and $g(t) = t^n$ (as in the proof of the Rogers-Shephard inequality) to write

$$\int_{K-L} \mathrm{vol}_n\big(K \cap (L+x)\big)\,dx = \int_{K-L} \Big[\mathrm{vol}_n\big(K \cap (L+x)\big)^{1/n}\Big]^n\,dx \geq \\ \geq n\mathrm{vol}_n(K-L) \int_0^1 \big[t\mathrm{vol}_n(K \cap L)^{1/n}\big]^n (1-t)^{n-1}\,dt = \\ = n\mathrm{vol}_n(K-L)\mathrm{vol}_n(K \cap L) \int_0^1 t^n (1-t)^{n-1}\,dt = \binom{2n}{n}^{-1} \mathrm{vol}_n(K-L)\mathrm{vol}_n(K \cap L),$$

where the last equality follows from standard integration by parts. Putting these together, we arrive at

$$\mathrm{vol}_n(K-L) \leq \binom{2n}{n} \frac{\mathrm{vol}_n(K)\mathrm{vol}_n(L)}{\mathrm{vol}_n(K \cap L)},$$

and with the assumption that L maximizes the intersection of the translates $L+x$ with K, we get the desired inequality.

□

Exercise 5.23 (Simultaneous convergence of symmetrizations)**.** *Let $K, L \subseteq \mathbb{R}^n$ be convex bodies, and recall that for each $u \in \mathbb{S}^{n-1}$, we write the Steiner symmetral of a given $M \in \mathcal{K}^n$ in $u^\perp$ by $S_u(M)$. Prove that there exists a sequence $(u_j)_{j\in\mathbb{N}}$ such that the iterations $S_{u_m} \circ \ldots \circ S_{u_1}(K)$ and $S_{u_m} \circ \ldots \circ S_{u_1}(L)$ converge to balls as $m \to \infty$.*

Solution 5.23. Of course, we may assume that $\mathrm{vol}_n(K) = \mathrm{vol}_n(L) = \mathrm{vol}_n(B^n)$. In this case, Theorem 2.7.1 (and Remark 2.7.3) from the main book guarantees that K and L converge "separately" to B^n for (a priori distinct) iterations of Steiner symmetrizations. Given an arbitrary number $\varepsilon > 0$, we may let $u_1, \ldots, u_{m_1} \in \mathbb{S}^{n-1}$ be such that

$$S_{u_{m_1}} \circ \ldots \circ S_{u_1}(K) \subseteq (1+\varepsilon)B^n.$$

Now, we notice that since $S_{u_{m_1}} \circ \ldots \circ S_{u_1}(L)$ is a convex body, there are $u_{m_1+1}, \ldots, u_{m_2} \in \mathbb{S}^{n-1}$ such that

$$S_{u_{m_2}} \circ \ldots \circ S_{u_{m_1+1}}(S_{u_{m_1}} \circ \ldots \circ S_{u_1}(L)) \subseteq (1+\varepsilon)B^n.$$

We observe that we also have

$$S_{u_{m_2}} \circ \ldots \circ S_{u_{m_1+1}}(S_{u_{m_1}} \circ \ldots \circ S_{u_1}(K)) \subseteq S_{u_{m_2}} \circ \ldots \circ S_{u_{m_1+1}}((1+\varepsilon)B^n) = (1+\varepsilon)B^n.$$

This is sufficient to show the desired statement, since we may repeat the argument for $\varepsilon/2$, and so on. Alternatively, one can proceed by contradiction.

□

Exercise 5.24. *Using Exercise 5.23, prove, for convex bodies, the version of the Brunn-Minkowski inequality given in Corollary 5.1.1 of the main book.*

Solution 5.24. For any $u \in \mathbb{S}^{n-1}$, we have that the Steiner symmetrization S_u preserves volume. Using this fact together with item **(iii)** of Proposition 2.7.2 (main book), we write

$$\mathrm{vol}_n(K+T)^{1/n} = \mathrm{vol}_n(S_u(K+T))^{1/n} \geq \mathrm{vol}_n(S_u(K) + S_u(T))^{1/n}.$$

From Exercise 5.23, we may consider iterated Steiner symmetrizations such that K and T converge simultaneously to balls, and from Remark 2.7.3 in the main book, we get that these balls must have the same volume as K and T, respectively. Thus, we arrive at

$$\mathrm{vol}_n(K+T)^{\frac{1}{n}} \geq \mathrm{vol}_n\left[\left(\frac{\mathrm{vol}_n(K)}{\kappa_n}\right)^{\frac{1}{n}} B_n + \left(\frac{\mathrm{vol}_n(T)}{\kappa_n}\right)^{\frac{1}{n}} B_n\right]^{\frac{1}{n}} =$$

$$= \left[\left(\frac{\mathrm{vol}_n(K)}{\kappa_n}\right)^{\frac{1}{n}} + \left(\frac{\mathrm{vol}_n(T)}{\kappa_n}\right)^{\frac{1}{n}}\right] \mathrm{vol}_n(B^n)^{\frac{1}{n}} = \mathrm{vol}_n(K)^{\frac{1}{n}} + \mathrm{vol}_n(T)^{\frac{1}{n}},$$

and this is the Brunn-Minkowski inequality.

□

Exercise 5.25. **(a)** *Let $u, v \in \mathbb{S}^{n-1}$ be orthogonal. Assume that $K \subseteq \mathbb{R}^n$ is a convex body symmetric with respect to $v^\perp$, that is,* $\mathrm{ref}_{v^\perp}(K) = K$. *Show that $S_u(K)$ is also symmetric with respect to $v^\perp$.*

(b) *Let $K \subseteq \mathbb{R}^n$ be any convex body. Prove that $S_{e_n} \circ \ldots \circ S_{e_1}(K)$ is symmetric with respect to the origin.*

(c) *Show that if $L \subseteq \mathbb{R}^n$ is a convex body symmetric with respect to the origin, then*

$$L \subseteq \frac{\mathrm{diam}L}{2} B^n.$$

(d) *Let $K \subseteq \mathbb{R}^n$ be any convex body. Prove the inequality*

$$\mathrm{vol}_n(K) \leq \frac{\kappa_n}{2^n} (\mathrm{diam}K)^n.$$

This is known as the isodiametric inequality.

Remark 5.3. Notice that the isodiametric inequality gives that the maximum ratio between the volume of a convex body in $\mathbb{R}^n$ and the n-th power of its diameter is attained for B^n. Actually, this is attained for any ball, since this ratio is clearly invariant under translations and dilations.

Solution 5.25. (a) Let $x \in S_u(K)$ be arbitrarily given, and let $z = \mathrm{proj}_{u^\perp}(x)$. Thus, we may assume that the line $z + \lambda u$ intersects K in the segment $z + \mathrm{seg}[\alpha u, \beta u]$, $\alpha \leq \beta$, and writing $x = z + tu$, we must have

$$\frac{\alpha - \beta}{2} \leq t \leq \frac{\beta - \alpha}{2}.$$

Since K is symmetric with respect to $v^\perp$, we have

$$K \supseteq \mathrm{ref}_{v^\perp}\big(z + \mathrm{seg}[\alpha u, \beta u]\big) = \mathrm{ref}_{v^\perp}(z) + \mathrm{seg}[\alpha u, \beta u].$$

To check the last equality, we notice that the reflection is linear, and because u and v are orthogonal, we have that

$$\mathrm{ref}_{v^\perp}(\lambda u) = \lambda u - 2\langle \lambda u, v\rangle v = \lambda u,$$

for every $\lambda \in \mathbb{R}$. Observe carefully that we also have $\mathrm{ref}_{v^\perp}(z) \in u^\perp$ because $z \in u^\perp$, and hence, $\mathrm{ref}_{v^\perp}(z) \in \mathrm{proj}_{u^\perp}(K)$.

In the process of obtaining $S_u(K)$, the segment $\mathrm{ref}_{v^\perp}(z) + \mathrm{seg}[\alpha u, \beta u]$ is translated in such a way that $\mathrm{ref}_{v^\perp}(z)$ becomes its midpoint. As a consequence, we get that

$$\mathrm{ref}_{v^\perp}(z) + tu \in S_u(K).$$

Finally, it is immediate to see that $\mathrm{ref}_{v^\perp}(x) = \mathrm{ref}_{v^\perp}(z) + tu$, and this leads to

$$\mathrm{ref}_{v^\perp}(x) \in S_u(K).$$

This shows that $S_u(K)$ is symmetric with respect to $v^\perp$.

(b) It follows immediately from item **(a)** that $L := S_{e_n} \circ \ldots \circ S_{e_1}(K)$ is symmetric with respect to all hyperplanes $e_1^\perp, \ldots, e_n^\perp$. Thus, if $x \in L$, we have the sequence of implications:

$$x \in L \Rightarrow \mathrm{ref}_{e_1^\perp}(x) \in L \Rightarrow \mathrm{ref}_{e_2^\perp}(\mathrm{ref}_{e_1^\perp}(x)) \in L \Rightarrow \ldots \Rightarrow \mathrm{ref}_{e_n^\perp}(\ldots \mathrm{ref}_{e_2^\perp}(\mathrm{ref}_{e_1^\perp}(x))\ldots) \in L.$$

Since it is clear that

$$\mathrm{ref}_{e_n^\perp}(\ldots \mathrm{ref}_{e_2^\perp}(\mathrm{ref}_{e_1^\perp}(x))\ldots) = -x,$$

we get that L is symmetric with respect to the origin.

(c) Assume that there is a point $x \in L$ such that $|x| > (\mathrm{diam}L)/2$. Since L is symmetric with respect to $0_{\mathbb{R}^n}$, we must have $-x \in L$. Thus,

$$\mathrm{diam}L \geq |x-(-x)| = 2|x| > \mathrm{diam}L,$$

a contradiction.

(d) From item **(c)** above and item **(iv)** of Proposition 2.7.2 (main book), we have

$$S_{e_n} \circ \ldots \circ S_{e_1}(K) \subseteq \frac{\mathrm{diam}(S_{e_n} \circ \ldots S_{e_1}(K))}{2} B^n \subseteq \frac{\mathrm{diam}K}{2} B^n.$$

This, together with the fact that Steiner symmetrizations preserve volume, yields

$$\mathrm{vol}_n(K) = \mathrm{vol}_n(S_{e_n} \circ \ldots \circ S_{e_1}(K)) \leq \mathrm{vol}_n\left(\frac{\mathrm{diam}K}{2} B^n\right) = \frac{\kappa_n}{2^n}(\mathrm{diam}K)^n,$$

as we wanted to prove.

□

Recall that the *regular n-dimensional cross-polytope* is the polytope

$$P_\Delta^n := \mathrm{conv}\left\{\pm e_1, \ldots, \pm e_n\right\} := \left\{(x_1, \ldots, x_n) \in \mathbb{R}^n : |x_1| + \ldots + |x_n| \leq 1\right\}.$$

A *cross-polytope of* $\mathbb{R}^n$ is an affine image of the regular n-dimensional cross-polytope. That is, a cross-polytope of $\mathbb{R}^n$ is a set having the form $x_0 + T(P_\Delta^n)$ for some $x_0 \in \mathbb{R}^n$ and some $T \in \mathrm{GL}(n)$.

Exercise 5.26. **(a)** *Let $K \subseteq \mathbb{R}^n$ be a convex body. Prove that there is a unique cross-polytope P_0 contained in K maximizing the volume of the cross-polytopes contained in K.*

(b) *Assume that $K \subseteq \mathbb{R}^n$ is a convex body symmetric with respect to a point $x_0 \in K$. Show that the cross-polytope of maximal volume contained in K is also centered at x_0.*

(c) *Let $K \subseteq \mathbb{R}^n$ be a convex body centered at the origin, and assume that the regular n-dimensional cross-polytope is the cross-polytope of maximum volume contained in K. Prove that $K \subseteq [-1,1]^n$.*

Hint: For item **(a)**, use the same strategy as in the proof of Theorem 9.1.2 in the main book. The solution of item **(b)** is similar to the proof of Lemma 9.2.1 (main book).

Solution 5.26. **(a)** The proof is completely analogous to that of Theorem 9.1.2 (main book). The only substantial difference is the construction of a cross-polytope of greater volume contained in the convex hull of two translated copies of the same cross-polytope, but this construction is trivial. Indeed, up to an orthogonal map, we may assume that the translation is made in the direction of e_1. Then, we consider a linear map that "stretches" e_1 while maintaining all other vectors of canonical basis unaltered.

(b) Of course, we may assume that $x_0 = 0_{\mathbb{R}^n}$. Let $P = z_0 + T(P_\Delta^n)$ be any cross-polytope contained in K. Notice that

$$Q := \frac{1}{2}(P-P) = \frac{1}{2}(z_0 + T(P_\Delta^n) - z_0 - T(P_\Delta^n)) = \frac{1}{2}T(P_\Delta^n - P_\Delta^n) = T(P_\Delta^n).$$

That is, Q is a cross-polytope with same volume as P. Since K is origin-symmetric, we have from $P \subseteq K$ that $-P \subseteq -K = K$, and this gives that $Q \subseteq K$. It follows that any cross-polytope contained in K corresponds to an origin-symmetric cross-polytope of the same volume also contained in K.

Therefore, the cross-polytope of maximum volume contained in K is symmetric with respect to the origin.

(c) Assume that $K \not\subseteq [-1,1]^n$. This means that $|x_j| > 1$ for some point $x_0 = (x_1, \ldots, x_n) \in K$ and some $j \in \{1, \ldots, n\}$. Since K is centered at the origin, it holds that $-x_0 \in K$. The regular n-dimensional cross-polytope is contained in K, and hence, $\pm e_1, \ldots, \pm e_n$ are points of K. By convexity, we have that

$$P := \operatorname{conv}\left\{ \pm e_1, \ldots, e_{j-1}, \pm x_0, \pm e_{j+1}, \ldots, \pm e_n \right\} \subseteq K.$$

Now, notice that P is a cross-polytope, since $P = T(P_\Delta^n)$ for $T \in \mathrm{GL}(n)$ defined in such a way that $T(e_i) = e_i$ for $i \neq j$ and $T(e_j) = x_0$. We have that

$$\mathrm{vol}_n(P) = |\det(T)| \mathrm{vol}_n(P_\Delta^n),$$

and it is easy to compute the determinant of T. For that sake, just notice that the standard matricial representation of T is

$$T = \begin{bmatrix} 1 & 0 & \ldots & x_1 & \ldots & 0 \\ 0 & 1 & \ldots & x_2 & \ldots & 0 \\ \vdots & \vdots & \ldots & \vdots & \ldots & \vdots \\ 0 & 0 & \ldots & x_n & \ldots & 1 \end{bmatrix},$$

from where we easily get that $\det(T) = x_j$. It follows that

$$\mathrm{vol}_n(P) = |x_j| \mathrm{vol}_n(P_\Delta^n) > \mathrm{vol}_n(P_\Delta^n).$$

This contradicts the fact that P_Δ^n is the cross-polytope of maximum volume contained in K. Thus, we indeed have that $K \subseteq [-1,1]^n$.

□

Exercise 5.27 (Mahler)**.** *Let $K \subseteq \mathbb{R}^n$ be a convex body symmetric with respect to the origin. Prove that*

$$\mathrm{vol}_n(K) \mathrm{vol}_n(K^\circ) \geq \frac{4^n}{(n!)^2}.$$

Notice that this inequality is much weaker than that of Mahler's conjecture.

Solution 5.27. We know that $\mathrm{vol}_n(K)\mathrm{vol}_n(K^\circ)$ is invariant under linear transformations. Also, since K is origin-symmetric, it follows from item **(b)** of Exercise 5.26 that the cross-polytope of maximal volume contained in K is centered at $0_{\mathbb{R}^n}$. Therefore, up to a linear transformation, we may assume that the regular n-dimensional cross-polytope P_Δ^n is the cross-polytope of maximal volume contained in K. From item **(c)** of Exercise 5.26, it follows that $K \subseteq [-1,1]^n$. Thus,

$$\left([-1,1]^n\right)^\circ \subseteq K^\circ,$$

and from Exercise 4.5, we obtain that $\left([-1,1]^n\right)^\circ = P_\Delta^n$. Thus,

$$\mathrm{vol}_n(K) \mathrm{vol}_n(K^\circ) \geq \mathrm{vol}_n(P_\Delta^n)^2 = \frac{4^n}{(n!)^2},$$

where the last equality follows from item **(b)** of Exercise 4.3.

□

Exercise 5.28. *Let $f:\mathbb{R}^n\to\mathbb{R}$ be a continuous function that is strictly positive over $\mathbb{S}^{n-1}$. Use Jensen's inequality to prove that*

$$\int_{\mathbb{S}^{n-1}} f(u)\,d\mathcal{H}^{n-1}(u)\geq(\omega_n)^{(n+1)/n}\left(\int_{\mathbb{S}^{n-1}} f(u)^{-n}\,d\mathcal{H}^{n-1}(u)\right)^{-1/n}.$$

Show that if f is sublinear, then this inequality can be derived from inequality (5.5.3) of the main book.

Solution 5.28. For the first claim, we just have to use Jensen's inequality to the convex function $\lambda\mapsto\lambda^{-1/n}$ considering the spherical probability measure:

$$\sigma(\cdot)=\frac{1}{\omega_n}\mathcal{H}^{n-1}|_{\mathbb{S}^{n-1}}(\cdot).$$

Hence, all we have to prove is that the inequality also follows from (5.5.3) in the main book in the case where f is sublinear. In this case, we know that $f=h_K$ for some $K\in\mathcal{CC}^n$. Since f is strictly positive over $\mathbb{S}^{n-1}$, we get that $K\in\mathcal{K}^n_0$. Thus, we use inequality (5.5.3) to write

$$\int_{\mathbb{S}^{n-1}} f(u)\,d\mathcal{H}^{n-1}(u)=\frac{\omega_n}{2}\int_{\mathbb{S}^{n-1}} 2h_K(u)\,d\sigma(u)=\frac{\omega_n}{2}w(K)\geq\omega_n(\kappa_n)^{1/n}\big(\mathrm{vol}_n(K^\circ)\big)^{-1/n}.$$

Now we use (4.3.8) in the main book to write the volume of K°, and this yields

$$\begin{aligned}\int_{\mathbb{S}^{n-1}} f(u)\,d\mathcal{H}^{n-1}(u)&\geq\omega_n(\kappa_n)^{1/n}\left(\kappa_n\int_{\mathbb{S}^{n-1}} h_K(u)^{-n}\,d\sigma(u)\right)^{-1/n}=\\&=\omega_n(\kappa_n)^{1/n}\left(\frac{1}{n}\int_{\mathbb{S}^{n-1}} f(u)^{-n}\,d\mathcal{H}^{n-1}(u)\right)^{-1/n}.\end{aligned}$$

□

For the next exercise, recall that we denote by f^* the *conjugate* of a given function f. See Section 1.4 from the main book for the definition and properties.

Exercise 5.29. *Let $f:\mathbb{R}^n\to\mathbb{R}$ be a sublinear function that is strictly positive over $\mathbb{S}^{n-1}$. Prove that*

$$\int_{\mathbb{S}^{n-1}} f(u)\,d\mathcal{H}^{n-1}\geq n\kappa_n^{(n-1)/n}\left(\int_{\mathbb{R}^n} e^{-f^*(x)}\,dx\right)^{1/n}.$$

Solution 5.29. Since f is sublinear and strictly positive over $\mathbb{S}^{n-1}$, we have that $f=h_K$ for some $K\in\mathcal{K}^n_0$. We recall that $1_K=e^{-I^\infty_K}=e^{-h^*_K}=e^{-f^*}$. Thus, using Urysohn's inequality, we write

$$\begin{aligned}\int_{\mathbb{S}^{n-1}} f(u)\,d\mathcal{H}^{n-1}(u)&=\frac{\omega_n}{2}\int_{\mathbb{S}^{n-1}} 2h_K(u)\,d\sigma(u)=\frac{\omega_n}{2}w(K)\geq\\\geq\omega_n\left(\frac{\mathrm{vol}_n(K)}{\kappa_n}\right)^{1/n}&=\frac{n\kappa_n}{\kappa_n^{1/n}}\left(\int_{\mathbb{R}^n} 1_K\,dx\right)^{1/n}=n\kappa_n^{(n-1)/n}\left(\int_{\mathbb{R}^n} e^{-f^*(x)}\,dx\right)^{1/n}.\end{aligned}$$

□

A set $K\in\mathcal{CC}^n$ can be represented by its characteristic function 1_K, which is log-concave. This observation leads to extensions of geometric quantities defined for convex bodies to log-concave

functions. In [218], Rotem introduced the following notion of *mean width for a log-concave function* f:

$$\tilde{w}(f) := \frac{4}{n(2\pi)^{n/2}} \int_{\mathbb{R}^n} h_f(x) e^{-\frac{|x|^2}{2}}\, dx, \tag{5.4}$$

where h_f is the support function of f, defined in Exercise 1.33 as $h_f := (-\log f)^*$. It is worth mentioning here that we actually use a normalization different to that of [218] (to be specific, we multiply theirs by 2). Also, in the exercises below, we restrict ourselves to the family

$$\mathrm{LC}_\infty(\mathbb{R}^n) :=$$
$$:= \left\{ f = e^{-\varphi} : \varphi \text{ is closed, convex, proper, and such that } \lim_{|x|\to+\infty} \varphi(x) = +\infty \right\},$$

in order to avoid technicalities. In the next few exercises, we investigate this definition. Our first task is to show that there exists a constant $c_n > 0$ (which only depends on the dimension) such that

$$\tilde{w}(1_K) = c_n w(K)$$

for any $K \in CC^n$. This justifies why we may see $\tilde{w}(\cdot)$ as an extension of the "geometric" mean width to log-concave functions.

Exercise 5.30. *Let $K \in CC^n$. Prove that*

$$\tilde{w}(1_K) = \frac{2(2\pi)^{\frac{1}{2}} \kappa_n}{(n+1)\kappa_{n+1}} w(K).$$

Hint: You can use the equality

$$\int_0^{+\infty} s^k e^{-as^2}\, ds = \frac{\Gamma((k+1)/2))}{2a^{(k+1)/2}}, \tag{5.5}$$

which holds for any $a > 0$ and every non-negative integer k. Use also (4.3.3) from the main book.

Solution 5.30. From item **(c)** of Exercise 1.33, we have that $h_{1_K} = h_K$. Using integration by polar coordinates, Fubini's theorem, and equality (5.5), we compute

$$\begin{aligned}\int_{\mathbb{R}^n} h_{1_K}(x) e^{-\frac{|x|^2}{2}}\, dx &= \int_{\mathbb{R}^n} h_K(x) e^{-\frac{|x|^2}{2}}\, dx = \int_0^{+\infty} \int_{\mathbb{S}^{n-1}} h_K(tu) e^{\frac{-|tu|^2}{2}} t^{n-1}\, d\mathcal{H}^{n-1}(u) dt = \\ &= \left(\int_0^{+\infty} t^n e^{-\frac{t^2}{2}}\, dt \right) \left(\int_{\mathbb{S}^{n-1}} h_K(u)\, d\mathcal{H}^{n-1}(u) \right) = \\ &= \frac{\Gamma\left(\frac{n+1}{2}\right)}{2\left(\frac{1}{2}\right)^{\frac{n+1}{2}}} \frac{\mathcal{H}^{n-1}(\mathbb{S}^{n-1})}{2} \left(2 \int_{\mathbb{S}^{n-1}} h_K(u)\, d\sigma(u) \right) = \frac{\Gamma\left(\frac{n+1}{2}\right)}{2\left(\frac{1}{2}\right)^{\frac{n+1}{2}}} \frac{n\kappa_n}{2} w(K).\end{aligned}$$

Notice that we used the fact that the probability measure σ is just $\mathcal{H}^{n-1}|_{\mathbb{S}^{n-1}}$ normalized and also the fact that $\mathcal{H}^{n-1}(\mathbb{S}^{n-1}) = n\kappa_n$; see (4.3.3) in the main book. Thus,

$$\tilde{w}(1_K) = \frac{4}{n(2\pi)^{n/2}} \int_{\mathbb{R}^n} h_{1_K}(x) e^{-\frac{|x|^2}{2}}\, dx = \frac{4}{n(2\pi)^{n/2}} \frac{\Gamma\left(\frac{n+1}{2}\right)}{2\left(\frac{1}{2}\right)^{\frac{n+1}{2}}} \frac{n\kappa_n}{2} w(K).$$

Now we deal with the scalar multiplying $w(K)$ in the right-hand side. We use (4.3.3) to rewrite $\Gamma((n+1)/2)$. It follows that

$$\frac{4}{n(2\pi)^{n/2}} \frac{\Gamma\left(\frac{n+1}{2}\right)}{2\left(\frac{1}{2}\right)^{\frac{n+1}{2}}} \frac{n\kappa_n}{2} = \frac{1}{(2\pi)^{n/2}} 2^{\frac{n+1}{2}} \frac{2\pi^{\frac{n+1}{2}}}{\omega_{n+1}} \kappa_n = \frac{(2\pi)^{\frac{n+1}{2}}}{(2\pi)^{n/2}} \frac{2\kappa_n}{(n+1)\kappa_{n+1}} = \frac{2(2\pi)^{\frac{1}{2}}\kappa_n}{(n+1)\kappa_{n+1}},$$

and this concludes the proof.

□

Next, we investigate some other properties of $\tilde{w}$. Among them there is a type of linearity for "reasonable" operations in $\mathrm{LC}_\infty(\mathbb{R}^n)$. We also compute the mean width of the Gaussian function

$$G(x) := e^{-\frac{|x|^2}{2}},$$

which, in some sense, "plays the role" of the unit ball. Indeed, from item **(c)** of Exercise 1.39, we have that G is the unique self-polar function of $\mathrm{LC}_\infty(\mathbb{R}^n)$. To compute the integral of G, notice that the change of variables $y = x/\sqrt{2}$ leads to

$$\int_{\mathbb{R}^n} G \, d\mathcal{L}^n = \int_{\mathbb{R}^n} e^{-\frac{|x|^2}{2}} \, dx = \int_{\mathbb{R}^n} e^{-|y|^2} 2^{n/2} \, dy = (2\pi)^{n/2}, \tag{5.6}$$

where the last equality follows from (4.3.1) in the main book. For Exercises 5.31, 5.33 and 5.34, we use heavily the definitions introduced and investigated in the exercises of Chapter 1. In most parts, the precise references are given. However, if there is some confusion regarding notations, we advise the reader to take a look at those exercises.

Exercise 5.31. **(a)** *Recall that $f \star g$ denotes the* Asplund product *defined in Exercise 1.26 and that $\lambda \cdot f$ stands for the scalar multiplication introduced in Exercise 1.28. Verify that*

$$\tilde{w}(f \star (\lambda \cdot g)) = \tilde{w}(f) + \lambda \tilde{w}(g)$$

for any $f, g \in \mathrm{LC}_\infty(\mathbb{R}^n)$ and every $\lambda > 0$.

(b) *Show that $\tilde{w}(G) = 2$.*

(c) *Let $x_0 \in \mathbb{R}^n$, $U \in \mathrm{O}(n)$, and put $T(\cdot) := x_0 + U(\cdot)$. Show that for any $f \in \mathrm{LC}_\infty(\mathbb{R}^n)$, we have*

$$h_{f \circ T}(x) = -\langle U(x), x_0 \rangle + h_f(U(x))$$

for every $x \in \mathbb{R}^n$. Conclude that $\tilde{w}(f \circ T) = \tilde{w}(f)$. In other words, $\tilde{w}$ is translation-invariant and $\mathrm{O}(n)$-invariant, in some sense.

Solution 5.31. **(a)** This follows immediately from the fact that

$$h_{f \star (\lambda \cdot g)} = h_f + \lambda h_g,$$

as it is shown in Exercise 1.33.

(b) Since $G(x) = e^{-|x|^2/2}$, it is clear from Exercise 1.32 that $h_G = |x|^2/2$. Thus, using integration by parts, recalling that $\mathcal{H}^{n-1}(\mathbb{S}^{n-1}) = \omega_n = n\kappa_n$, and from (5.5), we get

$$\tilde{w}(G) = \frac{4}{n(2\pi)^{n/2}} \int_{\mathbb{R}^n} \frac{|x|^2}{2} e^{-\frac{|x|^2}{2}}\, dx = \frac{4}{n(2\pi)^{n/2}} \int_0^{+\infty} \int_{\mathbb{S}^{n-1}} \frac{t^2}{2} e^{-\frac{t^2}{2}} t^{n-1}\, d\mathcal{H}^{n-1} dt =$$
$$= \frac{2\mathcal{H}^{n-1}(\mathbb{S}^{n-1})}{n(2\pi)^{n/2}} \int_0^{+\infty} t^{n+1} e^{-\frac{t^2}{2}}\, dt = \frac{2n\kappa_n}{n(2\pi)^{n/2}} \frac{\Gamma((n+2)/2)}{2(1/2)^{\frac{n+2}{2}}}.$$

Now we use (4.3.3) to compute $\Gamma((n+2)/2)$. We arrive at

$$\tilde{w}(G) = \frac{2n\kappa_n}{n(2\pi)^{n/2}} \frac{\Gamma((n+2)/2)}{2(1/2)^{\frac{n+2}{2}}} = \frac{\kappa_n}{(2\pi)^{n/2}} 2^{\frac{n+2}{2}} \Gamma((n+2)/2) =$$
$$= \frac{2\kappa_n}{\pi^{n/2}} \Gamma((n+2)/2) = \frac{2\kappa_n}{\pi^{n/2}} \frac{2\pi^{\frac{n+2}{2}}}{\omega_{n+2}} = \frac{4\kappa_n \pi}{(n+2)\kappa_{n+2}}.$$

Finally, from item **(a)** of Exercise 4.7, we obtain

$$\tilde{w}(G) = \frac{4\kappa_n \pi}{(n+2)\kappa_{n+2}} = \frac{4\kappa_n \pi}{(n+2)\frac{2\pi}{n+2}\kappa_n} = 2.$$

(c) Notice that T is a bijection of $\mathbb{R}^n$. Write $f = e^{-\varphi}$. Thus, $f \circ T = e^{-\varphi \circ T}$, and hence,

$$h_{f\circ T}(x) = (\varphi \circ T)^*(x) = \sup\left\{\langle x, y\rangle - \varphi(T(y)) : y \in \mathbb{R}^n\right\} =$$
$$= \sup\left\{\langle x, y\rangle - \varphi(Uy + x_0) : y \in \mathbb{R}^n\right\} =$$
$$= \sup\left\{\langle Ux, Uy + x_0\rangle - \langle Ux, x_0\rangle - \varphi(Uy + x_0) : y \in \mathbb{R}^n\right\} =$$
$$= -\langle Ux, x_0\rangle + \sup\left\{\langle Ux, z\rangle - \varphi(z) : z \in \mathbb{R}^n\right\} = -\langle Ux, x_0\rangle + \varphi^*(Ux) =$$
$$= -\langle Ux, x_0\rangle + h_f(Ux),$$

for any $x \in \mathbb{R}^n$. The proof is complete.

□

Our next objective is to prove a version of Urysohn's inequality for $\tilde{w}$. This is taken from [218]. We need some auxiliary results. The first one is known as *Shannon's inequality*, and it is a corollary of Jensen's inequality.

Exercise 5.32 (Shannon's inequality)**.** *Let $p, q\colon \mathbb{R}^n \to [0, +\infty)$ be non-negative Lebesgue integrable functions. Assume that p is strictly positive, that*

$$\int_{\mathbb{R}^n} p(x)\, dx = 1,$$

and that

$$\int_{\mathbb{R}^n} q(x)\, dx > 0.$$

Prove that

$$\log\left(\int_{\mathbb{R}^n} q(x)\, dx\right) \geq \int_{\mathbb{R}^n} \log\left(\frac{q(x)}{p(x)}\right) p(x)\, dx,$$

where we assume that $\log(0) = -\infty$. *Show that equality holds if and only if there exists* $\alpha > 0$ *such that* $q = \alpha p$ *Lebesgue-almost everywhere.*

Hint: Use Jensen's inequality (Theorem 1.2.2 and Remark 1.2.3 of the main book).

Solution 5.32. Let μ be the measure on $\mathbb{R}^n$ whose density with respect to the Lebesgue measure is p. That is,

$$\mu(E) := \int_E p(x)\,dx$$

for every Borel set $E \subseteq \mathbb{R}^n$. Since q is Lebesgue integrable, it follows immediately that $f := q/p$ is μ-integrable. Let $\varphi = -\log\colon [0, +\infty) = (-\infty, +\infty]$. Notice that $\operatorname{int}(\operatorname{dom}(\varphi)) = (0, +\infty)$, and hence,

$$\int_{\mathbb{R}^n} f(x)\,d\mu = \int_{\mathbb{R}^n} \frac{q(x)}{p(x)} p(x)\,dx = \int_{\mathbb{R}^n} q(x)\,dx \in (0, +\infty) = \operatorname{int}(\operatorname{dom}(\varphi)).$$

Applying Jensen's inequality (in the version given in Remark 1.2.3 of the main book) for μ and f as defined above, we get

$$-\log\left(\int_{\mathbb{R}^n} \frac{q(x)}{p(x)} p(x)\,dx\right) \leq \int_{\mathbb{R}^n} -\log\left(\frac{q(x)}{p(x)}\right) p(x)\,dx.$$

Multiplying both sides by -1, we get the inequality. If the equality case holds, then since $-\log$ is not an affine function, we get that $f = q/p$ is constant Lebesgue-almost everywhere. Writing $q/p = \alpha$, since the integral of q is positive, we get that $\alpha > 0$.

□

We will also need an important functional version of the Blaschke-Santaló inequality. The proof is not easy enough to be inserted as an exercise. We state it below.

Theorem 5.1 (Theorem 1.3 in [22])**.** *Let* $f\colon \mathbb{R}^n \to [0, +\infty)$ *be any measurable function such that*

$$0 < \int_{\mathbb{R}^n} f < +\infty.$$

There exists a point $x_0 \in \mathbb{R}^n$ *such that letting* $\bar{f} := f(x - x_0)$, *the inequality*

$$\int_{\mathbb{R}^n} \bar{f} \int_{\mathbb{R}^n} \bar{f}^\circ \leq (2\pi)^n \tag{5.7}$$

holds, where $\bar{f}^\circ$ *is the polar function of* $\bar{f}$, *as defined in Exercise 1.39. That is,* $\bar{f}^\circ = e^{-h_{\bar{f}}}$.

Despite not being directly necessary for Urysohn's inequality, it is also useful to understand what happens when f is of the form $f(x) = Ce^{-\frac{|x-x_0|^2}{2}}$. This is done next.

Exercise 5.33. *Let* $a \in \mathbb{R}$ *and put*

$$\bar{\varphi}(x) = a + \frac{|x - x_0|^2}{2}.$$

Prove that

$$\int_{\mathbb{R}^n} e^{-\bar{\varphi}} \int_{\mathbb{R}^n} e^{-\bar{\varphi}^*} \geq (2\pi)^n,$$

with equality if and only if $x_0 = 0_{\mathbb{R}^n}$.

Solution 5.33. Since $|x|^2/2$ is self-conjugate, from Exercise 1.19, we get that

$$\bar{\varphi}^*(x) = -a + \langle x, x_0\rangle + \frac{|x|^2}{2} = -a + \frac{|x+x_0|^2}{2} - \frac{|x_0|^2}{2}.$$

Also, by standard change of variables and from (5.6), it is clear that

$$\int_{\mathbb{R}^n} e^{-\bar{\varphi}} = e^{-a}\int_{\mathbb{R}^n} e^{-\frac{|x|^2}{2}} = e^{-a}(2\pi)^{n/2}.$$

Thus,

$$\int_{\mathbb{R}^n} e^{-\bar{\varphi}}\int_{\mathbb{R}^n} e^{-\bar{\varphi}^*} = e^{-a}(2\pi)^{n/2}\int_{\mathbb{R}^n} e^{a-\frac{|x+x_0|^2}{2}+\frac{|x_0|^2}{2}} = (2\pi)^{n/2}e^{\frac{|x_0|^2}{2}}\int_{\mathbb{R}^n} e^{-\frac{|x+x_0|^2}{2}} =$$
$$= (2\pi)^n e^{\frac{|x_0|^2}{2}},$$

where in the last equality, we use change of variables and (5.6) again. Now, we simply notice that

$$e^{\frac{|x_0|^2}{2}} \geq 1$$

for every $x_0 \in \mathbb{R}^n$, and equality holds if and only if $x_0 = 0_{\mathbb{R}^n}$. This concludes the proof.

□

Now, we are in the position to state Urysohn's inequality for $\tilde{w}$.

Exercise 5.34 (Urysohn's inequality for $\tilde{w}$)**.** *Let* $f \in \mathrm{LC}_\infty(\mathbb{R}^n)$ *be such that*

$$\int_{\mathbb{R}^n} f^\circ > 0,$$

where we remind the reader that $f^\circ := e^{-h_f}$. *Prove that*

$$\tilde{w}(f) \geq 4\log\left(\frac{1}{(2\pi)^{\frac{n}{2}}}\int_{\mathbb{R}^n} f\right)^{\frac{1}{n}} + 2.$$

Show that equality holds if and only if

$$f(x) = Ce^{-\frac{|x+x_0|^2}{2}}$$

for some $C > 0$ *and some* $x_0 \in \mathbb{R}^n$.

Solution 5.34. Since $f \in \mathrm{LC}_\infty(\mathbb{R}^n)$, we know that

$$\int_{\mathbb{R}^n} f < +\infty.$$

On the other hand, if

$$\int_{\mathbb{R}^n} f = 0,$$

then there is nothing to prove. Thus, we assume that f has positive and finite integral.

Write $f = e^{-\varphi}$ for $\varphi \in \mathrm{Cvx}_\infty(\mathbb{R}^n)$. Then, $f^\circ = e^{-\varphi^*}$. In view of Exercise 1.35, the condition that

$$\int_{\mathbb{R}^n} f^\circ > 0$$

is equivalent to $\mathrm{int}(\mathrm{dom}(\varphi^*)) \neq \emptyset$. Let $\bar{\varphi}(x) := \varphi(x - x_0)$ be such that

$$\int_{\mathbb{R}^n} e^{-\bar{\varphi}} \int_{\mathbb{R}^n} e^{-\bar{\varphi}^*} \leq (2\pi)^n. \tag{5.8}$$

Letting $\bar{f} := e^{-\bar{\varphi}} = f(x - x_0)$, we have immediately that

$$\int_{\mathbb{R}^n} \bar{f} = \int_{\mathbb{R}^n} f \in (0, +\infty).$$

From the inequality above, we get that

$$\int_{\mathbb{R}^n} e^{-\bar{\varphi}^*} < +\infty,$$

and we claim that we also have

$$\int_{\mathbb{R}^n} e^{-\bar{\varphi}^*} > 0.$$

In fact, we know from Exercise 1.19 that $\bar{\varphi}^*(\cdot) = \langle \cdot, x_0 \rangle + \varphi^*(\cdot)$, and this gives that $\mathrm{dom}(\bar{\varphi}^*) = \mathrm{dom}(\varphi^*)$. Since we already know that $\mathrm{int}(\mathrm{dom}(\varphi^*)) \neq \emptyset$, it follows that $\mathrm{int}(\mathrm{dom}(\bar{\varphi}^*)) \neq \emptyset$, and using Exercise 1.35 again, we get that the integral of $e^{-\bar{\varphi}^*}$ is positive.

Now, we apply Shannon's inequality (Exercise 5.32) for $p(x) = (2\pi)^{-\frac{n}{2}} e^{-\frac{|x|^2}{2}}$ and $q(x) = e^{-\bar{\varphi}^*}$ (notice that all the previous argumentation was made to prove that $q(x)$ has the hypotheses needed to use Shannon's inequality). We arrive at

$$\begin{aligned}
\log\left(\int_{\mathbb{R}^n} e^{-\bar{\varphi}^*}\right) &\geq \int_{\mathbb{R}^n} \log\left((2\pi)^{n/2} e^{-\bar{\varphi}^* + \frac{|x|^2}{2}}\right) \frac{1}{(2\pi)^{n/2}} e^{-\frac{|x|^2}{2}} = \\
&= \frac{1}{(2\pi)^{n/2}} \int_{\mathbb{R}^n} \left(\log(2\pi)^{n/2} - \bar{\varphi}^* + \frac{|x|^2}{2}\right) e^{-\frac{|x|^2}{2}} = \\
&= -\frac{1}{(2\pi)^{n/2}} \int_{\mathbb{R}^n} \bar{\varphi}^* e^{\frac{-|x|^2}{2}} + \frac{\log(2\pi)^{n/2}}{(2\pi)^{n/2}} \int_{\mathbb{R}^n} e^{\frac{-|x|^2}{2}} + \frac{1}{(2\pi)^{n/2}} \int_{\mathbb{R}^n} \frac{|x|^2}{2} e^{-\frac{|x|^2}{2}}.
\end{aligned}$$

We use integration by polar coordinates (5.5) and (4.3.3) to compute the last integral, and observe that the integral of $\bar{\varphi}^* e^{-\frac{|x|^2}{2}}$ is a multiple of $\tilde{w}(\bar{f})$. We get

$$\log\left(\int_{\mathbb{R}^n} e^{-\bar{\varphi}^*}\right) \geq -\frac{1}{(2\pi)^{n/2}} \frac{n(2\pi)^{n/2}}{4} \tilde{w}(\bar{f}) + \log(2\pi)^{n/2} + \frac{1}{(2\pi)^{n/2}} \frac{n\kappa_n}{2} \frac{(2\pi)^{\frac{n+2}{2}}}{(n+2)\kappa_{n+2}}.$$

Using item **(a)** of Exercise 4.7 to rewrite κ_{n+2}, we arrive at

$$\log\left(\int_{\mathbb{R}^n} e^{-\bar{\varphi}^*}\right) \geq -\frac{n}{4} \tilde{w}(\bar{f}) + \log(2\pi)^{n/2} + \frac{n}{2}.$$

Now, we use (5.8) again to bound the left-hand side. This gives

$$-\log\left(\frac{\int_{\mathbb{R}^n}\bar{f}}{(2\pi)^n}\right) \geq -\frac{n}{4}\tilde{w}(\bar{f}) + \frac{n}{2} + \log(2\pi)^{n/2},$$

and rearranging the terms, we finally obtain

$$\frac{n}{4}\tilde{w}(\bar{f}) \geq \frac{n}{2} + \log(2\pi)^{n/2} + \log\left(\frac{\int_{\mathbb{R}^n}\bar{f}}{(2\pi)^n}\right) =$$

$$= \frac{n}{2} + \log\left((2\pi)^{n/2}\frac{\int_{\mathbb{R}^n}\bar{f}}{(2\pi)^n}\right) = \frac{n}{2} + \log\left(\frac{1}{(2\pi)^{n/2}}\int_{\mathbb{R}^n}\bar{f}\right).$$

Multiplying both sides by $4/n$, we get the desired inequality, but for $\bar{f}$. Since $\tilde{w}(\cdot)$ and the integral are translation-invariant, we get that the same inequality holds if we replace $\bar{f}$ by f.

If equality holds, then we have the equality case in Shannon's inequality. As a consequence, we have that $q = \alpha p$ Lebesgue-almost everywhere for some $\alpha > 0$. This reads

$$e^{-h_{\bar{f}}(x)} = \alpha(2\pi)^{n/2}e^{-\frac{|x|^2}{2}},$$

for almost all x, where $\bar{f}(x) := f(x - x_0)$ is such that

$$\int_{\mathbb{R}^n}\bar{f}\int_{\mathbb{R}^n}\bar{f}^\circ \leq (2\pi)^n. \tag{5.9}$$

This leads to

$$\bar{\varphi}^*(x) = \frac{|x|^2}{2} - a \tag{5.10}$$

for almost all x, where $a = \log(\alpha(2\pi)^{n/2})$. We claim that, in this case, the equality actually holds everywhere. In fact, in particular, we have that $\mathrm{dom}(\bar{\varphi}^*)$ has a total measure. Since $\bar{\varphi}^*$ is convex, it follows that $\mathrm{dom}(\bar{\varphi}^*) = \mathbb{R}^n$, because it is easy to see that the unique convex subset of $\mathbb{R}^n$, which has total Lebesgue measure, is $\mathbb{R}^n$ itself (if $C \subseteq \mathbb{R}^n$ is proper and convex, then $\mathbb{R}^n \setminus C$ contains a ball).

We have that $\bar{\varphi} \in \mathrm{Cvx}_\infty(\mathbb{R}^n)$, and this gives that $\bar{\varphi}$ is convex and closed. Thus, $\bar{\varphi} = \bar{\varphi}^{**}$, and we write

$$\bar{\varphi}(x) = \left(\frac{|\cdot|}{2} - a\right)^*(x) = \sup\left\{\langle x, y\rangle - \frac{|y|^2}{2} + a : y \in \mathbb{R}^n\right\} =$$

$$= a + \sup\left\{\langle x, y\rangle - \frac{|y|^2}{2} : y \in \mathbb{R}^n\right\} = a + \left(\frac{|\cdot|}{2}\right)^*(x) = a + \frac{|x|^2}{2},$$

where the last equality follows from Exercise 1.32 (the function $|\cdot|^2/2$ is self-conjugate). It follows that

$$\bar{f}(x) = e^{-a-\frac{|x|^2}{2}} = Ce^{-\frac{|x|^2}{2}}.$$

Thus,

$$f(x) = Ce^{-\frac{|x+x_0|^2}{2}}, \tag{5.11}$$

where $x_0 \in \mathbb{R}^n$ is such that (5.9) holds for $\bar{f}(x) := f(x - x_0)$. Notice that for f as above, from Exercise 5.33, we have that x_0 is the unique point with this property.

Conversely, assume that f is as (5.11) for some $C > 0$ and some $x_0 \in \mathbb{R}^n$. We prove that equality holds in Urysohn's equality for f. Since $\tilde{w}(\cdot)$ and the integral are translation-invariant, we may assume that $x_0 = 0_{\mathbb{R}^n}$. By doing that, we get that equality holds in Shannon's inequality and also in (5.8). Hence, equality holds in Urysohn's inequality.

□

For Further Reading

For short descriptions of corresponding literature background, we select from the keywords given above Exercise 5.1 the following three topics: the *inequality of Rogers and Shephard, Mahler's conjecture*, and *Urysohn's inequality*.

The Rogers-Shephard Inequality

We start with the *Rogers-Shephard inequality* comparing the volume of an n-dimensional convex body K and the volume of its *difference body* $DK := K + (-K)$. Note that the homothetical copy $(K + (-K))/2$ of DK is called the *central symmetral* of K, having many applications (such as DK). Recall that the sharp lower bound for the volume of DK, namely, the 2^n-fold volume of K, follows directly from the Brunn-Minkowski inequality and is satisfied precisely by all centrally symmetric bodies K. The Rogers-Shephard inequality yields the corresponding upper bound, and there equality holds exactly when K is an n-simplex. Comprehensive representations with proofs, interesting variants, important comments, relations to other notions, and extensions of this famous inequality can be found in the following selection of monographs: Leichtweiss (see § 20 in [161] and Section 3.13 in [164]), Groemer [116] (Section 2.6), Gardner [97] (Section 3.2, Notes 3.3 and 8.12), Gruber [117] (Sections 9.5 and 30.1), Schneider [228] (Sections 3.1 and 10.1), Toth [245] (Section 3.8), Montejano et al. [188] (Chapters 10 and 13), and the main book [29] (Section 5.3 and subsection 5.7.3). Surveys discussing also the *geometry of difference bodies* and related inequalities are [187] and [1]. And in the *Handbook of Convex Geometry* [118], difference bodies and central symmetrals are mainly discussed in the expositions of Sangwine-Yager (on symmetrizations in connection with mixed volumes), Groemer (stability questions on geometric inequalities), Lutwak (in view of affine isoperimetric inequalities), and Heil and Martini (on special types of convex bodies, mainly in connection with characterizations of simplices or central symmetry).

We continue with a few *extensions* of the Rogers-Shephard inequality, which was originally proved in [216] and shortly after this extended in [217] to any pair K_1, K_2 of convex bodies, with equality if and only if $K_1 = -K_2$ is an n-simplex. With a new proof of the Rogers-Shephard inequality, Chakerian [74] clarified its relation to the *Brunn-Minkowski theorem* and gave further generalizations. An interesting extension of the Rogers-Shephard inequality in terms of *mixed volumes* is the so-called Godbersen conjecture, which is still unsolved (see [107]). Artstein-Avidan et al. replaced this conjecture in [23] by a new conjecture whose confirmation would imply Godbersen's conjecture. Various generalizations of the Rogers-Shephard inequality and the Godbersen conjecture are given by Böröczky in [59]. For more related results, we refer to the above given book parts and surveys.

Mahler's Inequality and Related Topics

Our next topic is *Mahler's inequality*. The *Blaschke-Santaló inequality* states that if a convex body K in Euclidean n-space has its centroid at the origin, then the *volume product* of K and its *polar body* is bounded from above by the squared volume of the unit ball, with equality if and only if K is an ellipsoid. Together with the knowledge about corresponding lower bounds (in particular, the still unsettled conjecture of Mahler, see his first contribution [183]), this inequality has *deep applications* in convex geometry, Banach space theory, differential geometry, geometry of numbers, stochastic geometry, and various other fields. *Mahler's conjecture*, sometimes called *the reverse of the Blaschke-Santaló inequality*, is open even for n-dimensional centrally symmetric convex bodies. It says that the conjectured lower bound is $4^n/n!$, reached by hypercubes; it is confirmed for the family of zonoids. For example, this conjecture and related topics are discussed in the following convexity books: Leichtweiss [164] (Chapter 3.12), Gardner [97] (Notes 9.4), Thompson [243] (Chapters 2.3 and 2.6), Gruber [117] (Section 9.2), Schneider [228] (Section 10.7), Balestro et al. [29] (Section 5.4 and subsection 5.7.4), and, in particular, in Sections 3 and 4 of the Lecture Notes [219] of Ryabogin and Zvavitch. Books from asymptotic geometric analysis and Banach space theory, in which the volume product is deeply discussed, are Pisier [206], Brazitikos et al. [70], and Artstein-Avidan et al. [24]. Important survey-like contributions to Mahler's conjecture and related topics are given by Lutwak [178] (Section 12.1) and (with comprehensively collected material for the planar case) Böröczky et al. [63]. See also the nice exposition [244] of Thompson. Further important related papers are [210] of Reisner et al. and [144] of Iriyeh and Shibata; in the latter paper, the authors confirm Mahler's conjecture for three-dimensional centrally symmetric convex bodies by proving that the volume product has, for this dimension, the sharp lower bound of 32/3, with equality precisely if the original body (or its polar) is a parallelepiped.

Urysohn's Inequality

We continue with basic references treating *Urysohn's inequality* between *volume* and *mean width* of convex bodies (see the original paper [246]). It yields that, if the volume of a convex body is fixed, the balls minimize mean width. Also this inequality is cited, proved, or extended in several convexity books, namely, in Section 56 of Bonnesen and Fenchel [58], Note 9.6 in Gardner's book [97], Section 8.3 in Gruber's monograph [117], Section 7.2 in Schneider's book [228], and Chapter 1 of Artstein-Avidan et al. [24]. In the main book, Urysohn's inequality is discussed in Section 5.5 and subsection 5.7.2.

There are several interesting *extensions and variants of Urysohn's inequality*. For example, using *dual mixed volumes*, Lutwak [174] derives *a dual of Urysohn's inequality*, and also that among all convex bodies of given mean width, only the ball has the property that its polar has minimal quermass-integrals (of any index). Further on, this yields a sharp lower bound for the *product of the mean widths* of a convex body and its polar body. Here, one subcase gives a new proof of Urysohn's inequality. Groemer [114] derives a stability version of Minkowski's inequality on mixed volumes, which implies *a stability version for Urysohn's inequality*; here, we also refer to subsection 5.3 in Groemer's handbook survey [115]. Further stability versions of Urysohn's inequality are presented in [235] and [100]. Gao et al. [95] prove that among all (spherically) convex bodies in *spherical space* and with fixed volume, the total measure of great hyperspheres hitting them is minimal if and only if these bodies are spherical caps, yielding a *spherical analogue of Urysohn's inequality*. Hyperbolic Urysohn-type inequalities are also obtained. Continuing the paper [95], Hack and Pivovarov [125] obtain *randomized versions of Urysohn-type inequalities in spherical and hyperbolic spaces*.

PLATE 6

Reduced convex bodies

A reduced compact cone

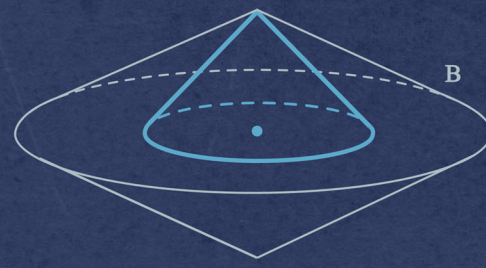
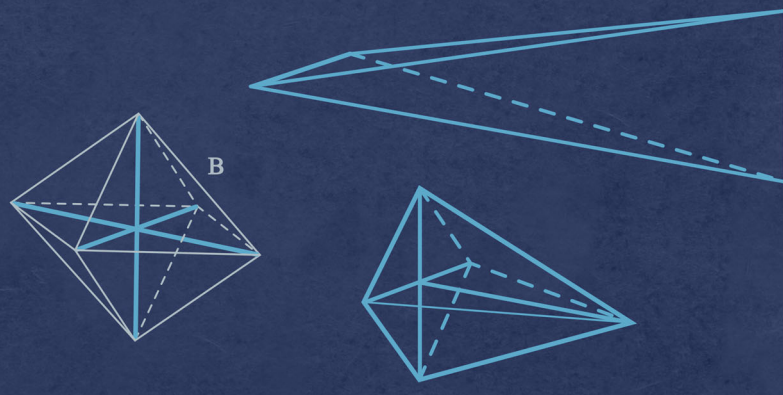


Figure 13

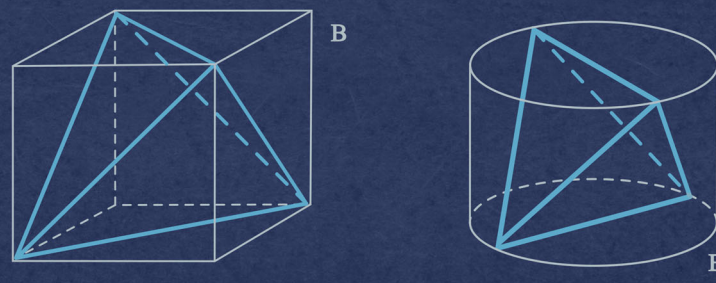
Reduced polyhedra in the Manhattan norm

Figure 14



Reduced tetrahedra

Figure 15



Reduced convex bodies in normed spaces with unit ball B ▸ (see, e.g., 2.9.1.10)

Chapter 6
Mixed Volumes



"God used beautiful mathematics in creating the world."

— Paul Dirac

Notions and topics treated in the exercises of Chapter 6 of the main book and here are: *mixed volumes, quermassintegrals*, and (as normalization of them) *intrinsic volumes*, the *width function, mean width*, and *bodies of constant width*, related inequalities combined with *in- and circumradius*, the *Wills functional*, and the famous *characterization theorem of Hadwiger*.

Plate 6:

A convex body K in an n-dimensional normed space is said to be *reduced* if the minimum width of each convex body properly contained in K is strictly smaller than that of K. For example, bodies of constant width are reduced. In Plate 6 one sees some examples of reduced bodies in three-dimensional normed spaces and the corresponding unit balls B. The most comprehensive references treating the knowledge on reduced bodies (until 2014) are the two expository papers [156] and [157] of Lassak and Martini. The first one treats the Euclidean case, and the second discusses the notion of reducedness in finite-dimensional normed spaces. For the two strongly related notions of *constant width sets* and (diametrically) *complete sets* we refer to the book [188] and to the survey [138], respectively. In the main book, the reducedness concept is discussed in parts 2.9.1.1 and 2.9.1.10, as well as in subsection 12.3.8.

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V. Balestro et al., *Convexity from the Geometric Point of View: Exercises and Solutions*, Cornerstones, https://doi.org/10.1007/978-3-031-83937-5_6

Exercises and Solutions

The volume of a Minkowski combination of non-empty compact convex sets of $\mathbb{R}^m$ is polynomial in the multiplying scalars. That is, if $C_1, \ldots, C_m \in CC^n$, then we may write

$$\mathrm{vol}_n(t_1C_1 + \ldots + t_mC_m) = \sum_{j_1,\ldots,j_m=1}^{m} V(C_{j_1}, \ldots, C_{j_n})t_{j_1} \ldots t_{j_n}$$

for $t_1, \ldots, t_m \geq 0$, and the coefficient $V(C_{j_1}, \ldots, C_{j_n})$ is the *mixed volume* of $C_{j_1}, \ldots, C_{j_n}$.

The *i-th quermassintegral* of a non-empty compact convex set $C \in CC^n$ $(0 \leq i \leq n)$ is the mixed volume

$$W_i(C) := V\big(\underbrace{C, \ldots, C}_{n-i}, \underbrace{B^n, \ldots, B^n}_{i}\big),$$

where B^n is the usual Euclidean unit ball.

Exercise 6.1. **(a)** *Let $Q^2 := [-1, 1]^2 \subseteq \mathbb{R}^2$, and let B^2 be the unit ball of $\mathbb{R}^2$. Compute the mixed volume $V(Q^2, B^2)$.*

(b) *Compute the quermassintegrals of $Q^3 := [-1, 1]^3 \subseteq \mathbb{R}^3$.*

(c) *Let $u \in \mathbb{S}^{n-1}$. Compute the quermassintegrals of the segment $\mathrm{seg}[-u, u] \subseteq \mathbb{R}^n$.*

Solution 6.1. **(a)** For $t > 0$, the sum $Q^2 + tB^2$ is formed by Q^2, four rectangles corresponding to the sides of Q^2 (whose sides have lengths 1 and t), and a quarter of B^2 corresponding to each vertex of Q^2. Hence

$$V(Q^2, Q^2) + 2tV(Q^2, B^2) + t^2V(B^2, B^2) = \mathrm{vol}_2(Q^2 + tB^2) = 1 + 4t + \pi t^2.$$

Since this holds for every $t \geq 0$, we get that the polynomials on the left-hand and right-hand side must be the same. It follows that $V(Q^2, B^2) = 2$.

(b) For $t \geq 0$, the sum $Q^3 + tB^3$ consists of Q^3, parallelepipeds corresponding to the facets of Q^3 (having sides 1, 1 and t), quarters of cylinders corresponding to the edges of Q^3 (with height 1 and t as radius of the basis), and octants of B^3 corresponding to the vertices of Q^3. Thus,

$$W_0(Q^3) + 3tW_1(Q^3) + 3t^2W_2(Q^3) + t^3W_3(Q^3) = \mathrm{vol}_3(Q^3 + tB^3) = \\ = 1 + 6t + 3\pi t^2 + \frac{4}{3}\pi t^3.$$

Comparing the coefficients of the polynomials we get the quermassintegrals of Q^3.

(c) The sum $\mathrm{seg}[-u, u] + tB^n$ consists of a cylinder whose height is 2 and having t as the radius of the basis, together with two half-spheres (of radius t) attached to the basis. Then,

$$\sum_{i=0}^{n} \binom{n}{i} W_i(\mathrm{seg}[-u, u])t^i = \mathrm{vol}_n\big(\mathrm{seg}[-u, u] + tB^n\big) = \\ = 2t^{n-1}\mathrm{vol}_{n-1}(B^{n-1}) + t^n\mathrm{vol}_n(B^n).$$

Therefore, it follows that $W_0(\text{seg}[-u,u]) = \ldots = W_{n-2}(\text{seg}[-u,u]) = 0$, and

$$W_{n-1}(\text{seg}[-u,u]) = 2\binom{n}{n-1}^{-1}\text{vol}_{n-1}(B^{n-1}) = \frac{2\kappa_{n-1}}{n}.$$

Of course, $W_n(\text{seg}[-u,u]) = \kappa_n$ (this can be seen on the equality above, but we also know that the n-th quermassintegral of any set of CC^n is κ_n).

□

Exercise 6.2. *Let $u \in \mathbb{S}^{n-1}$ be arbitrarily given. Compute the mean width of* $\text{seg}[-u,u]$.

Hint: Use the relation between the mean width and the $(n-1)$-th quermassintegral given in the beginning of Section 6.4 from the main book.

Solution 6.2. From (6.4.2) in the main book, we can describe the mean width in terms of the $(n-1)$-th quermassintegral. Using item **(c)** of Exercise 6.1, we get

$$w(\text{seg}[-u,u]) = \frac{2}{\kappa_n}W_{n-1}(\text{seg}[-u,u]) = \frac{4\kappa_{n-1}}{n\kappa_n} = 4\frac{\kappa_{n-1}}{\omega_{n-1}}.$$

□

Exercise 6.3. *Let $K \in CC^n$ and $i \in \{0,\ldots,n\}$ be arbitrarily given. Show that*

$$W_i(-K) = W_i(K).$$

Solution 6.3. It is clear that $B^n = -\text{Id}B^n$. Then, we just write

$$\begin{aligned} W_i(-K) = V(-K[n-i], B^n[i]) = V(-\text{Id}K[n-i], -\text{Id}B^n[i]) = \\ = |\det(-\text{Id})|V(K[n-i], B^n[i]) = W_i(K). \end{aligned}$$

□

Exercise 6.4. *Verify that*

$$V_j(B^n) = \binom{n}{n-j}\frac{\kappa_n}{\kappa_{n-j}}$$

for each $0 \le j \le n$.

Solution 6.4. This follows from the definition of V_j:

$$V_j(B^n) = \binom{n}{n-j}\frac{1}{\kappa_{n-j}}W_{n-j}(B^n) = \binom{n}{n-j}\frac{\kappa_n}{\kappa_{n-j}}.$$

We also recall that $W_i(B^n) = V(B^n[n-i], B^n[i]) = \text{vol}_n(B^n) = \kappa_n$ for any $i = 0,\ldots,n$.

□

Exercise 6.5. *Prove that for any $K_1,\ldots,K_n \in CC^n$ we have*

$$V(K_1,\ldots,K_n) = \frac{1}{n!}\frac{\partial^n}{\partial t_1\partial t_2\ldots\partial t_n}\text{vol}_n(t_1K_1+\ldots+t_nK_n)\Big|_{t_1=\ldots=t_n=0}.$$

Solution 6.5. For any $1 \leq i_1 \leq \ldots \leq i_n \leq n$, we have that the derivative

$$\frac{\partial^n}{\partial t_1 \ldots \partial t_n}(t_{i_1} \ldots t_{i_n})\Big|_{t_1=\ldots=t_n=0}$$

equals 1 if and only if $\{i_1, \ldots, i_n\} = \{1, \ldots, n\}$ (otherwise it equals 0, of course). Since we have $n!$ possible orders, and because $\mathrm{vol}(t_1K_1 + \ldots + t_nK_n)$ is a homogeneous polynomial of degree n in the variables $t_1, \ldots, t_n \geq 0$ whose coefficients are the mixed volumes, the desired follows.

□

Exercise 6.6. *Let $C \subseteq \mathbb{R}^n$ be a compact convex set such that* $\dim(C) = n - 1$. *Prove that*

$$nW_1(C) = 2\mathrm{vol}_{n-1}(C). \tag{6.1}$$

Hint: Use Proposition 6.4.2 from the main book.

Remark 6.1. Recall that for convex bodies the functional $nW_1(\cdot)$ yields the surface area. According to the exercise above, in the $(n-1)$-dimensional case we obtain twice the $(n-1)$-dimensional volume. One way to interpret that is the following: if we approximate C by orthogonal cylinders over C, then the surface area of each cylinder equals twice the area of C plus the area of the "lateral" part. The latter goes to 0 as the height of the cylinders goes to 0.

Solution 6.6. Up to an orthogonal transformation (which does not modify quermassintegrals or volumes), we may assume that C is contained in e_n^+. In the notation of Proposition 6.4.2 (main book), this gives that TC is naturally identified with C. Thus,

$$W_1(C) = \frac{\kappa_1}{n\kappa_0} w_0(C) = \frac{2}{n}\mathrm{vol}_{n-1}(C),$$

since w_0 stands for the 0-th quermassintegral of $\mathbb{R}^{n-1}$, which is the $(n-1)$-dimensional volume.

□

Remark 6.2. Alternatively, one can use item **(b)** of Exercise 6.7 for $i = n - 1$.

In item **(b)** of the exercise below we provide an extension of equality (6.1).

Exercise 6.7. *Let $0 \leq i \leq n$, and assume that $C \in CC^n$ is such that* $\dim(C) = i$.

(a) *Verify that $V_i(C) = \mathrm{vol}_i(C)$.*

(b) *Prove that*

$$\binom{n}{i} W_{n-i}(C) = \kappa_{n-i}\mathrm{vol}_i(C).$$

Solution 6.7. (a) Let $W_j^{(i)}$ denote the j-th quermassintegral of $\mathbb{R}^i$ (for $0 \leq j \leq i$, of course). It is clear that $W_0^{(i)}(\cdot)$ is the i-dimensional volume. We may compute the intrinsic volume $V_i(C)$ identifying C as a subset of any $\mathbb{R}^k$ for $k \geq i$. Computing it in $\mathbb{R}^i$, by (6.4.8) (main book) we get

$$V_i(C) = V_{i-0}(C) = \binom{i}{0} \frac{1}{\kappa_0} W_0^{(i)}(C) = \mathrm{vol}_i(C).$$

Observe that the numbers n and i of (6.4.8) were replaced by i and 0, respectively.

(b) We use item **(a)** and (6.4.8) (main book) for i replaced by $n-i$ to write

$$\mathrm{vol}_i(C) = V_i(C) = \binom{n}{i}\frac{1}{\kappa_{n-i}}W_{n-i}(C),$$

and this gives the desired result.

□

Exercise 6.8. *Let $K, L, M \in CC^n$ be such that $K + L = M$. Prove that the equality*

$$\mathrm{vol}_n(K) = \sum_{i=0}^{n}\binom{n}{i}(-1)^i V(M[n-i], L[i])$$

holds.

Solution 6.8. For each $i = 0, \ldots, n$, write $V_{(i)} := V(M[n-i], L[i])$, and observe that from the Minkowski additivity of mixed volumes we have the expressions

$$V_{(i)} = V((K+L)[n-i], L[i]) = \sum_{k=0}^{n-i}\binom{n-i}{k}V(K[n-i-k], L[i+k]). \tag{6.2}$$

Thus, it follows that

$$\begin{aligned}\sum_{i=0}^{n}\binom{n}{i}(-1)^i V(M[n-i], L[i]) &= \sum_{i=0}^{n}\binom{n}{i}(-1)^i V_{(i)} = \\ = \sum_{i=0}^{n}\left[\binom{n}{i}(-1)^i \sum_{k=0}^{n-i}\binom{n-i}{k}V(K[n-i-k], L[i+k])\right]&.\end{aligned} \tag{6.3}$$

Therefore, the expression in the right-hand side of the above is a combination of the mixed volumes $V(K[n-j], L[j])$, $0 \le j \le n$. Fixing any of these j's, it is clear that the term $V(K[n-j], L[j])$ appears in the expression (6.2) of $V_{(i)}$ if and only if $0 \le i \le j$, and it appears with coefficient $\binom{n-i}{j-i}$, since we must have $i + k = j$. Hence the coefficient of $V(K[n-j], L[j])$ in the right-hand side of (6.3) is

$$c_j := \sum_{i=0}^{j}(-1)^i\binom{n}{i}\binom{n-i}{n-j}.$$

Hence we may write

$$\sum_{i=0}^{n}\binom{n}{i}(-1)^i V(M[n-i], L[i]) = \sum_{j=0}^{n} c_j V(K[n-j], L[j]),$$

with each c_j given as above. It is obvious that $c_0 = 1$, and we claim that c_j is zero for $1 \le j \le n$. To verify that, we first write

$$\begin{aligned}\binom{n}{i}\binom{n-i}{n-j} = \frac{n!}{i!(n-i)!}\frac{(n-i)!}{(n-j)!(j-i)!} = \frac{n!}{(n-j)!i!(j-i)!} = \frac{n!}{(n-j)!j!}\frac{j!}{i!(j-i)!} = \\ = \binom{n}{j}\binom{j}{i}.\end{aligned}$$

Thus,

$$c_j = \sum_{i=0}^{j}(-1)^i\binom{n}{i}\binom{n-i}{n-j} = \sum_{i=0}^{j}(-1)^i\binom{n}{j}\binom{j}{i} = \binom{n}{j}\sum_{i=0}^{j}(-1)^j\binom{j}{i} =$$
$$= \binom{n}{j}\sum_{i=0}^{j}(-1)^i(1)^{j-i}\binom{j}{i} = \binom{n}{j}(1-1)^j = 0.$$

Finally, this leads to

$$\sum_{i=0}^{n}\binom{n}{i}(-1)^i V(M[n-i], L[i]) = \sum_{j=0}^{n} c_j V(K[n-j], L[j]) = c_0 V(K[n], L[0]) = \mathrm{vol}_n(K),$$

and the proof is complete.

□

Exercise 6.9. *Let $K, L \subseteq \mathbb{R}^2$ be convex bodies whose boundaries are C^2 curves, and parametrize their support functions as in Exercise 4.14. Prove the following expression for the mixed volume of K and L:*

$$V(K, L) = \frac{1}{2}\int_0^{2\pi} h_K(\theta)h_L(\theta) - h'_K(\theta)h'_L(\theta)\,d\theta.$$

Solution 6.9. Using Exercise 4.14 and the linearity of the support function **in the bodies**, we may write

$$\mathrm{vol}_2(sK + tL) = \frac{1}{2}\int_0^{2\pi} h_{sK+tL}(\theta)^2 - h'_{sK+tL}(\theta)^2\,d\theta =$$
$$= \frac{1}{2}\int_0^{2\pi} s^2h_K^2 + 2sth_Kh_L + t^2h_L^2 - s^2(h'_K)^2 - 2sth'_Kh'_L + t^2(h'_L)^2\,d\theta =$$
$$= \frac{s^2}{2}\int_0^{2\pi} h_K^2 - (h'_K)^2\,d\theta + st\int_0^{2\pi} h_Kh_L - h'_Kh'_L\,d\theta + \frac{t^2}{2}\int_0^{2\pi} h_L^2 - (h'_L)^2\,d\theta.$$

Now the desired statement comes directly from Exercise 6.5. Indeed,

$$V(K, L) = \frac{1}{2!}\frac{\partial^2}{\partial s\partial t}\mathrm{vol}_2(sK + tL)\Big|_{t=s=0} = \frac{1}{2}\int_0^{2\pi} h_Kh_L - h'_Kh'_L\,d\theta.$$

□

Exercise 6.10. *Prove that if $K, L \in CC^n$ are such that $K \cup L \in CC^n$, then*

$$V(K, L, C_3, \ldots, C_n) = V(K \cup L, K \cap L, C_3, \ldots, C_n)$$

for any $C_3, \ldots, C_n \in CC^n$.

Remark 6.3. The formulation of the exercise may suggest that we must have $n \geq 3$, but the result also holds true in dimension 2. It reads

$$V(K, L) = V(K \cup L, K \cap L)$$

whenever $K, L \in CC^2$ are such that $K \cup L \in CC^2$.

Solution 6.10. First notice that since $K \cup L$ is convex, we have that $K \cap L \neq \emptyset$, and hence the mixed volumes in the exercise are well-defined. For simplicity, denote $\mathbf{C} = (C_3, \ldots, C_n)$. From Proposition 6.4.1 in the main book, the map $M \mapsto V(M[2], \mathbf{C})$ has the valuation property. Thus, we write (also using the additivity of mixed volumes in each coordinate)

$$\begin{aligned} &V(K, K, \mathbf{C}) + V(L, L, \mathbf{C}) + 2V(K \cup L, K \cap L, \mathbf{C}) = \\ &= V(K \cup L, K \cup L, \mathbf{C}) + V(K \cap L, K \cap L, \mathbf{C}) + 2V(K \cup L, K \cap L, \mathbf{C}) = \\ &= V(K \cup L + K \cap L, K \cup L + K \cap L, \mathbf{C}). \end{aligned}$$

From Exercise 2.9, we have that $K \cup L + K \cap L = K + L$ (because we are assuming that $K \cup L$ is convex). Therefore, replacing this equality in the right-hand side of the above we arrive at

$$\begin{aligned} V(K, K, \mathbf{C}) + V(L, L, \mathbf{C}) + 2V(K \cup L, K \cap L, \mathbf{C}) &= V(K + L, K + L, \mathbf{C}) = \\ &= V(K, K, \mathbf{C}) + V(L, L, \mathbf{C}) + 2V(K, L, \mathbf{C}), \end{aligned}$$

and this yields the wished equality.

□

Exercise 6.11 (Rademacher)**.** *Let K be a planar convex body. Show that*

$$V(K, -K) \leq 2\mathrm{vol}_2(K).$$

Hint: Use the Rogers-Shephard inequality.

Solution 6.11. Notice that

$$\mathrm{vol}_2(K - K) = V(K - K, K - K) = 2V(K, K) + 2V(K, -K).$$

Using the Rogers-Shephard inequality, we get that

$$2V(K, -K) = \mathrm{vol}_2(K - K) - 2V(K, K) \leq \binom{4}{2}\mathrm{vol}_2(K) - 2\mathrm{vol}_2(K) = 4\mathrm{vol}_2(K),$$

and this gives the desired inequality.

□

Exercise 6.12. *Let $K \subseteq \mathbb{R}^n$ be a convex body. Prove that*

$$V(-K, K, \ldots, K) \geq \mathrm{vol}_n(K).$$

Solution 6.12. We recall that, as a consequence of the Brunn-Minkowski inequality, we have

$$\mathrm{vol}_n(K - K) \geq 2^n \mathrm{vol}_n(K).$$

From the 1st Minkowski inequality, it follows that

$$V(K - K, K, \ldots, K)^n \geq \mathrm{vol}_n(K - K)\mathrm{vol}_n(K)^n \geq 2^n \mathrm{vol}_n(K)^n,$$

which yields

$$V(K - K, K, \ldots, K) \geq 2\mathrm{vol}_n(K).$$

Since the mixed volume is Minkowski additive, we finally arrive at

$$2\mathrm{vol}_n(K) \le V(K-K,K,\dots,K) = V(K,\dots,K)+V(-K,K,\dots,K) = \\ = \mathrm{vol}_n(K)+V(-K,K,\dots,K).$$

The desired result follows.

□

Exercise 6.13. *Let $K_1,\dots,K_n \in CC^n$. Prove that $V(K_1,\dots,K_n) > 0$ if and only if for every $k \in \{1,\dots,n\}$ and any choice of indices $1 \le i_1 < \dots < i_k \le n$ the inequality $\dim(K_{i_1}+\dots+K_{i_k}) \ge k$ holds.*

Solution 6.13. If $V(K_1,\dots,K_n) > 0$, we may choose segments $S_i \subseteq K_i$ such that the directions of $S_1,\dots,S_n$ are linearly independent. Then, for each choice of indices $1 \le i_1 < \dots < i_k \le n$ the sum $K_{i_1}+\dots K_{i_k}$ contains the k linearly independent segments $S_{i_1},\dots,S_{i_k}$. This gives that $\dim(K_{i_1}+\dots+K_{i_k}) \ge k$.

Now we prove the converse. Let $A_i := \mathrm{aff}(K_i)$ for each i. For any m-tuple $\{i_1,\dots,i_m\}$ in $\{2,\dots,n\}$, we either have that $A_1 \subseteq A_{i_1}+\dots+A_{i_m}$ or $\dim(A_1 \cap (A_{i_1}+\dots+A_{i_m}) < \dim(A_1)$. Since it is clear that an affine subspace is not covered by finitely many lower dimensional affine subspaces, we may choose a line $L_1 \subseteq A_1$ such that if the second condition holds, then

$$L_1 \nsubseteq A_{i_1}+\dots+A_{i_m}.$$

It follows that for any $(k-1)$-tuple $\{i_2,\dots,i_k\}$ from $\{2,\dots,n\}$, we have the cases:

(i) $A_1 \subseteq A_{i_2}+\dots+A_{i_k}$. If this holds true, then from the hypothesis, we arrive at

$$\dim(L_1+A_{i_2}+\dots+A_{i_k}) = \dim(A_1+A_{i_2}+\dots+A_{i_k}) \ge k.$$

(ii) $A_1 \nsubseteq A_{i_2}+\dots+A_{i_k}$. In this case,

$$\dim(L_1+A_{i_2}+\dots+A_{i_k}) = 1+\dim(A_{i_2}+\dots+A_{i_k}) \ge 1+(k-1) = k.$$

Repeating the argument for $i = 2,\dots,n$, we get lines $L_1,\dots,L_n$ such that $\dim(L_1+\dots+L_n) = n$. Hence the directions of these lines are linearly independent. Since it is clear that each K_i contains a segment in the direction of the respective line L_i we get what we wanted.

□

Exercise 6.14. *Show that the Brunn-Minkowski inequality (Theorem 5.1.2 in the main book) can be obtained from the 1st Minkowski inequality (Theorem 6.3.1, main book).*

Solution 6.14. Let $K, L \in \mathcal{K}^n$. Due to the 1st Minkowski inequality, we write

$$\mathrm{vol}_n(K+L) = V(K,K+L[n-1])+V(L,K+L[n-1]) \ge \\ \ge \mathrm{vol}_n(K)^{\frac{1}{n}}\mathrm{vol}_n(K+L)^{\frac{n-1}{n}}+\mathrm{vol}_n(L)^{\frac{1}{n}}\mathrm{vol}_n(K+L)^{\frac{n-1}{n}},$$

and this gives the Brunn-Minkowski inequality.

□

For the next exercise, recall that $r_{\mathrm{in}}(K)$ and $r_{\mathrm{out}}(K)$ denote the inradius and circumradius of K, respectively. These concepts are defined and studied in Exercise 2.13.

Exercise 6.15. *Let $K \subseteq \mathbb{R}^n$ be a convex body. Prove that for any $i \in \{1, \dots, n\}$ the inequalities*

$$r_{\text{in}}(K) \leq \frac{W_{i-1}(K)}{W_i(K)} \leq r_{\text{out}}(K)$$

hold.

Solution 6.15. Up to a translation, which does not modify mixed volumes, we may assume that $K \subseteq r_{\text{out}} B^n$. By the monotonicity of mixed volumes, it follows that

$$\begin{aligned} W_{i-1}(K) &= V(K[n-i+1], B^n[i-1]) = V(K[n-i], K, B^n[i-1]) \leq \\ &\leq V(K[n-i], r_{\text{out}} B^n, B^n[i-1]) = r_{\text{out}} V(K[n-i], B^n[i]) = r_{\text{out}} W_i(K), \end{aligned}$$

and this shows the inequality in the right-hand side. For the remaining inequality, up to a translation assume that $r_{\text{in}}(K) B^n \subseteq K$ and repeat the argument.

□

Exercise 6.16. *Let $K \subseteq \mathbb{R}^n$ be a convex body, and assume that $\mathbf{g}$ is a k-dimensional vector subspace of $\mathbb{R}^n$ for some $k \in \{1, \dots, n-1\}$. Prove that for any $j \in \{0, \dots, k\}$ the reduction formula*

$$V\big(K[n-j], B^k[j]\big) = \binom{n}{j}^{-1} \binom{k}{j} \int_{\text{proj}_{\mathbf{g}^\perp}(K)} W_j^{(k)}\big(K \cap (\mathbf{g} + z)\big)\, d\mathcal{H}^{n-k}(z)$$

holds, where $W_j^{(k)}$ denotes the j-th quermassintegral of $\mathbb{R}^k$ and B^k is the unit ball of $\mathbb{R}^k$ identified with $B^n \cap \mathbf{g}$.

Solution 6.16. Let $L = B^k$ as in the proof of Theorem 6.2.4 (main book). We have the following equality:

$$\begin{aligned} \sum_{i=0}^{n} \binom{n}{i} \lambda^i V\big(K[n-i], B^k[i]\big) &= \text{vol}_n(K + \lambda B^k) = \\ = \sum_{i=0}^{k} \binom{k}{i} \lambda^i \int_{\text{proj}_{\mathbf{g}^\perp}(K)} & V^{(k)}\big(K \cap (\mathbf{g} + z)[k-i], B^k[i]\big)\, d\mathcal{H}^{n-k}(z), \end{aligned}$$

for any $\lambda > 0$. Hence in the left-hand side and in the right-hand side we have the same polynomial. Comparing the coefficients of the term λ^j we have

$$\begin{aligned} \binom{n}{j} V\big(K[n-j], B^k[j]\big) &= \binom{k}{j} \int_{\text{proj}_{\mathbf{g}^\perp}(K)} V^{(k)}\big(K \cap (\mathbf{g} + z)[k-j], B^k[j]\big)\, d\mathcal{H}^{n-k}(z) = \\ &= \binom{k}{j} \int_{\text{proj}_{\mathbf{g}^\perp}(K)} W_j^{(k)}\big(K \cap (\mathbf{g} + z)\big)\, d\mathcal{H}^{n-k}(z), \end{aligned}$$

and this concludes the proof.

□

Exercise 6.17. *Let $1 \leq j \leq n$, and let $u_1, \ldots, u_j \in \mathbb{S}^{n-1}$. Write $E := \operatorname{span}\{u_1, \ldots, u_j\}$. Prove that for any $C_1, \ldots, C_{n-j} \in CC^n$ we have*

$$\begin{aligned}\binom{n}{j} V(\operatorname{seg}[-u_1, u_1], \ldots, \operatorname{seg}[-u_j, u_j], C_1, \ldots, C_{n-j}) = \\ = \frac{2^j}{j!} D_j(u_1, \ldots, u_j) V^{(n-j)}(\operatorname{proj}_{E^\perp}(C_1), \ldots, \operatorname{proj}_{E^\perp}(C_{n-j})),\end{aligned}$$

where

$$D_j(u_1, \ldots, u_j) := \operatorname{vol}_j\big(\operatorname{seg}[0_{\mathbb{R}^n}, u_1] + \ldots + \operatorname{seg}[0_{\mathbb{R}^n}, u_j]\big).$$

Solution 6.17. For this solution, the reader may have in mind the statement of Proposition 6.2.3 from the main book, which, roughly speaking, states that a mixed volume always vanishes if we cannot choose a segment in each entry in such a way that the chosen segments are linearly independent.

First assume that $u_1, \ldots, u_j$ are linearly independent. We use Theorem 6.2.4 (main book) for $\mathbf{g} = E$, and $L_i = \operatorname{seg}[-u_i, u_i]$ for each $1 \leq i \leq j$. We have immediately

$$\begin{aligned}\binom{n}{j} V(\operatorname{seg}[-u_1, u_1], \ldots, \operatorname{seg}[-u_j, u_j], C_1, \ldots, C_{n-j}) = \\ = V^{(j)}(\operatorname{seg}[-u_1, u_1], \ldots, \operatorname{seg}[-u_j, u_j]) V^{(n-j)}(\operatorname{proj}_{E^\perp}(C_1), \ldots, \operatorname{proj}_{E^\perp}(C_{n-j})).\end{aligned} \tag{6.4}$$

Using the polynomiality of $\operatorname{vol}_j(\cdot)$ for Minkowski combinations, the symmetry of mixed volumes, and the fact that

$$V^{(j)}(\operatorname{seg}[-u_{i_1}, u_{i_1}], \ldots, \operatorname{seg}[-u_{i_j}, u_{i_j}]) = 0$$

whenever we repeat some of the indices, we arrive at

$$\begin{aligned}\operatorname{vol}_j(\operatorname{seg}[-u_1, u_1] + \ldots + \operatorname{seg}[-u_j, u_j]) = j! V^{(j)}(\operatorname{seg}[-u_1, u_1], \ldots, \operatorname{seg}[-u_j, u_j]) = \\ = j! 2^j V^{(j)}(\operatorname{seg}[0_{\mathbb{R}^n}, u_1], \ldots, \operatorname{seg}[0_{\mathbb{R}^n}, u_j]) = 2^j D_j(u_1, \ldots, u_j).\end{aligned}$$

Hence

$$V^{(j)}(\operatorname{seg}[-u_1, u_1], \ldots, \operatorname{seg}[-u_j, u_j]) = \frac{2^j}{j!} D_j(u_1, \ldots, u_j).$$

Replacing this equality in (6.4) settles the case where $u_1, \ldots, u_j$ are linearly independent.

If $u_1, \ldots, u_j$ are linearly dependent, then we clearly have that

$$V^{(j)}(\operatorname{seg}[-u_1, u_1], \ldots, \operatorname{seg}[-u_j, u_j]) = 0,$$

hence $D_j(u_1, \ldots, u_j) = 0$ (geometrically, if $u_1, \ldots, u_j$ are not linearly independent, then they do not span a parallelotope with positive j-dimensional volume). Since we also have that

$$V(\operatorname{seg}[-u_1, u_1], \ldots, \operatorname{seg}[-u_j, u_j], C_1, \ldots, C_{n-j}) = 0,$$

we get that the equality still holds in this case.

□

Exercise 6.18. *Verify that for any $K_1, \ldots, K_{n-1} \in CC^n$ and every $u \in \mathbb{S}^{n-1}$ we have*

$$V^{(n-1)}\big(\mathrm{proj}_{u^\perp}(K_1), \ldots, \mathrm{proj}_{u^\perp}(K_{n-1})\big) = \frac{n}{2} V(\mathrm{seg}[-u, u], K_1, \ldots, K_{n-1}).$$

Remark 6.4. The particular case where $K_1 = \ldots = K_{n-1} = K$ is derived later in the proof of Theorem 7.2.2 (see also Remark 7.21, both in the main book).

Solution 6.18. This is immediate by Exercise 6.17. Just let $j = 1$, $u_1 = u$, $K_i = C_i$ for each $i = 1, \ldots, n-1$, and notice that $D_j(u_1) = D_j(u) = 1$.

□

Exercise 6.19. *Let $K_1, \ldots, K_{n-1} \in CC^n$, and define the function $f : \mathbb{R}^n \to \mathbb{R}$ by*

$$f(\lambda u) = \lambda V^{(n-1)}\big(\mathrm{proj}_{u^\perp}(K_1), \ldots, \mathrm{proj}_{u^\perp}(K_{n-1})\big),$$

for any $\lambda \geq 0$ and every $u \in \mathbb{S}^{n-1}$. Prove that f is sublinear.

Hint: Use Exercise 6.18 and consider first the case where $K_1, \ldots, K_{n-1}$ are polytopes (for this case, use Proposition 6.2.2 from the main book).

Remark 6.5. Another approach for the sublinearity of f is given later in Exercise 10.17 (see also the definition of the *mixed projection body* above this exercise). There, we use *mixed surface area measures* to rewrite f. In the case where $K_1 = \ldots = K_{n-1} = K$, this is related to the important definition of the *projection body* of K, which is studied in Section 10.3.

Solution 6.19. Let $K_1 = P_1, \ldots, K_{n-1} = P_{n-1}$ be polytopes, and assume that $N \subseteq \mathbb{S}^{n-1}$ is a finite set containing all normals to the facets of $t_1 P_1 + \ldots + t_{n-1} P_{n-1}$ (see Proposition 6.2.2 in the main book). For the sake of simplicity of the notation, write

$$V^{(n-1)}(P_1 \cap \mathbf{h}(P_1, v), \ldots, P_{n-1} \cap \mathbf{h}(P_{n-1}, v)) = V^{(n-1)}(\mathbf{P}(v))$$

for each $v \in N$. Let $x \in \mathbb{R}^n$. Writing $x = \lambda u$ for some $\lambda \geq 0$ and some $u \in \mathbb{S}^{n-1}$, we get from Exercise 6.18 and Proposition 6.2.2 (main book) that

$$f(x) = f(\lambda u) = \frac{\lambda n}{2} V\big(\mathrm{seg}[-u, u], P_1, \ldots, P_{n-1}\big) = \frac{\lambda}{2} \sum_{v \in N} h_{\mathrm{seg}[-u,u]}(v) V^{(n-1)}(\mathbf{P}(v)) =$$

$$= \frac{\lambda}{2} \sum_{v \in N} |\langle u, v \rangle| V^{(n-1)}(\mathbf{P}(v)) = \frac{1}{2} \sum_{v \in N} |\langle x, v \rangle| V^{(n-1)}(\mathbf{P}(v)).$$

Thus,

$$f(x + y) = \frac{1}{2} \sum_{v \in N} |\langle x + y, v \rangle| V^{(n-1)}(\mathbf{P}(v)) \leq \frac{1}{2} \sum_{v \in N} \big(|\langle x, v \rangle| + |\langle y, v \rangle|\big) V^{(n-1)}(\mathbf{P}(v)) =$$

$$= f(x) + f(y)$$

for any $x, y \in \mathbb{R}^n$. This shows that f is subadditive. The fact that f is positively homogeneous is immediate due to the definition, and hence the case for polytopes is settled. The general case follows straightforwardly when one considers approximations by polytopes.

□

In the next exercises we investigate some facts concerning mixed volumes (and quermassintegrals) involving polar bodies. For that sake, we remind the reader that $\mathcal{K}_0^n$ denotes the family of the convex bodies of $\mathbb{R}^n$ which contain the origin as an interior point.

Exercise 6.20. *Let $K \in \mathcal{K}_0^n$, and let $C_2, \ldots, C_n \in CC^n$ be arbitrarily given. Prove that*

$$V(K, C_2, \ldots, C_n)V(K^\circ, C_2, \ldots, C_n) \geq V(B^n, C_2, \ldots, C_n)^2.$$

Hint: Consider first the case where $C_2, \ldots, C_n$ are polytopes. For this case, use Proposition 6.2.2 of the main book. Then, derive the general case by approximations.

Solution 6.20. First, assume that $C_2, \ldots, C_n$ are polytopes $P_2, \ldots, P_n$, respectively. As in Proposition 6.2.2 (main book), let $N \subseteq \mathbb{S}^{n-1}$ be the finite set containing the unit normal vectors of $t_2P_2 + \ldots + t_nP_n$. Then, we write

$$V(K, P_2, \ldots, P_n) = \frac{1}{n} \sum_{u \in N} h_K(u) V^{(n-1)}\big(P_2 \cap \mathbf{h}(P_2, u), \ldots, P_n \cap \mathbf{h}(P_n, u)\big),$$

and

$$V(K^\circ, P_2, \ldots, P_n) = \frac{1}{n} \sum_{u \in N} h_{K^\circ}(u) V^{(n-1)}\big(P_2 \cap \mathbf{h}(P_2, u), \ldots, P_n \cap \mathbf{h}(P_n, u)\big).$$

For simplicity, we denote $V^{(n-1)}\big(P_2 \cap \mathbf{h}(P_2, u), \ldots, P_n \cap \mathbf{h}(P_n, u)\big) = V^{(n-1)}(\mathbf{P}(u))$. The usual Cauchy-Schwarz inequality can be written as

$$\sum c_i \sum d_i \geq \left(\sum (c_i d_i)^{1/2}\right)^2,$$

for $c_i, d_i \geq 0$. Observing that h_K and h_{K° are positive, and remembering that mixed volumes are non-negative, using this inequality for the summations above we arrive at

$$V(K, P_2, \ldots, P_n)V(K^\circ, P_2, \ldots, P_n) \geq \left(\frac{1}{n} \sum_{u \in N} \left[h_K(u) h_{K^\circ}(u)\right]^{1/2} V^{(n-1)}(\mathbf{P}(u))\right)^2.$$

Now, from Proposition 2.3.6 (main book) we get

$$h_K(u) h_{K^\circ}(u) = ||u||_{K^\circ} ||u||_K \geq \langle u, u \rangle = 1$$

for each $u \in \mathbb{S}^{n-1}$ and, in particular, for any $u \in N$. Therefore, we arrive at

$$V(K, P_2, \ldots, P_n)V(K^\circ, P_2, \ldots, P_n) \geq \left(\frac{1}{n} \sum_{u \in N} V^{(n-1)}(\mathbf{P}(u))\right)^2.$$

Because $h_{B^n}(u) = 1$, using Proposition 6.2.2 from the main book again we have that

$$\frac{1}{n} \sum_{u \in N} V^{(n-1)}(\mathbf{P}(u)) = V(B^n, P_2, \ldots, P_n).$$

It follows that

$$V(K, P_2, \ldots, P_n)V(K^\circ, P_2, \ldots, P_n) \geq V(B^n, P_2, \ldots, P_n)^2,$$

and the result is proved for polytopes. For the general case we just need to consider approximations by polytopes.

□

Exercise 6.21. *Prove that for any $K \in \mathcal{K}_0^n$ the inequality*

$$W_{n-1}(K^\circ)W_{n-1}(K) \geq \kappa_n^2$$

holds. Conclude that

$$w(K)w(K^\circ) \geq 4,$$

where we remind the reader that $w(\cdot)$ stands for the mean width.

Remark 6.6. The latter inequality is obtained in a different way in Corollary 11.1.2 (main book).

Solution 6.21. Just use Exercise 6.20 for $C_2 = \ldots = C_n = B^n$. For the other inequality, just use the equality

$$w(\cdot) = \frac{2}{\kappa_n} W_{n-1}(\cdot),$$

given in (6.4.2) of the main book.

□

Exercise 6.22. **(a)** *Let $K \in \mathcal{K}_0^n$. Show that*

$$B^n \subseteq \frac{K + K^\circ}{2}.$$

Verify that equality holds if and only if $K = B^n$.

(b) *If $K \in \mathcal{K}_0^2$, prove that*

$$V(K, K^\circ) \geq \kappa_2.$$

Check that this inequality is sharp.

Hint: For any $\lambda > 0$, let

$$M_\lambda := \frac{1}{2}\left(\lambda K + \frac{1}{\lambda} K^\circ\right).$$

Show that

$$\min\left\{W_0(M_\lambda) : \lambda > 0\right\} = \frac{1}{2}\left(V(K, K^\circ) + V(K, K)^{\frac{1}{2}} V(K^\circ, K^\circ)^{\frac{1}{2}}\right),$$

and use the 1st Minkowski inequality to bound the last term. Finally, apply item **(a)** to verify that

$$W_0(M_\lambda) \geq \kappa_2$$

for all $\lambda > 0$.

Solution 6.22. (a) Let $u \in \mathbb{S}^{n-1}$ be arbitrarily given. Since $u/||u||_K \in K$ (this is the point where the ray $[0, u\rangle$ intersects ∂K), we have that

$$h_K(u) \geq \left\langle \frac{u}{||u||_K}, u \right\rangle = \frac{1}{||u||_K}.$$

On the other hand, Proposition 2.3.6 (main book) gives that $h_{K^\circ}(u) = ||u||_K$. Hence, we have

$$h_K(u) + h_{K^\circ}(u) \geq \frac{1}{||u||_K} + ||u||_K.$$

From standard calculus we get immediately that the function $t \mapsto t + 1/t$ (for $t > 0$) attains the minimum value 2 for (and only for) $t = 1$. Thus, from the above we get

$$h\left(\frac{K + K^\circ}{2}, u\right) \geq 1 = h_{B^n}(u),$$

for every $u \in \mathbb{S}^{n-1}$. This gives the desired inclusion. Equality holds in the above for all $u \in \mathbb{S}^{n-1}$ if and only if $||u||_K = 1$ for any $u \in \mathbb{S}^{n-1}$. This holds if and only if $h_{K^\circ}(\cdot) = 1 = h_{B^n}(\cdot)$, and this is equivalent to K° (and hence K) being the unit ball B^n.

(b) Let

$$M_\lambda := \frac{1}{2}\left(\lambda K + \frac{1}{\lambda}K^\circ\right) = \frac{\lambda K + (\lambda K)^\circ}{2}.$$

From item **(a)** we have that $B^n \subseteq M_\lambda$ for any $\lambda > 0$. Using the monotonicity of mixed volumes we get

$$W_0(M_\lambda) = V(M_\lambda, M_\lambda) \geq V(B^2, B^2) = \kappa_2$$

for every $\lambda > 0$. As a consequence, we get that

$$\inf\left\{W_0(M_\lambda) : \lambda > 0\right\} \geq \kappa_2. \tag{6.5}$$

This infimum is actually a minimum, as it will be clear below. Now notice that

$$\begin{aligned} W_0(M_\lambda) = V(M_\lambda, M_\lambda) &= \frac{1}{4}V\left(\lambda K + \frac{1}{\lambda}K^\circ, \lambda K + \frac{1}{\lambda}K^\circ\right) = \\ &= \frac{1}{4}\left[\lambda^2 V(K,K) + 2V(K,K^\circ) + \frac{1}{\lambda^2}V(K^\circ,K^\circ)\right] = \\ &= \frac{1}{2}V(K,K^\circ) + \frac{1}{4}\left[\lambda^2 V(K,K) + \frac{1}{\lambda^2}V(K^\circ,K^\circ)\right]. \end{aligned}$$

Letting $\lambda^2 = t > 0$, standard calculus arguments give immediately that the function

$$tV(K,K) + \frac{1}{t}V(K^\circ,K^\circ)$$

is minimized for

$$t = \left(\frac{V(K^\circ,K^\circ)}{V(K,K)}\right)^{\frac{1}{2}},$$

and hence

$$\min\left\{W_0(M_\lambda) : \lambda > 0\right\} = \frac{1}{2}\left[V(K,K^\circ) + V(K,K)^{\frac{1}{2}}V(K^\circ,K^\circ)^{\frac{1}{2}}\right].$$

From the 1st Minkowski inequality, it follows immediately that

$$V(K,K)^{\frac{1}{2}}V(K^\circ,K^\circ)^{\frac{1}{2}} \leq V(K,K^\circ),$$

and then it follows that

$$\min\left\{W_0(M_\lambda) : \lambda > 0\right\} \leq V(K,K^\circ).$$

Together with (6.5) we are done. This inequality is sharp because for $K = B^2$ we obviously have that $V(K,K^\circ) = \kappa_2$.

□

Exercise 6.23. *Show that in dimension $n > 3$ the unit ball B^n does **not** minimize the product*

$$W_1(K)W_1(K^\circ)$$

over $K \in \mathcal{K}_0^n$.

Hint: Recall that $nW_1(\cdot)$ is the surface area, and compare the cases where $K = B^n$ and where $K = P_\Delta^n$ (the regular n-dimensional cross-polytope). See Exercises 4.3 and 4.5.

Solution 6.23. Let $K = P_\Delta^n$. Then, from Exercise 4.5 we have that $K^\circ = [-1,1]^n = Q^n$. The surface areas of P_Δ^n and Q^n were computed in Exercises 4.3 and 4.5. Using that, we write

$$W_1(K)W_1(K^\circ) = \frac{1}{n^2}S(P_\Delta^n)S(Q^n) = \frac{1}{n^2}\frac{2^n\sqrt{n}}{(n-1)!}n2^n = \frac{4^n\sqrt{n}}{n!}.$$

In the case where $K = B^n$, we clearly have that $W_1(K)W_1(K^\circ) = \kappa_n^2$ (because the surface area of B^n is $n\kappa_n$). For n even, the expression of κ_n is

$$\kappa_n = \frac{\pi^n}{(n/2)!},$$

and it is not difficult to see that the inequality

$$\frac{4^n\sqrt{n}}{n!} < \left(\frac{\pi^n}{(n/2)!}\right)^2$$

holds for $n \geq 4$. This gives the desired counterexample.

□

The objective of the next exercise is to construct a multilinear (for Minkowski combinations) function $F\colon (CC^n)^n \to \mathbb{R}$ which has "good" properties but is not the same as the mixed volume (in this direction, see also Exercise 6.26 ahead). To do that, we first notice that since $\mathrm{GL}(n)$ has a topology endowed by the norm of $\mathrm{L}(n) \simeq \mathbb{R}^{n^2}$, we can endow it with a finite Borel measure.

Exercise 6.24. *Let μ be a finite Borel measure on* $\mathrm{GL}(n)$ *with the property that*

$$\int_{\mathrm{GL}(n)} ||T||_{\mathrm{L}(n)}\, d\mu(T) < +\infty.$$

(a) *For each $K \in CC^n$, $u \in \mathbb{S}^{n-1}$ and $\lambda \geq 0$ prove that the map*

$$\mathrm{GL}(n) \ni T \mapsto h(TK, \lambda u)$$

is continuous, and define

$$h(K^\mu, \lambda u) := \int_{\mathrm{GL}(n)} h(TK, \lambda u)\, d\mu(T).$$

(b) *Verify that $h(K^\mu, \lambda u) < +\infty$ always.*

(c) *Prove that $h(K^\mu, \cdot)\colon \mathbb{R}^n \to \mathbb{R}$ is a sublinear function. Conclude that $h(K^\mu, \cdot)$ is the support function of some set of CC^n (which we denote by K^μ, thus justifying the notation).*

Let $F_\mu\colon (CC^n)^n \to \mathbb{R}$ be defined as

$$F_\mu(K_1, \ldots, K_n) = V(K_1^\mu, \ldots, K_n^\mu), \quad K_1, \ldots, K_n \in CC^n.$$

(d) *Show that F_μ is monotone for inclusions and linear for Minkowski combinations in each entry.*

(e) *Show that F_μ is symmetric in the entries.*

(f) *Prove that if μ is* even *(i.e., if $A^\mu = (-A)^\mu$ for every Borel set $A \subseteq \mathrm{GL}(n)$), then F^μ is translation-invariant.*

(g) *Give an example of a finite even Borel measure μ on* $\mathrm{GL}(n)$ *such that F_μ is not a multiple of the mixed volume of $\mathbb{R}^n$.*

Solution 6.24. (a) If $T_1, T_2 \in \mathrm{GL}(n)$, and $K \in CC^n$, $\lambda \geq 0$ and $u \in \mathbb{S}^{n-1}$ are fixed, then we have

$$|h(T_1K, u) - h(T_2K, u)| = |h(K, T_1^* u) - h(K, T_2^* u)|.$$

Since the adjoint map ($T \mapsto T^*$) and $h(K, \cdot)$ are continuous, we get that the latter becomes arbitrarily small provided T_1 is close enough to T_2 in the norm of $\mathrm{L}(n)$.

(b) Observe that the classical Cauchy-Schwarz inequality yields the estimate

$$h(TK, u) = \sup\{\langle u, Tz\rangle : z \in K\} \leq ||T||_{\mathrm{L}(n)} \sup\{|z| : z \in K\}.$$

Hence

$$\int_{\mathrm{GL}(n)} h(TK, u)\, d\mu(u) \leq \int_{\mathrm{GL}(n)} ||T||_{\mathrm{L}(n)} \sup\{|z| : z \in K\}\, d\mu(T) =$$
$$= \sup\{|z| : z \in K\} \int_{\mathrm{GL}(n)} ||T||_{\mathrm{L}(n)}\, d\mu(T),$$

and the latter is finite by our hypothesis.

(c) This follows immediately from the sublinearity of $h(TK, \cdot)$ and from the linearity of the integral.

(d) It suffices to show that the mapping $K \mapsto K^\mu$ is monotone for inclusions and linear for Minkowski combinations. The monotonicity comes straightforwardly from the fact that $K \subseteq L$ implies $TK \subseteq TL$ for any $T \in \mathrm{GL}(n)$, and hence $h(TK, u) \leq h(TL, u)$ for every $T \in \mathrm{GL}(n)$ and any $u \in \mathbb{S}^{n-1}$. Thus,

$$h(K^\mu, u) = \int_{\mathrm{GL}(n)} h(TK, u)\, d\mu(u) \leq \int_{\mathrm{GL}(n)} h(TL, u)\, d\mu(u) = h(L^\mu, u)$$

for any $u \in \mathbb{S}^{n-1}$. This gives the inclusion $K^\mu \subseteq L^\mu$. For the linearity, we just have to use the linearity of T, of the support function (on the involved bodies), and of the integral.

(e) The symmetry follows immediately from the symmetry of the mixed volume.

(f) Let $K \in CC^n$, $u \in \mathbb{S}^{n-1}$ and $x_0 \in \mathbb{R}^n$ be arbitrarily given. We have

$$h((x_0 + K)^\mu, u) = \int_{\mathrm{GL}(n)} h(Tx_0 + TK, u)\, d\mu(u) =$$

$$= \int_{\mathrm{GL}(n)} h(Tx_0, u)\, d\mu(u) + \int_{\mathrm{GL}(n)} h(TK, u)\, d\mu = \int_{\mathrm{GL}(n)} \langle Tx_0, u\rangle\, d\mu(u) + h(K^\mu, u).$$

The function $T \mapsto \langle Tx_0, u\rangle$ is odd, meaning that $\langle -Tx_0, u\rangle = -\langle Tx_0, u\rangle$. If μ is even, this gives that

$$\int_{\mathrm{GL}(n)} \langle Tx_0, u\rangle\, d\mu(u) = 0.$$

Thus, $(x_0 + K)^\mu = K^\mu$.

(g) Let μ be a finite Borel measure on $\mathrm{GL}(n)$ which is strictly positive in any non-empty open subset of $\mathrm{GL}(n)$. For example, we may let $\mathcal{L}^{n^2}$ be the Lebesgue measure of $\mathbb{R}^{n^2}$ identified as a measure on $\mathrm{L}(n)$, and restrict it to $\mathrm{GL}(n)$ with the (continuous) density

$$d\mu := \frac{e^{-||T||^2_{\mathrm{L}(n)}/2}}{||T||_{\mathrm{L}(n)}} d\mathcal{L}^{n^2}.$$

This is similar to the Gaussian measure, but we divide it by $||T||_{\mathrm{L}(n)}$ in order that our initial condition on μ is easily verified. Since $\mathrm{GL}(n)$ is open in $\mathrm{L}(n)$, every open subset A of $\mathrm{GL}(n)$ contains a closed ball. It follows from a compactness argument that $\mu(A) > 0$ (the density attains a positive minimum in this closed ball).

Let $v_0 \in \mathbb{S}^{n-1}$. Letting $S := \mathrm{seg}[-v_0, v_0]$, we claim that S^μ is a convex body. In fact, for any given $u \in \mathbb{S}^{n-1}$, let $T_0 \in \mathrm{GL}(n)$ be such that $Tv_0 = u$. Thus,

$$h(T_0 S, u) = 1,$$

and hence, by continuity there is an open ball $B := B_{\mathrm{GL}(n)}(T_0, \varepsilon)$ such that

$$h(TS, u) > \frac{1}{2}$$

for every $T \in B$. Moreover, it is clear that, for any $T \in \mathrm{GL}(n)$, TS is a non-degenerate segment centered at the origin. Hence $h(TS, u) > 0$ for any $T \in \mathrm{GL}(n)$. As a consequence, we get that

$$h(S^\mu, u) = \int_{\mathrm{GL}(n)} h(TS, u)\, d\mu(u) \geq \int_B h(TS, u)\, d\mu(u) \geq \frac{\mu(B)}{2} > 0.$$

This shows that $h(S^\mu, u) > 0$ for any $u \in \mathbb{S}^{n-1}$, and gives that S^μ is a convex body (see, e.g., Exercise 2.31). If $n \geq 2$, then $V(S, \dots, S) = 0$, but $F_\mu(S, \dots, S) = V(S^\mu, \dots, S^\mu) > 0$. Therefore, in this case we have that F_μ is not a multiple of the mixed volume.

□

Of course, identifying each set of CC^n with its support function, it follows that the mixed volume can be seen as a functional defined for support functions (over $\mathbb{S}^{n-1}$, say). For the next exercise, we will use a theorem from [227] which states, roughly speaking, that the two-dimensional mixed volume (defined for support functions) does not admit a bilinear continuous extension to $C(\mathbb{S}^1) \times C(\mathbb{S}^1)$. Here, "bilinear" has the usual linear algebra sense: we are considering also combinations with negative scalars.

Theorem 6.1. [227, Theorem 5.2.2] *There is no bilinear function $F\colon C(\mathbb{S}^1)^2 \to \mathbb{R}$ which is continuous in each variable (with respect to the maximum norm) and satisfies $F(h_K, h_L) = V(K, L)$ for every $K, L \in CC^n$.*

Exercise 6.25. *Let $n \geq 3$. Prove that there is no multilinear (in the usual linear algebra sense) function $F\colon C(\mathbb{S}^{n-1})^n \to \mathbb{R}$ which is continuous in each variable (with respect to the maximum norm) and satisfies*

$$F(h_{K_1}, \dots, h_{K_n}) = V(K_1, \dots, K_n)$$

for every $K_1, \dots, K_n \in CC^n$.

Solution 6.25. Let $\mathbf{g}$ be a two-dimensional vector subspace of $\mathbb{R}^n$. Then $\mathbf{g}$ is naturally identified with $\mathbb{R}^2$, and if $K, L \in CC^2$, then identifying them with subsets of $\mathbf{g}$, using Theorem 6.2.4 from the main book we may write

$$\binom{n}{2} V(K, L, B^n[n-2]) = V^{(2)}(K, L) V^{(n-2)}(\mathrm{proj}_{\mathbf{g}^\perp}(B^n)[n-2]),$$

from where we get that

$$V^{(2)}(K, L) = \frac{1}{\kappa_{n-2}} \binom{n}{2} V(K, L, B^n[n-2]).$$

If there exists a function $F\colon C(\mathbb{S}^{n-1})^n \to \mathbb{R}$ which is multilinear, continuous in each entry, and agrees with the mixed volume for support functions, then the function $G\colon C(\mathbb{S}^1)^2 \to \mathbb{R}$ defined as

$$G(f, g) := \frac{1}{\kappa_{n-2}} \binom{n}{2} F(f, g, h_{B^n}[n-2])$$

is bilinear, continuous in each variable, and agrees with $V^{(2)}$ for support functions. This contradicts Theorem 6.1.

□

Exercise 6.26. *Let $f\colon (\mathbb{S}^{n-1})^n \to \mathbb{R}$ be a function which is continuous, non-negative, symmetric, and even in each entry. Define $H\colon (CC^n)^n \to \mathbb{R}$ by*

$$H(K_1, \dots, K_n) :=$$

$$:= \int_{\mathbb{S}^{n-1}} \dots \int_{\mathbb{S}^{n-1}} h_{K_1}(u_1) \dots h_{K_n}(u_n) f(u_1, \dots, u_n)\, d\mathcal{H}^{n-1}(u_1) \dots d\mathcal{H}^{n-1}(u_n).$$

(a) *Verify that H is symmetric, Minkowski multilinear, and translation-invariant.*

(b) *Show that H is non-decreasing for inclusions in each variable.*

(c) *Prove that H has the* valuation property *in each entry. That is, show that if $K, L \in CC^n$ are such that $K \cup L \in CC^n$, then*

$$H(K \cup L, K_2, \dots, K_n) + H(K \cap L, K_2, \dots, K_n) = H(K, K_2, \dots, K_n) + H(L, K_2, \dots, K_n)$$

for every $K_2, \dots, K_n \in CC^n$.

(d) *Verify that H is not a scalar multiple of the mixed volume of $\mathbb{R}^n$.*

Solution 6.26. (a) The symmetry is obvious, and the positive multilinearity follows from the positive linearity of the support functions in the bodies. To prove that H is translation-invariant, let $x \in \mathbb{R}^n$ be arbitrarily given, and notice that

$$H(K_1 + x, K_2, \dots, K_n) - H(K_1, \dots, K_n) =$$
$$= \int_{\mathbb{S}^{n-1}} \dots \int_{\mathbb{S}^{n-1}} \langle x, u_1 \rangle h_{K_2}(u_2) \dots h_{K_n}(u_n) f(u_1, \dots, u_n)\, d\mathcal{H}^{n-1}(u_1) \dots d\mathcal{H}^{n-1}(u_n).$$

Since f is even in each entry, we get that the function $u_1 \mapsto \langle x, u_1 \rangle f(u_1, \dots, u_n)$ is odd for any fixed $(u_2, \dots, u_n) \in (\mathbb{S}^{n-1})^{n-1}$. Since $\mathcal{H}^{n-1}$ is an even measure, we get

$$\int_{\mathbb{S}^{n-1}} \langle x, u_1 \rangle f(u_1, \dots, u_n)\, d\mathcal{H}^{n-1}(u_1) = 0.$$

Finally, by Fubini's theorem we may write

$$H(K_1 + x, K_2, \dots, K_n) - H(K_1, \dots, K_n) =$$
$$= \int_{\mathbb{S}^{n-1}} \dots \left(\int_{\mathbb{S}^{n-1}} \langle x, u_1 \rangle f(u_1, \dots, u_n)\, d\mathcal{H}^{n-1}(u_1) \right) h_{K_2}(u_2) \dots h_{K_n}(u_n)\, d\mathcal{H}^{n-1}(u_2) \dots d\mathcal{H}^{n-1}(u_n) =$$
$$= 0,$$

and this shows that H is translation-invariant in the first entry. For the other entries, we simply use the symmetry property.

(b) This follows immediately from the fact that $h_K \leq h_L$ whenever $K \subseteq L$. We also need the fact that f is non-negative, of course.

(c) This is a straightforward consequence of Lemma 6.2.2 (main book), which states that support functions have the valuation property in the involved bodies.

(d) If H is a scalar multiple of the mixed volume, then up to rescaling f we may assume that H is the mixed volume itself. The function $\tilde{H} \colon C(\mathbb{S}^{n-1})^n \to \mathbb{R}$ defined as

$$\tilde{H}(f_1, \dots, f_n) :=$$
$$:= \int_{\mathbb{S}^{n-1}} \int_{\mathbb{S}^{n-1}} f_1(u_1) \dots f_n(u_n) f(u_1, \dots, u_n)\, d\mathcal{H}^{n-1}(u_1) \dots d\mathcal{H}^{n-1}(u_n)$$

is clearly continuous in each entry (in the maximum norm) and multilinear. The hypothesis that H is the mixed volume gives that

$$\tilde{H}(h_{K_1}, \ldots, h_{K_n}) = H(K_1, \ldots, K_n) = V(K_1, \ldots, K_n),$$

and this contradicts the result that we proved in Exercise 6.25.

□

The objective of the next exercise is to prove that the mixed volume of $\mathbb{R}^n$ cannot be "suitably" extended to a functional defined in a family of compact sets larger than CC^n. In what follows, for the convenience of the reader we recall that C^n stands for the family of non-empty compact subsets of $\mathbb{R}^n$. Also, if $\mathcal{A}$ is a family of subsets of $\mathbb{R}^n$ which is closed under Minkowski addition, then we say that a functional $\varphi\colon \mathcal{A} \to \mathbb{R}$ is *Minkowski additive* if

$$\varphi(A + B) = \varphi(A) + \varphi(B)$$

for any $A, B \in \mathcal{A}$. Notice that the mixed volume (seen as a functional defined for n-tuples of sets from CC^n) is Minkowski additive in each entry.

Exercise 6.27. **(a)** *Let $A \subseteq \mathbb{R}^n$ be non-empty. Prove that*

$$A + n\mathrm{conv}(A) = (n + 1)\mathrm{conv}(A).$$

(b) *Let $\varphi\colon \mathcal{A} \to \mathbb{R}$ be a Minkowski additive functional, where $\mathcal{A}$ is a family of subsets of $\mathbb{R}^n$ which is closed for Minkowski addition. Show that*

$$\varphi(A) = \varphi(\mathrm{conv}(A))$$

for any $A \in \mathcal{A}$.

(c) *Let $C \subseteq C^n$ be full-dimensional and non-convex. Verify that*

$$\mathrm{vol}_n(C) < \mathrm{vol}_n(\mathrm{conv}(C)).$$

(d) *Assume that $\mathcal{D}$ is a family of subsets of $\mathbb{R}^n$ which is closed for Minkowski addition, and such that $CC^n \subseteq \mathcal{D} \subseteq C^n$. Prove that if there exists a functional $\tilde{V}\colon \mathcal{D}^n \to \mathbb{R}$ which is Minkowski additive in each entry and such that*

$$\tilde{V}(D, \ldots, D) = \mathrm{vol}_n(D)$$

for each $D \in \mathcal{D}$, then $\mathcal{D} = CC^n$.

Hint: Compare the volumes of D and $\mathrm{conv}(D)$. Notice that if D is full-dimensional, then these sets have the same volume if and only if D is convex. In the case where D is not full-dimensional, let C be a non-empty, compact, and convex subset of an affine subspace complementary to $\mathrm{aff}(D)$, and compare the volumes of $C + D$ and $\mathrm{conv}(C + D)$.

Solution 6.27. **(a)** Since $A \subseteq \mathrm{conv}(A)$, we get the inclusion

$$A + \mathrm{conv}(A) \subseteq (n + 1)\mathrm{conv}(A).$$

The non-trivial part is the reverse inclusion. Let $x \in \mathrm{conv}(A)$. From Carathéodory's theorem (Theorem 2.1.2 in the main book), we have that x can be written as a convex combination of $n + 1$ or less points

of A. Thus, we may write

$$x = \lambda_1 a_1 + \ldots + \lambda_{n+1} a_{n+1},$$

where $\lambda_j \geq 0$ for each $j = 1, \ldots, n+1$, and $\lambda_1 + \ldots + \lambda_{n+1} = 1$. The λ_j's are $n+1$ non-negative numbers summing up to 1, and hence at least one of them must be greater than or equal to $1/(n+1)$. Up to renumbering them, we may assume that

$$\lambda_1 \geq \frac{1}{n+1}.$$

Thus, we simply write

$$(n+1)x = (n+1)\lambda_1 a_1 + \ldots + (n+1)\lambda_{n+1} a_{n+1} =$$
$$= a_1 + n\left[\frac{(n+1)\lambda_1 - 1}{n} a_1 + \frac{(n+1)\lambda_2}{n} a_2 + \ldots + \frac{(n+1)\lambda_{n+1}}{n} a_{n+1}\right],$$

and the expression between the brackets is obviously a convex combination of points of A. This gives that $(n+1)x \in A + n\mathrm{conv}(A)$, and since x was arbitrarily taken in $\mathrm{conv}(A)$ we get what we wanted to have.

(b) From item **(a)** combined with the Minkowski additivity property of φ we get

$$\varphi(A) + \varphi(n\mathrm{conv}(A)) = \varphi(A + n\mathrm{conv}(A)) = \varphi((n+1)\mathrm{conv}(A)) = \varphi(\mathrm{conv}(A)) + \varphi(n\mathrm{conv}(A)).$$

This gives that $\varphi(A) = \varphi(\mathrm{conv}(A))$.

(c) We have that C is properly contained in $\mathrm{conv}(C)$, and these sets are compact (the convex hull of a compact set is compact). From item **(b)** of Exercise 2.23, we get that $\mathrm{int}(\mathrm{conv}(C))$ is not contained in C. In fact, otherwise we would have

$$\mathrm{conv}(C) = \mathrm{cl}(\mathrm{conv}(C)) = \mathrm{cl}(\mathrm{int}(\mathrm{conv}(C))) \subseteq \mathrm{cl}(C) = C,$$

which would contradict the fact that C is properly contained in $\mathrm{conv}(C)$.

It follows that we may choose a point x_0 of the interior of $\mathrm{conv}(C)$ such that $x_0 \notin C$. From compactness, x_0 has a positive distance to C, and since x_0 is an interior point of $\mathrm{conv}(C)$, we may pick a (non-degenerate) closed ball B centered in x_0 such that $B \subseteq \mathrm{conv}(C) \setminus C$. Since any non-degenerate closed ball has positive volume we get the desired inequality.

(d) Because $\tilde{V}$ is Minkowski additive in each variable, from item **(b)** we have

$$\mathrm{vol}_n(D) = \tilde{V}(D, \ldots, D) = \tilde{V}(\mathrm{conv}(D), D, \ldots, D) = \tilde{V}(\mathrm{conv}(D), \mathrm{conv}(D), D, \ldots, D) = \ldots$$
$$= \ldots = \tilde{V}(\mathrm{conv}(D), \ldots, \mathrm{conv}(D)) = \mathrm{vol}_n(\mathrm{conv}(D))$$

for any $D \in \mathcal{D}$. If D has maximal affine dimension, then from item **(c)** we get that $D = \mathrm{conv}(D)$. It follows that all sets of $\mathcal{D}$ with maximal affine dimension are convex.

Now assume that $D \in \mathcal{D}$ is not full-dimensional. If we choose any compact convex set C within (and full-dimensional in) an affine subspace complementary to $\mathrm{aff}(D)$, then $C + D$ is convex and n-dimensional. Since $C \in \mathcal{CC}^n \subseteq \mathbb{D}$, and because $\mathcal{D}$ is closed for Minkowski addition, we get that $C + D \in \mathcal{D}$. From the previous argument it follows that $C + D$ is convex, and hence

$$\mathrm{conv}(C) + D = C + D = \mathrm{conv}(C + D) = \mathrm{conv}(C) + \mathrm{conv}(D) = C + \mathrm{conv}(D),$$

where the third inequality justifies since C and D lie in complementary affine subspaces. Thus, $D = \mathrm{conv}(D)$, and this shows that all sets of $\mathcal{D}$ are convex.

□

Recall that a convex body K has *constant width* if $h_K(u) + h_K(-u)$ is a constant function of $u \in \mathbb{S}^{n-1}$ (see [188] for more on bodies of constant width). In the next exercise, we investigate the *Barbier theorem* for planar bodies of constant width. Observe that the perimeter of a planar convex body can be calculated in terms of mixed volumes by (6.1.2) from the main book.

Exercise 6.28 (Barbier's Theorem)**.** *Let $K \subseteq \mathbb{R}^2$ be a convex body having constant width $c > 0$. Prove that the perimeter of K equals πc.*

Solution 6.28. We first notice that

$$V(-K, B^2) = V(-\mathrm{Id}K, -\mathrm{Id}B^2) = |\det(-\mathrm{Id})|V(K, B^2) = V(K, B^2).$$

From Exercise 2.18 we know that $K - K$ is the ball cB^2. Hence

$$V(cB^2, B^2) = V(K - K, B^2) = V(K, B^2) + V(-K, B^2) = 2V(K, B^2),$$

and then

$$V(K, B^2) = \frac{1}{2}V(cB^2, B^2) = \frac{c}{2}\mathrm{vol}_2(B^2) = \frac{\pi c}{2}.$$

The perimeter of K is its surface area $S(K)$. Using (6.1.2), we get

$$S(K) = 2V(K, B^2) = \pi c.$$

□

Exercise 6.29 (Blaschke)**.** *Let $K \subseteq \mathbb{R}^3$ be a convex body of constant width $c > 0$.*

(a) *Prove that*

$$3cS(K) - 6\mathrm{vol}_3(K) = 2\pi c^3.$$

(b) *Show that*

$$W_2(K) = \frac{2}{3}\pi c.$$

Solution 6.29. (a) We know that $K - K = cB^3$. Notice that

$$V(K, K, K) = V(-\mathrm{Id}(-K), -\mathrm{Id}(-K), -\mathrm{Id}(-K)) = V(-K, -K, -K) \text{ and}$$
$$V(K, -K, -K) = V(-\mathrm{Id}(-K), -\mathrm{Id}K, -\mathrm{Id}K) = V(-K, K, K),$$

and hence

$$c^3V(B^3, B^3, B^3) = V(K - K, K - K, K - K) = 2V(K, K, K) + 6V(-K, K, K),$$

which can be rewritten as

$$\frac{4}{3}\pi c^3 = 2\mathrm{vol}_3(K) + 6V(-K, K, K). \tag{6.6}$$

To compute the term $V(-K, K, K)$ we just observe that

$$cV(K, K, B^3) = V(K, K, K - K) = V(K, K, K) + V(K, K, -K),$$

and then, since $S(K) = 3V(K, K, B^3)$, we get

$$\frac{c}{3}S(K) = \mathrm{vol}_3(K) + V(-K, K, K). \tag{6.7}$$

Multiplying (6.7) by -6 and summing up with (6.6), we get

$$\frac{4}{3}\pi c^3 - 2cS(K) = 2\mathrm{vol}_3(K) - 6\mathrm{vol}_3(K) = -4\mathrm{vol}_3(K).$$

Multiplying both sides by $3/2$ we get the desired equality.

(b) By the same argument used in Exercise 6.28 and in item **(a)** above, it is clear that $V(K, B^3, B^3) = V(-K, B^3, B^3)$ and $V(K, K, B^3) = V(-K, -K, B^3)$. Since $K - K = cB^3$, we may write

$$c^2V(B^3, B^3, B^3) = V(K - K, K - K, B^3) = 2V(K, K, B^3) + 2V(-K, K, B^3).$$

Also,

$$cV(K, B^3, B^3) = V(K, K - K, B^3) = V(K, K, B^3) + V(-K, K, B^3).$$

Since $V(B^3, B^3, B^3) = \mathrm{vol}_3(B^3) = \frac{4}{3}\pi$, we may summarize these equalities as

$$\begin{aligned} \frac{4}{3}\pi c^2 &= 2V(K, K, B^3) + 2V(-K, K, B^3), \quad \text{and} \\ cV(K, B^3, B^3) &= V(K, K, B^3) + V(-K, K, B^3). \end{aligned}$$

The right-hand side of the second equality is half the right-hand side of the first one. Thus,

$$2cV(K, B^3, B^3) = \frac{4}{3}\pi c^2,$$

and this gives that

$$V(K, B^3, B^3) = \frac{2}{3}\pi c.$$

Since $W_2(K) = V(K, B^3, B^3)$ the proof is complete.

□

The *Wills functional* is defined as the functional $\mathcal{W}\colon CC^n \to \mathbb{R}$ which associates each set of CC^n to the sum of its intrinsic volumes. That is,

$$\mathcal{W}(K) := \sum_{j=0}^{n} V_j(K)$$

for each $K \in CC^n$. For motivations, further results, and applications around this definition, we refer the reader to [17] and references therein.

Exercise 6.30. *Prove that the Wills functional has the following integral representation:*

$$\mathcal{W}(K) = 2\pi \int_0^{+\infty} te^{-\pi t^2} \mathrm{vol}_n(K + tB^n)\, dt.$$

Remark 6.7. Another integral representation of the Wills functional is given in Exercise 11.19. For that, we need *Hadwiger's theorem* on the characterization of *valuations* (see Section 11.3 and, in particular, Theorem 11.3.1 from the main book), and this is why we do not formulate it here.

Solution 6.30. We compute the integral in the right-hand side using Steiner's formula for the volume of $K + tB^n$:

$$2\pi \int_0^{+\infty} te^{-\pi t^2} \mathrm{vol}_n(K + tB^n)\, dt = 2\pi \int_0^{+\infty} te^{-\pi t^2} \sum_{j=0}^{n} \binom{n}{j} t^j W_j(K)\, dt =$$

$$= 2\pi \sum_{j=0}^{n} \binom{n}{j} W_j(K) \int_0^{+\infty} t^{j+1} e^{-\pi t^2}\, dt.$$

To compute the integrals in the right-hand side of the above, we may use (5.5) from above and the expression for $\kappa_j := \mathrm{vol}_j(B^j)$ given in Proposition 4.3.1 (main book) to write

$$\int_0^{+\infty} t^{j+1} e^{-\pi t^2}\, dt = \frac{\Gamma((j+2)/2))}{2\pi^{(j+2)/2}} = \frac{\Gamma\left(\frac{j}{2}+1\right)}{\pi^{\frac{j}{2}}} \frac{1}{2\pi} = \frac{1}{2\pi\kappa_j}.$$

Thus,

$$2\pi \int_0^{+\infty} te^{-\pi t^2} \mathrm{vol}_n(K + tB^n)\, dt = 2\pi \sum_{j=0}^{n} \binom{n}{j} W_j(K) \frac{1}{2\pi\kappa_j} = \sum_{j=0}^{n} \binom{n}{j} \frac{1}{\kappa_j} W_j(K) =$$

$$= \sum_{j=0}^{n} V_{n-j}(K) = \sum_{i=0}^{n} V_i(K) = \mathcal{W}(K).$$

□

Remember that (5.4) above is a notion of mean width for log-concave functions introduced in [218]. In the next exercises we investigate a different notion of mean width for log-concave functions which is based on the characterization of the mean width as a multiple of the $(n-1)$-th quermassintegral. This was introduced by Klartag and Milman in [153]. As it was the case in the exercises in which we deal with definition (5.4), here we will restrict ourselves to log-concave functions of the family

$$\mathrm{LC}_\infty(\mathbb{R}^n) := \left\{ f = e^{-\varphi} : \varphi \text{ is closed, convex, proper, and such that } \lim_{|x|\to+\infty} \varphi(x) = +\infty \right\}.$$

For our purposes, the family $\mathrm{LC}_\infty(\mathbb{R}^n)$ is the adequate family of log-concave functions to work with. We note that $1_K \in \mathrm{LC}_\infty(\mathbb{R}^n)$ for every $K \in CC^n$, and from Exercise 1.34 we get that

$$\int_{\mathbb{R}^n} f\, d\mathcal{L}^n < \infty.$$

Since the integral of a log-concave function plays the role of the "volume," we have that the functions of $\mathrm{LC}_\infty(\mathbb{R}^n)$ have "finite volume." In what follows, we remember that

$$G(x) := e^{-\frac{|x|^2}{2}}$$

is the standard Gaussian function defined for $x \in \mathbb{R}^n$. From Exercise 1.36 we get that this is the unique self-polar function of $\mathrm{LC}_\infty(\mathbb{R}^n)$, and hence it "plays the role" of the unit ball, in some sense. The integral of G is computed in (5.6) above.

In what follows, **all** integrations are taken with respect to the Lebesgue measure of $\mathbb{R}^n$. Recall that from (6.4.2) (main book) the mean width $w(K)$ of a convex body $K \subseteq \mathbb{R}^n$ can be computed in terms of mixed volumes as

$$\begin{aligned} w(K) = \frac{2}{\kappa_n} W_{n-1}(K) = \frac{2}{\kappa_n} V(B^n[n-1], K) = \\ = \frac{2}{\kappa_n} \lim_{\varepsilon \to 0^+} \frac{\mathrm{vol}_n(B^n + \varepsilon K) - \mathrm{vol}_n(B^n)}{\varepsilon} = \frac{2}{\kappa_n} \lim_{\varepsilon \to 0^+} \frac{1}{\varepsilon} \left[\int_{\mathbb{R}^n} 1_{B^n + \varepsilon K} - \int_{\mathbb{R}^n} 1_{B^n} \right] = \\ = \frac{2}{\kappa_n} \lim_{\varepsilon \to 0^+} \frac{1}{\varepsilon} \left[\int_{\mathbb{R}^n} 1_{B^n} \star (\varepsilon \cdot 1_K) - \int_{\mathbb{R}^n} 1_{B^n} \right]. \end{aligned}$$

In the last equality, we remember that $\star$ denotes the *Asplund product* introduced in Exercise 1.26 and that the scalar multiplication $\varepsilon \cdot 1_K$ is as defined in Exercise 1.28. The equality $1_{B^n + \varepsilon K} = 1_{B^n} \star (\varepsilon \cdot 1_K)$ follows from these exercises (see item **(a)** of both).

To extend the definition of mean width for arbitrary log-concave functions, it is reasonable to assume that the standard Gaussian function $G(x)$ plays the role of the unit ball, since it is the unique self-polar function in $\mathrm{LC}_\infty(\mathbb{R}^n)$ (see item **(c)** of Exercise 1.39). Hence we define the *mean width* of a function $f \in \mathrm{LC}_\infty(\mathbb{R}^n)$ as

$$w(f) := \frac{4}{n(2\pi)^{n/2}} \lim_{\varepsilon \to 0^+} \frac{1}{\varepsilon} \left[\int_{\mathbb{R}^n} G \star (\varepsilon \cdot f) - \int_{\mathbb{R}^n} G \right], \tag{6.8}$$

where all the integrations are taken with respect to the standard Lebesgue measure, as it was mentioned above. The normalization constant is chosen so that we have $w(G) = 2$. We use a normalization different to the one used in the references cited above (the reason is that, for us, the mean width of a convex body is really the mean of the width function over $\mathbb{S}^{n-1}$, whereas some authors take half of this mean). Notice that it is not a priori clear that the limit in the definition above exists. This was formally proved (in greater generality) in [80, Section 3.1], though we may have $w(f) = +\infty$.

It is important to mention that Rotem proved in [218] that the notions of mean width for log-concave functions defined in (5.4) and (6.8) actually coincide (later we prove this for characteristic functions; see Exercise 11.22).

Exercise 6.31. **(a)** *Verify that indeed we have that* $w(G) = 2$.

(b) *Let* $T \colon \mathbb{R}^n \to \mathbb{R}^n$ *be an affine transformation having the form* $T(\cdot) = x_0 + U(\cdot)$ *for some* $x_0 \in \mathbb{R}^n$ *and some* $U \in \mathrm{O}(n)$. *Prove that*

$$w(f \circ T) = w(f).$$

In other words, w *is translation-invariant and* $\mathrm{O}(n)$*-invariant, in some sense.*

Solution 6.31. (a) From Exercise 1.29 we have

$$[G \star (\varepsilon \cdot G)](x) = [(1+\varepsilon)\cdot G](x) = G\left(\frac{x}{1+\varepsilon}\right)^{\varepsilon} = e^{-\frac{1}{1+\varepsilon}\frac{|x|^2}{2}},$$

and making the change of variables $y = x/\sqrt{1+\varepsilon}$ we arrive at

$$\int_{\mathbb{R}^n} G \star (\varepsilon \cdot G) = \int_{\mathbb{R}^n} e^{-\frac{1}{1+\varepsilon}\frac{|x|^2}{2}}\, dx = \int_{\mathbb{R}^n} e^{-\frac{|y|^2}{2}}(1+\varepsilon)^{n/2}\, dy = [2\pi(1+\varepsilon)]^{n/2}.$$

It follows that

$$\begin{aligned} w(G) &= \frac{4}{n(2\pi)^{n/2}} \lim_{\varepsilon\to 0^+} \frac{1}{\varepsilon}\left[\int_{\mathbb{R}^n} G\star(\varepsilon\cdot G) - \int_{\mathbb{R}^n} G\right] = \\ &= \frac{4}{n(2\pi)^{n/2}} \lim_{\varepsilon\to 0^+} \frac{(2\pi)^{n/2}(1+\varepsilon)^{n/2} - (2\pi)^{n/2}}{\varepsilon} = \\ &= \frac{4}{n} \lim_{\varepsilon\to 0^+} \frac{(1+\varepsilon)^{n/2}-1}{\varepsilon} = \frac{4}{n}\frac{d}{d\varepsilon}(1+\varepsilon)^{n/2}\Big|_{\varepsilon=0} = \frac{4}{n}\frac{n}{2} = 2. \end{aligned}$$

(b) For any given $x \in \mathbb{R}^n$, we have

$$\begin{aligned} &[G\star(\varepsilon\cdot(f\circ T))](x) = \sup\left\{G(x-z)[\varepsilon\cdot(f\circ T)](z) : z\in\mathbb{R}^n\right\} = \\ &= \sup\left\{e^{-\frac{|x-z|^2}{2}} f\circ T\left(\frac{z}{\varepsilon}\right)^{\varepsilon} : z\in\mathbb{R}^n\right\} = \sup\left\{e^{-\frac{|x-z|^2}{2}} f\left(x_0+\frac{1}{\varepsilon}Uz\right)^{\varepsilon} : z\in\mathbb{R}^n\right\}. \end{aligned}$$

Letting

$$\frac{y}{\varepsilon} = x_0 + \frac{1}{\varepsilon}Uz,$$

we have that $z = U^{-1}(y-\varepsilon x_0)$. Since this establishes a bijection of $\mathbb{R}^n$, we may change the variable in the supremum above to get

$$[G\star(\varepsilon\cdot(f\circ T))](x) = \sup\left\{e^{-\frac{|x-U^{-1}(y-\varepsilon x_0)|^2}{2}} f\left(\frac{y}{\varepsilon}\right)^{\varepsilon} : y\in\mathbb{R}^n\right\}.$$

Since U is orthogonal, it is an isometry. Thus,

$$|x-U^{-1}(y-\varepsilon x_0)| = |Ux+\varepsilon x_0 - y|.$$

Hence

$$[G\star(\varepsilon\cdot(f\circ T))](x) = \sup\left\{e^{-\frac{|Ux+\varepsilon x_0-y|^2}{2}} f\left(\frac{y}{\varepsilon}\right)^{\varepsilon} : y\in\mathbb{R}^n\right\} = [G\star(\varepsilon\cdot f)](Ux+\varepsilon x_0).$$

Since the Lebesgue measure of $\mathbb{R}^n$ is invariant under orthogonal transformations and translations, we get that

$$\int_{\mathbb{R}^n} [G\star(\varepsilon\cdot f\circ T)](x)\, dx = \int_{\mathbb{R}^n} G\star(\varepsilon\cdot f)(Ux+\varepsilon x_0)\, dx = \int_{\mathbb{R}^n} G\star(\varepsilon\cdot f)(y)\, dy,$$

and the desired statement follows.

□

Exercise 6.32. *Let $f \in \mathrm{LC}_\infty(\mathbb{R}^n)$ be such that*

$$\int_{\mathbb{R}^n} G \star (\varepsilon \cdot f) \to \int_{\mathbb{R}^n} G \tag{6.9}$$

as $\varepsilon \to 0$. Prove that for every $a > 0$ we have

$$w(af) = \frac{4}{n}\log(a) + w(f),$$

*where af denotes the **usual** scalar multiplication. Show that $w(\cdot)$ is not necessarily positive.*

Remark 6.8. According to [218], $f = e^{-\varphi}$ has the property (6.9) if and only if $\varphi^*(x) < +\infty$ for every $x \in \mathbb{R}^n$ (recall that φ^* is the conjugate function of φ; see Section 1.4).

Solution 6.32. It is immediate to verify that

$$[G \star (\varepsilon \cdot (af))](x) = a^\varepsilon [G \star (\varepsilon \cdot f)](x)$$

for any $x \in \mathbb{R}^n$ and every $\varepsilon > 0$. The Taylor expansion of a^ε at $\varepsilon = 0$ reads

$$a^\varepsilon = 1 + \varepsilon \log(a) + o(\varepsilon^2),$$

where, in particular,

$$\lim_{\varepsilon \to 0} \frac{o(\varepsilon^2)}{\varepsilon} = 0.$$

Thus, using (twice) the hypothesis on the convergence of the integral of $G \star (\varepsilon \cdot f)$, we get

$$\begin{aligned}
w(af) &= \frac{4}{n(2\pi)^{n/2}} \lim_{\varepsilon \to 0^+} \frac{1}{\varepsilon} \left[\int_{\mathbb{R}^n} G \star (\varepsilon \cdot (af)) - \int_{\mathbb{R}^n} G \right] = \\
&= \frac{4}{n(2\pi)^{n/2}} \lim_{\varepsilon \to 0^+} \frac{1}{\varepsilon} \left[\left(1 + \varepsilon \log(a) + o(\varepsilon^2)\right) \int_{\mathbb{R}^n} G \star (\varepsilon \cdot f) - \int_{\mathbb{R}^n} G \right] = \\
&= \frac{4}{n(2\pi)^{n/2}} \lim_{\varepsilon \to 0^+} \left[\int_{\mathbb{R}^n} G \star (\varepsilon \cdot f) - \int_{\mathbb{R}^n} G \right] + \frac{4}{n(2\pi)^{n/2}} \log(a) \lim_{\varepsilon \to 0^+} \int_{\mathbb{R}^n} G \star (\varepsilon \cdot f) = \\
&= w(f) + \frac{4}{n(2\pi)^{n/2}} \log(a) \int_{\mathbb{R}^n} G = w(f) + \frac{4}{n} \log(a).
\end{aligned}$$

Notice that in the last equality we also used (5.6).

□

Exercise 6.33. *Let $K \in CC^n$ and $\varepsilon > 0$. Prove that*

$$[G \star (\varepsilon \cdot 1_K)](x) = e^{-\frac{1}{2} d(\varepsilon K, x)^2}$$

for every $x \in \mathbb{R}^n$.

Remark 6.9. This equality will be used later, in Exercise 11.21, to prove that the function

$$\varepsilon \mapsto \int_{\mathbb{R}^n} G \star (\varepsilon \cdot 1_K) \tag{6.10}$$

is a polynomial on $\varepsilon > 0$. As a consequence, we will obtain that

$$w(1_K) = c_n w(K)$$

for any $K \in CC^n$, where c_n is a constant which only depends on the dimension. In some sense, this confirms that the definition (6.8) gives a reasonable extension of the concept of mean width to log-concave functions. Moreover, an equality which is very similar to (6.10) will be used to obtain a lower bound for the Wills functional (see Exercise 11.23).

Solution 6.33. Recalling that $\varepsilon \cdot 1_K = 1_{\varepsilon K}$, from the definition of the Asplund product we have

$$[G \star (\varepsilon \cdot 1_K)](x) = \sup \left\{ e^{-\frac{|x-z|^2}{2}} 1_{\varepsilon K}(z) : z \in \mathbb{R}^n \right\}.$$

Since $1_{\varepsilon K}(z) = 0$ whenever $z \notin \varepsilon K$ and because $e^{-|x-z|^2/2}$ is always positive, we have that the supremum above is attained for $z \in \varepsilon B^n$. Hence,

$$[G \star (\varepsilon \cdot 1_K)](x) = \sup \left\{ e^{-\frac{|x-z|^2}{2}} 1_{\varepsilon K}(z) : z \in \varepsilon K \right\} = \sup \left\{ e^{-\frac{|x-z|^2}{2}} : z \in \varepsilon K \right\} =$$
$$= \exp\left(-\frac{1}{2} \inf \left\{ |x-z|^2 : z \in \varepsilon K \right\} \right) = e^{-\frac{1}{2} d(\varepsilon K, x)^2},$$

as we wanted to prove.

□

Exercise 6.34 (Urysohn's Inequality)**.** *Let $f \in \mathrm{LC}_\infty(\mathbb{R}^n)$ be such that*

$$\int_{\mathbb{R}^n} f = \int_{\mathbb{R}^n} G \quad \left(= (2\pi)^{n/2} \right).$$

Prove that $w(f) \geq w(G)$ $(= 2)$. In other words, G minimizes the mean width among all functions having the same volume as it.

Hint: Use the version of the Prékopa-Leindler inequality given in Exercise 1.28.

Solution 6.34. From the version of the Prékopa-Leindler given in Exercise 1.28 we may write

$$\int_{\mathbb{R}^n} G \star (\varepsilon \cdot f) = \int_{\mathbb{R}^n} (1-\varepsilon) \cdot \left((1-\varepsilon)^{-1} \cdot G \right) \star (\varepsilon \cdot f) \geq \left(\int_{\mathbb{R}^n} (1-\varepsilon)^{-1} \cdot G \right)^{1-\varepsilon} \left(\int_{\mathbb{R}^n} f \right)^{\varepsilon} =$$
$$= (2\pi)^{\frac{n\varepsilon}{2}} \left(\int_{\mathbb{R}^n} \left(e^{-\frac{|(1-\varepsilon)x|^2}{2}} \right)^{\frac{1}{1-\varepsilon}} \right)^{1-\varepsilon} = (2\pi)^{\frac{n\varepsilon}{2}} \left(\int_{\mathbb{R}^n} e^{-\frac{(1-\varepsilon)}{2}|x|^2} \right)^{1-\varepsilon}.$$

We compute the integral in the right-hand side similarly as we did to calculate the integral of G. We consider the change of variables $y = [(1-\varepsilon)/2]^{1/2} x$ and use (4.3.1) from the main book to write

$$\int_{\mathbb{R}^n} e^{-\frac{(1-\varepsilon)}{2}|x|^2} \, dx = \int_{\mathbb{R}^n} e^{-|y|^2} \left(\frac{1-\varepsilon}{2} \right)^{-\frac{n}{2}} dy = \left(\frac{2\pi}{1-\varepsilon} \right)^{n/2}.$$

Therefore, we get

$$\int_{\mathbb{R}^n} G \star (\varepsilon \cdot f) \geq (2\pi)^{\frac{n\varepsilon}{2}} \left(\int_{\mathbb{R}^n} e^{-\frac{(1-\varepsilon)}{2}|x|^2} \right)^{1-\varepsilon} = (2\pi)^{\frac{n\varepsilon}{2}} \left(\left(\frac{2\pi}{1-\varepsilon} \right)^{n/2} \right)^{1-\varepsilon} = \\ = (2\pi)^{n/2} (1-\varepsilon)^{-\frac{n}{2}(1-\varepsilon)},$$

and hence

$$\int_{\mathbb{R}^n} G \star (\varepsilon \cdot f) - \int_{\mathbb{R}^n} G \geq (2\pi)^{n/2} (1-\varepsilon)^{-\frac{n}{2}(1-\varepsilon)} - (2\pi)^{n/2} = (2\pi)^{n/2} \Big[(1-\varepsilon)^{-\frac{n}{2}(1-\varepsilon)} - 1 \Big].$$

Finally, we estimate

$$w(f) = \frac{4}{n(2\pi)^{n/2}} \lim_{\varepsilon \to 0^+} \frac{1}{\varepsilon} \left[\int_{\mathbb{R}^n} G \star (\varepsilon \cdot f) - \int_{\mathbb{R}^n} G \right] \geq \\ \geq \frac{4}{n(2\pi)^{n/2}} \lim_{\varepsilon \to 0^+} \frac{1}{\varepsilon} (2\pi)^{n/2} \Big[(1-\varepsilon)^{-\frac{n}{2}(1-\varepsilon)} - 1 \Big] = \\ = \frac{4}{n} \lim_{\varepsilon \to 0^+} \frac{1}{\varepsilon} \Big[(1-\varepsilon)^{-\frac{n}{2}(1-\varepsilon)} - 1 \Big] = \frac{4}{n} \frac{d}{d\varepsilon} \Big[(1-\varepsilon)^{-\frac{n}{2}(1-\varepsilon)} \Big] \Big|_{\varepsilon=0}.$$

We use standard calculus to compute the derivative in the right-hand side (to derivate x^x, simply write $x^x = e^{x \log(x)}$ and use the chain rule). We easily get the equality

$$\Big((1-\varepsilon)^{-\frac{n}{2}(1-\varepsilon)} \Big)' = \frac{n}{2} (1-\varepsilon)^{-\frac{n}{2}(1-\varepsilon)} \big(\log(1-\varepsilon) + 1 \big),$$

and hence for $\varepsilon = 0$ the value of the derivative is $n/2$. Then,

$$w(f) \geq \frac{4}{n} \frac{d}{d\varepsilon} \Big[(1-\varepsilon)^{-\frac{n}{2}(1-\varepsilon)} \Big] \Big|_{\varepsilon=0} = \frac{4}{n} \frac{n}{2} = 2 = w(G),$$

and the proof is complete.

□

For Further Reading

Since the notions of *mixed volumes* and of *quermassintegrals* are visibly important (one can find them in almost all essential books on convex sets), we decided to choose them for a deeper study of related references. This decision implies that also *intrinsic volumes* are automatically in our focus.

Mixed Volumes

As a letter to Hilbert verifies, the notion of *mixed volumes* goes back to Minkowski; see pp. 133–136 in [198]. Further early and comprehensive representations of this basic notion and its applications were given by Bonnesen and Fenchel (see Sections 29 and 37 in [58]) and Busemann (within the framework of the Brunn-Minkowski theory, cf. Section 6 of his book [73]). Several later books contain complete chapters or sections on mixed volumes, extensions, and applications; e.g., we refer to Hadwiger (see [130] and [131]), Eggleston [86], Leichtweiss [161], and Burago and Zalgaller [72]. The last two

books give already a very detailed representation: Leichtweiss [161] presents in Section 15 the two-dimensional case and in Section 24 results on general mixed quermassintegrals, and Burago and Zalgaller [72] offer fundamental properties in Section 19, *dual mixed volumes* in subsection 24.3, several *analogues of mixed volumes* (e.g., vector analogues) in Section 25, and mixed volumes for *non-convex sets* in Section 26.

Bridges from convexity to *algebraic geometry* (in particular, to *toric varieties*) are shown in the two monographs [203] and [89] of Oda and Ewald, respectively. In the first book, mixed volumes of several polytopes are expressed via the intersection indices of certain line bundles, and Chapter 4 of [89] treats Minkowski sums and mixed volumes. In a "dictionary" between convexity and the corresponding part of algebraic geometry, mixed volumes are contrasted with intersection numbers of the Cartier divisors, and the Alexandrov-Fenchel inequality with the Hodge inequality.

We continue with more modern monographs on convexity. Webster [256] dedicates his complete Chapter 6 to *mixed volumes and related extremum problems*, using classical tools, e.g., Steiner symmetrization. Groemer [116] studies various (more analytical) aspects and applications of mixed volumes, such as: *mean projection measures* (Section 2.4) as well as *relations to Fourier series* (Chapter 4) and *to harmonic expansions* (Section 5.1). Thompson [243] studies fundamental results around mixed volumes and extended notions (Section 2.3), their role regarding area and content as well as isoperimetrices in Minkowski spaces (Chapter 5), and special relations to the Holmes-Thompson and the Busemann definition of area in normed spaces (Chapters 6 and 7). In Appendix A of [97], Gardner studies mixed volumes and *mixed area measures*, relations to quermassintegrals and intrinsic volumes, and also dual mixed volumes. In Section 6.1 of his monograph [117], Gruber treats mixed volumes, Minkowski's corresponding theorem, *Steiner's formula*, and several basic properties of mixed volumes including the *valuation property*.

Schneider [228] gives the most comprehensive representation of mixed volumes, their extensions and applications. For example, in Chapter 5 he gives basic definitions and related notions like mixed area measures, generalizations, curvature centroids, Minkowski tensors, and mixed discriminants, and in Chapter 7 inequalities for mixed volumes (including isoperimetric inequalities and the Alexandrov-Fenchel inequality) are presented, with discussions of the *equality cases, stability versions*, and related properties of *Wulff shapes*. In Section 6.3 *mixed valuations* are discussed, Section 7.3 treats *mixed quermassintegrals*, and in the notes to Section 8.2 *mixed bodies* occur. Section 9.4, refers to *mixed width integrals*, and Chapter 10 deals with mixed affine and mixed p-affine surface area, dual mixed volumes, *mixed projection bodies*, and *mixed intersection bodies*.

Two very recent books discussing also mixed volumes are [188] and [142]. In the first one, having bodies of constant width and related notions as main subjects, the whole Chapter 12 is devoted to *mixed volumes as a tool*, used to prove classical theorems (e.g.) on bodies of constant width. Hug and Weil [142] deal with mixed volumes in Section 3.3 (basic properties), Section 4.1 (mixed surface area measures), and Section 4.3 (the notion of mixed discriminant and a local Steiner formula). In our main book, mixed volumes are the central subject of Chapter 6, and the next chapter deals with the closely related *mixed surface area measures*. In addition, *dual mixed volumes* are mainly studied in Section 11.1 and in subsection 11.7.1 of [29].

Quermassintegrals

Now we turn to the notion of *quermassintegral*, which is a special case of that of mixed volume (therefore we can be briefer here). The name "quermassintegral" uses the German word "quermass," meaning practically "measure of a cross-section" for a prescribed direction. A brief introduction to types of such *cross-section measures* is given by Bonnesen and Fenchel in §7 of [58]; see also the beginning of subsection 6.7.3 in the main book [29]. Defined for fixed directions, examples of such measures are the *width* (yielding the support function of the difference body of a convex body), the *maximal chord length* (giving the radial function of the difference body), and also the *brightness*

(being the support function of the projection body). Further references on cross-section measures are Hadwiger's monograph [131] (cf. Sections 4.4.2 and 4.4.3, slightly different to other sources in the case of section measures), Gardner's book [97] (Chapters 3, 4, 6, 7, and 8), Schneider's monograph [228] (Sections 4.1 and 4.2), and the survey [187].

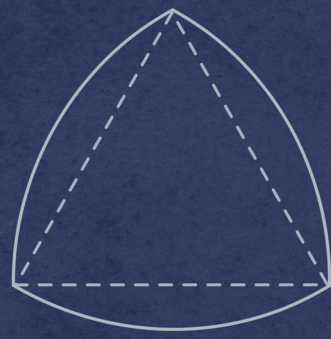
Quermassintegrals are *integral means* of such (projection) measures. One of the most popular examples in this direction is the correspondence between brightness and surface area of a convex body; the connecting link is given by *Cauchy's projection formula* (or *Cauchy's surface area formula*). A first comprehensive reference is again [58] (see Sections 30, 31, and 32 there) and the longest and integral-geometric chapter of Hadwiger's monograph [131], containing Sections 6.1.5 (with Cauchy's projection formula), 6.1.6 (getting quermassintegrals via Kubota's recursion formula), 6.1.7 (mean width), 6.1.10 (the famous characterization of quermassintegrals (which got new proofs due to Klain and Chen in [151] and [76], respectively), 6.3.4 (quermassintegrals in the so-called convex ring), 6.4.8 (harmonic quermassintegrals, based on harmonic means), 6.5.7 (generalizing the isoperimetric problem), and 6.5.8 (mixed quermassintegrals).

The story goes on with Santaló; he investigates quermassintegrals and related topics in Chapter 13 of his famous monograph [221]. Also Cauchy's projection formula is discussed there. The complete Section 16 of Leichtweiss' monograph [161] gives a detailed study of quermassintegrals, where besides Steiner's formula, Cauchy's projection formula, and Kubota's recursion formula also the dependence on the dimension of the space (which plays no role for intrinsic volumes) is studied. In Section 24 mixed quermassintegrals are used to generalize the *Brunn-Minkowski theorem for quermassintegrals*. In Gardner's monograph [97], also *affine and harmonic quermassintegrals* are discussed, namely, in Section 9.3 and the corresponding notes 9.6. Appendix A.5 there (also treating intrinsic volumes) shows interesting basic properties, Steiner's formula and further formulas, as well as the valuation property. In Appendix A.7 quermassintegrals are interestingly studied in the setting of *dual mixed volumes* combined with *star bodies*. Further basic material about quermassintegrals is contained in Section 6.4 of Gruber's monograph [117], Section 4.2 in Schneider's book [228], Section 3.3 (also on mixed volumes) and Chapter 5 of the monograph [142] of Hug and Weil, and Appendix B.5 of the monograph [24] of Artstein-Avidan et al.; see also subsection 9.1.4. in the second part [25] of [24]. In our main book, quermassintegrals are mainly studied in Sections 6.4 and 6.5 as well as in subsection 6.7.3.

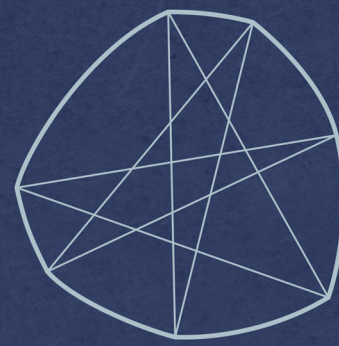
PLATE 7

Bodies of constant width

Reuleaux triangle



Reuleaux 7-gon



Figures 16 - 17

Rotationally symmetric body of constant width

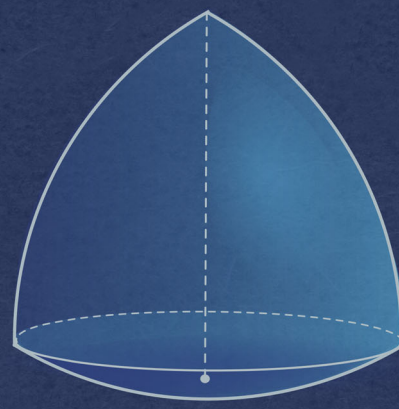


Figure 18

Bodies of constant width ► (see, e.g., 2.9.1.8)

Chapter 7
Mixed Surface Area Measures



"Mathematics is the cheapest science. All one needs for mathematics is a pencil and a paper."

— George Pólya

In the exercises to Chapter 7 the following notions are mainly discussed: *mixed surface area measures* and *mixed bodies*, their relations to *quermassintegrals* and *projection bodies*, a characterization of *centrally symmetric polytopes*, the *Gauss map* and also *Minkowski symmetrization*, the important concept of *Blaschke addition* (including also the Kneser-Süss inequality), and the notion of *affine surface area*.

Plate 7:

Bodies of constant width are those convex bodies whose width function is constant for all directions or, equivalently, whose *minimum width* (also called *thickness*) equals its *maximum width* (*diameter*). Having very many properties, applications, and generalizations, this type of convex bodies represents even a popular topic of *classical geometry*, not only of convex geometry. There is an impressively extensive list of references about them, whose permanent growth can be conveniently seen by comparing the well-known survey [75] of Chakerian and Groemer (1983) and the book [188] of Martini, Montejano, and Oliveros (2019). Plate 7 shows three Euclidean examples of such bodies, which are also complete and reduced.

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V. Balestro et al., *Convexity from the Geometric Point of View: Exercises and Solutions*, Cornerstones, https://doi.org/10.1007/978-3-031-83937-5_7

Exercises and Solutions

The *mixed surface area measure* of $n-1$ sets $C_1,\dots,C_{n-1}\in CC^n$ is the unique Borel measure $S(C_1,\dots,C_{n-1},\cdot)$ on $\mathbb{S}^{n-1}$ having the property that

$$V(C_1,\dots,C_{n-1},L)=\frac{1}{n}\int_{\mathbb{S}^{n-1}} h_L(u)\,dS(C_1,\dots,C_{n-1},u)$$

for every $L\in CC^n$.

Exercise 7.1. *Prove that for any $K_1,\dots,K_n\in CC^n$ the mixed surface area measure $S(K_1,\dots,K_n,\cdot)$ has its centroid at the origin. That is,*

$$\int_{\mathbb{S}^{n-1}} v\,dS(K_1,\dots,K_n,v)=0_{\mathbb{R}^n}.$$

Remark 7.1. In particular, this shows that $S_{n-1}(K,\cdot)$ has its centroid at the origin also in the case when $K\in CC^n$ is not a convex body. Notice that in Section 7.3 of the main book only convex bodies were considered. See also Exercise 7.27 ahead.

Solution 7.1. Let $x\in\mathbb{R}^n$ be arbitrarily given. From the translation invariance of mixed volumes, we have that

$$V(x+C,K_1,\dots,K_{n-1})=V(C,K_1,\dots,K_{n-1})$$

for any $C\in CC^n$. Since $h_{C+x}(v)=h_C(v)+\langle x,v\rangle$ for any $v\in\mathbb{S}^{n-1}$, the definition of $S(K_1,\dots,K_{n-1},\cdot)$ leads to

$$\begin{aligned}0&=V(C+x,K_1,\dots,K_{n-1})-V(C,K_1,\dots,K_{n-1})=\\&=\frac{1}{n}\int_{\mathbb{S}^{n-1}} h_{C+x}(v)\,dS(K_1,\dots,K_{n-1},v)-\frac{1}{n}\int_{\mathbb{S}^{n-1}} h_C(v)\,dS(K_1,\dots,K_{n-1},v)=\\&=\frac{1}{n}\int_{\mathbb{S}^{n-1}}\langle x,v\rangle\,dS(K_1,\dots,K_{n-1},v)=\frac{1}{n}\left\langle x,\int_{\mathbb{S}^{n-1}} v\,dS(K_1,\dots,K_{n-1},v)\right\rangle.\end{aligned}$$

Since this holds for any $x\in\mathbb{R}^n$, the desired statement follows.

□

Exercise 7.2. *Let $K_1,\dots,K_{n-1}\subseteq\mathbb{R}^n$ be convex bodies. Prove that the mixed surface area measure $S(K_1,\dots,K_{n-1},\cdot)$ is not concentrated on a great subsphere of $\mathbb{S}^{n-1}$.*

Solution 7.2. For every $u\in\mathbb{S}^{n-1}$, recalling that $|\langle u,\cdot\rangle|=h(\mathrm{seg}[-u,u],\cdot)$, we have

$$\int_{\mathbb{S}^{n-1}}|\langle u,v\rangle|\,dS(K_1,\dots,K_{n-1},v)=nV(\mathrm{seg}[-u,u],K_1,\dots,K_{n-1})>0,$$

where the last equality follows from Proposition 6.2.3 in the main book (remember that $K_1,\dots,K_{n-1}$ are convex bodies). From Remark 7.3.2 (main book) we get that $S(K_1,\dots,K_{n-1},\cdot)$ is not concentrated on a great subsphere.

□

We introduce some terminology for the next exercises. A *translation family of convex bodies* is the family of all translates of a given convex body $K\subseteq\mathbb{R}^n$. Notice carefully that, as a consequence of

Theorem 7.2.1 in the main book, we have that $S_{n-1}(K,\cdot) = S_{n-1}(L,\cdot)$ if and only if K and L belong to the same translation family of convex bodies.

From now on, we denote the *translation family* of K by $[K]$. The concepts of Minkowski sum and scalar multiplication extend naturally for translation families of convex bodies: we just let $[K]+[L] = [K+L]$ and $\lambda[K] = [\lambda K]$ for any $K, L \in \mathcal{K}^n$ and every $\lambda > 0$. In view of the comment above, we also write $S_{n-1}([K],\cdot) := S_{n-1}(K,\cdot)$ for any convex body $K \subseteq \mathbb{R}^n$. In the same way, volumes and mixed volumes can be seen as functionals defined for translation families of convex bodies.

Exercise 7.3. *From the previous exercises, conclude that for any $K_1,\dots,K_{n-1} \in \mathcal{K}^n$ the mixed surface area measure $S(K_1,\dots,K_{n-1},\cdot)$ is the surface area measure of some convex body and hence of a unique translation family of convex bodies.*

Remark 7.2. For given $K_1,\dots,K_{n-1} \in \mathcal{K}^n$, denote by $[K_1,\dots,K_{n-1}]$ the translation family of convex bodies satisfying

$$S_{n-1}([K_1,\dots,K_{n-1}],\cdot) = S(K_1,\dots,K_{n-1},\cdot).$$

Then $[K_1,\dots,K_{n-1}]$ is called the *mixed body* of $K_1,\dots,K_{n-1}$. Next we investigate some properties of mixed bodies. These are mostly taken from [176].

Solution 7.3. Since $S(K_1,\dots,K_{n-1},\cdot)$ has its centroid at the origin and is not concentrated on a great subsphere, it follows from the Minkowski existence theorem that it is the surface area measure of a convex body.

□

Exercise 7.4. *Let $L_1, K_1,\dots,K_{n-1} \in \mathcal{K}^n$ and $\lambda_1,\dots,\lambda_{n-1} > 0$ be arbitrarily given. Verify that:*

(a) $[\lambda_1 K_1,\dots,\lambda_{n-1}K_{n-1}] = (\lambda_1\dots\lambda_{n-1})^{1/(n-1)}[K_1,\dots,K_{n-1}]$.

(b) $V\big([K_1,\dots,K_{n-1}][n-1], L_1\big) = V(K_1,\dots,K_{n-1},L_1)$.

Solution 7.4. **(a)** We simply observe that

$$\begin{aligned} S_{n-1}([\lambda_1K_1,\dots,\lambda_{n-1}K_{n-1}],\cdot) &= S(\lambda_1K_1,\dots,\lambda_{n-1}K_{n-1},\cdot) = \\ = \lambda_1\dots\lambda_{n-1}S(K_1,\dots,K_{n-1},\cdot) &= \lambda_1\dots\lambda_{n-1}S_{n-1}([K_1,\dots,K_{n-1}],\cdot) = \\ &= S_{n-1}((\lambda_1\dots\lambda_{n-1})^{1/(n-1)}[K_1,\dots,K_{n-1}],\cdot), \end{aligned}$$

and the desired equality follows.

(b) We simply write

$$\begin{aligned} V([K_1,\dots,K_{n-1}][n-1],L_1) &= \frac{1}{n}\int_{\mathbb{S}^{n-1}} h_{L_1}(u)\,dS_{n-1}([K_1,\dots,K_{n-1}],u) = \\ &= \frac{1}{n}\int_{\mathbb{S}^{n-1}} h_{L_1}(u)\,dS(K_1,\dots,K_{n-1},u) = V(K_1,\dots,K_{n-1},L_1). \end{aligned}$$

□

Exercise 7.5. *Prove that for any $K_1,\dots,K_{n-1} \in \mathcal{K}^n$, the inequality*

$$\mathrm{vol}_n([K_1,\dots,K_{n-1}])^{n-1} \le \frac{1}{\kappa_n}V(B^n,K_1,\dots,K_{n-1})^n$$

holds, with equality if and only if $[K_1,\dots,K_{n-1}]$ is homothetic to a ball.

Hint: Use the 1st Minkowski inequality (Theorem 6.3.1 in the main book).

Solution 7.5. We use item **(b)** of Exercise 7.4 for $L_1 = B^n$ and the 1st Minkowski inequality:

$$V(B^n, K_1, \dots, K_{n-1})^n = V([K_1, \dots, K_{n-1}][n-1], B^n)^n \geq$$
$$\geq \mathrm{vol}_n([K_1, \dots, K_{n-1}])^{n-1} \mathrm{vol}_n(B^n),$$

and dividing both sides by $\mathrm{vol}_n(B^n) = \kappa_n$ we get the desired inequality.

□

Exercise 7.6. *Let $n \geq 2$ and $K \in CC^n$ be such that $\dim(K) \leq n-2$. Verify that $S_{n-1}(K, \cdot)$ is the zero measure on $\mathbb{S}^{n-1}$.*

Solution 7.6. Since $\dim(K) \leq n-2$, we have that $K = \partial K$, and $\mathcal{H}^{n-1}(\partial K) = 0$. We get that $S_{n-1}(K, \mathbb{S}^{n-1}) = 0$.

There is yet another way to see this. If $\dim(K) \leq n-2$, then it is clear that $V(L, K[n-1]) = 0$ for every $L \in CC^n$. Indeed, one cannot find $n-1$ linearly independent segments in K. Thus,

$$0 = nV(B^n, K[n-1]) = \int_{\mathbb{S}^{n-1}} h_{B^n}(v)\, dS_{n-1}(K, v) = \int_{\mathbb{S}^{n-1}} dS_{n-1}(K, v) = S_{n-1}(K, \mathbb{S}^{n-1}).$$

□

Exercise 7.7. **(a)** *Prove that in the case where $n = 2$ we may replace $\mathcal{K}^n$ by CC^n in Theorem 7.2.1 of the main book.*

(b) *Explain why for $n \geq 3$ we cannot replace $\mathcal{K}^n$ by CC^n in Theorem 7.2.1, even if we assume the additional hypothesis that the involved mixed surface area measures are non-zero.*

Solution 7.7. **(a)** In Theorem 7.2.1 of the main book, let $n = 2$ and assume that $K \in CC^2$ is a segment whose unit outer normal is u. Then,

$$S_{n-1}(K, \cdot) = \mathrm{length}(K)\delta_u.$$

If $L \in CC^2$ is such that $S_{n-1}(K, \cdot) = S_{n-1}(L, \cdot)$, then $S_{n-1}(L, \cdot)$ is a measure concentrated at u, which means that L is a segment whose unit outer normal is u. Hence, L is parallel to K. Finally, since $S_{n-1}(L, \mathbb{S}^1) = \mathrm{length}(K)$, we get that $\mathrm{length}(L) = \mathrm{length}(K)$. It follows that K and L are translates.

If K is a point, then $S_{n-1}(K, \cdot)$ vanishes. Thus, $S_{n-1}(L, \cdot)$ also vanishes, and hence L is a point as well.

(b) Now let $n > 2$ and assume that $K \in CC^n$ is not a convex body. Then, $K \subseteq u^\perp$ for some $u \in \mathbb{S}^{n-1}$. Since $S_{n-1}(K, \cdot) = \mu_K$, if $\dim(K) < n-1$, then $S_{n-1}(K, \cdot)$ vanishes. In this case, any other $L \in CC^n$ with $\dim(L) < n-1$ has the same surface area measure as K. On the other hand, if $\dim(K) = n-1$, then

$$S_{n-1}(K, \cdot) = \mathrm{vol}_{n-1}(K)\delta_u.$$

Thus, we may replace K by any $L \in CC^n$ such that $L \subseteq u^\perp$ and $\mathrm{vol}_{n-1}(L) = \mathrm{vol}_{n-1}(K)$ to have $S_{n-1}(K, \cdot) = S_{n-1}(L, \cdot)$.

□

Exercise 7.8. *Let $K \subseteq \mathbb{R}^n$ be a convex body with the property that*

$$V(K[n-1], L) = V(K[n-1], -L)$$

for every $L \in CC^n$. Prove that K is centrally symmetric.

Solution 7.8. By standard change of variables, it is clear that

$$\int_{\mathbb{S}^{n-1}} h_{-L}(u)\, dS_{n-1}(K,u) = \int_{\mathbb{S}^{n-1}} h_L(-u)\, dS_{n-1}(K,u) = \int_{\mathbb{S}^{n-1}} h_L(v)\, d(-\mathrm{Id}_* S_{n-1}(K,v)).$$

From $V(K[n-1], L) = V(K[n-1], -L)$ we get that

$$\int_{\mathbb{S}^{n-1}} h_L(v)\, dS_{n-1}(K,v) = \int_{\mathbb{S}^{n-1}} h_L(v)\, d(-\mathrm{Id}_* S_{n-1}(K,v)),$$

and of course this also holds if we replace h_L by any function having the form $h_L - h_M$, $L, M \in CC^n$. Since these functions are dense in $C(\mathbb{S}^{n-1})$ in the maximum norm, we arrive at

$$\int_{\mathbb{S}^{n-1}} f(v)\, dS_{n-1}(K,v) = \int_{\mathbb{S}^{n-1}} f(v)\, d(-\mathrm{Id}_* S_{n-1}(K,v))$$

for any $f \in C(\mathbb{S}^{n-1})$. From Proposition A.2.4 it follows that

$$S_{n-1}(K,\cdot) = -\mathrm{Id}_* S_{n-1}(K,\cdot),$$

and this means that $S_{n-1}(K,\cdot)$ is even. From Proposition 7.2.4 (main book) we get that K is centrally symmetric.

□

Exercise 7.9. *Let $P \subseteq \mathbb{R}^n$ be a proper polytope. Prove that P is centrally symmetric if and only if for each facet F of P there is a facet $\tilde{F}$ of P having opposite unit outer normal vector as F, and such that* $\mathrm{vol}_{n-1}(F) = \mathrm{vol}_{n-1}(\tilde{F})$.

Solution 7.9. Let $u_1, -u_1, \ldots, u_m, -u_m$ be the unit outer normal vectors of the facets of P, and let $a_1, a_1, \ldots, a_m, a_m$ be the corresponding areas of these facets. The facets of the polytope $-P$ have precisely the same unit outer normal vectors and areas. From Theorem 7.3.2 (main book) we get that P and $-P$ are translates, and hence P is centrally symmetric.

□

Exercise 7.10. **(a)** *Show that for any $L, C_1, \ldots, C_{n-1} \in CC^n$, and every $T \in \mathrm{GL}(n)$ we have that*

$$\int_{\mathbb{S}^{n-1}} h_{TL}(u)\, dS(TC_1, \ldots, TC_{n-1}, u) = |\det(T)| \int_{\mathbb{S}^{n-1}} h_L(u)\, dS(C_1, \ldots, C_{n-1}, u).$$

(b) *Conclude that for any $f \in C(\mathbb{S}^{n-1})$, all $T \in \mathrm{GL}(n)$ and every $C_1, \ldots, C_{n-1} \in CC^n$, we have the equality*

$$\int_{\mathbb{S}^{n-1}} f(u)\, dS(TC_1, \ldots, TC_{n-1}, u) =$$
$$= |\det(T)| \int_{\mathbb{S}^{n-1}} f \circ (T^*)^{-1}(u)\, dS(C_1, \ldots, C_{n-1}, u).$$

Solution 7.10. **(a)** We simply write

$$\begin{aligned}\int_{\mathbb{S}^{n-1}} h_{TL}(u)\, dS(TC_1,\dots,TC_{n-1},u) =\\ = nV(TL,TC_1,\dots,TC_{n-1}) = n|\det(T)|V(L,C_1,\dots,C_{n-1}) =\\ = |\det(T)|\int_{\mathbb{S}^{n-1}} h_L(u)\, dS(C_1,\dots,C_{n-1},u).\end{aligned}$$

(b) For any $K, L \in CC^n$, recalling that $h_{TA}(u) = h_A(T^*u)$, item **(a)** gives straightforwardly that

$$\begin{aligned}\int_{\mathbb{S}^{n-1}} h_K(T^*u) - h_L(T^*u)\, dS(TC_1,\dots,TC_{n-1},u) =\\ = |\det(T)|\int_{\mathbb{S}^{n-1}} h_K(u) - h_L(u)\, dS(C_1,\dots,C_{n-1},u).\end{aligned}$$

Let $f \in C(\mathbb{S}^{n-1})$. Since the family $\left\{h_K - h_L : K, L \in CC^n\right\}$ is dense in $C(\mathbb{S}^{n-1})$ with respect to the maximum norm, by approximations we get that

$$\int_{\mathbb{S}^{n-1}} f(T^*u)\, dS(TC_1,\dots,TC_{n-1},u) = |\det(T)|\int_{\mathbb{S}^{n-1}} f(u)\, dS(C_1,\dots,C_{n-1},u).$$

Replacing f by $f \circ (T^*)^{-1}$ (which is also a function of $C(\mathbb{S}^{n-1})$, of course) we conclude the proof.

□

Exercise 7.11. *Verify that the following expression for the surface area of a convex body $K \subseteq \mathbb{R}^n$ holds:*

$$S(K) = \frac{1}{2\kappa_{n-1}}\int_{\mathbb{S}^{n-1}}\int_{\partial K} |\langle u_K(x), v\rangle|\, d\mathcal{H}^{n-1}(x) d\mathcal{H}^{n-1}(v),$$

where u_K stands for the Gauss map of ∂K (defined almost everywhere; see Remark 7.2.6 in the main book).

Solution 7.11. First we use Cauchy's surface area formula to write

$$S(K) = \frac{1}{\kappa_{n-1}}\int_{\mathbb{S}^{n-1}} \mathrm{vol}_{n-1}\left(\mathrm{proj}_{v^\perp}(K)\right) d\mathcal{H}^{n-1}(v).$$

Using Theorem 7.2.2 and (7.2.9) from the main book we get

$$\mathrm{vol}_{n-1}\left(\mathrm{proj}_{v^\perp}(K)\right) = \frac{1}{2}\int_{\mathbb{S}^{n-1}} |\langle v,u\rangle|\, dS_{n-1}(K,u) = \frac{1}{2}\int_{\partial K} |\langle v, u_K(x)\rangle|\, d\mathcal{H}^{n-1}(x),$$

and replacing this in the equality above yields the desired expression for $S(K)$.

□

Exercise 7.12. *Recall that we denote by $M_u(K)$ the Minkowski symmetrization of $K \in CC^n$ in the direction of $u \in \mathbb{S}^{n-1}$, introduced in Exercise 2.58. Prove that*

$$W_i(M_u(K)) \geq W_i(K) \tag{7.1}$$

for any $K \in CC^n$, $u \in \mathbb{S}^{n-1}$ and $0 \leq i \leq n-1$.

Remark 7.3. From Exercise 5.4 we have that Minkowski symmetrization preserves mean width, and hence equality in (7.1) always holds in the case where $i = n - 1$. In Exercise 8.12 we characterize the equality case for $i \in \{0, \dots, n-2\}$.

Solution 7.12. We just have to use the Brunn-Minkowski inequality for quermassintegrals given in Exercise 8.6:

$$W_i(M_u(K))^{\frac{1}{n-i}} = W_i\left(\frac{1}{2}\mathrm{ref}_{u^\perp}(K) + \frac{1}{2}K\right)^{\frac{1}{n-i}} \geq W_i\left(\frac{1}{2}\mathrm{ref}_{u^\perp}(K)\right)^{\frac{1}{n-i}} + W_i\left(\frac{1}{2}K\right)^{\frac{1}{n-i}} = \\ = \frac{1}{2}W_i(\mathrm{ref}_{u^\perp}(K))^{\frac{1}{n-i}} + \frac{1}{2}W_i(K)^{\frac{1}{n-i}}.$$

Since $\mathrm{ref}_{u^\perp}$ is volume-preserving (this is highly intuitive, but for formalities see the solution of Exercise 5.4) and $\mathrm{ref}_{u^\perp}(B^n) = B^n$, we have

$$W_i(\mathrm{ref}_{u^\perp}(K)) = V(\mathrm{ref}_{u^\perp}(K)[n-i], B^n[i]) = V(\mathrm{ref}_{u^\perp}(K)[n-i], \mathrm{ref}_{u^\perp}(B^n)[i]) = \\ = |\det(\mathrm{ref}_{u^\perp})|V(K[n-i], B^n[i]) = W_i(K),$$

and this concludes the proof.

□

Exercise 7.13. *Prove the following extension of Theorem 7.2.2 (main book): for any $K_1, \dots, K_{n-1} \in CC^n$ and every $u \in \mathbb{S}^{n-1}$, show that the equality*

$$V^{(n-1)}\big(\mathrm{proj}_{u^\perp}(K_1), \dots, \mathrm{proj}_{u^\perp}(K_{n-1})\big) = \frac{1}{2}\int_{\mathbb{S}^{n-1}} |\langle u, v\rangle|\, dS(K_1, \dots, K_{n-1}, v)$$

holds. Conclude that

$$\int_{\mathbb{S}^{n-1}} V^{(n-1)}\big(\mathrm{proj}_{u^\perp}(K_1), \dots, \mathrm{proj}_{u^\perp}(K_{n-1})\big)\, d\mathcal{H}^{n-1}(u) = \\ = n\kappa_{n-1}V(K_1, \dots, K_{n-1}, B^n).$$

Remark 7.4. Notice that we may reobtain Theorem 6.5.2 (main book) directly from the equality above.

Solution 7.13. First of all, from the definition of the mixed surface area measure $S(K_1, \dots, K_{n-1}, \cdot)$ and since the support function of $\mathrm{seg}[-u, u]$ is given as $|\langle u, \cdot\rangle|$, we get the equality

$$V\big(K_1, \dots, K_{n-1}, \mathrm{seg}[-u, u]\big) = \frac{1}{n}\int_{\mathbb{S}^{n-1}} |\langle u, v\rangle|\, dS(K_1, \dots, K_{n-1}, v). \tag{7.2}$$

Now we apply Theorem 6.2.4 (main book) for $\mathbf{g} = \mathrm{span}\{u\}$ (and, consequently, $\mathbf{g}^\perp = u^\perp$) to write

$$\binom{n}{1}V\big(K_1, \dots, K_{n-1}, \mathrm{seg}[-u, u]\big) = V^{(1)}\big(\mathrm{seg}[-u, u]\big)V^{(n-1)}\big(\mathrm{proj}_{u^\perp}(K_1), \dots, \mathrm{proj}_{u^\perp}(K_{n-1})\big).$$

Because $V^{(1)}$ is just the (Euclidean) length, we arrive at

$$V\big(K_1, \dots, K_{n-1}, \mathrm{seg}[-u, u]\big) = \frac{2}{n}V^{(n-1)}\big(\mathrm{proj}_{u^\perp}(K_1), \dots, \mathrm{proj}_{u^\perp}(K_{n-1})\big).$$

Replacing this equality in (7.2) we get the first desired equality. For the other equality, just integrate both sides of the equality that we already derived, and use Fubini's theorem and Cauchy's surface area formula:

$$\begin{aligned}
&\int_{\mathbb{S}^{n-1}} V^{(n-1)}\big(\mathrm{proj}_{u^\perp}(K_1),\dots,\mathrm{proj}_{u^\perp}(K_{n-1})\big)\,d\mathcal{H}^{n-1}(u) = \\
&= \frac{1}{2}\int_{\mathbb{S}^{n-1}}\int_{\mathbb{S}^{n-1}} |\langle u,v\rangle|\,dS(K_1,\dots,K_{n-1},v)\,d\mathcal{H}^{n-1}(u) = \\
&= \frac{1}{2}\int_{\mathbb{S}^{n-1}}\left(\int_{\mathbb{S}^{n-1}} |\langle u,v\rangle|\,d\mathcal{H}^{n-1}(u)\right) dS(K_1,\dots,K_{n-1},v) = \\
&= \kappa_{n-1}\int_{\mathbb{S}^{n-1}} dS(K_1,\dots,K_{n-1},v) = n\kappa_{n-1}V(K_1,\dots,K_{n-1},B^n),
\end{aligned}$$

where in the last equality we used the definition of $S(K_1,\dots,K_{n-1},\cdot)$ and the fact that $h_{B^n}=1$.

□

Exercise 7.14. **(a)** *Let $i\in\{1,\dots,n-1\}$, $K\in CC^n$, and $u\in\mathbb{S}^{n-1}$. Show that*

$$\frac{1}{2}\int_{\mathbb{S}^{n-1}} |\langle u,v\rangle|\,dS_i(K,v) = W_i^{(n-1)}\big(\mathrm{proj}_{u^\perp}(K)\big),$$

where $W_i^{(n-1)}$ denotes the i-th quermassintegral of $\mathbb{R}^{n-1}$.

Fix $K\in CC^n$ and $i\in\{1,\dots,n-1\}$. For each $\lambda\geq 0$ and $u\in\mathbb{S}^{n-1}$, let

$$h(\Pi_i K,\lambda u) := \frac{1}{2}\int_{\mathbb{S}^{n-1}} |\langle \lambda u,v\rangle|\,dS_i(K,v).$$

Of course, this defines a function $h(\Pi_i K,\cdot)\colon \mathbb{R}^n\to\mathbb{R}$.

(b) *Prove that the function $h(\Pi_i K,\cdot)\colon \mathbb{R}^n\to\mathbb{R}$ is sublinear. Conclude that it is the support function of a set from CC^n (which we denote by $\Pi_i K$, hence justifying the notation that we introduced above).*

(c) *Show that if K is a convex body and $i\in\{1,\dots,n-1\}$, then $\Pi_i K$ is a convex body symmetric with respect to the origin.*

Remark 7.5. In the case $i=n-1$, we obtain the so-called projection body of K. This construction is investigated in Section 10.3 of the main book.

Solution 7.14. **(a)** This follows immediately from the definition of $S_i(K,\cdot)$, from Exercise 7.13, and from the definition of the quermassintegral $W_i^{(n-1)}$. We just write

$$\begin{aligned}
\frac{1}{2}\int_{\mathbb{S}^{n-1}} |\langle u,v\rangle|\,dS_i(K,v) &= \frac{1}{2}\int_{\mathbb{S}^{n-1}} |\langle u,v\rangle|\,dS(K[i],B^n[n-i-1],v) = \\
= V^{(n-1)}\big(\mathrm{proj}_{u^\perp}(K)[i],\mathrm{proj}_{u^\perp}(B^n)[n-i-1]\big) &= V^{(n-1)}\big(\mathrm{proj}_{u^\perp}(K),B^{n-1}[n-i-1]\big) = \\
&= W_i^{(n-1)}\big(\mathrm{proj}_{u^\perp}(K)\big).
\end{aligned}$$

Of course, we also used the fact that $\mathrm{proj}_{u^\perp}(B^n)$ is naturally identified with B^{n-1}.

(b) This is a direct consequence of the fact that the function $|\langle\cdot,v\rangle|$ is sublinear for each $v\in\mathbb{S}^{n-1}$.

(c) From Exercise 7.12, we have that if K is a convex body, then $S_i(K,\cdot)$ is not concentrated on a great subsphere of $\mathbb{S}^{n-1}$. Thus, Remark 7.3.2 from the main book guarantees that

$$\int_{\mathbb{S}^{n-1}} |\langle u, v\rangle|\, dS_i(K, v) > 0$$

for every $u \in \mathbb{S}^{n-1}$. It follows that $h(\Pi_i K, u) > 0$ for all $u \in \mathbb{S}^{n-1}$. From Exercise 2.31, we get that $\Pi_i K$ is a convex body having the origin as an interior point. Since it is clear that $h(\Pi_i K,\cdot)$ is an even function, we obtain from the mentioned exercise that $\Pi_i K$ is symmetric with respect to the origin.

□

Exercise 7.15. *Let $K \in CC^n$, where $n \geq 2$.*

(a) *Verify that for any $u \in \mathbb{S}^{n-1}$ and every $v \in u^\perp$ one has that*

$$h\big(\mathrm{proj}_{u^\perp}(K), v\big) = h(K, v).$$

(b) *Prove that*

$$h(\Pi_1 K, u) = \frac{1}{n-1} \int_{\mathbb{S}^{n-1}\cap u^\perp} h_K(v)\, d\mathcal{H}^{n-2}(v)$$

for any $u \in \mathbb{S}^{n-1}$.

(c) *Show that $\Pi_1(K + L) = \Pi_1(K) + \Pi_1(L)$ for any $K, L \in CC^n$.*

Solution 7.15. (a) Let $x_0 \in K$ be such that $h(K, v) = \langle x_0, v\rangle$, and write

$$x_0 = \mathrm{proj}_{u^\perp}(x_0) + \langle x_0, u\rangle u.$$

Then,

$$h(K, v) = \langle x_0, v\rangle = \langle \mathrm{proj}_{u^\perp}(x_0) + \langle x_0, u\rangle u, v\rangle = \langle \mathrm{proj}_{u^\perp}(x_0), v\rangle \leq h(\mathrm{proj}_{u^\perp}(K), v).$$

If the inequality is strict, then there exists a point $z_0 \in \mathrm{proj}_{u^\perp}(K)$ such that

$$\langle z_0, v\rangle > h(K, v).$$

Since z_0 is a point of the projection of K onto $u^\perp$, we get that there exists a number $\alpha \in \mathbb{R}$ such that $z_0 + \alpha u \in K$. Thus,

$$h(K, v) \geq \langle z_0 + \alpha u, v\rangle = \langle z_0, v\rangle > h(K, v),$$

which is a contradiction. Hence we get the desired equality.

(b) First, observe that

$$h(\Pi_1 K, u) = \frac{1}{2} \int_{\mathbb{S}^{n-1}} |\langle v, u\rangle|\, dS_1(K, v) = \frac{1}{2} \int_{\mathbb{S}^{n-1}} h(\mathrm{seg}[-u, u], v)\, dS_1(K, v) =$$
$$= \frac{n}{2} V(\mathrm{seg}[-u, u], K, B^n[n-2]).$$

Using Exercise 6.18, it follows immediately that

$$\frac{n}{2}V(\mathrm{seg}[-u,u],K,B^n[n-2]) = V^{(n-1)}\big(\mathrm{proj}_{u^\perp}(K),\mathrm{proj}_{u^\perp}(B^n)[n-2]\big) = \\ = V^{(n-1)}\big(\mathrm{proj}_{u^\perp}(K),B^{n-1}_{u^\perp}[n-2]\big),$$

where the last equality follows because $\mathrm{proj}_{u^\perp}(B^n) = B^n \cap u^\perp$, and the latter is the unit ball $B^{n-1}_{u^\perp}$ of $u^\perp$. Since the surface area measure of $B^{n-1}_{u^\perp}$ is the restriction of the Hausdorff measure $\mathcal{H}^{n-2}$ to the unit sphere $\mathbb{S}^{n-1}\cap u^\perp$ of $u^\perp$, we get

$$h(\Pi_1 K,u) = V^{(n-1)}\big(\mathrm{proj}_{u^\perp}(K),B^{n-1}_{u^\perp}[n-2]\big) = \frac{1}{n-1}\int_{\mathbb{S}^{n-1}\cap u^\perp} h(\mathrm{proj}_{u^\perp}(K),v)\,d\mathcal{H}^{n-2}(v).$$

Because the integration is taken over $v\in u^\perp$, we get from item **(a)** that

$$h(\Pi_1 K,u) = \frac{1}{n-1}\int_{\mathbb{S}^{n-1}\cap u^\perp} h(K,v)\,d\mathcal{H}^{n-2}(v),$$

and the argument is complete.

(c) Since $h(K+L,\cdot) = h(K,\cdot)+h(L,\cdot)$, the equality

$$h(\Pi_1(K+L),\cdot) = h(\Pi_1 K,\cdot)+h(\Pi_1 L,\cdot) = h(\Pi_1 K+\Pi_1 L,\cdot)$$

follows immediately from item **(b)**. This gives the desired statement.

□

The next exercise yields a proof of the equality $\Pi_1(K+L) = \Pi_1(K)+\Pi_1(L)$ which does not rely on the integral expression given in item **(b)** above.

Exercise 7.16. *Show that*

$$S_1(\lambda K+L,\cdot) = \lambda S_1(K,\cdot)+S_1(L,\cdot)$$

for any $K,L\in CC^n$ and $\lambda>0$. In other words, the first-order surface area measure can be seen as a Minkowski linear map from CC^n to the space of finite Borel measures of $\mathbb{S}^{n-1}$.

Solution 7.16. Let $M\in CC^n$ be arbitrarily given. We have

$$\frac{1}{n}\int_{\mathbb{S}^{n-1}} h_M(v)\,dS_1(\lambda K+L,v) = V(M,\lambda K+L,B^n[n-2]) = \\ = \lambda V(M,K,B^n[n-2])+V(M,L,B^n[n-2]) = \\ = \frac{\lambda}{n}\int_{\mathbb{S}^{n-1}} h_M(v)\,dS_1(K,v)+\frac{1}{n}\int_{\mathbb{S}^{n-1}} h_M(v)\,dS_1(L,v),$$

and hence

$$\int_{\mathbb{S}^{n-1}} h_M(v)\,dS_1(\lambda K+L,v) = \int_{\mathbb{S}^{n-1}} h_M(v)\,d(\lambda S_1(K,v)+S_1(L,v)).$$

By linearity, this also holds if we replace h_M by $h_{M_1}-h_{M_2}$ for any $M_1,M_2\in CC^n$. Since these functions are dense in $C(\mathbb{S}^{n-1})$ with respect to the maximum norm, we get that

$$\int_{\mathbb{S}^{n-1}} f(v)\,dS_1(\lambda K+L,v) = \int_{\mathbb{S}^{n-1}} f(v)\,d(\lambda S_1(K,v)+S_1(L,v))$$

for every $f \in C(\mathbb{S}^{n-1})$. Proposition A.2.4 (main book) yields that

$$S_1(\lambda K + L, \cdot) = \lambda S_1(K, \cdot) + S_1(L, \cdot).$$

□

Exercise 7.17. *Let $K, L \subseteq \mathbb{R}^n$ be convex bodies. Prove that the measure*

$$\nu(\cdot) := S_{n-1}(K, \cdot) + S_{n-1}(L, \cdot)$$

has its centroid at the origin and is not concentrated on a great subsphere of $\mathbb{S}^{n-1}$. Conclude that ν is the surface area measure of some convex body and hence of a unique translation family of convex bodies.

Remark 7.6. This observation gives a new notion of addition in the space of convex bodies, which is called *Blaschke addition*. For $K, L \in \mathcal{K}^n$, we define the *Blaschke sum* of K and L as the unique convex body $K\#L$ having its centroid at the origin and such that

$$S_{n-1}(K\#L, \cdot) = S_{n-1}(K, \cdot) + S_{n-1}(L, \cdot).$$

This equality is satisfied for any body in the translation family of $K\#L$, and the condition on the centroid lying in the origin appears because we want to choose just one of these bodies.

Recall that to deal with mixed bodies, we worked with translation families rather than with convex bodies (the condition on the centroid was not adopted). Of course, this can be done here as well. The translation family $[K\#L]$ of $K\#L$ satisfies

$$S_{n-1}([K\#L], \cdot) = S_{n-1}([K], \cdot) + S_{n-1}([L], \cdot).$$

Hence we may write $[K]\#[L] = [K\#L]$, thus regarding the Blaschke addition as an operation between translation families of convex bodies. It is clear that both approaches are equivalent.

Solution 7.17. We notice that

$$\int_{\mathbb{S}^{n-1}} v \, d\nu(v) = \int_{\mathbb{S}^{n-1}} v \, dS_{n-1}(K, v) + \int_{\mathbb{S}^{n-1}} v \, dS_{n-1}(L, v) = 0_{\mathbb{R}^n},$$

and this means that ν has centroid at the origin. Moreover, we have

$$\int_{\mathbb{S}^{n-1}} |\langle u, v\rangle| \, d\nu(v) = \int_{\mathbb{S}^{n-1}} |\langle u, v\rangle| \, dS_{n-1}(K, v) + \int_{\mathbb{S}^{n-1}} |\langle u, v\rangle| \, dS_{n-1}(L, v) > 0$$

for every $u \in \mathbb{S}^{n-1}$. From Remark 7.3.2 (main book) we get that ν is not concentrated on a great subsphere.

□

Exercise 7.18. *Let $L_1, K_1, K_2, \ldots, K_{n-1}$ be convex bodies in $\mathbb{R}^n$, and recall the notation introduced in Remark 7.3.2 (main book). Verify that*

$$[K_1 + L_1, K_2, \ldots, K_{n-1}] = [K_1, K_2, \ldots, K_{n-1}]\#[L_1, K_2, \ldots, K_{n-1}].$$

Solution 7.18. From the definition, we have

$$\begin{aligned} S_{n-1}([K_1 + L_1, K_2, \ldots, K_{n-1}], \cdot) &= S(K_1 + L_1, K_2, \ldots, K_{n-1}, \cdot) = \\ &= S(K_1, K_2, \ldots, K_{n-1}, \cdot) + S(L_1, K_2, \ldots, K_{n-1}, \cdot) = \\ &= S_{n-1}([K_1, K_2, \ldots, K_{n-1}], \cdot) + S_{n-1}([L_1, K_2, \ldots, K_{n-1}], \cdot). \end{aligned}$$

The desired assertion follows.

□

Exercise 7.19. *Let $K, L \in \mathcal{K}^n$ and $T \in \mathrm{SL}(n)$. Prove that $TK\#TL = T(K\#L)$.*

Solution 7.19. Let $M \in CC^n$ be arbitrarily given. We write the mixed volume $V(M, T(K\#L)[n-1])$ in two ways. First, notice that

$$V(M, T(K\#L)[n-1]) = \frac{1}{n} \int_{\mathbb{S}^{n-1}} h_M(v)\, dS_{n-1}(T(K\#L), v).$$

Now, we observe that

$$\begin{aligned} V(M, T(K\#L)[n-1]) &= |\det(T)| V(T^{-1}M, K\#L[n-1]) = \\ &= \frac{|\det(T)|}{n} \int_{\mathbb{S}^{n-1}} h_{T^{-1}M}(v)\, dS_{n-1}(K\#L, v) = \\ &= \frac{|\det(T)|}{n} \left[\int_{\mathbb{S}^{n-1}} h_{T^{-1}M}(v)\, dS_{n-1}(K, v) + \int_{\mathbb{S}^{n-1}} h_{T^{-1}M}(v)\, dS_{n-1}(L, v) \right] = \\ &= |\det(T)| \Big(V(T^{-1}M, K[n-1]) + V(T^{-1}M, L[n-1]) \Big) = \\ &= V(M, TK[n-1]) + V(M, TL[n-1]) = \\ &= \frac{1}{n} \int_{\mathbb{S}^{n-1}} h_M(v)\, dS_{n-1}(TK, v) + \frac{1}{n} \int_{\mathbb{S}^{n-1}} h_M(v)\, dS_{n-1}(TL, v). \end{aligned}$$

Then, we have

$$\int_{\mathbb{S}^{n-1}} h_M(v)\, dS_{n-1}(T(K\#L), v) = \int_{\mathbb{S}^{n-1}} h_M(v)\, dS_{n-1}(TK, v) + \int_{\mathbb{S}^{n-1}} h_M(v)\, dS_{n-1}(TL, v)$$

for every $M \in CC^n$. By linearity, this also holds if we replace h_M by any function of the form $h_{M_1} - h_{M_2}$, $M_1, M_2 \in CC^n$. Since these functions are dense in $C(\mathbb{S}^{n-1})$, we may also replace h_M in the above by any function $f \in C(\mathbb{S}^{n-1})$. From Proposition A.2.4 (main book) we get that

$$S_{n-1}(T(K\#L), \cdot) = S_{n-1}(TK, \cdot) + S_{n-1}(TL, \cdot).$$

This leads to

$$S_{n-1}(T(K\#L), \cdot) = S_{n-1}(TK\#TL, \cdot),$$

and this means that $T(K\#L)$ and $TK\#TL$ are translates. We know that $K\#L$ has its centroid at the origin, and from item **(c)** of Exercise 4.10 we get that $T(K\#L)$ also has its centroid at the origin. Since $TK\#TL$ also has the origin as centroid, we obtain that $T(K\#L) = TK\#TL$ (since they are translates with the same centroid).

□

Exercise 7.20 (Kneser-Süss Inequality)**.** *Prove that for any $K, L \in \mathcal{K}^n$ we have the inequality*

$$\mathrm{vol}_n(K\#L)^{(n-1)/n} \geq \mathrm{vol}_n(K)^{(n-1)/n} + \mathrm{vol}_n(L)^{(n-1)/n}.$$

Show that equality holds if and only if K and L are homothetic.

Solution 7.20. From the definition of $K\#L$ we have

$$\begin{aligned} V(K\#L, K[n-1]) + V(K\#L, L[n-1]) = \\ = \frac{1}{n}\int_{\mathbb{S}^{n-1}} h_{K\#L}(v)\, dS_{n-1}(K, v) + \frac{1}{n}\int_{\mathbb{S}^{n-1}} h_{K\#L}(v)\, dS_{n-1}(L, v) = \\ = \frac{1}{n}\int_{\mathbb{S}^{n-1}} h_{K\#L}(v)\, dS_{n-1}(K\#L, v) = V(K\#L, \ldots, K\#L) = \mathrm{vol}_n(K\#L). \end{aligned}$$

On the other hand, from the 1st Minkowski inequality, we get

$$\begin{aligned} \mathrm{vol}_n(K\#L) = V(K\#L, K[n-1]) + V(K\#L, L[n-1]) \geq \\ \geq \mathrm{vol}_n(K\#L)^{\frac{1}{n}}\mathrm{vol}_n(K)^{\frac{n-1}{n}} + \mathrm{vol}_n(K\#L)^{\frac{1}{n}}\mathrm{vol}_n(L)^{\frac{n-1}{n}}. \end{aligned}$$

Dividing both sides by $\mathrm{vol}_n(K\#L)^{\frac{1}{n}}$ we arrive at the desired inequality.

□

Exercise 7.21. *Let $K, L \subseteq \mathbb{R}^n$ be convex bodies. Verify that*

$$\Pi_{n-1}(K\#L) = \Pi_{n-1}K + \Pi_{n-1}L,$$

where Π_{n-1} is defined as in Exercise 7.14.

Solution 7.21. For each $u \in \mathbb{S}^{n-1}$, we have

$$\begin{aligned} h\big(\Pi_{n-1}(K\#L), u\big) = \frac{1}{2}\int_{\mathbb{S}^{n-1}} |\langle u, v\rangle|\, dS_{n-1}(K\#L, v) = \\ = \frac{1}{2}\int_{\mathbb{S}^{n-1}} |\langle u, v\rangle|\, dS_{n-1}(K, v) + \frac{1}{2}\int_{\mathbb{S}^{n-1}} |\langle u, v\rangle|\, dS_{n-1}(L, v) = \\ = h\big(\Pi_{n-1}K, u\big) + h\big(\Pi_{n-1}L, u\big), \end{aligned}$$

and this gives the equality $\Pi_{n-1}(K\#L) = \Pi_{n-1}K + \Pi_{n-1}L$.

□

Exercise 7.22. *Let $K, L \subseteq \mathbb{R}^n$ be arbitrarily given convex bodies. Recall that $[K_1, \ldots, K_{n-1}]$ denotes the mixed body of $K_1, \ldots, K_{n-1} \in \mathcal{K}^n$, which, as we remember, is a translation family of convex bodies. Letting $[K + L]$ be the translation family of $K + L$, show that*

$$\begin{aligned} [K + L] = \\ = [K, \ldots, K]\#\binom{n-1}{1}^{\frac{1}{n-1}}\big[K[n-2], L\big]\#\ldots\#\binom{n-1}{n-2}^{\frac{1}{n-1}}\big[K, L[n-2]\big]\#[L, \ldots, L], \end{aligned}$$

where $[K[n-i-1], L[i]]$ denotes the mixed body with $n-i-1$ copies of K and i copies of L, of course.

Solution 7.22. This follows immediately from the linearity of mixed surface area measures. We simply write

$$S_{n-1}(K+L,\cdot) = S\big(\underbrace{K+L, K+L, \dots, K+L}_{n-1}, \cdot\big) = \sum_{j=0}^{n-1} \binom{n-1}{j} S(K[n-1-j], L[j], \cdot) =$$

$$= \sum_{j=0}^{n-1} \binom{n-1}{j} S_{n-1}\Big(\big[K[n-1-j], L[j]\big], \cdot\Big) =$$

$$= \sum_{j=0}^{n-1} S_{n-1}\left(\binom{n-1}{j}^{\frac{1}{n-1}} \big[K[n-1-j], L[j]\big], \cdot\right) =$$

$$= S_{n-1}\left(\big[K,\dots,K\big]\#\binom{n-1}{1}\big[K[n-2], L\big]\#\dots\#\binom{n-1}{n-2}\big[K, L[n-2]\big]\#\big[L,\dots,L\big], \cdot\right).$$

This gives what we wanted to show.

□

Let $K \subseteq \mathbb{R}^n$ be a convex body whose boundary is C^2_+. Denote by $f(K,\cdot)\colon \mathbb{S}^{n-1} \to \mathbb{R}$ the function which associates each unit vector $u \in \mathbb{S}^{n-1}$ to the inverse of the Gauss curvature of ∂K at the point where u is the unit outer normal vector. The *affine surface area* of K is defined as

$$\Omega(K) := \int_{\mathbb{S}^{n-1}} f(K,u)^{\frac{n}{n+1}}\, d\mathcal{H}^{n-1}(u).$$

For more on this important concept, we refer the reader to the recent survey [234] by Schütt and Werner.

Exercise 7.23. *Show that for any convex body with C^2_+ boundary we have the equality*

$$\Omega(K) = \int_{\partial K} \kappa(x)^{\frac{1}{n+1}}\, d\mathcal{H}^{n-1}(x),$$

where $\kappa(x)$ denotes the Gauss curvature of ∂K at x, as usual.

Solution 7.23. In the notation of Proposition 7.4.1 (main book), we clearly have $f(K,u) = \bar{\kappa}(u)^{-1}$. Hence we may write

$$\Omega(K) = \int_{\mathbb{S}^{n-1}} f(K,u)^{\frac{n}{n+1}}\, d\mathcal{H}^{n-1}(u) = \int_{\mathbb{S}^{n-1}} \bar{\kappa}(u)^{\frac{1}{n+1}} \frac{1}{\bar{\kappa}(u)}\, d\mathcal{H}^{n-1}(u) =$$

$$= \int_{\mathbb{S}^{n-1}} \bar{\kappa}(u)^{\frac{1}{n+1}}\, dS_{n-1}(K,u).$$

Let ξ_K denote the Gauss map of ∂K. Since we obviously have that $\kappa(x) = \bar{\kappa}(\xi_K(x))$, using the standard change of variables (7.4.1) (main book) we arrive at

$$\int_{\mathbb{S}^{n-1}} \bar{\kappa}(u)^{\frac{1}{n+1}}\, dS_{n-1}(K,u) = \int_{\partial K} \kappa(x)^{\frac{1}{n+1}}\, d\mathcal{H}^{n-1}(x).$$

This concludes the proof.

□

Let μ be a finite Borel measure on $\mathbb{S}^{n-1}$. We say that μ is *concentrated on a closed hemisphere* if there exists a vector $u_0 \in \mathbb{S}^{n-1}$ for which the hemisphere $H := \mathbb{S}^{n-1} \cap \mathbf{h}^+(u_0, 0)$ has the property that $\mu(\mathbb{S}^{n-1} \setminus H) = 0$.

Exercise 7.24. *Let μ be a finite Borel measure which is not concentrated on a great subsphere and has its centroid at the origin. Prove that μ is not concentrated on a closed hemisphere. Conclude that the surface area measure of any given convex body is not concentrated on a closed hemisphere.*

Solution 7.24. Let us assume that μ is concentrated on $H = \mathbb{S}^{n-1} \cap \mathbf{h}^+(u_0, 0)$. Since μ is not concentrated at the great subsphere $\mathbb{S}^{n-1} \cap u_0^{\perp}$, we get that $\mu(\mathrm{int}H) > 0$. Of course, we may write

$$\mathrm{int}H = \bigcup_{m=1}^{\infty} \big\{v \in \mathbb{S}^{n-1} : \langle u_0, v\rangle > 1/m\big\},$$

and it follows that there exists $m_0 \in \mathbb{N}$ such that

$$\mu\Big(\big\{v \in \mathbb{S}^{n-1} : \langle u_0, v\rangle > 1/m_0\big\}\Big) := c > 0.$$

Thus, since $\langle v, u_0\rangle \geq 0$ for every $v \in H$, and because μ is concentrated on H, we arrive at

$$\begin{aligned}\left\langle \int_{\mathbb{S}^{n-1}} v\, d\mu(v), u_0 \right\rangle = \left\langle \int_H v\, d\mu(v), u_0 \right\rangle = \int_H \langle u_0, v\rangle\, d\mu(v) \geq \\ \geq \int_{\left\{v \in \mathbb{S}^{n-1} : \langle u_0, v\rangle > 1/m_0\right\}} \langle u_0, v\rangle\, d\mu(v) \geq \frac{c}{m_0} > 0,\end{aligned}$$

and this contradicts the fact that μ has its centroid at the origin.

□

For the next exercise, we recall that for a given fixed $u \in \mathbb{S}^{n-1}$, we define and denote the *positive part* and the *negative part* of the function $x \mapsto \langle u, x\rangle$ as

$$\begin{aligned}\langle u, x\rangle_+ &:= \max\{\langle u, x\rangle, 0\} \quad \text{and} \\ \langle u, x\rangle_- &:= \min\{\langle u, x\rangle, 0\},\end{aligned}$$

respectively.

Exercise 7.25. *Let $K \in CC^n$ and $u \in \mathbb{S}^{n-1}$ be arbitrarily given.*

(a) *Verify that*

$$h(\mathrm{seg}[0_{\mathbb{R}^n}, u], x) = \langle u, x\rangle_+$$

for any $x \in \mathbb{R}^n$.

(b) *Show that*

$$\int_{\mathbb{S}^{n-1}} \langle u, v\rangle_+\, dS_{n-1}(K, v) = \mathrm{vol}_{n-1}\big(\mathrm{proj}_{u^{\perp}}(K)\big),$$

and conclude that

$$\int_{\mathbb{S}^{n-1}} \langle u, v\rangle_-\, dS_{n-1}(K, v) = -\mathrm{vol}_{n-1}\big(\mathrm{proj}_{u^{\perp}}(K)\big).$$

(c) *Prove the following estimate:*

$$\int_{\mathbb{S}^{n-1}} \langle u, v\rangle_+ \, dS_{n-1}(K, v) \geq \frac{\mathrm{vol}_n(K)}{2\max_{\mathbb{S}^{n-1}} h_K}.$$

Solution 7.25. **(a)** We have

$$h(\mathrm{seg}[0_{\mathbb{R}^n}, u], x) = \sup\{\langle x, \lambda u\rangle : \lambda \in [0, 1]\},$$

and it is clear that the supremum is 0 if $\langle u, x\rangle < 0$ and $\langle u, x\rangle$ otherwise. Thus, we are done.

(b) Notice that

$$\mathrm{seg}[-u, u] = 2\mathrm{seg}[0_{\mathbb{R}^n}, u] - u.$$

Thus, from linearity and translation invariance of mixed volumes we get that

$$V(\mathrm{seg}[-u, u], K[n-1]) = V(2\mathrm{seg}[0_{\mathbb{R}^n}, u] - u, K[n-1]) = 2V(\mathrm{seg}[0_{\mathbb{R}^n}, u], K[n-1]).$$

Therefore, using item **(a)** above, the definition of $S_{n-1}(K, \cdot)$, and the equality (7.2.6) (main book), we arrive at

$$\int_{\mathbb{S}^{n-1}} \langle u, v\rangle_+ \, dS_{n-1}(K, v) = \int_{\mathbb{S}^{n-1}} h(\mathrm{seg}[0_{\mathbb{R}^n}, u] \, dS_{n-1}(K, v)) =$$
$$= nV(\mathrm{seg}[0_{\mathbb{R}^n}, u], K[n-1]) = \frac{n}{2} V(\mathrm{seg}[-u, u], K[n-1]) = \mathrm{vol}_{n-1}\big(\mathrm{proj}_{u^\perp}(K)\big).$$

This proves the first equality. The remaining one follows immediately from (7.2.5) (main book) together with the fact that

$$|\langle u, v\rangle| = \langle u, v\rangle_+ - \langle u, v\rangle_-$$

for any $v \in \mathbb{S}^{n-1}$.

(c) It is immediate to verify that K is contained in some translate of the orthogonal cylinder whose basis is $\mathrm{proj}_{u^\perp}(K)$ and whose height has length $h_K(-u) + h_K(u)$. Thus,

$$\big(2\max_{\mathbb{S}^{n-1}} h_K\big) \int_{\mathbb{S}^{n-1}} \langle u, v\rangle_+ \, dS_{n-1}(K, v) \geq \big(h_K(-u) + h_K(u)\big)\mathrm{vol}_{n-1}\big(\mathrm{proj}_{u^\perp}(K)\big) \geq \mathrm{vol}_n(K),$$

and this gives the desired estimate.

□

Exercise 7.26. *Extend item* **(b)** *of the previous exercise showing that for any* $K_1, \ldots, K_{n-1} \in CC^n$ *and every* $u \in \mathbb{S}^{n-1}$ *one has*

$$\int_{\mathbb{S}^{n-1}} \langle u, v\rangle_+ \, dS(K_1, \ldots, K_{n-1}, v) = V^{(n-1)}\big(\mathrm{proj}_{u^\perp}(K_1), \ldots, \mathrm{proj}_{u^\perp}(K_{n-1})\big).$$

Solution 7.26. We use the fact that $\langle u, v\rangle_+ = h(\mathrm{seg}[0_{\mathbb{R}^n}, u], v)$, together with the definition of $S(K_1, \ldots, K_{n-1}, \cdot)$ and Theorem 6.2.4 from the main book to obtain

$$\begin{aligned}\int_{\mathbb{S}^{n-1}} \langle u, v\rangle_+ \, dS(K_1, \ldots, K_{n-1}, v) &= \\ = nV\big(\mathrm{seg}[0_{\mathbb{R}^n}, u], K_1, \ldots, K_{n-1}\big) &= \\ = V^{(1)}\big(\mathrm{seg}[0_{\mathbb{R}^n}, u]\big) V^{(n-1)}\big(\mathrm{proj}_{u^\perp}(K_1), \ldots, \mathrm{proj}_{u^\perp}(K_{n-1})\big) &= \\ = V^{(n-1)}\big(\mathrm{proj}_{u^\perp}(K_1), \ldots, \mathrm{proj}_{u^\perp}(K_{n-1})\big), &\end{aligned}$$

where we used the fact that $V^{(1)}$ is just the length of segments, of course.

□

As it was announced at the end of Section 7.3 from the main book, in the next exercise we give a complete characterization of the area measure $S_{n-1}(K, \cdot)$ in the case where K is an $(n-1)$-dimensional compact convex set.

Exercise 7.27. *Let $K \in CC^n$ and $u \in \mathbb{S}^{n-1}$. Assume that $S_{n-1}(K, \cdot)$ is non-zero. Prove that the following statements are equivalent:*

(a) *$S_{n-1}(K, \cdot)$ is* concentrated on $\{-u, u\}$*, that is,*

$$S_{n-1}(K, \mathbb{S}^{n-1} \setminus \{-u, u\}) = 0.$$

(b) $\dim(K) = n - 1$ *and $K \subseteq u^\perp$, up to a translation.*

(c) *The equality*

$$\int_{\mathbb{S}^{n-1}} |\langle u, v\rangle| \, dS_{n-1}(K, v) = S_{n-1}(K, \mathbb{S}^{n-1})$$

holds.

(d) *There exist numbers $\alpha, \beta \geq 0$ such that*

$$S_{n-1}(K, \cdot) = \alpha\delta_u + \beta\delta_{-u}.$$

In this case, we have that $\alpha = \beta = \mathrm{vol}_{n-1}(K)$.

Solution 7.27. We start proving that **(c)** is equivalent to **(a)**. If **(c)** holds, then we may write

$$\int_{\mathbb{S}^{n-1}} 1 - |\langle u, v\rangle| \, dS_{n-1}(K, v) = 0.$$

On the other hand, $1 - |\langle v, u\rangle| > 0$ whenever $v \in \mathbb{S}^{n-1} \setminus \{-u, u\}$. We claim that this gives that $S_{n-1}(K, \mathbb{S}^{n-1} \setminus \{-u, u\}) = 0$. In fact, if this does not hold, then since $S_{n-1}(K, \cdot)$ is a regular Borel measure and $\mathbb{S}^{n-1} \setminus \{-u, u\}$ is open in $\mathbb{S}^{n-1}$ (hence a Borel set), there exists a compact set $C \subseteq \mathbb{S}^{n-1} \setminus \{-u, u\}$ such that $S_{n-1}(K, C) > 0$ (the regularity gives that the measure of any Borel set equals the supremum of the measures of its compact subsets). By continuity, we get that the integral of $1 - |\langle v, u\rangle|$ over C would be strictly positive, a contradiction. It follows that $S_{n-1}(K, \cdot)$ is concentrated in $\{-u, u\}$.

Conversely, if **(a)** holds, then since $|\langle u,-u\rangle| = |\langle u,u\rangle| = 1$, we get that

$$\int_{\mathbb{S}^{n-1}} 1 - |\langle u,v\rangle|\, dS_{n-1}(K,v) = \int_{\mathbb{S}^{n-1}\setminus\{-u,u\}} 1 - |\langle u,v\rangle|\, dS_{n-1}(K,v) = 0,$$

where the last inequality follows from the fact that $S_{n-1}(K,\cdot)$ is concentrated on $\{-u,u\}$, of course. As a consequence, we get that

$$\int_{\mathbb{S}^{n-1}} |\langle u,v\rangle|\, dS_{n-1}(K,v) = \int_{\mathbb{S}^{n-1}} 1\, dS_{n-1}(K,v) = S_{n-1}(K,\mathbb{S}^{n-1}).$$

Now we prove that **(a)** implies **(b)**. Since $S_{n-1}(K,\cdot)$ is non-zero, we have that $\dim(K) \geq n-1$ (see Exercise 7.6). Let $v^\perp$ be any hyperplane containing u and $-u$. Then, $\{-u,u\} \subseteq \mathbb{S}^{n-1}\cap v^\perp$, and this leads to

$$\mathbb{S}^{n-1}\setminus(\mathbb{S}^{n-1}\cap v^\perp) \subseteq \mathbb{S}^{n-1}\setminus\{-u,u\}.$$

From the hypothesis, we arrive at

$$0 = S_{n-1}(K,\mathbb{S}^{n-1}\setminus\{-u,u\}) \geq S_{n-1}(K,\mathbb{S}^{n-1}\setminus(\mathbb{S}^{n-1}\cap v^\perp)),$$

and this yields $S_{n-1}(K,\mathbb{S}^{n-1}\setminus(\mathbb{S}^{n-1}\cap v^\perp)) = 0$. This means that $S_{n-1}(K,\cdot)$ is concentrated on the great subsphere $\mathbb{S}^{n-1}\cap v^\perp$, and hence K is not a convex body. It follows that $\dim(K) = n-1$.

Up to a translation, we may assume that $0_{\mathbb{R}^n} \in K$. To prove that $K \subseteq u^\perp$, first notice that if this is not the case, then there is a segment $\mathrm{seg}[-w,w] \subseteq u^\perp$ such that

$$V(\mathrm{seg}[-w,w],K[n-1]) > 0.$$

Indeed, just let w be any unit vector of $u^\perp$ which is not the direction of a segment of K. It follows that

$$0 < nV(\mathrm{seg}[-w,w],K[n-1]) = \int_{\mathbb{S}^{n-1}} |\langle w,v\rangle|\, dS_{n-1}(K,v).$$

From Lemma 7.3.2 (main book) we obtain that $S_{n-1}(K,\cdot)$ is not concentrated on the great subsphere $\mathbb{S}^{n-1}\cap w^\perp$. This means that

$$S_{n-1}(K,\mathbb{S}^{n-1}\setminus(\mathbb{S}^{n-1}\cap w^\perp)) > 0.$$

Since $\{-u,u\} \subseteq \mathbb{S}^{n-1}\cap w^\perp$, we get that $\mathbb{S}^{n-1}\setminus\{-u,u\} \supseteq \mathbb{S}^{n-1}\setminus(\mathbb{S}^{n-1}\cap w^\perp)$. This leads to

$$S_{n-1}(K,\mathbb{S}^{n-1}\setminus\{-u,u\}) \geq S_{n-1}(K,\mathbb{S}^{n-1}\setminus(\mathbb{S}^{n-1}\cap w^\perp)) > 0,$$

a contradiction. This shows that **(a)** implies **(b)**.

We move forward to prove that **(b)** implies **(a)**. If $K \subseteq u^\perp$ up to a translation, then for any $w \in u^\perp \cap B^n$ we have

$$0 = nV(\mathrm{seg}[-w,w],K[n-1]) = \int_{\mathbb{S}^{n-1}} |\langle w,v\rangle|\, dS_{n-1}(K,v),$$

and this means that

$$S_{n-1}(K,\mathbb{S}^{n-1}\setminus(\mathbb{S}^{n-1}\cap w^\perp)) = 0$$

for every $w \in u^{\perp} \cap B^n$ (see again Lemma 7.3.2 from the main book). The great subspheres $\mathbb{S}^{n-1} \cap w^{\perp}$ for $w \in u^{\perp} \cap B^n$ are precisely the great subspheres passing through the points $-u$ and u. Let $\{w_1, \ldots, w_{n-1}\}$ be an orthonormal basis of $u^{\perp}$. It is not difficult to see that

$$\mathbb{S}^{n-1} \setminus \{-u, u\} = \bigcup_{j=1}^{n-1} \left[\mathbb{S}^{n-1} \setminus (\mathbb{S}^{n-1} \cap w_j^{\perp})\right].$$

In fact, if $x \in \mathbb{S}^{n-1} \setminus \{-u, u\}$, then decomposing x in the orthonormal basis $\{u, w_1, \ldots, w_{n-1}\}$ we may write $x = \alpha u + \alpha_1 w_1 + \ldots + \alpha_{n-1} w_{n-1}$, and some α_j have to be non-zero (since otherwise $x = u$ or $x = -u$). In this case, $x \notin w_j^{\perp}$, and hence $x \in \mathbb{S}^{n-1} \setminus (\mathbb{S}^{n-1} \cap w_j^{\perp})$. It follows that

$$S_{n-1}(K, \mathbb{S}^{n-1} \setminus \{-u, u\}) = S_{n-1}\left(K, \bigcup_{j=1}^{n-1} \left[\mathbb{S}^{n-1} \setminus (\mathbb{S}^{n-1} \cap w_j^{\perp})\right]\right) \leq$$

$$\leq \sum_{j=1}^{n-1} S_{n-1}(K, \mathbb{S}^{n-1} \setminus (\mathbb{S}^{n-1} \cap w_j^{\perp})) = 0,$$

and this gives **(a)**.

Our last step is to prove that **(a)** is equivalent to **(d)**. If **(d)** holds, then it is immediate that $S_{n-1}(K, \mathbb{S}^{n-1} \setminus \{-u, u\}) = 0$, and this is **(a)**. Conversely, it is clear that the condition $S_{n-1}(K, \mathbb{S}^{n-1} \setminus \{-u, u\}) = 0$ given in **(a)** implies that $S_{n-1}(K, \cdot)$ vanishes for all Borel sets disjoint from $\{u, -u\}$. Also, if E is a Borel set containing u, but not $-u$, then

$$S_{n-1}(K, E) = S_{n-1}(K, \{u\} \cup (E \setminus \{u\})) = S_{n-1}(K, \{u\}),$$

and we have an analogous equality for Borel sets containing $-u$, but not u. Still using the same argument, we get that if E is a Borel set containing both u and $-u$, then

$$S_{n-1}(K, E) = S_{n-1}(K, \{u\}) + S_{n-1}(K, \{-u\}).$$

This shows that $S_{n-1}(K, \cdot)$ is a combination of the Dirac delta measures δ_u and δ_{-u}. Thus, **(d)** follows.

It remains to show that if $S_{n-1}(K, \cdot) = \alpha\delta_u + \beta\delta_{-u}$, then $\alpha = \beta = \mathrm{vol}_{n-1}(K)$. Recall that we are assuming that $S_{n-1}(K, \cdot)$ is non-zero. Also, we already know that $\dim(K) = n-1$ and that K is contained in $u^{\perp}$ (up to a translation). Thus,

$$\mathrm{vol}_{n-1}(K) = \mathrm{vol}_{n-1}\left(\mathrm{proj}_{u^{\perp}}(K)\right) = \frac{n}{2} V(\mathrm{seg}[-u, u], K[n-1]),$$

where the last equality follows from (7.2.6) in the main book. Since

$$\mathrm{seg}[-u, u] = 2\mathrm{seg}[0_{\mathbb{R}^n}, u] - u,$$

we get from linearity and translation invariance of the mixed volume that

$$\mathrm{vol}_{n-1}(K) = nV(\mathrm{seg}[0_{\mathbb{R}^n}, u], K[n-1]) = \int_{\mathbb{S}^{n-1}} \langle u, v\rangle_+ \, dS_{n-1}(K, v),$$

since by Exercise 7.25 we know that the support function of the segment $\mathrm{seg}[0_{\mathbb{R}^n}, u]$ is the function $\langle u, \cdot\rangle_+$. Because $S_{n-1}(K, \cdot) = \alpha\delta_u + \beta\delta_{-u}$, it is easy to compute the integral in the right-hand side of the above. By doing that, we obtain

$$\mathrm{vol}_{n-1}(K) = \alpha\langle u, u\rangle_+ + \beta\langle u, -u\rangle_+ = \alpha.$$

Finally, to prove that $\beta = \mathrm{vol}_{n-1}(K)$, we write

$$\mathrm{seg}[-u,u] = 2\mathrm{seg}[-u,0_{\mathbb{R}^n}] + u$$

and repeat the argument.

□

Exercise 7.28. *Let $f, h \in C(\mathbb{S}^{n-1})$. Prove that*

$$\int_{\mathbb{S}^{n-1}} f(v)\, dS_{n-1}(K,v) = \int_{\mathbb{S}^{n-1}} h(v)\, dS_{n-1}(K,v)$$

for every $K \in CC^n$ if and only if $f - h$ is the restriction of a linear function to $\mathbb{S}^{n-1}$.

Solution 7.28. If $f - h$ is the restriction of a linear function to $\mathbb{S}^{n-1}$, then there exists $y \in \mathbb{R}^n$ such that $f(v) - h(v) = \langle v, y\rangle$ for every $v \in \mathbb{S}^{n-1}$. For every $K \in CC^n$, the measure $S_{n-1}(K,\cdot)$ has the centroid at the origin, and hence

$$\int_{\mathbb{S}^{n-1}} f(v) - h(v)\, dS_{n-1}(K,v) = \int_{\mathbb{S}^{n-1}} \langle v, y\rangle\, dS_{n-1}(K,v) =$$
$$= \left\langle y, \int_{\mathbb{S}^{n-1}} v\, dS_{n-1}(K,v)\right\rangle = \langle y, 0_{\mathbb{R}^n}\rangle = 0.$$

This gives the desired equality.

Conversely, assume that

$$\int_{\mathbb{S}^{n-1}} f(v)\, dS_{n-1}(K,v) = \int_{\mathbb{S}^{n-1}} h(v)\, dS_{n-1}(K,v)$$

for any $K \in CC^n$. For simplicity, denote $g := f - h$. We first prove that g is odd. For that sake, notice that for any $u \in \mathbb{S}^{n-1}$, letting $K \in CC^n$ be such that $K \subseteq u^\perp$ and $\mathrm{vol}_{n-1}(K) = 1$, we get that $S_{n-1}(L,\cdot) = \delta_u + \delta_{-u}$. As a consequence, we get that

$$f(u) + f(-u) = \int_{\mathbb{S}^{n-1}} f(v)\, dS_{n-1}(K,v) = \int_{\mathbb{S}^{n-1}} h(v)\, dS_{n-1}(K,v) = h(u) + h(-u),$$

from where it follows that

$$g(u) = f(u) - h(u) = -(f(-u) - h(-u)) = -g(-u).$$

Thus, g is odd, in fact. Let $x \in \mathbb{S}^{n-1}$, and write $x = \alpha_1 e_1 + \ldots + \alpha_n e_n$. Suppose that all α_j's are non-zero. Notice that

$$x + (\mathrm{sgn}(\alpha_1)\alpha_1)[\mathrm{sgn}(\alpha_1)(-e_1)] + \ldots + (\mathrm{sgn}(\alpha_n)\alpha_n)[\mathrm{sgn}(\alpha_n)(-e_n)] = 0_{\mathbb{R}^n}.$$

The numbers $1, \mathrm{sgn}(\alpha_1)\alpha_1, \ldots, \mathrm{sgn}(\alpha_n)\alpha_n$ are all positive, and the vectors

$$x, \mathrm{sgn}(\alpha_1)(-e_1), \ldots, \mathrm{sgn}(\alpha_n)(-e_n)$$

are pairwise distinct and span $\mathbb{R}^n$. From the Minkowski existence theorem for polytopes (Theorem 7.3.2 from the main book), we get that the measure

$$\delta_x + \mathrm{sgn}(\alpha_1)\alpha_1\delta_{\mathrm{sgn}(\alpha_1)(-e_1)} + \ldots + \mathrm{sgn}(\alpha_n)\alpha_n\delta_{\mathrm{sgn}(\alpha_n)(-e_n)}$$

equals $S_{n-1}(P,\cdot)$ for some proper polytope $P \subseteq \mathbb{R}^n$. Thus,

$$g(x) + \operatorname{sgn}(\alpha_1)\alpha_1 g(\operatorname{sgn}(\alpha_1)(-e_1)) + \ldots + \operatorname{sgn}(\alpha_n)\alpha_n g(\operatorname{sgn}(\alpha_n)(-e_n)) = 0.$$

Since g is odd, we get

$$g(x) = \operatorname{sgn}(\alpha_1)\alpha_1 g(\operatorname{sgn}(\alpha_1)e_1) + \ldots + \operatorname{sgn}(\alpha_n)\alpha_n g(\operatorname{sgn}(\alpha_n)e_n).$$

Using again the fact that g is odd, we observe that $g(\operatorname{sgn}(\alpha_j)e_j) = \operatorname{sgn}(\alpha_j)g(e_j)$ either if α_j is positive or negative. Then, we may rewrite the equality above as

$$g(x) = \alpha_1 g(e_1) + \ldots + \alpha_n g(e_n) = \langle x, (g(e_1), \ldots, g(e_n))\rangle.$$

By continuity, we get that the same holds in the case where some α_j's are zero. It follows that g is the restriction of the linear function $\langle \cdot, (g(e_1), \ldots, g(e_n))\rangle$ to $\mathbb{S}^{n-1}$.

□

In the next exercise, we give a reverse Minkowski inequality which provides an upper bound for the mixed volume $V(K, L, \ldots, L)$ in terms of *intrinsic volumes* of K and L. We remind the reader that V_j stands for the j-th intrinsic volume (see Section 6.4 in the main book for definitions). The reader may use the following theorem.

Theorem 7.1. [142, Theorem 4.6] *If $K \in CC^n$ has positive diameter, then*

$$V_1(K) \geq 2r_{\text{out}}(K).$$

Equality holds if and only if K is a segment.

Exercise 7.29. *Let $K, L \in CC^n$. Prove that*

$$V(K, L[n-1]) \leq \frac{1}{n} V_1(K) V_{n-1}(L).$$

Show that if $\dim(K) \geq 1$ *and* $\dim(L) \geq n-1$*, then equality holds if and only if K is a segment and L is contained in a hyperplane orthogonal to K.*

Solution 7.29. First, recall that

$$V_{n-1}(L) = \frac{n}{\kappa_1} V(L[n-1], B^n) = \frac{n}{2} V(L[n-1], B^n).$$

Up to a translation, we may assume that $r_{\text{out}}(K)B^n$ is the circumball of K. Then, $h_K(u) \leq r_{\text{out}}(K)$ for each $u \in \mathbb{S}^{n-1}$. It follows that

$$V(K, L[n-1]) = \frac{1}{n} \int_{\mathbb{S}^{n-1}} h_K(u)\, dS_{n-1}(L, u) \leq \frac{r_{\text{out}}(K)}{n} \int_{\mathbb{S}^{n-1}} dS_{n-1}(L, u) =$$
$$= r_{\text{out}}(K) V(L[n-1], B^n) \leq \frac{1}{2} V_1(K) V(L[n-1], B^n) = \frac{1}{n} V_1(K) V_{n-1}(L),$$

and this settles the inequality.

If $\dim(L) \geq n-1$, then it is clear that $V_{n-1}(L) > 0$. Then, if equality holds in the inequality above, we have that $V_1(K) = 2r_{\text{out}}(K)$. Since $\dim(K) \geq 1$, we get that K has positive diameter, and from

Theorem 7.1 we obtain that K is a segment ($K = \mathrm{seg}[-ru_0, ru_0]$ for some $u_0 \in \mathbb{S}^{n-1}$ and some $r > 0$, say). Moreover, by the above we have the equality

$$\frac{1}{n}\int_{\mathbb{S}^{n-1}} h_K(u)\, dS_{n-1}(L,u) = \frac{r_{\mathrm{out}}(K)}{n}\int_{\mathbb{S}^{n-1}} dS_{n-1}(L,u).$$

Since we obviously have $r_{\mathrm{out}}(K) = r$ and $h_K(u) = |\langle ru, u_0\rangle|$, we arrive at

$$\frac{r}{n}\int_{\mathbb{S}^{n-1}} |\langle u, u_0\rangle|\, dS_{n-1}(L,u) = \frac{r}{n}\int_{\mathbb{S}^{n-1}} dS_{n-1}(L,u) = \frac{r}{n}\int_{\mathbb{S}^{n-1}} dS_{n-1}(L,u). \tag{7.3}$$

Thus,

$$\int_{\mathbb{S}^{n-1}} |\langle u, u_0\rangle|\, dS_{n-1}(L,u) = S_{n-1}(L, \mathbb{S}^{n-1}).$$

From Exercise 7.27 we get that $L \subseteq u_0^{\perp}$.

Finally, if K is a segment and L is $(n-1)$-dimensional and contained in a hyperplane orthogonal to K, then the equalities $V_1(K) = 2r_{\mathrm{out}}(K)$ and (7.3) hold (from Theorem 7.1 and Exercise 7.27, respectively). Then we have the equality case.

□

For Further Reading

From the topics and notions mentioned before the exercises above, we select now, for presenting corresponding in-depth literature, (mixed) *surface area measures, Blaschke addition*, and *affine surface area*. We start with mixed surface area measures and surface area in general.

(Mixed) Surface Area Measures

A. D. Alexandrov [3] as well as Fenchel and Jessen [94] independently introduced the notion of *surface area measures*, e.g., in the form of *Alexandrov's integral curvature* (also called integral Gauss curvature). Busemann studied several types of curvature and their influence on the local shape of surfaces, their intrinsic metrics and curvature, and local realizations of metrics with non-negative curvature (cf., e.g., Chapters 2 and 3 and Section 21 of [73]). A next step was the work [207] of Pogorelov (see mainly Chapters 1, 2, 3, 4, 5, and 7), also summarizing Alexandrov's former monographs and articles on *area measures and the intrinsic geometry of convex surfaces*, the spherical map, and related uniqueness results.

In Hadwiger's monograph [131] the following related themes are treated: *Minkowski's relative surface area* and the isoperimetric theorem as an inequality for *relative surface areas and measures* of two arbitrary compact sets; see Chapters 5 and 6. Burago and Zalgaller [72] study *closed surfaces of bounded curvature*, various definitions of area, and isoperimetric inequalities and also for sets which are more general than compact ones (see Chapters 1 and 3). And Leichtweiss [161] treats surface area and *Cauchy's projection formula* in Section 16, and curvature properties in Section 24. Also Groemer [116] gives various approaches to the notion of surface area, e.g., using Cauchy's surface area formula and several related inequalities; see especially Sections 2.4, 5.2, and 5.3.

Surface area is also a central theme of Thompson's monograph [243], referring to the *geometry of finite-dimensional real Banach spaces* which is also called *Minkowski geometry* (not meaning the Minkowskian space-time continuum). Based on different definitions of area in n-dimensional

Minkowski spaces ($n > 2$, but mainly $n = 3$), Thompson comprehensively discusses the respective variants of the *isoperimetric problem* in such spaces. These differing concepts are called Choquet-Busemann area, Holmes-Thompson area, Benson area, etc.; it turns out that in the first two cases the construction of the corresponding *isoperimetrices* is based on convexity notions like *intersection body, projection body*, and *polar body*, applied to the respective unit balls.

In Appendix A.2 of his monograph [97], Gardner approaches *area measures via outer parallel sets*, treating also *j-th area measures* obtained via approximation by polytopes. He continues with mixed area measures in Appendix A.3. Various results in Chapters 3 and 9 need these and related notions and are based on theorems of A. D. Alexandrov, Blaschke, and others. More precisely, here we mean results on the *Blaschke body* (Section 3.3 of [97] and its Notes 3.4), *affine surface area* (Notes 9.3), and *affine isoperimetric inequalities* (Notes 9.4). We also refer to Appendix A.6, where, e.g., Cauchy's surface area formula is discussed. In the second part of her book [201], Moszyńska particularly deals with *curvature and surface area measures* (the whole Chapter 10 is dedicated to this theme). For example, curvature measures are used to study the relationship between the mean curvature considered in Section 9.1 and the integral of the mean curvatures in view of differential geometry. Although Morvan's monograph [200] is mainly devoted to differential geometry, it brings together some concepts close to convexity, among them also various aspects of related *curvature measures* (see Parts VI, VII, and VIII).

Section 4.2 of Schneider's monograph [228] contains the most comprehensive and deepest representation of *curvature measures and area measures of order j* ($j = 0, 1, \ldots, n-1$) for general n-dimensional convex bodies (various important surveys, showing the development of this subfield, are covered by this important section). Schneider shows how they are derived from the corresponding *support measures* (or generalized curvature measures, concentrated on the corresponding support elements) by using also the local Steiner formula and the notion of intrinsic volumes. For $j = n - 1$, the area measures become the so-called *surface area measures*; in the smooth and in the strictly convex case, the more general support measures can be replaced by the curvature measures and area measures. In the notes to this section various aspects and generalizations of used notions are discussed, e.g.: (local) Steiner-type formulae (also for non-Euclidean spaces or for non-convex sets), Steiner polynomials (expressing the volume of outer parallel bodies), *axiomatic characterizations of curvature and area measures, and also of support measures*, Hermitian curvature and area measures, as well as Hessian measures of convex functions. Chapter 8 of [228] is inspired by results on the existence, uniqueness, and stability of smooth, closed, and convex hypersurfaces by given curvature functions, such as Gauss curvature, mean curvature, and corresponding generalizations. Neglecting differentiability assumptions leads to general curvature and area measures. Here also *Minkowski's existence theorem* (for given surface area measure of order $n - 1$), Christoffel's problem, and area measures of intermediate order are studied, and related stability estimates are given. The notes in this chapter show impressively the developments around these basic topics. Further on, Section 9.1 of [228] treats the L_p surface area measure (as part of the L_p Brunn-Minkowski theory), and Section 10.5 discusses generalizations of curvature functions and *affine surface areas*; see here also Section 1.4 in the monograph [164] of Leichtweiss. Various related *affine inequalities* are collected there, and in the notes to this section of [164] further related or more general notions are surveyed.

Although mainly devoted to bodies of constant width, the book [188] of Martini, Montejano, and Oliveros contains also passages which refer to *surface area and curvature results of general convex bodies*. Section 11.2, where aspects of differential geometry are treated, is especially interesting in this direction, but also Sections 10.3 and 12.2 contain related material. The very recent monograph [142] of Hug and Weil refers to surface area in Section 3.2, and to *mixed area measures* in Section 4.1. In other sections of Chapter 4 further related topics appear, like the (local) Steiner formula, Minkowski's existence and uniqueness result, and (at the beginning of Chapter 5) Cauchy's projection formula. In the main book, Chapter 7 is dedicated to mixed surface area measures.

Blaschke Addition

We continue now with *Blaschke addition of convex bodies*. The *Blaschke sum* of two convex n-polytopes P_1 and P_2 is a polytope $P = P_1 \# P_2$ which has a facet parallel to every facet of P_1 and P_2, with the same measure. In case that P_1 and P_2 have parallel facets with the same outer normals, then the measure of the corresponding facet of P is the sum of the corresponding measures from P_1 and P_2. This operation (called *Blaschke addition*) was implicitly used already by Minkowski, and Blaschke succeeded in extending it to smooth convex bodies. An extension to arbitrary convex bodies is obtained by using measures on the Gauss map, i.e.: the Blaschke sum of two convex bodies K_1 and K_2 is that convex body $K_1 \# K_2$ whose surface area measure equals the sum of the surface area measures of K_1 and K_2. Thus, Blaschke sums of convex bodies exist and are unique up to translation. Basic references referring to Blaschke addition, its applications, and generalizations are Section 3.3 and its Notes 3.4 of Gardner's monograph [97], and subsection 8.2.2 of Schneider's book [228] with its corresponding Notes of Section 8.2. For the case of polytopes, we refer to Section 15.3 of Grünbaum's monograph [119]; see also p. 340c there. In the main book, Blaschke addition is introduced in Remark 7.5.6.

There are many interesting properties of Blaschke addition (see also the introduction to Gardner et al. [98] for various further references). For example, each convex polytope is a finite Blaschke sum of simplices, and every origin-symmetric convex polytope is a finite Blaschke sum of parallelotopes; each origin-symmetric convex body is a limit, in the Hausdorff metric, of finite Blaschke sums of ellipsoids. Haberl [122] gives a classification for all continuous symmetric *Blaschke valuations* on convex bodies being compatible with the general linear group. He succeeds in showing that there is a unique non-trivial such valuation, and that, for symmetric bodies, it is the curvature image operator. Minkowski addition and the projection body operator are used in [98] to get interesting characterizations of Blaschke and Minkowski addition. Guo and Jia [121] study the *dual of Blaschke addition*, namely, the so-called radial Blaschke addition as operation between centrally symmetric star bodies.

Affine Surface Area

Let us finally discuss the notion of *affine surface area* of convex bodies, whose origins go back to *affine differential geometry* (see Blaschke's monograph [53]). Leichtweiss [162] started to extend this notion to not necessarily smooth convex bodies, and the same author presented, at a relatively early stage, an interesting historical overview to related developments until 1991, cf. [163]. This notion is clearly invariant with respect to volume-preserving affine transformations, and its importance comes also from essential applications, such as *affine inequalities*, asymptotic *polytopal approximations* of general convex bodies, and *valuations*. An important contribution followed with Lutwak's part II of his *Brunn-Minkowski-Firey theory*; see [179]. Namely, Lutwak extended known inequalities treating (extended) affine surface areas of convex bodies to the concept of p-affine surface areas with $p < 1$, using so-called Firey linear combinations of convex bodies, created with the help of their support functions.

The following parts of monographs are dedicated to the notion of affine surface area and suitable generalizations: Chapter 2 of the monograph [164] of Leichtweiss, showing also nicely the *equivalence of several definitions*, and Sections 10.5 and 10.6 of Schneider's book [228], the latter section treating the related concept of floating bodies. In the monograph [29] the notion of affine surface area is introduced after Exercise 7.22 (see also the exercises after this remark) and studied also in subsection 7.6.2.

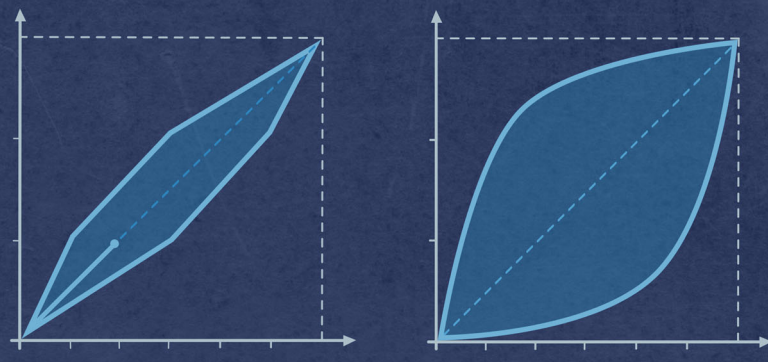
One interesting geometric aspect of affine surface area is its connection to problems and notions leading to certain geometrically *associated bodies*. Such associated bodies or concepts are: the *floating body problem* (see, e.g., [232] and [258]), the *illumination body* (cf. [258]), the concept of *surface bodies* (see [233], also in view of Lutwak's abovementioned concept introduced in [179]), and the so-called separation bodies, as a conceptual dual to floating bodies (see [230]). (The latter notion is not

related to affine surface area, but mentioned here since it presents, besides this duality, a modification of illumination bodies.)

We continue with a few essential publications on affine surface area, showing a bunch of interesting properties. Ludwig [170] surveys results on (equi-)affine invariant and further related *valuations on convex bodies*, discussing in particular also characterizations of affine surface area. Also related *characterizations of projection, difference, moment, and intersection bodies* are presented. The same author introduces in [171] families of general affine surface areas for collections of convex bodies with the origin as interior point. For certain convex bodies, these families are infinite (also for certain polytopes), and they are lower semicontinuous functionals on the considered families of convex bodies. Related *affine isoperimetric inequalities* and further topics are presented in [171]. Generalizing Lutwak's concept proposed in [179] (see above), Wang and He [254] introduce an L_p-dual affine surface area based on L_p dual mixed volumes (the latter also taken from [179]). Also here related affine isoperimetric inequalities are derived. The most recent survey-like publication on affine surface areas seems to be the paper [234] of Schütt and Werner. The authors present a comprehensive overview to *basic properties and the history of the notion of affine surface area*, discussing also interesting *applications* (e.g., to *random polytopes*) and giving various instructive *examples*.

PLATE 8

Zonoids

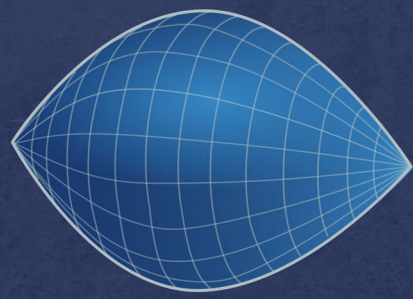


Lorenz zonoids
(statistical distributions)

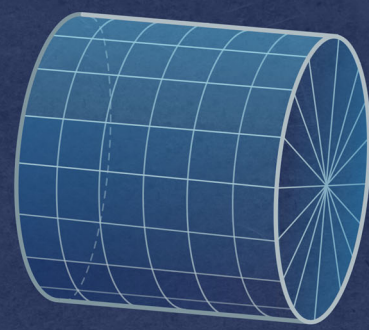
Figures 19 - 20

Figures 21 - 23

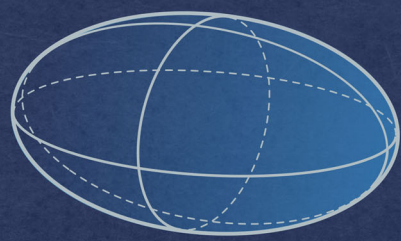
Spindle-shaped zonoids



Cylindrical zonoids



Ellipsoids



Zonoids ► (see, e.g., 2.9.11, 10.2, and 10.6.2)

Chapter 8
The Alexandrov-Fenchel Inequality



"Logic is the technique by which we add conviction to truth."
— Jean de la Bruyere

In Chapter 8 of the main book, *Wulff shapes* and the *Alexandrov-Fenchel inequality* are treated, and the exercises to this chapter refer directly to the *Alexandrov-Fenchel inequality* by using also the notions of *mixed bodies, quermassintegrals*, and *intrinsic volumes*. Special cases and related results are also discussed, yielding or using, for example, the *isoperimetric inequality* as such a special case, or the *Brunn-Minkowski inequality for quermassintegrals*. Thus, several characterization theorems based on homothetical copies or translates of convex bodies are taken into consideration.

Plate 8:

Zonotopes are known as finite sums of line segments, and *zonoids* are convex bodies which can be approximated in the Hausdorff metric by zonotopes. Examples of zonoids are the ellipsoids and certain spindle-shaped and cylindrical bodies as shown in Figures 21 and 22 in Plate 8. Like zonotopes, also zonoids have many applications. For example, full-dimensional zonoids centered at the origin are *projection bodies* of general convex bodies, and zonoids are also known as *ranges of vector measures*. Important surveys on zonoids are [231] by Schneider and Weil as well as the handbook article [109] of Goodey and Weil. Further (and also more recent) literature on zonoids is collected in part 2.9.1.11, Sections 10.2 and 10.3, as well as subsections 10.6.2 and 10.6.3 of the main book [29].

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V. Balestro et al., *Convexity from the Geometric Point of View: Exercises and Solutions*, Cornerstones, https://doi.org/10.1007/978-3-031-83937-5_8

Exercises and Solutions

For the exercise below, recall from Remark 7.2 that $[K_1, \ldots, K_{n-1}]$ denotes the mixed body of an $(n-1)$-tuple $K_1, \ldots, K_{n-1} \in \mathcal{K}^n$.

Exercise 8.1. *Let $K_1, \ldots, K_{n-1} \in \mathcal{K}^n$. Prove that*

$$\mathrm{vol}_n([K_1, \ldots, K_{n-1}])^{n-1} \geq \mathrm{vol}_n(K_1) \ldots \mathrm{vol}_n(K_{n-1}).$$

Hint: Use Corollary 8.3.1 (main book).

Solution 8.1. From item **(b)** of Exercise 7.4 for $L_1 = [K_1, \ldots, K_{n-1}]$ and Corollary 8.3.1 (main book), we write

$$\begin{aligned} \mathrm{vol}_n([K_1, \ldots, K_{n-1}])^n &= V([K_1, \ldots, K_{n-1}][n-1], [K_1, \ldots, K_{n-1}])^n = \\ &= V(K_1, \ldots, K_{n-1}, [K_1, \ldots, K_{n-1}])^n \geq \mathrm{vol}_n(K_1) \ldots \mathrm{vol}_n(K_{n-1}) \mathrm{vol}_n([K_1, \ldots, K_{n-1}]), \end{aligned}$$

and this gives the desired inequality.

□

Exercise 8.2. *Let $0 \leq i < n-1$ and $K \in CC^n$. Prove that*

$$W_{i+1}(K)^{n-i} \geq \kappa_n W_i(K)^{n-i-1}. \tag{8.1}$$

Hint: Use inequality (8.3.1) from the main book.

Solution 8.2. We use inequality (8.3.1) for $m = n-i$, $K_1 = \ldots = K_{n-i-1} = K$, $K_{n-i} = B^n$ and $K_{n-i+1} = \ldots = K_n = B^n$. This leads to

$$\begin{aligned} W_{i+1}(K)^{n-i} &= V(K[n-i-1], B^n[i+1])^{n-i} = V(K_1, \ldots, K_n)^{n-i} \geq \\ \geq \prod_{j=1}^{n-i} V(K_j[n-i], K_{n-i+1}, \ldots, K_n) &= \left[\prod_{j=1}^{n-i-1} V(K[n-i], B^n[i]) \right] V(K_{n-i}[n-i], B^n[i]) = \\ &= V(K[n-i], B^n[i])^{n-i-1} V(B^n[n-i], B^n[i]) = \kappa_n W_i(K)^{n-i-1}, \end{aligned}$$

and this settles the inequality.

□

Inequality (8.1) relates consecutive quermassintegrals of a set $K \in CC^n$. We can actually do better, and relate any pair of quermassintegrals of K. This is made in the next exercise.

Exercise 8.3. *Let $0 \leq i < j < n$. Prove that*

$$W_j(K)^{n-i} \geq \kappa_n^{j-i} W_i(K)^{n-j} \tag{8.2}$$

for every $K \in CC^n$.

Solution 8.3. We consider the case of inequality (8.3.1) where $m = n-i$, $K_1, \ldots, K_{n-j} = K$, $K_{n-j+1}, \ldots, K_{n-i} = B^n$, and $K_{n-i+1}, \ldots, K_n = B^n$. That is, the $m = n-i$ bodies which we "re-

peat" in the right-hand side of the mentioned inequality consist of $n-j$ copies of K and $j-i$ copies of B^n. The remaining i bodies are all B^n. This leads to

$$W_j(K)^{n-i} = V(K[n-j], B^n[j])^{n-i} \geq V(K[n-i], B^n[i])^{n-j} V(B^n[n-i], B^n[i])^{j-i} = \\ = \kappa_n^{j-i} W_i(K)^{n-j}.$$

□

Of course, (8.1) is just (8.2) for $j = i+1$. In the next exercise we discuss the equality case in the more general inequality and, of course, the case where $j = i+1$ provides a version for the more restricted one.

Exercise 8.4. *Fix $0 \leq i < j < n$, and assume that $K \in CC^n$ is such that $\dim(K) \geq n-i$. Prove that in (8.2) equality holds if and only if the function $f : [0,1] \to \mathbb{R}$ given as*

$$f(t) = V((1-t)K + tB^n[n-i], B^n[i])^{1/(n-i)}$$

is linear.

Hint: Use Theorem 8.3.2 (main book).

Remark 8.1. For fixed $0 \leq i < j < n$, if $\dim(K) \geq n-i$, then the equality case in inequality (8.2) holds if and only if K is a ball. The reader is invited to prove it in Exercise 8.13.

Solution 8.4. We write

$$V_{(k)} := V(K[n-i-k], B^n[k], B^n[i])$$

for $k = 0, \ldots, n-i$ (recall that i is fixed). In this notation, the equality case in the inequality (8.2) reads

$$V_{(j-i)}^{n-i} = V_{(0)}^{n-j} V_{(n-i)}^{j-i}.$$

This is condition **(iii)** of Theorem 8.3.2 of the main book for $m = n-i$ and $k = j-i$, and by this theorem this condition is equivalent to f being linear. Notice that the fact that we are assuming that $\dim(K) \geq n-i$ guarantees that $V_{(k)} > 0$ for every $k = 0, \ldots, n-i$.

□

Exercise 8.5. *Verify that the case $i = 0$ of inequality (8.1) is the isoperimetric inequality (Theorem 5.2.1 in the main book).*

Solution 8.5. Recall that $nW_1(K) = S(K)$ and $W_0(K) = \mathrm{vol}_n(K)$. Thus, the case $i = 0$ reads

$$\frac{S(K)^n}{n^n} \geq \kappa_n \mathrm{vol}_n(K)^{n-1}.$$

We rewrite it as

$$\left(\frac{S(K)}{n\kappa_n}\right)^n \geq \left(\frac{\mathrm{vol}_n(K)}{\kappa_n}\right)^{n-1},$$

and since $n\kappa_n = S(B^n)$ this is the isoperimetric inequality.

□

Exercise 8.6 (Brunn-Minkowski Inequality for Quermassintegrals)**.** *Let $1 \leq m \leq n$. Verify that for any $K, L, K_{m+1}, \ldots, K_n \in CC^n$ we have that*

$$V(K + L[m], K_{m+1}, \ldots, K_n)^{\frac{1}{m}} \geq$$
$$\geq V(K[m], K_{m+1}, \ldots, K_n)^{\frac{1}{m}} + V(L[m], K_{m+1}, \ldots, K_n)^{\frac{1}{m}}.$$

Conclude that for all $0 \leq i \leq n-1$ and every $K, L \in CC^n$ we have

$$W_i(K + L)^{\frac{1}{n-i}} \geq W_i(K)^{\frac{1}{n-i}} + W_i(L)^{\frac{1}{n-i}}.$$

Remark 8.2. The equality cases in these inequalities are discussed later in Exercise 8.11.

Solution 8.6. This follows immediately from letting $t = 1/2$ in (8.3.7) from the main book. The inequality for the quermassintegrals is obtained simply by letting $m = n-i$ and $K_{m+1} = \ldots = K_n = B^n$.

□

Exercise 8.7. *Let $K \in \mathcal{K}_0^3$. Prove that*

$$V(K, K^\circ, B^3) \geq \kappa_3.$$

Verify that this inequality is sharp.

Hint: Use the same strategy as in Exercise 6.22, but taking the Alexandrov-Fenchel inequality instead of the 1st Minkowski inequality.

Solution 8.7. In the same notation as in the solution of Exercise 6.22, the inclusion $B^n \subseteq M_\lambda$ yields that

$$W_1(M_\lambda) = V(M_\lambda[2], B^3) \geq V(B^3, B^3, B^3) = \kappa_3$$

for any $\lambda > 0$. On the other hand, we have

$$W_1(M_\lambda) = \frac{1}{4}\left[\lambda^2 V(K, K, B^3) + 2V(K, K^\circ, B^3) + \frac{1}{\lambda^2}V(K^\circ, K^\circ, B^3)\right] =$$
$$= \frac{1}{2}V(K, K^\circ, B^3) + \frac{1}{4}\left[\lambda^2 V(K, K, B^3) + \frac{1}{\lambda^2}V(K^\circ, K^\circ, B^3)\right].$$

Similarly to what we have done in Exercise 6.22, we minimize $W_1(M_\lambda)$ by minimizing the function

$$t \mapsto tV(K, K, B^3) + \frac{1}{t}V(K^\circ, K^\circ, B^3).$$

By standard calculus, this function is minimized for

$$t = \left(\frac{V(K, K, B^3)}{V(K^\circ, K^\circ, B^3)}\right)^{\frac{1}{2}}.$$

We arrive at

$$\min\left\{W_1(M_\lambda) : \lambda > 0\right\} = \frac{1}{2}\left[V(K, K^\circ, B^3) + V(K, K, B^3)^{\frac{1}{2}} V(K^\circ, K^\circ, B^3)^{\frac{1}{2}}\right].$$

From the Alexandrov-Fenchel inequality we get that

$$V(K, K, B^3)^{\frac{1}{2}} V(K^\circ, K^\circ, B^3)^{\frac{1}{2}} \leq V(K, K^\circ, B^3),$$

and since κ_3 is a lower bound for $W_1(M_\lambda)$, we finally arrive at

$$\kappa_3 \leq \min\left\{W_1(M_\lambda), \lambda > 0\right\} =$$
$$= \frac{1}{2}\left[V(K, K^\circ, B^3) + V(K, K, B^3)^{\frac{1}{2}} V(K^\circ, K^\circ, B^3)^{\frac{1}{2}}\right] \leq V(K, K^\circ, B^3),$$

and the proof is complete. The inequality is sharp because for $K = B^3$ we obviously have that $V(K, K^\circ, B^3) = \kappa_3$.

□

The exercise below gives an alternative proof of Theorem 8.3.1 (see also Remark 8.3.2) in the main book.

Exercise 8.8. *Let $m \in \{2, \ldots, n\}$, and let $K_1, K_2, K_{m+1}, \ldots, K_n \in \mathcal{K}^n$. Define $g_m : [0, +\infty) \to \mathbb{R}$ by*

$$g_m(t) = V(K_1 + tK_2[m], K_{m+1}, \ldots, K_n)^{\frac{1}{m}}, \quad t \geq 0.$$

Denote $\mathbf{K}(t) := (K_1 + tK_2[m-2], K_{m+1}, \ldots, K_n)$. Prove that

$$g_m''(t) = -(m-1)g_m(t)^{1-2m}\left[V(K_1, K_2, \mathbf{K}(t))^2 - V(K_1, K_1, \mathbf{K}(t))V(K_2, K_2, \mathbf{K}(t))\right]$$

for any $t \geq 0$. Conclude that the Alexandrov-Fenchel inequality implies the concavity of g_m, and show how this implies Theorem 8.3.1 from the main book.

Solution 8.8. For simplicity, denote

$$V_m(t) := V(K_1 + tK_2[m], K_{m+1}, \ldots, K_n),$$

for $t \geq 0$. Thus, $V_m(t) = g_m(t)^m$. A direct computation gives

$$g_m''(t) = \frac{1-m}{m^2} V_m(t)^{\frac{1-2m}{m}} V_m'(t)^2 + \frac{1}{m} V_m(t)^{\frac{1-m}{m}} V_m''(t) =$$
$$= \frac{1}{m} g_m(t)^{1-2m}\left(\frac{1-m}{m} V_m'(t)^2 + V_m(t)V_m''(t)\right).$$

Also, the linearity and the symmetry of mixed volumes give

$$V_m'(t) = mV(K_1 + tK_2[m-1], K_2, K_{m+1}, \ldots, K_n) = mV(K_1 + tK_2, K_2, \mathbf{K}(t))$$

and

$$V_m''(t) = m(m-1)V(K_1 + tK_2[m-2], K_2[2], K_{m+1}, \dots, K_n) = m(m-1)V(K_2[2], \mathbf{K}(t)).$$

Therefore, observing that we may write $V_m(t) = V(K_1 + tK_2[2], \mathbf{K}(t))$, it follows that

$$\begin{aligned}
\frac{1-m}{m} V_m'(t)^2 + V_m(t)V_m''(t) = \\
= (1-m)mV(K_1 + tK_2, K_2, \mathbf{K}(t))^2 + m(m-1)V_m(t)V(K_2[2], \mathbf{K}(t)) = \\
= (1-m)m\Big[\big(V(K_1, K_2, \mathbf{K}(t)) + tV(K_2[2], \mathbf{K}(t))\big)^2 - V(K_1 + tK_2[2], \mathbf{K}(t))V(K_2[2], \mathbf{K}(t))\Big] = \\
= (1-m)m\Big[V(K_1, K_2, \mathbf{K}(t))^2 - V(K_1[2], \mathbf{K}(t))V(K_2[2], \mathbf{K}(t))\Big].
\end{aligned}$$

Substituting this equality in the expression for $g_m''(t)$ derived above, we have what we wanted.

We explain how this implies that g_m is concave. Notice that the Alexandrov-Fenchel inequality applied for the bodies $K_1, K_2, \mathbf{K}(t)$ yields immediately that

$$V(K_1, K_2, \mathbf{K}(t))^2 - V(K_1[2], \mathbf{K}(t))V(K_2[2], \mathbf{K}(t)) \geq 0.$$

This inequality and the fact that

$$-(m-1)g_m(t)^{1-2m} < 0 \ \ \forall\, t \geq 0$$

lead to $g_m''(t) \leq 0$ for every $t \geq 0$. Hence g_m is concave. From Lemma 8.2.4 (main book), this gives that the function

$$f(t) := V((1-t)K_1 + tK_2[m], K_{m+1}, \dots, K_n)^{\frac{1}{m}}$$

is concave in $[0, 1]$, and this is Theorem 8.3.1 in the main book.

□

The next exercise can be seen as an extension of the Minkowski inequalities (see Section 6.3 of the main book).

Exercise 8.9. *Let $m \in \{2, \dots, n\}$, and let $K_1, K_2, K_{m+1}, \dots, K_n \in \mathcal{K}^n$. Prove that*

$$\begin{aligned}
V(K_1[m-1], K_2, K_{m+1}, \dots, K_n)^m \geq \\
\geq V(K_1[m], K_{m+1}, \dots, K_n)^{m-1} V(K_2[m], K_{m+1}, \dots, K_n).
\end{aligned}$$

Solution 8.9. For simplicity, write $\mathbf{K} = (K_{m+1}, \dots, K_n)$. From Theorem 8.3.1 (main book) we have that the function $f\colon [0, 1] \to \mathbb{R}$ given by

$$f(t) = V((1-t)K_1 + tK_2[m], \mathbf{K})^{\frac{1}{m}}$$

is concave. Hence it holds that

$$f_+'(0) \geq f(1) - f(0) = V(K_2[m], \mathbf{K})^{\frac{1}{m}} - V(K_1[m], \mathbf{K})^{\frac{1}{m}}. \tag{8.3}$$

Using the linearity of mixed volumes, we compute $f'_+(0)$:

$$\begin{aligned} f'_+(0) &= \frac{1}{m} V(K_1[m], \mathbf{K})^{\frac{1}{m}-1} \frac{d}{dt} V((1-t)K_1 + tK_2[m], \mathbf{K})\Big|_{t=0} = \\ &= \frac{1}{m} V(K_1[m], \mathbf{K})^{\frac{1}{m}-1} \frac{d}{dt} \left(\sum_{j=0}^{m} \binom{m}{j} (1-t)^j t^{m-j} V(K_1[j], K_2[m-j], \mathbf{K}) \right)\Big|_{t=0} = \\ &= \frac{1}{m} V(K_1[m], \mathbf{K})^{\frac{1}{m}-1} \left[-\binom{m}{m} m V(K_1[m], \mathbf{K}) + \binom{m}{m-1} V(K_1[m-1], K_2, \mathbf{K}) \right] = \\ &= V(K_1[m], \mathbf{K})^{\frac{1}{m}-1} \Big[V(K_1[m-1], K_2, \mathbf{K}) - V(K_1[m], \mathbf{K}) \Big] = \\ &= V(K_1[m], \mathbf{K})^{\frac{1}{m}-1} V(K_1[m-1], K_2, \mathbf{K}) - V(K_1[m], \mathbf{K})^{\frac{1}{m}}, \end{aligned}$$

where the second equality follows immediately by differentiating the polynomial and letting $t = 0$ in the obtained expression. Replacing this equality in (8.3), we get

$$V(K_1[m], \mathbf{K})^{\frac{1}{m}-1} V(K_1[m-1], K_2, \mathbf{K}) - V(K_1[m], \mathbf{K})^{\frac{1}{m}} \geq V(K_2[m], \mathbf{K})^{\frac{1}{m}} - V(K_1[m], \mathbf{K})^{\frac{1}{m}}.$$

Cancelling the term $V(K_1[m], \mathbf{K})^{\frac{1}{m}}$ on both sides, dividing everything by $V(K_1[m], \mathbf{K})^{\frac{1}{m}-1}$, and elevating both sides to the m-th power, we obtain the desired inequality.

□

Exercise 8.10. *Give an example to show that Proposition 8.4.4 from the main book does not hold in the case where the dimension is* 2.

Solution 8.10. Let T be an equilateral triangle and, for each of its edges, draw an arc through its endpoints and with center at the respective opposite vertex. The convex body bounded by the three constructed arcs is a *Reuleaux triangle* (R, say), which is a figure of constant width (d, say). All orthogonal projections of R are segments whose length is d. Now compare with a ball B having diameter d. All hypotheses of Proposition 8.4.4 (main book) are satisfied, but R and B are not translates.

□

Remember that in Exercise 8.6 the inequality

$$\begin{aligned} V(K + L[m], K_{m+1}, \ldots, K_n)^{\frac{1}{m}} \geq \\ \geq V(K[m], K_{m+1}, \ldots, K_n)^{\frac{1}{m}} + V(L[m], K_{m+1}, \ldots, K_n)^{\frac{1}{m}} \end{aligned} \tag{8.4}$$

has been stated for every $1 \leq m \leq n$ and any $K, L, K_{m+1}, \ldots, K_n \in CC^n$. In the case $m = 1$ the linearity of mixed volumes gives equality. Below, we discuss the case where $m > 1$.

Exercise 8.11. **(a)** *Let $m \in \{2, \ldots, n\}$. Show that if $K_{m+1}, \ldots, K_n$ are smooth convex bodies, and if K, L are convex bodies, then we have equality in (8.4) if and only if K and L are homothetical.*

(b) *Show that for any convex bodies $K, L \subseteq \mathcal{K}^n$ and every $i \in \{0, \ldots, n-2\}$ the equality*

$$W_i(K + L)^{\frac{1}{n-i}} = W_i(K)^{\frac{1}{n-i}} + W_i(L)^{\frac{1}{n-i}}$$

holds if and only if K and L are homothetical.

Solution 8.11. **(a)** It is clear that the equality holds if K and L are homothetical; hence we only prove the converse. If equality holds, then equality holds in (8.3.7) (main book) for $t = 1/2$. From Remark 8.3.3 there, we get that the function

$$f(t) = V\big((1-t)K + tL[m], K_{m+1}, \ldots, K_n\big)^{\frac{1}{m}}$$

is linear. And from Theorem 8.4.4 there it follows that K and L are homothetical.

(b) This is the case of item **(a)** with $m = n - i$ and $K_{m+1} = \ldots = K_n = B^n$. Since B^n is smooth, the desired follows.

□

Exercise 8.12. *Let $K \subseteq \mathbb{R}^n$ be a convex body. Prove that equality holds in (7.1) for some $0 \leq i \leq n-2$ and $u \in \mathbb{S}^{n-1}$ if and only if $\mathrm{ref}_{u^\perp}(K)$ is a translate of K.*

Solution 8.12. Equality in (7.1) is equivalent to

$$W_i(\mathrm{ref}_{u^\perp}(K) + K)^{\frac{1}{n-i}} = W_i(\mathrm{ref}_{u^\perp}(K))^{\frac{1}{n-i}} + W_i(K)^{\frac{1}{n-i}},$$

and due to Exercise 8.11 this holds for some $i \in \{0, \ldots, n-2\}$ if and only if $\mathrm{ref}_{u^\perp}$ and K are homothetical. Since $\mathrm{ref}_{u^\perp}$ is volume-preserving (for a formal proof of this intuitive fact, see Exercise 5.4), these bodies are homothetical if and only if they are translates.

□

Exercise 8.13. *Fix $0 \leq i < j < n$, and let $K \in CC^n$ be such that $\dim(K) \geq n - i$. Prove that equality in (8.2) holds for K if and only if K is a ball.*

Solution 8.13. From Exercise 8.2, we know that in the case where $\dim(K) \geq n - i$ equality (8.1) holds if and only if the function

$$f(t) = V((1-t)K + tB^n[n-i], B^n[i])^{1/(n-i)}$$

is linear. Because B^n is smooth, from Theorem 8.4.4 in the main book, we get that this function is smooth if and only if K and B^n are homothetical.

□

Exercise 8.14. *Where in the proof of Theorem 8.4.3 do we use the smoothness hypothesis?*

Solution 8.14. We use it so that we can apply Corollary 8.4.1 (main book).

□

For Further Reading

The *Alexandrov-Fenchel inequality* itself goes back to the contributions [3] and [92] of A. D. Alexandrov and Fenchel, respectively. Extensions, modifications, and applications of it are comprehensively discussed in Sections 7.3, 7.4, and 7.6 of Schneider's book [228], and in the appendices B3, B4, and B5 of Gardner's monograph [97]. Further on, the two known proofs of Alexandrov himself can be found in several further convexity books, such as those of Busemann [73], Leichtweiss [161], Hörmander

[140], as well as Martini, Montejano, and Oliveros [188]. Burago and Zalgaller present in [72] an algebraic approach. In our main book, the whole Chapter 8 is dedicated to the Alexandrov-Fenchel inequality; see particularly also the notes in subsection 8.6.2 (in the mail book). And further on, also subsections like 5.7.1 and 6.7.2 (main book) are related.

Survey-like papers on the Alexandrov-Fenchel inequality and extensions are Schneider [225] (with a conjecture to the general case of equality), the handbook articles [220] of Sangwine-Yager (discussing also the equality case) and [115] of Groemer (studying related stability questions), and Schneider's survey [227] (updating the knowledge about the *equality case*). In addition, the exposition [152] of Klartag should be mentioned here; the author discusses *functional versions of Alexandrov-Fenchel inequalities*. Ge et al. [99] use, to prove a Penrose-type inequality, a version of the Alexandrov-Fenchel inequality established for the *hyperbolic space*; this paper gives also a survey on related questions.

PLATE 9

Ellipsoids

General ellipsoid

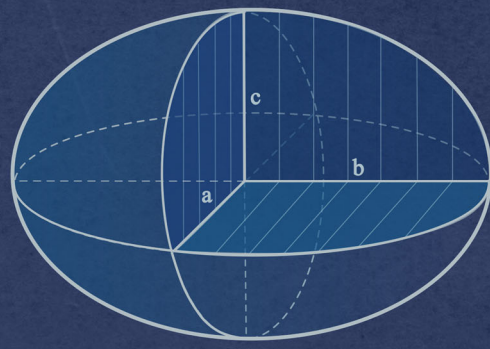
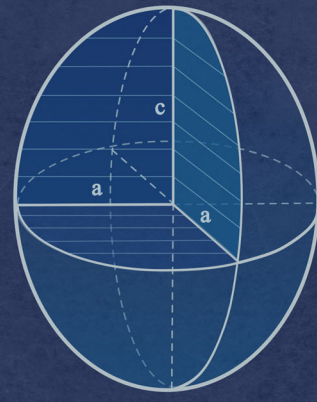


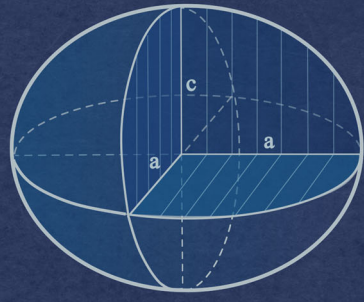
Figure 24

Figures 25 - 26

Prolate spheroid (a < c)



Oblate spheroid (a > c)



Ellipsoids ► (see, e.g., 2.9.1.6 and Chapter 9)

Chapter 9
Affine Convex Geometry—Part 1



"To the wise, life is a problem; to the fool, a solution."

— Marcus Aurelius

The exercises of Chapter 9 from the main book are clearly dominated by the notion of *positions*, with subcases like positions referring to *Löwner-John ellipsoids*. With this background, consequences of the *geometric Brascamp-Lieb inequality* and also properties of *Orlicz projection bodies* are studied in this chapter. Concrete further notions, discussed in these exercises, are *affine diameters* of convex bodies, *Minkowski asymmetry, John position, positions minimizing quermassintegrals*, the *isotropic position*, and the *reverse isoperimetric inequality*.

Plate 9:

Plate 9 shows kinds of *ellipsoids* which are important *special types of convex bodies*, for example, yielding the extremal cases for many deep inequalities and, considered as unit balls of normed spaces, essential characterization theorems. Surveys collecting results on ellipsoids (from the viewpoint of convexity) are, e.g., Section 2.12 of [188] and its notes, as well as part 2.9.1.6, Sections 5.4 and 9.1, and subsection 9.6.1 of the main book [29].

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Exercises and Solutions

Let $K \subseteq \mathbb{R}^n$ be a convex body. A *chord* of K is a segment joining two boundary points of K. A *metric diameter* of K is a chord whose length equals $\operatorname{diam}(K)$ (by a compactness argument, the diameter of K is indeed attained by a pair of points, and it is obvious that these points must be boundary points).

A chord $\operatorname{seg}[x, y] \subseteq K$ is an *affine diameter* of K if K is supported by parallel hyperplanes at x and y. In the exercise below, we explore some properties of affine diameters. These were mostly taken from Soltan's survey [239].

Exercise 9.1. *Let $K \subseteq \mathbb{R}^n$ be any convex body.*

(a) *Assume that* $\operatorname{seg}[x, y]$ *is a metric diameter of K. Let $\mathbf{h}_1$ be the hyperplane through x and orthogonal to $v := y - x$, and let $\mathbf{h}_2$ be the hyperplane through y and orthogonal to v. Prove that $\mathbf{h}_1$ and $\mathbf{h}_2$ are supporting hyperplanes of K.*

(b) *Prove that a chord* $\operatorname{seg}[x, y] \subseteq K$ *is an affine diameter of K if and only if it is the longest chord in a given direction of $\mathbb{R}^n$.*

(c) *Use item* **(b)** *to show that K has an affine diameter in each direction of $\mathbb{R}^n$.*

(d) *Prove that affine diameters are affine invariants in the following sense: if* $\operatorname{seg}[x, y]$ *is an affine diameter of K and $A(\,) = x_0 + T(\cdot)\colon \mathbb{R}^n \to \mathbb{R}^n$ is an affine transformation of $\mathbb{R}^n$, then* $\operatorname{seg}[Ax, Ay]$ *is an affine diameter of AK.*

Solution 9.1. **(a)** Assume (without loss of generality) that $y \in \mathbf{h}_1^-$. If there is a point $z \in K \cap \operatorname{int}(\mathbf{h}_1^+)$, then it is immediate to check that the length of the segment $\operatorname{seg}[y, z]$ is greater than the length of $\operatorname{seg}[x, y]$. This contradicts the fact that $\operatorname{seg}[x, y]$ is a metric diameter. It follows that $K \subseteq \mathbf{h}_1^-$, and hence $\mathbf{h}_1$ is a supporting hyperplane of K (observe that $x \in K \cap \mathbf{h}_1$). The argument for $\mathbf{h}_2$ is obviously the same.

(b) Let $\operatorname{seg}[x, y]$ be an affine diameter, and let $v := y - x$. We have to prove that $\operatorname{seg}[x, y]$ is the longest chord of K in the direction of v. We know that K is supported at x and y by parallel hyperplanes $\mathbf{h}_1$ and $\mathbf{h}_2$, respectively, with $K \subseteq \mathbf{h}_1^- \cap \mathbf{h}_2^-$, say. Since $x \in \mathbf{h}_1$ and $y \in \mathbf{h}_2$, it is immediate to see that the "slab" $\mathbf{h}_1^- \cap \mathbf{h}_2^-$ does not contain a chord in the direction of v whose length is greater than the length of $\operatorname{seg}[x, y]$. Hence K does not contain a chord in the direction of v which is longer than $\operatorname{seg}[x, y]$.

Now we prove the converse. Assume that v is a direction of $\mathbb{R}^n$ and that $\operatorname{seg}[x, y]$ is the longest chord of K in the direction of v. Consider the convex body $K' = x - y + K$. Then, it is clear that $x \in K'$ (because $x = x - y + y$), and we claim that $\operatorname{int}(K') \cap \operatorname{int}(K) = \emptyset$. To prove that, assume by contradiction that there is a point $z \in \operatorname{int}(K') \cap \operatorname{int}(K)$. Then, since $z \in \operatorname{int}(K')$ and $K = K' + (y - x)$, we get that $w := z + (y - x)$ is an interior point of K, and $w - z = x - y$. However, z and w are interior points of K, and hence the chord $\operatorname{seg}[z, w]$ can be extended to a chord in the direction of v with length greater than $||x - y||$. This contradiction gives that we in fact have $\operatorname{int}(K') \cap \operatorname{int}(K) = \emptyset$. From Theorem 2.2.3 (main book) we get that K and K' can be (properly) separated by a hyperplane $\mathbf{h}$. Since $x \in K \cap K'$, we must have that $x \in \mathbf{h}$. Thus, $\mathbf{h}$ supports both K and K' at x. The hyperplane $y - x + \mathbf{h}$ (which is parallel to $\mathbf{h}$) supports $y - x + K'$ at y. Since $y - x + K' = K$, we get what we wanted.

(c) Let v be any direction of $\mathbb{R}^n$, and define

$$C_v := \{\text{chords of } K \text{ in the direction of } v\}.$$

Since the interior of K is non-empty, it is obvious that $C_v \neq \emptyset$. Let

$$\ell_v := \sup\{||x - y|| : \text{seg}[x, y] \in C_v\}.$$

That is, ℓ_v is the supremum of the lengths of the chords of K in the direction of v. It is clear that $\ell_v \leq \text{diam}(K) < \infty$, and hence there exists a sequence of chords $\text{seg}[x_m, y_m]$ of K in the direction of v such that $||x_m - y_m|| \to \ell_v$ as $m \to \infty$. Because ∂K is compact, passing to subsequences if necessary we may assume that $x_m \to x_0$ and $y_m \to y_0$ for some $x_0, y_0 \in \partial K$. It is clear that $||x_0 - y_0|| = \ell_v$ and that $x_0 - y_0$ is in the direction of v. It follows that $\text{seg}[x_0, y_0]$ has the greatest length among all chords of K in the direction of v, and from item **(b)** we get that this chord is an affine diameter.

□

For given convex bodies $K, L \subseteq \mathbb{R}^n$, we define the *inner radius of K with respect to L* as the number

$$r_{\text{in}}(K, L) = \max\{\lambda \geq 0 : \lambda L + x \subseteq K \text{ for some } x \in \mathbb{R}^n\}.$$

Thus, r_{in} gives the largest homothetical copy of L which can be translated to lie within K. The number $r_{\text{in}}(K, -K)$ is called the *Minkowski asymmetry measure* of K. Recently, Brandenberg, von Dichter, and González Merino investigated this concept, finding interesting results. See, e.g., [68], [66], [67], and [252].

Exercise 9.2. **(a)** *Prove that $r_{\text{in}}(K, L)$ is an affine invariant in the following sense: if $A(\cdot) := x_0 + T(\cdot)\colon \mathbb{R}^n \to \mathbb{R}^n$ is any affine transformation of $\mathbb{R}^n$, then*

$$r_{\text{in}}(AK, AL) = r_{\text{in}}(K, L).$$

(b) *Show that $r_{\text{in}}(K, -K) \leq 1$ for any $K \in \mathcal{K}^n$.*

(c) *Show that $r_{\text{in}}(K, -K) = 1$ if and only if K is symmetric with respect to an interior point (hence the Minkowski asymmetry measure is indeed an asymmetry measure).*

(d) *Compute the Minkowski asymmetry of a triangle in $\mathbb{R}^2$.*

(e) *Extend item* **(d)** *for a (full-dimensional) simplex in arbitrary dimension $n \geq 2$.*

Solution 9.2. **(a)** If $\lambda L + x \subseteq K$, then

$$AK \supseteq A(\lambda L + x) = x_0 + T(\lambda L + x) = x_0 + Tx + \lambda TL = \lambda(x_0 + TL) + x_0 + Tx - \lambda x_0 = \\ = \lambda(AL) + (1 - \lambda)x_0 + Tx,$$

and this gives that

$$r_{\text{in}}(AK, AL) \geq r_{\text{in}}(K, L).$$

Using this same argument for A^{-1} and for the sets AK and AL, we get

$$r_{\text{in}}(K, L) = r_{\text{in}}(A^{-1}AK, A^{-1}AL) \geq r_{\text{in}}(AK, AL),$$

and this concludes the proof.

(b) This follows from volume comparison. If $\lambda(-K) + x \subseteq K$, then

$$\mathrm{vol}_n(K) \geq \mathrm{vol}_n(\lambda(-K) + x) = \lambda^n \mathrm{vol}_n(K).$$

Thus, $\lambda^n \leq 1$. Since $\lambda \geq 0$, we get that $\lambda \leq 1$.

(c) If K is symmetric with respect to a point $x \in \mathrm{int}K$, then $K - x = -(K - x)$. This gives that $-K + 2x = K$, and hence $r_{\mathrm{in}}(K, -K) \geq 1$. Since we already have the reverse inequality, the equality follows.

Conversely, assume that $r_{\mathrm{in}}(K, -K) = 1$. Then there exists a point $x \in \mathbb{R}^n$ such that $-K + x \subseteq K$. Thus,

$$-\left(K - \frac{x}{2}\right) \subseteq K - \frac{x}{2}.$$

Multiplying both sides by -1, we get the reverse inclusion. It follows that $-(K - x/2) = K - x/2$, and hence K is symmetric with respect to $x/2$.

(d) First notice that from item **(a)** any triangle in $\mathbb{R}^2$ has the same Minkowski asymmetry measure. Let K be a triangle, and let m be its barycenter (i.e., the arithmetic mean of its vertices). It is easy to see that

$$L := \frac{1}{2}(-K) + \frac{3}{2}m \subseteq K.$$

In fact, L is simply the triangle obtained by joining the midpoints of the sides of K. It follows that $r_{\mathrm{in}}(K, -K) \geq 1/2$. Since the vertices of L lie in the sides of K, it is immediate to check that any translate or dilated copy (by a factor greater than 1) of L is not contained in K. Therefore, we get that

$$r_{\mathrm{in}}(K, -K) = \frac{1}{2}.$$

(e) Let S be a simplex in $\mathbb{R}^n$. The convex hull of the barycenters of the facets of S is a translate of $\frac{1}{2}(-S)$. The same argument as in item **(d)** yields that no dilated copy (with factor greater than 1) may be contained in S.

□

In what follows, remember that the *Banach-Mazur distance* between convex bodies K and L is defined as

$$d_{BM}(K, L) = \inf\{c \geq 1 : \exists\, T \in \mathrm{GL}(n) \text{ such that } ||x||_K \leq ||Tx||_L \leq c||x||_K \ \forall\, x \in \mathbb{R}^n\}.$$

Exercise 9.3. *Prove that the Banach-Mazur distance is symmetric in its entries.*

Solution 9.3. Consider the equivalent definition of the Banach-Mazur distance given in Proposition 9.2.3 of the main book. If $L \subseteq T(K) \subseteq cL$ for some $c \geq 1$ and some $T \in \mathrm{GL}(n)$, then letting $S = cT^{-1}$ we have that $S \in \mathrm{GL}(n)$, and $K \subseteq S(L) \subseteq cK$. Repeating the argument in the other direction, we get that, for a given number $c \geq 1$, there exists a transformation $T \in \mathrm{GL}(n)$ such that $L \subseteq T(K) \subseteq cL$ if and only if there exists a transformation $S \in \mathrm{GL}(n)$ such that $K \subseteq S(L) \subseteq cK$. This shows that $d_{BM}(K, L) = d_{BM}(L, K)$.

□

Exercise 9.4. *Let $K, L \subseteq \mathbb{R}^n$ be convex bodies.*

(a) *Verify that if K and L are in minimal mean width position, then $K + L$ is also in minimal mean width position.*

(b) *Show that if $K + L$ and K are in minimal mean width position, then L is also in minimal mean width position.*

Solution 9.4. (a) Let $T \in \mathrm{L}(n)$ be arbitrarily given. We use the equalities $h_{K+L} = h_K + h_L$, $w(K+L) = w(K) + w(L)$ and Theorem 9.3.4 (main book) to write

$$\int_{\mathbb{S}^{n-1}} \langle \nabla h_{K+L}(u), Tu \rangle \, d\sigma(u) = \int_{\mathbb{S}^{n-1}} \langle \nabla h_K(u), Tu \rangle \, d\sigma(u) + \int_{\mathbb{S}^{n-1}} \langle \nabla h_L(u), Tu \rangle \, d\sigma(u) =$$
$$= \frac{\mathrm{tr}(T)}{2n} w(K) + \frac{\mathrm{tr}(T)}{2n} w(L) = \frac{\mathrm{tr}(T)}{2n} w(K+L).$$

From the converse direction of Theorem 9.3.4, we get that $K + L$ is in minimal mean width position.

(b) Let $T \in \mathrm{L}(n)$ be arbitrarily given. Since $K + L$ is in minimal mean width position, we may write

$$\int_{\mathbb{S}^{n-1}} \langle \nabla h_{K+L}(u), Tu \rangle \, d\sigma(u) = \frac{\mathrm{tr}(T)}{2n} w(K+L).$$

Since $\nabla h_{K+L} = \nabla h_K + \nabla h_L$ and $w(K+L) = w(K) + w(L)$, and using the fact that K is in minimal mean width position, we arrive at

$$\int_{\mathbb{S}^{n-1}} \langle \nabla h_L(u), T(u) \rangle \, d\sigma(u) = \frac{\mathrm{tr}(T)}{2n} w(L).$$

This gives that L is in minimal mean width position.

□

The next exercise was taken from the paper [102] of Giannopoulos and Papadimitrakis.

Exercise 9.5. *Let $K \subseteq \mathbb{R}^n$ be a convex body in minimal surface area position. Prove that*

$$\frac{S(K)}{2n} \leq \mathrm{vol}_{n-1}\big(\mathrm{proj}_{u^\perp}(K)\big) \leq \frac{S(K)}{2\sqrt{n}}$$

for every $u \in \mathbb{S}^{n-1}$.

Hint: Use the integral representation for the volume of a projection given in the beginning of Section 10.3 in the main book. For the upper bound, use Hölder's inequality, and for the lower bound simply observe that $\langle u, v \rangle^2 \leq |\langle u, v \rangle|$ for any $u, v \in \mathbb{S}^{n-1}$.

Solution 9.5. Let $u \in \mathbb{S}^{n-1}$ be arbitrarily given. We know that

$$\mathrm{vol}_{n-1}\big(\mathrm{proj}_{u^\perp}(K)\big) = \frac{1}{2} \int_{\mathbb{S}^{n-1}} |\langle u, v \rangle| \, dS_{n-1}(K, v),$$

and since K is in minimal surface area position, the measure $S_{n-1}(K, \cdot)$ is isotropic. Hence

$$\int_{\mathbb{S}^{n-1}} \langle u, v \rangle^2 \, dS_{n-1}(K, v) = \frac{S(K)}{n}.$$

From the classical Hölder inequality, we get the bound

$$\mathrm{vol}_{n-1}\big(\mathrm{proj}_{u^\perp}(K)\big) = \frac{1}{2}\int_{\mathbb{S}^{n-1}} |\langle u,v\rangle|\, dS_{n-1}(K,v) \leq$$

$$\leq \frac{1}{2}\left(\int_{\mathbb{S}^{n-1}} dS_{n-1}(K,v)\right)^{\frac{1}{2}}\left(\int_{\mathbb{S}^{n-1}} \langle u,v\rangle^2\, dS_{n-1}(K,v)\right)^{\frac{1}{2}} = \frac{1}{2}S(K)^{\frac{1}{2}}\left(\frac{S(K)}{n}\right)^{\frac{1}{2}} = \frac{S(K)}{2\sqrt{n}}.$$

This settles the upper bound. For the lower bound, we notice that since

$$|\langle u,v\rangle| \leq |u||v| = 1$$

for any $u, v \in \mathbb{S}^{n-1}$, we have that

$$|\langle u,v\rangle|^2 \leq |\langle u,v\rangle|$$

for every $u, v \in \mathbb{S}^{n-1}$. This leads to

$$\mathrm{vol}_{n-1}\big(\mathrm{proj}_{u^\perp}(K)\big) = \frac{1}{2}\int_{\mathbb{S}^{n-1}} |\langle u,v\rangle|\, dS_{n-1}(K,v) \geq \frac{1}{2}\int_{\mathbb{S}^{n-1}} \langle u,v\rangle^2\, dS_{n-1}(K,v) = \frac{S(K)}{2n},$$

and the proof is complete.

□

Exercise 9.6. *Let $u_1, \ldots, u_m \in \mathbb{S}^{n-1}$ and $c_1, \ldots, c_m > 0$ be such that*

$$\mathrm{Id}_{\mathbb{R}^n} = \sum_{j=1}^m c_j u_j \otimes u_j.$$

Prove that the Borel measure μ on $\mathbb{S}^{n-1}$ given as

$$\mu = \sum_{j=1}^m c_j \delta_{u_j}$$

is isotropic. Conclude that if $K \subseteq \mathbb{R}^n$ is a convex body symmetric with respect to the origin and in John position, then there is an isotropic Borel measure on the sphere which is concentrated in $\partial K \cap \mathbb{S}^{n-1}$.

Remark 9.1. A converse of this result will be given later in Exercise 9.12: we show that if $K \subseteq \mathbb{R}^n$ is an $0_{\mathbb{R}^n}$-symmetric convex body containing B^n, then the existence of a non-zero isotropic Borel measure concentrated in $\partial K \cap \mathbb{S}^{n-1}$ guarantees that K is in John position. This means that, for origin-symmetric convex bodies, the John position is an isotropic position, in some sense.

Solution 9.6. By Remark 9.2.3 from the main book, we know that $c_1 + \ldots + c_m = n$. Thus,

$$\frac{\mu(\mathbb{S}^{n-1})}{n} = 1.$$

Let $u \in \mathbb{S}^{n-1}$ be arbitrarily given. We have

$$\int_{\mathbb{S}^{n-1}} \langle u,v\rangle^2\, d\mu(v) = \sum_{j=1}^m c_j \langle u,u_j\rangle^2 = \left\langle \sum_{j=1}^m c_j u_j \otimes u_j(u), u\right\rangle = \langle u,u\rangle = 1 = \frac{\mu(\mathbb{S}^{n-1})}{n},$$

and by Proposition 9.2.3 (main book) this means that μ is isotropic.

If $K \subseteq \mathbb{R}^n$ is a $0_{\mathbb{R}^n}$-symmetric convex body in John position, then from Theorem 8.3.1 (main book) we have that there exist contact points $u_1, \dots, u_m \in \partial K \cap \mathbb{S}^{n-1}$ and numbers $c_1, \dots, c_m > 0$ such that

$$\mathrm{Id}_{\mathbb{R}^n} = \sum_{j=1}^{m} c_j u_j \otimes u_j.$$

The measure $\mu = \sum_{j=1}^{m} c_j \delta_{u_j}$ is isotropic and concentrated at $\{u_1, \dots, u_m\} \subseteq \partial K \cap \mathbb{S}^{n-1}$.

□

Exercise 9.7. *Let $u_1, \dots, u_m \in \mathbb{S}^{n-1}$ and $c_1, \dots, c_m > 0$. Prove that*

$$\mathrm{Id}_{\mathbb{R}^n} = \sum_{j=1}^{m} c_j u_j \otimes u_j, \tag{9.1}$$

if and only if

$$\sum_{j=1}^{m} c_j \langle u, u_j \rangle^2 = 1 \quad \forall \ u \in \mathbb{S}^{n-1}. \tag{9.2}$$

Solution 9.7. The fact that (9.1) implies (9.2) is obvious, and hence all we have to prove is the converse implication. We use a polarization argument. First of all, it is clear that if (9.2) holds, then

$$\sum_{j=1}^{m} c_j \langle x, u_j \rangle = |x|^2$$

for every $x \in \mathbb{R}^n$. Let $\{e_1, \dots, e_n\}$ be the canonical basis of $\mathbb{R}^n$. Notice that for any distinct $i, k \in \{1, \dots, n\}$ we have

$$\begin{aligned} 2 = |e_i + e_k|^2 = \sum_{j=1}^{m} c_j \langle e_i + e_k, u_j \rangle^2 = \\ = \sum_{j=1}^{m} c_j \langle e_i, u_j \rangle^2 + 2 \sum_{j=1}^{m} c_j \langle e_i, u_j \rangle \langle e_k, u_j \rangle + \sum_{j=1}^{m} c_j \langle e_k, u_j \rangle^2 = \\ = 2 + 2 \sum_{j=1}^{m} c_j \langle e_i, u_j \rangle \langle e_k, u_j \rangle, \end{aligned}$$

and hence

$$\sum_{j=1}^{m} c_j \langle e_i, u_j \rangle \langle e_k, u_j \rangle = 0$$

whenever $i \neq k$. If $i = k$, then the sum in the left-hand side equals $|e_i|^2 = 1$. Therefore, we may summarize the above as

$$\sum_{j=1}^{m} c_j \langle e_i, u_j \rangle \langle e_k, u_j \rangle = \delta_{ik}.$$

As a consequence, we arrive at

$$\left\langle \sum_{j=1}^{m} c_j u_j \otimes u_j(e_i), e_k \right\rangle = \left\langle \sum_{j=1}^{m} c_j \langle u_j, e_i \rangle u_j, e_k \right\rangle = \sum_{j=1}^{m} c_j \langle u_j, e_i \rangle \langle u_j, e_k \rangle = \delta_{ik},$$

and this gives immediately that

$$\sum_{j=1}^{m} c_j u_j \otimes u_j(e_i) = e_i$$

for each $1 \leq i \leq n$. By linearity, we get (9.1). The proof is complete.

□

Exercise 9.8. *Let $P \subseteq \mathbb{R}^n$ be a proper polytope with unit outer normal vectors $u_1, \ldots, u_m$ and corresponding facets $F_1, \ldots, F_m$. Prove that P is in minimal surface area position if and only if*

$$\mathrm{Id}_{\mathbb{R}^n} = \sum_{j=1}^{m} c_j u_j \otimes u_j,$$

where

$$c_j = \frac{n \mathrm{vol}_{n-1}(F_j)}{S(P)}$$

for each $1 \leq j \leq m$.

Solution 9.8. We know that P is in minimal surface area position if and only if $S_{n-1}(P, \cdot)$ is isotropic. From Lemma 9.3.2 (main book) we have that this is equivalent to

$$\int_{\mathbb{S}^{n-1}} \langle u, v \rangle^2 \, dS_{n-1}(P, v) = \frac{S_{n-1}(P, \mathbb{S}^{n-1})}{n}$$

for every $u \in \mathbb{S}^{n-1}$. It is clear that $S_{n-1}(P, \mathbb{S}^{n-1})$ is the total surface area $S(P)$ of P, and since $S_{n-1}(P, \cdot) = \sum_{j=1}^{m} \mathrm{vol}_{n-1}(F_j) \delta_{u_j}$ we may rewrite the above and obtain that P is in minimal surface area position if and only if

$$\sum_{j=1}^{m} \langle u, u_j \rangle^2 \mathrm{vol}_{n-1}(F_j) = \frac{S(P)}{n}$$

for every $u \in \mathbb{S}^{n-1}$. Letting $c_j = n \mathrm{vol}_{n-1}(F_j)/S(P)$ for each $1 \leq j \leq m$, we have that

$$\sum_{j=1}^{m} c_j \langle u, u_j \rangle^2 = 1 \quad \forall \ u \in \mathbb{S}^{n-1}.$$

Due to Exercise 9.7, this is equivalent to

$$\mathrm{Id}_{\mathbb{R}^n} = \sum_{j=1}^{m} c_j u_j \otimes u_j,$$

and the proof is complete.

□

For the next exercise, we need an inequality known as the *geometric Brascamp-Lieb inequality*. This is stated below. In [34], Ball presented a proof which, according to him, is due to Barthe. For Exercises 9.10 and 9.11, we refer the reader to [32] and [30], respectively (also, we thank Julian Haddad for showing us the result in Exercise 9.11).

Theorem 9.1 (Geometric Brascamp-Lieb Inequality)**.** *Let* $u_1, \ldots, u_m \in \mathbb{S}^{n-1}$ *and* $c_1, \ldots, c_m > 0$ *be such that*

$$\mathrm{Id}_{\mathbb{R}^n} = \sum_{j=1}^{m} c_j u_j \otimes u_j.$$

For any integrable functions $f_1, \ldots, f_m \colon \mathbb{R} \to [0, +\infty)$ *it holds that*

$$\int_{\mathbb{R}^n} \prod_{j=1}^{m} f(\langle u_j, x\rangle)^{c_j}\, dx \leq \prod_{j=1}^{m} \left(\int_{\mathbb{R}} f_j(t)\, dt \right)^{c_j}.$$

Exercise 9.9. (a) *Verify that the cube* $Q^n = [-1, 1]^n$ *is in John position.*

(b) *Prove that* Q^n *has the maximal volume among all origin-symmetric convex bodies in John position.*

Hint: If K is origin-symmetric and in John position, then we may consider touching points $u_1, \ldots, u_m$ of $\mathbb{S}^{n-1} \cap \partial K$ as in Theorem 9.2.1 (main book). The points $-u_1, \ldots, -u_m$ are also touching points of $\mathbb{S}^{n-1} \cap \partial K$, and hence K is contained in the "strips" bounded by its (parallel) supporting hyperplanes at u_j and $-u_j$, for $j = 1, \ldots, m$. Use the geometric Brascamp-Lieb inequality to bound the volume of the intersection of these strips. It is also useful to notice that for any characteristic function 1_X and every $c > 0$ it holds that $(1_X)^c = 1_X$.

Solution 9.9. (a) Just notice that $e_1, \ldots, e_n \in Q^n \cap \mathbb{S}^{n-1}$, and

$$\mathrm{Id}_{\mathbb{R}^n} = \sum_{j=1}^{n} e_j \otimes e_j.$$

From Theorem 9.2.1 (main book) we get that Q^n is in John position.

(b) Let K be any origin-symmetric convex body which is in John position. We may let $u_1, \ldots, u_m \in \mathbb{S}^{n-1} \cap \partial K$ and $c_1, \ldots, c_m > 0$ be such that

$$\mathrm{Id}_{\mathbb{R}^n} = \sum_{j=1}^{m} c_j u_j \otimes u_j.$$

Notice carefully that Remark 9.2.3 (main book) gives

$$\sum_{j=1}^{m} c_j = n.$$

Since K is symmetric with respect to the origin, we get that $-u_1, \ldots, -u_m \in \partial K$. At each $\pm u_j$, K and B^n have the same supporting hyperplane, which is $\mathbf{h}(\pm u_j, 1)$. Thus, K is contained in the following intersection of "strips":

$$K \subseteq \bigcap_{j=1}^{m} \left\{ x \in \mathbb{R}^n : -1 \leq \langle x, u_j \rangle \leq 1 \right\}.$$

Let the intersection of "strips" in the right-hand side of the above be C. Then, $x \in C$ if and only if $1_{[-1,1]}(\langle x, u_j\rangle) = 1$ for every $j = 1, \ldots, m$. Of course, the latter happens if and only if $1_{[-1,1]}(\langle x, u_j\rangle)^{c_j} = 1$ for each $j = 1, \ldots, m$.

In other words, we may write

$$1_C(x) = \prod_{j=1}^{m} 1_{[-1,1]}(\langle x, u_j\rangle)^{c_j}.$$

Therefore, using the geometric Brascamp-Lieb inequality, we arrive at

$$\mathrm{vol}_n(K) \le \mathrm{vol}_n(C) = \int_{\mathbb{R}^n} 1_C(x)\,dx = \int_{\mathbb{R}^n} \prod_{j=1}^{m} 1_{[-1,1]}(\langle x, u_j\rangle)^{c_j}\,dx \le$$
$$\le \prod_{j=1}^{m} \left(\int_{\mathbb{R}} 1_{[-1,1]}(t)\,dt\right)^{c_j} = 2^{c_1+\ldots+c_n} = 2^n = \mathrm{vol}_n(Q^n).$$

□

Exercise 9.10 (Reverse Isoperimetric Inequality)**.** *Let $K \subseteq \mathbb{R}^n$ be a convex body. Prove that there is a position $\tilde{K}$ of K such that*

$$\mathrm{vol}_n\big(\tilde{K}\big) = \mathrm{vol}_n(Q^n) \text{ and}$$
$$S\big(\tilde{K}\big) \le S(Q^n).$$

Hint: Consider the scalar multiple of the John position of K which satisfies the volume condition.

Solution 9.10. Let $J(K)$ be the John position of K, where J is the due affine transformation. Then, $B^n \subseteq J(K)$. Let $c > 0$ be such that $\mathrm{vol}_n(cJ(K)) = \mathrm{vol}_n(Q^n) = 2^n$. Of course, we have

$$c^n = \frac{2^n}{\mathrm{vol}_n(J(K))},$$

and then we write

$$S(cJ(K)) = \lim_{t\to 0^+} \frac{\mathrm{vol}_n(cJ(K) + tB^n) - \mathrm{vol}_n(cJ(K))}{t} \le$$
$$\le \lim_{t\to 0^+} \frac{\mathrm{vol}_n((c+t)J(K)) - \mathrm{vol}_n(cJ(K))}{t} = \lim_{t\to 0^+} \frac{\big[(c+t)^n - c^n\big]\mathrm{vol}_n(J(K))}{t} =$$
$$= nc^{n-1}\mathrm{vol}_n(J(K)) = n\frac{2^{n-1}}{\mathrm{vol}_n(J(K))^{\frac{n-1}{n}}}\mathrm{vol}_n(J(K)) = n2^{n-1}\mathrm{vol}_n(J(K))^{\frac{1}{n}}.$$

Since $J(K)$ is in John position, we have from item **(b)** of Exercise 9.9 that $\mathrm{vol}_n(J(K)) \le \mathrm{vol}_n(Q^n) \le 2^n$. Hence

$$S(cJ(K)) \le n2^{n-1}(2^n)^{\frac{1}{n}} = n2^n = S(Q^n).$$

Therefore, $\tilde{K} = cJ(K)$ is the position that we were looking for.

□

Exercise 9.11. *Let* $Q_n = \left[-\frac{1}{2}, \frac{1}{2}\right]^n \subseteq \mathbb{R}^n$, *and let* $\mathbf{g}$ *be a* k*-dimensional subspace of* $\mathbb{R}^n$, *where* $k < n$. *Prove that*

$$\mathrm{vol}_k(\mathbf{g} \cap Q_n) \leq \left(\frac{n}{k}\right)^{\frac{k}{2}}.$$

Hint: Let $P\colon \mathbb{R}^n \to \mathbf{g}$ be the orthogonal projection onto $\mathbf{g}$ and, up to reordering the terms, assume that $Pe_i \neq 0_{\mathbf{g}}$ for $i = 1, \ldots, m$ (for some $m \leq n$). Letting $u_i = Pe_i/|Pe_i|$ and $c_i = |Pe_i|^2$ for each $i = 1, \ldots, m$, show that

$$\mathbf{g} \cap Q_n = \Big\{x \in \mathbf{g} : |\langle x, u_i\rangle| \leq 1/2\sqrt{c_i} \text{ for all } i = 1, \ldots, m\Big\}.$$

Letting f_i be the characteristic function of $[-1/2\sqrt{c_i}, 1/2\sqrt{c_i}]$ for each $i = 1, \ldots, m$, notice that we always have $f_i = f_i^{c_i}$ and conclude that

$$\mathrm{vol}_k(\mathbf{g} \cap Q_n) = \int_{\mathbf{g}} \prod_{i=1}^{m} f_i(\langle x, u_i\rangle)^{c_1}\, dx.$$

Further, prove that

$$\mathrm{Id}_{\mathbf{g}} = \sum_{j=1}^{m} c_i u_i \otimes u_i,$$

and use the Brascamp-Lieb inequality. Finally, use the well-known fact that the product $\prod c_i^{c_i}$ is minimized (for positive c_i's) when all the c_i's are the same.

Solution 9.11. Let $e_1, \ldots, e_n$ be the canonical basis of $\mathbb{R}^n$, and let $P\colon \mathbb{R}^n \to \mathbf{g}$ be the orthogonal projection onto $\mathbf{g}$. Up to reordering the vectors if necessary, we may assume that $Pe_i \neq 0_{\mathbf{g}}$ for $i = 1, \ldots, m$, for some $m \leq n$. Let $c_i = |Pe_i|^2$ for every $i = 1, \ldots, m$.

A point $x \in \mathbf{g}$ belongs to Q_n if and only if $|\langle x, e_i\rangle| \leq 1/2$ for all $i = 1, \ldots, n$. Of course, this happens trivially for all $x \in \mathbf{g}$ in the case where $Pe_i = 0$. Thus, we may write

$$\mathbf{g} \cap Q_n = \Big\{x \in \mathbf{g} : |\langle x, e_i\rangle| \leq 1/2 \text{ for all } i = 1, \ldots, m\Big\}.$$

Since it is easy to check that $\langle x, Pe_i\rangle = \langle x, e_i\rangle$ whenever $x \in \mathbf{g}$, we may write

$$\mathbf{g} \cap Q_n = \Big\{x \in \mathbf{g} : |\langle x, Pe_i\rangle| \leq 1/2 \text{ for all } i = 1, \ldots, m\Big\}.$$

Now write $u_i = Pe_i/|Pe_i|$ for all $i = 1, \ldots, m$, and also write $c_i = |Pe_1|^2$ (again for $i = 1, \ldots, m$). Then, each inequality $|\langle x, Pe_i\rangle| \leq 1/2$ can be rewritten as $|\langle x, u_i\rangle| \leq 1/2\sqrt{c_i}$. Hence

$$\mathbf{g} \cap Q_n = \Big\{x \in \mathbf{g} : |\langle x, u_i\rangle| \leq 1/2\sqrt{c_i} \text{ for all } i = 1, \ldots, m\Big\}.$$

For each $i = 1, \ldots, m$, let $f_i\colon \mathbb{R} \to \mathbb{R}$ be the characteristic function of the interval $[-1/2\sqrt{c_i}, 1/2\sqrt{c_i}]$. The equality above means that a point $x \in \mathbf{g}$ is in $\mathbf{g} \cap Q_n$ if and only if $\langle x, u_i\rangle \in [-1/2\sqrt{c_i}, 1/2\sqrt{c_i}]$ for each $i = 1, \ldots, m$. Thus, we may write

$$1_{\mathbf{g}\cap Q_n}(x) = \prod_{i=1}^{m} f_i(\langle x, u_i\rangle)$$

for each $x \in \mathbf{g}$. It follows that

$$\mathrm{vol}_k(\mathbf{g} \cap Q_n) = \int_{\mathbf{g}} 1_{\mathbf{g}\cap Q_n}(x)\, dx = \int_{\mathbf{g}} \prod_{i=1}^{m} f_i(\langle x, u_i\rangle)\, dx,$$

where the integrations can be considered under any natural identification of $\mathbf{g}$ with $\mathbb{R}^k$. Now we notice that since each f_i only assumes values 0 or 1, and because all the c_i's are non-zero, we may write $f_i = f_i^{c_i}$ for each $i = 1, \ldots, m$. Thus, we get

$$\mathrm{vol}_k(\mathbf{g} \cap Q_n) = \int_{\mathbf{g}} \prod_{i=1}^{m} f_i(\langle x, u_i\rangle)^{c_1}\, dx. \tag{9.3}$$

We want to use the Brascamp-Lieb inequality in the right-hand side of the above. To do that, just observe that for any $x \in \mathbf{g}$ it holds that $x = Px$. Then, writing $x = \sum_{i=1}^{n} \langle x, e_i\rangle e_i$ and remembering again that $\langle x, e_i\rangle = \langle x, Pe_i\rangle$ for all i and that $Pe_i = 0_{\mathbf{g}}$ for $i > m$, we arrive at

$$x = Px = P\left(\sum_{i=1}^{n}\langle x, e_i\rangle e_i\right) = \sum_{i=1}^{n}\langle x, e_i\rangle Pe_i = \sum_{i=1}^{m}\langle x, Pe_i\rangle Pe_i = \sum_{i=1}^{m}|Pe_i|^2\langle x, u_i\rangle u_i =$$
$$= \sum_{i=1}^{m} c_i u_i \otimes u_i(x).$$

Then, the c_i's and u_i's give a decomposition of the identity $\mathrm{Id}_{\mathbf{g}}$ (as in the statement of the Brascamp-Lieb inequality). It follows that we may use this inequality in the right-hand side of (9.3). We obtain

$$\mathrm{vol}_k(\mathbf{g} \cap Q_n) \le \prod_{i=1}^{m}\left(\int_{\mathbb{R}} f_i(t)\, dt\right)^{c_i} = \prod_{i=1}^{m}(c_i^{-1/2})^{c_i} = \left(\prod_{i=1}^{m} c_i^{c_i}\right)^{-1/2}.$$

It is well-known that the product $\prod (c_i)^{c_i}$ is minimized when all the c_i's are the same. Because the c_i's and u_i's form a decomposition of the identity of the k-dimensional subspace $\mathbf{g}$, we know that the sum of the c_i's is k. Hence, we have

$$\left(\prod_{i=1}^{m} c_i^{c_i}\right)^{-1/2} \le \left(\frac{k}{m}\right)^{-k/2}.$$

Since $m \le n$, it follows that $k/m \ge k/n$, and hence

$$\left(\frac{k}{m}\right)^{-k/2} \le \left(\frac{k}{n}\right)^{-k/2} = \left(\frac{n}{k}\right)^{k/2}.$$

This concludes the proof.

□

The objective of the next exercise is to prove a sort of converse of Exercise 9.6. We want to prove that if $K \subseteq \mathbb{R}^n$ is an origin-symmetric convex body such that $B^n \subseteq K$, then the existence of an isotropic Borel measure on $\mathbb{S}^{n-1}$ concentrated on $\partial K \cap \mathbb{S}^{n-1}$ guarantees that K is in John position. This result appeared originally in [101].

Exercise 9.12. *Let $K \subseteq \mathbb{R}^n$ be a convex body symmetric with respect to the origin such that $B^n \subseteq K$. Assume that there exists a non-zero isotropic Borel measure μ on $\mathbb{S}^{n-1}$ concentrated in $\partial K \cap \mathbb{S}^{n-1}$.*

(a) *For simplicity, write $A = \partial K \cap \mathbb{S}^{n-1}$. Show that* $\mathrm{pos}(A) = \mathbb{R}^n$.

(b) *Let L be the Wulff shape*

$$L = W(A, 1) = \bigcap_{u \in A} \mathbf{h}^-(u, 1).$$

Show that

$$L = \left\{ z \in \mathbb{R}^n : |\langle z, u \rangle| \leq 1 \ \ \forall\, u \in A \right\}.$$

Show that L is a convex body. Verify also that $K \subseteq L$.

(c) *Assume that for an orthonormal basis $\{v_1, \ldots, v_n\}$ of $\mathbb{R}^n$ and for numbers $\alpha_1, \ldots, \alpha_n > 0$ the ellipsoid*

$$E = \left\{ y \in \mathbb{R}^n : \sum_{i=1}^n \frac{\langle y, v_i \rangle^2}{\alpha_i^2} \leq 1 \right\}$$

is contained in L. Prove that

$$\sum_{j=1}^n \alpha_j \leq n.$$

(d) *Using the inequality of arithmetic and geometric means, verify that*

$$\mathrm{vol}_n(E) \leq \mathrm{vol}_n(B^n).$$

(e) *Show that B^n is the ellipsoid of maximal volume contained in K, that is, K is in John position.*

Hint for **(c)**: For an arbitrary $u \in A$, let

$$z(u) = \left(\sum_{j=1}^n \alpha_j^2 \langle u, v_j \rangle^2 \right)^{-\frac{1}{2}} \sum_{j=1}^n \alpha_j^2 \langle u, v_j \rangle v_j.$$

Show that $z(u) \in E$. From $E \subseteq L$, conclude that $|\langle u, z(u) \rangle| \leq 1$. Use this inequality to get

$$\sum_{j=1}^n \alpha_j^2 \langle u, v_j \rangle^2 \leq 1.$$

Using the classical Cauchy-Schwarz inequality for the vectors

$$(\alpha_1 \langle u, v_1 \rangle, \ldots, \alpha_n \langle u, v_n \rangle) \ \text{ and } \ (\langle u, v_1 \rangle, \ldots, \langle u, v_n \rangle),$$

conclude that

$$\sum_{j=1}^{n} \alpha_j \langle u, v_j \rangle^2 \leq 1.$$

Finally, recalling that μ is isotropic and concentrated in A, write

$$\sum_{j=1}^{n} \alpha_j = \frac{n}{\mu(\mathbb{S}^{n-1})} \int_A \sum_{j=1}^{n} \alpha_j \langle u, v_j \rangle^2 \, d\mu(x)$$

and notice that the right-hand side is bounded by the inequality above.

Remark 9.2. Recently, Baêta and Haddad considered the problem of constructing a non-negative centered isotropic measure supported on the contact points of a convex body in Löwner position; see [28].

Solution 9.12. (a) We first check that μ is not concentrated on a great subsphere. Indeed, since μ is non-zero and isotropic, we have that

$$\int_{\mathbb{S}^{n-1}} \langle u, v \rangle^2 \, d\mu(v) = \frac{\mu(\mathbb{S}^{n-1})}{n} > 0$$

for every $u \in \mathbb{S}^{n-1}$. If μ is concentrated on a great subsphere $\mathbb{S}^{n-1} \cap u_0^{\perp}$, then

$$\int_{\mathbb{S}^{n-1}} \langle u_0, v \rangle^2 \, d\mu(v) = \int_{\mathbb{S}^{n-1} \cap u_0^{\perp}} \langle u_0, v \rangle^2 \, d\mu(v) = 0,$$

a contradiction.

Since μ is concentrated in A, it follows that A is not contained on a great subsphere. Hence A contains n linearly independent vectors $v_1, \ldots, v_n$. Since A is the intersection of sets which are symmetric with respect to the origin, we get that A itself is also origin-symmetric. Thus, $-v_1, \ldots, -v_n \in A$. This gives that $\mathrm{pos}(A) = \mathbb{R}^n$.

(b) The equality

$$\bigcap_{u \in A} \mathbf{h}^{-}(u, 1) = \left\{ z \in \mathbb{R}^n : |\langle z, u \rangle| \leq 1 \ \ \forall \, u \in A \right\}$$

follows directly from the fact that A is symmetric with respect to $0_{\mathbb{R}^n}$. The fact that L is a convex body is an immediate consequence of Lemma 8.1.1 (main book).

To verify that $K \subseteq L$, we just observe that K is supported by $\mathbf{h}(u, 1)$ at each contact point $u \in \partial K \cap \mathbb{S}^{n-1} = A$ (see the comment below the proof of Proposition 9.2.1 in the main book). Then, $K \subseteq \mathbf{h}^{-}(u, 1)$ for all $u \in A$, and this gives that $K \subseteq L$.

(c) Let $u \in A$ be arbitrarily given. We claim that

$$z(u) = \left(\sum_{j=1}^{n} \alpha_j^2 \langle u, v_j \rangle^2 \right)^{-\frac{1}{2}} \sum_{j=1}^{n} \alpha_j^2 \langle u, v_j \rangle v_j$$

is a point of E. Indeed,

$$\sum_{i=1}^{n} \frac{\langle z(u), v_i\rangle^2}{\alpha_i^2} = \left(\sum_{j=1}^{n} \alpha_j^2 \langle u, v_j\rangle^2\right)^{-1} \sum_{i=1}^{n} \alpha_i^{-2} \left\langle \sum_{j=1}^{n} \alpha_j^2 \langle u, v_j\rangle v_j, v_i \right\rangle^2 =$$

$$= \left(\sum_{j=1}^{n} \alpha_j^2 \langle u, v_j\rangle^2\right)^{-1} \sum_{j=1}^{n} \alpha_j^{-2} \alpha_j^4 \langle u, v_j\rangle^2 = \left(\sum_{j=1}^{n} \alpha_j^2 \langle u, v_j\rangle^2\right)^{-1} \sum_{j=1}^{n} \alpha_j^2 \langle u, v_j\rangle^2 = 1.$$

Since $E \subseteq L$, we have that $z(u) \in L$. Since $u \in A$, this means in particular that $|\langle u, z(u)\rangle| \leq 1$. Therefore,

$$1 \geq \left\langle \left(\sum_{j=1}^{n} \alpha_j^2 \langle u, v_j\rangle^2\right)^{-\frac{1}{2}} \sum_{j=1}^{n} \alpha_j^2 \langle u, v_j\rangle v_j, u \right\rangle =$$

$$= \left(\sum_{j=1}^{n} \alpha_j^2 \langle u, v_j\rangle^2\right)^{-\frac{1}{2}} \sum_{j=1}^{n} \alpha_j^2 \langle u, v_j\rangle^2 = \left(\sum_{j=1}^{n} \alpha_j^2 \langle u, v_j\rangle^2\right)^{\frac{1}{2}},$$

and squaring both sides we arrive at

$$\sum_{j=1}^{n} \alpha_j^2 \langle u, v_j\rangle^2 \leq 1.$$

Now we use the Cauchy-Schwarz inequality for the vectors

$$(\alpha_1 \langle u, v_1\rangle, \ldots, \alpha_n \langle u, v_n\rangle) \text{ and } (\langle u, v_1\rangle, \ldots, \langle u, v_n\rangle).$$

This leads to

$$\sum_{j=1}^{n} \alpha_j \langle u, v_j\rangle^2 \leq \left(\sum_{j=1}^{n} \alpha_j^2 \langle u, v_j\rangle^2\right)^{\frac{1}{2}} \left(\sum_{j=1}^{n} \langle u, v_j\rangle^2\right)^{\frac{1}{2}} \leq \left(\sum_{j=1}^{n} \langle u, v_j\rangle^2\right)^{\frac{1}{2}} = |u| = 1,$$

where we used the fact that $\{v_1, \ldots, v_j\}$ is an orthonormal basis of $\mathbb{R}^n$.

Moving forward, since μ is isotropic and concentrated in A, we have that

$$1 = \frac{n}{\mu(\mathbb{S}^{n-1})} \int_{\mathbb{S}^{n-1}} \langle u, v_j\rangle^2 \, d\mu(u) = \frac{n}{\mu(A)} \int_A \langle u, v_j\rangle^2 \, d\mu(u)$$

for each $j = 1, \ldots, n$. Hence

$$\sum_{j=1}^{n} \alpha_j = \sum_{j=1}^{n} \alpha_j \frac{n}{\mu(A)} \int_A \langle u, v_j\rangle^2 \, d\mu(u) = \frac{n}{\mu(A)} \int_A \sum_{j=1}^{n} \alpha_j \langle u, v_j\rangle^2 \, d\mu(u) \leq$$

$$\leq \frac{n}{\mu(A)} \int_A d\mu(u) = n.$$

(d) From the inequality of arithmetic and geometric means (see Exercise 1.12) with "weights" $1/n$, we have

$$1 = \sum_{j=1}^{n} \frac{\alpha_j}{n} \geq \left(\prod_{j=1}^{n} \alpha_j \right)^{\frac{1}{n}},$$

and hence

$$\prod_{j=1}^{n} \alpha_j \leq 1.$$

Let O be the orthogonal transformation such that $O(e_j) = v_j$ for each $1 \leq j \leq n$, and let $D = \operatorname{diag}(\alpha_1, \ldots, \alpha_n)$. Then, it is clear that

$$E = DO(B^n).$$

It follows that

$$\operatorname{vol}_n(E) = |\det(DO)| \operatorname{vol}_n(B^n) = |\det(D)| \operatorname{vol}_n(B^n) = \left(\prod_{j=1}^{n} \alpha_j \right) \operatorname{vol}_n(B^n) \leq \operatorname{vol}_n(B^n),$$

where we used the fact that any orthogonal transformation has determinant 1 or -1.

(e) We have that $B^n \subseteq K \subseteq L$, and item **(d)** gives that any other ellipsoid contained in L has volume smaller than or equal to $\operatorname{vol}_n(B^n)$. Hence K cannot contain an ellipsoid with volume greater than $\operatorname{vol}_n(B^n)$. This shows that K is in John position.

□

Exercise 9.13. *Let $\phi \colon \mathbb{R} \to [0, +\infty)$ be a convex function such that $\phi(0) = 0$.*

(a) *Verify that ϕ is non-increasing in $(-\infty, 0]$ and non-decreasing in $[0, +\infty)$.*

From now on, we also assume that ϕ is strictly decreasing in $(-\infty, 0]$ or strictly increasing in $[0, +\infty)$. For a given convex body $K \subseteq \mathbb{R}^n$ having the origin as an interior point, we define

$$h(\Pi_\phi K, x) = \inf \left\{ \lambda > 0 : \frac{1}{n \operatorname{vol}_n(K)} \int_{\mathbb{S}^{n-1}} \phi \left(\frac{\langle x, v \rangle}{\lambda h_K(v)} \right) h_K(v) \, dS_{n-1}(K, v) \leq 1 \right\}$$

for each $x \in \mathbb{R}^n$.

(b) *Show that if ϕ is strictly decreasing in $(-\infty, 0]$, then we have $\lim_{t \to -\infty} \phi(t) = +\infty$, and if ϕ is strictly increasing in $[0, +\infty)$, then it holds that $\lim_{t \to +\infty} \phi(t) = +\infty$.*

(c) *Verify that for any given $x \in \mathbb{R}^n$ there exists a number $\lambda > 0$ for which we indeed have*

$$\frac{1}{n \operatorname{vol}_n(K)} \int_{\mathbb{S}^{n-1}} \phi \left(\frac{\langle x, v \rangle}{\lambda h_K(v)} \right) h_K(v) \, dS_{n-1}(K, v) \leq 1.$$

Conclude that $h(\Pi_\phi K, \cdot)$ defines a real-valued function.

(d) *Prove that if* $x \neq 0_{\mathbb{R}^n}$*, then the function*

$$\lambda \mapsto \int_{\mathbb{S}^{n-1}} \phi\left(\frac{\langle x, v\rangle}{\lambda h_K(v)}\right) h_K(v)\, dS_{n-1}(K, v)$$

is continuous and strictly decreasing in $(0, +\infty)$*. In particular, show that if* $x \neq 0_{\mathbb{R}^n}$*, then* $h(\Pi_\phi K, x) > 0$*. Conclude that in this case one has that*

$$\frac{1}{n\mathrm{vol}_n(K)} \int_{\mathbb{S}^{n-1}} \phi\left(\frac{\langle x, v\rangle}{\lambda_0 h_K(v)}\right) h_K(v)\, dS_{n-1}(K, v) = 1$$

if and only if $\lambda_0 = h(\Pi_\phi K, x)$*.*

(e) *Prove that the function* $h(\Pi_\phi K, \cdot)\colon \mathbb{R}^n \to \mathbb{R}$ *is sublinear.*

It follows that $h(\Pi_\phi K, \cdot)$ *is the support function of a set of* CC^n*, which we denote by* $\Pi_\phi K$*, of course. This is known as the* Orlicz projection body *of* K *with respect to* ϕ*. Notice that from item* **(c)** *we have that* $\Pi_\phi K \in \mathcal{K}_0^n$*.*

(f) *Let*

$$c_\phi := \max\left\{c > 0 : \max\{\phi(-c), \phi(c)\} \leq 1\right\}.$$

Verify that $c_\phi > 0$ *and prove that*

$$\frac{1}{2nc_\phi \max_{v\in\mathbb{S}^{n-1}} h_K(v)} \leq h(\Pi_\phi K, u) \leq \frac{1}{c_\phi \min_{v\in\mathbb{S}^{n-1}} h_K(v)}$$

for any $u \in \mathbb{S}^{n-1}$*.*

Hint for **(f)**: For the lower bound, use Jensen's inequality and the estimate derived in item **(c)** of Exercise 7.25.

Solution 9.13. (a) Let $0 < t_1 < t_2$. Letting $\lambda := t_1/t_2$, we have that $\lambda \in (0, 1)$, and hence

$$\phi(t_1) = \phi(\lambda t_2 + (1-\lambda)0) \leq \lambda\phi(t_2) + (1-\lambda)\phi(0) = \lambda\phi(t_2).$$

Since $\phi \geq 0$ always, and because $\lambda < 1$, we get that $\lambda\phi(t_2) \leq \phi(t_2)$. Consequently, $\phi(t_1) \leq \phi(t_2)$. This shows that ϕ is non-decreasing in $[0, +\infty)$. The proof of the fact that ϕ is non-increasing in $(-\infty, 0]$ is completely analogous.

(b) Assume that ϕ is strictly increasing in $[0, +\infty)$. For any $t > 1$, letting $\lambda = 1/t$ we have that $0 < \lambda < 1$, and by convexity it follows that

$$\phi(1) = \phi((1-\lambda)0 + \lambda t) \leq (1-\lambda)\phi(0) + \lambda\phi(t) = \lambda\phi(t).$$

Hence

$$\phi(t) \geq t\phi(1).$$

Since ϕ is strictly increasing in $[0, +\infty)$, we get that $\phi(1) > 0$. Hence the right-hand side of the above goes to $+\infty$ as $t \to +\infty$. This shows what we wanted to confirm. The case of ϕ strictly decreasing in $(-\infty, 0]$ is completely analogous.

(c) If $x = 0_{\mathbb{R}^n}$, then it is clear that the integral is zero for every $\lambda > 0$. Thus, we assume that x is non-zero.

Using the fact that $\phi(0) = 0$, continuity, and item **(a)**, we observe that the set $\phi^{-1}([0,1])$ contains an open interval $(-\delta, \delta)$ for some $\delta > 0$. For any $v \in \mathbb{S}^{n-1}$ and $\lambda > 0$, we have

$$\left| \frac{\langle x, v \rangle}{\lambda h_K(v)} \right| \leq \frac{|x|}{\lambda h_K(v)} \leq \frac{|x|}{\lambda \min_{v \in \mathbb{S}^{n-1}} h_K(v)}.$$

Here we used the Cauchy-Schwarz inequality and also recalled that $h_K(v) > 0$ for every $v \in \mathbb{S}^{n-1}$ since $K \in \mathcal{K}_0^n$. The fact that the minimum in the denominator of the right-hand side of the above is positive follows from a compactness argument, of course. As a consequence, we get that for λ large enough it holds that

$$\frac{\langle x, v \rangle}{\lambda h_K(v)} \in (-\delta, \delta) \tag{9.4}$$

for every $v \in \mathbb{S}^{n-1}$. In this case,

$$\phi\left(\frac{\langle x, v \rangle}{\lambda h_K(v)} \right) \leq 1$$

for each $v \in \mathbb{S}^{n-1}$, and this leads to

$$\frac{1}{n\mathrm{vol}_n(K)} \int_{\mathbb{S}^{n-1}} \phi\left(\frac{\langle x, v \rangle}{\lambda h_K(v)} \right) h_K(v)\, dS_{n-1}(K, v) \leq \frac{1}{n\mathrm{vol}_n(K)} \int_{\mathbb{S}^{n-1}} h_K(v)\, dS_{n-1}(K, v) = \\ = \frac{nV(K, \ldots, K)}{n\mathrm{vol}_n(K)} = 1$$

for any $\lambda > 0$ large enough such that (9.4) holds.

(d) The continuity is immediate from the fact that $S_{n-1}(K, \cdot)$ is a finite measure (we also use the continuity of ϕ, as well as the compactness of $\mathbb{S}^{n-1}$; the latter is to guarantee that the entry of ϕ is controlled under small perturbations of λ).

Assume that ϕ is strictly increasing in $[0, +\infty)$ (the proof for the case where ϕ is strictly decreasing in $(-\infty, 0]$ is completely analogous). Fix $x \in \mathbb{R}^n \setminus \{0_{\mathbb{R}^n}\}$ and let $0 < \lambda_1 < \lambda_2$ be arbitrarily given. For any given $v \in \mathbb{S}^{n-1}$, one of the following possibilities hold:

(i) $v \in \mathbb{S}^{n-1} \cap \mathrm{int}(\mathbf{h}^+(x, 0))$. In this case, it is clear that

$$\frac{\langle x, v \rangle}{\lambda_i h_K(v)} > 0$$

for $i = 1, 2$, and

$$\frac{\langle x, v \rangle}{\lambda_1 h_K(v)} > \frac{\langle x, v \rangle}{\lambda_2 h_K(v)}.$$

Since ϕ is strictly increasing in $[0, +\infty)$, it follows that

$$\phi\left(\frac{\langle x, v \rangle}{\lambda_1 h_K(v)} \right) > \phi\left(\frac{\langle x, v \rangle}{\lambda_2 h_K(v)} \right). \tag{9.5}$$

(ii) $v \in \mathbb{S}^{n-1} \cap \mathbf{h}^-(x,0)$. If this holds, then we clearly have

$$\frac{\langle x, v\rangle}{\lambda_i h_K(v)} \leq 0$$

for $i = 1, 2$, and

$$\frac{\langle x, v\rangle}{\lambda_1 h_K(v)} \leq \frac{\langle x, v\rangle}{\lambda_2 h_K(v)}.$$

From item **(a)**, we know that ϕ is non-increasing in $(-\infty, 0]$. Thus,

$$\phi\left(\frac{\langle x, v\rangle}{\lambda_1 h_K(v)}\right) \geq \phi\left(\frac{\langle x, v\rangle}{\lambda_2 h_K(v)}\right). \tag{9.6}$$

Now we notice that, according to Exercise 7.24, the open hemisphere $\mathbb{S}^{n-1} \cap \operatorname{int}(\mathbf{h}^+(x,0))$ has non-zero measure, that is,

$$S_{n-1}\big(K, \mathbb{S}^{n-1} \cap \mathbf{h}^+(x,0))\big) > 0.$$

Therefore, recalling also that $h_K > 0$, we have from (9.5) the strict inequality

$$\begin{aligned}\int_{\mathbb{S}^{n-1}\cap \operatorname{int}(\mathbf{h}^+(x,0)))} \phi\left(\frac{\langle x, v\rangle}{\lambda_1 h_K(v)}\right) h_K(v)\, dS_{n-1}(K, v) > \\ > \int_{\mathbb{S}^{n-1}\cap \operatorname{int}(\mathbf{h}^+(x,0)))} \phi\left(\frac{\langle x, v\rangle}{\lambda_2 h_K(v)}\right) h_K(v)\, dS_{n-1}(K, v).\end{aligned}$$

Integrating over the closed hemisphere $\mathbb{S}^{n-1} \cap \mathbf{h}^-(x,0)$, inequality (9.6) leads to

$$\begin{aligned}\int_{\mathbb{S}^{n-1}\cap \mathbf{h}^-(x,0))} \phi\left(\frac{\langle x, v\rangle}{\lambda_1 h_K(v)}\right) h_K(v)\, dS_{n-1}(K, v) \geq \\ \geq \int_{\mathbb{S}^{n-1}\cap \mathbf{h}^-(x,0))} \phi\left(\frac{\langle x, v\rangle}{\lambda_2 h_K(v)}\right) h_K(v)\, dS_{n-1}(K, v).\end{aligned}$$

Adding both integral inequalities, we finally arrive at

$$\int_{\mathbb{S}^{n-1}} \phi\left(\frac{\langle x, v\rangle}{\lambda_1 h_K(v)}\right) h_K(v)\, dS_{n-1}(K, v) > \int_{\mathbb{S}^{n-1}} \phi\left(\frac{\langle x, v\rangle}{\lambda_2 h_K(v)}\right) h_K(v)\, dS_{n-1}(K, v),$$

and this proves that the map

$$\lambda \mapsto \int_{\mathbb{S}^{n-1}} \phi\left(\frac{\langle x, v\rangle}{\lambda h_K(v)}\right) h_K(v)\, dS_{n-1}(K, v)$$

is strictly decreasing in $(0, +\infty)$. In what follows, let us denote this map by $\psi_K(\lambda)$.

Now we use this fact to prove that $h(\Pi_\phi K, x) > 0$ whenever x is non-zero. As in the solution of Exercise 7.24, we may let $m_0 \in \mathbb{N}$ be such that

$$S_{n-1}\big(K, \{v \in \mathbb{S}^{n-1} : \langle x, v\rangle > 1/m_0\}\big) := \beta > 0.$$

Also, let $m := \min_{\mathbb{S}^{n-1}} h_K$ and $M := \max_{\mathbb{S}^{n-1}} h_K$ (we clearly have $0 < m \leq M$, since K is a convex body having the origin as an interior point). Thus, if we continue assuming that ϕ is strictly increasing

in $[0,+\infty)$ (the other case is again completely analogous), we arrive at

$$\begin{aligned}\int_{\mathbb{S}^{n-1}} \phi\left(\frac{\langle x,v\rangle}{\lambda h_K(v)}\right) h_K(v)\, dS_{n-1}(K,v) \geq \\ \geq \int_{\left\{v\in\mathbb{S}^{n-1}:\langle x,v\rangle>1/m_0\right\}} \phi\left(\frac{\langle x,v\rangle}{\lambda h_K(v)}\right) h_K(v)\, dS_{n-1}(K,v) \geq \\ \geq \int_{\left\{v\in\mathbb{S}^{n-1}:\langle x,v\rangle>1/m_0\right\}} \phi\left(\frac{1}{m_0\lambda M}\right) m\, dS_{n-1}(K,v) = \\ = \beta\phi\left(\frac{1}{m_0\lambda M}\right) m.\end{aligned}$$

Since $\lim_{t\to\infty}\phi(t)=+\infty$, it follows immediately from the estimate above that

$$\int_{\mathbb{S}^{n-1}} \phi\left(\frac{\langle x,v\rangle}{\lambda h_K(v)}\right) h_K(v)\, dS_{n-1}(K,v) \to +\infty$$

as $\lambda\to 0$. This shows that $h(\Pi_\phi K,x)>0$. With this in mind, it follows from continuity that for $\lambda = h(\Pi_\phi K,x)$ we have the equality

$$\frac{1}{n\mathrm{vol}_n(K)}\int_{\mathbb{S}^{n-1}} \phi\left(\frac{\langle x,v\rangle}{\lambda h_K(v)}\right) h_K(v)\, dS_{n-1}(K,v) = 1.$$

Since we already know that the function in the left-hand side is strictly decreasing on $\lambda>0$, we get that this equality holds only for $\lambda = h(\Pi_\phi K,x)$.

(e) First, we observe that it is immediate to see that $h(\Pi_\phi K, 0_{\mathbb{R}^n})=0$. If $\alpha>0$, then

$$\frac{\langle \alpha x,v\rangle}{\lambda h_K(v)} = \frac{\langle x,v\rangle}{\frac{\lambda}{\alpha}h_K(v)}$$

for any $x\in\mathbb{R}^n$ and every $v\in\mathbb{S}^{n-1}$. Thus,

$$\begin{aligned}h(\Pi_\phi K,\alpha x) &= \inf\left\{\lambda>0 : \frac{1}{n\mathrm{vol}_n(K)}\int_{\mathbb{S}^{n-1}} \phi\left(\frac{\langle \alpha x,v\rangle}{\lambda h_K(v)}\right) h_K(v)\, dS_{n-1}(K,v)\leq 1\right\} = \\ &= \inf\left\{\lambda>0 : \frac{1}{n\mathrm{vol}_n(K)}\int_{\mathbb{S}^{n-1}} \phi\left(\frac{\langle x,v\rangle}{\frac{\lambda}{\alpha}h_K(v)}\right) h_K(v)\, dS_{n-1}(K,v)\leq 1\right\} = \\ &= \alpha\inf\left\{\lambda>0 : \frac{1}{n\mathrm{vol}_n(K)}\int_{\mathbb{S}^{n-1}} \phi\left(\frac{\langle x,v\rangle}{\lambda h_K(v)}\right) h_K(v)\, dS_{n-1}(K,v)\leq 1\right\} = \alpha h(\Pi_\phi K,x),\end{aligned}$$

and this shows that $h(\Pi_\phi K,\cdot)$ is positively homogeneous.

Now we prove that $h(\Pi_\phi K,\cdot)$ is sublinear. Let $x,y\in\mathbb{R}^n$ be non-zero (the subadditivity is trivial in the case where one of them is zero). Let $\lambda_1 := h(\Pi_\phi K,x)$ and $\lambda_2 := h(\Pi_\phi K,y)$. We already know that these numbers are positive, and that we have

$$\begin{aligned}\frac{1}{n\mathrm{vol}_n(K)}\int_{\mathbb{S}^{n-1}} \phi\left(\frac{\langle x,v\rangle}{\lambda_1 h_K(v)}\right) h_K(v)\, dS_{n-1}(K,v) = \\ = \frac{1}{n\mathrm{vol}_n(K)}\int_{\mathbb{S}^{n-1}} \phi\left(\frac{\langle y,v\rangle}{\lambda_2 h_K(v)}\right) h_K(v)\, dS_{n-1}(K,v) = 1.\end{aligned}$$

Notice that if $t_1, t_2, t_3, t_4 \in \mathbb{R}$ and $t_3, t_4 > 0$, then we have the following inequality arising from the convexity of ϕ:

$$\phi\left(\frac{t_1+t_2}{t_3+t_4}\right) = \phi\left(\frac{t_3}{t_3+t_4}\frac{t_1}{t_3} + \frac{t_4}{t_3+t_4}\frac{t_2}{t_4}\right) \leq$$
$$\leq \frac{t_3}{t_3+t_4}\phi\left(\frac{t_1}{t_3}\right) + \frac{t_4}{t_3+t_4}\phi\left(\frac{t_2}{t_4}\right).$$

For any given $v \in \mathbb{S}^{n-1}$, letting $t_1 = \langle x, v\rangle$, $t_2 = \langle y, v\rangle$, $t_3 = \lambda_1 h_K(v)$ and $t_4 = \lambda_2 h_K(v)$, we get

$$\frac{1}{n\mathrm{vol}_n(K)}\int_{\mathbb{S}^{n-1}} \phi\left(\frac{\langle x+y, v\rangle}{(\lambda_1+\lambda_2)h_K(v)}\right) h_K(v)\, dS_{n-1}(K,v) \leq$$
$$\leq \frac{1}{n\mathrm{vol}_n(K)}\int_{\mathbb{S}^{n-1}} \left[\frac{\lambda_1}{\lambda_1+\lambda_2}\phi\left(\frac{\langle x, v\rangle}{\lambda_1 h_K(v)}\right) + \frac{\lambda_2}{\lambda_1+\lambda_2}\phi\left(\frac{\langle y, v\rangle}{\lambda_2 h_K(v)}\right)\right] h_K(v)\, dS_{n-1}(K,v) =$$
$$= \frac{\lambda_1}{\lambda_1+\lambda_2}\frac{1}{n\mathrm{vol}_n(K)}\int_{\mathbb{S}^{n-1}} \phi\left(\frac{\langle x, v\rangle}{\lambda_1 h_K(v)}\right) h_K(v)\, dS_{n-1}(K,v) +$$
$$+\frac{\lambda_2}{\lambda_1+\lambda_2}\frac{1}{n\mathrm{vol}_n(K)}\int_{\mathbb{S}^{n-1}} \phi\left(\frac{\langle y, v\rangle}{\lambda_2 h_K(v)}\right) h_K(v)\, dS_{n-1}(K,v) =$$
$$= \frac{\lambda_1}{\lambda_1+\lambda_2} + \frac{\lambda_2}{\lambda_1+\lambda_2} = 1.$$

Thus,

$$h(\Pi_\phi K, x+y) \leq \lambda_1 + \lambda_2 = h(\Pi_\phi K, x) + h(\Pi_\phi K, y).$$

(f) By the continuity of ϕ, we must have that $\phi(-c_\phi) = 1$ or $\phi(c_\phi) = 1$. Assume first that $\phi(c_\phi) = 1$, and let

$$\lambda_0 := h(\Pi_\phi K, u).$$

From item **(d)** above we know that

$$\frac{1}{n\mathrm{vol}_n(K)}\int_{\mathbb{S}^{n-1}} \phi\left(\frac{\langle u, v\rangle}{\lambda_0 h_K(v)}\right) h_K(v)\, dS_{n-1}(K,v) = 1.$$

Since

$$\frac{1}{n}\int_{\mathbb{S}^{n-1}} h_K(v)\, dS_{n-1}(K,v) = V(K,\ldots,K) = \mathrm{vol}_n(K),$$

we get that the measure on $\mathbb{S}^{n-1}$ given (in terms of densities) by

$$d\mu := \frac{h_K(v)}{n\mathrm{vol}_n(K)} dS_{n-1}(K,v)$$

is a probability measure on $\mathbb{S}^{n-1}$. Hence we may use Jensen's inequality for this measure. Also, notice that

$$\phi\left(\frac{\langle u, v\rangle}{\lambda_0 h_K(v)}\right) \geq \phi\left(\frac{\langle u, v\rangle_+}{\lambda_0 h_K(v)}\right)$$

for each $v \in \mathbb{S}^{n-1}$. In fact, if $\langle u, v\rangle \geq 0$, then we have equality. On the other hand, if $\langle u, v\rangle < 0$, then the left-hand side of the above is non-negative, whereas the right-hand side equals 0, because $\phi(0) = 0$. This, together with the estimate given in item **(c)** of Exercise 7.25 and Jensen's inequality, yields

$$\begin{aligned}\phi(c_\phi) = 1 = \frac{1}{n\mathrm{vol}_n(K)} \int_{\mathbb{S}^{n-1}} \phi\left(\frac{\langle u, v\rangle}{\lambda_0 h_K(v)}\right) h_K(v)\, dS_{n-1}(K, v) \geq \\ \geq \int_{\mathbb{S}^{n-1}} \phi\left(\frac{\langle u, v\rangle_+}{\lambda_0 h_K(v)}\right) \frac{h_K(v)}{n\mathrm{vol}_n(K)}\, dS_{n-1}(K, v) \geq \\ \geq \phi\left(\int_{\mathbb{S}^{n-1}} \frac{\langle u, v\rangle_+}{\lambda_0 h_K(v)} \frac{h_K(v)}{n\mathrm{vol}_n(K)}\, dS_{n-1}(K, v)\right) = \\ = \phi\left(\frac{1}{\lambda_0 n\mathrm{vol}_n(K)} \int_{\mathbb{S}^{n-1}} \langle u, v\rangle_+\, dS_{n-1}(K, v)\right) \geq \\ \geq \phi\left(\frac{1}{2n\lambda_0 \max_{\mathbb{S}^{n-1}} h_K}\right).\end{aligned}$$

We also used the fact that ϕ is non-decreasing in $[0, +\infty)$. Using this fact once again, the inequality above leads to

$$c_\phi \geq \frac{1}{2n\lambda_0 \max_{\mathbb{S}^{n-1}} h_K},$$

and this gives the desired estimate. In the case where $\phi(-c_\phi) = 1$, we observe that we also have

$$\phi\left(\frac{\langle u, v\rangle}{\lambda_0 h_K(v)}\right) \geq \phi\left(\frac{\langle u, v\rangle_-}{\lambda_0 h_K(v)}\right)$$

for every $v \in \mathbb{S}^{n-1}$. To estimate the integral of $\langle u, v\rangle_-$, we rely on items **(b)** and **(c)** of Exercise 7.25 to get

$$\begin{aligned}\int_{\mathbb{S}^{n-1}} \langle u, v\rangle_-\, dS_{n-1}(K, v) = -\mathrm{vol}_{n-1}\big(\mathrm{proj}_{u^\perp}(K)\big) = \\ = -\int_{\mathbb{S}^{n-1}} \langle u, v\rangle_+\, dS_{n-1}(K, v) \leq -\frac{\mathrm{vol}_n(K)}{2\max_{\mathbb{S}^{n-1}} h_K}.\end{aligned}$$

Recalling also that ϕ is non-increasing in $(-\infty, 0]$, and using Jensen's inequality, we get

$$\begin{aligned}\phi(-c_\phi) = \frac{1}{n\mathrm{vol}_n(K)} \int_{\mathbb{S}^{n-1}} \phi\left(\frac{\langle u, v\rangle}{\lambda_0 h_K(v)}\right) h_K(v)\, dS_{n-1}(K, v) \geq \\ \geq \phi\left(\int_{\mathbb{S}^{n-1}} \frac{\langle u, v\rangle_-}{\lambda_0 h_K(v)} \frac{h_K(v)}{n\mathrm{vol}_n(K)}\, dS_{n-1}(K, v)\right) \geq \phi\left(-\frac{1}{2n\lambda_0 \max_{\mathbb{S}^{n-1}} h_K}\right),\end{aligned}$$

and the inequality for λ_0 comes if we use again the fact that ϕ is non-increasing in $(-\infty, 0]$.

Now we prove the upper bound for $\lambda_0 := h(\Pi_\phi K, u)$. For any $v \in \mathbb{S}^{n-1}$, we either have $\langle u, v\rangle = |\langle u, v\rangle|$ or $\langle u, v\rangle = -|\langle u, v\rangle|$. Thus, it follows that

$$\phi\left(\frac{\langle u, v\rangle}{\lambda_0 h_K(v)}\right) \leq \max\left\{\phi\left(\frac{|\langle u, v\rangle|}{\lambda_0 h_K(v)}\right), \phi\left(\frac{-|\langle u, v\rangle|}{\lambda_0 h_K(v)}\right)\right\}$$

for each $v \in \mathbb{S}^{n-1}$. From the Cauchy-Schwarz inequality, and remembering that ϕ is non-increasing in $(-\infty, 0]$ and non-decreasing in $[0, +\infty)$, we have

$$\phi\left(\frac{|\langle u, v\rangle|}{\lambda_0 h_K(v)}\right) \leq \phi\left(\frac{1}{\lambda_0 \min_{\mathbb{S}^{n-1}} h_K}\right) \quad \text{and}$$
$$\phi\left(\frac{-|\langle u, v\rangle|}{\lambda_0 h_K(v)}\right) \leq \phi\left(\frac{-1}{\lambda_0 h_K(v)}\right) \leq \phi\left(-\frac{1}{\lambda_0 \min_{\mathbb{S}^{n-1}} h_K}\right).$$

Consequently, writing $\min_{\mathbb{S}^{n-1}} h_K =: m_K$ for simplicity, we arrive at

$$\phi\left(\frac{\langle u, v\rangle}{\lambda_0 h_K(v)}\right) \leq \max\left\{\phi\left(\frac{1}{\lambda_0 m_K}\right), \phi\left(-\frac{1}{\lambda_0 m_K}\right)\right\}.$$

Hence

$$\begin{aligned}\max\{\phi(-c_\phi), \phi(c_\phi)\} = 1 = \frac{1}{n\mathrm{vol}_n(K)} \int_{\mathbb{S}^{n-1}} \phi\left(\frac{\langle u, v\rangle}{\lambda_0 h_K(v)}\right) h_K(v)\, dS_{n-1}(K, v) \leq \\ \leq \frac{1}{n\mathrm{vol}_n(K)} \int_{\mathbb{S}^{n-1}} \max\left\{\phi\left(\frac{1}{\lambda_0 m_K}\right), \phi\left(-\frac{1}{\lambda_0 m_K}\right)\right\} h_K(v)\, dS_{n-1}(K, v) = \\ = \max\left\{\phi\left(\frac{1}{\lambda_0 m_K}\right), \phi\left(-\frac{1}{\lambda_0 m_K}\right)\right\} \int_{\mathbb{S}^{n-1}} \frac{h_K(v)}{n\mathrm{vol}_n(K)}\, dS_{n-1}(K, v) = \\ = \max\left\{\phi\left(\frac{1}{\lambda_0 m_K}\right), \phi\left(-\frac{1}{\lambda_0 m_K}\right)\right\}.\end{aligned}$$

Since the function $0 \leq t \mapsto \max\{\phi(-t), \phi(t)\}$ is obviously non-decreasing, it follows that

$$c_\phi \leq \frac{1}{\lambda_0 m_K},$$

and the proof is complete.

□

In the same notation and under analogous conditions as in the previous exercise, we define the *Orlicz projection map* (associated to ϕ) as the map $\Pi_\phi : \mathcal{K}_0^n \to \mathcal{K}_0^n$ which relates each convex body $K \subseteq \mathbb{R}^n$ having the origin as an interior point to its Orlicz projection body $\Pi_\phi K$ (defined in Exercise 9.13).

Exercise 9.14. *Prove that the Orlicz projection map $\Pi_\phi : \mathcal{K}_0^n \to \mathcal{K}_0^n$ is continuous in the Hausdorff metric.*

Solution 9.14. Let $(K_j)_{j\in\mathbb{N}}$ be a sequence in $\mathcal{K}_0^n$ converging in the Hausdorff metric to a body $K \in \mathcal{K}_0^n$, and let $u \in \mathbb{S}^{n-1}$ be arbitrarily given. Write

$$\lambda_j := h(\Pi_\phi K_j, u)$$

for each $j \in \mathbb{N}$, and

$$\lambda_0 := h(\Pi_\phi K, u).$$

Write also $M_{K_j} := \max_{\mathbb{S}^{n-1}} h_{K_j}$ and $m_{K_j} := \min_{\mathbb{S}^{n-1}} h_{K_j}$ for each $j \in \mathbb{N}$, and also $M_K := \max_{\mathbb{S}^{n-1}} h_K$ and $m_K := \min_{\mathbb{S}^{n-1}} h_K$. From Exercise 9.13 we have

$$\frac{1}{2nc_\phi M_{K_j}} \leq \lambda_j \leq \frac{1}{c_\phi m_{K_j}}, \tag{9.7}$$

where $c_\phi := \max\{c > 0 : \max\{\phi(-c), \phi(c)\} \leq 1\}$. Because $K_j \to K$ we have that $h_{K_j} \to h_K$ in the maximum norm of $\mathbb{S}^{n-1}$. Thus, $M_{K_j} \to M_K$ and $m_{K_j} \to m_K$. It follows that the sequence $(\lambda_j)_{j\in\mathbb{N}}$ is bounded (away from zero). Therefore, there is a subsequence (λ_{j_k}) which converges to some $\lambda_0 \in \mathbb{R}$, and in this case we clearly have

$$0 < \frac{1}{2nc_\phi M_K} \leq \lambda_0 \leq \frac{1}{c_\phi m_K}.$$

Now observe that

$$\frac{1}{n\mathrm{vol}_n(K)} \int_{\mathbb{S}^{n-1}} \phi\left(\frac{\langle u, v\rangle}{\lambda_{j_k} h_{K_{j_k}}(v)}\right) h_{K_{j_k}}(v)\, dS_{n-1}(K_{j_k}, v) = 1.$$

Since $K_j \to K$, we get that $h_{K_j} \to h_K$ uniformly, and $S_{n-1}(K_j, \cdot) \to S_{n-1}(K, \cdot)$ weakly. Also, since $\lambda_{j_k} \to \lambda_0$ and ϕ is uniformly continuous in any compact interval (because it is convex, and hence Lipschitz in compact intervals), it is immediate to verify that

$$\phi\left(\frac{\langle u, \cdot\rangle}{\lambda_{j_k} h_{K_{j_k}}(\cdot)}\right) \to \phi\left(\frac{\langle u, \cdot\rangle}{\lambda_0 h_K(\cdot)}\right)$$

uniformly on $\mathbb{S}^{n-1}$ as $j \to \infty$. Hence, we have that

$$\frac{1}{n\mathrm{vol}_n(K)} \int_{\mathbb{S}^{n-1}} \phi\left(\frac{\langle u, v\rangle}{\lambda_0 h_K(v)}\right) h_K(v)\, dS_{n-1}(K, v) = 1,$$

and this shows that $\lambda_0 = h(\Pi_\phi K, u)$. Then,

$$\lim_{k\to\infty} h(\Pi_\phi K_{j_k}, u) = \lim_{k\to\infty} \lambda_{j_k} = \lambda_0 = h(\Pi_\phi K, u).$$

It follows that $h(\Pi_\phi K_j, u)$ has a subsequence which converges to $h(\Pi_\phi K, u)$. Even more, since we may replace (K_j) by any of its subsequences, our argument shows that **any** subsequence of $h(\Pi_\phi K_j, u)$ has a subsequence which converges to $h(\Pi_\phi K, u)$. By standard analysis arguments, it follows that

$$h(\Pi_\phi K_j, u) \to h(\Pi_\phi K, u),$$

and because $u \in \mathbb{S}^{n-1}$ was arbitrarily given, we get that $h(\Pi_\phi K_j, \cdot)$ converges pointwise to $h(\Pi_\phi K, \cdot)$ on $\mathbb{S}^{n-1}$.

To verify that the convergence is uniform, notice that the right-hand inequality in (9.7) gives that $(h(\Pi_\phi K_j, \cdot))$ is a uniformly bounded sequence of functions of $\mathbb{S}^{n-1}$ (recall that m_{K_j} converges to a positive number, and hence it is easy to derive a uniform bound). We also claim that this sequence of functions is equicontinuous. In fact, for any $j \in \mathbb{N}$ and $v, w \in \mathbb{S}^{n-1}$ (with $v \neq w$, say) sublinearity gives

$$|h(\Pi_\phi K_j, w) - h(\Pi_\phi K_j, v)| \leq \max\left\{h(\Pi_\phi K_j, v - w), h(\Pi_\phi K_j, w - v)\right\} =$$
$$= ||v - w|| \max\left\{h\left(\Pi_\phi K_j, \frac{v - w}{||v - w||}\right), h\left(\Pi_\phi K_j, \frac{w - v}{||v - w||}\right)\right\} \leq ||v - w|| M_{K_j}.$$

Since $M_{K_j} \to M_K$, we get that

$$|h(\Pi_\phi K_j, w) - h(\Pi_\phi K_j, v)| \leq M||v - w||$$

for some $M > 0$ which does not depend on j. This proves the equicontinuity. From the Ascoli-Arzelá theorem, it follows that $(h(\Pi_\phi K_j, \cdot))$ has a uniformly converging subsequence. Of course, the limit of such a subsequence must be $h(\Pi_\phi K, \cdot)$, because we already showed that this latter function is the pointwise limit of $(h(\Pi_\phi K_j, \cdot))$. All the arguments remain true if we originally replace the sequence (K_j) by some of its subsequences, and hence we conclude that every subsequence of $(h(\Pi_\phi K_j, \cdot))$ has a subsequence which converges uniformly (on $\mathbb{S}^{n-1}$) to $h(\Pi_\phi K, \cdot)$. It follows that

$$h(\Pi_\phi K_j, \cdot) \to h(\Pi_\phi K, \cdot)$$

uniformly on $\mathbb{S}^{n-1}$, and this is the same as saying that $\Pi_\phi K_j \to \Pi_\phi K$ in the Hausdorff metric.

□

The next exercise is dedicated to the behavior of the Orlicz projection body under volume-preserving linear transformations of $\mathbb{R}^n$. Hence we still consider the same conditions and notation as in Exercise 9.13 above.

Exercise 9.15. **(a)** *Prove that for any $K \in \mathcal{K}_0^n$ and every $T \in \mathrm{GL}(n)$ such that $|\det(T)| = 1$ one has*

$$\Pi_\phi(TK) = (T^*)^{-1}\Pi_\phi K. \tag{9.8}$$

Hint: Consider first the particular case where K is a polytope, and then settle the general case by approximation arguments.

(b) *Show that for every $c > 0$ and any $K \in \mathcal{K}_0^n$ we have*

$$\Pi_\phi(cK) = \frac{1}{c}\Pi_\phi K.$$

(c) *Verify that (9.8) holds for any $T \in \mathrm{GL}(n)$. Denoting by $\Pi_\phi^\circ K$ the polar body of $\Pi_\phi K$, conclude that*

$$\Pi_\phi^\circ(TK) = T\Pi_\phi^\circ K$$

for every $T \in \mathrm{GL}(n)$.

(d) *Prove that if ϕ is even, then $\Pi_\phi(-K) = \Pi_\phi K$ for every $K \in \mathcal{K}_0^n$.*

Solution 9.15. **(a)** We first prove the result for a proper polytope $P \subseteq \mathbb{R}^n$ having the origin as an interior point. Assume that $F_1, \ldots, F_m$ are the facets of P, having $u_1, \ldots, u_m$ as unit outer normal vectors, respectively. From Exercise 3.25 we get that $TF_1, \ldots, TF_m$ are the facets of TP, and the unit outer normal vectors are $(T^*)^{-1}u_1/|(T^*)^{-1}u_1|, \ldots, (T^*)^{-1}u_m/|(T^*)^{-1}u_m|$. For simplicity, we write

$$u_j^* := \frac{(T^{-1})^* u_j}{|(T^{-1})^* u_j|}$$

for each $j \in \{1, \ldots, m\}$. From (7.1.2) (main book) we know that

$$S_{n-1}(TP, \cdot) = \sum_{j=1}^m \mathrm{vol}_{n-1}(TF_j)\delta_{u_j^*}.$$

It is immediate that TP is also a proper polytope having the origin as an interior point. Since T is volume-preserving we have that for each j the volume of the pyramid having basis TF_j and apex at $0_{\mathbb{R}^n}$ equals the volume of the pyramid having basis F_j and apex at $0_{\mathbb{R}^n}$. It follows that

$$\frac{1}{n} h_{TP}(u_j^*)\mathrm{vol}_{n-1}(TF_j) = \frac{1}{n} h_P(u_j)\mathrm{vol}_{n-1}(F_j).$$

In case of any doubt regarding the volume of pyramids, we refer the reader to Example 4.4.1 in the main book. Let $u \in \mathbb{S}^{n-1}$ be arbitrarily given, and let $\lambda_0 := h(\Pi_\phi(TP), u)$. Since the integral of a function f with respect to a Dirac delta measure δ_v is just $f(v)$, and recalling that $h_{TA}(v) = h_A(T^* v)$, we get

$$\begin{aligned}
1 &= \frac{1}{n\mathrm{vol}_n(TP)} \int_{\mathbb{S}^{n-1}} \phi\left(\frac{\langle u, v\rangle}{\lambda_0 h_{TP}(v)}\right) h_{TP}(v)\, dS_{n-1}(TP, v) = \\
&= \frac{1}{n\mathrm{vol}_n(P)} \sum_{j=1}^{m} \phi\left(\frac{\langle u, u_j^*\rangle}{\lambda_0 h_{TP}(u_j^*)}\right) h_{TP}(u_j^*)\mathrm{vol}_{n-1}(TF_j) = \\
&= \frac{1}{n\mathrm{vol}_n(P)} \sum_{j=1}^{m} \phi\left(\frac{\langle u, u_j^*\rangle |(T^*)^{-1} u_j|}{\lambda_0 h_P(u_j)}\right) h_{TP}(u_j^*)\mathrm{vol}_{n-1}(TF_j) = \\
&= \frac{1}{n\mathrm{vol}_n(P)} \sum_{j=1}^{m} \phi\left(\frac{\langle u, (T^{-1})^* u_j\rangle}{\lambda_0 h_P(u_j)}\right) h_P(u_j)\mathrm{vol}_{n-1}(F_j) = \\
&= \frac{1}{n\mathrm{vol}_n(P)} \sum_{j=1}^{m} \phi\left(\frac{\langle T^{-1} u, u_j\rangle}{\lambda_0 h_P(u_j)}\right) h_P(u_j)\mathrm{vol}_{n-1}(F_j) = \\
&= \frac{1}{n\mathrm{vol}_n(P)} \int_{\mathbb{S}^{n-1}} \phi\left(\frac{\langle T^{-1} u, v\rangle}{\lambda_0 h_P(v)}\right) h_P(v)\, dS_{n-1}(P, v),
\end{aligned}$$

and hence it follows that $\lambda_0 = h(\Pi_\phi P, T^{-1} u)$. As a consequence,

$$h(\Pi_\phi(TP), u) = h(\Pi_\phi P, T^{-1} u) = h((T^{-1})^* \Pi_\phi P, u),$$

and this holds for any $u \in \mathbb{S}^{n-1}$. Therefore, we get that

$$\Pi_\phi(TP) = (T^{-1})^* \Pi_\phi P$$

for any proper polytope $P \subseteq \mathbb{R}^n$ having the origin as an interior point. The case for an arbitrary $K \in \mathcal{K}_0^n$ follows immediately by approximating K by polytopes. We use the continuity of the Orlicz projection map, and also the fact that convergence in the Hausdorff metric is preserved by linear isomorphisms.

(b) Let $\lambda_0 = h(\Pi_\phi(cK), u)$, for a given $u \in \mathbb{S}^{n-1}$. Then, we have

$$\begin{aligned}
1 &= \frac{1}{n\mathrm{vol}_n(cK)} \int_{\mathbb{S}^{n-1}} \phi\left(\frac{\langle u, v\rangle}{\lambda_0 h_{cK}(v)}\right) h_{cK}(v)\, dS_{n-1}(cK, v) = \\
&= \frac{1}{nc^n\mathrm{vol}_n(K)} \int_{\mathbb{S}^{n-1}} \phi\left(\frac{\langle u, v\rangle}{c\lambda_0 h_K(v)}\right) c h_K(v) c^{n-1}\, dS_{n-1}(K, v) = \\
&= \frac{1}{n\mathrm{vol}_n(K)} \int_{\mathbb{S}^{n-1}} \phi\left(\frac{\langle u, v\rangle}{c\lambda_0 h_K(v)}\right) h_K(v)\, dS_{n-1}(K, v),
\end{aligned}$$

and hence we get that $c\lambda_0 = h(\Pi_\phi K, u)$. Thus,

$$h(\Pi_\phi(cK), u) = \lambda_0 = \frac{1}{c} h(\Pi_\phi K, u) = h\left(\frac{1}{c}\Pi_\phi K, u\right),$$

and since this holds for every $u \in \mathbb{S}^{n-1}$, we get the desired equality.

(c) Let $T \in \mathrm{GL}(n)$. Setting

$$T_0 := \frac{T}{|\det(T)|^{1/n}},$$

we have that $|\det(T_0)| = 1$. Therefore,

$$\begin{aligned}\Pi_\phi(TK) = \Pi_\phi\left(|\det(T)|^{1/n} T_0 K\right) = \frac{1}{|\det(T)|^{1/n}} \Pi_\phi(T_0 K) = \frac{1}{|\det(T)|^{1/n}} (T_0^*)^{-1} \Pi_\phi K = \\ = \frac{1}{|\det(T)|^{1/n}} \left(\frac{T^*}{|\det(T)|^{1/n}}\right)^{-1} \Pi_\phi K = (T^*)^{-1} \Pi_\phi K,\end{aligned}$$

as we wanted to prove. For the other claim, just notice that

$$\Pi_\phi^\circ(TK) = \left[(T^*)^{-1} \Pi_\phi K\right]^\circ = T(\Pi_\phi K)^\circ = T\Pi_\phi^\circ K.$$

(d) Since $h_{-K}(v) = h_K(-v)$ and since $(-\mathrm{Id}_{\mathbb{R}^n})_* S_{n-1}(-K, \cdot) = S_{n-1}(K, \cdot)$, assuming that ϕ is even, we have

$$\begin{aligned}\frac{1}{n\mathrm{vol}_n(-K)} \int_{\mathbb{S}^{n-1}} \phi\left(\frac{\langle u, v\rangle}{\lambda h_{-K}(v)}\right) h_{-K}(v)\, dS_{n-1}(-K, v) = \\ = \frac{1}{n\mathrm{vol}_n(K)} \int_{\mathbb{S}^{n-1}} \phi\left(\frac{\langle u, -v\rangle}{\lambda h_K(-v)}\right) h_K(-v)\, dS_{n-1}(-K, v) = \\ = \frac{1}{n\mathrm{vol}_n(K)} \int_{\mathbb{S}^{n-1}} \phi\left(\frac{\langle u, v\rangle}{\lambda h_K(v)}\right) h_K(v)\, d(-\mathrm{Id}_{\mathbb{R}^n})_* S_{n-1}(-K, v) = \\ = \frac{1}{n\mathrm{vol}_n(K)} \int_{\mathbb{S}^{n-1}} \phi\left(\frac{\langle u, v\rangle}{\lambda h_K(v)}\right) h_K(v)\, dS_{n-1}(K, v),\end{aligned}$$

and this yields that $\Pi_\phi(-K) = \Pi_\phi K$.

□

Remark 9.3. We deal with some examples of Orlicz projection bodies in Exercise 11.14 below. In particular, we verify that the usual projection body can be reobtained (up to a normalization) if we use a suitable function ϕ.

The exercises about Orlicz projection bodies are taken from [181], which is the main reference for this subject. An important result concerning these bodies states that, denoting by $\Pi_\phi^\circ K$ the polar body of $\Pi_\phi K$, the ($\mathrm{GL}(n)$-invariant) ratio

$$\frac{\mathrm{vol}_n(\Pi_\phi^\circ K)}{\mathrm{vol}_n(K)}$$

is maximal (among all $K \in \mathcal{K}_0^n$) if and only if K is an ellipsoid symmetric with respect to the origin (this was proved in the mentioned reference only for the case where ϕ is strictly convex; the general case

was proved in [60]). Later, in Exercise 11.15, the reader is invited to use this result to give an alternative proof of the so-called Petty projection inequality, which we state and prove in Theorem 10.4.2 (main book). Even more, in Exercise 11.17 this result is applied to establish an L_p *extension* of the Petty projection inequality.

In Section 9.3 (main book), we dealt with *isotropic measures*, but above Exercise 4.34 we introduced the (related) notion of *isotropic body*. For the next exercises, which are basically results taken from [24], recall that a convex body $K \subseteq \mathbb{R}^n$ is said to be *isotropic* if it has its centroid at the origin and if there exists a constant $\rho \in \mathbb{R}$ such that

$$\int_K \langle x, u\rangle^2 \, dx = \rho^2$$

for every $u \in \mathbb{S}^{n-1}$.

As we have said above Exercise 4.34, most authors also demand that $\mathrm{vol}_n(K) = 1$, but we do not do it here. We do it in this way because the interesting features of isotropic bodies are invariant under rescaling. It is useful, however, to work with normalized volume when dealing with *isotropic constants* (which will be defined below), but in this case we may simply restrict ourselves to bodies of unit volume.

If $K \subseteq \mathbb{R}^n$ is a convex body, and $T \in \mathrm{GL}(n)$ and $x_0 \in \mathbb{R}^n$ are such that $x_0 + TK$ is isotropic, then we say that $x_0 + TK$ is an *isotropic position* of K.

Exercise 9.16. *Let $K \subseteq \mathbb{R}^n$ be a convex body. Define the linear map $M\colon \mathbb{R}^n \to \mathbb{R}^n$ as*

$$My = \int_K \langle x, y\rangle x \, dx$$

for $y \in \mathbb{R}^n$.

(a) *Prove that M is symmetric and positive-definite. Conclude that M has a symmetric and positive-definite square root (as it is defined in the proof of Lemma 9.1.1 in the main book).*

(b) *Let $T \in \mathrm{SL}(n)$. For a moment, denote by M_L the inertia map of the convex body L. Prove that $M_{TK} = TM_K T^*$.*

Remark 9.4. The linear map M is called the *inertia map* (or *moment map*) of K. Of course, in matrix form, we may write $M = (m_{ij})_{i,j=1,\ldots,n}$ as

$$m_{ij} = \int_K x_i x_j \, dx.$$

This matrix is called the *inertia matrix* (or *moment matrix*) of K.

Solution 9.16. (a) It is clear that M is symmetric. We prove that it is positive-definite. To verify that, we let $y \neq 0_{\mathbb{R}^n}$ and compute

$$\langle My, y\rangle = \int_K \langle x, y\rangle^2 \, dx.$$

Since K is a convex body, we have that K is not contained in $y^\perp$. Thus, $\langle x, y\rangle^2 > 0$ for some ball in K. Consequently, $\langle My, y\rangle > 0$.

It follows that we can write $M = V^{-1}DV$ for some orthogonal map V and some diagonal matrix D whose entries in the principal diagonal are precisely the eigenvalues of M (which are all positive, since M is positive-definite). The square root of M is then defined as $S = V^{-1}D^{1/2}V$, where $D^{1/2}$ is the

diagonal matrix whose entries in the principal diagonal are the respective square roots of the entries in the principal diagonal of D. It is clear that S is symmetric and positive-definite (formally, this follows from the facts that $D^{1/2}$ is diagonal and that $V^{-1} = V^*$).

(b) For any $y \in \mathbb{R}^n$ we have

$$M_{TK}y = \int_{TK} \langle x, y\rangle\, dx = \int_K \langle Tx, y\rangle Tx\, dx = T\left(\int_K \langle x, T^*y\rangle x\, dx\right) = T(M_K(T^*y)).$$

This gives the desired equality.

□

Exercise 9.17. *Let $K \subseteq \mathbb{R}^n$ be an arbitrarily given convex body. Prove that K has an isotropic position.*

Hint: Let $x_0 \in \mathbb{R}^n$ be such that $K_0 = x_0 + K$ has its centroid at the origin. Define the linear map $M\colon \mathbb{R}^n \to \mathbb{R}^n$ as

$$M(y) = \int_{K_0} \langle x, y\rangle x\, dx.$$

Use Exercise 9.16 and denote the square root of M by S. Then, prove that $K_1 = S^{-1}(K_0)$ is isotropic.

Solution 9.17. Let $K_0 = x_0 + K$ have its centroid at the origin, and define the map M for K_0 as in Exercise 9.16. Let S be the square root of M, and let $K_1 = S^{-1}(K_0)$. We claim that K_1 is isotropic. First, since K_0 has its centroid at the origin and S^{-1} is a linear isomorphism, we have that K_1 also has its centroid at the origin (the centroid is equivariant for linear isomorphisms; see Exercise 4.10). From (A.2.4) and Theorem A.3.3 from the main book (which are the standard formula for change of variables for integrations and the description of the push-forward of Lebesgue measures under linear maps, respectively), and using also the fact that S^{-1} is symmetric (because S is symmetric), we get

$$\begin{aligned}\int_{K_1} \langle x, y\rangle^2\, dx &= \int_{S^{-1}(K_0)} \langle x, y\rangle^2\, dx = \\ &= \int_{K_0} \langle S^{-1}x, y\rangle^2\, |\det(S^{-1})|\, dx = \int_{K_0} \langle x, S^{-1}y\rangle^2\, |\det(S^{-1})|\, dx = \\ &= |\det(S)|^{-1}\left\langle \int_{K_0} \langle x, S^{-1}y\rangle x\, dy, y\right\rangle = |\det(S)|^{-1}\langle MS^{-1}y, S^{-1}y\rangle = |\det(S)|^{-1}\langle Sy, S^{-1}y\rangle = \\ &= |\det(S)|^{-1}\langle S^{-1}Sy, y\rangle = |\det(S)|^{-1}|y|^2.\end{aligned}$$

Thus, for every $u \in \mathbb{S}^{n-1}$ we have

$$\int_{K_1} \langle x, y\rangle^2\, dx = |\det(S)|^{-1},$$

and the right-hand side is a constant which does not depend on u. This proves that K_1 is isotropic. Since $K_1 = S^{-1}x_0 + S^{-1}K$, we get that K_1 is an isotropic position of K.

□

Exercise 9.18. *Let $K \subseteq \mathbb{R}^n$ be a convex body having its centroid at the origin. Prove that the following assertions are equivalent:*

(i) *K is isotropic with*

$$\int_K \langle x, u\rangle^2 \, dx = \rho^2$$

for every $u \in \mathbb{S}^{n-1}$.

(ii) *Writing in coordinates $x = (x_1, \ldots, x_n)$, we have*

$$\int_K x_i x_j \, dx = \rho^2 \delta_{i,j}$$

for any $i, j = 1, \ldots, n$, where $\delta_{i,j}$ denotes the usual Kronecker delta.

(iii) *For every $T \in \mathrm{L}(n)$ we have the equality*

$$\int_K \langle x, Tx\rangle \, dx = \rho^2 \mathrm{tr}(T).$$

Remark 9.5. Notice that the equivalence between **(i)** and **(ii)** means that K is isotropic if and only if its inertia matrix is a multiple of the identity matrix.

Solution 9.18. Just proceed precisely as in the proof of Proposition 9.3.2 (main book).

□

It is easy to see that isotropic convex bodies remain isotropic under rescaling. Thus, in the next exercise we work with transformations of $\mathrm{SL}(n)$ instead of $\mathrm{GL}(n)$.

Exercise 9.19. *Let $K \subseteq \mathbb{R}^n$ be a convex body having its centroid at the origin. Define*

$$B(K) := \inf \left\{ \int_{TK} |x|^2 \, dx : T \in \mathrm{SL}(n) \right\}. \tag{9.9}$$

(a) *Prove that $T_0 K$ is isotropic for a given transformation $T_0 \in \mathrm{SL}(n)$ if and only if*

$$\int_{T_0 K} |x|^2 \, dx = B(K).$$

(b) *Show that if $T_0, T_1 \in \mathrm{SL}(n)$ are such that $K_0 = T_0 K$ and $K_1 = T_1 K$ are isotropic, then $T_1 = U T_0$ for some $U \in \mathrm{O}(n)$.*

Hint. If $T \in \mathrm{SL}(n)$, then we may use item **(i)** of Proposition 9.3.1 (main book) for $P = T^* T$.

Remark 9.6. Notice that this exercise gives that the isotropic position of a convex body is unique up to orthogonal transformations and rescalings.

Solution 9.19. Assume that $T_0 \in \mathrm{SL}(n)$ is such that $K_0 = T_0 K$ is isotropic (the existence of such T_0 is guaranteed by Exercise 9.17). From Exercise 9.18 we have that there exists a constant $\rho \in \mathbb{R}$ such that

$$\int_{K_0} \langle x, Sx\rangle \, dx = \rho^2 \mathrm{tr}(S)$$

for any $S \in \mathrm{L}(n)$. Letting $S = \mathrm{Id}_{\mathbb{R}^n}$, we get the equality

$$\int_{K_0} |x|^2 \, dx = n\rho^2.$$

For any $\bar{T} \in \mathrm{SL}(n)$, we have that $T = \bar{T}T_0^{-1} \in \mathrm{SL}(n)$. Thus, we may use standard change of variables and Proposition 9.3.1 from the main book (for $P = T^*T$), and compute:

$$\begin{aligned}\int_{\bar{T}K} |x|^2\,dx = \int_{TK_0} |x|^2\,dx = \int_{K_0} |Tx|^2\,dx = \int_{K_0} \langle x, T^*Tx\rangle\,dx = \\ = \rho^2 \mathrm{tr}(T^*T) \geq n\rho^2 = \int_{K_0} |x|^2\,dx.\end{aligned} \tag{9.10}$$

Since $\bar{T}$ is an arbitrary map in $\mathrm{SL}(n)$, this shows that

$$\int_{K_0} |x|^2\,dx = B(K).$$

Conversely, if $\bar{T} \in \mathrm{SL}(n)$ is such that

$$\int_{\bar{T}K} |x|^2\,dx = B(K),$$

then equality holds in (9.10), and from Proposition 9.3.1 we must have $T^*T = \mathrm{Id}_{\mathbb{R}^n}$, where $T = \bar{T}T_0^{-1}$. It follows that T is orthogonal. Hence we have that $\bar{T}K = TT_0K = T(T_0K)$, where T is orthogonal and T_0K is isotropic. Since isotropy is obviously invariant under orthogonal transformations, we get that $\bar{T}K$ is isotropic. This settles item **(a)**.

Finally, assume that $T_0, T_1 \in \mathrm{SL}(n)$ are such that $K_0 = T_0K$ and $K_1 = T_1K$ are isotropic. From the same argument as before (replacing $\bar{T}$ by T_1), the map $T_1T_0^{-1}$ is orthogonal. Letting $T_1T_0^{-1} = U$ we get item **(b)**.

□

Since we have the existence of an isotropic position for any convex body $K \subseteq \mathbb{R}^n$ having its centroid at the origin, item **(a)** of the previous exercise yields that the infimum in the definition of $B(K)$ given in (9.9) is actually a (positive) minimum. The number $L_K > 0$ such that

$$L_K^2 = \frac{1}{n}B(K)$$

is called the *isotropic constant* of K.

Exercise 9.20. **(a)** *In the same assumptions and notation as in Exercise 9.19, check that*

$$B(K) = \min\left\{\frac{1}{|\det(T)|^{1+\frac{2}{n}}}\int_{TK} |x|^2\,dx : T \in \mathrm{GL}(n)\right\}.$$

(b) *Verify that if K is isotropic, then*

$$\int_K \langle x, u\rangle^2\,dx = L_K^2$$

for every $u \in \mathbb{S}^{n-1}$.

(c) *Show that for any $\alpha \neq 0$ and every convex body $K \subseteq \mathbb{R}^n$ having its centroid at the origin we have*

$$L_{\alpha K} = |\alpha|^{1+\frac{n}{2}} L_K.$$

Remark 9.7. Item **(c)** of the previous exercise gives that our definition of the isotropic constant is not "invariant under rescaling" in a good way. In other words, rescaling K can make L_K become arbitrarily large or arbitrarily small. However, if we look at the class of convex bodies whose volume is unit, then we can ask about bounds for the isotropic constant. Alternatively, we may define the *normalized isotropic constant* as the number $\bar{L}_K > 0$ such that

$$\bar{L}_K^2 = \frac{1}{n} \min \left\{ \frac{1}{\mathrm{vol}_n(TK)^{1+\frac{2}{n}}} \int_{TK} |x|^2 \, dx : T \in \mathrm{GL}(n) \right\}.$$

Of course, we still work with convex bodies having their respective centroids at the origin. Also, it is easy to see that $\bar{L}_K$ is invariant in the linear class of K, and that $\bar{L}_K = L_K$ if K has unit volume.

Solution 9.20. **(a)** Since $\mathrm{SL}(n) \subseteq \mathrm{GL}(n)$, it is clear that

$$B(K) \geq \min \left\{ \frac{1}{|\det(T)|^{1+\frac{2}{n}}} \int_{TK} |x|^2 \, dx : T \in \mathrm{GL}(n) \right\}.$$

For the other inequality, we simply notice that any transformation $T \in \mathrm{GL}(n)$ can be written as $T = |\det(T)|^{1/n} \bar{T}$ for $\bar{T} \in \mathrm{SL}(n)$. Using standard change of variables, we arrive at

$$\frac{1}{|\det(T)|^{1+\frac{2}{n}}} \int_{TK} |x|^2 \, dx = \int_{\bar{T}K} |x|^2 \, dx,$$

and this settles **(a)**.

(b) If K is isotropic, then we know from Exercise 9.19 that

$$B(K) = \int_K |x|^2 \, dx,$$

from where we get that

$$\int_K |x|^2 \, dx = nL_K^2.$$

Also, letting $T = \mathrm{Id}_{\mathbb{R}^n}$ in item **(iii)** of Exercise 9.18, we get

$$\int_K |x|^2 \, dx = \rho^2 \mathrm{tr}(\mathrm{Id}_{\mathbb{R}^n}) = n\rho^2.$$

This leads to $\rho^2 = L_K^2$. Finally, from item **(i)** of the same exercise, we arrive at

$$\int_K \langle x, u \rangle^2 \, dx = \rho^2 = L_K^2.$$

(c) First, it is clear from the definition that $B(K) = B(-K)$, and hence $L_K = L_{-K}$. Thus, we may assume that $\alpha > 0$. In this case, we simply write $\alpha K = \alpha \mathrm{Id}_{\mathbb{R}^n} K$ and remember that $\det(\alpha \mathrm{Id}_{\mathbb{R}^n}) = \alpha^n$. Up to a linear transformation, we may assume that K is isotropic, and then αK is also isotropic. Hence, standard change of variables leads to

$$B(\alpha K) = \int_{\alpha K} |x|^2 \, dx = \int_K |\alpha x|^2 \alpha^n \, dx = \alpha^{n+2} \int_K |x|^2 \, dx = \alpha^{n+2} B(K).$$

Therefore,

$$L_{\alpha K}^2 = \frac{1}{n} B(\alpha K) = \alpha^{n+2} \frac{1}{n} B(K) = \alpha^{n+2} L_K^2.$$

Taking the square root on both sides yields the desired equality.

□

Exercise 9.21. **(a)** *Let $K \subseteq \mathbb{R}^n$ be a convex body having its centroid at the origin. Prove that*

$$\bar{L}_K \geq \bar{L}_{B^n}.$$

(b) *Prove that there exists a universal constant $c > 0$ such that*

$$\bar{L}_{B^n} \geq c$$

for every natural number $n \geq 1$.

Hint for **(b)**: Write $\bar{L}_{B^n}$ in terms of the volume κ_n of B^n and use Stirling's approximation (see (11.5.2) in the main book) for the Gamma function.

Solution 9.21. **(a)** Since B^n is isotropic, we have that

$$nL_{B^n}^2 = \int_{B^n} |x|^2 \, dx.$$

Let $r_n = \kappa_n^{-1/n}$. Because $r_n B^n$ has unit volume, we have

$$\bar{L}_{B^n}^2 = L_{r_n B^n}^2 = \frac{1}{n} \int_{r_n B^n} |x|^2 \, dx.$$

Since $\bar{L}_K$ is invariant under replacing K by some of its linear images, we may assume that K is isotropic and has unit volume. Thus, $\bar{L}_K = L_K$, and this leads to

$$n\bar{L}_K^2 = \int_K |x|^2 \, dx = \int_{K \cap r_n B^n} |x|^2 \, dx + \int_{K \setminus r_n B^n} |x|^2 \, dx. \tag{9.11}$$

Since both K and $r_n B^n$ have the same volume, we get that

$$\begin{aligned} \mathrm{vol}_n(K \cap r_n B^n) + \mathrm{vol}_n(K \setminus r_n B^n) = \mathrm{vol}_n(K) = \mathrm{vol}_n(r_n B^n) = \\ = \mathrm{vol}_n(K \cap r_n B^n) + \mathrm{vol}_n(r_n B^n \setminus K), \end{aligned}$$

from where we obtain the equality

$$\mathrm{vol}_n(K \setminus r_n B^n) = \mathrm{vol}_n(r_n B^n \setminus K).$$

Moreover, it is immediate to see that for $x \in K \setminus r_n B^n$ we have $|x| > r_n$, whereas if $x \in r_n B^n \setminus K$, then $|x| \leq r_n$. It follows that

$$\int_{K \setminus r_n B^n} |x|^2 \, dx \geq \int_{r_n B^n \setminus K} |x|^2 \, dx.$$

Using this inequality in (9.11), we arrive at

$$n\bar{L}_K^2 \geq \int_{K\cap r_n B^n} |x|^2\,dx + \int_{r_n B^n\setminus K} |x|^2\,dx = \int_{r_n B^n} |x|^2\,dx = n\bar{L}_{B^n}^2,$$

and this gives that $\bar{L}_K \geq \bar{L}_{B^n}$.

Now we prove the remaining bound. Since $r_n B^n$ has unit volume, and using item **(c)** of Exercise 9.20, we have that

$$\bar{L}_{B^n}^2 = L_{r_n B^n}^2 = r_n^{n+2} L_{B^n}^2 = \frac{r_n^{n+2}}{n}\int_{B^n} |x|^2\,dx.$$

The last integral can be easily computed using integration by polar coordinates. We write

$$\bar{L}_{B^n}^2 = \frac{r_n^{n+2}}{n}\int_0^1\int_{\mathbb{S}^{n-1}} t^{n+1}\,d\mathcal{H}^{n-1}dt = \frac{\kappa_n^{-(n+2)/n}}{n}\omega_n\left[\frac{t^{n+2}}{n+2}\right]\Bigg|_0^1 = \frac{\kappa_n^{-1-2/n}}{n}\frac{n\kappa_n}{n+2} = \frac{\kappa_n^{-2/n}}{n+2}.$$

Thus, from Proposition 4.3.1 and (11.5.2) in the main book, we get that

$$\bar{L}_{B^n}^2 \sim \frac{\left(\sqrt{2\pi}e^{-n/2}\left(\frac{n}{2}\right)^{(n+1)/2}\right)^{2/n}}{\pi(n+2)} = \frac{e^{-1}(2\pi)^{1/n}\left(\frac{n}{2}\right)^{1+1/n}}{\pi(n+2)} = \frac{ne^{-1}\left(2\pi\frac{n}{2}\right)^{1/n}}{2\pi(n+2)} =$$
$$= \frac{1}{2e\pi}\frac{n}{n+2}(\pi n)^{1/n},$$

and by standard calculus we have that the latter goes to 1 as $n\longrightarrow+\infty$. Since $\kappa_n > 0$ for every $n\in\mathbb{N}$, a trivial analysis argument gives that $\bar{L}_{B^n} \geq c^2$ for some constant $c > 0$ independent of the dimension n.

□

In the next exercise, we estimate $\bar{L}_K$ by the above in the case where K is symmetric with respect to the origin.

Exercise 9.22. *Prove that there exists a universal constant $c > 0$ such that*

$$\bar{L}_K \leq c\sqrt{n}$$

for every convex body $K\subseteq\mathbb{R}^n$ which is symmetric with respect to the origin.

Hint: Up to a linear transformation, we may assume that K is in John position, and up to a further rescaling, we may assume that K has volume 1 (the maximal ellipsoid inscribed in K becomes rB^n for some $r > 0$). In this case, check that $r \leq \kappa_n^{-1/n}$. Using Proposition 4.3.1 and (11.5.2) from the main book, show that $r \leq c\sqrt{n}$ for some universal constant c. Then, use Theorem 9.2.2 (main book).

Solution 9.22. Both the isotropic constant and the Banach-Mazur distance are invariant for affine transformations over K, and hence we may assume that K is in John position. Moving forward, up to a rescaling we may assume that $\mathrm{vol}_n(K) = 1$. In this case, we cannot guarantee anymore that B^n is the maximal ellipsoid inscribed in K; in fact, there exists a number $r > 0$ such that rB^n is the maximal ellipsoid inscribed in K. From Theorem 9.2.2 (main book) it follows that

$$K\subseteq r\sqrt{n}B^n.$$

Because $rB^n \subseteq K$ and $\mathrm{vol}_n(K) = 1$, we have

$$r^n \kappa_n \leq 1,$$

and hence

$$r \leq \kappa_n^{-1/n}.$$

From Proposition 4.3.1 and Stirling's approximation (11.5.2), both from the main book, we get

$$\kappa_n^{-1/n} = \frac{1}{\sqrt{\pi}} \Gamma\left(\frac{n}{2}+1\right)^{1/n} \sim \frac{1}{\sqrt{\pi}} \left(\sqrt{2\pi} e^{-n/2} \left(\frac{n}{2}\right)^{(n+1)/2}\right)^{1/n} =$$
$$= \frac{1}{\sqrt{\pi}} (2\pi)^{\frac{1}{2n}} e^{-1/2} \left(\frac{n}{2}\right)^{\frac{1}{2}+\frac{1}{2n}} = \frac{e^{-1/2}}{\sqrt{2\pi}} \sqrt{n} (\pi n)^{\frac{1}{2n}}.$$

By standard calculus, we have that $(\pi n)^{1/(2n)} \sim 1$, and hence it follows that

$$r \leq c\sqrt{n}$$

for some constant $c > 0$ which does not depend on the dimension (or on K). Finally, from the definition of $\bar{L}_K$ (and recalling that the volume of K is unit), we may write

$$\bar{L}_K^2 \leq \frac{1}{n} \int_K |x|^2 \, dx.$$

Because $K \subseteq r\sqrt{n}^n$, we have that $|x|^2 \leq r^2 n$ for any $x \in K$. Thus,

$$\bar{L}_K^2 \leq \frac{1}{n} r^2 n \mathrm{vol}_n(K) = r^2.$$

Therefore, we get that $\bar{L}_K \leq r \leq c\sqrt{n}$, and the proof is complete.

□

Exercise 9.23. *Let $K \subseteq \mathbb{R}^n$ be a convex body which is symmetric with respect to the origin and such that $\mathrm{vol}_n(K) = 1$. Let $M : \mathbb{R}^n \to \mathbb{R}^n$ be the inertia map of K, and let $S : \mathbb{R}^n \to \mathbb{R}^n$ be the square root of M (see Exercise 9.16 for the existence of this square root). Define the ellipsoid*

$$\mathcal{E}_{\mathcal{B}}(K) = S^{-1} B^n.$$

This is called the Binet ellipsoid of K.

(a) *Prove that K is isotropic if and only if*

$$\mathcal{E}_{\mathcal{B}}(K) = L_K^{-1} B^n.$$

(b) *Still assuming that the convex body K is symmetric with respect to the origin and has unit volume, but is not necessarily isotropic, show that*

$$\mathrm{vol}_n(\mathcal{E}_{\mathcal{B}}(K)) = \kappa_n L_K^{-n}.$$

Solution 9.23. **(a)** Let M_K be the inertia matrix of K, and let S_K be its square root. Assume that K is isotropic. From Remark 9.5 we have that M_K is a multiple of the identity, and from item **(b)** of

Exercise 9.20 we must have $M_K = L_K^2 \mathrm{Id}_{\mathbb{R}^n}$. Hence

$$\mathcal{E}_{\mathcal{B}}(K) = S_K^{-1} B^n = L_K^{-1} B^n.$$

On the other hand, if the equality above holds, then

$$S_K^{-1} B^n = L_K^{-1} B^n.$$

Let $T = L_K S_K^{-1}$. We know that T is an isomorphism, and we claim that $|Tx| = |x|$ for every $x \in \mathrm{R}^n$. In fact, the equality above gives that $TB^n = B^n$. If $|Tx| > |x|$, then $x \neq 0_{\mathbb{R}^n}$, and

$$\left| T\left(\frac{x}{|x|} \right) \right| > 1,$$

and hence $T(x/|x|) \notin B^n$, a contradiction. If $|Tx| < |x|$, then letting $x = T^{-1}y$ we have that $|T^{-1}y| > |y|$. Since $T^{-1}B^n = B^n$, repeating the previous argument we get a contradiction. It follows that $|Tx| = |x|$ for any $x \in \mathbb{R}^n$, indeed. From Lemma 2.6.5 (main book) we get that T is an orthogonal transformation. Consequently, $T^* = T^{-1}$. Thus,

$$(L_K S_K^{-1})^* = (L_K S_K^{-1})^{-1}.$$

Because S_K is symmetric, and since adjoining and inverting commute, this leads to

$$L_K S_K^{-1} = L_K^{-1} S_K.$$

Finally, we arrive at

$$M_K = S_K^2 = L_K^2 \mathrm{Id}_{\mathbb{R}^n}.$$

That is, the inertia matrix of K is a multiple of the identity. From Remark 9.5, we get that K is isotropic.

(b) We may write $K = TK_0$ for some $T \in \mathrm{SL}_n$ and some K_0 which is in isotropic position. For each convex body L, denote by M_L the inertia map of L and by S_L its respective square root. On the one hand, we have $\mathcal{E}_{\mathcal{B}}(K_0) = S_{K_0}^{-1} B^n$, and hence

$$\mathrm{vol}_n(\mathcal{E}_{\mathcal{B}}(K_0)) = |\det(S_{K_0})|^{-1} \kappa_n = |\det(M_{K_0})|^{-1/2} \kappa_n.$$

On the other hand, since K_0 is isotropic, item **(a)** gives that

$$\mathrm{vol}_n(\mathcal{E}_{\mathcal{B}}(K_0)) = L_K^{-n} \kappa_n.$$

Now we write

$$\mathcal{E}_{\mathcal{B}}(K) = S_K^{-1} B^n = S_{TK_0}^{-1} B^n.$$

Using Exercise 9.16 and the previous equalities, and recalling that $T \in \mathrm{SL}(n)$, we finally arrive at

$$\begin{aligned} \mathrm{vol}_n(\mathcal{E}_{\mathcal{B}}(K)) = |\det(S_{TK_0})|^{-1} \kappa_n = |\det(M_{TK_0})|^{-1/2} \kappa_n = |\det(TM_{K_0}T^*)|^{-1/2} \kappa_n = \\ = |\det(M_{K_0})|^{-1/2} \kappa_n = \mathrm{vol}_n(\mathcal{E}_{\mathcal{B}}(K)) = L_K^{-n} \kappa_n, \end{aligned}$$

and the solution is complete.

□

Exercise 9.24. *Let $K \subseteq \mathbb{R}^n$ be a convex body symmetric with respect to the origin, and assume that* $\mathrm{vol}_n(K) = 1$.

(a) *Show that*

$$||y||_{\mathcal{E}_{\mathcal{B}}(K)} = \left(\int_K \langle x, y \rangle^2 \, dx \right)^{1/2}$$

for every $y \in \mathbb{R}^n$.

(b) *Prove that there exists $u_0 \in \mathbb{S}^{n-1}$ such that*

$$\int_K \langle x, u_0 \rangle^2 \, dx \leq L_K^2.$$

Hint: Use the equality in item **(b)** of Exercise 9.23 and write the volume of $\mathcal{E}_{\mathcal{B}}(K)$ using formula (4.3.6) from the main book.

Solution 9.24. (a) We know that $\mathcal{E}_{\mathcal{B}}(K) = S^{-1}B^n$ and $S^2 = M$, where M is the inertia map of K. Both M and S are symmetric and positive-definite. Thus, we have

$$\begin{aligned} ||y||_{\mathcal{E}_{\mathcal{B}}(K)} = ||y||_{S^{-1}B^n} = ||Sy||_{B^n} = \langle Sy, Sy \rangle^{1/2} = \langle y, S^2 y \rangle^{1/2} = \langle y, My \rangle^{1/2} = \\ = \left\langle y, \int_K \langle x, y \rangle x \, dx \right\rangle^{1/2} = \left(\int_K \langle x, y \rangle^2 \, dx \right)^{1/2} \end{aligned}$$

for any $y \in \mathbb{R}^n$.

(b) From item **(b)** of Exercise 9.23, and using formula (4.3.6) in the main book, we may write

$$L_K^{-n} = \frac{\mathrm{vol}_n(\mathcal{E}_{\mathcal{B}}(K))}{\kappa_n} = \int_{\mathbb{S}^{n-1}} ||u||_{\mathcal{E}_{\mathcal{B}}(K)}^{-n} \, d\sigma(u),$$

where σ denotes the standard probability measure on $\mathbb{S}^{n-1}$. Now we trivially bound the integral in the right-hand side to arrive at

$$L_K^{-n} = \int_{\mathbb{S}^{n-1}} ||u||_{\mathcal{E}_{\mathcal{B}}(K)}^{-n} \, d\sigma(u) \geq \min_{u \in \mathbb{S}^{n-1}} ||u||_{\mathcal{E}_{\mathcal{B}}(K)}^{-n} = \left(\min_{u \in \mathbb{S}^{n-1}} ||u||_{\mathcal{E}_{\mathcal{B}}(K)} \right)^{-n},$$

and then

$$L_K \geq \min_{u \in \mathbb{S}^{n-1}} ||u||_{\mathcal{E}_{\mathcal{B}}(K)}.$$

By continuity, we may assume that the minimum is obtained for some $u_0 \in \mathbb{S}^{n-1}$. Hence, using item **(a)** we get

$$L_K \geq ||u_0||_{\mathcal{E}_{\mathcal{B}}(K)} = \left(\int_K \langle x, u_0 \rangle^2 \, dx \right)^{1/2},$$

and squaring both sides we get what we wanted to show.

□

For Further Reading

A *position* of an n-dimensional convex body K is the image of K obtained via a non-degenerate affine transformation (often restricted to the volume-preserving cases). A popular example is the *John position* of K, meaning that the John ellipsoid of K is then the Euclidean unit ball. Further on, K is said to be in *isotropic position* (or shortly *isotropic*) if its volume is 1, the origin coincides with its center of gravity, and the covariance matrix of a random vector uniformly distributed in K is the identity matrix times a constant, which is called the *isotropic constant of* K. Up to orthogonal transformations, each convex body K has a unique affine image that is isotropic, and the Euclidean ball has the smallest isotropic constant, thus giving a lower bound for the isotropic constant by an absolute positive constant. In this text part we restrict to literature (for further reading) only treating (isotropic) positions.

Positions as Subjects of Book Chapters and Surveys

In Section 10.13 of his book [228], Schneider summarizes the relations between the notions of positions, isotropic measures, and reverse inequalities in convexity. He discusses various references collected also in subsection 9.6.1 of our main book, e.g. treating John and Löwner ellipsoids, mean width, and basic (reverse) inequalities, as, for instance, the Ball-Barthe inequality for isotropic measures. Roughly at the same time as Schneider's second edition [228] appeared, the book [70] of Brazitikos et al. came to be, mainly dedicated to *isotropic positions* and *isotropic convex bodies*. In Chapter 1 *classical positions* (like that of John, minimal mean width, and minimal surface area) and reverse inequalities are studied in a detailed manner, and in Chapter 2 *isotropic log-concave measures*, their concentration inequalities, and the convex bodies associated with them are discussed. The known *isotropic constant conjecture* follows, and also convex bodies with maximal isotropic constants are investigated. Further on, the monograph [24] of Artstein-Avidan et al. contains a lot of material on positions, e.g. in Chapter 2 the classical positions also considered in [70] and reverse inequalities. Chapter 8 is dedicated to the M-position (see below), and Chapter 10 treats the isotropic position and isotropic log-concave measures. (Note that a convex body K of unit volume is said to be in *M-position* with constant $c > 0$ if the volume of $(K + B_n)^{1/n}$ is not larger than c, where B_n is a ball having also unit volume.) In their second part [25], Artstein-Avidan et al. study *isotropic log-concave probability measures* in Section 2.1, and *isotropic measures and positions* are studied in subsection 7.3.1, also in connection with *Steiner symmetrization* as useful tool. Finally, we cite here the main book [29], where longer discussions of *Löwner-John ellipsoids* and *positions* are given in subsections 9.6.1 and 9.6.2. We should mention also two essential surveys showing basic results on isotropic positions, related ellipsoids, and further such material close to convex geometry, namely, [196] by Milman and Pajor as well as [39] of Bastero, Bernués, and Romance.

Selected Research Results on Positions

We continue with a few selected results on (mainly isotropic) positions which are not discussed in the books mentioned above (except for [29]). The question whether every convex body of unit volume in minimal surface area position is also in M-position goes back to Giannopoulos and Milman. It was negatively answered by Saroglou [222], even already for *1-unconditional convex bodies* (which are symmetric to all coordinate hyperplanes), and further related results (also on isotropic measures) are obtained, too.

Results on *isotropy constants of convex polytopes* with given vertex or facet numbers (known before only for the symmetric case) are proved by Alonso-Gutiérrez [8]; he elegantly uses John and Löwner positions. Also the paper [13] treats polytopes. As for lower bounds on the isotropic constant (see above), one might ask whether there is an upper bound being also an absolute constant. Alonso-

Gutiérrez et al. [13] estimate the isotropic constant of a *random polytope in an isotropic convex body* K from above, using the symmetric convex hull of independent random vectors uniformly distributed in K. They succeed in getting an interesting upper bound.

Giannopoulos et al. [104] derive new inequalities comparing the surface area of a convex body K with the minimal, average, or maximal surface area of its hyperplanar (or lower dimensional) projections. Furthermore, the authors consider analogous problems for quermassintegrals of K, using as tools also the *minimal surface area position* and the volume of the projection body of K. The obtained results are nicely related to the famous *hyperplane conjecture*, with sections replaced by projections, and volume replaced by surface area.

González Merino and Schymura [108] obtain sharp estimates and asymptotics on the *isodiametric and isominwidth quotients* of a convex body K (i.e., the volume of K divided by the nth power of its diameter, and the nth power of its minimum width, respectively). For the isodiametric case (and for the isodiametric position) they get nice lower bounds and derive, for symmetric bodies, a nice characterization of the regular cross-polytope. Regarding the isominwidth case the cube occurs as extremal case, and the relation to John's and other positions are discussed, too. The authors present also a little survey, thus giving a nice view on related results and problems.

PLATE 10

Zonotopes and related polytope types

Translative space-fillers

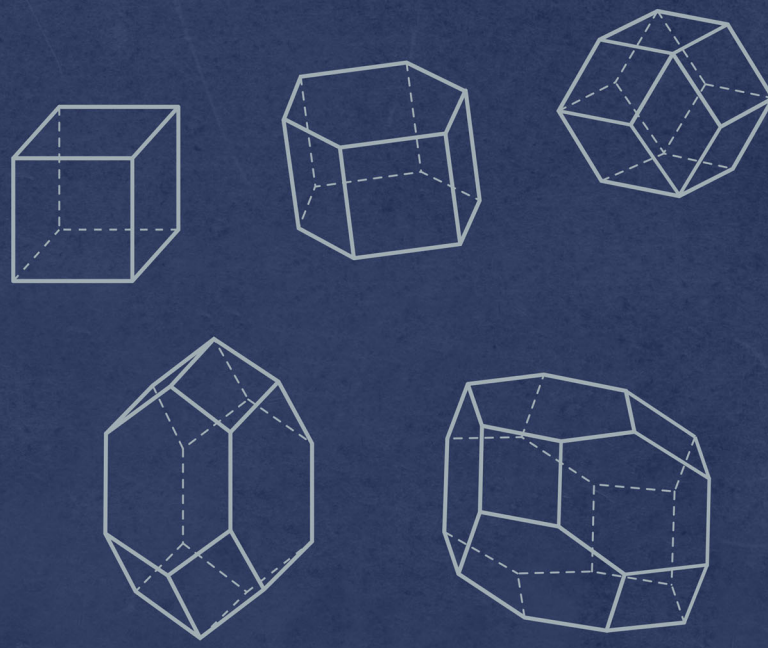
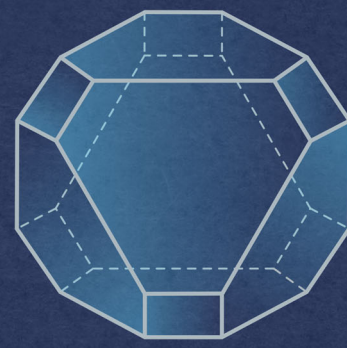


Figure 27

A belt polytope

Figure 28



Zonotopes and related polytope types ► (see, e.g., 2.9.1.11, 10.2, and 10.6.1)

Chapter 10
Affine Convex Geometry—Part 2



"If I feel unhappy, I do mathematics to become happy. If I am happy, I do mathematics to keep happy."

— Alfred Rényi

In this chapter, the following notions play the essential roles: *zonotopes* (finite sums of segments), *zonoids* (limits of zonotopes), *projection bodies* (zonotopes or zonoids whose support function values present the $(n-1)$-dimensional orthogonal projection areas of given n-dimensional convex polytopes or convex bodies), *polars of projection bodies, mixed projection bodies and projection maps, centroid bodies*, and *valuations* (for the latter, see also Chapter 11).

Plate 10:

Plate 10 (Figure 27) shows the five types of three-dimensional *translative space fillers* which are *special zonotopes*, i.e., special cases of finite vector sums of line segments. (Note that the translative space fillers in dimensions higher than three are no longer necessarily zonotopes.) The class of zonotopes is identical with the family of all *totally centrally symmetric polytopes*, i.e., of those convex polytopes all whose faces (of any dimension) are centrally symmetric. In Figure 28 a *belt polytope* can be seen, namely, a convex 3-polytope that has a segment summand parallel to each of its edges (a necessary, but not sufficient condition for yielding a zonotope). And for belt polytopes and the related, more general class of *belt bodies*, we refer to Chapter 7 of the monograph [57], where the combinatorial geometry of these types of convex polytopes and bodies is studied. Zonotopes and related types of polytopes are discussed in parts 2.9.1.11, 10.2, and 10.6.1 of the main book.

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V. Balestro et al., *Convexity from the Geometric Point of View: Exercises and Solutions*, Cornerstones, https://doi.org/10.1007/978-3-031-83937-5_10

Exercises and Solutions

Exercise 10.1. *Give a formal proof of the fact that the family described in (10.1.1) of the main book is a Minkowski class.*

Solution 10.1. The only non-trivial part is that the Hausdorff convergence is preserved under linear maps. This is proved in Exercise 2.50.

□

Exercise 10.2. *Prove that a set $Z \subseteq \mathbb{R}^n$ is a zonotope centered at the origin if and only if there exists a number $k \in \mathbb{N}$ and a linear map $T\colon \mathbb{R}^k \to \mathbb{R}^n$ such that $T[-1,1]^k = Z$.*

Remark 10.1. This exercise provides a classical characterization of zonotopes (see also Plate 10) as projections of cubes. It is enough to work with zonotopes which are centered at the origin, of course. Indeed, we already know that every zonotope can be translated to be centered at the origin.

Solution 10.2. Let $Z \subseteq \mathbb{R}^n$ be a zonotope given as

$$Z = \beta_1 \operatorname{seg}[-u_1, u_1] + \ldots + \beta_k \operatorname{seg}[-u_k, u_k]$$

for $\beta_1, \ldots, \beta_m > 0$ and $u_1, \ldots, u_m \in \mathbb{S}^{n-1}$. Then it is clear that

$$Z = \left\{\alpha_1\beta_1 u_1 + \ldots + \alpha_k\beta_k u_k : -1 \le \alpha_1, \ldots, \alpha_k \le 1\right\}.$$

Denote by $\{e_1^k, \ldots, e_k^k\}$ the canonical basis of $\mathbb{R}^k$, and let $T\colon \mathbb{R}^k \to \mathbb{R}^n$ be the linear map such that $Te_j^k = \beta_j u_j$ for each $j \in \{1, \ldots, k\}$. It is immediate to verify that $T[-1,1]^k = Z$.

On the other hand, if $T\colon \mathbb{R}^k \to \mathbb{R}^n$ is a linear map, then

$$\begin{aligned} T[-1,1]^k = \left\{\alpha_1 Te_1^k + \ldots + \alpha_k Te_k^k : -1 \le \alpha_1, \ldots, \alpha_k \le 1\right\} = \\ = \operatorname{seg}[-Te_1^k, Te_1^k] + \ldots + \operatorname{seg}[-Te_k^k, Te_k^k], \end{aligned}$$

and hence $T[-1,1]^k$ is a zonotope.

□

Exercise 10.3. *Let $P \subseteq \mathbb{R}^n$ be a polytope whose facets are $F_1, \ldots, F_m$ with corresponding unit outer normal vectors $u_1, \ldots, u_m$. Prove that*

$$\Pi P = \sum_{j=1}^{m} \operatorname{seg}\left[0, \operatorname{vol}_{n-1}(F_j)u_j\right].$$

That is, the projection body of a polytope is the sum of the outer normal vectors of the facets (considered as segments), respectively, rescaled to have their lengths equal to the areas of the corresponding facets. See Figure 10.1.

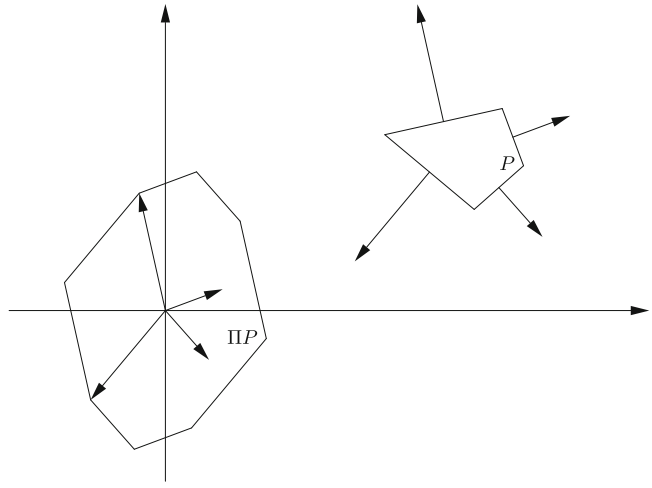


Fig. 10.1 The scaled outer normal vectors are $\mathrm{vol}_{n-1}(F_i)u_i$. The Minkowski sum of the corresponding segments (starting at the origin) is ΠP

Solution 10.3. Use (10.3.2) from the main book to write

$$\Pi P = \frac{1}{2}\sum_{j=1}^{m} \mathrm{seg}\big[-\mathrm{vol}_{n-1}(F_j)u_j, \mathrm{vol}_{n-1}(F_j)u_j\big] =$$

$$= \frac{1}{2}\sum_{j=1}^{m}\Big[-\mathrm{vol}_{n-1}(F_j)u_j + \mathrm{seg}\big[0, \mathrm{vol}_{n-1}(F_j)u_j\big]\Big] =$$

$$= -\frac{1}{2}\sum_{j=1}^{m}\mathrm{vol}_{n-1}(F_j)u_j + \frac{1}{2}\sum_{j=1}^{m}\mathrm{seg}\big[0, \mathrm{vol}_{n-1}(F_j)u_j\big].$$

Since the surface area measure of P has its centroid at the origin (as it is the case for the surface area measure of any convex body; see Section 7.3 of the main book), we get that

$$\sum_{j=1}^{m}\mathrm{vol}_{n-1}(F_j)u_j = 0_{\mathbb{R}^n},$$

and the proof is complete.

□

Exercise 10.4. *Let $Z \subseteq \mathbb{R}^n$ be a zonotope (centered at the origin) given as*

$$Z = \sum_{k=1}^{m}\beta_k \mathrm{seg}[-u_k, u_k],$$

for $\beta_1, \ldots, \beta_k > 0$ and $u_1, \ldots, u_k \in \mathbb{S}^{n-1}$, and denote by

$$\mu := \sum_{k=1}^{m}\frac{\beta_k}{2}\big(\delta_{u_k} + \delta_{-u_k}\big)$$

its generating measure. Given $C_1, \dots, C_{n-j} \in CC^n$, we write for simplicity

$$\mathbf{C}|\{v_1, \dots, v_j\}^\perp := \big(C_1|\{v_1, \dots, v_j\}^\perp, \dots, C_{n-j}|\{v_1, \dots, v_j\}^\perp\big),$$

where $A|\{v_1, \dots, v_j\}^\perp$ denotes the orthogonal projection of the set $A \subseteq \mathbb{R}^n$ onto the orthogonal complement of the vector subspace $\operatorname{span}\{v_1, \dots, v_j\}$*, that is,*

$$A|\{v_1, \dots, v_j\}^\perp := \operatorname{proj}_{(\operatorname{span}\{v_1, \dots, v_j\})^\perp}(A). \tag{10.1}$$

(a) *Show that for any $1 \le j < n$ and $C_1, \dots, C_{n-j} \in CC^n$ it holds that*

$$\frac{n!}{2^j(n-j)!} V(Z[j], C_1, \dots, C_{n-j}) =$$
$$= \int_{\mathbb{S}^{n-1}} \dots \int_{\mathbb{S}^{n-1}} D_j(v_1, \dots, v_j) V^{(n-j)}\big(\mathbf{C}|\{v_1, \dots, v_j\}^\perp\big)\, d\mu(v_1) \dots d\mu(v_j).$$

(b) *For a given $1 \le j < n$, let $Z_1, \dots, Z_j$ be zonotopes as above with generating measures $\mu_1, \dots, \mu_j$, respectively. Prove that*

$$\frac{n!}{2^j(n-j)!} V(Z_1, \dots, Z_j, C_1, \dots, C_{n-j}) =$$
$$= \int_{\mathbb{S}^{n-1}} \dots \int_{\mathbb{S}^{n-1}} D_j(v_1, \dots, v_j) V^{(n-j)}\big(\mathbf{C}|\{v_1, \dots, v_j\}^\perp\big)\, d\mu_1(v_1) \dots d\mu_j(v_j)$$

for any $C_1, \dots, C_{n-j} \in CC^n$.

(c) *Prove that the formula in item* **(b)** *still holds true if $Z_1, \dots, Z_j$ are zonoids with generating measures $\mu_1, \dots, \mu_j$, respectively.*

Solution 10.4. **(a)** Using the linearity of mixed volumes for Minkowski combinations, we write

$$V(Z[j], C_1, \dots, C_{n-j}) = V\left(\left(\sum_{i=1}^m \beta_i \operatorname{seg}[-u_i, u_i]\right)[j], C_1, \dots, C_{n-j}\right) =$$
$$= \sum_{1 \le i_1, \dots, i_j \le m} \beta_{i_1} \dots \beta_{i_j} V(\operatorname{seg}[-u_{i_1}, u_{i_1}], \dots, \operatorname{seg}[-u_{i_j}, u_{i_j}], C_1, \dots, C_{n-j}) =$$
$$= \sum_{1 \le i_1, \dots, i_j \le m} \beta_{i_1} \dots \beta_{i_j} V(\operatorname{seg}[-u_{i_1}, u_{i_1}], \dots, \operatorname{seg}[-u_{i_j}, u_{i_j}], C_1, \dots, C_{n-j}).$$

From Exercise 6.17, we have

$$V(\operatorname{seg}[-u_{i_1}, u_{i_1}], \dots, \operatorname{seg}[-u_{i_j}, u_{i_j}], C_1, \dots, C_{n-j}) =$$
$$= \frac{2^j(n-j)!}{n!} D_j(u_{i_1}, \dots, u_{i_j}) V^{(n-j)}(C_1|\{u_{i_1}, \dots, u_{i_j}\}^\perp, \dots, C_{n-j}|\{u_{i_1}, \dots, u_{i_j}\}^\perp).$$

Thus, we arrive at

$$V(Z[j], C_1, \dots, C_{n-j}) =$$
$$= \frac{2^j(n-j)!}{n!} \sum_{1 \le i_1, \dots, i_j \le m} \beta_{i_1} \dots \beta_{i_j} D_j(u_{i_1}, \dots, u_{i_j}) V^{(n-j)}(\mathbf{C}|\{u_{i_1}, \dots, u_{i_j}\}^\perp). \tag{10.2}$$

Now we compute the integral. First we observe that the function

$$(v_1, \ldots, v_j) \mapsto D_j(v_1, \ldots, v_j) V^{(n-j)}(\mathbf{C}|\{v_1, \ldots, v_j\}^\perp)$$

is clearly even in each entry, since D_j is obviously even and

$$\{v_1, \ldots, v_k, \ldots, v_j\}^\perp = \{v_1, \ldots, -v_k, \ldots, v_j\}^\perp.$$

Thus, integrating this function with respect to μ is the same as integrating it with respect to the measure

$$\bar{\mu} := \sum_{i=1}^{m} \beta_i \delta_{u_i},$$

and we recall that the integral of a function f with respect to that measure is just $\beta_1 f(u_1) + \ldots + \beta_m f(u_m)$. Hence, integrating iteratively we compute

$$\begin{aligned}
&\int_{\mathbb{S}^{n-1}} \cdots \int_{\mathbb{S}^{n-1}} D_j(v_1, \ldots, v_j) V^{(n-j)}(\mathbf{C}|\{v_1, \ldots, v_j\}^\perp)\, d\mu(v_1) \ldots d\mu(v_j) = \\
&= \int_{\mathbb{S}^{n-1}} \cdots \int_{\mathbb{S}^{n-1}} \left(\sum_{i_1=1}^{m} \beta_i D_j(u_{i_1}, v_2, \ldots, v_j) V^{(n-j)}(\mathbf{C}|\{u_{i_1}, v_2, \ldots, v_j\}^\perp) \right) d\mu(v_2) \ldots d\mu(v_j) = \\
&= \ldots = \sum_{i_j=1}^{m} \cdots \sum_{i_1=1}^{m} \beta_{i_1} \ldots \beta_{i_j} D_j(u_{i_1}, \ldots, u_{i_j}) V^{(n-j)}(\mathbf{C}|\{u_{i_1}, \ldots, u_{i_j}\}^\perp).
\end{aligned}$$

Comparing this and (10.2) yields the desired equality.

(b) Let $Z = t_1 Z_1 + \ldots + t_j Z_j$, $t_1, \ldots, t_j \geq 0$, in the equality of item **(a)**. It is clear that Z is a zonotope whose generating measure is

$$\mu := t_1 \mu_1 + \ldots + t_j \mu_j.$$

Clearly, both sides of the equality in item **(a)** become polynomials on $t_1, \ldots, t_j$. Comparing the coefficients of the term $t_1 \ldots t_j$, we get what we wanted to show.

(c) We just have to approximate each zonoid by a sequence of zonotopes. Then, we use the continuity of mixed volumes, Scholium 10.2.1 and Proposition 10.2.2 (main book).

□

Exercise 10.5. *Let $Z \subseteq \mathbb{R}^n$ be a proper zonotope given by*

$$Z = \sum_{j=1}^{m} \alpha_j \mathrm{seg}[-u_j, u_j],$$

for $u_1, \ldots, u_m \in \mathbb{S}^{n-1}$ and $\alpha_1, \ldots, \alpha_m > 0$. Prove that

$$\mathrm{vol}_n(Z) = \frac{2}{n} \sum_{j=1}^{m} \alpha_j \mathrm{vol}_{n-1}\left(\mathrm{proj}_{u_j^\perp}(Z)\right).$$

Solution 10.5. We simply use equality (7.2.6) from the main book and linearity of mixed volumes. We write

$$\mathrm{vol}_n(Z) = V(Z,\dots,Z) = V\left(\sum_{j=1}^m \alpha_j \mathrm{seg}[-u_j,u_j], Z,\dots,Z\right) =$$
$$= \sum_{j=1}^m \alpha_j V\big(\mathrm{seg}[-u_j,u_j],Z,\dots,Z\big) = \sum_{j=1}^m \alpha_j \frac{2}{n}\mathrm{vol}_{n-1}\big(\mathrm{proj}_{u_j^\perp}(Z)\big).$$

□

Exercise 10.6. *Let $u_1,\dots,u_m \in \mathbb{S}^{n-1}$ and $c_1,\dots,c_m > 0$ be such that*

$$\mathrm{Id}_{\mathbb{R}^n} = \sum_{j=1}^m c_j u_j \otimes u_j.$$

For any $\alpha_1,\dots,\alpha_m > 0$, consider the zonotope

$$Z = \sum_{j=1}^m \alpha_j \mathrm{seg}[-u_j,u_j].$$

Prove that

$$\mathrm{vol}_n(Z) \geq 2^n \prod_{j=1}^m \left(\frac{\alpha_j}{c_j}\right)^{c_j}.$$

Hint: Use Exercise 10.5 and induction on the dimension. A formal and detailed proof can be found in [31].

Solution 10.6. From Remark 9.2.3 (main book), the decomposition of the identity given in the formulation of the exercise yields

$$\sum_{j=1}^m c_j = n.$$

In the case where $n = 1$, Z is the interval

$$Z = [-(\alpha_1 + \dots + \alpha_m), \alpha_1 + \dots + \alpha_m],$$

and hence

$$\mathrm{vol}_1(Z) = 2\sum_{j=1}^m \alpha_j = 2\sum_{j=1}^m c_j\left(\frac{\alpha_j}{c_j}\right) \geq 2\prod_{j=1}^m \left(\frac{\alpha_j}{c_j}\right)^{c_j},$$

where the last inequality follows from the inequality of arithmetic and geometric means given in Exercise 1.12 (by the above, the sum of the c_j's is 1). Hence the result is proved for $n = 1$.

Now we assume that $n > 1$, and that the result holds true for dimension strictly smaller than n. For simplicity, we denote each projection $\mathrm{proj}_{u_j^\perp}$ by P_j. From Exercise 10.5, we have

$$\begin{aligned}\mathrm{vol}_n(Z) = \frac{2}{n}\sum_{i=1}^m \alpha_i \mathrm{vol}_{n-1}(P_i(Z)) = 2\sum_{j=1}^m \frac{c_i}{n}\frac{\alpha_i}{c_i}\mathrm{vol}_{n-1}(P_i(Z)) \geq \\ \geq 2\prod_{i=1}^m \left(\frac{\alpha_i}{c_i}\mathrm{vol}_{n-1}(P_i(Z))\right)^{\frac{c_i}{n}},\end{aligned} \tag{10.3}$$

where in the last inequality we used the inequality of the arithmetic and geometric means again (observe that the sum of the c_i's is n).

For each $i = 1, \ldots, m$, the projection $P_i(Z)$ is the zonotope

$$P_i(Z) = \sum_{j=1}^m \alpha_j \mathrm{seg}[-P_i(u_j), P_i(u_j)].$$

The vectors $P_i(u_j)$ are not necessarily unit. Thus, we write $\beta_{ij} = |P_i(u_j)|$, and we let $v_{ij} \in \mathbb{S}^{n-1}$ be the unit vector such that $P_i(u_j) = \beta_{ij} v_{ij}$. By doing that, we may rewrite the above as

$$P_i(Z) = \sum \alpha_j \beta_{ij} \mathrm{seg}[-v_{ij}, v_{ij}], \tag{10.4}$$

where the sum is taken over the indices $1 \leq j \leq m$ such that $\beta_{ij} \neq 0$. We also claim that

$$P_i = \sum_{j=1}^m c_j P_i u_j \otimes P_i u_j = \sum_j c_j \beta_{ij}^2 v_{ij} \otimes v_{ij}, \tag{10.5}$$

where in the last sum we discard again the indices $1 \leq j \leq m$ for which $\beta_{ij} = 0$. **From now on, all sums and products where the indices are not displayed are considered to be taken over all $j = 1, \ldots, m$ such that $\beta_{ij} \neq 0$.** The second equality is immediate, but we have to prove the first one. For that sake, let $x \in \mathbb{R}^n$ be arbitrarily given. We have

$$\begin{aligned}\sum_{j=1}^m c_j P_i u_j \otimes P_i u_j(x) = \sum_{j=1}^m c_j \langle P_i u_j, x\rangle P_i u_j = P_i\left(\sum_{j=1}^m c_j \langle P_i u_j, x\rangle u_j\right) = \\ = P_i\left(\sum_{j=1}^m c_j \langle u_j - \langle u_i, u_j\rangle u_i, x\rangle u_j\right) = \\ = P_i\left(\sum_{j=1}^m c_j \langle u_j, x\rangle u_j\right) - P_i\left(\sum_{j=1}^m c_j \langle u_i, u_j\rangle \langle x, u_i\rangle u_j\right) = \\ = P_i(x) - \langle x, u_i\rangle P_i\left(\sum_{j=1}^m c_j \langle u_i, u_j\rangle u_j\right) = P_i(x) - \langle x, u_i\rangle P_i(u_i) = P_i(x).\end{aligned}$$

Notice carefully that we used twice the decomposition of the identity given in the formulation of the exercise, and we also used the obvious fact that $P_i(u_i) = 0_{\mathbb{R}^n}$.

If we consider any natural identification between $u_i^\perp$ and $\mathbb{R}^{n-1}$, then P_i is obviously the identity map $\mathrm{Id}_{\mathbb{R}^{n-1}}$. Thus, the second equality in (10.5) is an $(n-1)$-dimensional decomposition of the identity. In view of that, we may use the induction hypothesis for the zonotope $P_i(Z)$ written as in (10.4). The

consideration of these bounds in (10.3) yields

$$\mathrm{vol}_{n-1}(P_i(Z)) \geq 2^{n-1} \prod_j \left(\frac{\alpha_j}{c_j \beta_{ij}} \right)^{c_j \beta_{ij}^2}.$$

It follows that

$$\mathrm{vol}_n(Z) \geq 2 \prod_{i=1}^m \left[\frac{\alpha_i}{c_i} 2^{n-1} \prod_j \left(\frac{\alpha_j}{c_j \beta_{ij}} \right)^{c_j \beta_{ij}^2} \right]^{\frac{c_i}{n}}. \tag{10.6}$$

As a particular consequence of the decomposition of the identity given in (10.5), we have that

$$\sum_j c_j \beta_{ij}^2 = n - 1$$

for each $1 \leq i \leq m$. Moreover, since each β_{ij} is an orthogonal projection of a unit vector onto a hyperplane, it is clear that we always have $\beta_{ij} \leq 1$. We can use this to bound the right-hand side of (10.6). Together with the equality above, this yields

$$\mathrm{vol}_n(Z) \geq 2 \prod_{i=1}^m \left[2^{n-1} \frac{\alpha_i}{c_i} \prod_j \left(\frac{\alpha_i}{c_i} \right)^{c_j \beta_{ij}^2} \right]^{\frac{c_i}{n}} = 2 \prod_{i=1}^m \left[2^{n-1} \left(\frac{\alpha_i}{c_i} \right)^n \right]^{\frac{c_i}{n}} = 2^n \prod_{i=1}^m \left(\frac{\alpha_i}{c_i} \right)^{c_i},$$

and the proof is complete.

□

Exercise 10.7. *Let $K_1, \ldots, K_n \in \mathcal{K}^n$ be arbitrarily given. Prove that*

$$V(\Pi K_1, \ldots, \Pi K_n) = \\ = \frac{1}{n!} \int_{\mathbb{S}^{n-1}} \cdots \int_{\mathbb{S}^{n-1}} D_n(v_1, \ldots, v_n)\, dS_{n-1}(K_1, v_1) \ldots dS_{n-1}(K_n, v_n).$$

Solution 10.7. From Proposition 10.3.2 (main book), we know that each projection body ΠK_i is the zonoid whose generating measure is

$$\nu_{K_i}(E) := \frac{1}{4} \big(S_{n-1}(K_i, E) + S_{n-1}(-K_i, E) \big).$$

From equality (10.2.8) in the main book, we get that

$$V(\Pi K_1, \ldots, \Pi K_n) = \frac{2^n}{n!} \int_{\mathbb{S}^{n-1}} \cdots \int_{\mathbb{S}^{n-1}} D_n(v_1, \ldots, v_n)\, d\nu_{K_1}(v_1) \ldots d\nu_{K_n}(v_n).$$

Now we proceed just as in the proof of Corollary 10.3.3 (main book): since D_n is even in each entry, the integrals with respect to $S_{n-1}(K_i, \cdot)$ and $S_{n-1}(-K_i, \cdot)$ are the same. Thus,

$$V(\Pi K_1, \ldots, \Pi K_n) = \frac{2^n}{n!} \int_{\mathbb{S}^{n-1}} \cdots \int_{\mathbb{S}^{n-1}} D_n(v_1, \ldots, v_n) \frac{2^n}{4^n}\, dS_{n-1}(K_1, v_1) \ldots dS_{n-1}(K_n, v_n).$$

□

The objective of the next exercise is to derive a formula for mixed volumes of projections of zonoids. We still denote the orthogonal complement of $\mathrm{span}\{v_1, \ldots, v_j\}$ by $\{v_1, \ldots, v_j\}^\perp$.

Let $\mathbf{g}_1, \mathbf{g}_2 \subseteq \mathbb{R}^n$ be k-dimensional vector subspaces, where $k \in \{1, \dots, n-1\}$. We denote by $[\mathbf{g}_1, \mathbf{g}_2]$ the factor by which the k-dimensional volume of a measurable subset of $\mathbf{g}_1$ is multiplied when orthogonally projected onto $\mathbf{g}_2$ (e.g., if $\mathbf{g}_1$ and $\mathbf{g}_2$ are lines, then $[\mathbf{g}_1, \mathbf{g}_2]$ is the cosine of the acute angle between them).

Exercise 10.8. *Let $1 \leq j \leq n-1$, and let $\mathbf{g} \subseteq \mathbb{R}^n$ be a j-dimensional vector subspace. Prove that if $Z_1, \dots, Z_j$ are zonoids with respective generating measures $\mu_1, \dots, \mu_j$, then*

$$V^{(j)}\big(\mathrm{proj}_{\mathbf{g}}(Z_1), \dots, \mathrm{proj}_{\mathbf{g}}(Z_j)\big) = \\ = \frac{2^j}{j!} \int_{\mathbb{S}^{n-1}} \dots \int_{\mathbb{S}^{n-1}} D_j(v_1, \dots, v_j)[\mathbf{g}^\perp, \{v_1, \dots, v_j\}^\perp]\, d\mu_1(v_1) \dots d\mu_j(v_j).$$

Solution 10.8. We use Theorem 6.2.4 (main book) replacing $\mathbf{g}$ by $\mathbf{g}^\perp$ (and vice versa, of course), and letting $k = n-j$, $L_1 = \dots = L_{n-j} = \mathbf{g}^\perp \cap B^n =: B_{\mathbf{g}^\perp}$ and $K_1 = Z_1, \dots, K_j = Z_j$. This leads to

$$\binom{n}{n-j} V(B_{\mathbf{g}^\perp}[n-j], Z_1, \dots, Z_j) = V^{(n-j)}(B_{\mathbf{g}^\perp}[n-j]) V^{(j)}(\mathrm{proj}_{\mathbf{g}}(Z_1), \dots, \mathrm{proj}_{\mathbf{g}}(Z_j)),$$

from where we get that

$$\frac{n!}{j!(n-j)!} V(B_{\mathbf{g}^\perp}, [n-j]Z_1, \dots, Z_j) = \mathrm{vol}_{n-j}(B_{\mathbf{g}^\perp}) V^{(j)}(\mathrm{proj}_{\mathbf{g}}(Z_1), \dots, \mathrm{proj}_{\mathbf{g}}(Z_j)) = \\ = \kappa_{n-j} V^{(j)}(\mathrm{proj}_{\mathbf{g}}(Z_1), \dots, \mathrm{proj}_{\mathbf{g}}(Z_j)), \tag{10.7}$$

since $B_{\mathbf{g}^\perp}$ is naturally identified with the unit ball of B^{n-j}. From Exercise 10.4, we have the following expression for the mixed volume appearing in the left-hand side of the above:

$$V(B_{\mathbf{g}^\perp}, [n-j]Z_1, \dots, Z_j) = \\ = \frac{2^j (n-j)!}{n!} \int_{\mathbb{S}^{n-1}} \dots \int_{\mathbb{S}^{n-1}} D_j(v_1, \dots, v_j) \mathrm{vol}_{n-j}\big(B_{\mathbf{g}^\perp} | \{v_1, \dots, v_j\}^\perp\big)\, d\mu_1(v_1) \dots d\mu_j(v_j) = \\ = \frac{2^j (n-j)!}{n!} \int_{\mathbb{S}^{n-1}} \dots \int_{\mathbb{S}^{n-1}} D_j(v_1, \dots, v_j) \kappa_{n-j}[\mathbf{g}^\perp, \{v_1, \dots, v_j\}^\perp]\, d\mu_1(v_1) \dots d\mu_j(v_j). \tag{10.8}$$

Here, the reader may notice that we used the equality

$$\mathrm{vol}_{n-j}\big(B_{\mathbf{g}^\perp} | \{v_1, \dots, v_j\}^\perp\big) = \kappa_{n-j}[\mathbf{g}^\perp, \{v_1, \dots, v_j\}^\perp],$$

because $B_{\mathbf{g}^\perp} | \{v_1, \dots, v_j\}^\perp$ is precisely the projection of $B_{\mathbf{g}^\perp}$ onto $\{v_1, \dots, v_j\}^\perp$, and the volume of $B_{\mathbf{g}^\perp}$ is κ_{n-j}.

Finally, replacing (10.8) in (10.7) concludes the proof.

□

Exercise 10.9. *Let $1 \leq j \leq n-1$, and let $Z_1, \dots, Z_j \subseteq \mathbb{R}^n$ be zonoids whose generating measures are $\mu_1, \dots, \mu_j$, respectively. Prove that*

$$\int_{\mathbb{S}^{n-1}} f(v)\, dS(Z_1, \dots, Z_j, B^n[n-j-1], v) = \\ = \frac{2^j (n-j-1)!}{(n-1)!} \int_{\mathbb{S}^{n-1}} \dots \int_{\mathbb{S}^{n-1}} \left[\int_{\mathbb{S}^{n-1} \cap \{v_1, \dots, v_j\}^\perp} f(v)\, d\mathcal{H}^{n-j-1}(v) \right] \times \\ \times D_j(v_1, \dots, v_j)\, d\mu_1(v_1) \dots d\mu_j(v_j),$$

for any $f \in C(\mathbb{S}^{n-1})$.

Solution 10.9. First assume that $f = h_K$ for some $K \in CC^n$. From the definition of the mixed surface area measure $S(Z_1, \dots, Z_j, B^n[n-j-1], \cdot)$ and from Exercise 10.4, we have

$$\int_{\mathbb{S}^{n-1}} h_K(v)\, dS(Z_1, \dots, Z_j, B^n[n-j-1], v) = nV(Z_1, \dots, Z_j, B^n[n-j-1], K) =$$
$$= \frac{2^j (n-j)!}{(n-1)!} \int_{\mathbb{S}^{n-1}} \dots \int_{\mathbb{S}^{n-1}} D_j(v_1, \dots, v_j) \times$$
$$\times V^{(n-j)}(B^n | \{v_1, \dots, v_j\}^\perp [n-j-1], K | \{v_1, \dots, v_j\}^\perp)\, d\mu_1(v_1) \dots d\mu_j(v_j).$$

Notice that if $v_1, \dots, v_j$ are linearly dependent, then $D_j(v_1, \dots, v_j) = 0$. Otherwise, $B^n | \{v_1, \dots, v_j\}^\perp$ is the unit ball of the $(n-j)$-dimensional subspace $\{v_1, \dots, v_j\}^\perp$, which is naturally identified with $\mathbb{R}^{n-j}$. Hence Corollary 7.1.1 from the main book gives that $S_{n-j-1}(B^n | \{v_1, \dots, v_j\}^\perp, \cdot) = \mathcal{H}^{n-j-1}$. It follows that

$$V^{(n-j)}(B^n | \{v_1, \dots, v_j\}^\perp [n-j-1], K | \{v_1, \dots, v_j\}^\perp) =$$
$$= \frac{1}{n-j} \int_{\mathbb{S}^{n-1} \cap \{v_1, \dots, v_j\}^\perp} h_K(v)\, dS_{n-j-1}(B^{n-j}, v) =$$
$$= \frac{1}{n-j} \int_{\mathbb{S}^{n-1} \cap \{v_1, \dots, v_j\}^\perp} h_K(v)\, d\mathcal{H}^{n-j-1}(v).$$

Replacing this in the previous equality, we get that the desired property holds for support functions. It is immediate to check that, as a consequence, the equality also holds for functions having the form $f = h_K - h_L$, for $K, L \in CC^n$. Since from Proposition 7.1.2 (main book) we know that these functions are dense in $C(\mathbb{S}^{n-1})$, and because all involved measures are finite, it follows that the equality in the formulation of the exercise holds for every $f \in C(\mathbb{S}^{n-1})$.

□

Let C be a closed convex cone of some (real, finite- or infinite-dimensional, topological) vector space. A subset having the form $R := \{\lambda v : \lambda \geq 0\}$ for some non-zero vector $v \in C$ is called an *extreme ray* of C if $x, y \in C$ and $x + y \in R$ imply $x, y \in R$.

Exercise 10.10. *Consider the family* $\mathcal{Z} \subseteq C(\mathbb{S}^{n-1})$ *given as*

$$\mathcal{Z} := \left\{ \text{support functions of zonoids centered at } 0_{\mathbb{R}^n} \right\}.$$

(a) *Verify that* $\mathcal{Z}$ *is a convex cone.*

(b) *Show that* $\mathcal{Z}$ *is closed in the maximum norm of* $C(\mathbb{S}^{n-1})$.

(c) *Let* $x \in \mathbb{R}^n \setminus \{0_{\mathbb{R}^n}\}$. *Prove that the subset*

$$\left\{ |\langle \cdot, \lambda x \rangle| : \mathbb{S}^{n-1} \to \mathbb{R} : \lambda \geq 0 \right\} \subseteq \mathcal{Z}$$

is an extreme ray of $\mathcal{Z}$.

Solution 10.10. (a) This follows immediately from the fact that the family of zonoids centered at the origin forms a Minkowski class. If Z_1 and Z_2 are zonoids centered at the origin, then $h_{Z_1} + h_{Z_2}$ is the support function of $Z_1 + Z_2$, which is a zonoid centered at the origin. Also, it is clear that λZ is a zonoid centered at the origin whenever Z is a zonoid centered at the origin and $\lambda \geq 0$.

(b) We recall that the convergence of support functions in the maximum norm of $C(\mathbb{S}^{n-1})$ is equivalent to the convergence of the respective bodies in the Hausdorff metric. The limit of zonoids centered at the origin is a zonoid centered at the origin because it is the limit (in the Hausdorff metric) of a sequence of zonotopes centered at the origin: if Z_m is a zonoid of the converging sequence, simply choose a zonotope W_m such that $d_{\mathcal{H}}(Z_m, W_m) < 1/2^m$ (we may assume that W_m is centered at the origin; see the proof of Theorem 10.2.2 in the main book). This yields convergence of the respective support functions in the maximum norm.

(c) For any $x \in \mathbb{R}^n \setminus \{0_{\mathbb{R}^n}\}$ and $\lambda \geq 0$, we have

$$|\langle \cdot, \lambda x\rangle| = h_{\mathrm{seg}[-\lambda x, \lambda x]}(\cdot).$$

That is, the function $|\langle \cdot, \lambda x\rangle|$ is the support function of a zonoid. Let $h_{Z_1}, h_{Z_2} \in \mathcal{Z}$ be such that

$$h_{Z_1} + h_{Z_2} \in \big\{|\langle \cdot, \lambda x\rangle| : \lambda \geq 0\big\}.$$

Thus, we have $h_{Z_1} + h_{Z_2} = |\langle \cdot, \lambda_0 x\rangle|$ for some $\lambda_0 \geq 0$. Thus, $Z_1 + Z_2 = \mathrm{seg}[-\lambda_0 x, \lambda_0 x]$. Since $Z_1, Z_2 \in CC^n$, this is only possible if Z_1 and Z_2 are (possibly degenerate) segments in the direction of x. Since Z_1 and Z_2 are symmetric with respect the origin, this gives that

$$Z_1, Z_2 \in \big\{|\langle \cdot, \lambda x\rangle| : \lambda \geq 0\big\},$$

and the proof is complete.

□

Exercise 10.11. **(a)** *Show that*

$$\Pi B^n = \kappa_{n-1} B^n.$$

(b) *Verify that the projection body of any ellipsoid is an ellipsoid.*

(c) *Let $E \subseteq \mathbb{R}^n$ be an ellipsoid centered at the origin. Prove that*

$$\Pi E = \mathrm{vol}_n(E) \frac{\kappa_{n-1}}{\kappa_n} E^\circ.$$

Solution 10.11. **(a)** From Corollary 7.1.1 in the main book, we have that $S_{n-1}(B^n, \cdot) = \mathcal{H}^{n-1}|_{\mathbb{S}^{n-1}}$. Thus, using the equality discussed in Remark 6.5.3 (main book), we obtain

$$h_{\Pi B^n}(x) = \frac{1}{2} \int_{\mathbb{S}^{n-1}} |\langle v, x\rangle| \, dS_{n-1}(B^n, v) = \frac{1}{2} \int_{\mathbb{S}^{n-1}} |\langle v, x\rangle| \, d\mathcal{H}^{n-1}(v) = \kappa_{n-1}$$

for any $x \in \mathbb{R}^n$. It follows that $\Pi B^n = \kappa_{n-1} B^n$.

(b) If $E \subseteq \mathbb{R}^n$ is an ellipsoid, then $E = x_0 + TB^n$ for some $x_0 \in \mathbb{R}^n$ and some $T \in \mathrm{GL}(n)$. Hence, from Proposition 10.3.1 (main book) and item **(a)** above, we get

$$\Pi E = \Pi(x_0 + TB^n) = \Pi(TB^n) = |\det(T)|(T^{-1})^* \Pi B^n = |\det(T)| \kappa_{n-1} (T^{-1})^* B^n.$$

Thus, ΠE is a linear image of the unit ball, that is, an ellipsoid.

(c) If $E \subseteq \mathbb{R}^n$ is an ellipsoid centered at the origin, then $E = TB^n$ for some $T \in \mathrm{GL}(n)$. From the calculations of item **(b)**, and recalling (from Lemma 5.4.1 of the main book) that $(TK)^\circ = (T^{-1})^* K^\circ$

for every convex body K having the origin as an interior point and any $T \in \mathrm{GL}(n)$, it follows that

$$\Pi E = |\det(T)|\kappa_{n-1}(T^{-1})^* B^n = |\det(T)|\kappa_{n-1}(TB^n)^\circ = |\det(T)|\kappa_{n-1}E^\circ.$$

Now we observe that

$$\mathrm{vol}_n(E) = \mathrm{vol}_n(TB^n) = |\det(T)|\mathrm{vol}_n(B^n) = |\det(T)|\kappa_n,$$

from where we get that $|\det(T)| = \mathrm{vol}_n(E)/\kappa_n$. This gives the desired statement.

□

Exercise 10.12. *Let* $\Pi\colon \mathcal{K}^n \to \mathcal{K}^n$ *be the* projection map*, which associates each* $K \in \mathcal{K}^n$ *to its projection body* ΠK*. Prove that this map is continuous in the Hausdorff metric.*

Solution 10.12. Let (K_j) be a sequence in $\mathcal{K}^n$ converging in the Hausdorff metric to a body $K \in \mathcal{K}^n$. From item **(v)** of Theorem 7.1.3 (main book), we get that $S_{n-1}(K_j, \cdot)$ converges weakly to $S_{n-1}(K, \cdot)$. This yields

$$\frac{1}{2}\int_{\mathbb{S}^{n-1}} |\langle x, u\rangle|\, dS_{n-1}(K_j, u) \to \frac{1}{2}\int_{\mathbb{S}^{n-1}} |\langle x, u\rangle|\, dS_{n-1}(K, u)$$

as $j \to \infty$ for each $x \in \mathbb{S}^{n-1}$. This means that $h_{\Pi K_j} \to h_{\Pi K}$ pointwise in $\mathbb{S}^{n-1}$.

Let $S(\cdot)$ denote the surface area measure. It is clear from the weak convergence of measures that $S(K_j) \to S(K)$, and hence the sequence $(S(K_j))_{j\in\mathbb{N}}$ is uniformly bounded (by m, say). For any $x \in \mathbb{S}^{n-1}$ we have $|\langle x, u\rangle| \le |x||u| = 1$, and this leads to

$$h_{\Pi K_j}(x) = \frac{1}{2}\int_{\mathbb{S}^{n-1}} |\langle x, u\rangle|\, dS_{n-1}(K_j, u) \le \frac{1}{2}\int_{\mathbb{S}^{n-1}} dS_{n-1}(K_j, u) = \frac{S(K_j)}{2} \le \frac{m}{2},$$

and hence $(h_{\Pi K_j})$ is a uniformly bounded sequence of functions. Also, for any $j \in \mathbb{N}$ and every $x, y \in \mathbb{S}^{n-1}$, we have

$$\begin{aligned} |h_{\Pi K_j}(x) - h_{\Pi K_j}(y)| &\le \frac{1}{2}\int_{\mathbb{S}^{n-1}} \big||\langle x, u\rangle| - |\langle y, u\rangle|\big|\, dS_{n-1}(K_j, u) \le \\ &\le \frac{1}{2}\int_{\mathbb{S}^{n-1}} \big|\langle x - y, u\rangle\big|\, dS_{n-1}(K_j, u) \le \frac{1}{2}\int_{\mathbb{S}^{n-1}} |x - y|\, dS_{n-1}(K_j, u) = \\ &= \frac{S(K_j)}{2}|x - y| \le \frac{m}{2}|x - y|, \end{aligned}$$

and this shows that the sequence $(h_{\Pi K_j})$ is equicontinuous. From the Ascoli-Arzelá theorem we get that $(h_{\Pi K_j})$ has a subsequence which converges uniformly. Since uniform convergence implies pointwise convergence, the limit of this subsequence must be $h_{\Pi K}$.

This argument can be repeated if we replace (K_j) by any subsequence of (K_j). Hence any subsequence of $h_{\Pi K_j}$ has a subsequence which converges uniformly to $h_{\Pi K}$. This clearly yields that $h_{\Pi K_j} \to h_{\Pi K}$ uniformly, and from item **(ii)** of Proposition 2.5.2 (main book) this is the same as saying that $\Pi K_j \to \Pi K$ in the Hausdorff metric.

□

Exercise 10.13. *Let* $K \subseteq \mathbb{R}^n$ *be a convex body in minimal surface area position. Prove that*

$$S(K) \le n\mathrm{vol}_n(\Pi K)^{\frac{1}{n}}. \tag{10.9}$$

Hint: Consider first polytopes. For this case, use (10.3.2) from the main book, and Exercises 9.8 and 10.6. Approach the general case using approximations by polytopes.

Remark 10.2. Inequality (10.9) appears in [102] and [103].

Solution 10.13. Let P be a polytope in minimal surface area position, having unit outer normal vectors $u_1, \ldots, u_m$ and corresponding facets $F_1, \ldots, F_j$. We know from Exercise 9.8 that

$$\mathrm{Id}_{\mathbb{R}^n} = \sum_{j=1}^{m} c_j u_j \otimes u_j$$

for $c_j = n\mathrm{vol}_{n-1}(F_j)/S(P)$. On the other hand, from (10.3.2), we know that ΠP is the zonotope

$$\Pi P = \frac{1}{2}\sum_{j=1}^{m} \mathrm{vol}_{n-1}(F_j)\mathrm{seg}[-u_j, u_j].$$

From Exercise 10.6 we get

$$\mathrm{vol}_n(\Pi P) \geq 2^n \prod_{i=1}^{m}\left(\frac{\mathrm{vol}_{n-1}(F_j)}{2c_j}\right)^{c_j} = 2^n \prod_{i=1}^{m}\left(\frac{S(P)}{2n}\right)^{c_j} = \left(\frac{S(P)}{n}\right)^n,$$

where we used the fact that $c_1 + \ldots + c_m = n$. Taking the n-th root in both sides, we get the inequality for polytopes. Since surface area, volume, and the projection map are continuous in the Hausdorff metric (see Exercise 10.12), the general case follows immediately from approximations by polytopes.

□

Exercise 10.14. *Consider the functional $\psi : \mathcal{K}^n \to \mathbb{R}$ given by*

$$\psi(K) = \mathrm{vol}_n(\Pi^\circ K)^{-1/n}, \quad K \in \mathcal{K}^n.$$

(a) *Verify that the expression*

$$\psi(K) = \left[\frac{1}{n}\int_{\mathbb{S}^{n-1}} \mathrm{vol}_{n-1}\left(\mathrm{proj}_{u^\perp}(K)\right)^{-n} d\mathcal{H}^{n-1}(u)\right]^{-1/n}$$

holds.

(b) *Show that ψ is (strictly) monotone for inclusion, that is, $\psi(K) \leq \psi(L)$ whenever $K \subseteq L$.*

(c) *Prove that ψ is homogeneous of degree $n-1$ and invariant under volume-preserving transformations.*

Remark 10.3. Notice that ψ "behaves" as an affine surface area concept, since it is $(n-1)$-homogeneous, invariant under volume-preserving transformations, and strictly monotone for inclusion. Recall that bounds for the volume of the polar projection body were derived in Section 10.4 (main book).

Solution 10.14. (a) From (4.3.8) in the main book, and remembering that $(\Pi^\circ K)^\circ = \Pi K$, we get

$$\mathrm{vol}_n(\Pi^\circ K) = \frac{1}{n}\int_{\mathbb{S}^{n-1}} h((\Pi^\circ K)^\circ, u)^{-n}\, d\mathcal{H}^{n-1}(u) =$$

$$= \frac{1}{n}\int_{\mathbb{S}^{n-1}} h(\Pi K, u)^{-n}\, d\mathcal{H}^{n-1}(u) = \frac{1}{n}\int_{\mathbb{S}^{n-1}} \mathrm{vol}_{n-1}\left(\mathrm{proj}_{u^\perp}(K)\right)^{-n} d\mathcal{H}^{n-1}(u).$$

The desired expression follows.

(b) If $K \subseteq L$, then $\Pi K \subseteq \Pi L$, and since polarity reverses inclusion, it follows that $\Pi^\circ L \subseteq \Pi^\circ K$. Thus,

$$\mathrm{vol}_n(\Pi^\circ L) \leq \mathrm{vol}_n(\Pi^\circ K).$$

Raising both sides to the power $-1/n$, we get that $\psi(K) \leq \psi(L)$. If the inclusion $K \subseteq L$ is proper, then it is clear that the inclusions $\Pi K \subseteq \Pi L$ and $\Pi^\circ L \subseteq \Pi^\circ K$ are also proper. Hence the volume inequality above is strict.

(c) From (10.3.3) in the main book, it follows that if T is a volume-preserving linear map, then $\Pi^\circ(TK) = T(\Pi^\circ K)$. Thus,

$$\mathrm{vol}_n(\Pi^\circ(TK)) = \mathrm{vol}_n(\Pi^\circ K).$$

It follows that ψ is invariant under volume-preserving transformations. For the homogeneity, simply use the computation below (10.3.3) which shows that, as a map, Π° is homogeneous of degree $n-1$.

□

Exercise 10.15. *Show that for any convex bodies $K, L \subseteq \mathbb{R}^n$ the equality*

$$V(\Pi K, L, \ldots, L) = V(\Pi L, K, \ldots, K)$$

holds.

Solution 10.15. The equality is a consequence of the expression (10.3.1), of the definition of mixed surface area measures (7.1.1) (both in the main book), and of Fubini's theorem. With these ingredients in mind, we write

$$\begin{aligned} V(\Pi K, L, \ldots, L) &= \frac{1}{n} \int_{\mathbb{S}^{n-1}} h_{\Pi K}(v)\, dS_{n-1}(L, v) = \\ &= \frac{1}{n} \int_{\mathbb{S}^{n-1}} \frac{1}{2} \int_{\mathbb{S}^{n-1}} |\langle u, v\rangle|\, dS_{n-1}(K, u) dS_{n-1}(L, v) = \\ &= \frac{1}{n} \int_{\mathbb{S}^{n-1}} \left(\frac{1}{2} \int_{\mathbb{S}^{n-1}} |\langle u, v\rangle|\, dS_{n-1}(L, v) \right) dS_{n-1}(K, u) = \frac{1}{n} \int_{\mathbb{S}^{n-1}} h_{\Pi L}(u)\, dS_{n-1}(K, u) = \\ &= V(\Pi L, K, \ldots, K). \end{aligned}$$

□

Exercise 10.16. *Let $K \subseteq \mathbb{R}^n$ be a convex body, and recall that ∇K denotes the Blaschke body of K. Prove that*

$$\mathrm{vol}_n(\nabla K) \geq \mathrm{vol}_n(K).$$

Show that equality holds if and only if K is centrally symmetric.

Hint: Use Exercise 7.20.

Solution 10.16. By the discussion above Theorem 10.3.2 (main book), we have

$$\nabla K = \frac{1}{2^{1/(n-1)}} K \# (-K).$$

From Exercise 7.20, we get that

$$\mathrm{vol}_n(\nabla K)^{\frac{n-1}{n}} = \mathrm{vol}_n\left(\frac{1}{2^{1/(n-1)}}K\#(-K)\right)^{\frac{n-1}{n}} = \frac{1}{2}\mathrm{vol}_n(K\#(-K))^{\frac{n-1}{n}} \geq$$
$$\geq \frac{1}{2}\left(\mathrm{vol}_n(K)^{\frac{n-1}{n}} + \mathrm{vol}_n(-K)^{\frac{n-1}{n}}\right) = \mathrm{vol}_n(K)^{\frac{n-1}{n}},$$

and the desired inequality follows immediately from elevating both sides to the power $n/(n-1)$. The equality case comes straightforwardly from the equality case in Exercise 7.20.

□

For the next exercise we introduce an extension of the notion of projection body. Let $K_1, \ldots, K_{n-1} \in \mathcal{K}^n$. The *mixed projection body* of $(K_1, \ldots, K_{n-1})$ is the (unique) convex body $\Pi(K_1, \ldots, K_{n-1})$ whose support function (over $\mathbb{S}^{n-1}$) is given as

$$h(\Pi(K_1, \ldots, K_{n-1}), u) := V^{(n-1)}\big(\mathrm{proj}_{u^\perp}(K_1), \ldots, \mathrm{proj}_{u^\perp}(K_{n-1})\big)$$

for any $u \in \mathbb{S}^{n-1}$. Of course, we extend this function to $\mathbb{R}^n$ by letting

$$h(\Pi(K_1, \ldots, K_{n-1}), \lambda u) = \lambda h(\Pi(K_1, \ldots, K_{n-1}), u)$$

for every $u \in \mathbb{S}^{n-1}$ and $\lambda \geq 0$. Notice that it is not a priori clear that this function is sublinear, and hence it is not clear that $\Pi(K_1, \ldots, K_{n-1})$ is a convex body; proving that is one of the objectives of the next exercise. It is immediate to check that in the case where $K_1 = \ldots = K_{n-1}$ we just reobtain the usual projection body. For more on mixed projection bodies, we refer the reader to [177].

Exercise 10.17. **(a)** *Prove that*

$$h(\Pi(K_1, \ldots, K_{n-1}), x) = \frac{1}{2}\int_{\mathbb{S}^{n-1}} |\langle x, v\rangle|\, dS(K_1, \ldots, K_{n-1}, v),$$

for every $x \in \mathbb{R}^n$.

(b) *Show that the mixed projection body* $\Pi(K_1, \ldots, K_{n-1})$ *is a convex body.*

(c) *Let* $K_1, \ldots, K_{n-1}, L_1, \ldots, L_{n-1} \subseteq \mathbb{R}^n$ *be convex bodies. Prove that*

$$V(\Pi(K_1, \ldots, K_{n-1}), L_1, \ldots, L_{n-1}) = V(\Pi(L_1, \ldots, L_{n-1}), K_1, \ldots, K_{n-1}).$$

(d) *Show that the* mixed projection map

$$\mathcal{K}^n \times \ldots \times \mathcal{K}^n \ni (K_1, \ldots, K_{n-1}) \mapsto \Pi(K_1, \ldots, K_{n-1})$$

is continuous in each entry with respect to the Hausdorff metric.

Solution 10.17. **(a)** The case where $u \in \mathbb{S}^{n-1}$ is given by Exercise 7.13 (see also Lemma 8.4.1 in the main book). For $x = \lambda u \in \mathbb{R}^n$ (with $\lambda \geq 0$ and $u \in \mathbb{S}^{n-1}$), we simply notice that the expression in the right-hand side is positively homogeneous, that is,

$$\frac{1}{2}\int_{\mathbb{S}^{n-1}} |\langle \lambda u, v\rangle|\, dS(K_1, \ldots, K_{n-1}, v) = \frac{\lambda}{2}\int_{\mathbb{S}^{n-1}} |\langle u, v\rangle|\, dS(K_1, \ldots, K_{n-1}, v).$$

(b) The function $h(\Pi(K_1,\dots,K_{n-1}),\cdot)\colon \mathbb{R}^n \to \mathbb{R}$ is positively homogeneous by definition. From the expression given in item **(a)**, this is also clearly subadditive. It follows from Theorem 2.3.1 (main book) that $\Pi(K_1,\dots,K_{n-1}) \in CC^n$.

It remains to prove that $\Pi(K_1,\dots,K_{n-1})$ has non-empty interior. Since $K_1,\dots,K_{n-1}$ are convex bodies, for any $u \in \mathbb{S}^{n-1}$ we have that the projections $\mathrm{proj}_{u^\perp}(K_1),\dots,\mathrm{proj}_{u^\perp}(K_{n-1})$ have dimension $n-1$. Thus, we get that

$$h(\Pi(K_1,\dots,K_{n-1}),u) = V^{(n-1)}\big(\mathrm{proj}_{u^\perp}(K_1),\dots,\mathrm{proj}_{u^\perp}(K_{n-1})\big) > 0$$

for each $u \in \mathbb{S}^{n-1}$. This gives that $\Pi(K_1,\dots,K_{n-1})$ is a convex body (see, e.g., item **(b)** of Exercise 2.31).

(c) We use the expression obtained in item **(a)** and Fubini's theorem:

$$\begin{aligned}
V(\Pi(K_1,\dots,K_{n-1}),L_1,\dots,L_{n-1}) &= \frac{1}{n}\int_{\mathbb{S}^{n-1}} h(\Pi(K_1,\dots,K_{n-1}),u)\,dS(L_1,\dots,L_{n-1},u) = \\
&= \frac{1}{2n}\int_{\mathbb{S}^{n-1}} |\langle u,v\rangle|\,dS(K_1,\dots,K_{n-1},v)\,dS(L_1,\dots,L_{n-1},u) = \\
&= \frac{1}{n}\int_{\mathbb{S}^{n-1}} \left(\frac{1}{2}\int_{\mathbb{S}^{n-1}} |\langle u,v\rangle|\,dS(L_1,\dots,L_{n-1},u)\right) dS(K_1,\dots,K_{n-1},v) = \\
&\quad \frac{1}{n}\int_{\mathbb{S}^{n-1}} h(\Pi(L_1,\dots,L_{n-1}),v)\,dS(K_1,\dots,K_{n-1},v) = \\
&\quad = V(\Pi(L_1,\dots,L_{n-1}),K_1,\dots,K_{n-1}).
\end{aligned}$$

This concludes the proof.

(d) Just adapt the proof of Exercise 10.12.

□

Exercise 10.18. *Prove that for any convex bodies $K, L \subseteq \mathbb{R}^n$ the Brunn-Minkowski type inequality*

$$\mathrm{vol}_n(\Pi(K+L))^{\frac{1}{n(n-1)}} \geq \mathrm{vol}_n(\Pi K)^{\frac{1}{n(n-1)}} + \mathrm{vol}_n(\Pi L)^{\frac{1}{n(n-1)}} \tag{10.10}$$

holds. Show that equality holds if and only if $\Pi(K+L)$, ΠK, and ΠL are homothetical.

Solution 10.18. For any convex bodies $M, K, L \subseteq \mathbb{R}^n$, item **(c)** of Exercise 10.17 yields

$$V(\Pi(K+L),M[n-1]) = V(K+L[n-1],\Pi(M,\dots,M)) = V(K+L[n-1],\Pi M). \tag{10.11}$$

Letting $m = n-1$ and $t = 1/2$ in (8.3.7) in the main book, we get that

$$V(K+L[n-1],\Pi M)^{\frac{1}{n-1}} \geq V(K[n-1],\Pi M)^{\frac{1}{n-1}} + V(L[n-1],\Pi M)^{\frac{1}{n-1}}. \tag{10.12}$$

Now we use (10.11), (10.12), item **(c)** of Exercise 10.17, and the 1st Minkowski inequality (Theorem 6.3.1 from the main book) to arrive at

$$\begin{aligned}
V(\Pi(K+L),M[n-1])^{\frac{1}{n-1}} &\geq V(K[n-1],\Pi M)^{\frac{1}{n-1}} + V(L[n-1],\Pi M)^{\frac{1}{n-1}} = \\
&= V(\Pi K,M[n-1])^{\frac{1}{n-1}} + V(\Pi L,M[n-1])^{\frac{1}{n-1}} \geq \\
&\geq \mathrm{vol}_n(\Pi K)^{\frac{1}{n(n-1)}}\mathrm{vol}_n(M)^{\frac{1}{n}} + \mathrm{vol}_n(\Pi L)^{\frac{1}{n(n-1)}}\mathrm{vol}_n(M)^{\frac{1}{n}}.
\end{aligned}$$

Now we simply let $M = \Pi(K + L)$ and divide both sides by $\mathrm{vol}_n(\Pi(K + L))^{\frac{1}{n}}$. The desired inequality follows. If equality holds, then equality holds in the 1st Minkowski inequality for the bodies ΠK and $\Pi(K + L)$, and ΠL and $\Pi(K + L)$. Thus, $\Pi(K + L)$, ΠK, and ΠL are homothetical.

Last, assume that $\Pi(K+L)$, ΠK, and ΠL are homothetical. These bodies are symmetric with respect to the origin, and hence there exist $\lambda, \alpha > 0$ such that

$$\Pi K = \lambda \Pi(K + L) \text{ and}$$
$$\Pi L = \alpha \Pi(K + L) = \frac{\alpha}{\lambda} \Pi K.$$

By volume comparison we easily get that $\lambda \leq 1$ and $\alpha \leq 1$. Since $\Pi(tM) = t^{n-1}\Pi M$ for every $t > 0$ (this follows from Proposition 10.3.1 in the main book), this leads to

$$\Pi K = \Pi(\lambda^{\frac{1}{n-1}}(K + L)) \text{ and}$$
$$\Pi L = \Pi\left(\frac{\alpha^{\frac{1}{n-1}}}{\lambda^{\frac{1}{n-1}}} K\right).$$

From Theorem 10.3.2 (main book), this yields

$$K - K = \lambda^{\frac{1}{n-1}}[K + L - (K + L)] = \lambda^{\frac{1}{n-1}}(K - K) + \lambda^{\frac{1}{n-1}}(L - L) \text{ and}$$
$$L - L = \frac{\alpha^{\frac{1}{n-1}}}{\lambda^{\frac{1}{n-1}}}(K - K).$$

Thus,

$$K - K = \lambda^{\frac{1}{n-1}}(K - K) + \alpha^{\frac{1}{n-1}}(K - K),$$

and this leads to

$$\lambda^{\frac{1}{n-1}} + \alpha^{\frac{1}{n-1}} = 1.$$

Finally, we write

$$\mathrm{vol}_n(\Pi K)^{\frac{1}{n(n-1)}} + \mathrm{vol}_n(\Pi L)^{\frac{1}{n-1}} = \mathrm{vol}_n(\lambda \Pi(K + L))^{\frac{1}{n(n-1)}} + \mathrm{vol}_n(\alpha \Pi(K + L))^{\frac{1}{n(n-1)}} =$$
$$= \left(\lambda^{\frac{1}{n-1}} + \alpha^{\frac{1}{n-1}}\right)\mathrm{vol}_n(\Pi(K + L))^{\frac{1}{n(n-1)}} = \mathrm{vol}_n(\Pi(K + L))^{\frac{1}{n(n-1)}},$$

and this is the equality that we wanted to prove.

□

Exercise 10.19. *Let $K, L \subseteq \mathbb{R}^n$ be convex bodies. Prove that if ΠK, ΠL, and $\Pi(K+L)$ are homothetical, then K and L are homothetical. Conclude that the equality case in (10.10) holds if and only if K and L are homothetical.*

Hint. Using the definition of projection body, compare the volumes of the projections of $K + L$, K, and L onto any given hyperplane $u^{\perp}$. Conclude from the equality case in the classical Brunn-Minkowski inequality that $\mathrm{proj}_{u^{\perp}}(K)$ and $\mathrm{proj}_{u^{\perp}}(L)$ are always homothetical. Prove that, up to dilating one of the bodies, one may assume that these projections are actually translates. Last, use Proposition 8.4.4 (main book).

Solution 10.19. As in the solution of Exercise 10.18, we write $\Pi K = \lambda\Pi(K+L)$ and $\Pi L = \alpha\Pi(K+L)$ for some $\lambda, \alpha > 0$. We already know that $\lambda, \alpha \leq 1$, and that

$$\lambda^{\frac{1}{n-1}} + \alpha^{\frac{1}{n-1}} = 1.$$

For simplicity of the notation, we will write orthogonal projections onto hyperplanes as $\mathrm{proj}_{u^\perp}(M) = M^u$. For any $u \in \mathbb{S}^{n-1}$, we have

$$\mathrm{vol}_{n-1}(K^u) = h(\Pi K, u) = h(\lambda\Pi(K+L), u) = \lambda \mathrm{vol}_{n-1}((K+L)^u),$$

and

$$\mathrm{vol}_{n-1}(L^u) = h(\Pi L, u) = h(\alpha\Pi(K+L), u) = \alpha \mathrm{vol}_{n-1}((K+L)^u).$$

Since $(K+L)^u = K^u + L^u$, it follows that

$$\mathrm{vol}_{n-1}(K^u)^{\frac{1}{n-1}} + \mathrm{vol}_{n-1}(L^u)^{\frac{1}{n-1}} = \left(\lambda^{\frac{1}{n-1}} + \alpha^{\frac{1}{n-1}}\right)\mathrm{vol}_{n-1}(K^u + L^u) = \mathrm{vol}_{n-1}(K^u + L^u),$$

and this is the equality case in the classical Brunn-Minkowski inequality (in dimension $n-1$, of course). Since K^u and L^u are obviously $(n-1)$-dimensional (because K and L are convex bodies of $\mathbb{R}^n$), we get that K^u and L^u are homothetical. For simplicity, write $\Pi K = t\Pi L$ (i.e., $t = \lambda/\alpha$). We have

$$\mathrm{vol}_{n-1}(K^u) = h(\Pi K, u) = h(t\Pi L, u) = t\mathrm{vol}_{n-1}(L^u)$$

for each $u \in \mathbb{S}^{n-1}$. If we let $\bar{K} = t^{-\frac{1}{n-1}}K$, we get that $\mathrm{vol}_{n-1}(\bar{K}^u) = \mathrm{vol}_{n-1}(L^u)$ for any given $u \in \mathbb{S}^{n-1}$. Since $\bar{K}^u$ and L^u are homothetical (because $\bar{K}^u$ is clearly homothetical to K^u), we get that $\bar{K}^u$ and L^u are translates. Since this holds for each $u \in \mathbb{S}^{n-1}$, from Proposition 8.4.4 (main book) we get that $\bar{K}$ and L are translates. As a consequence, it follows that K and L are homothetical.

□

For the next exercise, we extend the definition of *projection body* for a set $C \in CC^n$ naturally by letting

$$h(\Pi C, \lambda u) = \lambda \mathrm{vol}_{n-1}\left(\mathrm{proj}_{u^\perp}(C)\right)$$

for every $u \in \mathbb{S}^{n-1}$ and $\lambda \geq 0$.

In the same manner, we extend the notion of *mixed projection body* for each $(n-1)$-tuple $C_1, \ldots, C_{n-1}$ in CC^n by letting

$$h(\Pi(C_1, \ldots, C_{n-1}), \lambda u) = \lambda V^{(n-1)}\left(\mathrm{proj}_{u^\perp}(C_1), \ldots, \mathrm{proj}_{u^\perp}(C_{n-1})\right)$$

for any $u \in \mathbb{S}^{n-1}$ and $\lambda \geq 0$. By these definitions, it is immediate to check that $\Pi(C, \ldots, C) = \Pi C$.

Exercise 10.20. **(a)** *Prove that the functions* $h(\Pi C, \cdot)$ *and* $h(\Pi(C_1, \ldots, C_{n-1}), \cdot)$ *defined above are indeed support functions of sets in* CC^n.

(b) *Verify that*

$$h(\Pi C, x) = \frac{1}{2}\int_{\mathbb{S}^{n-1}} |\langle x, v\rangle| \, dS_{n-1}(C, v) = \frac{n}{2} V\left(C[n-1], \mathrm{seg}[-x, x]\right)$$

and that

$$h(\Pi(C_1,\dots,C_{n-1}),x)=\frac{1}{2}\int_{\mathbb{S}^{n-1}}|\langle x,v\rangle|\,dS(C_1,\dots,C_{n-1},v)=\frac{n}{2}V\big(C_1,\dots,C_{n-1},\mathrm{seg}[-x,x]\big)$$

for any $x\in\mathbb{R}^n$.

(c) *Assume that* $C\in\mathcal{CC}^n$ *is such that* $\dim(C)\leq n-2$. *Show that* $\Pi C=\{0_{\mathbb{R}^n}\}$.

(d) *Prove that if* $C\in\mathcal{CC}^n$ *is such that* $\dim(C)=n-1$, *then* ΠC *is a segment.*

Solution 10.20. (a) We just have to repeat the proof given for the case of convex bodies.

(b) The first equality for $h(\Pi C,\cdot)$ follows straightforwardly from (7.2.5) in the main book, which also holds for sets from $\mathcal{CC}^n$ (and not only for convex bodies). The second equality for $h(\Pi C,\cdot)$ is an immediate consequence of the definition of $S_{n-1}(C,\cdot)$, together with the fact that the function $v\mapsto|\langle v,x\rangle|$ is the support function of $\mathrm{seg}[-x,x]$.

As in Exercise 10.17, the first equality for $h(\Pi(C_1,\dots,C_{n-1}),\cdot)$ follows from Exercise 7.13 (or Lemma 8.4.1 (main book); notice that these are stated for sets in $\mathcal{CC}^n$, and not only for convex bodies). The second equality comes from the definition of the surface area measure $S(C_1,\dots,C_{n-1},\cdot)$.

(c) For every $u\in\mathbb{S}^{n-1}$, we have that

$$h(\Pi C,u)=\frac{n}{2}V\big(C[n-1],\mathrm{seg}[-u,u]\big).$$

The mixed volume in the right-hand side vanishes because we cannot pick $n-1$ linearly independent segments in C (see Proposition 6.2.3 from the main book). Hence $h(\Pi C,\cdot)=0$, and this gives that $\Pi C=\{0_{\mathbb{R}^n}\}$.

(d) Since C is $(n-1)$-dimensional, we may assume without loss of generality that $C\subseteq u_0^\perp$ for some $u_0\in\mathbb{S}^{n-1}$ (we also recall that the projection body of C is invariant under translations of C). This leads to

$$c_0:=\mathrm{vol}_{n-1}\big(\mathrm{proj}_{u_0^\perp}(C)\big)=\mathrm{vol}_{n-1}(C)>0.$$

Thus, $h(\Pi C,u_0)=c_0$, and it is clear that we also have that $h(\Pi C,-u_0)=c_0$. This gives the inclusion

$$\mathrm{seg}[-c_0u_0,c_0u_0]\subseteq\Pi C.$$

Moving forward, we have that $h(\Pi C,\cdot)$ vanishes in $\mathbb{S}^{n-1}\cap u_0^\perp$. In fact, if $u\in u_0^\perp$, then writing

$$h(\Pi C,u)=\frac{n}{2}V\big(C[n-1],\mathrm{seg}[-u,u]\big)$$

we have from Proposition 6.2.3 that the mixed volume in the right-hand side vanishes. As a consequence, we get that ΠC is contained in the intersection of the family of hyperplanes $\{u^\perp:u\in u_0^\perp\}$. Hence it is intuitive that ΠC has to be a segment in the direction of u_0, and now we formalize this idea.

Consider the direct sum $u_0^\perp\oplus\mathrm{span}\{u_0\}$, and write an arbitrary point of $\mathbb{R}^n$ as $\lambda u_0+\alpha z$, for some $z\in\mathbb{S}^{n-1}\cap u_0^\perp$ and some $\lambda,\alpha\in\mathbb{R}$. If $\lambda u_0+\alpha z\in\Pi C$, then in particular we have that $\lambda u_0+\alpha z\in z^\perp$. Thus,

$$0=\langle z,\lambda u_0+\alpha z\rangle=\alpha.$$

It follows that all points of ΠC have the form λu_0 for some $\lambda \in \mathbb{R}$. If $|\lambda| > c_0$, then we either have $\langle \lambda u_0, u_0 \rangle > c_0$ or $\langle \lambda u_0, -u_0 \rangle > c_0$. In this case, we have $\lambda u_0 \notin \Pi C$, since otherwise we would have $h(\Pi C, u_0) > c_0$ or $h(\Pi C, -u_0) > c_0$. Hence we have the reverse inclusion

$$\Pi C \subseteq \operatorname{seg}[-c_0 u_0, c_0 u_0],$$

and this concludes the proof.

□

Now we may see the mixed projection map as a map which associates each $(n-1)$-tuple $(C_1, \ldots, C_{n-1})$ of sets of CC^n (instead of $\mathcal{K}^n$) to the mixed projection body $\Pi(C_1, \ldots, C_{n-1})$. Below we investigate some properties of this map. As a first step, the reader may verify that items **(c)** and **(d)** of Exercise 10.17 still hold true when we consider the larger family CC^n.

Exercise 10.21. **(a)** *Let $C_1, \ldots, C_{n-1} \in CC^n$ and $x_1, \ldots, x_{n-1} \in \mathbb{R}^n$. Show that*

$$\Pi(x_1 + C_1, \ldots, x_{n-1} + C_{n-1}) = \Pi(C_1, \ldots, C_{n-1}).$$

(b) *Prove that for any $T \in \mathrm{GL}(n)$ and all $C_1, \ldots, C_{n-1} \in CC^n$ we have*

$$\Pi(TC_1, \ldots, TC_{n-1}) = |\det(T)|(T^{-1})^* \Pi(C_1, \ldots, C_{n-1}).$$

Solution 10.21. **(a)** This is immediate due to the fact that mixed surface area measures are invariant under translations.

(b) We use the expression for the support function of the mixed projection body in terms of mixed volumes (and the behavior of mixed volumes under linear transformations; see Corollary 6.2.1 in the main book):

$$\begin{aligned} h(\Pi(TC_1, \ldots, TC_{n-1}), u) &= \frac{n}{2} V(TC_1, \ldots, TC_{n-1}, \operatorname{seg}[-u, u]) = \\ &= \frac{n}{2} |\det(T)| V(C_1, \ldots, C_{n-1}, T^{-1} \operatorname{seg}[-u, u]) = \\ &= \frac{n}{2} |\det(T)| V(C_1, \ldots, C_{n-1}, \operatorname{seg}[-T^{-1}u, T^{-1}u]) = \\ &= |\det(T)| h(\Pi(C_1, \ldots, C_{n-1}), T^{-1}u), \end{aligned}$$

for each $u \in \mathbb{S}^{n-1}$. Now we recall that $\lambda h_A = h_{\lambda A}$ for any $\lambda \geq 0$ and $h_A(T(\cdot)) = h_{T^* A}(\cdot)$. Hence,

$$\begin{aligned} h(\Pi(TC_1, \ldots, TC_{n-1}), u) &= |\det(T)| h(\Pi(C_1, \ldots, C_{n-1}), T^{-1}u) = \\ &= h\big(|\det(T)|(T^{-1})^* \Pi(C_1, \ldots, C_{n-1}), u\big) \end{aligned}$$

for any $u \in \mathbb{S}^{n-1}$. This proves the desired equality.

□

Exercise 10.22. **(a)** *Let $C_1, \ldots, C_{n-1}, L_1, \ldots, L_{n-1} \in CC^n$ be such that $C_j \subseteq L_j$ for each $j = 1, \ldots, n-1$. Verify that*

$$\Pi(C_1, \ldots, C_{n-1}) \subseteq \Pi(L_1, \ldots, L_{n-1}).$$

(b) *Prove that if $C, L \in CC^n$ are such that $C \cup L \in CC^n$ and $C \cap L \neq \emptyset$, then for every $C_2, \ldots, C_{n-1} \in CC^n$ we have*

$$\Pi(C \cup L, C_2, \ldots, C_{n-1}) + \Pi(C \cap L, C_2, \ldots, C_{n-1}) = \Pi(C, C_2, \ldots, C_{n-1}) + \Pi(L, C_2, \ldots, C_{n-1}).$$

In other words, the mixed projection map has the valuation property.

Solution 10.22. For each $u \in \mathbb{S}^{n-1}$, we have

$$h(\Pi(C_1, \ldots, C_{n-1}), u) = \frac{n}{2} V(C_1, \ldots, C_{n-1}, \mathrm{seg}[-u, u]) \leq \frac{n}{2} V(L_1, \ldots, L_{n-1}, \mathrm{seg}[-u, u]) = \\ = h(\Pi(L_1, \ldots, L_{n-1}), u),$$

where the inequality follows from the monotonicity of mixed volumes. This gives the inequality that we wanted to prove.

(b) We use the valuation property of mixed volumes (see Theorem 6.2.3 in the main book) and the additivity of support functions (on the involved bodies) to get

$$\begin{aligned} h(\Pi(C \cup L, C_2, \ldots, C_{n-1}) + \Pi(C \cap L, C_2, \ldots, C_{n-1}), u) = \\ = h(\Pi(C \cup L, C_2, \ldots, C_{n-1}), u) + h(\Pi(C \cap L, C_2, \ldots, C_{n-1}), u) = \\ = \frac{n}{2}\Big[V(C \cup L, C_2, \ldots, C_{n-1}, \mathrm{seg}[-u, u]) + V(C \cap L, C_2, \ldots, C_{n-1}, \mathrm{seg}[-u, u])\Big] = \\ = \frac{n}{2}\Big[V(C, C_2, \ldots, C_{n-1}, \mathrm{seg}[-u, u]) + V(L, C_2, \ldots, C_{n-1}, \mathrm{seg}[-u, u])\Big] = \\ = h(\Pi(C, C_2, \ldots, C_{n-1}), u) + h(\Pi(L, C_2, \ldots, C_{n-1}), u) = \\ = h(\Pi(C, C_2, \ldots, C_{n-1}) + \Pi(L, C_2, \ldots, C_{n-1}), u) \end{aligned}$$

for each $u \in \mathbb{S}^{n-1}$. The equality follows.

□

The objective of the next exercise is to prove that the mixed projection map is polynomial for Minkowski combinations in a similar way as the mixed volume.

Exercise 10.23. *Let $C_1, \ldots, C_m \in CC^n$ be arbitrarily given. Prove that for any $\lambda_1, \ldots, \lambda_m \geq 0$ we have the equality*

$$\Pi(\lambda_1 C_1 + \ldots + \lambda_m C_m) = \sum_{1 \leq i_1, \ldots, i_{n-1} \leq m} \lambda_{i_1} \ldots \lambda_{i_m} \Pi(C_{i_1}, \ldots, C_{i_m}).$$

Solution 10.23. This is a consequence of the linearity (for Minkowski combinations) and symmetry of surface area measures proved in Theorem 7.1.3 (main book). We have

$$S(\lambda_1 C_1 + \ldots + \lambda_m C_m[n-1], \cdot) = \sum_{1 \leq i_1, \ldots, i_{n-1} \leq m} \lambda_{i_1} \ldots \lambda_{i_m} S(C_{i_1}, \ldots, C_{i_{n-1}}, \cdot).$$

Thus,

$$h(\Pi(\lambda_1 C_1 + \ldots + \lambda_m C_m), u) = \frac{1}{2}\int_{\mathbb{S}^{n-1}} |\langle u, v\rangle|\, dS(\lambda_1 C_1 + \ldots + \lambda_m C_m[n-1], v) =$$
$$= \sum_{1\le i_1,\ldots,i_{n-1}\le m} \lambda_{i_1}\ldots\lambda_{i_m}\frac{1}{2}\int_{\mathbb{S}^{n-1}} |\langle u, v\rangle|\, dS(C_{i_1},\ldots,C_{i_{n-1}}, v) =$$
$$= \sum_{1\le i_1,\ldots,i_{n-1}\le m} \lambda_{i_1}\ldots\lambda_{i_{n-1}} h(\Pi(C_{i_1},\ldots,C_{i_{n-1}}), u) =$$
$$= h\left(\sum_{1\le i_1,\ldots,i_{n-1}\le m} \lambda_{i_1}\ldots\lambda_{i_{n-1}}\Pi(C_{i_1},\ldots,C_{i_{n-1}}), u\right)$$

for each $u \in \mathbb{S}^{n-1}$. Notice that in the last step we used the linearity of support functions for Minkowski combinations of the sets.

□

Exercise 10.24. **(a)** *Prove that the functional $\phi\colon \mathcal{K}^n \to \mathbb{R}$ defined as*

$$\phi(K) = \mathrm{vol}_n(K)^{1-n}\mathrm{vol}_n(\Pi K)$$

is invariant under affine transformations.

(b) *Show that ϕ attains a minimum over $\mathcal{K}^n$.*

(c) *Show that for any convex body $K \subseteq \mathbb{R}^n$ we have the inequality*

$$\mathrm{vol}_n(\Pi K)\mathrm{vol}_n(K)^{1-n} \ge \mathrm{vol}_n(\Pi^2 K)\mathrm{vol}_n(\Pi K)^{1-n}.$$

(d) *Prove that if ϕ attains a minimum at $K_0 \in \mathcal{K}^n$, then K_0 and ΠK_0 are homothetical. Conclude that any minimizer of ϕ is a proper zonoid.*

Solution 10.24. **(a)** This is immediate from Proposition 10.3.1 (main book).

(b) Since ϕ is invariant under affine transformations, we have that ϕ attains a minimum over $\mathcal{K}^n$ if and only if ϕ attains a minimum over the family

$$\mathcal{V} := \left\{K \in CC^n : \mathrm{vol}_n(K) = 1\right\},$$

which is obviously contained in $\mathcal{K}^n$ (this observation is important to guarantee that ϕ is well-defined over $\mathcal{V}$). It is clear that $\mathcal{V}$ is bounded, and since the volume is continuous in CC^n, we get that $\mathcal{V}$ is closed in CC^n. Thus, $\mathcal{V}$ is compact (see Corollary 2.5.1, main book). From the continuity of the projection map (see Exercise 10.12), we obtain that ϕ is continuous in the Hausdorff metric. Thus, ϕ attains a minimum over $\mathcal{V}$.

(c) We use Exercise 10.15 with $L = \Pi K$ and the 1st Minkowski inequality to get

$$\mathrm{vol}_n(\Pi K)^n = V(\Pi K, \Pi K, \ldots, \Pi_K)^n = V(\Pi^2 K, K, \ldots, K)^n \ge \mathrm{vol}_n(K)^{n-1}\mathrm{vol}_n(\Pi^2 K),$$

and dividing both sides by $\mathrm{vol}_n(K)^{n-1}\mathrm{vol}_n(\Pi K)^{n-1}$, we get the desired inequality.

(d) If ϕ attains a minimum at K_0, then we have equality in the inequality of item **(c)**. As a consequence, we have equality in the 1st Minkowski inequality for ΠK_0 and K_0, and this implies that these bodies are homothetical. It follows that K_0 is a projection body, and from Proposition 10.3.2 (main book) we get that it is a proper zonoid.

□

From Exercise 10.14 (see also Remark 10.3), we have that the functional $\psi : \mathcal{K}^n \to \mathbb{R}$ given as $\psi(K) := \mathrm{vol}_n(\Pi^\circ K)^{-1/n}$ behaves as an affine surface area concept for convex bodies. The computations in the next exercise lead to bounds for $\psi(K)$ in terms of $S(K)$ and $\mathrm{vol}_n(K)$.

Exercise 10.25. **(a)** *Verify that*

$$\frac{\kappa_{n-1}}{\kappa_n}\mathrm{vol}_n(K)^{\frac{n-1}{n}} \le \psi(K) \le \frac{\kappa_{n-1}}{n\kappa_n^{(n+1)/n}}S(K)$$

for every convex body $K \subseteq \mathbb{R}^n$.

(b) *Show that*

$$\inf\left\{S(TK) : T \in \mathrm{SL}(n)\right\} \ge \frac{n\kappa_n^{(n+1)/n}}{\kappa_{n-1}}\psi(K).$$

Solution 10.25 **(a)** From the previous exercise, we have

$$\kappa_{n-1}\kappa_n^{(1-n)/n}\mathrm{vol}_n(K)^{(n-1)/n} \le S_{-n}(K) \le S_1(K) = \frac{\kappa_{n-1}}{n\kappa_n}S(K).$$

Multiplying by $\kappa_n^{-1/n}$, we get

$$\frac{\kappa_{n-1}}{\kappa_n}\mathrm{vol}_n(K)^{(n-1)/n} \le \mathrm{vol}_n(\Pi^\circ K)^{(n-1)/n} \le \frac{\kappa_{n-1}}{n\kappa_n^{(n+1)/n}}S(K).$$

(b) From Exercise 10.14, we have that ψ is $\mathrm{SL}(n)$-invariant. Hence

$$S(TK) \ge \frac{n\kappa_n^{(n+1)/n}}{\kappa_{n-1}}\psi(TK) = \frac{n\kappa_n^{(n+1)/n}}{\kappa_{n-1}}\psi(K)$$

for every $T \in \mathrm{SL}(n)$. This gives the desired.

□

For Further Reading

Here we concentrate on *zonotopes, zonoids*, and *projection bodies*. Due to the large variety of their geometric properties and characterizations, zonotopes and zonoids have also many applications. For basic surveys and expositions, reflecting almost the whole literature on them, we refer to Schneider and Weil [231], Goodey and Weil [109], and our main book [29], in the latter monograph meaning Section 10.2, subsections 10.6.1 and 10.6.2, as well as the notes 2.9.1.11. In these parts of [29] also various book chapters, which treat zonotopes and zonoids comprehensively, are collected. We mention some examples.

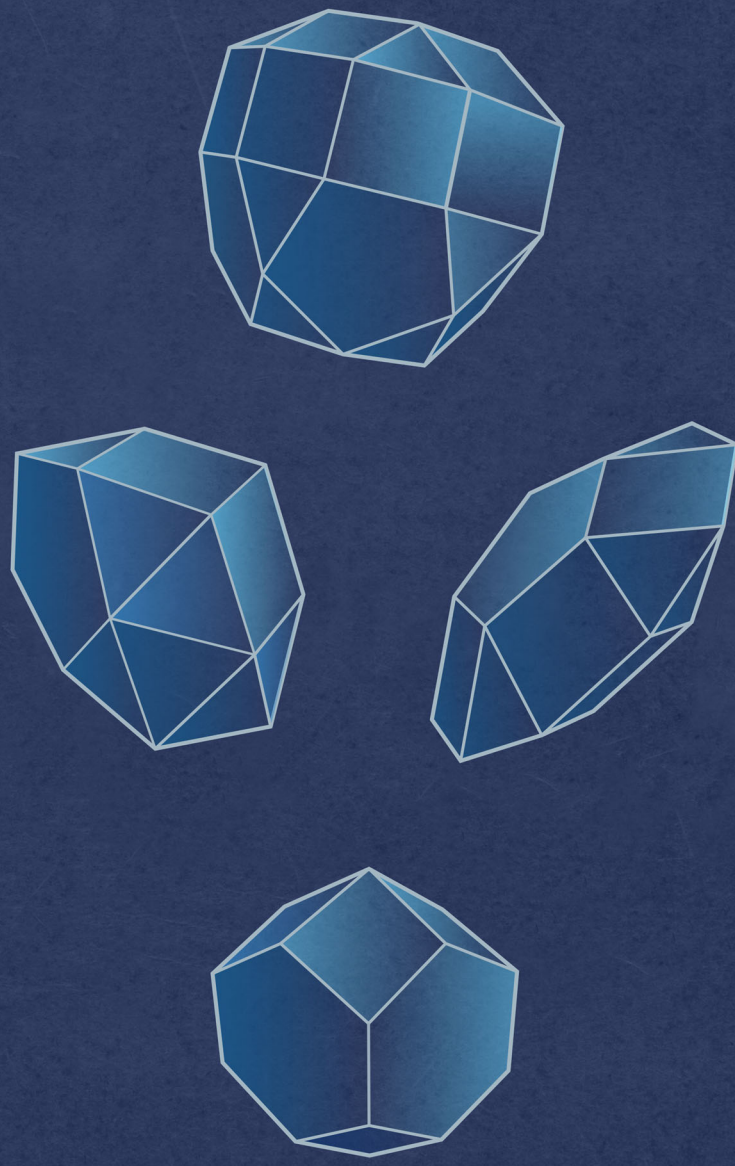
Zonotopes are discussed by Björner et al. [51] in view of *oriented matroids* and by Ziegler [264] regarding *zonotopal geometry*. Schneider [228] studies zonotopes in Section 3.5 and its notes, also in view of *Minkowski classes*, and de Concini and Procesi [85] in their Section 2.3 as *shadows of cubes*, regarding *tilings* and *lattices*. Also the exposition [185] refers particularly to zonotopes.

In Thompson's monograph [243] on Minkowski geometry, *zonoids* occur at various places, e.g. regarding the *isoperimetric problem* in finite-dimensional Banach spaces, and in Gardner's book [97] zonoids are mainly studied as *projection bodies*, as *centroid bodies*, and as extreme cases of *affine isoperimetric inequalities*. In Schneider's book [228] Section 3.5 deals comprehensively with zonoids, and they are also treated by Brazitikos et al. [70] regarding the *hyperplane conjecture* and as *centroid bodies*. For *projection bodies* we refer again to the expositions [231] and [109] and, in [29], to Section 10.3 and subsection 10.6.3, where also centroid bodies are discussed (since they are also zonoids). Here we recall related book parts, too. Namely, good examples are again the monographs [243] and [97], and the survey [187] collects also material on projection bodies.

PLATE 11

Figure 29

Regular-faced polyhedra



Four of the 92 Johnson-Zalgaller polyhedra ► (see, e.g., 2.9.1.4)

Chapter 11
Further Selected Topics



"Gödel gave a formal proof of the inadequacy of formal ideas."

— C. N. Yang

The exercises in Chapter 11 of the main book [29] mainly treat the following topics: *surface area measures, Petty's projection inequality, valuations in convexity, rotation means of convex bodies, mean width, star bodies, Orlicz projection bodies*, and the *Wills functional*.

Plate 11:

Johnson [148] published in 1966 a list of convex 3-polytopes having only regular polygons as facets and not being Platonic, Archimedean, prisms, or antiprisms. His list showed 92 such polyhedra, and Zalgaller [262] verified that this collection is complete. Therefore, these *regular-faced polyhedra* are often called *Johnson-Zalgaller polyhedra*, and Plate 11 presents four of them. In the main book, these polyhedra occur (as it is written also in Plate 11) in part 2.9.1.4.

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V. Balestro et al., *Convexity from the Geometric Point of View: Exercises and Solutions*, Cornerstones, https://doi.org/10.1007/978-3-031-83937-5_11

Exercises and Solutions

The objective of the first exercise is to show that the Petty projection inequality (cf. (10.4.10) in the main book) is, in some sense, stronger than the isoperimetric inequality for convex bodies (which is given in Theorem 5.2.1 of the main book). Our main source for this exercise is Lutwak [175]. For $p \neq 0$ and $K \in \mathcal{K}^n$, we define the *p-mean of the projection function of* K to be the number

$$\begin{aligned} S_p(K) &:= \left(\int_{\mathbb{S}^{n-1}} \mathrm{vol}_{n-1}\left(\mathrm{proj}_{u^\perp}(K)\right)^p \, d\sigma(u) \right)^{\frac{1}{p}} \\ &= \left(\frac{1}{n\kappa_n} \int_{\mathbb{S}^{n-1}} \mathrm{vol}_{n-1}\left(\mathrm{proj}_{u^\perp}(K)\right)^p \, d\mathcal{H}^{n-1}(u) \right)^{\frac{1}{p}}, \end{aligned}$$

where we recall that σ is the standard probability measure on $\mathbb{S}^{n-1}$, which is just the normalized Hausdorff measure on $\mathbb{S}^{n-1}$, and that κ_n is the volume of B^n. Of course, the term $n\kappa_n$ appears as the normalization factor because from (4.3.3) in the main book we have $\mathcal{H}^{n-1}(\mathbb{S}^{n-1}) = n\kappa_n$.

Exercise 11.1. *Let $K \subseteq \mathbb{R}^n$ be an arbitrarily given convex body, and recall that we denote its usual surface area measure by $S(K)$.*

(a) *Verify that*

$$S(K) = \frac{n\kappa_n}{\kappa_{n-1}} S_1(K).$$

(b) *Show that*

$$S_{-n}(K) = \kappa_n^{\frac{1}{n}} \mathrm{vol}_n(\Pi^\circ K)^{-\frac{1}{n}}.$$

(c) *Prove that $S_1(K) \geq S_{-n}(K)$.*

(d) *Using the Petty projection inequality, confirm that*

$$S_{-n}(K) \geq \kappa_{n-1} \kappa_n^{(1-n)/n} \mathrm{vol}_n(K)^{(n-1)/n}.$$

(e) *Derive the isoperimetric inequality for convex bodies from the previous items.*

Solution 11.1. **(a)** This equality follows straightforwardly from the Cauchy surface area formula (Theorem 6.5.1 from the main book).

(b) This is immediate due to Exercise 10.14.

(c) We use Jensen's inequality for the function $\varphi(t) = t^{-1/n}$, which is convex for $t \in (0, +\infty)$. We have

$$\begin{aligned} S_{-n}(K) &= \left(\int_{\mathbb{S}^{n-1}} \mathrm{vol}_{n-1}\big(\mathrm{proj}_{u^\perp}(K)\big)^{-n} \, d\sigma(u) \right)^{-1/n} \\ &= \varphi\left(\int_{\mathbb{S}^{n-1}} \mathrm{vol}_{n-1}\big(\mathrm{proj}_{u^\perp}(K)\big)^{-n} \, d\sigma(u) \right) \leq \\ &\leq \int_{\mathbb{S}^{n-1}} \varphi\Big(\mathrm{vol}_{n-1}\big(\mathrm{proj}_{u^\perp}(K)\big)^{-n} \Big) \, d\sigma(u) \\ &= \int_{\mathbb{S}^{n-1}} \mathrm{vol}_{n-1}\big(\mathrm{proj}_{u^\perp}(K)\big) \, d\sigma(u) = S_1(K). \end{aligned}$$

(d) The Petty projection inequality can be rewritten as

$$\mathrm{vol}_n(\Pi^\circ K) \leq \frac{\kappa_n^n}{\kappa_{n-1}^n} \mathrm{vol}_n(K)^{1-n}.$$

Raising both sides to the power $-1/n$, we arrive at

$$\mathrm{vol}_n(\Pi^\circ K)^{-1/n} \geq \frac{\kappa_{n-1}}{\kappa_n} \mathrm{vol}_n(K)^{(n-1)/n}$$

Thus,

$$S_{-n}(K) = \kappa_n^{1/n} \mathrm{vol}_n(\Pi^\circ K)^{-1/n} \geq \kappa_{n-1} \kappa_n^{1/n-1} \mathrm{vol}_n(K)^{\frac{n-1}{n}}.$$

(e) We just write

$$\begin{aligned} S(K) = \frac{n\kappa_n}{\kappa_{n-1}} S_1(K) \geq \frac{n\kappa_n}{\kappa_{n-1}} S_{-n}(K) \geq \frac{n\kappa_n}{\kappa_{n-1}} \kappa_{n-1} \kappa_n^{1/n-1} \mathrm{vol}_n(K)^{\frac{n-1}{n}} = \\ = n\kappa_n^{1/n} \mathrm{vol}_n(K)^{\frac{n-1}{n}}. \end{aligned}$$

Dividing the first and the last terms by $S(B^n) = n\kappa_n$, we get

$$\frac{S(K)}{S(B^n)} \geq \frac{1}{\kappa_n^{(n-1)/n}} \mathrm{vol}_n(K)^{(n-1)/n}.$$

Raising both sides to $1/(n-1)$ we finally arrive at the isoperimetric inequality.

□

Exercise 11.2. *Give an example of a valuation $\varphi : \mathcal{S} \to \mathbb{R}$ which is not a fully additive functional.*

Hint: Let $\mathcal{S} = \Big\{ \emptyset, \{a\}, \{b\}, \{c\}, \{a, b, c\} \Big\}$.

Solution 11.2. Simply let $\varphi(\{a\}) = \varphi(\{b\}) = \varphi(\{c\}) = 1$, and $\varphi(\{a, b, c\}) = 4$ (or any value distinct from 3). Of course, we also set $\varphi(\emptyset) = 0$. It is clear that φ is a valuation, but

$$\varphi(\{a, b, c\}) \neq \varphi(\{a\}) + \varphi(\{b\}) + \varphi(\{c\}).$$

Hence, φ is not fully additive.

□

A map $\varphi\colon CC^n \to CC^n$ is called a *Minkowski valuation* if whenever $K, L \in CC^n$ are such that $K \cup L \in CC^n$, it holds that

$$\varphi(K \cup L) + \varphi(K \cap L) = \varphi(K) + \varphi(L),$$

where we are considering the Minkowski sum in CC^n, of course. In the exercises below we briefly investigate some Minkowski valuations.

Exercise 11.3. *Let the* difference body map *be the map* $D\colon CC^n \to CC^n$ *defined as*

$$D(K) = K - K$$

for each $K \in CC^n$. *Show that* D *is a Minkowski valuation.*

Solution 11.3. If $K \cup L$ is convex, then from Exercise 2.9 we get that $K \cup L + K \cap L = K + L$. Thus,

$$\begin{aligned} D(K \cup L) + D(K \cap L) &= (K \cup L - K \cup L) + (K \cap L - K \cap L) = \\ = (K \cup L + K \cap L) - (K \cup L + K \cap L) &= K + L - (K + L) = K - K + L - L = D(K) + D(L). \end{aligned}$$

This shows that D is a Minkowski valuation.

□

Exercise 11.4. **(a)** *Let* $0 \le j \le n-1$ *and* $K, L, C_{j+1}, \ldots, C_{n-1} \in CC^n$, *and assume that* $K \cup L \in CC^n$. *Show that*

$$\begin{aligned} S(K \cup L[j], C_{j+1}, \ldots, C_{n-1}, \cdot) + S(K \cap L[j], C_{j+1}, \ldots, C_{n-1}\cdot, \cdot) = \\ = S(K[j], C_{j+1}, \ldots, C_{n-1}, \cdot) + S(L[j], C_{j+1}, \ldots, C_{n-1}, \cdot). \end{aligned}$$

That is, the mixed surface area measures have the valuation property.

(b) *For each* $1 \le i \le n-1$, *let* $\Pi_i : CC^n \to CC^n$ *be the map which associates each* $K \in CC^n$ *to the set* $\Pi_i K$ *as defined in Exercise 7.14. Prove that* Π_i *is a Minkowski valuation.*

Solution 11.4. We consider Proposition 6.4.1 (main book). From this proposition, and from the definition of mixed surface area measures, we get immediately that

$$\begin{aligned} \int_{\mathbb{S}^{n-1}} h_A(v)\, d\big(S(K \cup L[j], C_{j+1}, \ldots, C_{n-1}, v) + S(K \cap L[j], C_{j+1}, \ldots, C_{n-1}, v)\big) = \\ = \int_{\mathbb{S}^{n-1}} h_A(v)\, d\big(S(K[j], C_{j+1}, \ldots, C_{n-1}, v) + S(L[j], C_{j+1}, \ldots, C_{n-1}, v)\big) \end{aligned}$$

for every $A \in CC^n$. By linearity of the integral, we get that the same holds for functions $h_A - h_B$ for $A, B \in CC^n$. From Proposition 7.1.2 (main book), the space of such functions is dense in $C(\mathbb{S}^{n-1})$, and hence the equality above holds for any continuous function $f\colon \mathbb{S}^{n-1} \to \mathbb{R}$. Consequently, Proposition A.2.4 (main book) gives that the measures on both sides are the same.

□

Exercise 11.5. **(a)** *Let* $A, B \subseteq \mathbb{R}^n$. *Show that*

$$1_{A \cup B}(x) + 1_{A \cap B}(x) = 1_A(x) + 1_B(x)$$

for any $x \in \mathbb{R}^n$.

(b) *Let $M : CC^n \to CC^n$ be the map which associates each $K \in CC^n$ to its moment body MK, defined in (10.3.8) of the main book. Prove that M is a Minkowski valuation.*

Solution 11.5. (a) The left- and right-hand sides of the equality have three possible values: 0, 1, or 2. If it is 0, then $x \notin A \cup B$, in particular. Thus, $x \notin A$ and $x \notin B$, and hence the right-hand side is 0, too. If the left-hand side is 1, then we must have $x \in A \cup B$ but $x \notin A \cap B$ (because $A \cap B \subseteq A \cup B$). This means that x is contained in either A or B, and it follows that the right-hand side is also 1. Finally, if the left-hand side is 2, then $x \in A \cap B$. Therefore, $x \in A$ and $x \in B$, and the right-hand side equals 2.

(b) Assume that $K \cup L$ is convex. We notice that

$$\begin{aligned} h_{M(K\cup L)}(x) + h_{M(K\cap L)}(x) &= \int_{K\cup L} |\langle x, z\rangle|\, dz + \int_{K\cap L} |\langle x, z\rangle|\, dz = \\ = \int_{\mathbb{R}^n} |\langle x, z\rangle| (1_{K\cup L}(z) + 1_{K\cap L}(z))\, dz &= \int_{\mathbb{R}^n} |\langle x, z\rangle| (1_K(z) + 1_L(z))\, dz = \\ = \int_K |\langle x, z\rangle|\, dz + \int_L |\langle x, z\rangle|\, dz &= h_{M(K)}(x) + h_{M(L)}(x), \end{aligned}$$

for any $x \in \mathbb{R}^n$. Thus,

$$h_{M(K\cup L)+M(K\cap L)}(x) = h_{M(K)+M(L)}(x)$$

for all $x \in \mathbb{R}^n$. It follows that $M(K \cup L) + M(K \cap L) = M(K) + M(L)$.

□

Recall that CC_0^n denotes the family of sets $C \in CC^n$ such that $0_{\mathbb{R}^n} \in C$. Let $p > 1$. A map $\varphi : CC_0^n \to CC_0^n$ is an *L_p Minkowski valuation* if it has the valuation property for the p-sum. That is, if

$$\varphi(K \cup L) +_p \varphi(K \cap L) = \varphi(K) +_p \varphi(L)$$

for every $K, L \in CC_0^n$ such that $K \cup L \in CC_0^n$.

Exercise 11.6. *Let $p > 1$.*

(a) *Show that*

$$(K \cup L) +_p (K \cap L) = K +_p L$$

whenever $K, L, K \cup L \in CC_0^n$.

(b) *Let $D_p : CC_0^n \to CC_0^n$ be the L_p* difference body map*, defined as*

$$D_p(K) := K +_p (-K)$$

for each $K \in CC_0^n$. Prove that D_p is an L_p Minkowski valuation.

Hint: See the proof of Lemma 6.2.2 (main book).

Solution 11.6. (a) If $K \cup L$ is convex, then from the proof of Lemma 6.2.2 (main book), we know that

$$\begin{aligned} h_{K\cup L}(x) &= \max\{h_K(x), h_L(x)\} \quad \text{and} \\ h_{K\cap L}(x) &= \min\{h_K(x), h_L(x)\}, \end{aligned}$$

for every $x \in \mathbb{R}^n$. Thus,

$$h_{K\cup L}(x)^p + h_{K\cap L}(x)^p = \max\{h_K(x), h_L(x)\}^p + \min\{h_K(x), h_L(x)\}^p = \\ = \max\{h_K(x)^p, h_L(x)^p\} + \min\{h_K(x)^p, h_L(x)^p\} = h_K(x)^p + h_L(x)^p$$

for any $x \in \mathbb{R}^n$. Therefore,

$$h((K \cup L) +_p (K \cap L), x) = (h_{K\cup L}(x)^p + h_{K\cap L}(x)^p)^{1/p} = (h_K(x)^p + h_L(x)^p)^{1/p} = h(K +_p L, x)$$

for every $x \in \mathbb{R}^n$, and this gives that $(K \cup L) +_p (K \cap L) = K +_p L$.

(b) We use item **(a)** to compute:

$$\begin{aligned} h(D_p(K \cup L) +_p D_p(K \cap L), x)^p &= h(D_p(K \cup L), x)^p + h(D_p(K \cap L), x)^p = \\ &= h((K \cup L) +_p (-K \cup L), x)^p + h((K \cap L) +_p (-K \cap L), x)^p = \\ &= h_{K\cup L}(x)^p + h_{-K\cup L}(x)^p + h_{K\cap L}(x)^p + h_{-K\cap L}(x)^p = \\ &= h_K(x)^p + h_L(x)^p + h_{-K}(x)^p + h_{-L}(x)^p = \\ &= h(K +_p (-K), x)^p + h(L +_p (-L), x)^p = h(D_p(K), x)^p + h(D_p(L), x)^p = \\ &= h(D_p(K) +_p D_p(L), x)^p, \end{aligned}$$

and this gives the desired equality.

□

Exercise 11.7. **(a)** *For $p \geq 1$ and $K \in CC_0^n$, let*

$$h(M_p K, x) := \left(\int_K |\langle x, z\rangle|^p \, dz \right)^{1/p}$$

for each $x \in \mathbb{R}^n$. Show that the function $h(M_p K, \cdot)$ is sublinear, and conclude that this is the support function of a set $M_p K \in CC^n$ (we call it the L_p moment body of *K).*

(b) *Fix $p > 1$. Let $M_p : CC_0^n \to CC_0^n$ be the L_p* moment map*, which associates each $K \in CC_0^n$ to its L_p moment body $M_p K$. Prove that M_p is an L_p Minkowski valuation.*

Solution 11.7. **(a)** The positive homogeneity of M_p is trivial. To prove the subadditivity, we use Minkowski's classical inequality to write

$$\begin{aligned} h(M_p K, x + y) &= \left(\int_K |\langle x + y, z\rangle|^p \, dz \right)^{1/p} = \left(\int_K |\langle x, z\rangle + \langle y, z\rangle|^p \, dz \right)^{1/p} \leq \\ &\leq \left(\int_K \left(|\langle x, z\rangle| + |\langle y, z\rangle| \right)^p dz \right)^{1/p} \leq \left(\int_K |\langle x, z\rangle|^p \, dz \right)^{1/p} + \left(\int_K |\langle y, z\rangle|^p \, dz \right)^{1/p} = \\ &= h(M_p K, x) + h(M_p K, y), \end{aligned}$$

for any $x, y \in \mathbb{R}^n$.

(b) We simply use item **(a)** of Exercise 11.5 and write

$$\begin{aligned} h(M_p(K \cup L) +_p M_p(K \cap L), x)^p &= h(M_p(K \cup L), x)^p + h(M_p(K \cap L), x)^p = \\ = \int_{K \cup L} |\langle x, z\rangle|^p \, dz + \int_{K \cap L} |\langle x, z\rangle|^p \, dz &= \int_{\mathbb{R}^n} |\langle x, z\rangle|^p (1_{K \cup L}(z) + 1_{K \cap L}(z)) \, dz = \\ = \int_{\mathbb{R}^n} |\langle x, z\rangle|^p (1_K(z) + 1_L(z)) \, dz &= \int_K |\langle x, z\rangle|^p \, dz + \int_L |\langle x, z\rangle|^p \, dz = \\ = h(M_p(K), x)^p + h(M_p(L), x)^p &= h(M_p(K) +_p M_p(L), x)^p, \end{aligned}$$

for any $x \in \mathbb{R}^n$. This yields that

$$M_p(K \cup L) +_p M_p(K \cap L) = M_p(K) +_p M_p(L).$$

□

Let $K \in CC^n$. A *rotation mean* of K is a set $K' \in CC^n$ having the form

$$K' = \frac{1}{m}(T_1 K + \ldots + T_m K),$$

where $m \in \mathbb{N}$ and $T_1, \ldots, T_m \in \mathrm{SO}(n)$. The objective of the next exercise is to prove a theorem (due to Hadwiger) which states that for any $K \in CC^n$ there is a sequence of rotation means of K converging in the Hausdorff metric to the ball centered at the origin and whose radius is half the mean width of K.

Exercise 11.8. *Let $K \in CC^n$, and assume that $\dim K > 0$. Denote by $\mathcal{R}(K)$ the family of rotation means of K.*

(a) *Show that $\mathcal{R}(K)$ is a bounded family in $\mathbb{R}^n$.*

(b) *Prove that $\mathrm{cl}(\mathcal{R}(K))$ is compact in the Hausdorff metric.*

(c) *Verify that the map $r_0 \colon CC^n \to \mathbb{R}$ defined as*

$$r_0(L) := \min\{r \geq 0 : L \subseteq r B^n\}$$

is well-defined (meaning that the minimum indeed exists), and that it is continuous in the Hausdorff metric.

(d) *Prove that $w(K') = w(K)$ for any $K \in \mathcal{R}(K)$, where we remind the reader that $w(C)$ stands for the mean width of $C \in CC^n$.*

(e) *In view of* **(b)** *and* **(c)**, *we may let*

$$m_0 := \min\{r_0(L) : L \in \mathrm{cl}(\mathcal{R}(K))\}.$$

Use item **(d)** *to verify that $m_0 > 0$. Assume that this minimum is attained for $L_0 \in \mathrm{cl}(\mathcal{R}(K))$. Prove that $L_0 = m_0 B^n$.*

(f) *Show that there is a sequence of rotation means of K converging to $\frac{w(K)}{2}B^n$.*

Solution 11.8. **(a)** Let $\lambda_0 > 0$ be such that $K \subseteq \lambda_0 B^n$. Since $T(\lambda_0 B^n) = \lambda_0 B^n$ for every $T \in \mathrm{SO}(n)$, we get that any given rotation mean of K obeys

$$\frac{1}{m}(T_1K + \ldots + T_mK) \subseteq \frac{1}{m}(\lambda_0 B^n + \ldots + \lambda_0 B^n) = \lambda_0 B^n.$$

Hence $K' \subseteq \lambda_0 B^n$ for every $K' \in \mathcal{R}(K)$.

(b) From Proposition 2.5.3 and item **(a)** above, we get that $\mathcal{R}(K)$ is bounded in the Hausdorff metric. Hence $\mathrm{cl}(\mathcal{R}(K))$ is closed and bounded. Corollary 2.5.1 (main book) gives that $\mathrm{cl}(\mathcal{R}(K))$ is compact.

(c) For a given $L \in CC^n$, let $c_0 := \inf\{r \geq 0 : L \subseteq rB^n\}$. Then there exists a sequence $(r_j)_{j\in\mathbb{N}}$ such that $L \subseteq r_j B^n$ for each $j \in \mathbb{N}$, and $r_j \to c_0$ as $j \to \infty$. Hence $|x| \leq r_j$ for each $x \in L$ and $j \in \mathbb{N}$, and letting $j \to \infty$ we get $|x| \leq c_0$ for every $x \in L$. Therefore, $c_0 \in \{r \geq 0 : L \subseteq rB^n\}$, and this shows that the minimum is well-defined.

For the continuity, fix any $\varepsilon > 0$ and let $L, K \in CC^n$ be such that $d_{\mathcal{H}}(K, L) \leq \varepsilon$. Then, $K \subseteq L + \varepsilon B^n$ and $L \subseteq K + \varepsilon B^n$. Thus,

$$K \subseteq L + \varepsilon B^n \subseteq r_0(L)B^n + \varepsilon B^n = (r_0(L) + \varepsilon)B^n,$$

from which we get that

$$r_0(K) \leq r_0(L) + \varepsilon.$$

Analogously, we get that $r_0(L) \leq r_0(K) + \varepsilon$. This proves that r_0 is continuous.

(d) Let $K' \in \mathcal{R}(K)$, and write

$$K' = \frac{1}{m}(T_1K + \ldots + T_mK).$$

As it is shown in (5.5.1) of the main book, the mean width is $\mathrm{O}(n)$-invariant, and hence also $\mathrm{SO}(n)$-invariant. Thus, $w(T_jK) = w(K)$ for every $j = 1, \ldots, m$. Therefore,

$$w(K') = 2\int_{\mathbb{S}^{n-1}} h_{K'}(u)\,d\sigma(u) = 2\int_{\mathbb{S}^{n-1}} \frac{1}{m}\left(h_{T_1K}(u) + \ldots + h_{T_jK}(u)\right)d\sigma(u) =$$

$$\frac{1}{m}\left(2\int_{\mathbb{S}^{n-1}} h_{T_1K}(u)\,d\sigma(u) + \ldots + 2\int_{\mathbb{S}^{n-1}} h_{T_mK}(u)\,d\sigma(u)\right) = \frac{1}{m}\left(w(T_1K) + \ldots + w(T_mK)\right) =$$

$$= w(K).$$

(e) Since $\dim K > 0$, we have that $w(K) > 0$. It follows that $w(K')$ equals the positive constant $w(K)$ for any $K' \in \mathbb{R}(K)$. Since the mean width is continuous in the Hausdorff metric (as it was proved in Section 5.5 of the main book), we get that $w(L) = w(K)$ for any $L \in \mathrm{cl}(\mathcal{R}(K))$. Let $L_0 \in \mathrm{cl}(\mathcal{R}(K))$ be such that $r_0(L_0) = m_0$. From the definition of r_0, we have that $L_0 \subseteq m_0 B^n$. Because the mean width is obviously monotone for inclusion, and observing that the mean width of $m_0 B^n$ equals $2m_0$, we get that

$$2m_0 \geq w(L_0) = w(K) > 0,$$

and hence $m_0 > 0$.

Still assuming that the minimum m_0 is attained for $L_0 \in \mathrm{cl}(\mathcal{R}(K))$, suppose by contradiction that $L_0 \neq m_0 B^n$. Then, since $L_0 \subseteq m_0 B^n$, there exists a vector $u_0 \in \mathbb{S}^{n-1}$ such that $h_L(u_0) < h_{m_0 B^n}(u_0) = m_0$. By continuity of support functions, there exists a neighborhood $U \subseteq \mathbb{S}^{n-1}$ of u_0 such that $h_L(u) < m_0$ for any $u \in U$. It is not difficult to see that we may fix a finite family $T_1, \ldots, T_m$ of maps in $\mathrm{SO}(n)$ such that

$$\bigcup_{j=1}^{m} T_j U = \mathbb{S}^{n-1}.$$

In fact, for any $v \in \mathbb{S}^{n-1}$, there is a map $T_v \in \mathrm{SO}(n)$ such that $T_v u_0 = v$. The family $\{T_v U : v \in \mathbb{S}^{n-1}\}$ is an open cover of $\mathbb{S}^{n-1}$, and hence it has a finite subcover.

Define

$$L_1 := \frac{1}{m}(T_1 L_0 + \ldots + T_m L_0).$$

For a given $u \in \mathbb{S}^{n-1}$, we let $i \in \{1, \ldots, m\}$ be such that $u \in T_i U$. Hence $T_i^{-1} u \in U$, and this yields

$$h_{L_0}(T_i^{-1} u) < m_0.$$

For the indices $j \neq i$, we have that $h_{L_0}(T_j^{-1} u) \leq m_0$, because $T_j^{-1} u \in \mathbb{S}^{n-1}$ and $L_0 \subseteq m_0 B^n$. Since each T_j is orthogonal, it holds that $T_j^* = T_j^{-1}$ (see Remark 2.6.3). This, together with (2.6.5) (both in the main book), leads to

$$h_{L_1}(u) = \frac{1}{m}\sum_{j=1}^{m} h_{T_j L_0}(u) = \frac{1}{m}\sum_{j=1}^{m} h_{L_0}(T_j^* u) = \frac{1}{m}\sum_{j=1}^{m} h_{L_0}(T_j^{-1} u) < m_0.$$

Since $\mathbb{S}^{n-1}$ is compact and h_L is continuous, we may let $c_1 := \min\{m_0 - h_L(u) : u \in \mathbb{S}^{n-1}\}$, and we have that $c_1 > 0$. It follows that

$$h_L(u) \leq m_0 - c_1 < m_0$$

for any $u \in \mathbb{S}^{n-1}$. This gives that $L_1 \subseteq (m_0 - c_1) B^n$, and hence

$$r_0(L_1) \leq m_0 - c_1 < m_0.$$

Finally, let $(K_j)_{j \in \mathbb{N}}$ be a sequence in $\mathcal{R}(K)$ converging to L_0. Because the number $m \in \mathbb{N}$ and the maps $T_1, \ldots, T_m$ are fixed, we get that

$$C_j := \frac{1}{m}(T_1 K_j + \ldots + T_m K_j) \to \frac{1}{m}(T_1 L_0 + \ldots + T_m L_0) = L_1$$

as $j \to \infty$. We claim that each C_j is a rotation mean of K. In fact, since each K_j is a rotation mean of K, we may write

$$K_j = \frac{1}{m(j)}(T_1^j K + \ldots + T_{m(j)}^j K),$$

for some $m(j) \in \mathbb{N}$ and some maps $T_1^j, \ldots, T_{m(j)}^j \in \mathrm{SO}(n)$. Thus,

$$C_j = \frac{1}{m}\frac{1}{m(j)}\Big(T_1(T_1^j K + \ldots + T_{m(j)}^j K) + \ldots + T^m(T_1^j K + \ldots + T_{m(j)}^j K)\Big),$$

and since $\mathrm{SO}(n)$ is closed for composition, we get that the left-hand side is indeed a rotation mean of K. It follows that L_1 is the limit of the sequence (C_j) of sets of $\mathcal{R}(K)$, and hence $L_1 \in \mathrm{cl}(\mathcal{R}(K))$. The fact that $r_0(L_1) < m_0$ contradicts the minimality of m_0, and this contradiction concludes the proof.

(f) From item **(d)**, the set $L_0 \in \mathrm{cl}(\mathcal{R}(K))$ is the limit of a sequence of rotation means of K. Since we know that $L_0 = m_0 B^n$, and because $w(L_0) = w(K)$, we get that

$$2m_0 = w(m_0 B^n) = w(L_0) = w(K),$$

and the proof is complete.

□

Exercise 11.9. *Assume that $n \geq 2$, and let $\varphi\colon CC^n \to \mathbb{R}$ be a functional which is Minkowski additive, $\mathrm{SO}(n)$-invariant and continuous at B^n. Prove that φ is a multiple of the mean width functional.*

Solution 11.9. We first claim that $\varphi(\{0_{\mathbb{R}^n}\}) = 0$. Indeed, since φ is Minkowski additive, we have

$$\varphi(\{0_{\mathbb{R}^n}\}) = \varphi(\{0_{\mathbb{R}^n}\} + \{0_{\mathbb{R}^n}\}) = 2\varphi(\{0_{\mathbb{R}^n}\}),$$

and this yields the claim. Now for any given $x \in \mathbb{R}^n$, choosing $T \in \mathrm{SO}(n)$ such that $Tx = -x$, we get

$$0 = \varphi(\{0_{\mathbb{R}^n}\}) = \varphi(\{x\} + \{-x\}) = \varphi(\{x\} + \{Tx\}) = \varphi(\{x\}) + \varphi(\{Tx\}) = 2\varphi(\{x\}),$$

and this shows that $\varphi(\{x\}) = 0$ for all $x \in \mathbb{R}^n$.

Applying the Minkowski additivity property iteratively, we get immediately that $\varphi(mK) = m\varphi(K)$ for any positive integer m and every $K \in CC^n$. Moreover, we claim that $\varphi(\lambda K) = \lambda\varphi(K)$ for any $K \in CC^n$ and every positive rational number λ. In fact, if $\lambda = p/q$ for positive integers p and q (with $q \neq 0$, of course), then

$$p\varphi(K) = \varphi(pK) = \varphi(q(\lambda K)) = q\varphi(\lambda K),$$

and this gives what we wanted to get.

Let λ be any positive *real* number, and choose a sequence $(\lambda_j)_{j\in\mathbb{N}}$ of positive rational numbers converging to λ. If $(K_j)_{j\in\mathbb{N}}$ is any sequence in CC^n converging to λB^n, then K_j/λ_j converges to B^n. Thus,

$$\lim_{j\to\infty} \varphi(K_j) = \lim_{j\to\infty} \varphi\left(\lambda_j \frac{K_j}{\lambda_j}\right) = \lim_{j\to\infty} \lambda_j \varphi\left(\frac{K_j}{\lambda_j}\right) = \lambda\varphi(B^n). \tag{11.1}$$

We hold this information for a while. Let $K \in CC^n$, and remember that $\mathcal{R}(K)$ denotes the family of rotation means of K. Let

$$K' = \frac{1}{m}(T_1 K + \ldots + T_m K)$$

be any rotation mean of K. Since φ is additive and $\mathrm{SO}(n)$-invariant, we have

$$\varphi(K') = \varphi\left(\frac{1}{m}(T_1 K + \ldots + T_m K)\right) = \frac{1}{m}\left[\varphi(T_1 K) + \ldots + \varphi(T_m K)\right] = \varphi(K).$$

According to Exercise 11.8, we may let $(K_j)_{j\in\mathbb{N}}$ be a sequence in $\mathcal{R}(K)$ converging to $\frac{w(K)}{2}B^n$. Since $\varphi(K_j) = \varphi(K)$ for any $j \in \mathbb{N}$, and letting $\lambda = w(K)/2$ in (11.1), we get that

$$\varphi(K) = \lim_{j\to\infty} \varphi(K_j) = \frac{w(K)}{2}\varphi(B^n).$$

Because $K \in CC^n$ was arbitrarily given, this shows that

$$\varphi(K) = \frac{\varphi(B^n)}{2}w(K)$$

for every $K \in CC^n$. Since $\varphi(B^n)/2$ is a constant the proof is complete.

□

We introduce some new concepts for the next exercise. A *dissection* of a polytope $P \in \mathcal{P}^n$ is a (finite) family of polytopes $\{P_1, \ldots, P_m\}$ such that $P = P_1 \cup \ldots \cup P_m$ and $\dim(P_i \cap P_j) < n$ whenever $i \neq j$. We say that polytopes P and Q are *T-equidissectable* if there exist dissections $\{P_1, \ldots, P_m\}$ of P and $\{Q_1, \ldots, Q_m\}$ of Q with the property that each Q_i is a translate of P_i.

Exercise 11.10. *Let $\varphi\colon \mathcal{P}^n \cup \{\emptyset\} \to \mathbb{R}$ be a simple and translation-invariant valuation. Prove that if $P, Q \in \mathcal{P}^n$ are T-equidissectable, then $\varphi(P) = \varphi(Q)$.*

Solution 11.10 As commented above Theorem 11.3.3, Lemma 11.3.2, and Theorem 11.3.2 (all of them in the main book) yield that any valuation defined on $\mathcal{P}^n \cup \{\emptyset\}$ is fully additive. Hence, if $\{P_1, \ldots, P_m\}$ and $\{Q_1, \ldots, Q_m\}$ are dissections of P and Q, respectively, such that $Q_i = P_i + x_i$ for all $1 \leq i \leq m$, equality (11.3.2) (main book) and the assumption that φ is simple and translation-invariant give

$$\begin{aligned}\varphi(P) = \varphi(P_1 \cup \ldots \cup P_m) = \varphi(P_1) + \ldots + \varphi(P_m) = \\ = \varphi(P_1 + x_1) + \ldots + \varphi(P_m + x_m) = \varphi((P_1 + x_1) \cup \ldots \cup (P_m + x_m)) = \\ = \varphi(Q_1 \cup \ldots \cup Q_m) = \varphi(Q).\end{aligned}$$

This concludes the proof.

□

Recall that $\mathbf{Val}_n$ is the *vector space of valuations* $\varphi\colon CC^n \cup \{\emptyset\} \to \mathbb{R}$ which are continuous (when restricted to CC^n, of course) and invariant under proper rigid motions.

Exercise 11.11. *Let $\varphi \in \mathbf{Val}_n$ be simple. Prove that $\varphi = cV_n$ for some constant $c \in \mathbb{R}$. In other words, φ is homogeneous of degree n.*

Solution 11.11. Let $C \in CC^n$ be such that $\dim(C) = n - 1$. Since φ is simple, we must have $\varphi(tC) = 0$ for every $t > 0$. Thus, observing that $V_n(C) = 0$ and using Hadwiger's theorem we may write

$$0 = \varphi(tC) = \sum_{j=0}^{n-1} c_j V_j(tC) = \sum_{j=0}^{n-1} t^j c_j V_j(C),$$

and this equality holds for all $t > 0$. It follows that the polynomial in the right-hand side must be the zero polynomial. From that, we get that $c_0 = c_1 = \ldots = c_{n-1} = 0$, and hence $\varphi = cV_n$ for some constant $c \in \mathbb{R}$.

□

For the next exercise, we recall that a set $S \subseteq \mathbb{R}^n$ is called a *star body* if $\operatorname{seg}[0_{\mathbb{R}^n}, x] \subseteq S$ for every $x \in S$ and if the radial function $\rho_S \colon \mathbb{R}^n \setminus \{0_{\mathbb{R}^n}\} \to \mathbb{R}$ of S is positive and continuous. In what follows, we denote the family of star bodies of $\mathbb{R}^n$ by $\mathcal{S}^n$. (A comprehensive survey on star bodies and related notions is [135].) Notice that the definition of dual mixed volumes given in (11.1.1), from the main book, can be naturally extended to star bodies.

We define the *radial sum* of vectors $x, y \in \mathbb{R}^n$ as

$$x \tilde{+} y := \begin{cases} x + y, & \text{if } x \text{ and } y \text{ lie in a same ray starting at } 0_{\mathbb{R}^n} \\ 0, & \text{otherwise} \end{cases}.$$

The *radial sum of star bodies* $K, L \in \mathcal{S}^n$ is defined as

$$K \tilde{+} L := \left\{ x \tilde{+} y : x \in K \text{ and } y \in L \right\}.$$

Exercise 11.12. **(a)** *Let $K, L \in \mathcal{S}^n$. Prove that*

$$\rho_{K \tilde{+} L} = \rho_K + \rho_L,$$

and show that $K \tilde{+} L \in \mathcal{S}^n$.

(b) *Prove that the dual mixed volume is additive for $\tilde{+}$ in each argument; that is, show that*

$$\tilde{V}(K \tilde{+} L, K_2, \ldots, K_n) = \tilde{V}(K, K_2, \ldots, K_n) + \tilde{V}(L, K_2, \ldots, K_n)$$

for any $K, L, K_2, \ldots, K_n \in \mathcal{S}^n$.

(c) *Verify that equality (4.3.7) from the main book also holds for star bodies.*

Solution 11.12. **(a)** It suffices to consider $u \in \mathbb{S}^{n-1}$. In this case, $\rho_K(u)u \in K$ and $\rho_L(u)u \in L$. Hence

$$\big(\rho_K(u) + \rho_L(u)\big)u \in K \tilde{+} L,$$

and it is clear that $\alpha u \notin K \tilde{+} L$ if $\alpha > \rho_K(u) + \rho_L(u)$. It follows that

$$\rho_{K \tilde{+} L} = \rho_K + \rho_L.$$

Hence $\rho_{K \tilde{+} L}$ is continuous and positive in $\mathbb{R}^n \setminus \{0_{\mathbb{R}^n}\}$. It is clear that $K \tilde{+} L$ is compact, and that $\operatorname{seg}[0_{\mathbb{R}^n}, x] \subseteq K \tilde{+} L$ whenever $x \in K \tilde{+} L$. Thus, $K \tilde{+} L$ is a star body.

(b) This follows straightforwardly from the fact that $\rho_{K \tilde{+} L} = \rho_K + \rho_L$ and from the definition of $\tilde{V}$.

(c) We just have to repeat the proof of Proposition 4.3.2 (main book) verbatim, but not writing ρ_K as $\|\cdot\|_K$.

□

Exercise 11.13. ***(a)*** *Using Hölder's inequality, prove the* Minkowski inequality for integrals*: let X be a topological space, and let μ be a finite Borel measure on X. For any positive continuous functions $f, g\colon X \to \mathbb{R}$ and every $p \in (1, +\infty)$ it holds that*

$$\left(\int_X (f+g)^p \, d\mu\right)^{1/p} \leq \left(\int_X f^p \, d\mu\right)^{1/p} + \left(\int_X g^p \, d\mu\right)^{1/p},$$

with equality if and only if $f = \lambda g$ for some $\lambda > 0$.

(b) *Prove the* dual Brunn-Minkowski inequality*, which states that*

$$\mathrm{vol}_n(K \tilde{+} L)^{1/n} \leq \mathrm{vol}_n(K)^{1/n} + \mathrm{vol}_n(L)^{1/n},$$

for any $K, L \in \mathcal{S}^n$. Show also that equality holds if and only if K and L are dilates.

Solution 11.13. (a) We write

$$\int_X (f+g)^p \, d\mu = \int_X (f+g)(f+g)^{p-1} \, d\mu = \int_X f(f+g)^{p-1} \, d\mu + \int_X g(f+g)^{p-1} \, d\mu.$$

Now we apply Hölder's inequality in both integrals of the right-hand side for the conjugate numbers p and $p/(p-1)$ to obtain

$$\int_X f(f+g)^{p-1} \, d\mu \leq \left(\int_X f^p \, d\mu\right)^{1/p} \left(\int_X \left((f+g)^{p-1}\right)^{p/(p-1)} d\mu\right)^{(p-1)/p},$$

and

$$\int_X g(f+g)^{p-1} \, d\mu \leq \left(\int_X g^p \, d\mu\right)^{1/p} \left(\int_X \left((f+g)^{p-1}\right)^{p/(p-1)} d\mu\right)^{(p-1)/p}.$$

Thus,

$$\int_X (f+g)^p \, d\mu \leq \left(\int_X (f+g)^p \, d\mu\right)^{(p-1)/p} \left[\left(\int_X f^p \, d\mu\right)^{1/p} + \left(\int_X g^p \, d\mu\right)^{1/p}\right],$$

and the desired inequality follows immediately. Equality holds if and only if we have equality in Hölder's inequality for f and $f+g$, and this gives $f = \alpha(f+g)$ for some $\alpha > 0$. Since f and g are positive, we have that $\alpha < 1$, and $f = \lambda g$ for $\lambda = \alpha/(1-\alpha)$.

(b) Since $\rho_{K \tilde{+} L} = \rho_K + \rho_L$, we get (using (4.3.7) from the main book and item **(a)**) that

$$\mathrm{vol}_n(K \tilde{+} L)^{1/n} = \left(\frac{1}{n} \int_{\mathbb{S}^{n-1}} (\rho_K + \rho_L)^n \, d\mathcal{H}^{n-1}\right)^{1/n} \leq$$

$$\leq \left(\frac{1}{n} \int_{\mathbb{S}^{n-1}} \rho_K^n \, d\mathcal{H}^{n-1}\right)^{1/n} + \left(\frac{1}{n} \int_{\mathbb{S}^{n-1}} \rho_L^n \, d\mathcal{H}^{n-1}\right)^{1/n} = \mathrm{vol}_n(K)^{1/n} + \mathrm{vol}_n(L)^{1/n}.$$

Equality holds if and only if we have equality in Minkowski's inequality given in item **(a)**. This happens if and only if $\rho_K = \lambda \rho_L$ for some $\lambda > 0$. This is the same as $\rho_K = \rho_{\lambda L}$, which is equivalent to $K = \lambda L$.

□

In what follows, the reader must keep in mind the notations, hypotheses, and definitions of Exercise 9.13, where the *Orlicz projection bodies* were defined.

Exercise 11.14. **(a)** *Let $\phi_1 : \mathbb{R} \to [0, +\infty)$ be given as $\phi_1(t) = |t|$. Show that for any $K \in \mathcal{K}_0^n$ we have*

$$\Pi_{\phi_1} K = \frac{2}{n\mathrm{vol}_n(K)} \Pi K.$$

(b) *Verify that if $p > 1$, and letting $\phi_p(t) = |t|^p$, we have*

$$h(\Pi_{\phi_p} K, u)^p = \frac{1}{n\mathrm{vol}_n(K)} \int_{\mathbb{S}^{n-1}} |\langle u, v\rangle|^p \, dS_{p,0}(K, v). \tag{11.2}$$

Solution 11.14. **(a)** For a given $u \in \mathbb{S}^{n-1}$, let $\lambda_0 = h(\Pi_{\phi_1} K, u)$. Thus, using equality (7.2.5) from the main book, we get

$$\begin{aligned} 1 = \frac{1}{n\mathrm{vol}_n(K)} \int_{\mathbb{S}^{n-1}} \frac{|\langle u, v\rangle|}{\lambda_0 h_K(v)} h_K(v) \, dS_{n-1}(K, v) = \frac{1}{\lambda_0 n\mathrm{vol}_n(K)} \int_{\mathbb{S}^{n-1}} |\langle u, v\rangle| \, dS_{n-1}(K, v) = \\ = \frac{2}{\lambda_0 n\mathrm{vol}_n(K)} \mathrm{vol}_{n-1}\left(\mathrm{proj}_{u^\perp}(K)\right) = \frac{2}{\lambda_0 n\mathrm{vol}_n(K)} h(\Pi K, u), \end{aligned}$$

and hence

$$\lambda_0 = h(\Pi_{\phi_1} K, u) = \frac{2}{n\mathrm{vol}_n(K)} h(\Pi K, u) = h\left(\frac{2}{n\mathrm{vol}_n(K)} \Pi K, u\right).$$

Since this holds for any $u \in \mathbb{S}^{n-1}$ the desired property follows.

(b) This is immediate due to the definitions of $\Pi_{\phi_p} K$, ϕ_p and $S_{p,0}(K, \cdot)$.

□

Exercise 11.15. *Use item* **(a)** *of Exercise 11.14, the result stated in Remark 9.3, and Exercise 10.11 to give an alternative proof of the Petty projection inequality (Theorem 10.4.2 in the main book).*

Solution 11.15. From Remark 9.3, we know that

$$\frac{\mathrm{vol}_n(\Pi^\circ_{\phi_1} K)}{\mathrm{vol}_n(K)} \leq \frac{\mathrm{vol}_n(\Pi^\circ_{\phi_1} B^n)}{\mathrm{vol}_n(B^n)} = \frac{\mathrm{vol}_n(\Pi^\circ_{\phi_1} B^n)}{\kappa_n}. \tag{11.3}$$

From item **(a)** of Exercise 11.14, we have

$$\Pi^\circ_{\phi_1} K = (\Pi_{\phi_1} K)^\circ = \left(\frac{2}{n\mathrm{vol}_n(K)} \Pi K\right)^\circ = \frac{n\mathrm{vol}_n(K)}{2} \Pi^\circ K,$$

and, similarly,

$$\Pi_{\phi_1} B^n = \frac{n\kappa_n}{2} \Pi^\circ B^n.$$

Now, using Exercise 10.11, we may rewrite $\Pi^\circ B^n$ in the above as

$$\Pi_{\phi_1} B^n = \frac{n\kappa_n}{2} \Pi^\circ B^n = \frac{n\kappa_n}{2} (\kappa_{n-1} B^n)^\circ = \frac{n\kappa_n}{2\kappa_{n-1}} B^n.$$

Hence,

$$\mathrm{vol}_n(\Pi^\circ_{\phi_1} K) = \left(\frac{n\mathrm{vol}_n(K)}{2}\right)^n \mathrm{vol}_n(\Pi^\circ K), \tag{11.4}$$

and

$$\mathrm{vol}_n(\Pi^\circ_{\phi_1} B^n) = \left(\frac{n\kappa_n}{2\kappa_{n-1}}\right)^n \mathrm{vol}_n(B^n) = \left(\frac{n\kappa_n}{2\kappa_{n-1}}\right)^n \kappa_n. \tag{11.5}$$

Replacing (11.4) and (11.5) in (11.3) we obtain the Petty projection inequality.

□

For $n \in \mathbb{N}$, $p > 1$, and $u \in \mathbb{S}^{n-1}$, define

$$c_{n,p} := \int_{\mathbb{S}^{n-1}} |\langle u, v\rangle|^p \, d\mathcal{H}^{n-1}(v).$$

Since $\mathcal{H}^{n-1}|_{\mathbb{S}^{n-1}}$ is invariant under orthogonal transformations, we have that this number is a constant which does not depend on $u \in \mathbb{S}^{n-1}$. Having this in mind, and for $K \in \mathcal{K}^n_0$, we then define the L_p *projection body* of K as the convex body $\Pi_p K$ whose support function over $\mathbb{S}^{n-1}$ is given by

$$h(\Pi_p K, u)^p = \frac{1}{c_{n,p}} \int_{\mathbb{S}^{n-1}} |\langle u, v\rangle|^p \, dS_{p,0}(K, v) \tag{11.6}$$

for each $u \in \mathbb{S}^{n-1}$. Of course, we extend it to $\mathbb{R}^n$ by positive homogeneity. Our main reference for the L_p projection bodies is [180].

Exercise 11.16. **(a)** *For $p > 1$, check that*

$$h(\Pi_p K, \cdot) = \left(\frac{n\mathrm{vol}_n(K)}{c_{n,p}}\right)^{1/p} h(\Pi_{\phi_p} K, \cdot)$$

for any $K \in \mathcal{K}^n_0$, where $\Pi_{\phi_p} K$ is the Orlicz projection body whose support function is given in (11.2). Show that the function defined in (11.6) is indeed the support function of a convex body having the origin as an interior point.

(b) *Verify that $\Pi_p B^n = B^n$.*

Solution 11.16. **(a)** For any $u \in \mathbb{S}^{n-1}$, we have that

$$\int_{\mathbb{S}^{n-1}} |\langle u, v\rangle|^p \, dS_{p,0}(K, v) = n\mathrm{vol}_n(K) h(\Pi_{\phi_p} K, u)^p,$$

and hence

$$h(\Pi_p K, u)^p = \frac{1}{c_{n,p}} \int_{\mathbb{S}^{n-1}} |\langle u, v\rangle|^p \, dS_{p,0}(K, v) = \frac{n\mathrm{vol}_n(K)}{c_{n,p}} h(\Pi_{\phi_p} K, u)^p.$$

This gives the desired equality.

By Exercise 9.13, we know that $h(\Pi_{\phi_p} K, \cdot)$ is a sublinear function which is strictly positive in $\mathbb{S}^{n-1}$. Since $h(\Pi_p K, \cdot)$ is a (non-zero) scalar multiple of $h(\Pi_{\phi_p} K, \cdot)$, we get that $h(\Pi_p K, \cdot)$ is also sublinear

and strictly positive in $\mathbb{S}^{n-1}$. Therefore, $h(\Pi_p K, \cdot)$ is the support function of a convex body having the origin as an interior point.

(b) For each $u \in \mathbb{S}^{n-1}$, from the definition of $S_{p,0}(B^n, \cdot)$ and recalling that $S_{n-1}(B^n, \cdot) = \mathcal{H}^{n-1}|_{\mathbb{S}^{n-1}}$, we have

$$h(\Pi_p B^n, u)^p = \frac{1}{c_{n,p}} \int_{\mathbb{S}^{n-1}} |\langle u, v\rangle|^p \, dS_{p,0}(B^n, v) = \frac{1}{c_{n,p}} \int_{\mathbb{S}^{n-1}} |\langle u, v\rangle|^p \, h_{B^n}(v)^{1-p} \, dS_{n-1}(B^n, v) =$$
$$= \frac{1}{c_{n,p}} \int_{\mathbb{S}^{n-1}} |\langle u, v\rangle|^p \, d\mathcal{H}^{n-1}(v) = 1,$$

and hence $h(\Pi_p B^n, u) = 1 = h(B^n, u)$. The desired equality follows.

□

The objective of the next exercise is to give a proof (based on the result stated in Remark 9.3) of the so-called L_p Petty projection inequality, which can be seen as an extension of the Petty projection inequality for polar L_p projection bodies.

Exercise 11.17 (L_p Petty projection inequality)**.** *Prove that for any $p > 1$ and every $K \in \mathcal{K}_0^n$ it holds that*

$$\mathrm{vol}_n(K)^{\frac{n}{p}-1} \mathrm{vol}_n(\Pi_p^\circ K) \le \kappa_n^{\frac{n}{p}},$$

where $\Pi_p^\circ K := (\Pi_p K)^\circ$. Verify that the equality case holds if and only if K is an ellipsoid symmetric with respect to the origin.

Solution 11.17. For simplicity, write

$$c_1 := \left(\frac{n \mathrm{vol}_n(K)}{c_{n,p}} \right)^{1/p}.$$

Thus, from item **(a)** of Exercise 11.16, we have that $\Pi_p K = c_1 \Pi_{\phi_p} K$. It follows that

$$c_1 \Pi_p^\circ K = \Pi_{\phi_p}^\circ K. \tag{11.7}$$

Similarly, let

$$c_2 := \left(\frac{n \mathrm{vol}_n(B^n)}{c_{n,p}} \right)^{1/p} = \left(\frac{n\kappa_n}{c_{n,p}} \right)^{1/p}.$$

We have that $\Pi_p B^n = c_2 \Pi_{\phi_p} B^n$, and hence, using again item **(b)** of Exercise 11.16, we obtain

$$\Pi_{\phi_p} B^n = c_2 \Pi_p^\circ B^n = c_2 B^n. \tag{11.8}$$

From Remark 11.16, it follows that

$$\frac{\mathrm{vol}_n(\Pi_{\phi_p}^\circ K)}{\mathrm{vol}_n(K)} \le \frac{\mathrm{vol}_n(\Pi_{\phi_p}^\circ B^n)}{\kappa_n},$$

with equality if and only if K is an ellipsoid symmetric with respect to the origin. Now we just have to rewrite this inequality using (11.7) and (11.8). We arrive at

$$\frac{c_1^n \mathrm{vol}_n(\Pi_p^\circ K)}{\mathrm{vol}_n(K)} \leq c_2^n,$$

and from the definitions of c_1 and c_2 the desired inequality follows.

□

Exercise 11.18. *Prove that the definitions of the separation number given in Definition 11.4.2 (main book) are indeed equivalent.*

Solution 11.18. This follows immediately from the obvious fact that $||x - y||_B > 1$ if and only if $x \notin y + B$.

□

Recall that $\mathcal{W}\colon CC^n \to \mathbb{R}$ stands for the *Wills functional* defined as

$$\mathcal{W}(K) := \sum_{j=0}^{n} V_j(K),$$

for each $K \in CC^n$. The next exercise is an integral representation for the Wills functional which was taken from [132].

Exercise 11.19. *Prove that the Wills functional has the integral representation*

$$\mathcal{W}(K) = \int_{\mathbb{R}^n} e^{-\pi d(K,x)^2}\, dx,$$

where $d(K, x)$ represents the distance from a point $x \in \mathbb{R}^n$ to K, and the integration is taken with respect to the Lebesgue measure of $\mathbb{R}^n$.

Hint: Show that

$$\varphi(K) := \int_{\mathbb{R}^n} e^{-\pi d(K,x)^2}\, dx$$

defines a real-valued, continuous valuation on $CC^n \cup \{\emptyset\}$ which is invariant under proper rigid motions (extended as $\varphi(\emptyset) = 0$, of course). Then, use Hadwiger's theorem to show that φ is a combination of intrinsic volumes. To find the coefficients, compute $\varphi(\varepsilon B^n)$ for $\varepsilon > 0$. For that computation you can use equality (5.5).

Solution 11.19. We start by proving that φ is invariant under proper rigid motions. This follows easily from the facts that $d(K + y, x) = d(K, x - y)$ for any $x, y \in \mathbb{R}^n$, and $d(TK, Tx) = d(K, x)$ for any $x \in \mathbb{R}^n$ if T is an orthogonal transformation (recall that orthogonal transformations are isometries). Thus, for $y \in \mathbb{R}^n$ and $T \in \mathrm{SO}(n)$ we have

$$\varphi(y + TK) = \int_{\mathbb{R}^n} e^{-\pi d(y+TK,x)^2}\, dx = \int_{\mathbb{R}^n} e^{-\pi d(TK,x-y)^2}\, dx = \int_{\mathbb{R}^n} e^{-\pi d(K,T^{-1}(x-y))^2}\, dx.$$

Now all we have to do is to consider the change of variables $z = T^{-1}x + T^{-1}y$, which is the composition of a translation with a linear map whose determinant is 1. It follows that

$$\varphi(y + TK) = \int_{\mathbb{R}^n} e^{-\pi d(K,z)^2}\, dz = \varphi(K).$$

Moving forward, we prove that φ is real-valued, which is the same as showing that the integral converges for any $K \in CC^n$. To do that, let ρB^n be any ball containing K. We write

$$\varphi(K) = \int_{\rho B^n} e^{-\pi d(K,x)^2}\, dx + \int_{\mathbb{R}^n \setminus \rho B^n} e^{-\pi d(K,x)^2}\, dx.$$

The first integral is bounded because ρB^n is compact. To bound the other integral, we notice that since $K \subseteq \rho B^n$ we have that $d(K, x) \geq d(\rho B^n, x)$ for every $x \in \mathbb{R}^n \setminus \rho B^n$. Hence, for these points we have that $e^{-\pi d(K,x)^2} \leq e^{-\pi d(\rho B^n,x)^2}$. Moreover, for $x \in \mathbb{R}^n \setminus \rho B^n$, we clearly have that $d(\rho B^n, x) = |x| - \rho$. It follows that

$$\int_{\mathbb{R}^n \setminus \rho B^n} e^{-\pi d(K,x)^2}\, dx \leq \int_{\mathbb{R}^n \setminus \rho B^n} e^{-\pi d(\rho B^n,x)^2}\, dx = \int_{\mathbb{R}^n \setminus \rho B^n} e^{-\pi(|x|-\rho)^2}\, dx,$$

and the latter is a typical Gaussian integral, which we already know that converges (if there are doubts regarding the "extra" factor $-\rho$, simply use polar coordinates and change of variables).

Now we prove that φ is continuous in the Hausdorff metric. For that sake, let $(K_m)_{m\in\mathbb{N}}$ be a sequence in CC^n converging in the Hausdorff metric to $K \in CC^n$. It is clear that $d(K_m, x) \to d(K, x)$ for each $x \in \mathbb{R}^n$ as $m \to +\infty$. Thus, $e^{-\pi d(K_m,\cdot)^2}$ converge pointwise to $e^{-\pi d(K,\cdot)^2}$ as $m \to +\infty$. Since it is obvious that $0 < e^{-\pi d(K_m,x)^2} \leq 1$ for every $m \in \mathbb{N}$ and any $x \in \mathbb{R}^n$, we may use the dominated convergence theorem (Theorem A.2.4 in the main book) and write

$$\lim_{m\to\infty} \varphi(K_m) = \lim_{m\to\infty} \int_{\mathbb{R}^n} e^{-\pi d(K_m,x)^2}\, dx = \int_{\mathbb{R}^n} e^{-\pi d(K,x)^2}\, dx = \varphi(K),$$

and this proves that φ is continuous in the Hausdorff metric.

Finally, the fact that φ is a valuation follows straightforwardly from Exercise 2.6. Since we have proved that φ is a continuous valuation which is invariant under proper rigid motions, we can apply Hadwiger's theorem. From this theorem, we may write

$$\int_{\mathbb{R}^n} e^{-\pi d(K,x)^2}\, dx = \sum_{j=0}^{n} c_j V_j(K)$$

for coefficients $c_0, \ldots, c_n \in \mathbb{R}$. Therefore, to prove that the left-hand side equals $\mathcal{W}(K)$ all we have to do is to show that $c_0 = \ldots = c_n = 1$. To do that, we look at the case where $K = \varepsilon B^n$, for $\varepsilon > 0$. Since V_j is homogeneous of degree j, we have

$$\sum_{j=0}^{n} c_j V_j(B^n)\varepsilon^j = \int_{\mathbb{R}^n} e^{-\pi d(\varepsilon B^n,x)^2}\, dx. \tag{11.9}$$

Hence we must write the right-hand side as a polynomial on ε and compare the coefficients. To do that, we first observe that

$$d(\varepsilon B^n, x) = \begin{cases} 0, & \text{if } x \in \varepsilon B^n, \\ |x| - \varepsilon, & \text{if } x \in \mathbb{R}^n \setminus \varepsilon B^n. \end{cases}$$

Having this in mind, and also using integration by polar coordinates, change of variables, and the Gaussian integral (5.5), we compute

$$\begin{aligned}\int_{\mathbb{R}^n} e^{-\pi d(\varepsilon B^n,x)^2}\,dx &= \int_{\varepsilon B^n} e^{-\pi d(\varepsilon B^n,x)^2}\,dx + \int_{\mathbb{R}^n\setminus\varepsilon B^n} e^{-\pi d(\varepsilon B^n,x)^2}\,dx = \\ &= \int_{\varepsilon B^n} 1\,dx + \int_{\varepsilon}^{+\infty}\int_{\mathbb{S}^{n-1}} e^{-\pi(t-\varepsilon)^2} t^{n-1}\,d\mathcal{H}^{n-1}(z)dt = \\ &= \mathrm{vol}_n(\varepsilon B^n) + \mathcal{H}^{n-1}(\mathbb{S}^{n-1})\int_{\varepsilon}^{+\infty} e^{-\pi(t-\varepsilon)^2} t^{n-1}\,dt = \\ &= \varepsilon^n \mathrm{vol}_n(B^n) + n\kappa_n \int_0^{+\infty} e^{-\pi s^2}(s+\varepsilon)^{n-1}\,ds = \\ &= \varepsilon^n \mathrm{vol}_n(B^n) + n\kappa_n \sum_{j=0}^{n-1}\binom{n-1}{j}\varepsilon^j \int_0^{+\infty} e^{-\pi s^2} s^{n-j-1}\,ds = \\ &= \varepsilon^n \mathrm{vol}_n(B^n) + n\kappa_n \sum_{j=0}^{n-1}\binom{n-1}{j}\varepsilon^j \frac{\Gamma\left(\frac{n-j}{2}\right)}{2\pi^{\frac{n-j}{2}}}.\end{aligned}$$

Due to (4.3.3) in the main book we write

$$\frac{\Gamma\left(\frac{n-j}{2}\right)}{2\pi^{\frac{n-j}{2}}} = \frac{1}{\omega_{n-j}} = \frac{1}{(n-j)\kappa_{n-j}},$$

and then we rewrite the equality above as

$$\begin{aligned}\int_{\mathbb{R}^n} e^{-\pi d(\varepsilon B^n,x)^2}\,dx &= \varepsilon^n \mathrm{vol}_n(B^n) + n\kappa_n \sum_{j=0}^{n-1}\binom{n-1}{j}\frac{1}{(n-j)\kappa_{n-j}}\varepsilon^j = \\ &= \varepsilon^n \mathrm{vol}_n(B^n) + \sum_{j=0}^{n-1}\binom{n-1}{j}\frac{n}{n-j}\frac{\kappa_n}{\kappa_{n-j}}\varepsilon^j = \\ &= \varepsilon^n \mathrm{vol}_n(B^n) + \sum_{j=0}^{n-1}\binom{n}{n-j}\frac{\kappa_n}{\kappa_{n-j}}\varepsilon^j.\end{aligned}$$

Since

$$V_j(B^n) = V_{n-(n-j)}(B^n) = \binom{n}{n-j}\frac{1}{\kappa_{n-j}} W_{n-j}(B^n) = \binom{n}{n-j}\frac{\kappa_n}{\kappa_{n-j}}$$

for all $0 \le j \le n$ (and, in particular, $\mathrm{vol}_n(B^n) = V_n(B^n)$), we finally arrive at

$$\int_{\mathbb{R}^n} e^{-\pi d(\varepsilon B^n,x)^2}\,dx = \sum_{j=0}^{n} V_j(B^n)\varepsilon^n.$$

Comparing with (11.9) we get that $c_0 = \ldots = c_n = 1$, and the proof is complete.

□

Exercise 11.20. *Extend the previous exercise and prove that for any $a > 0$ the functional*

$$\varphi_a(K) = \int_{\mathbb{R}^n} e^{-ad(K,x)^2}\, dx$$

defined for $K \in CC^n$ (and extended as 0 to $K = \emptyset$) is a real-valued, continuous valuation invariant under proper rigid motions. Show that

$$\varphi_a(K) = \sum_{j=0}^{n} \left(\frac{\pi}{a}\right)^{\frac{n-j}{2}} V_j(K)$$

for any $K \in CC^n$.

Solution 11.20. To prove that φ_a is a real-valued, continuous, and invariant under proper rigid motions valuation we just use precisely the same arguments as in Exercise 11.19. From Hadwiger's theorem, it follows that

$$\varphi_a(K) = \sum_{j=0}^{n} c_j V_j(K),$$

and hence all we have to do is to compute the coefficients c_j. To compute these coefficients, we follow again the solution of Exercise 11.19. Letting $K = \varepsilon B^n$, for $\varepsilon > 0$, we have

$$\varphi_a(\varepsilon B^n) = \sum_{j=0}^{n} c_j V_j(B^n)\varepsilon^j = \sum_{j=0}^{n} c_j \binom{n}{n-j} \frac{\kappa_n}{\kappa_{n-j}} \varepsilon^j.$$

On the other hand, to write $\varphi_a(\varepsilon B^n)$ as a polynomial, we repeat the exact computation as in Exercise 11.19, but modifying the Gaussian integral (5.5): in that exercise we had $a = \pi$, and now we have to perform the calculations for arbitrary $a > 0$. For the convenience of the reader, we write down everything below.

$$\begin{aligned}
\varphi_a(\varepsilon B^n) = \int_{\mathbb{R}^n} e^{-ad(\varepsilon B^n,x)^2}\, dx &= \int_{\varepsilon B^n} e^{-ad(\varepsilon B^n,x)^2}\, dx + \int_{\mathbb{R}^n \setminus \varepsilon B^n} e^{-ad(\varepsilon B^n,x)^2}\, dx = \\
&= \int_{\varepsilon B^n} 1\, dx + \int_{\varepsilon}^{+\infty} \int_{\mathbb{S}^{n-1}} e^{-a(t-\varepsilon)^2} t^{n-1}\, d\mathcal{H}^{n-1}(z) dt = \\
&= \mathrm{vol}_n(\varepsilon B^n) + \mathcal{H}^{n-1}(\mathbb{S}^{n-1}) \int_{\varepsilon}^{+\infty} e^{-a(t-\varepsilon)^2} t^{n-1}\, dt = \\
&= \varepsilon^n \mathrm{vol}_n(B^n) + n\kappa_n \int_0^{+\infty} e^{-as^2} (s+\varepsilon)^{n-1}\, ds = \\
&= \varepsilon^n \mathrm{vol}_n(B^n) + n\kappa_n \sum_{j=0}^{n-1} \binom{n-1}{j} \varepsilon^j \int_0^{+\infty} e^{-as^2} s^{n-j-1}\, ds = \\
&= \varepsilon^n \mathrm{vol}_n(B^n) + n\kappa_n \sum_{j=0}^{n-1} \binom{n-1}{j} \frac{\Gamma\left(\frac{n-j}{2}\right)}{2a^{\frac{n-j}{2}}} \varepsilon^j.
\end{aligned}$$

Now we compare the coefficients of both polynomial expressions for $\varphi_a(\varepsilon B^n)$ that we derived. Since $V_n(B^n) = \mathrm{vol}_n(B^n)$, we get that $c_n = 1$. For $0 \le j \le n-1$, we have

$$c_j \binom{n}{n-j} \frac{\kappa_n}{\kappa_{n-j}} = n\kappa_n \binom{n-1}{j} \frac{\Gamma\left(\frac{n-j}{2}\right)}{2a^{\frac{n-j}{2}}},$$

and finally

$$c_j = n\kappa_{n-j}\binom{n}{n-j}^{-1}\binom{n-1}{j}\frac{\Gamma\left(\frac{n-j}{2}\right)}{2a^{\frac{n-j}{2}}} = (n-j)\kappa_{n-j}\frac{\Gamma\left(\frac{n-j}{2}\right)}{2a^{\frac{n-j}{2}}} = \omega_{n-j}\frac{\Gamma\left(\frac{n-j}{2}\right)}{2a^{\frac{n-j}{2}}} =$$
$$= \frac{2\pi^{\frac{n-j}{2}}}{\Gamma\left(\frac{n-j}{2}\right)}\frac{\Gamma\left(\frac{n-j}{2}\right)}{2a^{\frac{n-j}{2}}} = \left(\frac{\pi}{a}\right)^{\frac{n-j}{2}}.$$

Noticing that this equality also holds for $j = n$, we summarize as

$$\varphi_a(K) = \int_{\mathbb{R}^n} e^{-ad(K,x)^2}\,dx = \sum_{j=0}^{n}\left(\frac{\pi}{a}\right)^{\frac{n-j}{2}} V_j(K)$$

for any $K \in CC^n$. This concludes the proof.

□

Exercise 11.21. *Using Exercise 6.33 in the definition (6.8), write*

$$w(1_K) := \frac{4}{n(2\pi)^{n/2}}\lim_{\varepsilon\to 0^+}\frac{1}{\varepsilon}\left[\int_{\mathbb{R}^n} e^{-\frac{1}{2}d(\varepsilon K,x)^2} - \int_{\mathbb{R}^n} e^{-\frac{|x|^2}{2}}\right].$$

From Exercise 11.20, conclude that

$$w(1_K) = \frac{2\kappa_n}{(2\pi)^{\frac{1}{2}}\kappa_{n-1}}w(K)$$

for every $K \in CC^n$.

Solution 11.21. Let $K \in CC^n$ be arbitrarily given. From Exercise 6.33, we know that

$$[G \star (\varepsilon \cdot 1_K)](x) = e^{-\frac{1}{2}d(K,x)^2}.$$

Thus, definition (6.8) reads

$$w(1_K) = \frac{4}{n(2\pi)^{n/2}}\lim_{\varepsilon\to 0^+}\frac{1}{\varepsilon}\left[\int_{\mathbb{R}^n} G\star(\varepsilon\cdot 1_K) - \int_{\mathbb{R}^n} G\right] =$$
$$= \frac{4}{n(2\pi)^{n/2}}\lim_{\varepsilon\to 0^+}\frac{1}{\varepsilon}\left[\int_{\mathbb{R}^n} e^{-\frac{1}{2}d(K,x)^2}\,dx - \int_{\mathbb{R}^n} e^{-\frac{|x|^2}{2}}\right].$$

As a consequence, using Exercise 11.20 for $a = 1/2$ (and having also (5.6) in mind) we arrive at

$$w(1_K) = \frac{4}{n(2\pi)^{n/2}}\lim_{\varepsilon\to 0^+}\frac{1}{\varepsilon}\left[\sum_{j=0}^{n}(2\pi)^{\frac{n-j}{2}}V_j(\varepsilon K) - (2\pi)^{n/2}\right].$$

Since V_j is homogeneous of degree j, and because $V_0 = 1$ for every set of CC^n, we get that

$$\frac{1}{\varepsilon}\left[\sum_{j=0}^{n}(2\pi)^{\frac{n-j}{2}}V_j(\varepsilon K) - (2\pi)^{n/2}\right] = \frac{1}{\varepsilon}\sum_{j=1}^{n}(2\pi)^{\frac{n-j}{2}}V_j(K)\varepsilon^j = \sum_{j=1}^{n}(2\pi)^{\frac{n-j}{2}}V_j(K)\varepsilon^{j-1},$$

and hence

$$w(1_K) = \frac{4}{n(2\pi)^{n/2}} \lim_{\varepsilon \to 0^+} \sum_{j=1}^{n} (2\pi)^{\frac{n-j}{2}} V_j(K) \varepsilon^{j-1} = \frac{4}{n(2\pi)^{n/2}} (2\pi)^{\frac{n-1}{2}} V_1(K) =$$

$$= \frac{4}{n(2\pi)^{\frac{1}{2}}} V_1(K).$$

Now we use the definition of V_1 in terms of W_{n-1} and the relation between this quermassintegral and mean width given in (6.4.2) from the main book to get

$$w(1_K) = \frac{4}{n(2\pi)^{\frac{1}{2}}} V_1(K) = \frac{4}{n(2\pi)^{\frac{1}{2}}} \binom{n}{n-1} \frac{1}{\kappa_{n-1}} W_{n-1}(K) = \frac{4}{n(2\pi)^{\frac{1}{2}}} \frac{n}{\kappa_{n-1}} \frac{\kappa_n}{2} w(K) =$$

$$= \frac{2\kappa_n}{(2\pi)^{\frac{1}{2}} \kappa_{n-1}} w(K),$$

and this concludes the proof.

□

Exercise 11.22. *Let $\tilde{w}(\cdot)$ be as defined in (5.4). Show that*

$$\tilde{w}(1_K) = w(1_K)$$

for any $K \in CC^n$.

Hint: Use Exercises 11.21 and 5.30 and item **(a)** of Exercise 4.7.

Solution 11.22. In view of Exercises 11.21 and 5.30, we only need to verify that

$$\frac{2(2\pi)^{\frac{1}{2}} \kappa_n}{(n+1)\kappa_{n+1}} = \frac{2\kappa_n}{(2\pi)^{\frac{1}{2}} \kappa_{n-1}}.$$

This follows immediately applying the identity in item **(a)** of Exercise 4.7 for κ_{n+1} and κ_{n-1}.

□

Exercise 11.23. *Let $H \colon \mathbb{R}^n \to \mathbb{R}$ be defined as $H(x) := e^{-\pi |x|^2}$ for each $x \in \mathbb{R}^n$. Similarly to Exercise 6.33, show that*

$$[H \star (\varepsilon \cdot K)](x) = e^{-\pi d(\varepsilon K, x)^2}$$

for any $K \in CC^n$, $\varepsilon > 0$ and $x \in \mathbb{R}^n$. Using this equality, show that

$$\mathcal{W}(K) \geq \frac{(1-\varepsilon)^{-\frac{n}{2}(1-\varepsilon)}}{\varepsilon^{n\varepsilon}} \mathrm{vol}_n(K)^{\varepsilon} \tag{11.10}$$

for every $0 < \varepsilon < 1$ and any $K \in CC^n$.

Hint: Using Exercise 11.19, write the Wills functional of εK as the integral of $H \star (\varepsilon \cdot K)$. Then, use the version of the Prékopa-Leindler inequality given in Exercise 1.28.

Remark 11.1. The inequality (11.10) was originally proved in [17].

Solution 11.23. The proof of the first equality is exactly the same as in Exercise 6.33. From Exercise 11.19, and using the version of the Prékopa-Leindler given in Exercise 1.28 we write, for $0 < \varepsilon < 1$,

$$\begin{aligned}\mathcal{W}(\varepsilon K) = \int_{\mathbb{R}^n} e^{-\pi d(\varepsilon K, x)^2} = \int_{\mathbb{R}^n} H \star (\varepsilon \cdot 1_K) = \int_{\mathbb{R}^n} (1-\varepsilon) \cdot (1-\varepsilon)^{-1} \cdot H \star (\varepsilon \cdot 1_K) \geq \\ \geq \left(\int_{\mathbb{R}^n} (1-\varepsilon)^{-1} \cdot H \right)^{1-\varepsilon} \left(\int_{\mathbb{R}^n} 1_K \right)^{\varepsilon} = \left(\int_{\mathbb{R}^n} \left(e^{-\pi |(1-\varepsilon)x|^2} \right)^{\frac{1}{1-\varepsilon}} \right)^{1-\varepsilon} \mathrm{vol}_n(K)^{\varepsilon} = \\ = \left(\int_{\mathbb{R}^n} e^{-\pi(1-\varepsilon)|x|^2} \right)^{1-\varepsilon} \mathrm{vol}_n(K)^{\varepsilon}.\end{aligned}$$

The change of variables $y = \sqrt{\pi(1-\varepsilon)}x$ and (4.3.1) from the main book lead to

$$\int_{\mathbb{R}^n} e^{-\pi(1-\varepsilon)|x|^2} = \int_{\mathbb{R}^n} e^{-|y|^2} (\pi(1-\varepsilon))^{-n/2} \, dy = \pi^{n/2} ((\pi(1-\varepsilon))^{-n/2} = (1-\varepsilon)^{-n/2}.$$

Therefore,

$$\mathcal{W}(\varepsilon K) \geq (1-\varepsilon)^{-\frac{n(1-\varepsilon)}{2}} \mathrm{vol}_n(K)^{\varepsilon}.$$

Since this holds for every $K \in CC^n$ and any $\varepsilon > 0$, we may replace εK by K. This gives

$$\mathcal{W}(K) \geq (1-\varepsilon)^{-\frac{n(1-\varepsilon)}{2}} \mathrm{vol}_n \left(\frac{K}{\varepsilon} \right)^{\varepsilon} = \frac{(1-\varepsilon)^{-\frac{n(1-\varepsilon)}{2}}}{\varepsilon^{n\varepsilon}} \mathrm{vol}_n(K)^{\varepsilon},$$

and this concludes the proof.

□

Exercise 11.24. *Let $A, B \subseteq \mathbb{R}^n$ be convex bodies, and assume that B is centered at the origin. Let $x_1, \ldots, x_N \in A$ be such that $x_i \notin x_j + B$ whenever $i \neq j$. Prove that*

$$N \leq \frac{\mathrm{vol}_n(2A + B)}{\mathrm{vol}_n(B)}.$$

Show that the separation number $M(A, B)$ is well-defined as a natural number.

Solution 11.24. First, notice that

$$\left(x_i + \frac{B}{2} \right) \cap \left(x_j + \frac{B}{2} \right) = \emptyset$$

whenever $i \neq j$. In fact, if z is a point of this intersection, then we may write $z = x_i + u/2$ and $z = x_j + v/2$ for some $u, v \in B$. Thus,

$$x_i = x_j + \frac{v-u}{2}$$

and, since B is centered at the origin, we have that $(v-u)/2 \in B$. This contradicts the fact that $x_i \notin x_j + B$.

Using again the fact that B is centered at the origin, we have that $B/2 \subseteq B$. Thus, it is clear that we have the inclusion

$$\bigcup_{j=1}^{N} \left(x_j + \frac{B}{2} \right) \subseteq A + \frac{B}{2}.$$

From the previous discussion, the union in the left-hand side is disjoint. Therefore,

$$\mathrm{vol}_n \left(A + \frac{B}{2} \right) \geq N \mathrm{vol}_n \left(\frac{B}{2} \right),$$

and multiplying both sides by 2^n we arrive at the desired inequality.

□

For Further Reading

For presenting some literature background and in view of themes treated already in other chapters, we reduce the list of subjects proposed for further reading to *Petty's projection inequality, valuations in convexity*, and the *Wills functional*.

On Petty's Projection Inequality

The two classical inequalities which combine the volume of a convex body with that of the polar of its projection body are the *Petty projection inequality* (giving a sharp upper bound characterizing the ellipsoids) and *Zhang's projection inequality* (giving a sharp lower bound characterizing the simplices). (It seems that, for $n > 2$, the minimum for centrally symmetric convex bodies is still unknown.) More precisely, the projection inequality of Petty (which interests us here) says that, among all convex bodies of fixed volume, the ones whose polar projection bodies have maximal volume are precisely the ellipsoids, thus also strengthening and implying the classical *isoperimetric inequality*. Connecting a pair of geometric functionals whose product is invariant under affine transformations, Petty's inequality is an *affine isoperimetric inequality* (like Zhang's inequality, too). Basic references discussing Petty's projection inequality are Lutwak's elegant survey [178] (see Sections 3 and 5 there), Gardner's monograph [97] (cf. Section 9.2, in particular Theorem 9.2.9), and Schneider's book [228] (see Section 10.9). In the main book [29] Petty's projection inequality is treated in subsection 6.7.2, Theorem 10.4.2, subsection 10.6.3, and Exercises 11.15, 11.16, and 11.17. Also the inequality of Zhang (Theorem 10.4.3) is proved there, with discussion of the equality case (Remark 10.4.6).

We continue with some *research contributions to Petty's projection inequality*. Lutwak et al. extend in [180] Petty's inequality (as well as its equality case) to the L_p *setting*. Haberl and Schuster [124] establish sharp L_p affine isoperimetric inequalities for all L_p *projection bodies*. These new inequalities strengthen and imply the L_p Petty projection inequality from [180]. The authors show in [124] that the new inequalities for the asymmetric L_p *projection bodies* are the strongest inequalities within the correspondingly considered family of inequalities. Other generalizations of the L_p *Petty projection inequality* are derived by Fang et al. [90].

In [181], Lutwak et al. derive the *Orlicz-Petty projection inequality*, with origin-symmetric ellipsoids reaching the maximum. That they are even characterized by this property is shown only for the strictly convex case. Böröczky [60] confirms this for the general case, getting also *related stability results*. Haberl [123] obtains *complex extensions* of Petty's projection inequality. This is continued by Wang and Liu [255], who also successfully extend Haberl's results (from [123]) to the *complex* L_p *setting*.

Inspired by ideas of Schneider, leading to so-called higher-order difference bodies and a higher-order extension of the Rogers-Shephard inequality, Haddad et al. [129] apply these ideas also to projection bodies and derive an associated generalization of Petty's projection inequality.

Berg and Schuster [45] as well as E. Milman and Yehudayoff [197] study affine isoperimetric inequalities referring to affine invariant analogues of the classical intrinsic volumes, introduced by Lutwak and called k-th *affine quermassintegrals*. In [45], Lutwak's volume inequalities for polar projection bodies of all orders are generalized to polarizations of *Minkowski valuations* (see also the next paragraph) generated by even, zonal measures on the Euclidean unit sphere. One fundament of this is given by analogues of mixed projection bodies for such Minkowski valuations and a conjecture on affine quermassintegrals. Namely, Lutwak's conjecture that for all convex bodies of fixed volume the k-th affine quermassintegral is minimized precisely for the ellipsoids is confirmed in [197]. This extends the known results for the borderline cases $k = 1$ and $k = n - 1$ (Blaschke-Santaló inequality and *Petty projection inequality*) to all cases $k = 1, \ldots, n - 1$ in a single and unified way.

The authors of [168] use gnomonic projection and a suitable Poincaré model to define first the spherical and then the hyperbolic projection body. Further on, related analogues of the Steiner symmetrization are established, which yield finally *spherical and hyperbolic analogues of Petty's projection inequality*.

Valuations in Convexity

Valuations in convexity are additive functionals on the family of compact, convex subsets of a finite-dimensional real vector space. Well-known examples are *Lebesgue measure* and *Euler characteristic*. Further examples of valuations on finite unions of convex bodies are *surface area* and *mean width*. A historical starting point was the *dissection theory of polytopes* (in particular, Hilbert's third problem), and Blaschke obtained in the 1930s *a first classification of real-valued, SL(n) invariant valuations*. Hadwiger continued with his famous *classification of continuous and rigid motion invariant valuations*. For all these (earlier) steps of development we refer to Hadwiger's classical book [131] and the two essential surveys [193] and [190] of McMullen and Schneider.

In her much more recent survey [170], Ludwig collects further interesting results, mainly on *affine invariant, equi-affine invariant, and SL(n)-covariant valuations on convex bodies*. This is also investigated in Chapter 6 of Schneider's monograph [228] and his survey [229], nicely discussing basic notions, geometric methods, and also interesting relations to stochastic geometry. Also in further recent and summarizing publications basic notions and methods as well as relations to neighboring disciplines (e.g., integral and stochastic geometry) are presented. See, for example, the proceedings [146] edited by Jensen and Kiderlen, and Alesker's important books [7] as well as [6] (the first one jointly with Fu).

For *convex polytopes* we refer to the survey [191] of McMullen on *valuations and the polytope algebra* and the useful notes [147] of Jochemko on *lattice polytopes*. Regarding lattice polytopes, also the paper [62] is essential. Namely, the authors present a nice survey on valuations on lattice polytopes, in which Ehrhart polynomials and classification results for Minkowskian, vector, and tensor valuations are discussed. In their expository paper [48], Bernig and Hug give an interesting overview of new algebraic structures on the space of translation-invariant, smooth tensor valuations, and related kinematic formulas (see also their research paper [49] and further papers cited below).

We finish this list of book- and survey-like publications with the very recent and essential survey [172] of Ludwig, updating almost all aspects of valuations in convexity, showing the stormy development around this notion, and discussing also various more general settings (e.g., *valuations on function spaces*). In our book [29], valuations are mainly discussed in subsection 11.7.2 and in Chapter 10, where several papers treating valuations are mentioned and where it is also discussed how they are related to notions like *projection bodies, intersection bodies*, and *centroid bodies*.

We continue with a few *recent research results on valuations*. In a series of papers, Alesker develops a theory of *valuations on manifolds*. Related results obtained by him and other mathematicians are

summarized in his survey [4]; see also his further paper [5]. Studying *local Minkowski tensors* (which are valuations on the space of convex bodies with values in a space of tensor measures), Hug and Schneider [141] generalize notions like intrinsic volumes and curvature measures of convex bodies. In [49] (already mentioned above), Bernig and Hug derive kinematic formulae for continuous and translation-invariant tensor valuations on convex bodies and extend some of the algebraic notions of valuation theory from the scalar-valued to the tensor-valued setting. Ludwig and Reitzner [173] give a classification of $\mathrm{SL}(n)$ invariant valuations on the space of convex polytopes and of polytopes which contain the origin.

In [79] Colesanti gives an up-to-date representation of the geometric theory of log-concave functions, discussing the main points of this theory and its applications. Topics discussed are, e.g., *functional versions of area measures and mixed volumes, and valuations on such functions*. Real-valued, translation-invariant valuations in relation to *covariograms* are studied by Averkov and Bianchi [27].

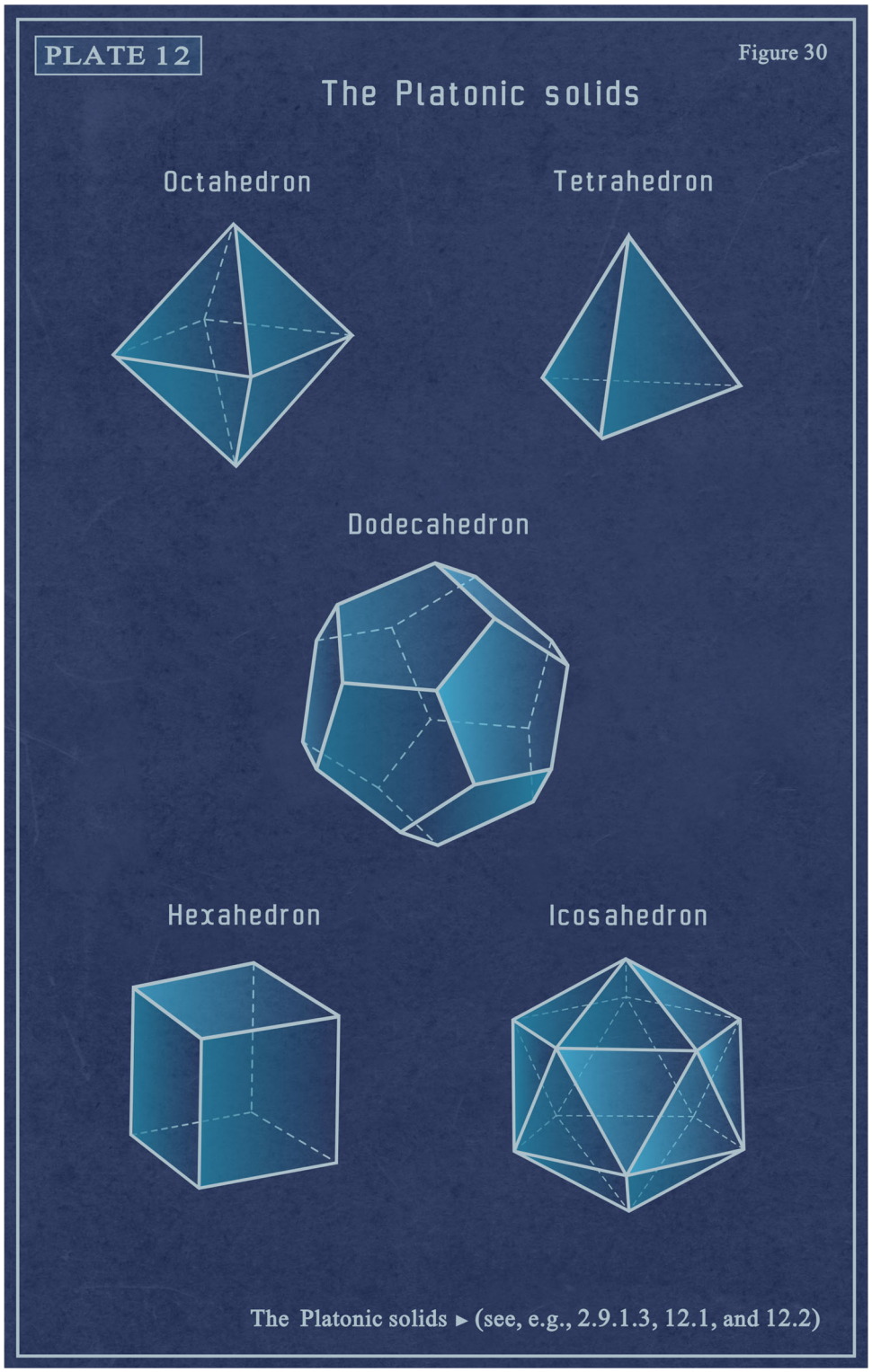
We finish this representation of results by mentioning a recent series of papers due to Colesanti, Ludwig, and Mussnig referring to valuations and related topics on convex functions under the common title "The Hadwiger theorem on convex functions." They are completely cited in the last, fourth part which is [82]. For example, in this part new proofs of Hadwiger's theorem for smooth and for continuous valuations on convex functions as well as the valuation-related theorem of Klain-Schneider on convex functions are derived.

On the Wills Functional

The *Wills functional* $W(K)$ of a convex body K, defined as the sum of the intrinsic volumes of K, has many interesting properties and applications. For example, it occurs in the sixth note to Section 5.3 of Schneider's monograph [228] and in the Exercises 6.30 , 11.19, and 11.20 of the main book [29]. It is nicely discussed in the exposition [260] of Wills, and in [259] Wills posed the interesting conjecture that $G(K) \leq W(K)$, holding for n-dimensional convex bodies K. (Here $G(K)$ denotes the number of integer points which are, regarding an orthonormal basis, contained in K.) For $n = 3$ this is true (see [204]), and for $n \geq 441$ Hadwiger found counterexamples in [133]. Further informations about this nice problem are collected at the end of Section 3.1 of the handbook article [113] of Gritzmann and Wills.

Vitale and Wang [251] obtain a *Poisson analogue of the Wills functional for Gaussian processes*. Various related papers are cited, among them [250]. In [17], the authors use the fact that $W(K)$ can be represented as the integral of a log-concave function (being also the Asplund product of two other log-concave functions) to derive new properties of $W(K)$. For example, they obtain upper bounds on $W(K)$ in terms of the volume of K, and additionally they get *inequalities of Brunn-Minkowski and Rogers-Shephard type* for that functional. It turns out that, among the origin-symmetric convex bodies in John position, the maximum of $W(K)$ is reached by the cube of edge-length 2. Using a *distance notion based on intrinsic volumes* of convex bodies, and another one based on the Wills functional, Besau et al. [50] derive estimates for an *asymptotically best approximation of Euclidean balls by polytopes* and use the Wills functional for this.

Alonso-Gutiérrez and Brazitikos [10] present a thorough study of lower dimensional sections or projections of convex bodies in special *positions* (like those of John or minimal surface area) that maximize or minimize the volume, the mean width, and the Wills functional. In this way they generalize several former results restricted to hyperplane sections and special types of convex bodies. Zhou and Li [263] dualize (in the sense of the *dual Brunn-Minkowski theory*) the definition of the Wills functional considering it for star bodies and dual intrinsic volumes.


PLATE 12
Figure 30
The Platonic solids
Octahedron
Tetrahedron
Dodecahedron
Hexahedron
Icosahedron
The Platonic solids ▸ (see, e.g., 2.9.1.3, 12.1, and 12.2)

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